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Data Quality Management SDK

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Developer Guide

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1 Data Quality Management SDK

The Data Quality Management SDK provides a framework and APIs that enable you to write applications that use SAP Data Quality technology.

Include SAP Data Quality technology in your applications. Include parsing, standardization, correction, and data matching. Use the SDK to create applications that target specific Data Quality functionalities with an in-process integration.

Relationship to Data Services

Data Quality Management SDK provides functionality similar to SAP Data Services, but deploys that technology as an API.

The Data Quality Management SDK provides a lighter footprint than Data Services. The SDK product does not require server components (either from SAP or a third party) or a user interface to access the Data Quality functionality.

Many customers choose to use this product in conjunction with Data Services. For example, users configure transform options in Data Services and create a configuration XML file to use with the SDK.

To create a configuration XML file, configure the transform in Data Services, right-click the transform icon in Data Services workspace, and select *Save settings for DQM SDK*. For more information about using Data Services Designer, see the [SAP Data Services documentation](#).

When you use Data Services as a configuration tool for the Data Quality Management SDK, Data Services does not support the creation of a change log for changes to the configuration. You can employ the Data Services central repository to manage changes to the Data Quality transforms, but Data Services does not create a change log. Instead, implement a change log within a custom application that you create using the SDK.

For more information about the central repository concept in Data Services, see the *Designer Guide*.

EmDQ

In many aspects of this product, we often use the letters “emdq” or “EmDQ” in naming conventions. You can see this convention in namespaces, folder names, and file names. Because the Data Quality Management SDK is an embedded, in-line processing, data quality solution, you might think of the letters “emdq” meaning Embedded Data Quality.

2 Install Data Quality Management SDK

Installing Data Quality Management SDK is as simple as running a self-extracting executable.

Obtain the installer from the SAP Software Download Center. Make sure that you choose the installer that is appropriate for your platform and version.

If you install the SDK to a network location, install only one Data Quality Management SDK per directory based on the platform type. If you plan to install more than one platform version of the software to a network, create a separate directory for each platform.

❖ Example

For example, you plan to install both an AIX and a Solaris SDK on your network. Create two directories: One for AIX, and one for Solaris.

[Additional requirements for Red Hat Linux \[page 18\]](#)

Before installing on Red Hat Linux, you must ensure all the required libraries are installed.

[Upgrade install \[page 19\]](#)

To upgrade Data Quality Management SDK, install a newer version to the same machine as an existing installation, but to a different directory.

[Installing the SDK on Windows \[page 19\]](#)

Install SAP Data Quality Management SDK on Windows by downloading and running the executable file.

[Installing the SDK on Unix \[page 20\]](#)

Install SAP Data Quality Management SDK on Unix by downloading and running the executable file.

2.1 Additional requirements for Red Hat Linux

Before installing on Red Hat Linux, you must ensure all the required libraries are installed.

If you are using Red Hat Linux version 8, use the *Yum* software installation tool to ensure that the following are installed:

- `yum install libnsl.i686`
- `yum install libxcrypt.i686`
- `yum install xz-libs.i686`
- `yum install libnsl`

Parent topic: [Install Data Quality Management SDK \[page 18\]](#)

Related Information

[Upgrade install \[page 19\]](#)

[Installing the SDK on Windows \[page 19\]](#)

[Installing the SDK on Unix \[page 20\]](#)

2.2 Upgrade install

To upgrade Data Quality Management SDK, install a newer version to the same machine as an existing installation, but to a different directory.

Installing a newer version to a different directory prevents files from the previous version from being overwritten.

Data Quality Management SDK provides a method to the TransformFactory class called `UpgradeTransformSettings()`. This TransformFactory class makes the transform settings built from the previous version of this product compatible with the newer version.

Parent topic: [Install Data Quality Management SDK \[page 18\]](#)

Related Information

[Additional requirements for Red Hat Linux \[page 18\]](#)

[Installing the SDK on Windows \[page 19\]](#)

[Installing the SDK on Unix \[page 20\]](#)

2.3 Installing the SDK on Windows

Install SAP Data Quality Management SDK on Windows by downloading and running the executable file.

Download the installation package that is applicable to your Windows platform from the SAP Software Download Center.

1. Run the executable file.
The *Welcome* dialog box appears.

→ Tip

If the installer does not start by running the executable, you can begin the installation routine by running `setup.exe`, which is in the installation package.

2. Click [Next](#).
The [License Agreement](#) dialog box opens.
3. Read the license agreement, indicate that you accept the license agreement, and click [Next](#).
The [Specify the destination folder](#) dialog box opens.
4. Select an installation folder for this product and click [Next](#).
The [Start Installation](#) dialog box opens.
5. Click [Next](#).
The installation routine extracts and places the files for this product in the folder you specified. After the install routine finishes installing the software, it displays a final dialog box.
6. Click [Finish](#) to close the installer.

After successfully installing the software, install the addressing directories and cleansing package. Ensure that the addressing directories and cleansing packages are installed before you use the following features in the SDK:

- Address correction
- Data Cleanse
- Sample applications

Task overview: [Install Data Quality Management SDK \[page 18\]](#)

Related Information

[Additional requirements for Red Hat Linux \[page 18\]](#)

[Upgrade install \[page 19\]](#)

[Installing the SDK on Unix \[page 20\]](#)

2.4 Installing the SDK on Unix

Install SAP Data Quality Management SDK on Unix by downloading and running the executable file.

Download the installation package that is applicable to your Unix platform from the SAP Software Download Center.

1. Unpack the *.tgz file.
The program copies the required installation files to your system.
2. Run `setup.sh`.
The [Destination Path](#) dialog box opens.
3. Type a destination path for the installation.

i Note

Choose a different path than the default, which is the current working directory.

The [Welcome](#) dialog box opens.

4. Press `Enter` to close the [Welcome](#) dialog box
The [License Agreement](#) dialog box opens.
5. Press `Enter` to accept the license agreement.
The installation routine places the files for this product in the path you specified.

The files for the SDK are now installed.

Also install the addressing directories and cleansing package. Install the addressing directories and cleansing package before you use the following features in the SDK:

- Address correction
- Data Cleanse
- Sample applications

i Note

If you are running on Solaris 11, add the following line to the environment:
`LD_PRELOAD_64=<install_path>/bin/libnanosleep.so` where `<install_path>` is the actual installation destination.

Task overview: [Install Data Quality Management SDK \[page 18\]](#)

Related Information

[Additional requirements for Red Hat Linux \[page 18\]](#)

[Upgrade install \[page 19\]](#)

[Installing the SDK on Windows \[page 19\]](#)

3 Directory data

The USA Regulatory Address Cleanse transform and the Global Address Cleanse transform and engines rely on several directories to cleanse your data.

If you prepare addresses for mailing with postal discounts, you use the directories to standardize addresses and assign postal codes. The applicable postal authority requires that you use the most up-to-date directories, which requires you to update your files periodically with new directories. When you purchase the directories, we keep you informed about current directory dates, and alert you when new directories are ready to download.

The directories contain specific types of information related to the processes you perform on your addresses. For example, the United States Postal Service has directories with address information about every address in the United States and its territories. And the software uses many other directories issued by other postal authorities such as Canada, New Zealand, and Japan.

i Note

Address cleanse transforms support many countries, however, the Global Address Cleanse transform supports only some countries with official directories.

Besides the basic address directories, there are many specialized directories. For example, the USA Regulatory Address Cleanse transform uses the following specialized directories:

- DPV
- Early Warning System (EWS)
- eLOT
- GeoCensus
- LACSLink
- NCOALink
- RDI
- SuiteLink
- Z4Change

The specialized directories help extend US address cleansing beyond the basic parsing and standardizing.

[Install address directories \[page 23\]](#)

All information about installing address directories is contained in the Address and Geocoding Directories documentation.

[Directory file locations \[page 23\]](#)

Include the locations of the directory files in your job setup.

[Installing SAP Download Manager \[page 24\]](#)

Install and configure SAP Download Manager before you download the directory files.

3.1 Install address directories

All information about installing address directories is contained in the Address and Geocoding Directories documentation.

Open the Address and Geocoding Directories documentation page in the User Assistance Customer Portal at https://help.sap.com/viewer/p/ADDRESSING_DIRECTORIES. Find the applicable directory installation and upgrade information instructions under “Installation and Upgrade.”

Install and configure the SAP Download Manager first. Then use the SAP Download Manager to download and extract the directory files.

Parent topic: [Directory data \[page 22\]](#)

Related Information

[Directory file locations \[page 23\]](#)

[Installing SAP Download Manager \[page 24\]](#)

3.2 Directory file locations

Include the locations of the directory files in your job setup.

We recommend that you install the directory files in the `reference_data` folder for each transform created when you install Data Quality Management SDK. By default, the software looks for directories in the following location:

- Windows: `SAPBusinessObjects\Data Quality Mgmt SDK\windows_32\DataQuality\reference_data`
- UNIX: `<DQM SDK install location>/DataQuality/reference_data`

If you place your directories in a different location, change the individual reference file option values in the transform XML files.

⚠ Caution

Incompatible or out-of-date directories can render the software unusable. Ensure that you install directories for each country engine as follows:

- USA Regulatory Address Cleanse transform: Weekly, monthly, or bimonthly directory updates.
- Australia and Canada engines: Monthly directory updates.
- Global Address engine: Quarterly updates.

Ensure that the directories are compatible with the current software.

Parent topic: [Directory data \[page 22\]](#)

Related Information

[Install address directories \[page 23\]](#)

[Installing SAP Download Manager \[page 24\]](#)

3.3 Installing SAP Download Manager

Install and configure SAP Download Manager before you download the directory files.

1. Access the SAP Support Portal at <https://support.sap.com/en/my-support/software-downloads.html>.

You must have software download authorization to access the download.

2. Scroll down to the SAP Download Manager section and select [Download now](#).

The install package is in a Zip file.

3. In the [Downloads](#) bar, select to save the Zip file to a local folder.
4. Extract the Zip file.

The Zip file contains a folder named `dlmanager <version>` that contains the file named `DLManager.jar`. The JAR file is a runnable file.

5. Run the `DLManager.jar` file in one of the following ways:
 - a. Open the `dlmanager <version>` folder in the local directory to which you extracted the installation package. Double-click the `DLManager.jar` file to run the Configuration Assistant.
 - b. If the option to double-click and run the `DLManager.jar` file is not available, open a shell and change directories to the location of the `DLManager.jar` file. Enter the following command to start the Configuration Assistant:

```
java -classpath DLManager.jar dlmanager.Application
```

6. Follow the steps in the Configuration Assistant to install and set up the SAP Download Manager.

Access the online help document at https://support.sap.com/content/dam/support/en_us/library/ssp/my-support/help-for-sap-support-applications/online_help-download_manager.html. The online help document details the installation steps if you need further assistance.

Download and extract the directory files using the SAP Download Manager.

Task overview: [Directory data \[page 22\]](#)

Related Information

[Install address directories \[page 23\]](#)

[Directory file locations \[page 23\]](#)

4 Get started with samples

The best way to get started with SAP Data Quality Management SDK is to examine, build, and run the provided sample programs.

i Note

Any software coding and/or code snippets are examples. They are not for productive use. The example code is only intended to better explain and visualize the syntax and phrasing rules. SAP does not warrant the correctness and completeness of the example code. SAP shall not be liable for errors or damages caused by the use of example code.

When you install SAP Data Quality Management SDK, the installer places folders containing sample program files in each supported operating system and language (for example, `<install_path>\windows_32\Java\samples`). The `\samples` folder contains files for integrating the public API and running Data Quality transforms built using this SDK.

[Sample program files \[page 26\]](#)

The sample program files contain the elements to demonstrate the Data Quality transforms.

[About environment settings for compiler paths \[page 28\]](#)

All the build scripts included in the samples folder assume that you are running from a command prompt and that you have your compiler paths set up correctly.

[Building samples \[page 29\]](#)

Use the provided scripts to build sample application programs.

[About running sample program files \[page 29\]](#)

All of the sample run scripts included with the software assume you are running from a command prompt.

[Running samples \[page 30\]](#)

After you build the samples, run the sample transforms from a command prompt.

4.1 Sample program files

The sample program files contain the elements to demonstrate the Data Quality transforms.

Each folder and subfolder contain the source code, build scripts, or run scripts that the SDK samples use to demonstrate the Data Quality transforms.

Content of each sample program folder

Folder	Description
cpp\	Contains files that integrate this SDK into a C++ environment. Subfolders include the following: inc\ Contains the SDK public headers that you include in your code. lib\ Contains the SDK public libraries to link against for C++ code. Note Do not link the certifiedreportgenerator.lib library in the lib\ subfolder unless you intend to use the certified report generator. samples\ Contains C++ sample drivers used by several Data Quality transforms.
dotNet\	Contains files to integrate the SDK into a .Net environment. Subfolders include the following: bin\ Contains the SDK public library to link against for .Net code. samples\ Contains .Net sample drivers for several Data Quality transforms.
Java\	Contains files to integrate the SDK into a Java environment. Subfolders include the following: bin\ Contains the SDK public library to link against for Java code. samples\ Contains Java sample drivers for several Data Quality transforms.
bin\	Contains many of the shared libraries and binaries that run the Data Quality transforms. Ensure that you enter the bin\ directory in your PATH and shared library load environment variables. Including the bin\ directory in the environment variables ensures that the software properly loads the shared objects and other required files. The run scripts for the included samples set these variables for you.
DataQuality\	Contains many files required for the Data Quality transforms to run.

Folder	Description
redist\	Contains the MSVC VS 2005 redistributable package that you install to run the windows executables.
xsd\	Contains all Data Quality transform configuration file XSD files.

i Note
 The `xsi:schemaLocation` element in your configuration xml files has to be able to locate these XSD files.

Parent topic: [Get started with samples \[page 26\]](#)

Related Information

[About environment settings for compiler paths \[page 28\]](#)

[Building samples \[page 29\]](#)

[About running sample program files \[page 29\]](#)

[Running samples \[page 30\]](#)

4.2 About environment settings for compiler paths

All the build scripts included in the samples folder assume that you are running from a command prompt and that you have your compiler paths set up correctly.

On Windows, for C++ and .Net builds, the `devenv` and `dumpbin` executable from the VisualStudio 2005 SP1 or greater are in your PATH environment variable. Likewise on Unix platforms, the appropriate compiler for that platform should be available.

For Java projects, set JAVA_HOME to a compatible JDK location so that the software can find the `javac` executable.

For .Net projects, ensure that you add the `<install_path>\<platform>\bin` folder to your PATH environment variable before you launch Visual Studio. The samples require these libraries to create an instance of an SDK transform object.

Parent topic: [Get started with samples \[page 26\]](#)

Related Information

[Sample program files \[page 26\]](#)

[Building samples \[page 29\]](#)

[About running sample program files \[page 29\]](#)

[Running samples \[page 30\]](#)

4.3 Building samples

Use the provided scripts to build sample application programs.

1. From a command prompt, open the samples folder in the applicable language folder.
2. Run the `build.bat` file to build all Windows sample programs.
3. Run the `build.sh` file to build all Unix sample programs.

Example

To build all C++ samples on Windows 32 bit: Open a command prompt and type
`<install_path>\windows_32\cpp\samples\build.bat`.

To build a specific sample, open the transform subfolder and run the `build.bat` for Windows or `build.sh` for Unix script within that subdirectory.

Task overview: [Get started with samples \[page 26\]](#)

Related Information

[Sample program files \[page 26\]](#)

[About environment settings for compiler paths \[page 28\]](#)

[About running sample program files \[page 29\]](#)

[Running samples \[page 30\]](#)

4.4 About running sample program files

All of the sample run scripts included with the software assume you are running from a command prompt.

For Java projects, set `JAVA_HOME` to a compatible JDK or JRE location so that the script can find the Java executable.

Each run script takes at least the configuration XML file as the first command line argument. Multi-record transforms such as Match also require you to list the input .txt file as the second argument to the run script.

Parent topic: [Get started with samples \[page 26\]](#)

Related Information

[Sample program files \[page 26\]](#)

[About environment settings for compiler paths \[page 28\]](#)

[Building samples \[page 29\]](#)

[Running samples \[page 30\]](#)

4.5 Running samples

After you build the samples, run the sample transforms from a command prompt.

Before you can run the samples, you must first install the address directory reference files and cleansing package.

1. In a command prompt, open the sample transform directory location.
2. Include the run script for your platform with the applicable command line argument. Windows, `run.bat`. Unix, `run.sh`.

The scripts set up your environment.

3. Run the sample program. Use the following examples as a guide.

❖ Example

To run the Global Address Cleanse C++ sample on Windows 32 bit:

1. Type the following in a command prompt and press `Enter`:
`<install_path>\windows_32\cpp\samples\gac.`
2. Type `run.bat EmDQ_GlobalAddressCleanse.xml` and press `Enter`.

To run the Match C++ sample on Windows 32 bit:

1. Type the following in a command prompt and press `Enter`:
`<install_path>\windows_32\cpp\cpp\samples\match.`
2. Type `run.bat EmDQ_NameAddressMatch.xml MatchNameAddrUSSingleSource.txt` and press `Enter`.

Task overview: [Get started with samples \[page 26\]](#)

Related Information

[Sample program files \[page 26\]](#)

[About environment settings for compiler paths \[page 28\]](#)

[Building samples \[page 29\]](#)

[About running sample program files \[page 29\]](#)

5 Message types

Each transform has message types that you can include in log files.

The following table contains the message types for the transforms in SAP Data Quality Management SDK.

Message types for transforms

Type	Description
MESSAGETYPE_ERROR	Error messages
MESSAGETYPE_WARNING	Warning messages
MESSAGETYPE_INFO	Information messages
MESSAGETYPE_TRACE	Tracing messages

6 API reference for C++

Each element for building Data Quality transforms in C++ contains specific syntax and elements to use in the syntax.

Use the following define for the C++ namespace: `EmDQ`.

Transform logging is not enabled by default. Enable logging to use the logging capability of the methods in this API. We recommend that you set a log handler to see specific warnings or error messages. Each applicable method contains details about the information included in log files.

[ToLatin1 \[page 34\]](#)

Use the `ToLatin1` method to convert Unicode data to standard Latin1 data.

[Class `CertifiedReportGenerator` \[page 35\]](#)

Class `CertifiedReportGenerator` enables certified reports that are required by specific postal authorities.

[Class `DataRecordSchema` \[page 36\]](#)

Class `DataRecordSchema` defines the schema of a data record.

[Class `Date` \[page 42\]](#)

Class `Date` holds date values for record fields.

[Class `DateTime` \[page 49\]](#)

Class `DateTime` holds date and time values for record fields.

[Class `EmdqException` \[page 51\]](#)

Class `EmdqException` is the exception class thrown by all public interfaces of this product. This class is required for processing.

[Class `InputDataRecord` \[page 53\]](#)

Class `InputDataRecord` is the main interface to the Input Data Record functionality.

[Class `MessageHandler` \[page 61\]](#)

Class `MessageHandler` is a callback class to handle messages from a transform.

[Class `MultiRecordTransform` \[page 62\]](#)

Class `MultiRecordTransform` processes multiple records.

[Class `MultiRecordTransformHelper` \[page 73\]](#)

Class `MultiRecordTransformHelper` is the helper class for the `MultiRecordTransform` class.

[Class `OutputDataRecord` \[page 77\]](#)

Class `OutputDataRecord` is the main interface to the Output Data Record functionality.

[Class `ProgressHandler` \[page 88\]](#)

Class `ProgressHandler` shows the progress of a `MultiRecordTransform` and allows the handler to end processing.

[Class `RecordTransform` \[page 90\]](#)

Class `RecordTransform` is the record processing Transform class for processing single records.

[Class `RecordTransformHelper` \[page 99\]](#)

Class `RecordTransformHelper` is the helper class for the `RecordTransform` class.

[Class `StatisticsHandler` \[page 101\]](#)

Class `StatisticsHandler` is a callback class to handle statistics records.

[Class `StatisticsSchema` \[page 104\]](#)

Class `StatisticsSchema` defines the layout of a statistics table.

[Class `Time` \[page 109\]](#)

Class `Time` holds a time value for a record field.

[Class `TransformFactory` \[page 116\]](#)

Class `TransformFactory` contains the methods for creating Transforms.

6.1 ToLatin1

Use the `ToLatin1` method to convert Unicode data to standard Latin1 data.

Syntax

```
char* ToLatin1(const uint16_t* str, char* dstBuf, int32_t dstBufLength, char
invalidCharReplacement = '?');
```

SAP Data Quality Management SDK defines the `ToLatin1` method in the file `utility.h` for C++ development.

Use this method to convert the UCS2 characters defined in the [str](#) parameter to Latin1 characters. The method further copies the Latin1 characters into the passed-in buffer. Data Quality Management SDK always adds a Null terminator to allow room in the buffer for the entire string plus a Null terminator.

Return value

Returns the buffer in the [dstBuf](#) parameter.

Where

Parameter	Description
str [IN]	String to convert to Latin1.

Parameter	Description
<i>dstBuf [OUT]</i>	Buffer to receive Latin1 characters and a Null terminator. i Note The SDK adds a Null terminator by default. Ensure that there is room in the buffer for the entire string plus the Null terminator.
<i>dstBufLength [IN]</i>	Buffer size in bytes.
<i>invalidCharReplacement [IN]</i>	Character that replaces the invalid Latin1 characters. Enter a 0 to drop invalid characters. The SDK considers a UCS2 value greater than 255 to be an invalid Latin 1 character. If you set the <i>invalidCharReplacement</i> value to 0, then the SDK deletes the invalid Latin1 character. The SDK does not convert the locale.

6.2 Class CertifiedReportGenerator

Class `CertifiedReportGenerator` enables certified reports that are required by specific postal authorities.

Class `CertifiedReportGenerator` is an implementation of the public `StatisticsEventHandler` interface. The interface generates the certified mailing Statement of Address Accuracy (SERP), Address Matching Processing Summary (AMAS), and Coding Accuracy Support System (CASS) 3553 report.

Class `CertifiedReportGenerator` contains the following method for C++ development:

```
void SetReportFile(const char* fileName, REPORT_TYPE report);
```

[SetReportFile \[page 35\]](#)

Use the `SetReportFile` method to set the path and file name to use for the specified report type.

6.2.1 SetReportFile

Use the `SetReportFile` method to set the path and file name to use for the specified report type.

Syntax

```
void SetReportFile(const char* fileName, REPORT_TYPE report);
```

`SetReportFile` is a part of the `CertifiedReportGenerator` class for C++ development.

Return

This method implements the [Statistics Handler](#) interface, and generates the specified report.

Where

Parameter	Description
fileName [IN]	Valid path and file name for the generated report. The path must exist. If the path does not exist, the method issues an error. If the specified file exists, the method overwrites the existing version.
report [IN]	Name of the report. Valid values are: • REPORT_3553 • REPORT_AMAS • REPORT_SERP Call this method for each report type to generate. Call this method before you use any transform.

Related Information

[Class CertifiedReportGenerator](#) [page 35]

6.3 Class DataRecordSchema

Class DataRecordSchema defines the schema of a data record.

Methods

Class DataRecordSchema has the following methods for C++ development:

- `int GetFieldCount();`
- `int GetFieldIndex(const uint16_t* fieldName);`

- `int GetFieldIndex(const char* fieldName);`
- `int GetFieldLength(int fieldIndex);`
- `const uint16_t* GetFieldName(int fieldIndex);`
- `bool GetFieldName(int fieldIndex, char* buffer, int bufferSize, char unicodeReplacement = '?');`
- `DATATYPE GetDataType(int fieldIndex, bool& status);`

[GetFieldCount \[page 37\]](#)

Use the `GetFieldCount` method to return the number of fields defined in the data record.

[GetFieldIndex \[page 38\]](#)

Use the `GetFieldIndex` method to return the field index for the field specified in the *fieldName* parameter.

[GetFieldLength \[page 38\]](#)

Use the `GetFieldLength` method to get the length of the field defined by the *fieldIndex* parameter.

[GetFieldName Constant \[page 39\]](#)

Use the `GetFieldName` method to get the name of the field specified by the *fieldIndex* parameter.

[GetFieldName Boolean \[page 40\]](#)

Use the `GetFieldName` method to get the name of the field specified in the *fieldIndex* parameter.

[GetDataType \[page 41\]](#)

Use the `GetDataType` method to get the data type of the field specified in the *fieldIndex* parameter.

6.3.1 GetFieldCount

Use the `GetFieldCount` method to return the number of fields defined in the data record.

≡, Syntax

```
int GetFieldCount();
```

`GetFieldCount` is a part of the `DataRecordSchema` class for C++ development.

Return value

The `GetFieldCount` method returns the number of fields defined in the data record.

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.3.2 GetFieldIndex

Use the `GetFieldIndex` method to return the field index for the field specified in the *fieldName* parameter.

Syntax

```
int GetFieldIndex(const uint16_t* fieldName);  
  
int GetFieldIndex(const char* fieldName);
```

`GetFieldIndex` is a part of the `DataRecordSchema` class for C++ development. Use the syntax that is compatible with Unicode or Latin1 characters as applicable.

Return value

The `GetFieldIndex` method returns the field index (0-based). If the specified field in *fieldName* is invalid, returns -1. Use the returned field index in other methods that have a field index as a parameter.

Where

Parameter	Description
<i>fieldName</i> [IN]	The name of the field for which to get the field index value. The Data Quality Management SDK treats field names as case insensitive. For example, a field named "ADDRESS" is equivalent to a field named "address".

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.3.3 GetFieldLength

Use the `GetFieldLength` method to get the length of the field defined by the *fieldIndex* parameter.

Syntax

```
int GetFieldLength(int fieldIndex);
```

GetFieldLength is a part of the DataRecordSchema class for C++ development.

Return value

The GetFieldLength method returns the length of the *fieldIndex* parameter (0-based). It returns 0 if it cannot return the field length, and—or on a non-fatal error.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get the field length (0-based).

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.3.4 GetFieldName Constant

Use the GetFieldName method to get the name of the field specified by the *fieldIndex* parameter.

≡, Syntax

```
const uint16_t* GetFieldName(int fieldIndex);
```

GetFieldName is a part of the DataRecordSchema class for C++ development.

Return value

The GetFieldName method returns the name of the field specified by the *fieldIndex* parameter. Returns 0 if it does not return the field name, and—or on a non-fatal error.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get the field name (0-based).

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.3.5 GetFieldName Boolean

Use the `GetFieldName` method to get the name of the field specified in the *fieldIndex* parameter.

Syntax

```
bool GetFieldName(int fieldIndex, char* buffer, int bufferSize, char  
unicodeReplacement = '?');
```

`GetFieldName` is a part of the `DataRecordSchema` class for C++ development.

Return value

The `GetFieldName` method returns `True` if it successfully gets the name of the field for the specified value in the *fieldIndex* parameter. Returns `False` if it does not successfully get the name of the field, and—or on a non-fatal error.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get the field name (0-based).
<i>buffer</i> [OUT]	Buffer for the field name location.

Parameter	Description
<i>bufferSize</i> [IN]	Size of the buffer.
<i>unicodeReplacement</i> [IN]	The character that replaces a character that cannot be represented as Latin1. Alternately, set to remove the character.

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.3.6 GetDataType

Use the `GetDataType` method to get the data type of the field specified in the *fieldIndex* parameter.

≡ Syntax

```
DATATYPE GetDataType(int fieldIndex, bool& status);
```

`GetDataType` is a part of the `DataRecordSchema` class for C++ development.

Return value

The `GetDataType` method populates *status* parameter with `True` when it successfully gets the data type. It returns `False` if it does not successfully return the data type, and—or there are no fatal errors.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	Index of the field for which to get data type (0-based).
<i>status</i> [OUT]	True or False.

Related Information

[Class DataRecordSchema \[page 36\]](#)

6.4 Class Date

Class `Date` holds date values for record fields.

Use the class `Date` with the following methods for C++ development:

- `bool SetDate(const char* dateStr);`
- `bool SetDate(int year, int month, int day);`
- `bool SetDay(int day);`
- `bool SetMonth(int month);`
- `bool SetYear(int year);`
- `void GetDate(char* dateStr, int bufferSize) const;`
- `int GetDay() const;`
- `int GetMonth() const;`
- `int GetYear() const;`

Constructors for class `Date`

Default Constructor	<code>Date();</code>
Copy Constructor	<code>Date(const Date& rhs);</code>

[SetDate Boolean \[page 43\]](#)

Use the `SetDate` method to set the date of the object based on the value in the *dateStr* parameter.

[SetDate Boolean Year, Month, Day \[page 43\]](#)

Use the `SetDate` method to set the date of this object based on the parameter values for *year*, *month*, and *day*.

[SetDay Boolean \[page 44\]](#)

Use the `SetDay` method to set the day of this object.

[SetMonth Boolean \[page 45\]](#)

Use the `SetMonth` method to set the month of this object.

[SetYear Boolean \[page 46\]](#)

Use the `SetYear` method to set the year of this object.

[GetDate \[page 46\]](#)

Use the `GetDate` method to get the date value of this object.

[GetDay \[page 47\]](#)

Use the `GetDay` method to return the numeric representation of the day of this object.

[GetMonth \[page 48\]](#)

Use the `GetMonth` method to return the numeric representation of the month of this object.

[GetYear \[page 48\]](#)

Use the `GetYear` method to return the numeric representation of the year of this object.

6.4.1 SetDate Boolean

Use the `SetDate` method to set the date of the object based on the value in the [dateStr](#) parameter.

≡ Syntax

```
bool SetDate(const char* dateStr);
```

`SetDate` is a part of the `Date` class for C++ development.

Return value

The `SetDate` method returns `True` if the date is valid. It returns `False` if the date is invalid. If you do not format the value in the [dateStr](#) parameter correctly, the method does not set the date value.

Where

Parameter	Description
dateStr [IN]	Date using the format YYYYMMDD: • YYYY: Numeric year value from 0 to 9999. • MM: Numeric month value from 1 to 12. • DD: Numeric day value from 1 to 31.

Related Information

[Class Date](#) [page 42]

6.4.2 SetDate Boolean Year, Month, Day

Use the `SetDate` method to set the date of this object based on the parameter values for [year](#), [month](#), and [day](#).

≡ Syntax

```
bool SetDate (int year, int month, int day);
```

`SetDate` is a part of the `Date` class for C++ development.

Return value

The `SetDate` method returns `True` if the date is valid. It returns `False` if the date is invalid.

Where

Parameter	Description
<i>year</i> [IN]	Enter the year as a numeric value from 0 to 9999.
<i>month</i> [IN]	Enter the month as a numeric value from 1 to 12.
<i>day</i> [IN]	Enter the day as a numeric value from 1 to 31

Related Information

[Class Date \[page 42\]](#)

6.4.3 SetDay Boolean

Use the `SetDay` method to set the day of this object.

≡, Syntax

```
bool SetDay(int day);
```

`SetDay` is a part of the `Date` class for C++ development.

Return value

The `SetDay` method returns `True` if the day is valid for the month and year. It returns `False` if the day is not valid for the month and year.

Where

Parameter	Description
<i>day</i> [IN]	Numeric value for the day, from 1 to 31.

Related Information

[Class Date \[page 42\]](#)

6.4.4 SetMonth Boolean

Use the `SetMonth` method to set the month of this object.

≡ Syntax

```
bool SetMonth(int month);
```

`SetMonth` is a part of the `Date` class for C++ development.

Return value

The `SetMonth` method returns `True` if the month is valid for the day and year. It returns `False` if the month is not valid for the day and year.

Where

Parameter	Description
<i>month</i> [IN]	Numeric value for the month, from 1 to 12.

Related Information

[Class Date \[page 42\]](#)

6.4.5 SetYear Boolean

Use the SetYear method to set the year of this object.

≡ Syntax

```
bool SetYear(int year);
```

SetYear is a part of the Date class for C++ development.

Return value

The SetYear method returns True if the year is valid for the day and month. It returns False if the year is not valid for the day and month.

Where

Parameter	Description
year [IN]	Numeric value for the year, from 0 to 9999.

Related Information

[Class Date \[page 42\]](#)

6.4.6 GetDate

Use the GetDate method to get the date value of this object.

≡ Syntax

```
void GetDate(char* dateStr, int bufferSize) const;
```

GetDate is part of the Date class for C++ development.

Return value

The `GetDate` method returns the date as a string with the format `YYYYMMDD`.

Where

Parameter	Description
<code>bufferSize [IN]</code>	Size of the <code>dateStr</code> destination buffer.
<code>dateStr [OUT]</code>	Date value in <code>YYYYMMDD</code> format.

Related Information

[Class Date \[page 42\]](#)

6.4.7 GetDay

Use the `GetDay` method to return the numeric representation of the day of this object.

≡ Syntax

```
int GetDay() const;
```

`GetDay` is a part of the `Date` class for C++ development.

Return value

The `GetDay` method returns the numeric value for the given day, from 1 to 31.

Related Information

[Class Date \[page 42\]](#)

6.4.8 GetMonth

Use the `GetMonth` method to return the numeric representation of the month of this object.

≡ Syntax

```
int GetMonth() const;
```

`GetMonth` is a part of the `Date` class for C++ development.

Return value

The `GetMonth` method returns the numeric value for the given month, from 1 to 12.

Related Information

[Class Date \[page 42\]](#)

6.4.9 GetYear

Use the `GetYear` method to return the numeric representation of the year of this object.

≡ Syntax

```
int GetYear() const;
```

`GetYear` is a part of the `Date` class for C++ development.

Return value

The `GetYear` method returns the numeric value for the given year, from 0 to 9999.

Related Information

[Class Date \[page 42\]](#)

6.5 Class DateTime

Class `DateTime` holds date and time values for record fields.

This class inherits from both the `Date` and the `Time` class. Therefore, the methods of those classes are also available.

The class `DateTime` includes the following methods for C++ development:

- `bool SetDateTime(const char* dateTimeStr);`
- `void GetDateTime(char* dateTimeStr, int bufferSize) const;`

Constructors for class `DateTime`

Default Constructor	<code>DateTime();</code>
Copy Constructor	<code>DateTime(const DateTime& rhs);</code>

[SetDateTime \[page 49\]](#)

Use the `SetDateTime` method to verify the format of the given date and time.

[GetDateTime \[page 50\]](#)

Use the `GetDateTime` method to get the date and time value of the object.

6.5.1 SetDateTime

Use the `SetDateTime` method to verify the format of the given date and time.

≡, Syntax

```
bool SetDateTime(const char* dateTimeStr);
```

`SetDateTime` is a part of the `DateTime` class for C++ development.

Return value

The `SetDateTime` method returns `True` if the date and time formats are valid. It returns `False` if the date and time formats are invalid.

Date and time formats consist of the following valid values:

- **Year (YYYY):** Numeric value from 0 to 9999.
- **Month (MM):** Numeric value from 1 to 12.
- **Day (DD):** Numeric value from 1 to 31.
- **Hours (HH):** Numeric value from 0 to 23.
- **Minutes (MM):** Numeric value from 0 to 59.

- **Seconds (SS):** Numeric value from 0 to 59.
- **Fraction (F):** Optional. Can be repeated. Numeric value from 0 to 9.

Where

Parameter	Description
<i>dateTimeStr</i> [IN]	Date and time in the following format: YYYYMMDDHHMMSS.F.

Related Information

[Class DateTime \[page 49\]](#)

6.5.2 GetDateTime

Use the `GetDateTime` method to get the date and time value of the object.

≡ Syntax

```
void GetDateTime(char* dateTimeStr, int bufferSize) const;
```

`GetDateTime` is a part of the `DateTime` class for C++ development.

Return value

The `GetDateTime` method returns the date and time value of the object. It returns the date and time as a string in the following format: YYYYMMDDHHMMSS.F.

i Note

F is an optional one-digit portion of a fraction, which can be repeated.

Where

Parameter	Description
<i>dateTimeStr</i> [OUT]	DateTime in the format YYYYMMDDHHMMSS.F.
<i>bufferSize</i> [IN]	Buffer in number of characters for the value in the <i>dateTimeStr</i> parameter.

Related Information

[Class DateTime \[page 49\]](#)

6.6 Class EmdqException

Class `EmdqException` is the exception class thrown by all public interfaces of this product. This class is required for processing.

The `EmdqException` class is defined in the `exception.h` file. It contains the following methods for C++ development:

- `virtual const uint16_t* GetMessage() const = 0;`
- `virtual const char* GetMessageId() const = 0;`

[GetMessage \[page 51\]](#)

Use the `GetMessage` method to return the message for this object.

[GetMessageID \[page 52\]](#)

Use the `GetMessageID` method to return the message ID of this exception object.

6.6.1 GetMessage

Use the `GetMessage` method to return the message for this object.

Syntax

```
virtual const uint16_t* GetMessage() const = 0;
```

`GetMessage` is a part of the `EmdqException` class for C++ development.

Return value

The `GetMessage` method returns the exception message for this object in UCS2 characters.

Related Information

[Class EmdqException \[page 51\]](#)

6.6.2 GetMessageID

Use the `GetMessageID` method to return the message ID of this exception object.

Syntax

```
virtual const char* GetMessageId() const = 0;
```

`GetMessageID` is a part of the `EmdqException` class for C++ development.

Return value

The `GetMessageID` method returns the exception message ID in the format CCCNNN:

- **CCC**: Alpha characters that represent the source of the error.
- **NNN**: Numeric characters that represent the message number.

Example

REC001 is the first message for the `DataRecord` class.

Related Information

[Class EmdqException \[page 51\]](#)

6.7 Class InputDataRecord

Class `InputDataRecord` is the main interface to the Input Data Record functionality.

The `InputDataRecord` class inherits from the super class `DataRecord`. The class `InputDataRecord` includes the following methods for C++ development:

- `void Clear();`
- `void SetFieldData(int fieldIndex, const uint16_t* fieldData, int fieldDataLength = -1);`
- `void SetFieldData(int fieldIndex, const char* fieldData, int fieldDataLength = -1);`
- `void SetFieldData(int fieldIndex, const Date& fieldData);`
- `void SetFieldData(int fieldIndex, const DateTime& fieldData);`
- `void SetFieldData(int fieldIndex, const Time& fieldData);`
- `void SetFieldData(int fieldIndex, double fieldData);`
- `void SetFieldData(int fieldIndex, int fieldData);`
- `void SetFieldNull(int fieldIndex);`

[Clear \[page 54\]](#)

Use the `Clear` method to clear all data record fields.

[SetFieldData UCS2 \[page 54\]](#)

Use the `SetFieldData` method to set the data for the data record field to UCS2 characters.

[SetFieldData Latin1 \[page 55\]](#)

Use the `SetFieldData` method to set the data for the data record field to Latin1 characters.

[SetFieldData Date \[page 56\]](#)

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

[SetFieldData DateTime \[page 57\]](#)

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

[SetFieldData Time \[page 58\]](#)

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

[SetFieldData for Double \[page 58\]](#)

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

[SetFieldData int \[page 59\]](#)

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

[SetFieldNull \[page 60\]](#)

Use the `SetFieldNull` method to set the record field to Null.

6.7.1 Clear

Use the `Clear` method to clear all data record fields.

≡ Syntax

```
void Clear();
```

`Clear` is a part of the `InputDataRecord` class for C++ development.

Return

The `Clear` method clears all data from record fields, leaving character fields with a length of 0.

Related Information

[Class `InputDataRecord` \[page 53\]](#)

6.7.2 SetFieldData UCS2

Use the `SetFieldData` method to set the data for the data record field to UCS2 characters.

≡ Syntax

```
void SetFieldData(int fieldIndex, const uint16_t* fieldData, int  
fieldDataLength = -1);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method copies characters from the buffer for the value in the [fieldData](#) parameter to the specified data record field using UCS2 characters. The data record field is indicated by the value in the [fieldIndex](#) parameter.

- If the field is set to Null, the method clears the Null.
- If the [fieldDataLength](#) parameter is set to the default -1, the method assumes that [fieldData](#) is to be NULL terminated and the length calculated.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to set (0-based).
<i>fieldData</i> [IN]	The field data to set.
<i>fieldDataLength</i> [IN]	The number of UCS2 characters in <i>fieldData</i> . The default is -1.

Related Information

[Class InputDataRecord \[page 53\]](#)

6.7.3 SetFieldData Latin1

Use the `SetFieldData` method to set the data for the data record field to Latin1 characters.

Syntax

```
void SetFieldData(int fieldIndex, const char* fieldData, int fieldDataLength = -1);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method copies characters from the buffer for the value in the *fieldData* parameter to the specified data record field using Latin1 characters. The data record field is indicated by the value in the *fieldIndex* parameter.

- If the data is longer than the specified field length, the method truncates the data.
- If the data is shorter than the specified field length, the method copies the field into the specified record field, left justified.
- If the field is set to Null, the method clears the Null.
- If the *fieldDataLength* parameter is set to the default -1, the method uses *strlen()* to determine the length of *fieldData*.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to set (0-based).
<i>fieldData</i> [IN]	The field data to set.
<i>fieldDataLength</i> [IN]	Number of Latin1 characters in <i>fieldData</i> . The default is -1.

Related Information

[Class InputDataRecord \[page 53\]](#)

6.7.4 SetFieldData Date

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

Syntax

```
void SetFieldData(int fieldIndex, const Date& fieldData);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method returns the data if it is compatible with `Date` data type:

- If the field is set as `Null`, the method clears the `Null`.
- If the field data type is not compatible with `Date` data type, the method issues an exception.

Where

Parameter	Description
<code>fieldIndex</code> [IN]	The index of the field to set (0-based).

Parameter	Description
<code>fieldData</code> [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 53\]](#)

6.7.5 SetFieldData DateTime

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

≡ Syntax

```
void SetFieldData(int fieldIndex, const DateTime& fieldData);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method returns the data if it is compatible with `DateTime` data type:

- If the field is set as Null, the method clears the Null.
- If the field data type is not compatible with `DateTime` data type, the method issues an exception.

Where

Parameter	Description
<code>fieldIndex</code> [IN]	The index of the field to set (0-based).
<code>fieldData</code> [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 53\]](#)

6.7.6 SetFieldData Time

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

≡ Syntax

```
void SetFieldData(int fieldIndex, const Time& fieldData);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method returns the data if it is compatible with `Time` data type.

- If the field is set as `Null`, the method clears the `Null`.
- If the field data type is not compatible with `Time`, the method issues an exception.

Where

Parameter	Description
<code>fieldIndex</code> [IN]	The index of the field to set (0-based).
<code>fieldData</code> [IN]	The field data to set.

Related Information

[Class `InputDataRecord` \[page 53\]](#)

6.7.7 SetFieldData for Double

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

≡ Syntax

```
void SetFieldData(int fieldIndex, double fieldData);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method returns field data if the data is compatible with the `Double` data type:

- If the field is set as null, the method clears the null.
- If the field data type is not compatible with the `Double` data type, the method throws an exception.

Where

Parameter	Description
<code>fieldIndex</code> [IN]	The index of the field to set (0-based).
<code>fieldData</code> [IN]	The field data to set.

Related Information

[Class `InputDataRecord` \[page 53\]](#)

6.7.8 `SetFieldData` int

Use the `SetFieldData` method to set the data of the `fieldIndex` field of the data record.

Syntax

```
void SetFieldData(int fieldIndex, int fieldData);
```

`SetFieldData` is a part of the `InputDataRecord` class for C++ development.

Return value

The `SetFieldData` method returns field data if the data is compatible with the `int` data type:

- If the field is set as null, the method clears the null.
- If the field data type is not compatible with the `int` data type, the method throws an exception.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).
fieldData [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 53\]](#)

6.7.9 SetFieldNull

Use the `SetFieldNull` method to set the record field to Null.

≡ Syntax

```
void SetFieldNull(int fieldIndex);
```

`SetFieldNull` is a part of the `InputDataRecord` class for C++ development.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set to Null (0-based).

Related Information

[Class InputDataRecord \[page 53\]](#)

6.8 Class MessageHandler

Class `MessageHandler` is a callback class to handle messages from a transform.

Class `MessageHandler` is required for processing. The integrating application implements the interface.

The class `MessageHandler` includes the following method for C++ development:

```
virtual bool HandleMessage(MESSAGETYPE messageType, const char* messageId, const
uint16_t* message) = 0;
```

[HandleMessage \[page 61\]](#)

Use the `HandleMessage` method to accept a message for the implementor to handle.

6.8.1 HandleMessage

Use the `HandleMessage` method to accept a message for the implementor to handle.

≡ Syntax

```
virtual bool HandleMessage(MESSAGETYPE messageType, const char* messageId,
const uint16_t* message) = 0;
```

`HandleMessage` is a part of the `MessageHandler` class for C++ development.

Return value

The `HandleMessage` method returns `True` when it successfully accepts a message. It returns `False` and stops processing when it does not successfully accept a message.

Where

Parameter	Description
<i>messageType</i> [IN]	The type of message to handle.
<i>messageId</i> [IN]	The ID of the message to handle.
<i>message</i> [IN]	The message to handle.

Related Information

[Class MessageHandler \[page 61\]](#)

6.9 Class MultiRecordTransform

Class `MultiRecordTransform` processes multiple records.

`MultiRecordTransform` is the record processing transform class. It is required for processing.

You cannot instantiate instances of class `MultiRecordTransform`. To create valid `MultiRecordTransform` instances, use the `CreateMultiRecordTransform` method in the `TransformFactory` class.

Use class `MultiRecordTransform` objects to represent Match.

Some of the methods for the `MultiRecordTransform` class are inherited from the `Transform` class. The class `MultiRecordTransform` includes the following methods for C++ development:

- `MultiRecordTransformHelper* CreateHelper();`
- `void DestroyHelper(MultiRecordTransformHelper* helper);`
- `void LoadInputDataRecord(InputDataRecord* record);`
- `void Process();`
- `void SetProgressHandler(ProgressHandler* handler);`
- `const OutputDataRecord* UnloadOutputDataRecord();`
- `void ClearRecords();`
- `ProgressHandler* GetProgressHandler();`
- `void SetStatisticsHandler(StatisticsHandler* handler);`
- `int GetExtraInfoLength(uint16_t* request);`
- `void GetExtraInfo(uint16_t* request, uint16_t* response, int responseBufferLength);`
- `InputDataRecord* GetInputDataRecord();`
- `StatisticsHandler* GetStatisticsHandler();`
- `const DataRecordSchema* GetInputSchema() const;`
- `const DataRecordSchema* GetOutputSchema() const;`
- `const StatisticsSchema* GetStatisticsSchema(int schemaIndex) const;`
- `int GetStatisticsSchemaCount() const;`

[CreateHelper \[page 63\]](#)

Use the `CreateHelper` method to create a helper to process records for this transform.

[DestroyHelper \[page 64\]](#)

Use the `DestroyHelper` method to destroy a helper that you no longer need.

[LoadInputDataRecord \[page 65\]](#)

Use the `LoadInputDataRecord` method to load an input data record into this transform.

[Process \[page 65\]](#)

Use the `Process` method to process the input data records that are loaded into this transform.

[SetProgressHandler \[page 66\]](#)

Use the `SetProgressHandler` method to set the progress event handler.

[OutputDataRecord \[page 67\]](#)

Use the `OutputDataRecord` method to unload the next available output data record from this transform.

[ClearRecords \[page 67\]](#)

Use the `ClearRecords` method to clear all input and output records from this transform, and to ready the transform to process again.

[GetProgressHandler \[page 68\]](#)

Use the `GetProgressHandler` method to return a pointer to the current progress event handler.

[SetStatisticsHandler \[page 68\]](#)

Use the `SetStatisticsHandler` method to set the statistics event handler.

[GetExtraInfoLength \[page 69\]](#)

Use the `GetExtraInfoLength` method to determine the length of the buffer to pass into the method `GetExtraInfo()`.

[GetExtraInfo \[page 70\]](#)

Use the `GetExtraInfo` method to get extra information from a transform.

[GetStatisticsHandler \[page 71\]](#)

Use the `GetStatisticsHandler` to return a pointer to the current statistics event handler.

[GetInputSchema \[page 71\]](#)

Use the `GetInputSchema` method to get the schema of the input data record.

[GetOutputSchema \[page 72\]](#)

Use the `GetOutputSchema` method to get the schema of the output data record.

[GetStatisticsSchema \[page 72\]](#)

Use the `GetStatisticsSchema` method to get the *schemaIndex* statistics schema.

[GetStatisticsSchemaCount \[page 73\]](#)

Use the `GetStatisticsSchemaCount` method to get the number of statistics schemas defined for this transform.

6.9.1 CreateHelper

Use the `CreateHelper` method to create a helper to process records for this transform.

Syntax

```
MultiRecordTransformHelper* CreateHelper();
```

`CreateHelper` is a part of the `MultiRecordTransform` class for C++ development.

The `CreateHelper` method is not thread safe.

Use a helper to process records. Run a helper in a different thread than this transform. A helper is like a transform in the following ways:

- Helper has all the same settings as a transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `CreateHelper` method returns a pointer to a newly created helper object.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.2 DestroyHelper

Use the `DestroyHelper` method to destroy a helper that you no longer need.

Syntax

```
void DestroyHelper(MultiRecordTransformHelper* helper);
```

`DestroyHelper` is a part of the `MultiRecordTransform` class for C++ development.

Ensure that you run the `DestroyHelper` method to destroy all helpers of a transform before you destroy the transform.

Where

Parameter	Description
<i>helper</i> [IN]	The helper to destroy.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.3 LoadInputDataRecord

Use the `LoadInputDataRecord` method to load an input data record into this transform.

Syntax

```
void LoadInputDataRecord(InputDataRecord* record);
```

`LoadInputDataRecord` is a part of the `MultiRecordTransform` class for C++ development.

Return value

The `LoadInputDataRecord` method returns the input data record. The method copies the input data record so that the passed-in input data record is available for use upon return from this method.

Caution

Do not attempt to use an input data record that belongs to a different transform.

Where

Parameter	Description
<i>record</i> [IN]	The input data record to load.

Related Information

[Class `MultiRecordTransform` \[page 62\]](#)

6.9.4 Process

Use the `Process` method to process the input data records that are loaded into this transform.

Syntax

```
void Process();
```

`Process` is a part of the `MultiRecordTransform` class for C++ development.

Return value

When the `Process` method returns successfully, the output data records with the posted results are ready to be unloaded.

Related Information

[Class `MultiRecordTransform` \[page 62\]](#)

6.9.5 `SetProgressHandler`

Use the `SetProgressHandler` method to set the progress event handler.

Syntax

```
void SetProgressHandler(ProgressHandler* handler);
```

`SetProgressHandler` is a part of the `MultiRecordTransform` class for C++ development.

A progress event handler is optional.

The `SetProgressHandler` method sets the progress event handler, which is called by this transform as it processes the loaded input records. The `SetProgressHandler` method saves a shallow copy of the passed-in progress event handler. Your application is responsible for deleting the shallow copy. However, ensure that your application does not delete the shallow copy of the passed-in progress event handler until after you destroy this transform.

Where

Parameter	Description
<i>handler</i> [IN]	The progress event handler.

Related Information

[Class `MultiRecordTransform` \[page 62\]](#)

6.9.6 OutputDataRecord

Use the `OutputDataRecord` method to unload the next available output data record from this transform.

Syntax

```
const OutputDataRecord* UnloadOutputDataRecord();
```

`OutputDataRecord` is a part of the `MultiRecordTransform` class for C++ development.

This method requires that there be one output data record for each input data record that the transform loads. An output data record becomes available after the transform has finished processing the input data record and has posted results to the output data record.

Output records are available for unloading after the application calls `Process()`. But it is possible for a transform to make an output record available for unloading immediately after the call to `LoadInputDataRecord()`. The application is free to call this method at any time to see if there are any output records available for unloading.

Once the transform calls `Process()`, the transform unloads all current output records before it loads any new input records for processing.

Return value

The `OutputDataRecord` method returns the next available output record. It returns 0 when no records are available.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.7 ClearRecords

Use the `ClearRecords` method to clear all input and output records from this transform, and to ready the transform to process again.

Syntax

```
void ClearRecords();
```

`ClearRecords` is a part of the `MultiRecordTransform` class for C++ development.

Call this method only after you have loaded your input records and processed and extracted your output records.

Return value

The `ClearRecords` method clears the transform and redies it to process again.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.8 GetProgressHandler

Use the `GetProgressHandler` method to return a pointer to the current progress event handler.

≡ Syntax

```
ProgressHandler* GetProgressHandler();
```

`GetProgressHandler` is a part of the `MultiRecordTransform` class for C++ development.

Return value

The `GetProgressHandler` method returns a pointer to the progress event handler.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.9 SetStatisticsHandler

Use the `SetStatisticsHandler` method to set the statistics event handler.

≡ Syntax

```
void SetStatisticsHandler(StatisticsHandler* handler);
```

`SetStatisticsHandler` is a part of the `MultiRecordTransform` class for C++ development.

A statistics event handler is optional. However, if you do not use the `SetStatisticsHandler` method, the transform does not output statistics.

The `SetStatisticsHandler` method sets the statistics handler, which is called by the transform when it outputs statistics. The transform outputs statistics when it terminates.

The `SetStatisticsHandler` method saves a shallow copy of the passed-in statistics event handler. Your application is responsible for deleting the copy. However, ensure that your application does not delete the copy of the passed-in statistics handler until you destroy this transform.

i Note

When you destroy a transform using the `DestroyTransform` method, the transform may pass final statistics to the statistics event handler. Read about the `DestroyTransform` method in the `TransformFactory` section at [DestroyTransform \[page 120\]](#).

Where

Parameter	Description
<i>handler</i> [IN]	The statistics event handler.

Related Information

[Class `MultiRecordTransform` \[page 62\]](#)

6.9.10 `GetExtraInfoLength`

Use the `GetExtraInfoLength` method to determine the length of the buffer to pass into the method `GetExtraInfo()`.

Syntax

```
int GetExtraInfoLength(uint16_t* request);
```

`GetExtraInfoLength` is a part of the `MultiRecordTransform` class for C++ development.

Where

Parameter	Description
<i>request</i> [IN]	A buffer with the request string.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.11 GetExtraInfo

Use the `GetExtraInfo` method to get extra information from a transform.

Syntax

```
void GetExtraInfo(uint16_t* request, uint16_t* response, int  
responseBufferLength);
```

`GetExtraInfo` is a part of the `MultiRecordTransform` class for C++ development.

Return value

By default, the `GetExtraInfo` method returns an empty string. However, you could customize a transform to determine valid requests from this method and define appropriate responses. The method would then return the appropriate response.

Example

For example, customize the Global Address Cleanse transform to have this method return a list of available address cleanse engines.

Where

Parameter	Description
<i>request</i> [IN]	A buffer with the request string.

Parameter	Description
<i>response</i> [OUT]	A buffer with a response string.
<i>responseBufferLength</i> [IN]	The length of the response string.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.12 GetStatisticsHandler

Use the `GetStatisticsHandler` to return a pointer to the current statistics event handler.

≡ Syntax

```
StatisticsHandler* GetStatisticsHandler();
```

`GetStatisticsHandler` is a part of the `MultiRecordTransform` class for C++ development.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.13 GetInputSchema

Use the `GetInputSchema` method to get the schema of the input data record.

≡ Syntax

```
const DataRecordSchema* GetInputSchema() const;
```

`GetInputSchema` is a part of the `MultiRecordTransform` class for C++ development.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.14 GetOutputSchema

Use the `GetOutputSchema` method to get the schema of the output data record.

≡ Syntax

```
const DataRecordSchema* GetOutputSchema() const;
```

`GetOutputSchema` is a part of the `MultiRecordTransform` class for C++ development.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.15 GetStatisticsSchema

Use the `GetStatisticsSchema` method to get the *schemaIndex* statistics schema.

≡ Syntax

```
const StatisticsSchema* GetStatisticsSchema(int schemaIndex) const;
```

`GetStatisticsSchema` is a part of the `MultiRecordTransform` class for C++ development.

Where

Parameter	Description
<i>schemaIndex</i> [IN]	The handler of the statistics schema for which to get information.

Related Information

[Class MultiRecordTransform \[page 62\]](#)

6.9.16 GetStatisticsSchemaCount

Use the `GetStatisticsSchemaCount` method to get the number of statistics schemas defined for this transform.

Syntax

```
int GetStatisticsSchemaCount() const;
```

`GetStatisticsHandlerCount` is a part of the `MultiRecordTransform` class for C++ development.

Return value

The `GetStatisticsSchemaCount` method returns the number of statistics schemas defined for this transform. If statistics are not enabled, or if the transform does not provide statistics, the method returns a count of 0.

Related Information

[Class `MultiRecordTransform` \[page 62\]](#)

6.10 Class `MultiRecordTransformHelper`

Class `MultiRecordTransformHelper` is the helper class for the `MultiRecordTransform` class.

The class `MultiRecordTransformHelper` includes the following methods for C++ development:

- `void LoadInputDataRecord(InputDataRecord* record);`
- `void Process();`
- `const OutputDataRecord* UnloadOutputDataRecord();`
- `void ClearRecords();`
- `InputDataRecord* GetInputDataRecord();`

[LoadInputDataRecord \[page 74\]](#)

Use the `LoadInputDataRecord` method to load an input data record into this transform.

[Process \[page 74\]](#)

Use the `Process` method to process the input data records that were loaded into this transform.

[UnloadOutputDataRecord \[page 75\]](#)

Use the `UnloadOutputDataRecord` method to unload the next available output data record from this transform.

[ClearRecords \[page 76\]](#)

Use the `ClearRecords` method to clear all input and output records from this transform and to ready the transform to process again.

[GetInputDataRecord \[page 76\]](#)

Use the `GetInputDataRecord` method to return the input data record that holds the input fields from this transform.

6.10.1 LoadInputDataRecord

Use the `LoadInputDataRecord` method to load an input data record into this transform.

Syntax

```
void LoadInputDataRecord(InputDataRecord* record);
```

`LoadInputDataRecord` is a part of the `MultiRecordTransformHelper` class for C++ development.

Because the `LoadInputDataRecord` method copies the input data record, the passed-in input data record is available for further use upon return from this method.

Caution

Do not process an input data record from a different transform.

Where

Parameter	Description
record [IN]	The input data record to load.

Related Information

[Class MultiRecordTransformHelper \[page 73\]](#)

6.10.2 Process

Use the `Process` method to process the input data records that were loaded into this transform.

Syntax

```
void Process();
```

Process is a part of the `MultiRecordTransformHelper` class for C++ development.

Return value

When the `Process` method returns, the output data records with the posted results are ready to be unloaded.

Related Information

[Class `MultiRecordTransformHelper` \[page 73\]](#)

6.10.3 `UnloadOutputDataRecord`

Use the `UnloadOutputDataRecord` method to unload the next available output data record from this transform.

≡, Syntax

```
const OutputDataRecord* UnloadOutputDataRecord();
```

`UnloadOutputDataRecord` is a part of the `MultiRecordTransformHelper` class for C++ development.

This method requires that there be one output data record for each input data record that is loaded. An output data record becomes available after the transform has finished processing the input data record and posted results to the output data record.

Output records are available for unloading after `Process()` has been called. However, a transform can make an output record available for unloading immediately after the call to `LoadInputDataRecord()`. The transform is free to call this method at any time to check if there are any output records available for unloading.

Once the transform calls `Process()`, the transform unloads all current output records before it loads any new input records for processing.

Return value

The `UnloadOutputDataRecord` method returns the next available output record. It returns 0 if no records are available.

Related Information

[Class `MultiRecordTransformHelper` \[page 73\]](#)

6.10.4 ClearRecords

Use the `ClearRecords` method to clear all input and output records from this transform and to ready the transform to process again.

≡ Syntax

```
void ClearRecords();
```

`ClearRecords` is a part of the `MultiRecordTransformHelper` class for C++ development.

Call the `ClearRecords` method only after you have loaded your input records, processed them, and extracted your output.

Related Information

[Class `MultiRecordTransformHelper` \[page 73\]](#)

6.10.5 GetInputDataRecord

Use the `GetInputDataRecord` method to return the input data record that holds the input fields from this transform.

≡ Syntax

```
InputDataRecord* GetInputDataRecord();
```

`GetInputDataRecord` is a part of the `MultiRecordTransformHelper` class for C++ development.

Return value

The `GetInputDataRecord` method returns the input data record that holds the input fields from this transform. An application can use the returned record to pass data to this transform. The application gets the input record only once. However, the application can use the same input record repeatedly.

Related Information

[Class `MultiRecordTransformHelper` \[page 73\]](#)

6.11 Class OutputDataRecord

Class `OutputDataRecord` is the main interface to the Output Data Record functionality.

The `OutputDataRecord` class is required for processing. It inherits methods from the superclass `DataRecord`.

The class `OutputDataRecord` includes the following methods for C++ development:

- `void GetFieldData(int fieldIndex, uint16_t* buffer, int bufferSize;`
- `void GetFieldData(int fieldIndex, char* buffer, int bufferSize, char`
`unicodeReplacement = '?');`
- `void GetFieldData(int fieldIndex, uint16_t* buffer, int bufferSize, int&`
`numCopied);`
- `void GetFieldData(int fieldIndex, Date& output);`
- `void GetFieldData(int fieldIndex, Time& output);`
- `void GetFieldData(int fieldIndex, DateTime& output);`
- `void GetFieldData(int fieldIndex, double& output);`
- `void GetFieldData(int fieldIndex, int& output);`
- `int GetFieldDataLength(int fieldIndex);`
- `bool IsFieldNull(int fieldIndex);`

[GetFieldData UCS2 \[page 78\]](#)

Use the `GetFieldData` method to get the data record field

[GetFieldData Latin1 \[page 79\]](#)

Use the `GetFieldData` method to get the data record field.

[GetFieldData Not Null-terminated \[page 80\]](#)

Use the `GetFieldData` method to get the `fieldIndex` field of the data record.

[GetFieldData Date \[page 81\]](#)

Use the `GetFieldData` method to get the data record field when it is compatible with the `Date` data type.

[GetFieldData Time \[page 82\]](#)

Use the `GetFieldData` method to get the data record field when it is compatible with the `Time` data type.

[GetFieldData DateTime \[page 83\]](#)

Use the `GetFieldData` method to get the data record field when the field data type is compatible with the `DateTime` data type.

[GetFieldData Double \[page 84\]](#)

Use the `GetFieldData` method to get the data record field when the data type is compatible with the `Double` data type.

[GetFieldData int \[page 85\]](#)

Use the `GetFieldData` method to get the `fieldIndex` field of the data record when the data type is compatible with the `int` data type.

[GetFieldDataLength \[page 86\]](#)

Use the `GetFieldDataLength` method to return the number of characters in the data record field.

[IsFieldNull](#) [page 87]

Use the `IsFieldNull` method to determine if the data record field is Null.

6.11.1 GetFieldData UCS2

Use the `GetFieldData` method to get the data record field

Syntax

```
void GetFieldData(int fieldIndex, uint16_t* buffer, int bufferSize;
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same [fieldIndex](#) parameter value. Use the results of `IsFieldNull` to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method returns the data record field indicated in the [fieldIndex](#) parameter. The transform copies the data to the buffer, including a Null terminator. The value in the [bufferSize](#) parameter indicates the number of UCS2 characters to fit into the buffer.

If the data and Null terminator are longer than the specified buffer length, the transform truncates the data. The transform always copies the Null terminator.

If the field indicated by the [fieldIndex](#) parameter is Null, there is no processing.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to get (0-based).
buffer [OUT]	The UCS2 data and a NULL terminator.
bufferSize [IN]	The size of the buffer.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.2 GetFieldData Latin1

Use the `GetFieldData` method to get the data record field.

Syntax

```
void GetFieldData(int fieldIndex, char* buffer, int bufferSize, char  
unicodeReplacement = '?');
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of `IsFieldNull` to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method returns the data record field indicated in the *fieldIndex* parameter.

The transform copies the data to the buffer, including a Null terminator. The value in *bufferSize* parameter indicates the number of Latin1 characters to fit into the buffer.

The transform converts or drops all UCS2 values above 255 characters. The transform saves all UCS2 values less than or equal to 255 characters. The transform does not consider locality and—or codepage.

If the data and Null terminator are longer than the specified buffer length, the transform truncates the data. The transform always copies the Null terminator.

If the field indicated by the *fieldIndex* parameter is Null, there is no processing.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Parameter	Description
<i>buffer</i> [OUT]	The Latin1 data and a NULL terminator.
<i>bufferSize</i> [IN]	The size of the buffer.
<i>unicodeReplacement</i> [IN]	The character that is substituted for a character that the transform cannot recognize as a Latin1 character. Set this parameter to 0 to remove characters that cannot be recognized as Latin1.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.3 GetFieldData Not Null-terminated

Use the `GetFieldData` method to get the `fieldIndex` field of the data record.

Syntax

```
void GetFieldData(int fieldIndex, uint16_t* buffer, int bufferSize, int& numCopied);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call `IsFieldNull` using the same `fieldIndex` parameter value. Use the results of `IsFieldNull` to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` function call because they are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` function call because they are reliable.

Return value

The `GetFieldData` method returns the `fieldIndex` field of the data record.

The transform copies the data to the buffer. The value in `bufferSize` indicates the number of UCS2 characters to fit into the buffer. The value in `numCopied` indicates the number of UCS2 characters copied into the buffer. The buffer is not NULL-terminated.

If the data is longer than the specified buffer length, the transform truncates the data.

If the `fieldIndex` field is `Null`, there is no processing.

Where

Parameter	Description
<code>fieldIndex</code> [IN]	The index of the field to get (0-based).
<code>buffer</code> [OUT]	Holds Latin1 data.
<code>bufferSize</code> [IN]	The size of buffer.
<code>numCopied</code> [OUT]	The number of UCS2 characters copied to buffer.

Related Information

[Class `OutputDataRecord` \[page 77\]](#)

6.11.4 `GetFieldData` Date

Use the `GetFieldData` method to get the data record field when it is compatible with the Date data type.

Syntax

```
void GetFieldData(int fieldIndex, Date& output);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method returns the data record field indicated in the *fieldIndex* parameter. The method copies the data to the *output* parameter. The transform throws an exception if the data record field data type is not compatible with the Date data type.

If the data record field is `Null`, there is no processing.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).
<i>output</i> [OUT]	The resulting date data type data.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.5 GetFieldData Time

Use the `GetFieldData` method to get the data record field when it is compatible with the Time data type.

Syntax

```
void GetFieldData(int fieldIndex, Time& output);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method gets the data record field indicated by the *fieldIndex* parameter. The method copies the data to the *output* parameter.

If the field data type is not compatible with the Time data type, the transform throws an exception.

If the field indicated by the *fieldIndex* parameter is `Null`, there is no processing.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to get (0-based).
output [OUT]	The resulting time.

Related Information

[Class OutputDataRecord](#) [page 77]

6.11.6 GetFieldData DateTime

Use the `GetFieldData` method to get the data record field when the field data type is compatible with the `DateTime` data type.

Syntax

```
void GetFieldData(int fieldIndex, DateTime& output);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same [fieldIndex](#) parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method gets the data record field indicated by the [fieldIndex](#) parameter. The method copies the data to the [output](#) parameter.

If the field data type is not compatible with `DateTime`, the transform throws an exception.

If the field indicated by the [fieldIndex](#) parameter is `Null`, there is no processing.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).
<i>output</i> [OUT]	The resulting DateTime data.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.7 GetFieldData Double

Use the `GetFieldData` method to get the data record field when the data type is compatible with the Double data type.

Syntax

```
void GetFieldData(int fieldIndex, double& output);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method gets the data record field indicated by the *fieldIndex* parameter. The method copies the data to the *output* parameter.

If the field data type is not compatible with the Double data type, the transform throws an exception.

If the field data is Null, there is no processing.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).
<i>output</i> [OUT]	The resulting Double data type data.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.8 GetFieldData int

Use the `GetFieldData` method to get the `fieldIndex` field of the data record when the data type is compatible with the `int` data type.

Syntax

```
void GetFieldData(int fieldIndex, int& output);
```

`GetFieldData` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldData` method gets the data record field indicated by the *fieldIndex* parameter. The method copies the data to the *output* parameter.

If the field data type is not compatible with the `int` data type, the transform throws an exception.

If the field data is `Null`, there is no processing.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).
<i>output</i> [OUT]	The resulting int data type data.

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.9 GetFieldDataLength

Use the `GetFieldDataLength` method to return the number of characters in the data record field.

Syntax

```
int GetFieldDataLength(int fieldIndex);
```

`GetFieldDataLength` is a part of the `OutputDataRecord` class for C++ development.

Before you call `GetFieldData`, call the `IsFieldNull` method using the same *fieldIndex* parameter value. Use the results of the `IsFieldNull` method to determine whether to use the results of `GetFieldData`:

- If `IsFieldNull` returns `True`, do not use the results of the `GetFieldData` method because the results are undefined.
- If `IsFieldNull` returns `False`, use the results of the `GetFieldData` method because the results are reliable.

Return value

The `GetFieldDataLength` method returns the number of characters in the data record indicated by the *fieldIndex* parameter. It returns 0 if the field is `Null`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.11.10 IsFieldNull

Use the `IsFieldNull` method to determine if the data record field is Null.

≡ Syntax

```
bool IsFieldNull(int fieldIndex);
```

`IsFieldNull` is a part of the `OutputDataRecord` class for C++ development.

Return value

The `IsFieldNull` method returns `True` if the data record field indicated in the *fieldIndex* parameter is Null. It returns `False` if the data record field is not Null.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to check (0-based).

Related Information

[Class OutputDataRecord \[page 77\]](#)

6.12 Class ProgressHandler

Class `ProgressHandler` shows the progress of a `MultiRecordTransform` and allows the handler to end processing.

Class `ProgressHandler` is a callback class. The class `ProgressHandler` includes the following methods for C++ development:

- `virtual bool HandleProgress(double percentDone) = 0;`
- `bool SetProgressInterval(int interval);`
- `int GetProgressInterval();`

[HandleProgress \[page 88\]](#)

Use the `HandleProgress` method to see the percentage of completion for the current set of records in progress.

[SetProgressInterval \[page 89\]](#)

Use the `SetProgressInterval` method to specify the number of seconds the transform waits between calls to the `Progress()` method.

[GetProgressInterval \[page 90\]](#)

Use the `GetProgressInterval` method to get the current progress interval in seconds.

6.12.1 HandleProgress

Use the `HandleProgress` method to see the percentage of completion for the current set of records in progress.

≡, Syntax

```
virtual bool HandleProgress(double percentDone) = 0;
```

`HandleProgress` is a part of the `ProgressHandler` class for C++ development.

Return value

The `HandleProgress` method returns `True` when the method is successful. It returns `False` and stops processing when the method is not successful.

Where

Parameter	Description
<i>percentDone</i> [IN]	The percent done from 0.0 to 100.0.

Related Information

[Class `ProgressHandler`](#) [page 88]

6.12.2 `SetProgressInterval`

Use the `SetProgressInterval` method to specify the number of seconds the transform waits between calls to the `Progress()` method.

Syntax

```
bool SetProgressInterval(int interval);
```

`SetProgressInterval` is part of the `ProgressHandler` class for C++ development.

The method reports the interval in seconds.

Return value

The `SetProgressInterval` method returns `True` on successfully returning an interval value greater than 0. It returns `False` when the interval value is invalid.

Where

Parameter	Description
<i>interval</i> [IN]	The number of seconds to wait between calls to <code>Progress()</code> . The method ignores values less than or equal to 0 because they are invalid.

Related Information

[Class ProgressHandler \[page 88\]](#)

6.12.3 GetProgressInterval

Use the `GetProgressInterval` method to get the current progress interval in seconds.

≡ Syntax

```
int GetProgressInterval();
```

`GetProgressInterval` is a part of the `ProgressHandler` class for C++ development.

Return value

The `GetProgressInterval` method returns the current progress interval in seconds.

Related Information

[Class ProgressHandler \[page 88\]](#)

6.13 Class RecordTransform

Class `RecordTransform` is the record processing `Transform` class for processing single records.

Class `RecordTransform` is required for processing.

Instances of `RecordTransform` cannot be instantiated. Use the `CreateRecordTransform` methods in the `TransformFactory` class to create valid `RecordTransform` instances.

Use the `RecordTransform` objects to represent Data Cleanse, USA Regulatory Address Cleanse, Global Address Cleanse, and Geocoder.

Some of the methods in class `RecordTransform` are inherited from the `Transform` class. The following methods for C++ development are included in the `RecordTransform` class:

- `RecordTransformHelper* CreateHelper();`
- `void DestroyHelper(RecordTransformHelper* helper);`
- `const OutputDataRecord* Process(InputDataRecord* record);`

- `void SetStatisticsHandler(StatisticsHandler* handler);`
- `int GetExtraInfoLength(uint16_t* request);`
- `void GetExtraInfo(uint16_t* request, uint16_t* response, int responseBufferLength);`
- `InputDataRecord* GetInputDataRecord();`
- `StatisticsHandler* GetStatisticsHandler();`
- `const DataRecordSchema* GetInputSchema() const;`
- `const DataRecordSchema* GetOutputSchema() const;`
- `const StatisticsSchema* GetStatisticsSchema(int schemaIndex) const;`
- `int GetStatisticsSchemaCount() const;`

[RecordTransformHelper \[page 92\]](#)

Use the `RecordTransformHelper` method to create a helper to process records for this transform.

[DestroyHelper \[page 92\]](#)

Use the `DestroyHelper` method to destroy a helper that you no longer need.

[OutputDataRecord \[page 93\]](#)

Use the `OutputDataRecord` method to read the fields of the input data record and post results to an output data record.

[SetStatisticsHandler \[page 94\]](#)

Use the `SetStatisticsHandler` method to set the statistics event handler to output statistics.

[GetExtraInfoLength \[page 95\]](#)

Use the `GetExtraInfoLength` method to determine the length of the buffer to pass into the `GetExtraInfo` method.

[GetExtraInfo \[page 95\]](#)

Use the `GetExtraInfo` method to get extra information from a transform.

[GetInputDataRecord \[page 96\]](#)

Use the `GetInputDataRecord` method to get the input data record that holds the input fields for this transform.

[GetStatisticsHandler \[page 97\]](#)

Use the `GetStatisticsHandler` method to return a pointer to the current statistics event handler.

[GetInputSchema \[page 97\]](#)

Use the `GetInputSchema` method to return the schema of the input data record.

[GetOutputSchema \[page 98\]](#)

Use the `GetOutputSchema` method to return the schema of the output data record.

[GetStatisticsSchema \[page 98\]](#)

Use the `GetStatisticsSchema` method to return the statistics schema related to the index value in the *schemaIndex* parameter.

[GetStatisticsSchemaCount \[page 99\]](#)

Use the `GetStatisticsSchemaCount` method to return the number of statistics schemas defined for this transform.

Related Information

[Class TransformFactory \[page 116\]](#)

6.13.1 RecordTransformHelper

Use the `RecordTransformHelper` method to create a helper to process records for this transform.

Syntax

```
RecordTransformHelper* CreateHelper();
```

`RecordTransformHelper` is a part of the `RecordTransform` class for C++ development.

Use a helper to process records. Run a helper in a different thread than the transform. A helper is like a transform in the following ways:

- Helper has all the same settings as this transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `RecordTransformHelper` method returns a pointer to a newly created helper object.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.2 DestroyHelper

Use the `DestroyHelper` method to destroy a helper that you no longer need.

Syntax

```
void DestroyHelper(RecordTransformHelper* helper);
```

`DestroyHelper` is a part of the `RecordTransform` class for C++ development.

Ensure that you run the `DestroyHelper` method to destroy all helpers of a transform before you destroy the transform.

Where

Parameter	Description
<code>helper [IN]</code>	The helper to be destroyed.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.3 OutputDataRecord

Use the `OutputDataRecord` method to read the fields of the input data record and post results to an output data record.

Syntax

```
const OutputDataRecord* Process(InputDataRecord* record);
```

`OutputDataRecord` is a part of the `RecordTransform` class for C++ development.

The `OutputDataRecord` method processes the input data record owned by this transform. The method obtains the input data record by calling the `GetInputDataRecord()` method. Ensure that the input data record is loaded with data before it is passed to the `OutputDataRecord` method.

Return value

The `OutputDataRecord` method returns the output data record. The method returns 0 when a non-fatal error occurs. You can query results from the output data record.

Caution

Do not process an input data record from a different transform.

Where

Parameter	Description
record [IN]	The input data record to process

Related Information

[Class RecordTransform](#) [page 90]

6.13.4 SetStatisticsHandler

Use the `SetStatisticsHandler` method to set the statistics event handler to output statistics.

Syntax

```
void SetStatisticsHandler(StatisticsHandler* handler);
```

`SetStatisticsHandler` is a part of the `RecordTransform` class for C++ development.

A statistics event handler is optional. However, if you do not include the `SetStatisticsHandler` method, the transform does not output statistics.

The transform calls a statistics event handler to output statistics. The transform also outputs statistics when it is terminated.

The `SetStatisticsHandler` method saves a shallow copy of the passed-in statistics event handler. Your application is responsible for deleting the copy. Ensure that your application does not delete the copy of the passed-in statistics handler until you destroy this transform.

Note

When you destroy a transform using the `DestroyTransform` method, the transform may pass final statistics to the statistics event handler. Read about the `DestroyTransform` in the `TransformFactory` section at [DestroyTransform](#) [page 120].

Where

Parameter	Description
handler [IN]	The statistics event handler.

Related Information

[Class TransformFactory \[page 116\]](#)

[Class RecordTransform \[page 90\]](#)

6.13.5 GetExtraInfoLength

Use the `GetExtraInfoLength` method to determine the length of the buffer to pass into the `GetExtraInfo` method.

Syntax

```
int GetExtraInfoLength(uint16_t* request);
```

`GetExtraInfoLength` is a part of the `RecordTransform` class for C++ development.

Where

Parameter	Description
<i>request</i> [IN]	A buffer with the request string.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.6 GetExtraInfo

Use the `GetExtraInfo` method to get extra information from a transform.

Syntax

```
void GetExtraInfo(uint16_t* request, uint16_t* response, int  
responseBufferLength);
```

`GetExtraInfo` is a part of the `RecordTransform` class for C++ development.

Return value

The `GetExtraInfo` method returns an empty string by default. However, customize a transform to determine valid requests from this method, and define appropriate responses. The information can be returned using the `GetExtraInfo` method.

❖ Example

Customize the Global Address Cleanse transform to work with this method and return a list of available address cleanse engines.

Where

Parameter	Description
<i>request [IN]</i>	A buffer with the request string.
<i>response [OUT]</i>	A buffer with a response string.
<i>responseBufferLength [IN]</i>	The length of the response string.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.7 GetInputDataRecord

Use the `GetInputDataRecord` method to get the input data record that holds the input fields for this transform.

≡ Syntax

```
InputDataRecord* GetInputDataRecord();
```

`GetInputDataRecord` is a part of the `RecordTransform` class for C++ development.

An application can use the input data record to pass data to this transform. The application only has to get the input record once. Then it can use the input record many times.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.8 GetStatisticsHandler

Use the `GetStatisticsHandler` method to return a pointer to the current statistics event handler.

≡ Syntax

```
StatisticsHandler* GetStatisticsHandler();
```

`GetStatisticsHandler` is a part of the `RecordTransform` class for C++ development.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.9 GetInputSchema

Use the `GetInputSchema` method to return the schema of the input data record.

≡ Syntax

```
const DataRecordSchema* GetInputSchema() const;
```

`GetInputSchema` is a part of the `RecordTransform` class for C++ development.

Related Information

[Class RecordTransform \[page 90\]](#)

6.13.10 GetOutputSchema

Use the `GetOutputSchema` method to return the schema of the output data record.

≡ Syntax

```
const DataRecordSchema* GetOutputSchema() const;
```

`GetOutputSchema` is a part of the `RecordTransform` class for C++ development.

Related Information

[Class `RecordTransform` \[page 90\]](#)

6.13.11 GetStatisticsSchema

Use the `GetStatisticsSchema` method to return the statistics schema related to the index value in the *schemaIndex* parameter.

≡ Syntax

```
const StatisticsSchema* GetStatisticsSchema(int schemaIndex) const;
```

`GetStatisticsSchema` is part of the `RecordTransform` class for C++ development.

Where

Parameter	Description
<i>schemaIndex</i> [IN]	The index of the handler of the statistics schema for which to get information.

Related Information

[Class `RecordTransform` \[page 90\]](#)

6.13.12 GetStatisticsSchemaCount

Use the `GetStatisticsSchemaCount` method to return the number of statistics schemas defined for this transform.

Syntax

```
int GetStatisticsSchemaCount() const;
```

`GetStatisticsSchemaCount` is a part of the `RecordTransform` class for C++ development.

Return value

The `GetStatisticsSchemaCount` method returns the number of statistics schemas define for this transform. It returns 0 when statistics are not enabled or when the transform does not provide statistics.

Related Information

[Class RecordTransform \[page 90\]](#)

6.14 Class RecordTransformHelper

Class `RecordTransformHelper` is the helper class for the `RecordTransform` class.

The class `RecordTransformHelper` includes the following methods for C++ development:

- `const OutputDataRecord* Process(InputDataRecord* record);`
- `InputDataRecord* GetInputDataRecord();`

[Process \[page 100\]](#)

Use the `Process` method to process the input data record owned by this transform.

[GetInputDataRecord \[page 100\]](#)

Use the `GetInputDataRecord` method to obtain the input data record that contains the input fields for this transform.

6.14.1 Process

Use the `Process` method to process the input data record owned by this transform.

Syntax

```
const OutputDataRecord* Process(InputDataRecord* record);
```

The `Process` method is a part of the `RecordTransformHelper` class for C++ development.

Obtain the input data record by calling the `GetInputDataRecord` method. Ensure that the input data record is loaded with data before you pass it to the `Process` method.

Caution

Do not process an input data record from a different transform.

Return value

The `Process` method reads the fields of the input data record and posts the results to an output data record. The output data record can be queried.

Where

Parameter	Description
<i>record</i> [IN]	The input data record to process.

Related Information

[Class `RecordTransformHelper` \[page 99\]](#)

6.14.2 GetInputDataRecord

Use the `GetInputDataRecord` method to obtain the input data record that contains the input fields for this transform.

Syntax

```
InputDataRecord* GetInputDataRecord();
```

`GetInputDataRecord` is part of the `RecordTransformHelper` class for C++ development.

Before you pass data to the `GetInputDataRecord` method, ensure that the input data record is loaded with data. Your application passes the input fields from the input data record into this transform. Your application can get the input data record only once. However, your application can use the same input records repeatedly. Before you reuse the same input records, ensure that you call `Clear()` from the `InputDataRecord` class to clear all data record fields from the transform before reloading the records.

Return value

The `GetInputDataRecord` method returns the input data record that holds the input fields from the transform.

⚠ Caution

Do not use this method on an input data record owned by a different transform.

Where

Parameter	Description
<code>record [IN]</code>	The input data record to process.

Related Information

[Clear \[page 54\]](#)

[Class `RecordTransformHelper` \[page 99\]](#)

6.15 Class `StatisticsHandler`

Class `StatisticsHandler` is a callback class to handle statistics records.

An integrating application implements the interface for statistics. Class `StatisticsHandler` contains the following methods for C++ development:

- `virtual bool HandleStatistics(const OutputDataRecord* record) = 0;`
- `int GetRecordsRemainingCount() const;`
- `const StatisticsSchema* GetStatisticsSchema() const;`

[HandleStatistics \[page 102\]](#)

Use the `HandleStatistics` method to pass a pointer to an output record that holds statistics information.

[GetRecordsRemainingCount \[page 103\]](#)

Use the `GetRecordsRemainingCount` method to get the number of output records that remain to be passed to the `Output` method.

[GetStatisticsSchema \[page 103\]](#)

Use the `GetStatisticsSchema` method to get the statistics schema.

6.15.1 HandleStatistics

Use the `HandleStatistics` method to pass a pointer to an output record that holds statistics information.

Syntax

```
virtual bool HandleStatistics(const OutputDataRecord* record) = 0;
```

`HandleStatistics` is a part of the `StatisticsHandler` class for C++ development.

Program your application to query the record to find the table to which the record belongs. Delete the pointer after the method completes because it becomes invalid. Further, set your application to query the output record and save any field data from the record that it intends to keep.

Return value

The `HandleStatistics` method returns `True` after the method passes a pointer. If the method doesn't pass a pointer, it returns `False`, throws an error, and stops processing.

Where

Parameter	Description
<i>record</i> [IN]	The output statistics record to output.

Related Information

[Class StatisticsHandler \[page 101\]](#)

6.15.2 GetRecordsRemainingCount

Use the `GetRecordsRemainingCount` method to get the number of output records that remain to be passed to the `Output` method.

Syntax

```
int GetRecordsRemainingCount() const;
```

`GetRecordsRemainingCount` is part of the `StatisticsHandler` class for C++ development.

If the transform has a block of records to send, the transform calls `SetRecordsRemainingCount` before each call to `Output` to indicate how many records are left to send. The application can use this information to buffer the records instead of processing them individually.

Return value

The `GetRecordsRemainingCount` method returns the number of additional output records ready to be passed to the `Output` method.

Related Information

[Class `StatisticsHandler` \[page 101\]](#)

6.15.3 GetStatisticsSchema

Use the `GetStatisticsSchema` method to get the statistics schema.

Syntax

```
const StatisticsSchema* GetStatisticsSchema() const;
```

`GetStatisticsSchema` is a part of the `StatisticsHandler` class for C++ development.

Return value

The `GetStatisticsSchema` method returns the statistics schema.

Related Information

[Class StatisticsHandler \[page 101\]](#)

6.16 Class StatisticsSchema

Class StatisticsSchema defines the layout of a statistics table.

Class StatisticsSchema contains the following methods for C++ development:

- `DATATYPE GetFieldDataType(int fieldIndex, bool& status) const = 0`
- `const uint16_t* GetTableName();`
- `bool GetTableName(char* buffer, int bufferSize, char unicodeReplacement = '?');`
- `bool AllowNull(int fieldIndex;`
- `bool IsPrimaryKey(int fieldIndex);`
- `DataRecordSchema::DATATYPE GetDataType(int fieldIndex, bool& status);`

[GetFieldDataType \[page 104\]](#)

Use the `GetFieldDataType` method to get the data type of the field indicated in the *fieldIndex* parameter.

[GetTableName Constant \[page 105\]](#)

Use the `GetTableName` method to get the name of the table that this schema describes.

[GetTableName Boolean \[page 106\]](#)

Use the `GetTableName` method to get the name of the table that this schema describes.

[AllowNull \[page 106\]](#)

Use the `AllowNull` method to indicate whether the field indicated by the *fieldIndex* parameter allows a Null value.

[IsPrimaryKey \[page 107\]](#)

Use the `IsPrimaryKey` method to indicate whether the field indicated in the *fieldIndex* parameter is a primary key.

[GetDataType \[page 108\]](#)

Use the `GetDataType` method to get the data type for the field indicated in the *fieldIndex* parameter.

6.16.1 GetFieldDataType

Use the `GetFieldDataType` method to get the data type of the field indicated in the *fieldIndex* parameter.

≡ Syntax

```
DATATYPE GetFieldDataType(int fieldIndex, bool& status) const = 0
```

`GetFieldDataType` is a part of the `StatisticsSchema` class for C++ development.

Return value

The `GetFieldDataType` method returns `True` when it successfully gets the data type. It returns `False` if it does not successfully get the data type and—or on a non-fatal error.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index for the field on which to get information (0-based).
<i>status</i> [OUT]	True or False.

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.16.2 GetTableName Constant

Use the `GetTableName` method to get the name of the table that this schema describes.

≡ Syntax

```
const uint16_t* GetTableName();
```

`GetTableName` is a part of the `StatisticsSchema` class for C++ development.

Return value

The `GetTableName` method returns the name of the table described by the schema.

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.16.3 GetTableName Boolean

Use the `GetTableName` method to get the name of the table that this schema describes.

Syntax

```
bool GetTableName(char* buffer, int bufferSize, char unicodeReplacement = '?');
```

`GetTableName` is a part of the `StatisticsSchema` class for C++ development.

Return value

The `GetTableName` method returns the name of the table described by this schema.

Where

Parameter	Description
<i>buffer</i> [OUT]	The buffer to hold the table name.
<i>bufferSize</i> [IN]	The size of the buffer.
<i>unicodeReplacement</i> [IN]	The character substituted when a character is encountered that cannot be represented as Latin1. Set to 0 to have characters removed that cannot be converted.

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.16.4 AllowNull

Use the `AllowNull` method to indicate whether the field indicated by the *fieldIndex* parameter allows a Null value.

Syntax

```
bool AllowNull(int fieldIndex;
```

AllowNull is a part of the StatisticsSchema class for C++ development.

Return value

The AllowNull method returns True if the field indicated by the [fieldIndex](#) parameter allows a Null value or if the field is invalid. It returns False if the field indicated by the [fieldIndex](#) parameter does not allow a Null value.

Where

Parameter	Description
fieldIndex [IN]	The index of the field for which to get information (0-based).

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.16.5 IsPrimaryKey

Use the IsPrimaryKey method to indicate whether the field indicated in the [fieldIndex](#) parameter is a primary key.

 Syntax

```
bool IsPrimaryKey(int fieldIndex);
```

IsPrimaryKey is a part of the StatisticsSchema class for C++ development.

Return value

The IsPrimaryKey method returns True if the field indicated in the [fieldIndex](#) parameter is a primary key. It returns False if the field is not a primary key.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information (0-based).

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.16.6 GetDataType

Use the `GetDataType` method to get the data type for the field indicated in the [*fieldIndex*](#) parameter.

≡ Syntax

```
DataRecordSchema::DATATYPE GetDataType(int fieldIndex, bool& status);
```

`GetDataType` is part of the `StatisticsSchema` class for C++ development.

Return value

The `GetDataType` method returns the data type for the field indicated in the [*fieldIndex*](#) parameter.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information (0-based)
<i>status</i> [OUT]	FALSE on a non-fatal error.

Related Information

[Class StatisticsSchema \[page 104\]](#)

6.17 Class Time

Class `Time` holds a time value for a record field.

The time value must be of a data type that is compatible with the `Time` data type. The `Time` class has the following methods for C++ development:

- `bool SetTime(const char* timeStr);`
- `bool SetHours(int hours);`
- `bool SetMinutes(int minutes);`
- `bool SetSeconds(int seconds);`
- `bool SetFractionOfSeconds(double fraction);`
- `void GetTime(char* timeStr, int bufferLength) const;`
- `int GetHours() const;`
- `int GetMinutes() const;`
- `int GetSeconds() const;`
- `double GetFractionOfSeconds() const;`

[SetTime \[page 110\]](#)

Use the `SetTime` method to set the time of this object.

[SetHours \[page 111\]](#)

Use the `SetHours` method to set the hours of this object.

[SetMinutes \[page 111\]](#)

Use the `SetMinutes` method to set the minutes of this object.

[SetSeconds \[page 112\]](#)

Use the `SetSeconds` method to set the seconds of this object.

[SetFractionOfSeconds \[page 113\]](#)

Use the `SetFractionOfSeconds` method to set the fractional seconds of this object.

[GetTime \[page 114\]](#)

Use the `GetTime` method to get the time value of this object.

[GetHours \[page 114\]](#)

Use the `GetHours` method to get the hours value of this object.

[GetMinutes \[page 115\]](#)

Use the `GetMinutes` method to get the minutes value of this object.

[GetSeconds \[page 115\]](#)

Use the `GetSeconds` method to get the seconds value of this object.

[GetFractionOfSeconds \[page 116\]](#)

Use the `GetFractionOfSeconds` method to get the fractional seconds of this object.

6.17.1 SetTime

Use the `SetTime` method to set the time of this object.

≡, Syntax

```
bool SetTime(const char* timeStr);
```

`SetTime` is a part of the `Time` class for C++ development.

Return value

The `SetTime` method returns `True` when the [timeStr](#) parameter value is valid. It returns `False` when the [timeStr](#) parameter value is not valid.

Where

Parameter	Description
timeStr [IN]	The time in the format HHMMSSF. Use the following format information to enter the string: • HH = Hours. Enter a two-digit integer from 0 to 23. For example, 01, 02, 20, 21. • MM = Minutes. Enter a two-digit integer from 0 to 59. • SS = Seconds. Enter a two-digit integer from 0 to 59. • F = Fraction of a second. Optional. Enter a one-digit integer from 0 to 9. You can repeat F as necessary to complete the fraction.

Related Information

[Class Time \[page 109\]](#)

6.17.2 SetHours

Use the `SetHours` method to set the hours of this object.

Syntax

```
bool SetHours(int hours);
```

`SetHours` is a part of the `Time` class for C++ development.

Return value

The `SetHours` method returns `True` when the [hours](#) parameter value is valid. It returns `False` when the [hours](#) parameter value is not valid.

Where

Parameter	Description
hours [IN]	The hours value from 0 to 23.

Related Information

[Class Time \[page 109\]](#)

6.17.3 SetMinutes

Use the `SetMinutes` method to set the minutes of this object.

Syntax

```
bool SetMinutes(int minutes);
```

`SetMinutes` is a part of the `Time` class for C++ development.

Return value

The `SetMinutes` method returns `True` when the *minutes* parameter value is valid. It returns `False` when the *minutes* parameter value is not valid.

Where

Parameter	Description
<i>minutes</i> [IN]	The minutes value from 0 to 59.

Related Information

[Class Time \[page 109\]](#)

6.17.4 SetSeconds

Use the `SetSeconds` method to set the seconds of this object.

≡ Syntax

```
bool SetSeconds(int seconds);
```

`SetSeconds` is a part of the `Time` class for C++ development.

Return value

The `SetSeconds` method returns `True` when the *seconds* parameter value is valid. It returns `False` when the *seconds* parameter value is not valid.

Where

Parameter	Description
<i>seconds</i> [IN]	The seconds value from 0 to 59.

Related Information

[Class Time \[page 109\]](#)

6.17.5 SetFractionOfSeconds

Use the `SetFractionOfSeconds` method to set the fractional seconds of this object.

Syntax

```
bool SetFractionOfSeconds(double fraction);
```

`SetFractionOfSeconds` is a part of the `Time` class for C++ development.

Return value

The `SetFractionOfSeconds` method returns `True` when the *fraction* parameter value is valid. It returns `False` when the *fraction* parameter value is not valid.

Where

Parameter	Description
<i>fraction</i> [IN]	The fractional seconds value. Enter a value of $0.0 \leq \textit{fraction} < 1.0$.

Related Information

[Class Time \[page 109\]](#)

6.17.6 GetTime

Use the `GetTime` method to get the time value of this object.

Syntax

```
void GetTime(char* timeStr, int bufferLength) const;
```

`GetTime` is a part of the `Time` class for C++ development.

Return value

The `GetTime` method returns the time as a string in the format HHMMSSF:

- **HH:** Hours
- **MM:** Minutes
- **SS:** Seconds
- **F:** Fraction of a second. Optional and can be repeated.

Where

Parameter	Description
<i>bufferLength</i> [IN]	The size of the <i>timeStr</i> buffer.
<i>timeStr</i> [OUT]	The time value in HHMMSSF format.

Related Information

[Class Time \[page 109\]](#)

6.17.7 GetHours

Use the `GetHours` method to get the hours value of this object.

Syntax

```
int GetHours() const;
```

`GetHours` is a part of the `Time` class for C++ development.

Return value

The `GetHours` method returns the `hours` value of this object, from 0 to 23.

Related Information

[Class `Time` \[page 109\]](#)

6.17.8 `GetMinutes`

Use the `GetMinutes` method to get the `minutes` value of this object.

≡ Syntax

```
int GetMinutes() const;
```

`GetMinutes` is a part of the `Time` class for C++ development.

Return value

The `GetMinutes` method returns the `minutes` value from 0 to 59.

Related Information

[Class `Time` \[page 109\]](#)

6.17.9 `GetSeconds`

Use the `GetSeconds` method to get the `seconds` value of this object.

≡ Syntax

```
int GetSeconds() const;
```

GetSeconds is a part of the `Time` class for C++ development.

Return value

The `GetSeconds` method gets the seconds value of this object from 0 to 59.

Related Information

[Class `Time` \[page 109\]](#)

6.17.10 GetFractionOfSeconds

Use the `GetFractionOfSeconds` method to get the fractional seconds of this object.

≡ Syntax

```
double GetFractionOfSeconds() const;
```

`GetFractionOfSeconds` is a part of the `Time` class for C++ development.

Return value

The `GetFractionOfSeconds` method returns the fractional seconds ranging from 0 to less than 1.

Related Information

[Class `Time` \[page 109\]](#)

6.18 Class TransformFactory

Class `TransformFactory` contains the methods for creating `Transforms`.

`TransformFactory` class is required for processing. It contains the following methods for C++ development:

- `MultiRecordTransform* CreateMultiRecordTransform(const char* transformSettings, int transformSettingsBufferSize);`

- `RecordTransform* CreateRecordTransform(const char* transformSettings, int transformSettingsBufferSize);`
- `void DestroyTransform(Transform* transform);`
- `const char* UpgradeTransformSettings(const char* transformSettings, int transformSettingsLength, int& upgradedSettingsLength);`
- `bool ValidateTransformSettings(const char* transformSettings, int transformSettingsBufferSize);`
- `void SetMessageHandler(MessageHandler* handler);`
- `void SetLocale(const char* locale);`
- `MessageHandler* GetMessageHandler();`
- `const char* GetLocale() const;`
- `static const char* GetVersion();`

[CreateMultiRecordTransform \[page 118\]](#)

Use the `CreateMultiRecordTransform` method to create a multi-record transform, using the XML found in the parameter *transformSettingsBuffer*.

[CreateRecordTransform \[page 119\]](#)

Use the `CreateRecordTransform` method to create a record transform.

[DestroyTransform \[page 120\]](#)

Use the `DestroyTransform` method to destroy a transform or a multi-record transform.

[UpgradeTransformSettings \[page 120\]](#)

Use the `UpgradeTransformSettings` method to upgrade the XML settings of a transform.

[ValidateTransformSettings \[page 121\]](#)

Use the `ValidateTransformSettings` method to validate the XML in the *transformSettingsBuffer* parameter for the transform.

[SetMessageHandler \[page 122\]](#)

Use the `SetMessageHandler` method to set the log event handler.

[SetLocale \[page 123\]](#)

Use the `SetLocale` method to set the locale for messages produced by the transform.

[GetMessageHandler \[page 123\]](#)

Use the `GetMessageHandler` method to return a pointer to the current log event handler.

[GetLocale \[page 124\]](#)

Use the `GetLocale` method to get the name of the locale that this transform uses.

[GetVersion \[page 124\]](#)

Use the `GetVersion` method to find out the current version of the Data Quality Management SDK.

6.18.1 CreateMultiRecordTransform

Use the `CreateMultiRecordTransform` method to create a multi-record transform, using the XML found in the parameter *transformSettingsBuffer*.

Syntax

```
MultiRecordTransform* CreateMultiRecordTransform(const char* transformSettings,  
int transformSettingsBufferSize);
```

`CreateMultiRecordTransform` is a part of the `TransformFactory` class for C++ development.

The method uses the value in the parameter *transformSettingsBuffer*, which is a buffer of bytes. The method determines the encoding automatically, or by the encoding XML attribute:

- If you pass UCS2 data (2-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UCS2 or UTF16.
- If you pass Latin1 data (1-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UTF-8 or Latin1.

Return value

The `CreateMultiRecordTransform` method returns a pointer to the created multi-record transform.

Where

Parameter	Description
<i>transformSettingsBuffer</i> [IN]	The transform settings buffer.
<i>transformSettingsBufferSize</i>	The number of bytes in <i>transformSettingsBuffer</i> .

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.2 CreateRecordTransform

Use the `CreateRecordTransform` method to create a record transform.

Syntax

```
RecordTransform* CreateRecordTransform(const char* transformSettings, int
transformSettingsBufferSize);
```

`CreateRecordTransform` is a part of the `TransformFactory` class for C++ development.

This method uses the value in the [transformSettingsBuffer](#) parameter, which is a buffer of bytes. The method determines the XML automatically, or by the encoding XML attribute:

- If you pass UCS2 data (2-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UCS2 or UTF16.
- If you pass Latin1 data (1-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UTF-8 or Latin1.

Return value

The `CreateRecordTransform` method returns a pointer to the created record transform.

Where

Parameter	Description
transformSettingsBuffer [IN]	The transform settings buffer.
transformSettingsBufferSize	The number of bytes in transformSettingsBuffer .

Related Information

[Class TransformFactory](#) [page 116]

6.18.3 DestroyTransform

Use the `DestroyTransform` method to destroy a transform or a multi-record transform.

≡, Syntax

```
void DestroyTransform(Transform* transform);
```

`DestroyTransform` is a part of the `TransformFactory` class for C++ development.

Destroying a transform may cause the application to pass final statistics to the statistics event handler.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.4 UpgradeTransformSettings

Use the `UpgradeTransformSettings` method to upgrade the XML settings of a transform.

≡, Syntax

```
const char* UpgradeTransformSettings(const char* transformSettings, int  
transformSettingsLength, int& upgradedSettingsLength);
```

`UpgradeTransformSettings` is a part of the `TransformFactory` class for C++ development.

The `UpgradeTransformSettings` method upgrades the XML settings of a transform. It uses the XML from the buffer that you specify in the [transformSettings](#) parameter. The [transformSettings](#) parameter value is a buffer of bytes. The method determines the encoding automatically, or by the encoding XML attribute:

- If you pass UCS2 data (2-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UCS2 or UTF16.
- If you pass Latin1 data (1-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UTF-8 or Latin1.

Once the method successfully parses the XML, it checks the XML version. If the XML is current, the method returns the pointer to the passed in buffer. If the XML is not current, the method upgrades the XML with the latest version string and other transform-specific changes. The method then stores the upgraded XML as a string in an internal buffer, and that internal buffer is returned. The number of bytes in the returned buffer is stored in the parameter [upgradedSettingsLength](#).

Return value

The `UpgradeTransformSettings` method returns the pointer to the passed in buffer when the XML is current. It returns the internal buffer that holds the updated XML when the XML is not current. If the method returns the internal buffer, ensure that you clear the data in the buffer before you make any other calls to this object.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.
<i>transformSettingsLength</i> [IN]	The number of bytes in the <i>transformSettings</i> parameter.
<i>upgradedSettingsLength</i> [OUT]	The actual length in bytes of the upgraded XML determined by the method.

Related Information

[Class TransformFactory](#) [page 116]

6.18.5 ValidateTransformSettings

Use the `ValidateTransformSettings` method to validate the XML in the *[transformSettingsBuffer](#)* parameter for the transform.

Syntax

```
bool ValidateTransformSettings(const char* transformSettings, int
transformSettingsBufferSize);
```

`ValidateTransformSettings` is a part of the `TransformFactory` class for C++ development.

The method uses the value in the parameter *[transformSettingsBuffer](#)*, which is a buffer of bytes. The method determines the encoding automatically, or by the encoding XML attribute:

- If you pass UCS2 data (2-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UCS2 or UTF16.
- If you pass Latin1 data (1-byte character) to this method, ensure that the encoding attribute in the XML does not exist, or that you set it to UTF-8 or Latin1.

Where

Parameter	Value
<i>transformSettingsBuffer</i> [IN]	The transform settings buffer.
<i>transformSettingsBufferSize</i>	The number of bytes in the <i>transformSettingsBuffer</i> parameter.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.6 SetMessageHandler

Use the `SetMessageHandler` method to set the log event handler.

≡, Syntax

```
void SetMessageHandler(MessageHandler* handler);
```

`SetMessageHandler` is a part of the `TransformFactory` class for C++ development.

A log event handler is required. Your application calls a log event handler when a transform outputs log information.

The `SetMessageHandler` method saves a shallow copy of the passed-in log event handler for logging messages. Each subsequent newly created transform uses the copy.

If your application requires a separate event handler for each new transform, include a new `SetMessageHandler` method for each new transform you create.

Your application is responsible to preserve the event handler until you destroy all transforms that use the event handler.

Where

Parameter	Description
<i>handler</i> [IN]	The log event handler.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.7 SetLocale

Use the `SetLocale` method to set the locale for messages produced by the transform.

Syntax

```
void SetLocale(const char* locale);
```

`SetLocale` is a part of the `TransformFactory` class for C++ development.

If the application does not support the locale that you set for this method, the transform logs a warning to the [MessageHandler](#) parameter using the `SetMessageHandler` method. If you enter a locale that is not supported by SAP Data Quality Management SDK, all messages default to the `en_US` locale.

Where

Parameter	Description
<i>locale</i> [IN]	The locale to use.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.8 GetMessageHandler

Use the `GetMessageHandler` method to return a pointer to the current log event handler.

Syntax

```
MessageHandler* GetMessageHandler();
```

`GetMessageHandler` is a part of the `TransformFactory` class for C++ development.

Return value

The `GetMessageHandler` method returns a pointer to the current log event handler.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.9 GetLocale

Use the `GetLocale` method to get the name of the locale that this transform uses.

```
⌵ Syntax  
const char* GetLocale() const;
```

`GetLocale` is a part of the `TransformFactory` class for C++ development.

Return value

The `GetLocale` method returns the locale value set in the `SetLocale` method when the locale is supported. If the locale is not supported, the method returns `en_US`, which is the default locale.

Related Information

[Class TransformFactory \[page 116\]](#)

6.18.10 GetVersion

Use the `GetVersion` method to find out the current version of the Data Quality Management SDK.

```
⌵ Syntax  
static const char* GetVersion();
```

`GetVersion` is a part of the `TransformFactory` class for C++ development.

Return value

The `GetVersion` method returns the version number for the current Data Quality Management SDK software.

Related Information

[Class TransformFactory \[page 116\]](#)

7 API reference for Java

Each element for building Data Quality transforms in Java contains specific syntax and elements to use in the syntax.

The package for the Java API is `com.sap.emdq`.

By default, logging is not enabled for Java. To be able to use the logging capability of the methods in JAVA API, set a log handler. If you don't set a log handler, the transforms that you create using Java do not output specific warnings or error messages.

For information about the errors and warnings included in the log files for transforms created using the Java API, see the applicable method details.

For information about message types that you can include in log files, see [Message types \[page 32\]](#).

[Class CertifiedReportGenerator \[page 127\]](#)

Class `CertifiedReportGenerator` enables certified reports that are required by specific postal authorities.

[Class DataRecordSchema \[page 130\]](#)

Class `DataRecordSchema` defines the layout of a data record.

[Class EmdqException \[page 134\]](#)

Class `EmdqException` is the exception class thrown by all public interfaces of this product.

[Class InputDataRecord \[page 135\]](#)

Class `InputDataRecord` is the main interface to the Input Data Record functionality.

[Class MessageHandler \[page 142\]](#)

Class `MessageHandler` is a call back class to handle messages from a transform.

[Class MultiRecordTransform \[page 144\]](#)

Class `MultiRecordTransform` is the record processing Transform class for processing multiple records.

[Class MultiRecordTransformHelper \[page 156\]](#)

Class `MultiRecordTransformHelper` is the helper class for the `MultiRecordTransform` class

[Class OutputDataRecord \[page 160\]](#)

Class `OutputDataRecord` is the main interface to the Output Data Record functionality.

[Class ProgressHandler \[page 168\]](#)

Class `ProgressHandler` is a callback class to show the progress of a multirecord transform and to enable the handler to end processing.

[Class RecordTransform \[page 171\]](#)

Class `RecordTransform` is the record processing transform class for processing single records.

[Class RecordTransformHelper \[page 181\]](#)

Class `RecordTransformHelper` is the helper class for the record processing transform class.

[Class StatisticsHandler \[page 182\]](#)

Class `StatisticsHandler` is a callback class to handle statistics records.

[Class StatisticsSchema \[page 185\]](#)

Class `StatisticsShema` defines the layout of a statistics table.

[Class `TransformFactory` \[page 188\]](#)

Class `TransformFactory` is the class to use to create transforms.

7.1 Class `CertifiedReportGenerator`

Class `CertifiedReportGenerator` enables certified reports that are required by specific postal authorities.

The `CertifiedReportGenerator` class is an implementation of the public `StatisticsEventHandler` interface. The interface generates the certified mailing Statement of Address Accuracy (SERP), Address Matching Processing Summary (AMAS), and Coding Accuracy Support System (CASS) 3553 report.

Class `CertifiedReportGenerator` contains the following methods for Java development:

- `CertifiedReportGenerator()`
- `void Destroy()`
- `boolean handleStatistics(OutputDataRecord record)`
- `void setReportFile(ReportType reportType, String reportFile)`

[CertifiedReportGenerator \[page 127\]](#)

Use the `CertifiedReportGenerator` method before you use the Certified Report Generator.

[Destroy \[page 128\]](#)

Use the `Destroy` method to invalidate the object.

[HandleStatistics \[page 128\]](#)

Use the `HandleStatistics` method to implement the *Statistics Handler* interface.

[SetReportFile \[page 129\]](#)

Use the `SetReportFile` method to set a report type, and to specify a path and file name for the report.

7.1.1 CertifiedReportGenerator

Use the `CertifiedReportGenerator` method before you use the Certified Report Generator.

≡ Syntax

```
CertifiedReportGenerator()
```

`CertifiedReportGenerator` is a part of the `CertifiedReportGenerator` class for Java development.

The `CertifiedReportGenerator` method is a constructor. Therefore, the software requires that you run this method before you use the Certified Report Generator.

Related Information

[Class CertifiedReportGenerator \[page 127\]](#)

7.1.2 Destroy

Use the `Destroy` method to invalidate the object.

≡ Syntax

```
void Destroy()
```

`Destroy` is a part of the `CertifiedReportGenerator` class for Java development.

`Destroy` is a required method. Call `Destroy` after all processing is complete. The object will no longer be valid after this call.

Return value

The `Destroy` method invalidates the object. Ensure that you call this method only after all processing is done.

Related Information

[Class CertifiedReportGenerator \[page 127\]](#)

7.1.3 HandleStatistics

Use the `HandleStatistics` method to implement the [Statistics Handler](#) interface.

≡ Syntax

```
boolean handleStatistics(OutputDataRecord record)
```

`HandleStatistics` is a part of the `CertifiedReportGenerator` class for Java development.

Return value

The `HandleStatistic` method implements the [Statistics Handler](#) interface.

Where

Parameter	Description
<i>record</i>	The statistics record to output.

Related Information

[Class CertifiedReportGenerator \[page 127\]](#)

7.1.4 SetReportFile

Use the `SetReportFile` method to set a report type, and to specify a path and file name for the report.

≡, Syntax

```
void setReportFile(ReportType reportType, String reportFile)
```

`SetReportFile` is a part of the `CertifiedReportGenerator` class for Java development.

Call the `SetReportFile` method before you use the transform. Data Quality Management SDK requires that you create a separate `SetReportFile` method for each report you generate.

Return value

If the `SetReportFile` method does not successfully generate the report and save it to the specified path and file name, the method throws an `EmdqException`.

Where

Parameter	Description
<i>reportType</i>	The report type to generate. Valid values are: • Cass3553Report • AmasReport • SerpReport

Parameter	Description
<i>reportFile</i>	The path and file name for the report. If the specified path does not exist, the method does not create the path. If the file exists, the method overwrites the contents of the existing file with the new report. If the file does not exist, the method creates the file in the specified path.

Related Information

[Class CertifiedReportGenerator \[page 127\]](#)

7.2 Class DataRecordSchema

Class `DataRecordSchema` defines the layout of a data record.

The `DataRecordSchema` class has the following methods for Java development:

- `int getFieldCount()`
- `int getFieldIndex(String fieldName)`
- `int getLength(int fieldIndex)`
- `String getFieldName(int fieldIndex)`
- `DataType getDataType(int fieldIndex)`

[getFieldCount \[page 131\]](#)

Use the `getFieldCount` method to return the field count for this data record.

[getFieldIndex \[page 131\]](#)

Use the `getFieldIndex` method to get the field index for the field in the *fieldName* parameter.

[getLength \[page 132\]](#)

Use the `getLength` method to get the field length for the field in the *fieldIndex* parameter.

[getFieldName \[page 133\]](#)

The `getFieldName` method gets the field name for the field in the *fieldIndex* parameter.

[getDataType \[page 134\]](#)

Use the `getDataType` method to get the data type for the field specified in *fieldIndex*.

7.2.1 getFieldCount

Use the `getFieldCount` method to return the field count for this data record.

≡ Syntax

```
int getFieldCount()
```

`getFieldCount` is a part of the `DataRecordSchema` class for Java development.

Return value

The `getFieldCount` method returns the number of fields defined in the data record. If the method does not successfully return the number of fields, it throws the `EmdqException`.

Related Information

[Class DataRecordSchema \[page 130\]](#)

7.2.2 getFieldIndex

Use the `getFieldIndex` method to get the field index for the field in the *fieldName* parameter.

≡ Syntax

```
int getFieldIndex(String fieldName)
```

`getFieldIndex` is a part of the `DataRecordSchema` class for Java development.

Return value

The `getFieldIndex` method returns the field index (0-based) for the field specified in the *fieldName* parameter. Use the returned field index value in other methods that have the *fieldIndex* parameter.

If the method does not return a field index because the *fieldName* parameter value is invalid, it returns -1 and throws the `EmdqException`.

Where

Parameter	Description
<i>fieldName</i> [IN]	The name of the field for which to return the index. Field names are not case-sensitive. For example, a field name of "NAME" is the same as a field name of "name".

Related Information

[Class DataRecordSchema](#) [page 130]

7.2.3 getFieldLength

Use the `getFieldLength` method to get the field length for the field in the [*fieldIndex*](#) parameter.

≡ Syntax

```
int getFieldLength(int fieldIndex)
```

`getFieldLength` is a part of the `DataRecordSchema` class for Java development.

Return value

This method gets the length of the field in [*fieldIndex*](#).

The `getFieldLength` method returns the length of the [*fieldIndex*](#) field. It returns 0 if it does not successfully return the field length and—or on a non-fatal error. The method throws an `EmdqException`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information (0-based).

Related Information

[Class DataRecordSchema \[page 130\]](#)

7.2.4 getFieldname

The `getFieldname` method gets the field name for the field in the *fieldIndex* parameter.

Syntax

```
String getFieldname(int fieldIndex)
```

`getFieldname` is a part of the `DataRecordSchema` class for Java development.

Return value

This method gets the name of the field *fieldIndex*.

The `getFieldname` method returns the name of the field for the value in *fieldIndex*. It returns 0 if it does not successfully return the field name and—or on a non-fatal error. It throws the `EmdqException`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information (0-based).

Related Information

[Class DataRecordSchema \[page 130\]](#)

7.2.5 getDataType

Use the `getDataType` method to get the data type for the field specified in [fieldIndex](#).

≡ Syntax

```
DataType getDataType(int fieldIndex)
```

`getDataType` is a part of the `DataRecordSchema` class for Java development.

Return value

The `getDataType` method returns the data type for the field referenced in the [fieldIndex](#) parameter. It returns 0 if it does not successfully return the field data type and—or on a non-fatal error. It throws the `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field for which to get information (0-based).

Related Information

[Class DataRecordSchema](#) [page 130]

7.3 Class EmdqException

Class `EmdqException` is the exception class thrown by all public interfaces of this product.

SAP Data Quality Management SDK requires the class `EmdqException` for processing. The `EmdqException` class includes the following method for Java development:

```
public String getMessageId()
```

[getMessageID](#) [page 135]

Use the `getMessageID` method to get the message ID of this exception object.

7.3.1 getMessageID

Use the `getMessageID` method to get the message ID of this exception object.

Syntax

```
public String getMessageId()
```

`getMessageID` is a part of the `EmdqException` class for Java development.

Return value

The `getMessageID` method returns the exception message ID in the format CCCNNN:

- **CCC**: Alpha characters that represent the source of the error.
- **NNN**: Numeric characters that represent the message number.

Example

REC001 is the first message for the `DataRecord` class.

Related Information

[Class EmdqException \[page 134\]](#)

7.4 Class InputDataRecord

Class `InputDataRecord` is the main interface to the Input Data Record functionality.

The `InputDataRecord` class inherits from the superclass `DataRecord`. This class is required for processing. The `InputDataRecord` class includes the following methods for Java development:

- `void clear()`
- `void setStringData(int fieldIndex, String fieldData)`
- `void setDateData(int fieldIndex, Calendar fieldData)`
- `void setDateTimeData(int fieldIndex, Calendar fieldData)`
- `void setTimeData(int fieldIndex, Calendar fieldData)`
- `void setDoubleData(int fieldIndex, double fieldData)`
- `void setIntegerData(int fieldIndex, int fieldData)`
- `void setFieldNull(int fieldIndex)`

[clear \[page 136\]](#)

Use the `clear` method to clear all of the fields of the data record.

[setStringData \[page 137\]](#)

Use the `setStringData` method to set the data for the specified data record field.

[setDateData \[page 137\]](#)

Use the `setDateData` method to set the data for the data record field.

[setDateTimeData \[page 138\]](#)

Use the `setDateTimeData` method to set the data of the data record field.

[setTimeData \[page 139\]](#)

Use the `setTimeData` method to set the data of the data record field.

[setDoubleData \[page 140\]](#)

Use the `setDoubleData` method to set the data of the data record field.

[setIntegerData \[page 141\]](#)

Use the `setIntegerData` method to set the data of the data record field.

[setFieldNull \[page 141\]](#)

Use the `setFieldNull` method to set the data record field to Null.

7.4.1 clear

Use the `clear` method to clear all of the fields of the data record.

≡ Syntax

```
void clear()
```

`clear` is a part of the `InputDataRecord` class for Java development.

Return value

The `clear` method returns each character field with a data length of 0. The method throws the `EmdqException`.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.2 setStringData

Use the `setStringData` method to set the data for the specified data record field.

Syntax

```
void setStringData(int fieldIndex, String fieldData)
```

`setStringData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setStringData` method sets the data for the record field specified in the [fieldIndex](#) parameter. If it is not successful, it throws an `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).
fieldData [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.3 setDateData

Use the `setDateData` method to set the data for the data record field.

Syntax

```
void setDateData(int fieldIndex, Calendar fieldData)
```

`setDateData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setDateData` method sets the data of the data record field specified in the [fieldIndex](#) parameter. If the field is set to Null, the method clears the Null. The method throws an `EmdqException` if the data type for the field is not compatible with the Date data type.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).
fieldData [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.4 setDateTimeData

Use the `setDateTimeData` method to set the data of the data record field.

Syntax

```
void setDateTimeData(int fieldIndex, Calendar fieldData)
```

`setDateTimeData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setDateTimeData` method sets the data of the data record field specified in the [fieldIndex](#) parameter. If the field is set as null, the method clears the Null.

If the field data type is not compatible with the DateTime data type, the method throws an `EmdqException`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to set (0-based).
<i>fieldData</i> [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.5 setTimeData

Use the `setTimeData` method to set the data of the data record field.

Syntax

```
void setTimeData(int fieldIndex, Calendar fieldData)
```

`setTimeData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setDateTimeData` method sets the data of the data record field specified in the *fieldIndex* parameter. If the field is set as null, the method clears the Null.

If the field data type is not compatible with the Time data type, the method throws an `EmdqException`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to set (0-based).
<i>fieldData</i> [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.6 setDoubleData

Use the `setDoubleData` method to set the data of the data record field.

Syntax

```
void setDoubleData(int fieldIndex, double fieldData)
```

`setDoubleData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setDoubleData` method sets the data of the data record field specified in the [fieldIndex](#) parameter. If the field is set as null, the method clears the Null.

If the field data type is not compatible with the Double data type, the method throws an `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).
fieldData [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.7 setIntegerData

Use the `setIntegerData` method to set the data of the data record field.

Syntax

```
void setIntegerData(int fieldIndex, int fieldData)
```

`setIntegerData` is a part of the `InputDataRecord` class for Java development.

Return value

The `setIntegerData` method sets the data of the data record field specified in the [fieldIndex](#) parameter. If the field is set as null, the method clears the Null.

If the field data type is not compatible with the Integer data type, the method throws an `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).
fieldData [IN]	The field data to set.

Related Information

[Class InputDataRecord \[page 135\]](#)

7.4.8 setFieldNull

Use the `setFieldNull` method to set the data record field to Null.

Syntax

```
void setFieldNull(int fieldIndex)
```

`setFieldNull` is a part of the `InputDataRecord` class for Java development.

Return value

The `setFieldNull` method sets the data of the data record field specified in the [fieldIndex](#) parameter to Null. If the method is not successful, it throws an `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to set (0-based).

Related Information

[Class InputDataRecord](#) [page 135]

7.5 Class MessageHandler

Class `MessageHandler` is a call back class to handle messages from a transform.

Class `MessageHandler` is required for processing. Your application implements the `MessageHandler` interface. The `MessageHandler` class has the following methods for Java development:

- `MessageHandler()`
- `abstract boolean handleMessage(MessageType type, String messageId, String message)`

[MessageHandler](#) [page 142]

Use the `MessageHandler` method before you use the extended class.

[handleMessage](#) [page 143]

Use the `handleMessage` method to handle log messages produced by SAP Data Quality Management SDK.

7.5.1 MessageHandler

Use the `MessageHandler` method before you use the extended class.

≡ Syntax

```
MessageHandler()
```

`MessageHandler` is a part of the `MessageHandler` class for Java development.

The `MessageHandler` method is the protected scope constructor.

Related Information

[Class `MessageHandler` \[page 142\]](#)

7.5.2 `handleMessage`

Use the `handleMessage` method to handle log messages produced by SAP Data Quality Management SDK.

Syntax

```
abstract boolean handleMessage(MessageType type, String messageId, String message)
```

`handleMessage` is a part of the `MessageHandler` class for Java development.

Set a delegate object on `TransformFactory` using the `LogHandler` property.

Return value

The `handleMessage` method returns `True` when it successfully handles logs for a produced message. It returns `False` and stops processing when it does not successfully handle logs for a produced message.

Where

Parameter	Description
<i>type</i>	The type of message to handle.
<i>messageId</i>	The ID of the message to handle.
<i>message</i>	The message to handle.

Related Information

[Class `MessageHandler` \[page 142\]](#)

7.6 Class MultiRecordTransform

Class `MultiRecordTransform` is the record processing Transform class for processing multiple records.

The `MultiRecordTransform` class is required for processing. Some of the methods listed for this class are inherited from the `Transform` class.

You cannot instantiate instances of class `MultiRecordTransform`. Instead, use the `CreateMultiRecordTransform` method in the `TransformFactory` class to create valid `MultiRecordTransform` class instances.

Use the class `MultiRecordTransform` objects to represent Match.

Class `MultiRecordTransform` has the following methods for Java development:

- `MultiRecordTransformHelper createHelper()`
- `void destroyHelper(MultiRecordTransformHelper helper)`
- `void loadInputDataRecord(InputDataRecord record)`
- `void process()`
- `void setProgressHandler(ProgressHandler handler)`
- `OutputDataRecord unloadOutputDataRecord()`
- `void clearRecords()`
- `ProgressHandler getProgressHandler()`
- `void setStatisticsHandler(StatisticsHandler handler)`
- `String getExtraInfo(String request)`
- `InputDataRecord getInputDataRecord()`
- `StatisticsHandler getStatisticsHandler()`
- `DataRecordSchema getInputSchema()`
- `DataRecordSchema getOutputSchema()`
- `StatisticsSchema getStatisticsSchema(int schemaIndex)`
- `int getStatisticsSchemaCount()`

[createHelper \[page 145\]](#)

Use the `createHelper` method to create a helper for this transform.

[destroyHelper \[page 146\]](#)

Use the `destroyHelper` method to destroy a helper that you no longer need.

[loadInputDataRecord \[page 147\]](#)

Use the `loadInputDataRecord` method to load an input data record into this transform.

[process \[page 147\]](#)

Use the `process` method to process the input data records that are loaded into this transform.

[setProgressHandler \[page 148\]](#)

Use the `setProgressHandler` method to set the progress event handler.

[unloadOutputDataRecord \[page 149\]](#)

Use the `unloadOutputDataRecord` method to unload the next available output data record from this transform.

[clearRecords \[page 150\]](#)

Use the `clearRecords` method to clear all input and output records and ready the transform to process again.

[setStatisticsHandler \[page 150\]](#)

Use the `setStatisticsHandler` to set the statistics event handler to output statistics.

[getExtraInfo \[page 151\]](#)

Use the `getExtraInfo` method to get extra information from this transform.

[getInputDataRecord \[page 152\]](#)

Use the `getInputDataRecord` method to return the input data record that holds the input fields from this transform.

[getStatisticsHandler \[page 153\]](#)

Use the `getStatisticsHandler` to get the current statistics event handler.

[getInputSchema \[page 153\]](#)

Use the `getInputSchema` to get the input data record schema.

[getOutputSchema \[page 154\]](#)

Use the `getOutputSchema` to get the output data record schema.

[getStatisticsSchema \[page 154\]](#)

Use the `getStatisticsSchema` to get the statistics schema specified in the [schemaIndex](#) parameter.

[getStatisticsSchemaCount \[page 155\]](#)

Use the `getStatisticsSchemaCount` method to get the number of statistics schemas defined for this transform.

7.6.1 createHelper

Use the `createHelper` method to create a helper for this transform.

Syntax

```
MultiRecordTransformHelper createHelper()
```

`createHelper` is a part of the `MultiRecordTransform` class for Java development.

The `createHelper` method is not thread safe.

Use a helper to process records. Run a helper in a different thread than this transform. A helper is like a transform in the following ways:

- Helper has all the same settings as a transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `createHelper` method returns a new helper object. If it does not return a new helper object, it throws an `EmdqException`.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.2 destroyHelper

Use the `destroyHelper` method to destroy a helper that you no longer need.

≡ Syntax

```
void destroyHelper(MultiRecordTransformHelper helper)
```

`destroyHelper` is a part of the `MultiRecordTransform` class for Java development.

Before you destroy a transform, ensure that you destroy all helpers of the transform.

Return value

The `destroyHelper` method throws an `EmdqException` when it does not successfully destroy the specified helper.

Where

Parameter	Description
<i>helper</i> [IN]	The helper to be destroyed.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.3 loadInputDataRecord

Use the `loadInputDataRecord` method to load an input data record into this transform.

Syntax

```
void loadInputDataRecord(InputDataRecord record)
```

`loadInputDataRecord` is a part of the `MultiRecordTransform` class for Java development.

The `loadInputDataRecord` method copies the input data record so it is available to use again upon return from this method.

Do not use an input data record from a different transform.

Return value

If the `loadInputDataRecord` method does not successfully load an input data record to this transform, it throws an `EmdqException`.

Where

Parameter	Description
<i>record</i> [IN]	The input data record to load.

Related Information

[Class `MultiRecordTransform` \[page 144\]](#)

7.6.4 process

Use the `process` method to process the input data records that are loaded into this transform.

Syntax

```
void process()
```

`process` is a part of the `MultiRecordTransform` class for Java development.

When the `Process` method returns, the processed input data records become output data records. The output data records, with the posted results, are ready to be unloaded.

Return value

When the `Process` method returns successfully, the output data records with the posted results are ready to be unloaded. The process method throws an `EmdqException` when there is an error.

Related Information

[Class `MultiRecordTransform` \[page 144\]](#)

7.6.5 `setProgressHandler`

Use the `setProgressHandler` method to set the progress event handler.

≡ Syntax

```
void setProgressHandler(ProgressHandler handler)
```

`setProgressHandler` is a part of the `MultiRecordTransform` class for Java development.

The transform calls the progress event handler as it processes the loaded input records. An event handler is optional.

The `setProgressHandler` method saves a shallow copy of the passed-in progress event handler. Your application is responsible to delete the event handler only after you destroy this transform.

Return value

If the `setProgressHandler` method does not successfully set the progress event handler, the method throws an `EmdqException`.

Where

Parameter	Description
<i>handler</i> [IN]	The progress event handler.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.6 unloadOutputDataRecord

Use the `unloadOutputDataRecord` method to unload the next available output data record from this transform.

Syntax

```
OutputDataRecord unloadOutputDataRecord()
```

`unloadOutputDataRecord` is a part of the `MultiRecordTransform` class for Java development.

There is one output data record for each input data record loaded into the transform. An output data record becomes available only after the transform has finished processing the input data record and posted the results to the output data record.

Output records are available for unloading after the application calls `Process`. But a transform can make an output record available for unloading immediately after the call to `LoadInputDataRecord`. Then the application is free to call this method at any time to see if there are any output records available for unloading.

Return value

The `unloadOutputDataRecord` returns the next available output record. It returns `Null` if no output records are available. It throws an `EmdqException` when there is an error.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.7 clearRecords

Use the `clearRecords` method to clear all input and output records and ready the transform to process again.

≡, Syntax

```
void clearRecords()
```

`clearRecords` is a part of the `MultiRecordTransform` class for Java development.

Call the `clearRecords` method only after you load and process your input records, and extract your output.

Return value

The `clearRecords` method throws an `EmdqException` when there is an error.

Related Information

[Class `MultiRecordTransform` \[page 144\]](#)

7.6.8 setStatisticsHandler

Use the `setStatisticsHandler` to set the statistics event handler to output statistics.

≡, Syntax

```
void setStatisticsHandler(StatisticsHandler handler)
```

`setStatisticsHandler` is a part of the `MultiRecordTransform` class for Java development.

Calling `setStatisticsHandler` is optional. The transform automatically outputs statistics when the application calls the method `DestroyTransform`. If omitted, the statistics are not output until the transform terminates.

The `setStatisticsHandler` method saves a shallow copy of the statistics event handler to use for handling statistics events. Ensure that your application does not delete the event handler until you destroy the transform.

Return value

The `setStatisticsHandler` method sets the statistics event handler. It throws an `EmdqException` when there is an error.

Where

Parameter	Description
<code>handler</code> [IN]	The statistics event handler.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.9 getExtraInfo

Use the `getExtraInfo` method to get extra information from this transform.

Syntax

```
String getExtraInfo(String request)
```

`getExtraInfo` is a part of the `MultiRecordTransform` class for Java development.

By default, the `getExtraInfo` method returns an empty string. You can customize a transform to determine valid requests from this method and output appropriate responses as a return from the `getExtraInfo` method.

Example

For example, customize the Global Address Cleanse transform to work with this method and return a list of available address cleanse engines.

Return value

The `getExtraInfo` method returns an empty string. It returns an `EmdqException` when there is an error.

Where

Parameter	Description
<i>request</i> [IN]	The request for the transform.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.10 getInputDataRecord

Use the `getInputDataRecord` method to return the input data record that holds the input fields from this transform.

Syntax

```
InputDataRecord getInputDataRecord()
```

`getInputDataRecord` is a part of the `MultiRecordTransform` class for Java development.

An application can use the input data record to pass data to this transform. The application obtains the input data record only once. However, it can use the same input data record repeatedly.

Return value

The `getInputDataRecord` method returns the input data record that holds the input fields from this transform. It throws an `EmdqException` when there is an error.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.11 getStatisticsHandler

Use the `getStatisticsHandler` to get the current statistics event handler.

≡, Syntax

```
StatisticsHandler getStatisticsHandler()
```

`getStatisticsHandler` is a part of the `MultiRecordTransform` class for Java development.

Return value

The `getStatisticsHandler` method throws an `EmdqException` when there is an error.

Related Information

[Class `MultiRecordTransform` \[page 144\]](#)

7.6.12 getInputSchema

Use the `getInputSchema` to get the input data record schema.

≡, Syntax

```
DataRecordSchema getInputSchema()
```

`getInputSchema` is a part of the `MultiRecordTransform` class for Java development.

Return value

The `getInputSchema` method throws an `EmdqException` when there is an error.

Related Information

[Class `MultiRecordTransform` \[page 144\]](#)

7.6.13 getOutputSchema

Use the `getOutputSchema` to get the output data record schema.

≡, Syntax

```
DataRecordSchema getOutputSchema()
```

`getOutputSchema` is a part of the `MultiRecordTransform` class for Java development.

Return value

The `getOutputSchema` method throws an `EmdqException` when there is an error.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.14 getStatisticsSchema

Use the `getStatisticsSchema` to get the statistics schema specified in the [schemaIndex](#) parameter.

≡, Syntax

```
StatisticsSchema getStatisticsSchema(int schemaIndex)
```

`getStatisticsSchema` is a part of the `MultiRecordTransform` class for Java development.

Return value

The `getStatisticsSchema` method returns the statistics schema specified in [schemaIndex](#). It throws an `EmdqException` when there is an error.

Where

Parameter	Description
schemaIndex [IN]	The index of the statistics schema for which to get information.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.6.15 getStatisticsSchemaCount

Use the `getStatisticsSchemaCount` method to get the number of statistics schemas defined for this transform.

≡ Syntax

```
int getStatisticsSchemaCount()
```

`getStatisticsSchemaCount` is a part of the `MultiRecordTransform` class for Java development.

Return value

The `getStatisticsSchemaCount` method returns 0 when statistics are not enabled, or when the transform does not provide statistics. It throws an `EmdqException` when there is an error.

Related Information

[Class MultiRecordTransform \[page 144\]](#)

7.7 Class MultiRecordTransformHelper

Class `MultiRecordTransformHelper` is the helper class for the `multiRecordTransform` class

Constructor—Destructor for `MultiRecordTransformHelper`

Default Constructor	<code>MultiRecordTransformHelper()</code>
Default Destructor	<code>virtual ~MultiRecordTransformHelper()</code>

The class `MultiRecordTransformHelper` has the following methods for Java development:

- `void loadInputDataRecord(InputDataRecord record)`
- `void process()`
- `OutputDataRecord unloadOutputDataRecord()`
- `void clearRecords()`
- `InputDataRecord getInputDataRecord()`

[loadInputDataRecord \[page 156\]](#)

Use the `loadInputDataRecord` method to load an input data record into this transform.

[process \[page 157\]](#)

Use the `process` method to process the input data records that are loaded into this transform.

[unloadOutputDataRecord \[page 158\]](#)

Use the `unloadOutputDataRecord` method to unload the next available output data record from this transform.

[clearRecords \[page 159\]](#)

Use the `clearRecords` method to clear all input and output records and make this transform ready to process again.

[getInputDataRecord \[page 159\]](#)

Use the `getInputDataRecord` method to get the input data record that contains input fields for this transform.

7.7.1 loadInputDataRecord

Use the `loadInputDataRecord` method to load an input data record into this transform.

≡ Syntax

```
void loadInputDataRecord(InputDataRecord record)
```

`loadInputDataRecord` is a part of the `MultiRecordTransformHelper` class for Java development.

The method saves a copy of the input data record so the record is available for use upon return from the method. The method cannot process an input data record owned by a different transform.

Return value

The `loadInputDataRecord` method returns an input data record. It throws an `EmdqException` when there is an error.

⚠ Caution

Do not use this method on an input data record owned by a different transform.

Where

Parameter	Description
record [IN]	The input data record to load.

Related Information

[Class `MultiRecordTransformHelper` \[page 156\]](#)

7.7.2 process

Use the `process` method to process the input data records that are loaded into this transform.

≡ Syntax

```
void process()
```

`process` is a part of the `MultiRecordTransformHelper` class for Java development.

Return value

The `process` method processes the input data records that are loaded into the transform. After this method returns, the output data records with the posted results are ready to be unloaded. The method throws an `EmdException` when there is an error.

Related Information

[Class MultiRecordTransformHelper \[page 156\]](#)

7.7.3 unloadOutputDataRecord

Use the `unloadOutputDataRecord` method to unload the next available output data record from this transform.

Syntax

```
OutputDataRecord unloadOutputDataRecord()
```

`unloadOutputDataRecord` is a part of the `MultiRecordTransformHelper` class for java development.

This method requires that there be one output data record for each input data record that the transform loads. An output data record becomes available after the transform has finished processing the input data record and has posted results to the output data record.

Output records are available for unloading after the application calls `process`. But it is possible for a transform to make an output record available for unloading immediately after the call to `LoadInputDataRecord`. The application is free to call this method at any time to see if there are any output records available for unloading.

Once the transform calls `process`, the transform unloads all current output records before it loads any new input records for processing.

Return value

The `unloadOutputDataRecord` method returns the next available output record if available. It returns `0` when no records are available. It throws an `EmdqException` when there is an error.

Related Information

[Class MultiRecordTransformHelper \[page 156\]](#)

7.7.4 clearRecords

Use the `clearRecords` method to clear all input and output records and make this transform ready to process again.

Syntax

```
void clearRecords()
```

`clearRecords` is a part of the `MultiRecordTransformHelper` class for Java development.

Call this method only after you have loaded your input records and processed and extracted your output records.

Throws `EmdqException`.

Return value

The `clearRecords` readies the transform to process again. It throws an `EmdqException` when there is an error.

Related Information

[Class `MultiRecordTransformHelper` \[page 156\]](#)

7.7.5 getInputDataRecord

Use the `getInputDataRecord` method to get the input data record that contains input fields for this transform.

Syntax

```
InputDataRecord getInputDataRecord()
```

`getInputDataRecord` is a part of the `MultiRecordTransformHelper` class for Java development.

Your application can use the returned input data record to pass data to this transform.

Return value

The `getInputDataRecord` method returns the input data record that contains the input fields for the transform. Your application can use the returned input data record to pass data to the transform. An application gets the input data record only once. It can then use the same input data record multiple times.

⚠ Caution

Do not use this method on an input data record owned by a different transform.

Related Information

[Class MultiRecordTransformHelper \[page 156\]](#)

7.8 Class OutputDataRecord

Class `OutputDataRecord` is the main interface to the Output Data Record functionality.

Class `OutputDataRecord` inherits its methods from the superclass `DataRecord`. This class is required for processing.

Class `OutputDataRecord` has the following methods for Java development:

- `OutputDataRecord(long internalPtr)`
- `String getStringData(int fieldIndex)`
- `Calendar getDateData(int fieldIndex)`
- `Calendar getTimeData(int fieldIndex)`
- `Calendar getDateTimeData(int fieldIndex)`
- `double getDoubleData(int fieldIndex)`
- `int getIntData(int fieldIndex)`
- `boolean isFieldNull(int fieldIndex)`
- `int getFieldDataLength(int fieldIndex)`

[OutputDataRecord\(long internalPtr\) \[page 161\]](#)

The `OutputDataRecord(long internalPtr)` method is the package scope constructor of the `OutputDataRecord` class.

[getStringData \[page 162\]](#)

The `getStringData` gets the field specified in *fieldIndex* and creates and returns a new string that contains the data from the field.

[getDateData \[page 162\]](#)

Use the `getDateData` method to get the date from the field specified in *fieldIndex* and return it as a copy in a calendar object.

[getTimeData \[page 163\]](#)

Use the `getTimeData` method to get the time data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

[getDateTimeData \[page 164\]](#)

Use the `getDateTimeData` method to get the datetime data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

[getDoubleData \[page 165\]](#)

Use the `getDoubleData` method to get the double data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

[getIntData \[page 166\]](#)

Use the `getIntData` method to get the integer data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

[isFieldNull \[page 167\]](#)

Use the `isFieldNull` method to determine if the field specified in the *fieldIndex* parameter is Null.

[getFieldDataLength \[page 167\]](#)

Use the `getFieldDataLength` method to get the length of the data in the field specified by the *fieldIndex* parameter.

7.8.1 OutputDataRecord(long internalPtr)

The `OutputDataRecord(long internalPtr)` method is the package scope constructor of the `OutputDataRecord` class.

Syntax

```
OutputDataRecord(long internalPtr)
```

Where

Parameter	Description
<i>internalPtr</i>	The long value representing C++ pointer to <code>OutputDataRecord</code> object in native code.

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.2 getStringData

The `getStringData` gets the field specified in *fieldIndex* and creates and returns a new string that contains the data from the field.

≡ Syntax

```
String getStringData(int fieldIndex)
```

`getStringData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getStringData` method returns the field data. It returns `Null` if there is no data. It throws an `EmdqException` error when there is an error.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class `OutputDataRecord` \[page 160\]](#)

7.8.3 getDateData

Use the `getDateData` method to get the date from the field specified in *fieldIndex* and return it as a copy in a calendar object.

≡ Syntax

```
Calendar getDateData(int fieldIndex)
```

`getDateData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getDateData` method returns date data from the field specified in [fieldIndex](#). If the data is not compatible with the Date data type, or if there is an error, the method throws an `EmdqException`.

If the data record field is Null, there is no processing.

→ Tip

To determine if the field is Null before you run `getDateData`, run `isFieldNull`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord](#) [page 160]

7.8.4 getTimeData

Use the `getTimeData` method to get the time data from the field specified in [fieldIndex](#) and return it as a copy in a calendar object.

≡ Syntax

```
Calendar getTimeData(int fieldIndex)
```

`getTimeData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getTimeData` method returns the time data from the specified field. If the data is not compatible with the Time data type, or if there is an error, the method throws an `EmdqException`.

If the data record field is Null, there is no processing.

→ Tip

To determine if the field is Null before you run `getTimeData`, run `isFieldNull`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.5 getDateTimeData

Use the `getDateTimeData` method to get the datetime data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

≡ Syntax

```
Calendar getDateTimeData(int fieldIndex)
```

`getDateTimeData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getDateTimeData` method returns datetime data from the field specified in *fieldIndex*. If the data is not compatible with the `DateTime` data type, or if there is an error, the method throws an `EmdqException`.

If the data record field is Null, there is no processing.

→ Tip

To determine if the field is Null before you run `getDateTimeData`, run `isFieldNull`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.6 getDoubleData

Use the `getDoubleData` method to get the double data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

Syntax

```
double getDoubleData(int fieldIndex)
```

`getDoubleData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getDoubleData` method returns date data from the field specified in *fieldIndex*. If the data is not compatible with the Double data type, or if there is an error, the method throws an `EmdqException`.

If the data record field is Null, there is no processing.

Tip

To determine if the field is Null before you run `getDoubleData`, run `isFieldNull`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.7 getIntData

Use the `getIntData` method to get the integer data from the field specified in *fieldIndex* and return it as a copy in a calendar object.

≡, Syntax

```
int getIntData(int fieldIndex)
```

`getIntData` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getIntData` method returns integer data from the field specified in *fieldIndex*. If the data is not compatible with the `Int` data type, or if there is an error, the method throws an `EmdqException`.

If the data record field is `Null`, there is no processing.

→ Tip

To determine if the field is `Null` before you run `getIntData`, run `isFieldNull`.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.8 isFieldNull

Use the `isFieldNull` method to determine if the field specified in the [fieldIndex](#) parameter is Null.

Syntax

```
boolean isFieldNull(int fieldIndex)
```

`isFieldNull` is part of the `OutputDataRecord` class for Java development.

Return value

The `isFieldNull` method returns `True` if the field is Null. It returns `False` if the field is not Null. If there is an error, it throws an `EmdqException`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to get (0-based).

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.8.9 getFieldDataLength

Use the `getFieldDataLength` method to get the length of the data in the field specified by the [fieldIndex](#) parameter.

Syntax

```
int getFieldDataLength(int fieldIndex)
```

`getFieldDataLength` is a part of the `OutputDataRecord` class for Java development.

Return value

The `getFieldDataLength` method returns the number of characters in the field specified by the [fieldIndex](#) parameter. If [fieldIndex](#) is invalid, it returns -1. It throws an `EmdqException` when there are errors.

→ Tip

To determine if the field is Null before you run `getFieldDataLength`, run `isFieldNull`.

Where

Parameter	Description
fieldIndex [IN]	The index of the field to get (0-based).
output [OUT]	The length of the data in the field.

Related Information

[Class OutputDataRecord \[page 160\]](#)

7.9 Class ProgressHandler

Class `ProgressHandler` is a callback class to show the progress of a multirecord transform and to enable the handler to end processing.

The `ProgressHandler` class has the following methods for Java development:

- `protected abstract boolean handleProgress(double percentDone)`
- `boolean setProgressInterval(int interval)`
- `int getProgressInterval()`

[handleProgress \[page 169\]](#)

Use the `handleProgress` method to get the percentage of completion for the current set of records being processed.

[setProgressInterval \[page 169\]](#)

Use the `setProgressInterval` method to specify the interval that the transform waits between calls to progress.

[getProgressInterval \[page 170\]](#)

Use the `getProgressInterval` method to get the number of seconds in the current progress interval.

7.9.1 handleProgress

Use the `handleProgress` method to get the percentage of completion for the current set of records being processed.

≡ Syntax

```
protected abstract boolean handleProgress(double percentDone)
```

`handleProgress` is a part of the `ProgressHandler` class for Java development.

Return value

The `handleProgress` method returns `True` when the percentage of completion is not 100%. It returns `False` and stops processing when the percentage reaches 100%.

Where

Parameter	Description
<code>percentDone</code> [IN]	The percent done (0.0 - 100.0).

Related Information

[Class `ProgressHandler`](#) [page 168]

7.9.2 setProgressInterval

Use the `setProgressInterval` method to specify the interval that the transform waits between calls to `progress`.

≡ Syntax

```
boolean setProgressInterval(int interval)
```

`setProgressInterval` is a part of the `ProgressHandler` class for Java development.

Return value

The `setProgressInterval` method throws an `EmdqException` when there is an error.

Where

Parameter	Description
<i>interval</i> [IN]	The number of seconds to wait between calls to <code>progress</code> . Enter a value greater than 0.

Related Information

[Class `ProgressHandler` \[page 168\]](#)

7.9.3 `getProgressInterval`

Use the `getProgressInterval` method to get the number of seconds in the current progress interval.

≡ Syntax

```
int getProgressInterval()
```

`getProgressInterval` is a part of the `ProgressHandler` class for Java development.

Return value

The `getProgressInterval` method returns the number of seconds in the current progress interval. It throws an `EmdqException` when there is an error.

Related Information

[Class `ProgressHandler` \[page 168\]](#)

7.10 Class RecordTransform

Class `RecordTransform` is the record processing transform class for processing single records.

Class `RecordTransform` is required for processing. Some of the class methods are inherited from the `Transform` class.

Instances of `RecordTransform` cannot be instantiated. Instead, use the `CreateRecordTransform` methods in the `TransformFactory` class to create valid `RecordTransform` instances.

Use class `RecordTransform` objects to represent Data Cleanse, USA Regulatory Address Cleanse, Global Address Cleanse, and Geocoder transforms.

The `RecordTransform` class has the following methods for Java development:

- `RecordTransform(long internalPtr)`
- `RecordTransformHelper createHelper()`
- `void destroyHelper(RecordTransformHelper helper)`
- `OutputDataRecord process(InputDataRecord record)`
- `void setStatisticsHandler(StatisticsHandler handler)`
- `String getExtraInfo(String request)`
- `InputDataRecord getInputDataRecord()`
- `StatisticsHandler getStatisticsHandler()`
- `DataRecordSchema getInputSchema()`
- `DataRecordSchema getOutputSchema()`
- `StatisticsSchema getStatisticsSchema(int schemaIndex)`
- `int getStatisticsSchemaCount()`

[RecordTransform\(long internalPtr\) \[page 172\]](#)

The `RecordTransform(long internalPtr)` method is the package scoped constructor for `RecordTransform`.

[createHelper \[page 173\]](#)

Use the `createHelper` method to create a helper for this transform.

[destroyHelper \[page 173\]](#)

Use the `destroyHelper` method to destroy a helper that you no longer need.

[process \[page 174\]](#)

Use the `process` method to process the input data record for this transform.

[setStatisticsHandler \[page 175\]](#)

Use the `setStatisticsHandler` method to set the statistics event handler to output statistics.

[getExtraInfo \[page 176\]](#)

Use the `getExtraInfo` method to get extra information from a transform.

[getInputDataRecord \[page 177\]](#)

Use the `getInputDataRecord` method to get the input data record that contains the input fields for this transform.

[getStatisticsHandler \[page 178\]](#)

Use the `getStatisticsHandler` method to return a pointer to the current statistics event handler.

[getInputSchema \[page 178\]](#)

Use the `getInputSchema` method to return the schema of the input data record.

[getOutputSchema \[page 179\]](#)

Use the `getOutputSchema` method to return the schema of the output data record.

[getStatisticsSchema \[page 179\]](#)

Use the `getStatisticsSchema` method to return the statistics schema related to the index value in the *schemaIndex* parameter.

[getStatisticsSchemaCount \[page 180\]](#)

Use the `getStatisticsSchemaCount` method to return the number of statistics schemas defined for this transform.

7.10.1 RecordTransform(long internalPtr)

The `RecordTransform(long internalPtr)` method is the package scoped constructor for `RecordTransform`.

Syntax

```
RecordTransform(long internalPtr)
```

Call `RecordTransform(long internalPtr)` only from the `TransformFactory`.

Where

Parameter	Description
<code>internalPt</code>	The pointer representing the C++ pointer to the C++ <code>RecordTransform</code> object in native code.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.2 createHelper

Use the `createHelper` method to create a helper for this transform.

≡ Syntax

```
RecordTransformHelper createHelper()
```

`createHelper` is part of the `RecordTransform` class for Java development.

Use a helper to process records. Run a helper in a different thread than this transform. A helper is like a transform in the following ways:

- Helper has all the same settings as a transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `createHelper` method returns a newly created helper object. It throws an `EmdqException` when there are errors.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.3 destroyHelper

Use the `destroyHelper` method to destroy a helper that you no longer need.

≡ Syntax

```
void destroyHelper(RecordTransformHelper helper)
```

`destroyHelper` is a part of the `RecordTransform` class for Java development.

Ensure that you run the `destroyHelper` method to destroy all helpers of a transform before you destroy the transform.

Return value

The `destroyHelper` method throws `EmdqException` when there is an error.

Where

Parameter	Description
<code>helper [IN]</code>	The helper to destroy.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.4 process

Use the `process` method to process the input data record for this transform.

≡ Syntax

```
OutputDataRecord process(InputDataRecord record)
```

`process` is a part of the `RecordTransform` class for Java development.

Before you call `process`, call the `getInputDataRecord` method to get the input data record. Ensure that the input data record is loaded with data before you pass it to the `process` method.

Return value

The `process` method returns the output data record. First, the method reads the fields of the input data record and posts results to an output data record. You can query the results in the output data record, however, values in the output data record are valid only until you call `process` again.

⚠ Caution

Do not process an input data record owned by a different transform.

The method returns `Null` for non-fatal errors. It throws an `EmdqException` when there is a fatal error.

Where

Parameter	Description
record [IN]	The input data record to process.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.5 setStatisticsHandler

Use the `setStatisticsHandler` method to set the statistics event handler to output statistics.

Syntax

```
void setStatisticsHandler(StatisticsHandler handler)
```

`setStatisticsHandler` is a part of the `RecordTransform` class for Java development.

A statistics event handler is optional. However, if you do not include the `setStatisticsHandler` method, the transform does not output statistics.

The transform calls a statistics event handler to output statistics. The transform also outputs statistics when it is destroyed.

Note

When you destroy a transform using the `destroyTransform` method, the transform may pass final statistics to the statistics event handler. See [destroyTransform \[page 190\]](#) for more information.

The `setStatisticsHandler` method saves a shallow copy of the passed-in statistics event handler. Your application is responsible for deleting the copy. Ensure that your application does not delete the copy of the passed-in statistics handler until you destroy this transform.

Return value

The `setStatisticsHandler` throws an `EmdqException` when there are errors.

Where

Parameter	Description
handler [IN]	The statistics event handler to call.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.6 getExtraInfo

Use the `getExtraInfo` method to get extra information from a transform.

Syntax

```
String getExtraInfo(String request)
```

`getExtraInfo` is a part of the `RecordTransform` class for Java development.

Return value

The `getExtraInfo` method returns an empty string by default. However, populate the empty string by customizing a transform to determine valid requests from this method and define appropriate responses. The information can be returned using the `getExtraInfo` method.

Example

Customize the Global Address Cleanse transform to work with this method and return a list of available address cleanse engines.

The `getExtraInfo` method throws an `EmdqException` when there are errors.

Where

Parameter	Description
request [IN]	The request for the transform.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.7 getInputDataRecord

Use the `getInputDataRecord` method to get the input data record that contains the input fields for this transform.

Syntax

```
InputDataRecord getInputDataRecord()
```

`getInputDataRecord` is a part of the `RecordTransform` class for Java development.

An application can use the input data record to pass data to this transform. The application only has to get the input record once. Then it can use the input record many times.

Return value

The `getInputDataRecord` method returns the input data record that contains the input fields for this transform. It throws an `EmdqException` when there are errors.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.8 getStatisticsHandler

Use the `getStatisticsHandler` method to return a pointer to the current statistics event handler.

≡ Syntax

```
StatisticsHandler getStatisticsHandler()
```

`getStatisticsHandler` is a part of the `RecordTransform` class for Java development.

Return value

The `getStatisticsHandler` method returns the current statistics event handler. It throws an `EmdqException` when there are errors.

Related Information

[Class `RecordTransform` \[page 171\]](#)

7.10.9 getInputSchema

Use the `getInputSchema` method to return the schema of the input data record.

≡ Syntax

```
DataRecordSchema getInputSchema()
```

`getInputSchema` is a part of the `RecordTransform` class for Java development.

Return value

the `getInputSchema` method returns the schema of the input data record. It throws an `EmdqException` when there are errors.

Related Information

[Class `RecordTransform` \[page 171\]](#)

7.10.10 getOutputSchema

Use the `getOutputSchema` method to return the schema of the output data record.

≡ Syntax

```
DataRecordSchema getOutputSchema()
```

`getOutputSchema` is a part of the `RecordTransform` class for Java development.

Return value

The `getOutputSchema` method returns the schema of the output data record. It throws an `EmdqException` when there are errors.

Related Information

[Class RecordTransform \[page 171\]](#)

7.10.11 getStatisticsSchema

Use the `getStatisticsSchema` method to return the statistics schema related to the index value in the *schemaIndex* parameter.

≡ Syntax

```
StatisticsSchema getStatisticsSchema(int schemaIndex)
```

`getStatisticsSchema` is a part of the `RecordTransform` class for Java development.

Return value

The `getStatisticsSchema` method returns the statistics schema. It throws an `EmdqException` when there are errors.

Where

Parameter	Description
schemaIndex [IN]	The index of the statistics schema for which to get information (0-based).

Related Information

[Class RecordTransform](#) [page 171]

7.10.12 getStatisticsSchemaCount

Use the `getStatisticsSchemaCount` method to return the number of statistics schemas defined for this transform.

≡ Syntax

```
int getStatisticsSchemaCount()
```

`getStatisticsSchemaCount` is a part of the `RecordTransform` class for Java development.

Return value

The `getStatisticsSchemaCount` method returns the number of statistics schemas defined for this transform. If statistics are not enabled, or if the transform does not provide statistics, it returns 0. It throws an `EmdqException` when there are errors.

Related Information

[Class RecordTransform](#) [page 171]

7.11 Class RecordTransformHelper

Class `RecordTransformHelper` is the helper class for the record processing transform class.

Class `RecordTransformHelper` contains the following methods for Java development:

- `OutputDataRecord process(InputDataRecord record)`
- `InputDataRecord getInputDataRecord()`

[process \[page 181\]](#)

Use the `process` method to process the input data record owned by this transform.

[getInputDataRecord \[page 182\]](#)

Use the `getInputDataRecord` method to obtain the input data record that contains the input fields for this transform.

7.11.1 process

Use the `process` method to process the input data record owned by this transform.

≡ Syntax

```
OutputDataRecord process(InputDataRecord record)
```

`process` is a part of the `RecordTransformHelper` class for Java development.

Obtain the input data record by calling the `getInputDataRecord` method. Ensure that the input data record is loaded with data before you pass it to the `process` method.

Return value

The `process` method reads the fields of the input data record and post results to an output data record. The output data record can be queried. It throws an `EmdqException` when there are errors.

Where

Parameter	Description
record [IN]	The input data record to process.

Related Information

[Class RecordTransformHelper \[page 181\]](#)

7.11.2 getInputDataRecord

Use the `getInputDataRecord` method to obtain the input data record that contains the input fields for this transform.

≡ Syntax

```
InputDataRecord getInputDataRecord()
```

`getInputDataRecord` is a part of the `RecordTransformHelper` class for Java development.

An application can use the input data record to pass data to this transform. Your application obtains the input data record only once. However, your application can use the same input data record multiple times.

Return value

The `getInputDataRecord` method returns the input data record from this transform. It throws an `EmdqException` when there are errors.

Related Information

[Class RecordTransformHelper \[page 181\]](#)

7.12 Class StatisticsHandler

Class `StatisticsHandler` is a callback class to handle statistics records.

Your application implements the `StatisticsHandler` interface.

The class `StatisticsHandler` has the following methods for Java development:

- `abstract boolean handleStatistics(OutputDataRecord record);`
- `int getRecordsRemainingCount();`
- `StatisticsSchema getStatisticsSchema();`

[handleStatistics \[page 183\]](#)

Use the `handleStatistics` method to pass a pointer to an output record that holds statistics information.

[getRecordsRemainingCount \[page 184\]](#)

Use the `getRecordsRemainingCount` method to get the number of output records that remain to be passed to the `Output ( )` method.

[getStatisticsSchema \[page 184\]](#)

Use the `getStatisticsSchema` method to get the statistics schema.

7.12.1 handleStatistics

Use the `handleStatistics` method to pass a pointer to an output record that holds statistics information.

Syntax

```
abstract boolean handleStatistics(OutputDataRecord record);
```

`handleStatistics` is a part of the `StatisticsHandler` class for Java development.

Program your application to query the record to find the statistics table to which the record belongs. Do not save the record pointer passed to this method because it becomes invalid after this method returns. Ensure that your application saves any field data from the record that it intends to keep.

Return value

The `handleStatistics` method returns `True` when it successfully passes a pointer to an output record that holds statistics. It returns `False`, produces an error, and stops processing when it is not successful.

Where

Parameter	Description
<i>record [IN]</i>	The output statistics record to output.

Related Information

[Class StatisticsHandler \[page 182\]](#)

7.12.2 getRecordsRemainingCount

Use the `getRecordsRemainingCount` method to get the number of output records that remain to be passed to the `Output()` method.

≡ Syntax

```
int getRecordsRemainingCount()
```

`getRecordsRemainingCount` is a part of the `StatisticsHandler` class for Java development.

If the transform has a block of records to send, the transform calls `setRecordsRemainingCount` before each call to `Output()` to indicate how many records are left to send. The application can use this information to buffer the records instead of processing them individually.

Return value

The `getRecordsRemainingCount` method returns the number of additional records ready to be output.

Related Information

[Class `StatisticsHandler` \[page 182\]](#)

7.12.3 getStatisticsSchema

Use the `getStatisticsSchema` method to get the statistics schema.

≡ Syntax

```
StatisticsSchema getStatisticsSchema()
```

`getStatisticsSchema` is a part of the `StatisticsHandler` class for Java development.

Return value

The `getStatisticsSchema` method returns the statistics schema.

Related Information

[Class StatisticsHandler](#) [page 182]

7.13 Class StatisticsSchema

Class `StatisticsSchema` defines the layout of a statistics table.

The `StatisticsSchema` class has the following methods for Java development:

- `DataType getFieldDataType(int fieldIndex)`
- `String getTableName()`
- `boolean allowNull(int fieldIndex)`
- `boolean isPrimaryKey(int fieldIndex)`

[getFieldDataType](#) [page 185]

Use the `getFieldDataType` method to get the data type for the specified field.

[getTableName](#) [page 186]

Use the `getTableName` method to get the name of the table that the schema describes.

[allowNull](#) [page 186]

Use the `allowNull` method to indicate whether the specified field allows a Null value or that it is invalid.

[isPrimaryKey](#) [page 187]

Use the `isPrimaryKey` method to indicate whether the specified field is a primary key.

7.13.1 getFieldDataType

Use the `getFieldDataType` method to get the data type for the specified field.

≡ Syntax

```
DataType getFieldDataType(int fieldIndex)
```

`getFieldDataType` is a part of the `StatisticsSchema` class for Java development.

Return value

The `getFieldDataType` method returns the data type of the field specified in the [fieldIndex](#) parameter. It throws an `EmdqException` when there are errors.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information.

Related Information

[Class StatisticsSchema \[page 185\]](#)

7.13.2 getTableName

Use the `getTableName` method to get the name of the table that the schema describes.

≡ Syntax

```
String getTableName()
```

`getTableName` is a part of the `StatisticsSchema` class for Java development.

Return value

the `getTableName` method returns the name of the table that the schema describes. It throws and `EmdqException` when there are errors.

Related Information

[Class StatisticsSchema \[page 185\]](#)

7.13.3 allowNull

Use the `allowNull` method to indicate whether the specified field allows a Null value or that it is invalid.

≡ Syntax

`allowNull` is a part of the `StatisticsSchema` class for Java development.

Return value

The `allowNull` method returns `True` if the field specified by the [fieldIndex](#) parameter allows a `Null` value. It also returns `True` if the field specified by the [fieldIndex](#) parameter is invalid. It returns `False` if the field does not allow a `Null` value or if the field is valid. It throws an `EmdqException` when there are errors.

Where

Parameter	Description
fieldIndex [IN]	The index of the field for which to get information (0-based).

Related Information

[Class StatisticsSchema](#) [page 185]

7.13.4 isPrimaryKey

Use the `isPrimaryKey` method to indicate whether the specified field is a primary key.

≡ Syntax

```
boolean isPrimaryKey(int fieldIndex)
```

`isPrimaryKey` is a part of the `StatisticsSchema` class for Java development.

Return value

The `isPrimaryKey` method returns `True` if the field specified in the [fieldIndex](#) parameter is a primary key. It returns `False` if the field is not a primary key. It throws an `EmdqException` when there are errors.

Where

Parameter	Description
<i>fieldIndex</i> [IN]	The index of the field for which to get information (0-based).

Related Information

[Class StatisticsSchema \[page 185\]](#)

7.14 Class TransformFactory

Class `TransformFactory` is the class to use to create transforms.

Class `TransformFactory` is required for processing. The `TransformFactory` class has the following methods for Java development:

- `synchronized MultiRecordTransform createMultiRecordTransform (String transformSettings)`
- `synchronized RecordTransform createRecordTransform(String transformSettings)`
- `void destroyTransform(Transform transform)`
- `synchronized String upgradeTransformSettings(String transformSettings)`
- `synchronized boolean validateTransformSettings(String transformSettings)`
- `void setMessageHandler(MessageHandler logEventHandler)`
- `public void setLocale(String locale)`
- `public MessageHandler getMessageHandler()`
- `String getLocale()`
- `static const char* getVersion();`

[createMultiRecordTransform \[page 189\]](#)

Use the `createMultiRecordTransform` method to create a multirecord transform using XML from the *transformSettings* parameter.

[createRecordTransform \[page 190\]](#)

Use the `createRecordTransform` method to create a record transform using the XML from the *transformSettings* parameter.

[destroyTransform \[page 190\]](#)

Use the `destroyTransform` method to delete a specified record transform or a multirecord transform.

[upgradeTransformSettings \[page 191\]](#)

Use the `upgradeTransformSettings` method to upgrade the transform XML settings in the specified in the *transformSettings* parameter.

[validateTransformSettings \[page 192\]](#)

Use the `validateTransformSettings` method to validate the XML for the transform specified in the *transformSettings* parameter.

[setMessageHandler \[page 193\]](#)

Use the `setMessageHandler` method to set the log event handler.

[createLocale \[page 193\]](#)

Use the `createLocale` method to set the locale for messages that this transform produces.

[getMessageHandler \[page 194\]](#)

Use the `getMessageHandler` method to get the current log event handler.

[getLocale \[page 194\]](#)

Use the `getLocale` method to get the current local.

[getVersion \[page 195\]](#)

Use the `getVersion` method to get the version of the SAP Data Quality Management SDK that you are using.

7.14.1 createMultiRecordTransform

Use the `createMultiRecordTransform` method to create a multirecord transform using XML from the *transformSettings* parameter.

≡ Syntax

```
synchronized MultiRecordTransform createMultiRecordTransform (String  
transformSettings)
```

`createMultiRecordTransform` is part of the `TransformFactory` class for Java development.

Return value

The `createMultiRecordTransform` method creates a multirecord transform and returns a handle to the created multirecord transform. If there are errors, the method throws an `EmdqException`.

Where

Parameter	Description
transformSettings [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.2 createRecordTransform

Use the `createRecordTransform` method to create a record transform using the XML from the *transformSettings* parameter.

≡, Syntax

```
synchronized RecordTransform createRecordTransform(String transformSettings)
```

`createRecordTransform` is part of the `TransformFactory` class for Java development.

Return value

The `createRecordTransform` method creates a record transform and returns a handle to the created record transform. If there are errors, it throws an `EmdqException`.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.3 destroyTransform

Use the `destroyTransform` method to delete a specified record transform or a multirecord transform.

≡, Syntax

```
void destroyTransform(Transform transform)
```

`destroyTransform` is part of the `TransformFactory` class for Java development.

Return value

The `destroyTransform` method destroys a record transform or a multirecord transform. The method may cause the transform to pass final statistics to the event handler.

Only call the `destroyTransform` method when you are finished using the transform. If there are errors, the method throws an `EmdqException`.

Where

Parameter	Description
<i>transform</i> [IN]	The transform to destroy.

Related Information

[Class `TransformFactory` \[page 188\]](#)

7.14.4 `upgradeTransformSettings`

Use the `upgradeTransformSettings` method to upgrade the transform XML settings in the specified in the [*transformSettings*](#) parameter.

≡, Syntax

```
synchronized String upgradeTransformSettings(String transformSettings)
```

`upgradeTransformSettings` is part of the `TransformFactory` class for Java development.

Return value

The `upgradeTransformSettings` method returns the upgraded XML settings as a string. If there are errors, the method throws an `EmdqException`.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.5 validateTransformSettings

Use the `validateTransformSettings` method to validate the XML for the transform specified in the *transformSettings* parameter.

≡ Syntax

```
synchronized boolean validateTransformSettings(String transformSettings)
```

`validateTransformSettings` is part of the `TransformFactory` class for Java development.

Return value

The `validateTransformSettings` method returns `True` when the transform settings do not have errors. It returns `False` if the transform settings have errors. If there are other errors, it throws an `EmdqException`.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.6 setMessageHandler

Use the `setMessageHandler` method to set the log event handler.

≡, Syntax

```
void setMessageHandler(MessageHandler logEventHandler)
```

`setMessageHandler` is part of the `TransformFactory` class for Java development.

The transform calls the log event handler whenever it outputs log information. A log event handler is required.

Where

Parameter	Description
logEventHandler [IN]	The log event handler to set.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.7 createLocale

Use the `createLocale` method to set the locale for messages that this transform produces.

≡, Syntax

```
public void setLocale(String locale)
```

`createLocale` is part of the `TransformFactory` class for Java development.

Return value

The `createLocale` method sets the locale.

If the locale is not supported, the transform logs a warning to the `MessageHandler` class using `setMessageHandler` and all messages default to `en_US`.

Where

Parameter	Description
<i>locale</i> [IN]	The locale to use.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.8 getMessageHandler

Use the `getMessageHandler` method to get the current log event handler.

≡ Syntax

```
public MessageHandler getMessageHandler()
```

`getMessageHandler` is part of the `TransformFactory` class for Java development.

Return value

The `getMessageHandler` method returns the current log event handler.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.9 getLocale

Use the `getLocale` method to get the current local.

≡ Syntax

```
String getLocale()
```

`getLocale` is part of the `TransformFactory` class for Java development.

Return value

If the locale set by a call to the `setLocale` method is supported, the `getLocale` method returns that value. If the locale set using `SetLocale` is not supported, the `getLocale` method returns the default locale of `en_US`.

Related Information

[Class TransformFactory \[page 188\]](#)

7.14.10 getVersion

Use the `getVersion` method to get the version of the SAP Data Quality Management SDK that you are using.

 Syntax

```
static const char* getVersion();
```

`getVersion` is part of the `TransformFactory` class for Java development.

Return value

The `getVersion` method returns the current Data Quality Management SDK.

Related Information

[Class TransformFactory \[page 188\]](#)

8 API reference for .Net

Each element for building Data Quality transforms in .Net contains specific syntax and elements to use in the syntax.

The namespace for the .Net API is Sap.Emdq.

Ensure that the `<install_location>\<platform>\bin` folder is added to your PATH environment variable before you launch Visual Studio.

In Visual Studio, set the application type within the application integrating this product to one of the following settings:

- 32-bit applications: x86
- 64-bit applications: x64

The default value of “Either” is not sufficient.

By default, logging is not enabled for .Net. To be able to use the logging capability of the methods in .Net API, set a log handler. If you don't set a log handler, the transforms that you create using .Net do not output specific warnings or error messages.

For information about the errors and warnings included in the log files for transforms created using the .Net API, see the applicable method details.

[Message types \[page 197\]](#)

Each transform has message types that you can include in log files.

[Class EmDQException \[page 197\]](#)

Class `EmDQException` is the exception class thrown by all public interfaces of this product.

[Class LogHandler \[page 199\]](#)

Class `LogHandler` is a delegate object that passes messages from the SAP Data Quality Management SDK to the integrating application.

[Class MultiRecordProgressHandler \[page 200\]](#)

Class `MultiRecordProgressHandler` is a delegate object for passing progress information from the SAP Data Quality Management SDK to the integrating application.

[Class MultiRecordTransform \[page 201\]](#)

Class `MultiRecordTransform` is the record processing `Transform` class for processing multiple records.

[Class MultiRecordTransformHelper \[page 207\]](#)

Class `MultiRecordTransformHelper` is a shared resource-based processing object that is a clone of a `MultiRecordTransform` class.

[Class RecordTransform \[page 208\]](#)

Class `RecordTransform` is the record processing `Transform` class for processing single records.

[Class RecordTransformHelper \[page 213\]](#)

Class `RecordTransformHelper` is the helper class for the record processing `Transform` class.

[Class TransformFactory \[page 214\]](#)

Class `TransformFactory` is required to create a transform.

8.1 Message types

Each transform has message types that you can include in log files.

The following table contains the message types for the transforms in SAP Data Quality Management SDK using the .Net API.

Message types for transforms created with the .Net API

Type	Description
ErrorMessage	Error messages
WarningMessage	Warning messages
InfoMessage	Information messages
TraceMessage	Tracing messages

8.2 Class EmDQException

Class `EmDQException` is the the exception class thrown by all public interfaces of this product.

The `EmDQException` class is derived from the `Exception` class. Find the text of the message in the `Message` method. This class is required for processing. Class `EmDQException` has the following methods for .Net development:

- property `System::String^ MessageID`
- property `System::String^ Message`

[MessageID \[page 197\]](#)

Use the `MessageID` method to get the message ID of this exception object.

[Message \[page 198\]](#)

Use the `Message` method to return the message content for this exception object.

8.2.1 MessageID

Use the `MessageID` method to get the message ID of this exception object.

⌵ Syntax

```
property System::String^ MessageID
```

`MessageID` is a part of the `EmDQException` class for .Net development.

Return value

The `MessageID` method returns the message ID of this exception object. The message ID is in the format CCCNNN:

- **CCC**: Alpha characters that represent the source of the error.
- **NNN**: Numeric characters that represent the record number.

❖ Example

For example, REC001 is the first message for the `DataRecord` class.

Related Information

[Class `EmDQException` \[page 197\]](#)

8.2.2 Message

Use the `Message` method to return the message content for this exception object.

≡ Syntax

```
property System::String^ Message
```

`Message` is a part of the `EmDQException` class for .Net development.

The `Message` method is a member of the base class.

Return value

The `Message` method returns the content of the message.

Related Information

[Class `EmDQException` \[page 197\]](#)

8.3 Class LogHandler

Class `LogHandler` is a delegate object that passes messages from the SAP Data Quality Management SDK to the integrating application.

The `LogHandler` class contains the following method for .Net development: `delegate bool LogHandler(LogMessageType type, string messageId, string message)`

[LogHandler \[page 199\]](#)

Use the `LogHandler` method to pass messages from the SAP Data Quality Management SDK to the integrating application.

8.3.1 LogHandler

Use the `LogHandler` method to pass messages from the SAP Data Quality Management SDK to the integrating application.

≡ Syntax

```
delegate bool LogHandler(LogMessageType type, string messageId, string message)
```

`LogHandler` is a part of the [LogHandler](#) class for .Net development.

Where

Parameter	Description
type	The type of message to handle.
messageId	The ID of the message to handle.
message	The message to handle.

Related Information

[Class LogHandler \[page 199\]](#)

8.4 Class MultiRecordProgressHandler

Class `MultiRecordProgressHandler` is a delegate object for passing progress information from the SAP Data Quality Management SDK to the integrating application.

Class `MultiRecordProgressHandler` has the following method for .Net development: `delegate bool MultiRecordProgressHandler(double percentDone)`

[MultiRecordProgressHandler \[page 200\]](#)

Use the `MultiRecordProgressHandler` method to see the percent of progress for this transform.

8.4.1 MultiRecordProgressHandler

Use the `MultiRecordProgressHandler` method to see the percent of progress for this transform.

Syntax

```
delegate bool MultiRecordProgressHandler(double percentDone)
```

`MultiRecordProgressHandler` is a part of the `MultiRecordProgressHandler` class for .Net development.

Return value

The `MultiRecordProgressHandler` method returns the percent of progress.

Where

Parameter	Description
<i>percentDone</i>	The percent of progress.

Related Information

[Class MultiRecordProgressHandler \[page 200\]](#)

8.5 Class MultiRecordTransform

Class `MultiRecordTransform` is the record processing `Transform` class for processing multiple records.

Class `MultiRecordTransform` is required for processing. Some of the methods listed for the `MultiRecordTransform` class are inherited from the `Transform` class.

You cannot instantiate class `MultiRecordTransform`. Instead, use the `CreateMultiRecordTransform` methods in `TransformFactory` class to create valid `MultiRecordTransform` instances.

Use `MultiRecordTransform` objects to represent the Match transform.

The `MultiRecordTransform` class has the following methods:

- `System::Data::DataTable^ Process(System::Data::DataTable^ input);`
- `MultiRecordTransformHelper^ CreateHelper();`
- `void DestroyHelper(MultiRecordTransformHelper^ helper);`
- `property System::Data::DataTable^ InputSchema`
- `property System::Data::DataTable^ OutputSchema`
- `property System::Data::DataSet^ StatisticsSchemas`
- `property MultiRecordProgressHandler^ ProgressHandler`
- `property int ProgressInterval`
- `System::String^ GetExtraInfo(System::String^ request);`

[Process \[page 202\]](#)

Use the `Process` method to process the input data records that are loaded into this transform.

[CreateHelper \[page 202\]](#)

Use the `CreateHelper` method to create a helper for this transform.

[DestroyHelper \[page 203\]](#)

Use the `DestroyHelper` method to destroy a helper that you no longer need.

[InputSchema \[page 204\]](#)

Use the `InputSchema` method to get the schema of the input data record.

[OutputSchema \[page 204\]](#)

Use the `OutputSchema` method to get the schema of the output data record.

[StatisticsSchemas \[page 205\]](#)

Use the `StatisticsSchemas` to get the set of statistics tables to be populated.

[ProgressHandler \[page 205\]](#)

Use the `ProgressHandler` to return a pointer to the progress handler that receives progress status information from the Data Quality Management SDK.

[ProgressInterval \[page 206\]](#)

Use the `ProgressInterval` method to get the current progress interval in seconds.

[GetExtraInfo \[page 206\]](#)

Use the `GetExtraInfo` method to get extra information from a transform.

8.5.1 Process

Use the `Process` method to process the input data records that are loaded into this transform.

≡, Syntax

```
System::Data::DataTable^ Process(System::Data::DataTable^ input);
```

`Process` is a part of the `MultiRecordTransform` class for .Net development.

Return value

The `Process` method returns the output data records with the posted results. After the `Process` method completes, the output data records are ready to be unloaded.

To monitor processing progress, register a delegate with the `DataTable` input, `RowChanged` event.

Where

Parameter	Description
<i>input</i>	The collection of input data records to process.

Related Information

[Class `MultiRecordTransform` \[page 201\]](#)

8.5.2 CreateHelper

Use the `CreateHelper` method to create a helper for this transform.

≡, Syntax

```
MultiRecordTransformHelper^ CreateHelper();
```

`CreateHelper` is a part of the `MultiRecordTransform` class for .Net development.

The `CreateHelper` method is not thread-safe.

Use a helper to process records just as you use transforms. Run a helper in a different thread than this transform. A helper is like a transform in the following ways:

- Helper has all the same settings as a transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `CreateHelper` method returns a newly created helper object.

Related Information

[Class MultiRecordTransform \[page 201\]](#)

8.5.3 DestroyHelper

Use the `DestroyHelper` method to destroy a helper that you no longer need.

Syntax

```
void DestroyHelper(MultiRecordTransformHelper^ helper);
```

`DestroyHelper` is a part of the `MultiRecordTransform` class for .Net development.

Ensure that you run the `DestroyHelper` method to destroy all helpers of a transform before you destroy the transform.

Where

Parameter	Description
Helper [IN]	The helper to destroy.

Related Information

[Class MultiRecordTransform \[page 201\]](#)

8.5.4 InputSchema

Use the `InputSchema` method to get the schema of the input data record.

≡ Syntax

```
property System::Data::DataTable^ InputSchema
```

`InputSchema` is a part of the `MultiRecordTransform` class for .Net development.

Related Information

[Class MultiRecordTransform \[page 201\]](#)

8.5.5 OutputSchema

Use the `OutputSchema` method to get the schema of the output data record.

≡ Syntax

```
property System::Data::DataTable^ OutputSchema
```

`OutputSchema` is a part of the `MultiRecordTransform` class for .Net development.

Return value

The `OutputSchema` method returns the schema of the output data record.

Related Information

[Class MultiRecordTransform \[page 201\]](#)

8.5.6 StatisticsSchemas

Use the `StatisticsSchemas` to get the set of statistics tables to be populated.

≡ Syntax

```
property System::Data::DataSet^ StatisticsSchemas
```

`StatisticsSchemas` is a part of the `MultiRecordTransform` class for .Net development.

Add a `DataRowChangeEventHandler` method to each of the statistics tables in the `StatisticsSchemas` data set to populate the statistics tables. The Data Quality Management SDK calls the `DataRowChangeEventHandler` method each time this transform generates statistics. The transform generates statistics when it terminates under most circumstances.

Return value

This method gets the set of statistics tables that to be populated.

Related Information

[Class `MultiRecordTransform` \[page 201\]](#)

8.5.7 ProgressHandler

Use the `ProgressHandler` to return a pointer to the progress handler that receives progress status information from the Data Quality Management SDK.

≡ Syntax

```
property MultiRecordProgressHandler^ ProgressHandler
```

`ProgressHandler` is a part of the `MultiRecordTransform` class for .Net development.

Related Information

[Class `MultiRecordTransform` \[page 201\]](#)

8.5.8 ProgressInterval

Use the `ProgressInterval` method to get the current progress interval in seconds.

≡, Syntax

```
property int ProgressInterval
```

`ProgressInterval` is a part of the `MultiRecordTransform` class for .Net development.

Related Information

[Class `MultiRecordTransform` \[page 201\]](#)

8.5.9 GetExtraInfo

Use the `GetExtraInfo` method to get extra information from a transform.

≡, Syntax

```
System::String^ GetExtraInfo(System::String^ request);
```

`GetExtraInfo` is a part of the `MultiRecordTransform` class for .Net development.

Return value

By default, the `GetExtraInfo` method returns an empty string. However, you could customize a transform to determine valid requests from this method and define appropriate responses. The method would then return the appropriate response.

❖ Example

For example, customize the Global Address Cleanse transform to have this method return a list of available address cleanse engines.

Where

Parameter	Description
request [IN]	The request for the transform.

Related Information

[Class MultiRecordTransform \[page 201\]](#)

8.6 Class MultiRecordTransformHelper

Class `MultiRecordTransformHelper` is a shared resource-based processing object that is a clone of a `MultiRecordTransform` class.

`MultiRecordTransformHelper` class has the following method: `System::Data::DataTable^ Process(System::Data::DataTable^ input);`

[Process \[page 207\]](#)

Use the `Process` method to process the input data records that are loaded into this transform.

8.6.1 Process

Use the `Process` method to process the input data records that are loaded into this transform.

≡ Syntax

```
System::Data::DataTable^ Process(System::Data::DataTable^ input);
```

`Process` is a part of the `MultiRecordTransformHelper` class for .Net development.

Return value

After the `Process` method returns, the output data records with the posted results can be unloaded.

Where

Parameter	Description
<i>input</i>	The collection of input data records to process.

Related Information

[Class MultiRecordTransformHelper \[page 207\]](#)

8.7 Class RecordTransform

Class `RecordTransform` is the record processing `Transform` class for processing single records.

Class `RecordTransform` is required for processing. Some of the methods listed here are inherited from the `Transform` class.

Instances of `RecordTransform` cannot be instantiated. Instead, use the `CreateRecordTransform` methods in the `TransformFactory` class to create valid `RecordTransform` instances.

Use class `RecordTransform` objects to represent Data Cleanse, USA Regulatory Address Cleanse, Global Address Cleanse, and Geocoder transforms.

The `RecordTransform` class has the following methods for .Net development:

- `void Process(System::Data::DataRow^ input, System::Data::DataRow^ output);`
- `RecordTransformHelper^ CreateHelper();`
- `void DestroyHelper(RecordTransformHelper^ helper);`
- `property System::Data::DataTable^ InputSchema;`
- `property System::Data::DataTable^ OutputSchema;`
- `property System::Data::DataSet^ StatisticsSchemas;`
- `System::String^ GetExtraInfo(System::String^ request);`

[Process \[page 209\]](#)

Use the `Process` method to process the input data record for this transform.

[CreateHelper \[page 210\]](#)

Use the `CreateHelper` method to create a helper for this transform.

[DestroyHelper \[page 210\]](#)

Use the `DestroyHelper` method to destroy a helper that you no longer need.

[InputSchema \[page 211\]](#)

Use the `InputSchema` method to get the input data record schema.

[OutputSchema \[page 211\]](#)

Use the `OutputSchema` method to get the output data record schema.

[StatisticsSchemas \[page 212\]](#)

Use the `StatisticsSchemas` method to get the set of statistics tables to populate.

[GetExtraInfo \[page 212\]](#)

Use the `GetExtraInfo` method to get extra information from a transform.

8.7.1 Process

Use the `Process` method to process the input data record for this transform.

Syntax

```
void Process(System::Data::DataRow^ input, System::Data::DataRow^ output);
```

`Process` is a part of the `RecordTransform` class for .Net development.

Before you call `Process`, call the `GetInputDataRecord` method to get the input data record. Ensure that the input data record is loaded with data before you pass it to the `Process` method.

Caution

Do not process an input data record owned by a different transform.

The method first reads the fields of the input data record and then posts results to an output data record. You can query the results in the output data record, however, values in the output data record are valid only until you call `Process` again.

Return value

The `Process` method returns the output data record. It returns a Null pointer when there are non-fatal errors.

Where

Parameter	Description
<i>record</i> [IN]	The input data record to process.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.2 CreateHelper

Use the `CreateHelper` method to create a helper for this transform.

≡ Syntax

```
RecordTransformHelper^ CreateHelper();
```

`CreateHelper` is a part of the `RecordTransform` class for .Net development.

Use a helper to process records. Run a helper in a different thread than this transform. A helper is like a transform in the following ways:

- Helper has all the same settings as a transform.
- Helper shares some of the transform resources.
- Helper shares the transform handlers such as logs and statistics.

Take advantage of the similarities between a helper and a transform. Use one or more helper objects instead of creating multiple, identical transforms. You save on resources and you produce only one set of statistics.

Return value

The `CreateHelper` method returns a newly created helper object.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.3 DestroyHelper

Use the `DestroyHelper` method to destroy a helper that you no longer need.

≡ Syntax

```
void DestroyHelper(RecordTransformHelper^ helper);
```

`DestroyHelper` is a part of the `RecordTransform` class for .Net development.

Ensure that you run the `DestroyHelper` method to destroy all helpers of a transform before you destroy the transform.

Where

Parameter	Description
helper [IN]	The helper to destroy.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.4 InputSchema

Use the `InputSchema` method to get the input data record schema.

≡ Syntax

```
property System::Data::DataTable^ InputSchema;
```

`InputSchema` is a part of the `RecordTransform` class for .Net development.

Return value

The `InputSchema` method returns the input data record schema.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.5 OutputSchema

Use the `OutputSchema` method to get the output data record schema.

≡ Syntax

```
property System::Data::DataTable^ OutputSchema;
```

OutputSchema is a part of the RecordTransform class for .Net development.

Return value

The OutputSchema method returns the output data record schema.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.6 StatisticsSchemas

Use the StatisticsSchemas method to get the set of statistics tables to populate.

≡ Syntax

```
property System::Data::DataSet^ StatisticsSchemas;
```

StatisticsSchemas is a part of the RecordTransform class for .Net development.

Return value

The StatisticsSchemas method returns the set of statistics tables to populate.

Related Information

[Class RecordTransform \[page 208\]](#)

8.7.7 GetExtraInfo

Use the GetExtraInfo method to get extra information from a transform.

≡ Syntax

```
System::String^ GetExtraInfo(System::String^ request);
```

`GetExtraInfo` is a part of the `RecordTransform` class for .Net development.

Return value

The `GetExtraInfo` method returns an empty string by default. However, populate the empty string by customizing a transform to determine valid requests from this method and define appropriate responses. The information can be returned using the `GetExtraInfo` method.

❖ Example

Customize the Global Address Cleanse transform to work with this method and return a list of available address cleanse engines.

Where

Parameter	Description
<i>request</i> [IN]	The request for the transform.

Related Information

[Class `RecordTransform` \[page 208\]](#)

8.8 Class `RecordTransformHelper`

Class `RecordTransformHelper` is the helper class for the record processing `Transform` class.

Class `RecordTransformHelper` has the following method for .Net development: `void Process(System::Data::DataRow^ input, System::Data::DataRow^ output);`

[Process \[page 214\]](#)

Use the `Process` method to process the input data records owned by this transform.

8.8.1 Process

Use the `Process` method to process the input data records owned by this transform.

Syntax

```
void Process(System::Data::DataRow^ input, System::Data::DataRow^ output);
```

`Process` is a part of the `RecordTransformHelper` class for .Net development.

Before you call `Process`, call the `GetInputDataRecord` method to get the input data record. Ensure that the input data record is loaded with data before you pass it to the `Process` method.

Caution

Do not process an input data record owned by a different transform.

The method first reads the fields of the input data record and then posts results to an output data record. You can query the results in the output data record, however, values in the output data record are valid only until you call `Process` again.

Return value

The `Process` method returns the output data record. It returns a Null pointer when there are non-fatal errors.

Related Information

[Class `RecordTransformHelper` \[page 213\]](#)

8.9 Class TransformFactory

Class `TransformFactory` is required to create a transform.

The `TransformFactory` class has the following methods:

- property `LogHandler^ LoggerHandler`;
- property `System::String^ Locale`;
- `MultiRecordTransform^ CreateMultiRecordTransform(System::String^ transformSettings)`;
- `RecordTransform^ CreateRecordTransform(System::String^ transformSettings)`;
- `bool ValidateTransformSettings(System::String^ transformSettings)`;
- `System::String^ UpgradeTransformSettings(System::String^ transformSettings)`;

- `void DestroyTransform(Transform^ transform);`
- `void SetSerpReport(System::String^ fileName);`
- `void SetAmasReport(System::String^ fileName);`
- `void SetCass3553Report(System::String^ fileName);`

In the .Net implementation, this class also contains the methods used by the Certified Report Generator.

[LogHandler \[page 215\]](#)

Use the `LogHandler` method to call the log event handler.

[Locale \[page 216\]](#)

Use the `Locale` method to set the locale for messages produced by this transform.

[CreateMultiRecordTransform \[page 217\]](#)

Use the `CreateMultiRecordTransform` method to create a multirecord transform using the specified XML in the *transformSettings* parameter.

[CreateRecordTransform \[page 218\]](#)

Use the `CreateRecordTransform` method to create a record transform using the XML specified in the *transformSettings* parameter.

[ValidateTransformSettings \[page 218\]](#)

Use the `ValidateTransformSettings` method to validate the transform settings in the XML specified in the *transformSettings* parameter.

[UpgradeTransformSettings \[page 219\]](#)

Use the `UpgradeTransformSettings` method to upgrade the transform settings in the XML specified in the *transformSettings* parameter.

[DestroyTransform \[page 220\]](#)

Use the `DestroyTransform` method to delete the specified record transform or multirecord transform.

[SetSerpReport \[page 221\]](#)

Use the `SetSerpReport` method to generate the Canada SERP (Software Evaluation and Recognition Program) certified mailing report.

[SetAmasReport \[page 221\]](#)

Use the `SetAmasReport` method to generate the Australia Post AMAS (Address Matching Processing Summary) report.

[SetCass3553Report \[page 222\]](#)

Use the `SetCass3553Report` method to generate the USPS CASS 3553 report.

8.9.1 LogHandler

Use the `LogHandler` method to call the log event handler.

Syntax

```
property LogHandler^ LoggerHandler;
```

`LogHandler` is a part of the `TransformFactory` class for .Net development.

Use the `LogHandler` method to have the transform output log information. The `LogHandler` method is required.

Where

Parameter	Description
<code>handler [IN]</code>	The log event handler.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.2 Locale

Use the `Locale` method to set the locale for messages produced by this transform.

≡ Syntax

```
property System::String^ Locale;
```

`Locale` is a part of the `TransformFactory` class for .Net development.

Return value

If the SAP Data Quality Management SDK does not support the set locale, it issues a warning to the `LogHandler` and outputs all messages using `en_US`.

Where

Parameter	Description
<code>locale [IN]</code>	The locale to use.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.3 CreateMultiRecordTransform

Use the `CreateMultiRecordTransform` method to create a multirecord transform using the specified XML in the [transformSettings](#) parameter.

Syntax

```
MultiRecordTransform^ CreateMultiRecordTransform(System::String^  
transformSettings);
```

`CreateMultiRecordTransform` is a part of the `TransformFactory` class for .Net development.

Return value

The `CreateMultiRecordTransform` method returns a pointer to the created multirecord transform.

Where

Parameter	Description
transformSettings [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.4 CreateRecordTransform

Use the `CreateRecordTransform` method to create a record transform using the XML specified in the *transformSettings* parameter.

≡ Syntax

```
RecordTransform^ CreateRecordTransform(System::String^ transformSettings);
```

`CreateRecordTransform` is a part of the `TransformFactory` class for .Net development.

Return value

The `CreateRecordTransform` method returns a pointer to the created record transform.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.5 ValidateTransformSettings

Use the `ValidateTransformSettings` method to validate the transform settings in the XML specified in the *transformSettings* parameter.

≡ Syntax

```
bool ValidateTransformSettings(System::String^ transformSettings);
```

`ValidateTransformSettings` is a part of the `TransformFactory` class for .Net development.

Return value

The `ValidateTransformSettings` method returns `True` when the transform settings have no errors. It returns `False` when the transform settings have errors.

Where

Parameter	Description
<i>transformSettings</i> [IN]	The transform settings buffer.

Related Information

[Class TransformFactory](#) [page 214]

8.9.6 UpgradeTransformSettings

Use the `UpgradeTransformSettings` method to upgrade the transform settings in the XML specified in the *[transformSettings](#)* parameter.

≡ Syntax

```
System::String^ UpgradeTransformSettings(System::String^ transformSettings);
```

`UpgradeTransformSettings` is a part of the `TransformFactory` class for .Net development.

Return value

The `UpgradeTransformSettings` method returns the upgraded XML as a string.

Where

Parameter	Description
transformSettings [IN]	The transform settings.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.7 DestroyTransform

Use the `DestroyTransform` method to delete the specified record transform or multirecord transform.

Syntax

```
void DestroyTransform(Transform^ transform);
```

`DestroyTransform` is a part of the `TransformFactory` class for .Net development.

`DestroyTransform` is required. Ensure that you call `DestroyTransform` only after you are done using the transform instance. `DestroyTransform` ensures that the SDK performs final operations such as generating final statistics, that are performed only when the transform is finished processing.

Where

Parameter	Description
transform [IN]	The transform to destroy.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.8 SetSerpReport

Use the `SetSerpReport` method to generate the Canada SERP (Software Evaluation and Recognition Program) certified mailing report.

≡ Syntax

```
void SetSerpReport(System::String^ fileName);
```

`SetSerpReport` is a part of the `TransformFactory` class for .Net development.

SERP is required only when you produce Canada Post-certified mailings. If you want to produce the SERP report, ensure that you call the `SetSerpReport` method before you create the Global Address Cleanse transform.

Return value

The `SetSerpReport` method returns the generated SERP report.

Where

Parameter	Description
fileName [IN]	The path and file name for the generated SERP report.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.9 SetAmasReport

Use the `SetAmasReport` method to generate the Australia Post AMAS (Address Matching Processing Summary) report.

≡ Syntax

```
void SetAmasReport(System::String^ fileName);
```

`SetAmasReport` is a part of the `TransformFactory` class for .Net development.

AMAS is required only when you produce Australia Post-certified mailings. If you want to produce the AMAS report, ensure that you call the `SetAmasReport` method before you create the Global Address Cleanse transform.

Return value

The `SetAmasReport` method generates the AMAS report.

Where

Parameter	Description
<code>fileName</code> [IN]	The path and file name for the generated AMAS report.

Related Information

[Class TransformFactory \[page 214\]](#)

8.9.10 SetCass3553Report

Use the `SetCass3553Report` method to generate the USPS CASS 3553 report.

Syntax

```
void SetCass3553Report(System::String^ fileName);
```

`SetCass3553Report` is a part of the `TransformFactory` class for .Net development.

The CASS 3553 report is required only when you produce a USPS-certified mailing. If you want to produce the CASS 3553 report, ensure that you call the `SetCass3553Report` method before you create the USA Regulatory Address Cleanse transform.

Return value

The `SetCass3553Report` method generates the CASS 3553 report.

Where

Parameter	Description
<i>fileName</i> [IN]	The path and file name for the generated CASS 3553 report.

Related Information

[Class TransformFactory \[page 214\]](#)

9 Address Cleanse

The Address Cleanse transforms provide options to generate a corrected, complete, and standardized form of your original address data that complies with postal authorities from countries, such as the U.S., France, and Japan.

SAP Data Quality Management SDK allows you to create applications that use many address cleanse features such as basic parsing and standardizing and more advanced concepts that are unique to specific transforms.

[How address cleanse works \[page 224\]](#)

The USA Regulatory Address Cleanse and Global Address Cleanse transforms cleanse your address data using reference files and postal directories.

[Input and output fields and field classes \[page 225\]](#)

Map your input and output data to SAP Data Services fields based on the field format and the type of data the field contains.

[Address Cleanse transforms \[page 227\]](#)

SAP Data Services provides address cleanse transforms with options to parse, cleanse, and standardize addresses from all over the world.

[Address cleanse reports \[page 229\]](#)

The USA Regulatory Address Cleanse and Global Address Cleanse transforms allow you to generate certified postal reports.

[Preparing your address-line input data \[page 230\]](#)

SAP Data Services has specific rules about the combination of input fields that you use for mapping input address-line data.

[Set up the reference files \[page 234\]](#)

The USA Regulatory Address Cleanse and Global Address Cleanse transforms and engines rely on several directories and specialized reference files to cleanse your data.

9.1 How address cleanse works

The USA Regulatory Address Cleanse and Global Address Cleanse transforms cleanse your address data using reference files and postal directories.

The following table describes the systematic method that the transforms use to cleanse your data.

Process	Description
Verifies code consistency	Uses postal directories to verify that a postal code matches the locality and region. Adds missing postal elements when other elements are present and verified. For example, if your data has a locality and region, adds the corresponding postal code depending on the country.

Process	Description
Standardizes address lines	Uses the address standardization options that you set in the transform to output addresses that include or exclude all punctuation, uses all upper or all lower case text, or abbreviates or spells out address types, for example.
Identifies undeliverable addresses	For USA records only, identifies when an address is undeliverable, such as a vacant lot or a condemned building.
Assigns diagnostic codes	Assigns established codes that indicate reasons for unsigned addresses, or how addresses were corrected.

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[Input and output fields and field classes \[page 225\]](#)

[Address Cleanse transforms \[page 227\]](#)

[Address cleanse reports \[page 229\]](#)

[Preparing your address-line input data \[page 230\]](#)

[Set up the reference files \[page 234\]](#)

9.2 Input and output fields and field classes

Map your input and output data to SAP Data Services fields based on the field format and the type of data the field contains.

Input and output field format types

Use multiline or discrete formats for your input and output fields, or use a combination of both. Also designate the type of output information for output fields by setting field classes.

The following table describes the address line formats supported by Data Services.

Address line format	Description
Multiline	Outputs address elements in the same order as they're input. When you use multiline address line format, Data Services applies the following processes: - Abbreviates data intelligently when it exceeds the maximum field length. - Capitalizes and standardizes output data based on options selected in transform.
Discrete	Breaks input address elements into smaller address elements for output. When you use discrete address line format, Data Services applies the following processes: - Outputs generated data to additional output fields, such as error and status codes. - Capitalizes and standardizes some output components based on options selected in transform.

i Note

Data Services doesn't apply intelligent abbreviation for data that exceeds the maximum field length.

Generated field address class and related field classes

Select a generated field address class type and a generated field class type when you map output field in either the USA Regulatory Address Cleanse or the Global Address Cleanse transforms:

- Field address class:** Specifies the address class for the generated field.
- Field class:** Specifies the type of information the transform includes in the output field.

The following table describes the possible field class types.

Field class type	Description
<i>Parsed</i>	Outputs the parsed value with some standardization applied to the following fields: <i>LOCALITY</i> , <i>REGION</i> , and <i>POSTCODE</i> .
<i>Best</i>	Outputs the parsed data when the transform can't assign the address. Outputs the corrected data when the transform can assign the address.
<i>Corrected</i>	Outputs the complete and correct value from the directories and standardizes the value based on set standardization options and other options. Output is blank when the transform can't find the address in the directory and the address is unassigned.

The following table describes the address class type and lists the applicable field class types.

Address class type	Description	Field class type
Delivery	Generates output fields related to the address used for address assignment.	• <i>Parsed</i> • <i>Best</i> • <i>Corrected</i>
Dual	Generates output fields related to the address elements not used for address assignment. Applicable for input records that contain both a street and postal address.	• <i>Parsed</i> • <i>Best</i> • <i>Corrected</i>: Contains the DUAL address details that are available on input.
Official	Generates output fields in the form of the data preferred by the Postal Authority.	• <i>Parsed</i> • <i>Best</i> • <i>Corrected</i>

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[How address cleanse works \[page 224\]](#)

[Address Cleanse transforms \[page 227\]](#)

[Address cleanse reports \[page 229\]](#)

[Preparing your address-line input data \[page 230\]](#)

[Set up the reference files \[page 234\]](#)

9.3 Address Cleanse transforms

SAP Data Services provides address cleanse transforms with options to parse, cleanse, and standardize addresses from all over the world.

The following table describes the transforms that provide address cleanse processing for your data. Transforms are listed in alphabetical order. For more complete details about the transforms and options, see the *Reference Guide*.

Transform	Description
Global Address Cleanse and engines	Cleanses and standardizes address data from any of the supported countries, excluding U.S. certification. Configure the Global Address Cleanse transform with one or more of the Global Address Cleanse engines: Canada, Global Address, or USA. Create mailings that comply with the following postal authority requirements: • Canada Post Software Evaluation and Recognition Program (SERP)—Statement of Address Accuracy Report • Australia Post Address Matching Processing Summary report (AMAS) • New Zealand Statement of Accuracy (SOA) report Additionally offers global suggestion lists and country ID options: • Global suggestion list offers suggestions for possible address matches for your USA, Canada, and global address data. • Country ID identifies the country of destination for the record and outputs an ISO code.
USA Regulatory Address Cleanse	Identifies, parses, validates, and corrects USA address data, within the Latin1 code page, according to the CASS (U.S. Coding Accuracy Support System). Generates the USPS Form 3553 and outputs many useful codes to your records. Choose to run in non-certified mode and utilize suggestion list functionality. Offers the following USPS postal service options: • DPV • eLOT • EWS • GeoCensus • LACSLink • NCOALink • RDI • SuiteLink • Z4Change

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[How address cleanse works \[page 224\]](#)

[Input and output fields and field classes \[page 225\]](#)

[Address cleanse reports \[page 229\]](#)

[Preparing your address-line input data \[page 230\]](#)

[Set up the reference files \[page 234\]](#)

9.4 Address cleanse reports

The USA Regulatory Address Cleanse and Global Address Cleanse transforms allow you to generate certified postal reports.

Many of the reports generated by the transforms include reports that are required by the US Postal Service or other global postal authorities.

The USA Regulatory Address Cleanse transform provides data for the creation of the USPS Form 3553, which is required for CASS, and the NCOALink Summary Report. The Global Address Cleanse transform provides data for the creation of the Canada Post SERP report, the Australia Post AMAS report, and the New Zealand Post Statement of Accuracy (SOA) Report.

Most postal and regulatory reports require that Data Quality Management SDK are certified by the respective postal authority on a periodic basis.

For information about all required address cleanse reports, see the *Management Console Guide*.

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[How address cleanse works \[page 224\]](#)

[Input and output fields and field classes \[page 225\]](#)

[Address Cleanse transforms \[page 227\]](#)

[Preparing your address-line input data \[page 230\]](#)

[Set up the reference files \[page 234\]](#)

9.5 Preparing your address-line input data

SAP Data Services has specific rules about the combination of input fields that you use for mapping input address-line data.

The USA Regulatory Address Cleanse transform and the USA engine in the Global Address Cleanse transform has a limit of 12 address input fields that you can use for address input mapping.

Caution

If your Unicode database has valid U.S. addresses from the Latin1 character set, the USA Regulatory Address Cleanse transform processes the data as usual. However, the transform doesn't process Unicode data with characters outside the Latin1 code page and character values greater than 255. Instead, the transform sends the input record to the corresponding standardized output field without any processing. Further, the transform doesn't populate any other output fields, such as component type output fields, for that record.

Data Services has a few address-type input fields that are supported only in the Global Address Cleanse transform. Address-type input fields supported only by the Global Address Cleanse transform include the following:

- *Locality[3-4]*
- *Region2*
- *Addr_Extra[1-4]*
- *Building[1-2]*
- *Floor*
- *Prim_Addr*
- *Prim_Nbr*
- *Secondary[1-3]*
- *Unit*

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[How address cleanse works \[page 224\]](#)

[Input and output fields and field classes \[page 225\]](#)

[Address Cleanse transforms \[page 227\]](#)

[Address cleanse reports \[page 229\]](#)

[Set up the reference files \[page 234\]](#)

9.5.1 About using fields with multiples

Map input address elements to numbered fields (multiples) so that SAP Data Services outputs address elements in a specific order.

In addition to the [Multiline\[1-12\]](#) fields, the address cleanse transform offers other discrete fields in multiples. Some multiples have specific purposes, so it's important that you read the description for each field.

When you use any multiple field, you don't have to use all of the fields. However, you can't skip a field.

❖ Example

For the [Multiline\[1-12\]](#) fields, you don't have to use all 12 of the fields. However, if you use [Multiline2](#), you have to include [Multiline1](#).

When you use [Multiline1](#) and [Multiline3](#), you must also include [Multiline2](#).

Some multiples are valid for both USA Regulatory Address Cleanse and Global Address Cleanse transforms, however, the valid content of the field is different.

i Note

Before you use any multiples, look at the field definition for the applicable transform to make sure you're using the field correctly. Field descriptions are in the *Reference Guide* listed under the applicable transform.

❖ Example

The following table contains descriptions for the [Locality\[1-4\]](#) fields for the USA Regulatory Address Cleanse transform and the Global Address Cleanse transform for Puerto Rico, China and Japan. However, you can use the Locality fields for other countries supported by the Global Address Cleanse transform.

Use of Locality[1-4]

Field	USA Regulatory Address Cleanse	Global Address Cleanse
Locality1	City, town, or suburb.	City, town, or suburb China: Prefecture-level localities. Japan: City, island, ward, and so on.
Locality2	Puerto Rico: Urbanization information.	Additional city, town, or suburb information. Puerto Rico: Urbanization information. China: County-level localities when present on input. Japan: Additional ward, district, village, and so on.

Field	USA Regulatory Address Cleanse	Global Address Cleanse
<i>Locality3</i>	N/A	Additional city, town, or suburb information. China: Township, Ethnic township, subdistrict, and so on. Japan: Additional district, village, sub-district, and so on.
<i>Locality4</i>	N/A	Additional city, town, or suburb information. China: Village and neighborhood level localities. Japan: Additional district, village, sub-district, and so on.

9.5.2 Accepted input field combinations for Multiline[1-12]

SAP Data Services has specific input field combinations to use when you include the *Multiline[1-12]* fields in your input field mapping.

The following table lists the valid input field combinations when you use *Multiline[1-12]* for your input mapping.

Multiline[1-12] address-line input fields

Format 1	Format 2	Format 3
<i>Firm</i> (optional)	<i>Firm</i> (optional)	<i>Firm</i> (optional)
<i>Multiline1</i>	<i>Multiline1</i>	<i>Multiline1</i>
<i>Multiline[2-12]</i> (optional)	<i>Multiline[2-12]</i> (optional)	<i>Multiline[2-12]</i> (optional)
<i>Country</i> (optional)	<i>Lastline</i>	<i>Locality1</i>
	<i>Country</i> (optional)	<i>Locality[2-4]</i> (optional)
		<i>Region[1-2]</i> (optional)
		<i>Postcode</i> (Global Address Cleanse) or <i>Postcode1</i> (USA Regulatory Address Cleanse)
		<i>Country</i> (optional)

9.5.3 Accepted input field combinations for Address_Line

SAP Data Services has specific input field combinations to use when you include the discrete input field [Address_Line](#) in your input field mapping.

The following table lists the valid input field combinations when you use the discrete input field [Address_Line](#).

Formats with Address_Line

Format 1	Format 2
Firm (optional)	Firm (optional)
Address_Line	Address_Line
Lastline	Locality1
Country (optional)	Locality[2-4] (optional)
	Region[1-2] (optional)
	Postcode (Global Address Cleanse)
	Postcode1 (USA Regulatory Address Cleanse)
	Country (optional)

9.5.4 Accepted input field combinations for Global Address Cleanse discrete input fields

The Global Address Cleanse transform has specific input field combinations using address-line input fields.

You can include any number of address-line input fields when you map your input address elements for the Global Address Cleanse transform. However, you can use only up to 12 of the following address-line input fields:

- [Extra\[1-4\]](#)
- [Building\[1-2\]](#)
- [Unit](#)
- [Floor](#)
- [Secondary\[1-2\]](#)
- [prim_nbr](#)
- [prim_addr](#)

The following table lists the valid input field combinations when you map address-line input fields with the Global Address Cleanse transform.

Discrete address-line fields

Format 1	Format 2
Firm (optional)	Firm (optional)
Addr_Extra[1-4] (optional)	Addr_Extra[1-4] (optional)
Building[1-2] (optional)	Building[1-2] (optional)

Format 1	Format 2
<i>Unit</i> (optional)	<i>Unit</i> (optional)
<i>Floor</i> (optional)	<i>Floor</i> (optional)
<i>Secondary[1-3]</i> (optional)	<i>Secondary[1-3]</i> (optional)
<i>Prim_Nbr</i> (optional)	<i>Prim_Nbr</i> (optional)
<i>Prim_Addr</i>	<i>Prim_Addr</i>
<i>Lastline</i>	<i>Locality1</i>
<i>Country</i> (optional)	<i>Locality[2-4]</i> (optional)
	<i>Region[1-2]</i> (optional)
	<i>Postcode</i>
	<i>Country</i> (optional)

i Note

If you include the *Prim_Nbr* field, you must include the *Prim_Addr* field. However, if you include the *Prim_Addr* field, you don't have to include the *Prim_Nbr* field.

9.6 Set up the reference files

The USA Regulatory Address Cleanse and Global Address Cleanse transforms and engines rely on several directories and specialized reference files to cleanse your data.

When you use directories to standardize addresses and assign postal codes, you must use the most up-to-date directories, which requires you to periodically update your files with new directories.

To correct addresses and assign postal codes, the address cleanse transforms rely on databases called postal directories.

Besides the basic address directories, there are many specialized directories that the USA Regulatory Address Cleanse transform uses:

- DPV
- Early Warning System (EWS)
- eLOT
- GeoCensus
- LACSLink
- NCOALink
- RDI
- SuiteLink
- Z4Change

The Global Address Cleanse transform relies on global directories based on the chosen domain.

Directories help extend address cleansing beyond the basic parsing and standardizing.

Each transform contains reference file option groups for directory file locations. Directory access is based on your software license. Some directories require postal authority permission or licenses. Users with applicable licenses receive regular directory updates accompanied by a letter that contains installation instructions.

Caution

Incompatible or out-of-date directories can render the software unusable. The system administrator is responsible for installing directory updates to ensure that they are compatible with the current software.

The following list contains the directory release schedule for the USA Regulatory Address Cleanse and the Global Address Cleanse directories:

- USA Regulatory Address Cleanse: Install weekly, monthly, or bimonthly directories.
- Global Address Cleanse: Install monthly directories for Australia and Canada engines.
- Global Address Cleanse: Install quarterly directories for the global engine.

Parent topic: [Address Cleanse \[page 224\]](#)

Related Information

[How address cleanse works \[page 224\]](#)

[Input and output fields and field classes \[page 225\]](#)

[Address Cleanse transforms \[page 227\]](#)

[Address cleanse reports \[page 229\]](#)

[Preparing your address-line input data \[page 230\]](#)

10 About data quality statistics

The data quality statistics tables enable you to analyze many aspects of the quality of your data.

The data quality statistics tables contain summary and record-level statistics about the cleansing and assignment processes performed on your data. Generate the tables by setting the Data Quality Statistics options in the applicable transforms: Data Cleanse, Geocoder, Global Address Cleanse.

[Data quality statistics common requirements \[page 236\]](#)

When you generate data quality statistics, there are several requirements that apply to more than one type of statistics tables.

[Data quality statistics input field requirements for Data Cleanse \[page 239\]](#)

When you generate data quality statistics, the Data Cleanse transform requires that you configure specific input fields.

[Requirements for Cleanse Component Info table \[page 239\]](#)

To generate the CLEANSE_COMPONENT_INFO_TABLE in Data Cleanse, ensure that you adhere to the output field configuration requirements for parsing person, firm, phone, and email fields.

[Requirements for Geocoder and data quality statistics \[page 241\]](#)

To generate data quality statistics for the Geocoder transform, ensure that you set up the transform with the supported mode.

[Global Address Cleanse and the Cleanse Component Info table \[page 241\]](#)

To generate the CLEANSE_COMPONENT_INFO_TABLE in Global Address Cleanse, ensure that you adhere to the configuration requirements.

10.1 Data quality statistics common requirements

When you generate data quality statistics, there are several requirements that apply to more than one type of statistics tables.

i Note

Generate data quality statistics tables in the Data Cleanse, Global Address Cleanse, and Geocoder transforms only.

Statistics tables and common requirements

Statistics table	Requirement
All data quality statistics tables: CLEANSE_ADDRESS_RECORD_INFO_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_INFO_CODES_TABLE CLEANSE_STATISTICS_TABLE GEOCODE_INFO_CODES_TABLE GEOCODE_STATISTICS_TABLE	Requires the Task Localization file, which you download from the SAP Customer Portal. The Task Localization file converts raw data that you extract from the data quality statistics tables into understandable language.
CLEANSE_STATISTICS_TABLE GEOCODE_STATISTICS_TABLE	Does not require the ROW_ID field output field. These statistic tables contain summary statistics and the ROW_ID field is required only for non-summary statistics tables. The software issues a warning when your job set up: - Includes the ROW_ID output field but is not configured to generate non-summary data quality statistics. - Excludes the ROW_ID output field but is configured to generate non-summary data quality statistics.
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_ADDRESS_RECORD_INFO_TABLE GEOCODE_INFO_CODES_TABLE	Requires the ROW_ID output field. These statistics tables contain non-summary statistics and the ROW_ID field is required to join the transform input or output with the output statistics. The software issues a warning when your job set up: - Excludes the ROW_ID output field but is configured to generate non-summary data quality statistics. - Includes the ROW_ID output field but is not configured to generate non-summary data quality statistics. You can use ROW_ID to join the transform input or output with the non-summary statistics table information. For example, use ROW_ID to drill into the before and after changes in a specific component that was cleansed.
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_ADDRESS_RECORD_INFO_TABLE GEOCODE_INFO_CODES_TABLE	Requires that you map an input file primary key to the output file. Use the mapped primary key to join the input with any of the data quality statistics reports.

Statistics table	Requirement
Applies to all data quality statistics tables: CLEANSE_ADDRESS_RECORD_INFO_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_INFO_CODES_TABLE CLEANSE_STATISTICS_TABLE GEOCODE_INFO_CODES_TABLE GEOCODE_STATISTICS_TABLE	If you generate data cleansing statistics, requires that you set up a field for data cleansing. If you do not set up a field for data cleansing, the software does not generate data cleansing statistics for that field. ❖ Example An input file contains firm, address, and phone information. You set up the firm and address fields for data cleansing, but you do not set up the phone field for data cleansing. The software generates statistics that include the firm name and address information, but not the phone information.
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_STATISTICS_TABLE CLEANSE_COMPONENT_INFO_TABLE	Configure to generate in both the Data Cleanse and Global Address Cleanse transforms. Not applicable for the Geocoder transform.

Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Related Information

[System group \[page 510\]](#)

[Data Cleanse Report and Analysis \[page 511\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Data Cleanse Cleansing Package \[page 519\]](#)

[Data Cleanse Input Word Breaker \[page 522\]](#)

[Data Cleanse person standardization options \[page 524\]](#)

[Data Cleanse gender options \[page 528\]](#)

[Data Cleanse Firm Standardization options \[page 530\]](#)

[Data Cleanse Other Standardization options \[page 531\]](#)

[Data Cleanse Date Options \[page 535\]](#)

[Data Cleanse Phone Options group \[page 538\]](#)

[Data Cleanse Parser configuration \[page 541\]](#)

[Data Cleanse editor options \[page 543\]](#)

10.2 Data quality statistics input field requirements for Data Cleanse

When you generate data quality statistics, the Data Cleanse transform requires that you configure specific input fields.

The following table lists the supported and unsupported input fields for generating data quality statistics in the Data Cleanse transform. It also includes specific input field configurations that are not supported for generating data quality statistics.

Input field requirements for data quality statistics in Data Cleanse

Supported input fields	Unsupported input fields	Unsupported input field configurations
- EMAIL# - FIRM_LINE# - DISCRETE_FIRM - NAME_LINE# - NAME_OR_FIRM_LINE# - DISCRETE_PERSON - PHONE# - TITLE_LINE#	- DATE# - MULTILINE# - SSN# - UDPM#	The software issues a warning when you enable data quality statistics and the following conditions exist: - You include multiline input fields. - You include fields associated with name or title. - You map DISCRETE_PERSON input fields and NAME_LINE input fields in the same job setup. - You map DISCRETE_FIRM input fields and FIRM_LINE input fields in the same job setup.

10.3 Requirements for Cleanse Component Info table

To generate the CLEANSE_COMPONENT_INFO_TABLE in Data Cleanse, ensure that you adhere to the output field configuration requirements for parsing person, firm, phone, and email fields.

The CLEANSE_COMPONENT_INFO statistics table contains the exact output field start position and length of individual data elements that the transform parsed during the cleansing process.

Person parsing

The software generates output field position data only for person-type output fields that have the following characteristics:

- Parent component is DUAL_NAME* or PERSON*
- FIELD_CLASS is STANDARDIZED

The software does not generate output field position data for person-type output fields that have the following field name values. These values are exceptions to the above-supported person field characteristics:

- NAME_SPECIAL
- GENDER
- GENDER_ID
- RULE_LABEL
- SCORE
- *_MATCH_STD*
- MATCH_*

Firm parsing

The software generates output field position data for firm-type output fields that have the following characteristics:

- Parent component is FIRM*
- FIELD_CLASS is STANDARDIZED

The software does not generate output field position data for firm-type output fields that have the following field name values. These values are exceptions to the above-supported firm field characteristics:

- RULE_LABEL
- SCORE
- *_MATCH_STD*
- MATCH_*

Phone parsing

The software generates output field position data for phone-type output fields that have the following characteristics:

- Parent component is PHONE*
- FIELD_CLASS is STANDARDIZED

The software does not generate output field position data for phone-type output fields that have the following field name values. These values are exceptions to the above-supported phone field characteristics:

- *_MATCH_STD*
- MATCH_*

Email parsing

The software generates output field position data for e-mail-type output fields that have the following characteristics:

- Parent component is EMAIL*
- FIELD_NAME is EMAIL
- FIELD_CLASS is STANDARDIZED

10.4 Requirements for Geocoder and data quality statistics

To generate data quality statistics for the Geocoder transform, ensure that you set up the transform with the supported mode.

Applicable Geocoder modes

Supported mode	Unsupported mode
Address Geocoding mode	• Reverse Geocoding mode • Point of Interest (POI) Textual Search mode The software issues a warning when you are in this mode and you select to generate data quality statistics.

10.5 Global Address Cleanse and the Cleanse Component Info table

To generate the CLEANSE_COMPONENT_INFO_TABLE in Global Address Cleanse, ensure that you adhere to the configuration requirements.

The CLEANSE_COMPONENT_INFO statistics table contains the exact output field location (start position and length) of individual data elements that the transform parses during the cleansing process. However, when you include any [NW_](#) input fields, the transform does not generate the input field start position and length.

The transform generates the output field start position and length for output fields with the following characteristics:

- FIELD_CLASS is BEST
- FIELD_CATEGORY is COMPONENT
- FIELD_ADDRCLASS is DELIVERY or DUAL.

Exceptions to the above-supported field characteristics include the following output fields:

- MULTILINE1-12
- MATCH_*

11 USA Regulatory Address Cleanse transform

Use the USA Regulatory Address Cleanse transform to identify, parse, validate, and correct U.S. address data according to the U.S. Post Office standards.

USA Regulatory Address Cleanse information

Characteristic	Description
Use	The USA Regulatory Address Cleanse transform identifies, parses, validates, and corrects U. S. address data according to the U.S. Coding Accuracy Support System (CASS). When you use the required options, this transform creates the USPS Form 3553 and enhances your records with many useful codes. When you run the transform in a non certification mode, you can use the suggestion list feature. Note If an input record has characters not included in the Latin1 code page, the USA Regulatory Address Cleanse transform does not process that data. Instead, the transform sends the mapped input record to the corresponding standardized output field, when applicable. The transform does not populate any other output fields for that record. If your Unicode database has valid U.S. addresses from the Latin1 character set, the transform processes as normal. If you perform both data cleansing and matching, place the USA Regulatory Address Cleanse transform before the Data Cleanse transform and any of the Match transforms in the data flow.
Content objects	Sample XML configurations for a regular USA Regulatory Address Cleanse job and one set for suggestion lists.

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

DPV is a USPS (United States Postal Service) product that assists users in validating the accuracy of their address information.

[USPS eLOT \[page 258\]](#)

eLOT narrows the mail carrier delivery route walk sequence to the house (delivery point) level.

[Early Warning System \(EWS\) \[page 259\]](#)

EWS helps reduce the amount of misdirected mail caused when the USPS adds valid delivery points between national directory updates.

[SuiteLink \[page 261\]](#)

SuiteLink adds suite numbers and other secondary data to high-rise business addresses for more accurate mail delivery.

[LACSLink \[page 265\]](#)

LACSLink updates addresses when the physical address doesn't move but the address is changed.

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

When you use RDI to determine the number of residential addresses in your mailing list, you can decide whether to use a commercial shipping service or to ship your parcels with the USPS and save money.

[Z4Change \[page 277\]](#)

The Z4Change feature in the USA Regulatory Address Cleanse transform enhances automation compatibility and improves deliverability.

[Suggestion lists \[page 279\]](#)

Use Suggestion List processing in transactional projects with the USA Regulatory Address Cleanse and the Global Address Cleanse transforms.

[USPS certifications \[page 282\]](#)

The software is certified for USPS address match, move update, and address quality processes that comply with various USPS certifications.

[NCOALink and Move Update \[page 292\]](#)

NCOALink (National Change of Address) is a part of USPS Move Update, which ensures that your mailing list contains up-to-date address information.

[Statistics for multiple data sources \[page 317\]](#)

The USA Regulatory Address Cleanse transform can report statistics for jobs that contain multiple data sources, based on logical groupings.

11.1 USPS DPV (Delivery Point Validation)

DPV is a USPS (United States Postal Service) product that assists users in validating the accuracy of their address information.

DPV identifies known addresses with problems, which can cause the addresses to become undeliverable, by comparing Postcode2 information against the DPV directories.

DPV is available for U.S. data in the USA Regulatory Address Cleanse transform only.

i Note

DPV processing is required for CASS certification. However, if you don't certify addresses for CASS, you can still benefit from DPV by running the transform in non-certified mode.

⚠ Caution

If you choose to disable DPV processing, SAP Data Services doesn't generate the CASS-required documentation and your mailing isn't eligible for postal discounts.

[Benefits of DPV \(Delivery Point Validation\) \[page 245\]](#)

Process your address data with DPV to reduce mailing costs, enhance information quality, increase assignment rate, and prevent mail-order fraud.

[DPV security \[page 245\]](#)

The USPS has processes in place that monitor your use of DPV data to prevent misuse.

[DPV monthly directories \[page 250\]](#)

SAP ships the DPV directories monthly along with the U.S. National Directory delivery according to USPS guidelines.

[Required USPS information \[page 251\]](#)

Complete the USPS-required components in the USA Regulatory Address Cleanse transform editor.

[DPV output fields \[page 251\]](#)

To obtain DPV output information from your job processing, include specific output fields in your output file setup.

[Add Postcode2 in non-certified mode \[page 252\]](#)

Assign Postcode2 data to addresses that don't DPV confirm.

[DPV performance \[page 252\]](#)

Processing time can increase with DPV processes, based on your operating system, system configuration, and other variables that are unique to your operating environment.

[DPV No Stats indicators \[page 253\]](#)

The USPS designates addresses as "No Stat" when the address is a vacant property, it receives mail as part of a drop, or it doesn't have an established delivery yet.

[DPV Vacant indicators \[page 254\]](#)

The USPS includes DPV Vacant indicators in the DPV directories to indicate when an address qualifies under the DPV Vacant rules.

[DPV PO Box Throwback indicator \[page 255\]](#)

The USPS uses the PO Box Throwback directory to identify delivery points that are street addresses but mail delivery is to a PO Box address instead.

[DPV Door Not Accessible indicator \[page 255\]](#)

The USPS uses Door Not Accessible (DNA) indicators to mark addresses where the carrier can't knock on the door for mail delivery.

[DPV No Secure Location indicator \[page 256\]](#)

The USPS uses No Secure Location (NSL) indicators to mark addresses where there's no security.

[R777 and R779 carrier route matching \[page 256\]](#)

When you process in CASS mode, SAP Data Quality Management SDK assigns addresses with "phantom routes" R777 and R779 as DPV No Stat records.

[DPV Drop indicator \[page 258\]](#)

The DPV Drop indicator identifies delivery points that serve multiple businesses or families, such as CMRA.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)
[LACSLink \[page 265\]](#)
[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)
[Z4Change \[page 277\]](#)
[Suggestion lists \[page 279\]](#)
[USPS certifications \[page 282\]](#)
[NCOALink and Move Update \[page 292\]](#)
[Statistics for multiple data sources \[page 317\]](#)

11.1.1 Benefits of DPV (Delivery Point Validation)

Process your address data with DPV to reduce mailing costs, enhance information quality, increase assignment rate, and prevent mail-order fraud.

The following list describes the benefits of DPV processing:

- **Reduced mailing costs:** DPV eliminates UAA (undeliverable-as-addressed) mail, which reduces the number of mail pieces that you pay for that don't get delivered.
- **Enhanced information quality:** DPV increases the level of data accuracy by verifying an address down to the individual house, suite, or apartment instead of only the block face.
- **Increased assignment rate:** DPV resolves an address “tie,” where other tie-breaking methods aren't conclusive. Tie breaking increases assignment rate.
- **Preventing mail-order-fraud:** DPV helps to eliminate shipping of merchandise to individuals who place fraudulent orders, by verifying valid delivery addresses and Commercial Mail Receiving Agencies (CMRA).

11.1.2 DPV security

The USPS has processes in place that monitor your use of DPV data to prevent misuse.

When you purchase the DPV functionality from the USPS, the USPS requires you to sign a legal agreement. The agreement states that you agree not to misuse the DPV product. If you abuse the DPV product, the USPS can prohibit you from using DPV in the future.

[DPV false positive addresses \[page 246\]](#)

The USPS includes false positive addresses in the DPV directories as an added security to prevent DPV data abuse.

[Stop Processing Alternative \[page 246\]](#)

To bypass any DPV or LACSLink directory locks, establish a Stop Processing Alternative agreement with the USPS and SAP.

[DPV false positive logs \[page 247\]](#)

During processing, SAP Data Services generates a false-positive log file anytime it encounters a DPV false positive record.

[DPV locking for end users \[page 248\]](#)

Without a Stop Processing Alternative agreement with the USPS, SAP Data Services stops DPV processing for the remainder of the data flow when it encounters a false positive record in your database.

[Obtaining DPV unlock code from SAP \[page 248\]](#)

To unlock DPV processing in a data flow, obtain an unlock code from SAP Customer Support.

[Obtaining a DPV list release from the USPS \[page 249\]](#)

To complete the DPV false positive process, you must send the DPV false positive log to the USPS and obtain a list release from the USPS before you use the list again.

11.1.2.1 DPV false positive addresses

The USPS includes false positive addresses in the DPV directories as an added security to prevent DPV data abuse.

A false positive address is a fake address that the USPS places in the DPV directories. When SAP Data Quality Management SDK encounters a false positive address in your data, it can stop DPV processing or continue processing based on your DPV user type. When SAP Data Quality Management SDK encounters a false positive, your required action also depends on your user type.

The following table explains the process for each user type when you encounter a DPV false positive.

User type	SAP Data Quality Management SDK action	User action
End users without a Stop Processing Alternative agreement	Locks the DPV directories and terminates DPV processing. Continues processing your job without DPV.	1. Obtain a DPV unlock code from SAP Support. 2. Use the code to unlock DPV processing. 3. Send false positive logs to the USPS to release the mailing list.
End user with Stop Processing Alternative agreement	Continues processing your job with DPV.	Send false positive logs to the USPS.
Service providers	Continues processing your job with DPV.	Send false positive logs to the USPS.

11.1.2.2 Stop Processing Alternative

To bypass any DPV or LACSLink directory locks, establish a Stop Processing Alternative agreement with the USPS and SAP.

The Stop Processing Alternative isn't an option in SAP Data Quality Management SDK, it's a key code that you obtain from SAP Customer Support. To obtain a key code from SAP Customer Support, perform the following tasks:

1. Obtain the proper permissions from the USPS.
2. Provide the proof of permission to SAP Customer Support.

3. Obtain a key code from SAP Customer Support that disables the DPV or LACSLink directory lock.

With the Stop Processing Alternative key code in place, SAP Data Quality Management SDK takes the following actions when it encounters a false positive:

- Marks the record as a false positive.
- Generates a log file containing the false positive address.
- Notes the path to the log files in the error log.
- Generates a US Regulatory Locking Report containing the path to the log file.
- Continues processing your job.

The Stop Processing Alternative saves you time. The DPV or LACSLink processing continues so you don't have to follow the process to unlock the directories and rerun the job. However, you still have to send the false positive log to the USPS to notify them that your database contained a false positive record. The USPS has to release the list before you can use the database for production.

11.1.2.3 DPV false positive logs

During processing, SAP Data Services generates a false-positive log file anytime it encounters a DPV false positive record.

Data Services creates a separate log file for each mailing list that contains a false positive record. If multiple false positives exist within one mailing list, it writes them all to the same log file.

Data Services stores DPV false positive log files in the directory that you specify in the transform editor. Set the DPV false positive log file location in the [USPS_LOG_PATH](#) option in the Reference Files group.

i Note

If the directory that you set in [USPS_LOG_PATH](#) isn't writable, Data Services issues an error.

Data Services uses the following naming convention when it generates a DPV false positive log: `dpv1####.log`, where `####` is a number between 0001 and 9999.

❖ Example

For the first database in which Data Services finds a false positive record, it names the log file `dpv10001.log`. For the next database in which Data Services finds a false positive record, it names the log file `dpv10002.log`.

i Note

When you set the degree of parallelism (DOP) to greater than 1, Data Services generates one log per thread that contains a false positive.

11.1.2.4 DPV locking for end users

Without a Stop Processing Alternative agreement with the USPS, SAP Data Services stops DPV processing for the remainder of the data flow when it encounters a false positive record in your database.

When Data Services stops DPV processing in a data flow, it performs the following tasks:

- Marks the record as a false positive address.
- Generates a lock code.
- Generates a false positive log.
- Issues a message in the error log stating that a DPV false positive address was encountered.
- Includes the false positive address and lock code in the error log.
- Continues processing your data flow without DPV processing.
- Generates a US Regulatory Locking Report that contains the false positive address and the lock code. (Applicable only when you enable report generation in the USA Regulatory Address Cleanse transform.)

To restore DPV functionality, obtain a DPV unlock code from SAP Customer Support.

11.1.2.5 Obtaining DPV unlock code from SAP

To unlock DPV processing in a data flow, obtain an unlock code from SAP Customer Support.

→ Remember

The following steps apply to end users who don't have a Stop Processing Alternative agreement with the USPS.

Before you perform the following steps, write down the lock code and the locking address from the `dpvx.txt` file. The file is located in the directory that you specified in the `DPV_PATH` option in the *Reference Files* settings in the transform editor.

1. Go to <https://support.sap.com/en/my-support/keys.html>.
2. Select the *USPS Unlock Code* link at the top of the page.
3. Select *Get your USPS Unlock Codes*.
4. Select *Search* and choose an applicable Data Quality Management SDK system from the list.
5. Enter the lock code that you wrote down from the `dpvx.txt` file.
6. Choose *DPV* for the lock type.
7. Choose *BOJ-EIM-DS* for the component.
8. Enter the address that caused the lock.

The address is from the `dpvx.txt` file.

9. Attach the `dpv1####.log` log file.
10. Select *Submit*.

The utility displays the unlock code.

i Note

If the utility couldn't generate the unlock code, the system sends a message to a Technical Customer Assurance engineer who generates an unlock code for you during regular business hours.

11. Open the `dpvw.txt` file, delete all content, and paste the unlock code into the file.

i Note

The `dpvw.txt` file in this step is different than the `dpvx.txt` file in previous steps.

12. Obtain a list release from the USPS.
13. Remove the record that caused the lock from the database.
14. Delete the applicable `dpvl####.log` file.

Use the unlock code only one time. If Data Services detects another false-positive, even if it's the same record, you have to retrieve a new DPV unlock code.

→ Remember

To avoid encountering the same false-positive record, make sure that you remove the record that caused the lock from the database, and you delete the applicable `dpvl####.log` file before you process the list again.

→ Remember

All DPV users must send the false positive log to the USPS to obtain release of the mailing list. Follow the steps in [Obtaining a DPV list release from the USPS \[page 249\]](#).

11.1.2.6 Obtaining a DPV list release from the USPS

To complete the DPV false positive process, you must send the DPV false positive log to the USPS and obtain a list release from the USPS before you use the list again.

This process applies to all DPV user types, regardless of whether they've a Stop Processing Alternative agreement with the USPS.

1. Send an email to the USPS NCSC (National Customer Support Center) at dsf2stop@usps.gov, and include the following information:

- **E-mail subject line:** "DPV False Positive"
- **Attachment:** The `dpvl####.log` file.

The USPS NCSC uses the information submitted with the e-mail to determine whether it can release the list to the mailer.

2. After the USPS NCSC has released the list that contained the locked or false positive record, perform the following tasks:
 - a. Delete the applicable `dpvl####.log` log file.
 - b. Remove the record that caused the lock from the database.
3. Reprocess the database.

End users who don't have a Stop Processing Alternative agreement can obtain a DPV unlock code from SAP by following the steps in [Obtaining DPV unlock code from SAP \[page 248\]](#)

11.1.3 DPV monthly directories

SAP ships the DPV directories monthly along with the U.S. National Directory delivery according to USPS guidelines.

Directory requirements

i Note

DPV directories are also included with the DSF2 tables. Only use the DPV directories delivered with the U.S. National Directory.

The directories expire in 105 days. Ensure that the DPV directories and the Address directory files have the same date. Don't process files with expired directories.

Don't rename any of the DPV files. If you change the file names, SAP Data Quality Management SDK doesn't run DPV processing.

DPV directory file names

The following are the DPV directory files:

- `dpva.dir`
- `dpvb.dir`
- `dpvc.dir`
- `dpvd.dir`
- `dpve.dir`
- `dpv_door_na.dir`
- `dpv_drop_ind.dir`
- `dpv_vacant.dir`
- `dpv_no_stats.dir`
- `dpv_no_stats_rc.dir`
- `dpv_non_delivery_days_flag.dir`
- `dpv_non_delivery_days.dir`
- `dpv_po_box_throwback.dir`
- `dpv_no_secure_loc.dir`

11.1.4 Required USPS information

Complete the USPS-required components in the USA Regulatory Address Cleanse transform editor.

The following list contains the USPS-required options located in the USPS License Information group:

CUSTOMER_COMPANY_NAME
CUSTOMER_COMPANY_ADDRESS
CUSTOMER_COMPANY_LOCALITY
CUSTOMER_COMPANY_REGION
CUSTOMER_COMPANY_POSTCODE1
CUSTOMER_COMPANY_POSTCODE2

11.1.5 DPV output fields

To obtain DPV output information from your job processing, include specific output fields in your output file setup.

Include the following DPV output fields for reporting DPV results:

DPV_CMRA
DPV_DOOR_NOT_ACCESSIBLE
DPV_DROP_INDICATOR
DPV_FOOTNOTE
DPV_NON_DELIVERY_DAYS
DPV_NON_DELIVERY_DAY_SUN
DPV_NON_DELIVERY_DAY_MON
DPV_NON_DELIVERY_DAY_TUE
DPV_NON_DELIVERY_DAY_WED
DPV_NON_DELIVERY_DAY_THU
DPV_NON_DELIVERY_DAY_FRI
DPV_NON_DELIVERY_DAY_SAT
DPV_NO_SECURE_LOCATION
DPV_NOSTATS
DPV_NOSTATS_REASON_CODE
DPV_PBSA
DPV_STATUS
DPV_STATUS_ENHANCED
DPV_THROWBACK_INDICATOR
DPV_VACANT

For complete descriptions of all output fields, see the *Developer Guide*.

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USA Regulatory Address Cleanse output fields \[page 796\]](#)

11.1.6 Add Postcode2 in non-certified mode

Assign Postcode2 data to addresses that don't DPV confirm.

When addresses don't DPV confirm, assign a Postcode 2 to your data by running in non-certified mode. The mailing doesn't qualify for CASS, but the addresses are more complete.

To add Postcode2 information to addresses that don't DPV confirm, ensure that you configure the transform as follows:

- Disable DPV.
- Enable the non-certified option [ASSIGN_POSTCODE2_NOT_DPV_VALIDATED](#).

⚠ Caution

When you disable DPV processing, SAP Data Services doesn't generate the CASS-required documentation, and the mailing isn't eligible for postal discounts.

11.1.7 DPV performance

Processing time can increase with DPV processes, based on your operating system, system configuration, and other variables that are unique to your operating environment.

You can decrease the time required for DPV processing by loading DPV directories into system memory before processing. Additionally, manage memory usage by caching the DPV directories.

[Memory usage \[page 252\]](#)

Make sure that your computer has enough memory available before performing DPV processing.

[Multiple DPV jobs with cached directories \[page 253\]](#)

If you configure multiple jobs to run with DPV consecutively, and you've cached the DPV directories into memory, add a 10-second pause between jobs.

11.1.7.1 Memory usage

Make sure that your computer has enough memory available before performing DPV processing.

SAP recommends a minimum of 768 MB to process with DPV enabled. However, base the amount of required memory for your system on the size of the DPV directories (recently about 1 GB) plus the amount of memory required to run SAP Data Services.

i Note

The size of the DPV directories vary depending on the amount of new data in each directory release.

Make sure that you have enough memory to cache the DPV directories.

To find the amount of disk space required to cache the directories, see the [Product Availability Matrix \(PAM\)](#)  on the SAP Support Portal.

In the PAM, find information about disk space for directories in the appendix “Disk Space Requirements for Referential Data & Cleansing Packages”.

Related Information

[DPV performance \[page 252\]](#)

11.1.7.2 Multiple DPV jobs with cached directories

If you configure multiple jobs to run with DPV consecutively, and you've cached the DPV directories into memory, add a 10-second pause between jobs.

When there's not enough time for your system to release the memory from the first job, the next job either produces an error, or accesses the directories from disk. The consequence is performance degradation.

The 10-second pause allows time for the system to release memory. For more information about setting pauses between jobs properly, see your operating system manual.

11.1.8 DPV No Stats indicators

The USPS designates addresses as “No Stat” when the address is a vacant property, it receives mail as part of a drop, or it doesn't have an established delivery yet.

DPV No Stats directory

The USPS includes the DPV No Stat directory file, `dpv_no_stats.dir`, with the DPV directories.

DPV No Stat indicator

When you enable DPV, SAP Data Quality Management SDK automatically checks for the DPV No Stat directory in the directory folder that you indicate in the transform. If the DPV No Stat directory is missing, SAP Data Quality Management SDK skips DPV No Stat processing and doesn't issue an error message.

To obtain DPV No Stat information from SAP Data Quality Management SDK output, enable DPV and include the `DPV_NOSTATS` output field in your output field setup. SAP Data Quality Management SDK populates the `DPV_NOSTATS` output field as follows:

- `Y`: Address is flagged as No Stat.
- `N`: Address isn't flagged as No Stat.

- *blank*: DPV isn't enabled, the DPV No Stat directory is missing, or there's an address correction error.

DPV No Stat Reason Codes

To output the reason that an address was flagged as No Stat, include the output field `DPV_NOSTATS_REASON_CODE` in your output fields. When the `DPV_NOSTATS` output field returns a *Y* value, the `DPV_NOSTATS_REASON_CODE` contains code describing why the record was a No Stat. The following list describes the possible reason codes:

- *01*: Input address was an Internal Drop Address (IDA), which is an address that doesn't receive mail delivery directly from the USPS. Instead, mail is delivered to a drop address that services the IDA.
- *02*: Input address is a Contract Delivery Service (CDS) address, which is either one of the following:
 - New construction for which delivery hasn't been established.
 - Rural/CDS/HCR where the delivery point has been unoccupied for more than 90 days.
- *04*: Input address is a College/Military Zone (CMZ) address, or other type of address where the ZIP+4 records have been incorporated into the data.
- *05*: Input address is a regular No Stat address, which is not a possible delivery address because of one of the following reasons:
 - Address is on an R777 or R779 phantom route.
 - The PO Box has never been rented.
 - The PO Box has been declared unrentable.
- *06*: Secondary input address required.
- *Other*: The No Stat reason is not one of the other reasons.

Related Information

[USA Regulatory Address Cleanse output fields \[page 796\]](#)

11.1.9 DPV Vacant indicators

The USPS includes DPV Vacant indicators in the DPV directories to indicate when an address qualifies under the DPV Vacant rules.

i Note

Vacancy doesn't apply to seasonal addresses.

The DPV Vacant indicator directory name is `dpv_vacant.dir`. The USPS assigns a DPV Vacant indicator to an address under the following circumstances:

- The address was an active delivery point in the past.
- The address has been vacant for over 90 days.

- The address doesn't currently receive mail deliveries.
- The address will receive mail delivery in the future, if eligible.

11.1.10 DPV PO Box Throwback indicator

The USPS uses the PO Box Throwback directory to identify delivery points that are street addresses but mail delivery is to a PO Box address instead.

The USPS introduced the PO Box Throwback indicator for the DPV program to help grow the package services business. PO Box Throwback indicators help carriers make sure that mail reaches the intended destinations.

SAP Data Quality Management SDK uses the PO Box Throwback directory named `dpv_po_box_throwback.dir`, which is included with the DPV directories.

The `dpv_po_box_throwback.dir` file is optional. Therefore, if you don't download the directory with your DPV files, Data Quality Management SDK doesn't issue an error.

To enable PO Box Throwback indication in Data Quality Management SDK, include the output field [DPV_THROWBACK_INDICATOR](#) in your output field configuration. Data Quality Management SDK populates the field as follows:

- **Y**: Address is flagged as PO Box Throwback.
- **N**: Address isn't flagged as PO Box Throwback.
- **blank**: DPV isn't enabled, you don't have the PO Box Throwback directory, or there's an error in address correction.

11.1.11 DPV Door Not Accessible indicator

The USPS uses Door Not Accessible (DNA) indicators to mark addresses where the carrier can't knock on the door for mail delivery.

For example, DNA indicates addresses with the following characteristics:

- Carrier can't physically access a residence or building because it is on a rural route or highway contract route with a long driveway.
- Address is in a gated community that the carrier can't access.

SAP Data Quality Management SDK uses the DNA table named `dpv_door_na.dir`, which is included with your DPV directories.

To enable DNA in Data Quality Management SDK, enable DPV in your job setup, and include the `dpv_door_na.dir` with your DPV directories. With DPV enabled, Data Quality Management SDK automatically checks for the DNA table in the directory folder that you indicate in the transform. If you don't install the table, Data Quality Management SDK skips DNA processing and doesn't issue an error message.

To obtain DNA information from Data Quality Management SDK, include the output field [DPV_DOOR_NOT_ACCESSIBLE](#) in your output field configuration. Data Quality Management SDK populates the field as follows:

- **Y**: Address is flagged as DNA.

- *N*: Address isn't flagged as DNA.
- *blank*: DPV isn't enabled, you don't have the DNA table, or there's an error in address correction.

11.1.12 DPV No Secure Location indicator

The USPS uses No Secure Location (NSL) indicators to mark addresses where there's no security.

For example, NSL indicates locations where the mail carrier can access the door, but can't leave the package because of security concerns. The NSL table also contains addresses under the following circumstances:

- There is no one at the address to pick up the mail.
- Business is closed on certain days.
- Storefront has no mail receptacle.

SAP Data Quality Management SDK uses the NSL table named `dpv_no_secure_loc.dir`, which is included with your DPV directories.

To enable NSL indicators in Data Quality Management SDK, enable DPV in your job setup and include the `dpv_no_secure_loc.dir` with your DPV directories. With DPV enabled, Data Quality Management SDK automatically checks for the NSL table in the DPV directory folder that you indicate in the USA Regulatory Address Cleanse transform. If you don't install the NSL directory, Data Quality Management SDK skips NSL processing, but doesn't issue an error message.

To obtain NSL information in the output from Data Quality Management SDK, include the output field `DPV_NO_SECURE_LOCATION` in your output field configuration. Data Quality Management SDK populates the field as follows:

- *Y*: Address is flagged as an NSL.
- *N*: Address isn't flagged as an NSL.
- *blank*: DPV isn't enabled, you don't have the NSL table, or there's an error in address correction.

Additionally, the US Addressing Report contains a row in the DPV Summary section. The row named **DPV No Secure Location** contains the record count and percentage of NSL addresses in your database.

11.1.13 R777 and R779 carrier route matching

When you process in CASS mode, SAP Data Quality Management SDK assigns addresses with "phantom routes" R777 and R779 as DPV No Stat records.

To obtain R777 and R779 carrier route information, include specific DPV output fields in your job setup. When Data Quality Management SDK encounters an R777 or R779 carrier route record, it outputs the codes described in the following table.

Output field	Phantom route code
DPV_FOOTNOTE	R7: Input address assigned to a phantom route of R777 or R779. i Note The output field returns the footnote R7 regardless of whether you have CASS mode enabled.
DPV_NOSTATS	Y : Address was flagged as No Stat.
DPV_NOSTATS_REASON_CODE	05: Input address is a regular No Stat address, which is not a possible delivery address because of one of the following reasons: • Address is on an R777 or R779 phantom route. • The PO Box has never been rented. • The PO Box has been declared unrentable.
DPV_STATUS	Phantom route addresses return either one of the following status codes: • Y: Both the primary and secondary (if present) input addresses DPV confirmed. • S: Input address was DPV confirmed for the primary number only. The secondary number was present but invalid, or a single trailing alpha on a primary number was dropped to make a DPV match. • D: Input address was DPV confirmed for the primary number only. The secondary number was missing but required.
DPV_STATUS_ENHANCED	R : Input address was DPV confirmed, but assigned to phantom route R777 or R779 and USPS delivery isn't provided.

When Data Quality Management SDK processes a phantom route record in CASS Mode, it removes the ZIP+4 code from the output address and assigns the error code E600, undeliverable.

i Note

Data Quality Management SDK removes the ZIP+4 from the address only after SuiteLink, LACSLink, and DSF processing.

When Data Quality Management SDK processes a phantom route record in non-CASS mode, it assigns R777 and R779 carrier route addresses as follows:

- Returns Error Code E601 under the following circumstances:
 - When DPV isn't enabled.
 - When you set [Disable Certification](#) to [Yes](#) to run in non-CASS mode.
- Outputs *blank* in the output field [DPV_NOSTATS](#) (when included in output configuration) because DPV wasn't enabled.
- Retains the ZIP+4 when you set the [Assign Postcode2 Not DPV Validated](#) to [Yes](#) in the Non Certified Option section.

Related Information

[USA Regulatory Address Cleanse output fields \[page 796\]](#)

11.1.14 DPV Drop indicator

The DPV Drop indicator identifies delivery points that serve multiple businesses or families, such as CMRA.

The USPS introduced the DPV Drop indicator for the DPV and DSF2 programs to help grow the package services business. DPV Drop indicators help carriers make sure that mail reaches the intended destinations.

SAP Data Quality Management SDK supports the DPV Drop indicator functionality using a directory named `dpv_drop_ind.dir`, which is included with the DPV directories.

To enable DPV Drop indicator in Data Quality Management SDK, include the output field [DPV_DROP_INDICATOR](#) in your output field configuration. Data Quality Management SDK populates the field as follows:

- **Y**: Input address was a DPV drop.
- **N**: Input address wasn't a DPV drop.
- *blank*: [Enable DPV](#) was set to **NO**, `dpv_drop_ind.dir` wasn't present, or there was an assignment error.

11.2 USPS eLOT

eLOT narrows the mail carrier delivery route walk sequence to the house (delivery point) level.

eLOT codes generated by the transform enable you to sort your mailings to a more precise level. eLOT is available for U.S. addresses in the USA Regulatory Address Cleanse transform only.

eLOT takes line of travel one step further. The original Line of Travel (LOT) product narrowed the mail carrier delivery route to the block face level (Postcode2 level) by discerning whether an address resided on the odd or even side of a street or thoroughfare.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

[USA Regulatory Address Cleanse transform \[page 242\]](#)

11.3 Early Warning System (EWS)

EWS helps reduce the amount of misdirected mail caused when the USPS adds valid delivery points between national directory updates.

EWS is available for U.S. records in the USA Regulatory Address Cleanse transform only.

The following is an example of how EWS helps with misdirected mail.

❖ Example

A new construction is completed on a house at 300 Main Avenue. The new owner signs up for utilities and mail. However, the delivery point isn't listed in the national directory yet.

Currently, the national directory has a valid delivery point of 300 Main Street. Therefore, all the mail intended for the new house at 300 Main Avenue is mis-directed to 300 Main Street.

With EWS, an additional directory informs CASS users of the existence of 300 Main Avenue long before it appears in the national directory. When using EWS processing, the previously mis-directed address now defaults to a 5-digit assignment.

[EWS directory \[page 260\]](#)

The EWS directory contains four months of rolling data, which the USPS updates weekly.

[EWS example \[page 260\]](#)

The EWS feature is the solution to the problem of misdirected mail caused by valid delivery points that appear between national directory updates.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

11.3.1 EWS directory

The EWS directory contains four months of rolling data, which the USPS updates weekly.

Each week, the USPS adds new data to the EWS directory, and drops data from the previous week. The USPS then publishes the latest EWS data. Each Friday, SAP converts the data to an SAP unique format and posts the directory in a Zip file (Ewyymmdd.zip) on the SAP support portal.

For more information about directories, and for instructions to access directories, see the latest directory letters at https://help.sap.com/viewer/p/ADDRESSING_DIRECTORIES.

Note

Ensure that you access the most recent version of the directory letter by checking the version in the upper right corner of the Help Portal.

Related Information

[Early Warning System \(EWS\) \[page 259\]](#)

11.3.2 EWS example

The EWS feature is the solution to the problem of misdirected mail caused by valid delivery points that appear between national directory updates.

Example

A new construction at 300 Main Avenue is completed and the new owner moves in. The new owner signs up for utilities and submits a change of address form, but it may take a couple of months before the delivery point is listed in the national directory. Because you do not have the EWS feature, all the mail intended for the new house at 300 Main Avenue is mis-directed to 300 Main Street until the delivery point is added to the national directory. The new owner does not receive their bills or any congratulatory cards from their friends.

The EWS feature solves this problem by using an additional directory which informs CASS users of the existence of 300 Main Avenue long before it appears in the national directory. When using EWS processing, the previously mis-directed address now defaults to a 5-digit assignment.

11.4 SuiteLink

SuiteLink adds suite numbers and other secondary data to high-rise business addresses for more accurate mail delivery.

When you use the SuiteLink directories, SAP Data Quality Management SDK adds secondary address information to your addresses, and sorts more addresses by delivery sequence. The additional data, and the delivery sequence sort ensures that the USPS delivers more mail pieces with accuracy and speed.

The USPS SuiteLink directory is multiple files of specially indexed address information, such as secondary numbers and unit designators. SAP Data Quality Management SDK uses the SuiteLink data for addresses identified as high-rise default buildings.

SuiteLink is required for CASS

SuiteLink is required when you process in CASS mode, and you set the [DISABLE_CERTIFICATION](#) option to *NO*.

i Note

If you disable SuiteLink in your job, but you are in CASS mode, SAP Data Quality Management SDK issues an error message and stops processing.

[Benefits of SuiteLink \[page 262\]](#)

The addition of secondary address information to high-rise addresses allows for the most efficient and cost-effective delivery sequencing and postage discounts.

[How SuiteLink works \[page 262\]](#)

SAP Data Quality Management SDK matches a company name, a known high-rise address, and the CASS-certified Postcode2 in your database to data in the SuiteLink directories.

[Retain input secondary address information \[page 263\]](#)

When an input record has a primary and secondary address, and SuiteLink returns a different secondary address, choose to retain the input secondary address information on output.

[Highrise default to street assignment \[page 263\]](#)

When an input record matches both a street address and a highrise default address, the USPS requires the record be assigned to the street address in specific situations.

[SuiteLink directory \[page 264\]](#)

Know and understand the requirements for the SuiteLink directory.

[Improve SuiteLink processing speed \[page 265\]](#)

To improve processing speed, load SuiteLink directories into memory.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)
[USPS eLOT \[page 258\]](#)
[Early Warning System \(EWS\) \[page 259\]](#)
[LACSLink \[page 265\]](#)
[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)
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[Suggestion lists \[page 279\]](#)
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[NCOALink and Move Update \[page 292\]](#)
[Statistics for multiple data sources \[page 317\]](#)

11.4.1 Benefits of SuiteLink

The addition of secondary address information to high-rise addresses allows for the most efficient and cost-effective delivery sequencing and postage discounts.

If your business depends on website, mail, or in-store orders from customers, use SuiteLink as a money-saving tool. Also know that your customers who reside in high-rises that house several businesses, receive their marketing materials, bank statements, and orders right at their door.

i Note

SuiteLink is required for CASS.

Related Information

[SuiteLink \[page 261\]](#)

11.4.2 How SuiteLink works

SAP Data Quality Management SDK matches a company name, a known high-rise address, and the CASS-certified Postcode2 in your database to data in the SuiteLink directories.

When there is a match in the SuiteLink directories, SAP Data Quality Management SDK creates a complete business address that includes the suite number.

❖ Example

Follow a data record through the SuiteLink process:

1. SAP Data Quality Management SDK standardizes the address elements based on the standardization options.
2. SuiteLink recognizes the address is a high-rise.
3. SuiteLink uses the high-rise information and the firm name to assign a correct suite number.
4. SuiteLink uses the high-rise information and the firm name to assign a correct ZIP+4 code.

Input record

Telera
910 E Hamilton Ave Fl2
Campbell CA 95008 0610

Output record

TELERA
910 E HAMILTON AVE STE 200
CAMPBELL CA 95008 0625

Related Information

[SuiteLink \[page 261\]](#)

11.4.3 Retain input secondary address information

When an input record has a primary and secondary address, and SuiteLink returns a different secondary address, choose to retain the input secondary address information on output.

i Note

If you set `ADD_FIRM_MATCH_SECONDARY` to `NO`, Data Quality Management SDK doesn't produce the USPS Form 3553 and your output isn't CASS-compliant.

11.4.4 Highrise default to street assignment

When an input record matches both a street address and a highrise default address, the USPS requires the record be assigned to the street address in specific situations.

An input address must have the following characteristics to be processed with the deliverable street address:

- No secondary address on input.
- Address has highrise-specific addresses.
- `DPV_STATUS` = `Y`, the address is a confirmed DPV.
- `DPV_NOSTATS` = `N`, the address isn't a No Stat.

SAP Data Quality Management assigns an address that qualifies as a highrise default to street assignment as follows:

- Outputs the deliverable street address.

- Outputs the deliverable street address ZIP+4.

❖ Example

The following address matches to both a high-rise default and a street address:

36 CAPITAL WAY
ATOKA TN 38004

The address has the following 4-digit ZIP Codes:

- 7924 at the street level
- 7985 at the highrise default level
- 7986 at the highrise level (with a range of secondary addresses)

Because the input address has no secondary information, is a confirmed DPV, and isn't a NoStat, CASS O requires Data Management SDK to assign at the street level using the street-level ZIP+4:

36 CAPITAL WAY
ATOKA TN 38004-7924

The output *DPV_FOOTNOTE* is *AABB*:

- *AA*: Input address matched to the ZIP+4 file.
- *BB*: Primary input address number matched to DPV/DSF2.

11.4.5 SuiteLink directory

Know and understand the requirements for the SuiteLink directory.

The SuiteLink directory is distributed monthly. Ensure that you use the SuiteLink directory with a ZIP+4 directory labeled for the same month.

❖ Example

Use the December 2018 SuiteLink directory only with the December 2018 `zip4us.dir` directory.

You can't use a SuiteLink directory that is older than 60 days based on its release date. SAP Data Quality Management SDK warns you 15 days before the directory expires. As with all directories, SAP Data Quality Management SDK doesn't process your records with an expired SuiteLink directory.

Related Information

[SuiteLink \[page 261\]](#)

11.4.6 Improve SuiteLink processing speed

To improve processing speed, load SuiteLink directories into memory.

To load SuiteLink directories into memory, set `CACHE_SUITELINK_DIRECTORIES` in the Transform Performance group to `YES`.

Related Information

[SuiteLink \[page 261\]](#)

11.5 LACSLink

LACSLink updates addresses when the physical address doesn't move but the address is changed.

An address change can happen with municipality address requirements, street renaming, or PO Box renumbering.

❖ Example

To make it easier for police, fire, ambulance, and postal personnel to locate rural addresses, a municipality requires that all rural addresses be assigned a street name.

LACSLink technology ensures that the data remains private and secure, and at the same time provides licensed users easy access to the data. LACSLink is an integrated part of address processing; it isn't an extra step. You must have the old address data before you can obtain new addresses.

LACSLink is a USPS product that is available for U.S. records with the USA Regulatory Address Cleanse transform only. LACSLink processing is required for CASS certification.

[How LACSLink works \[page 266\]](#)

When SAP Data Quality Management SDK processes data with LACSLink enabled, it follows a specific process.

[LACSLink security \[page 267\]](#)

To ensure that the LACSLink data is not misused, the USPS has processes that monitor the use of LACSLink.

[Conditions for address processing \[page 271\]](#)

To save processing time, SAP Data Quality Management SDK performs LACSLink processing only under certain conditions.

[LACSLink directory files \[page 271\]](#)

SAP ships the LACSLink directory files with the U.S. National Directory update.

[Required information in the job setup \[page 272\]](#)

Because CASS certification requires LACSLink, enter the required USPS license information in the USA Regulatory Address Cleanse transform.

[Reasons for LACSLink processing errors \[page 273\]](#)

You can fix some LACSLink processing errors in the transform configuration.

[LACSLink output fields \[page 273\]](#)

Several output fields are available for generated LACSLink processing results.

[Cache LACSLink directories \[page 273\]](#)

The transform performance improves when you cache the LACSLink directories.

[USPS Form 3553 \[page 274\]](#)

The USPS Form 3553 contains LACSLink record counts.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

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[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

11.5.1 How LACSLink works

When SAP Data Quality Management SDK processes data with LACSLink enabled, it follows a specific process.

LACSLink provides a new address when one is available. SAP Data Quality Management SDK performs the following steps when you enable LACSLink:

1. The USA Regulatory Address Cleanse transform standardizes the input address.
2. The transform looks for a matching address in the LACSLink data.
3. If the transform finds a match, it outputs the LACSLink-converted address and other LACSLink information.

11.5.2 LACSLink security

To ensure that the LACSLink data is not misused, the USPS has processes that monitor the use of LACSLink.

When a company purchases the LACSLink functionality, the USPS requires them to sign a legal agreement. The agreement states that the company agrees not to misuse the LACSLink product. If the company abuses the LACSLink product, the USPS can prohibit the user from using LACSLink in the future.

[LACSLink false positive addresses \[page 267\]](#)

The USPS includes false positive addresses in the LACSLink directories as an added security to prevent LACSLink abuse.

[LACSLink false positive logs \[page 268\]](#)

SAP Data Quality Management SDK generates a false-positive log file any time it encounters a LACSLink false positive record.

[LACSLink locking for end users \[page 268\]](#)

If you are an end user without a Stop Processing Agreement with the USPS and SAP Data Quality Management SDK encounters a false positive record in your database, it stops LACSLink processing for the remainder of the data flow.

[Obtaining LACSLink unlock code from SAP \[page 269\]](#)

When SAP Data Quality Management SDK locks LACSLink processing because of a false positive address in your data, you unlock LACSLink processing by obtaining an unlock code from SAP.

[Obtaining a LACSLink list release from the USPS \[page 270\]](#)

To complete the LACSLink false positive process, you must send the LACSLink false positive log to the USPS and obtain a list release from the USPS before you use the list again.

11.5.2.1 LACSLink false positive addresses

The USPS includes false positive addresses in the LACSLink directories as an added security to prevent LACSLink abuse.

A false positive address is a fake address that the USPS places in the LACSLink directories. When SAP Data Quality Management SDK encounters a false positive address in your data, it can stop LACSLink processing or continue processing based on your LACSLink user type. Your action when SAP Data Quality Management SDK encounters a false positive also depends on your user type.

The following table explains the behaviors for each user type.

User type	Software behavior	Action
End users without a Stop Processing Alternative agreement	Locks LACSLink directories and terminates LACSLink processing. Job continues to run without LACSLink.	1. Unlock LACSLink processing by obtaining a LACSLink unlock code from SAP Support. 2. Send false positive logs to the USPS to release the mailing list.
End user with Stop Processing Alternative agreement	LACSLink processing continues.	Send false positive logs to the USPS to release the mailing list

User type	Software behavior	Action
Service providers	LACSLink processing continues.	Send false positive logs to the USPS to release the mailing list.

11.5.2.2 LACSLink false positive logs

SAP Data Quality Management SDK generates a false-positive log file any time it encounters a LACSLink false positive record.

SAP Data Quality Management SDK creates a separate log file for each mailing list that contains a false positive record. If multiple false positives exist within one mailing list, it writes them all to the same log file.

SAP Data Quality Management SDK stores LACSLink log files in the directory specified for the [USPS_LOG_PATH](#) in the Reference Files group.

i Note

Ensure that the directory that you enter is writable. SAP Data Quality Management SDK issues an error if it can't write to the specified path.

SAP Data Quality Management SDK automatically names LACSLink false positive logs `lacs l###.log`, where `###` is a number between 001 and 999.

❖ Example

the first log file generated is `lacs l001.log`, the next one is `lacs l002.log`, and so on.

i Note

When you set the degree of parallelism (DOP) to greater than 1, SAP Data Quality Management SDK generates 1 log per thread that contains a false positive.

Related Information

[LACSLink security \[page 267\]](#)

11.5.2.3 LACSLink locking for end users

If you are an end user without a Stop Processing Agreement with the USPS and SAP Data Quality Management SDK encounters a false positive record in your database, it stops LACSLink processing for the remainder of the data flow.

In addition to terminating LACSLink processing, SAP Data Quality Management SDK also takes the following actions:

- Marks the record as a false positive address.
- Issues a message in the error log that a LACSLink false positive address was encountered.
- Includes the false positive address and lock code in the error log.
- Continues processing your data flow without LACSLink processing.
- Generates a lock code.
- Generates a false positive log.
- Generates a US Regulatory Locking Report that contains the false positive address and the lock code. (Report generation must be enabled in the USA Regulatory Address Cleanse transform.)

To restore LACSLink functionality, obtain a LACSLink unlock code from SAP Support following the steps in [Obtaining LACSLink unlock code from SAP \[page 269\]](#).

Related Information

[LACSLink security \[page 267\]](#)

11.5.2.4 Obtaining LACSLink unlock code from SAP

When SAP Data Quality Management SDK locks LACSLink processing because of a false positive address in your data, you unlock LACSLink processing by obtaining an unlock code from SAP.

→ Remember

The following steps apply to end users who don't have a Stop Processing Alternative agreement with the USPS.

Before you perform the following steps, write down the lock code and the locking address from the `lacsx.txt` file. The file is located in the directory you specify in the `LACSLINK_PATH` option in the Reference Files settings of the transform.

1. Log into the SAP Support Portal and go to <https://support.sap.com/en/my-support/keys.html>.
2. Select the **USPS Unlock Code** link at the top of the page.
3. Select *Get your USPS Unlock Codes*.
4. Select *Search* and select an applicable Data Quality Management SDK system from the list.
5. Enter the lock code that you wrote down from the `lacsx.txt` file.
6. Choose *LACSLink* as the lock type.
7. Choose *BOJ-EIM-DS* as the component.
8. Enter the address that caused the lock.

The address is from the `lacsx.txt` file.

9. Attach the `lacs1####.log` file.
 10. Select *Submit*.
- The utility displays the unlock code.

i Note

If the utility couldn't generate the unlock code, the system sends a message to a Technical Customer Assurance engineer who generates an unlock code for you during regular business hours.

11. Open the `lacs w . txt` file, delete all content, and paste the unlock code into the file.

i Note

The `lacs w . txt` file in this step is different than the `lacs x . txt` file in previous steps.

12. Obtain a list release from the USPS.
13. Remove the record that caused the lock from the applicable database.
14. Delete the `lacs l### . log` file before processing the list again.

Use the unlock code only one time. If Data Services detects another false-positive, even if it's the same record, you have to retrieve a new LACSLink unlock code.

→ Remember

To avoid encountering the same false-positive record, make sure that you remove the record that caused the lock from the database, and you delete the applicable `lacs l### . log` file before you process the list again.

→ Remember

All LACSLink users must send the LACSLink false positive log to the USPS to obtain a list release following the process in [Obtaining a LACSLink list release from the USPS \[page 270\]](#).

Related Information

[LACSLink security \[page 267\]](#)

11.5.2.5 Obtaining a LACSLink list release from the USPS

To complete the LACSLink false positive process, you must send the LACSLink false positive log to the USPS and obtain a list release from the USPS before you use the list again.

This process applies to all LACSLink user types, regardless of whether they've a Stop Processing Alternative agreement with the USPS.

1. Send an email to the USPS at `dsf2stop@usps.gov`. Include the following information:
 - **E-mail subject line:** "LACSLink False Positive"
 - Attachment: `lacs l### . log` file or files.

The USPS NCSC uses the information to determine whether it can release the list to the mailer.

2. After the USPS NCSC has released the list that contained the locked or false positive record, perform the following tasks:

- a. Delete the applicable lacs1###.log file.
 - b. Remove the record that caused the lock from the database.
3. Reprocess the database.

End users who don't have a Stop Processing Alternative agreement can obtain a LACSLink unlock code from SAP by following the steps in [Obtaining LACSLink unlock code from SAP \[page 269\]](#)

11.5.3 Conditions for address processing

To save processing time, SAP Data Quality Management SDK performs LACSLink processing only under certain conditions.

When you enable LACSLink, SAP Data Quality Management SDK doesn't process all of your addresses with LACSLink. The following list contains conditions under which SAP Data Quality Management SDK passes your data to LACSLink processing:

- The USPS flagged the address in the address directories as a LACS-convertible record.
- The USPS hasn't flagged the address for LACSLink, but it assigned the address for a rural route or highway contract by default
- The address isn't in the address directory, but the record contains enough information to qualify the address for LACSLink processing.

❖ Example

The following table shows an original address with a rural route. Therefore, the software processes the record with LACSLink and converts the address to the LACSLink street address.

Original address	After LACSLink conversion
RR2 BOX 204	463 SHOWERS RD
DU BOIS PA 15801	DU BOIS PA 15801-66675

Related Information

[LACSLink \[page 265\]](#)

11.5.4 LACSLink directory files

SAP ships the LACSLink directory files with the U.S. National Directory update.

The LACSLink directory files require about 600 MB of additional hard drive space.

i Note

The directory size changes over time. See the [Product Availability Matrix \(PAM\)](#) for the most recent directory size.

The LACSLink directories include the following files:

- lacsx.txt
- lacsx.txt
- lacsy.ll
- lacsx.ll

⚠ Caution

The LACSLink directories must reside on the hard drive in the same directory as the LACSLink supporting files. Don't rename any of the files. LACSLink doesn't run if the file names are changed.

LACSLink directories expire in 105 days. LACSLink directories must have the same date as the other directories that you are using from the U.S. National Directories.

Related Information

[LACSLink \[page 265\]](#)

11.5.5 Required information in the job setup

Because CASS certification requires LACSLink, enter the required USPS license information in the USA Regulatory Address Cleanse transform.

The following list contains the USPS-required options located in the USPS License Information group:

CUSTOMER_COMPANY_NAME
CUSTOMER_COMPANY_ADDRESS
CUSTOMER_COMPANY_LOCALITY
CUSTOMER_COMPANY_REGION
CUSTOMER_COMPANY_POSTCODE1
CUSTOMER_COMPANY_POSTCODE2
CUSTOMER_COMPANY_PHONE

Related Information

[LACSLink \[page 265\]](#)

11.5.6 Reasons for LACSLink processing errors

You can fix some LACSLink processing errors in the transform configuration.

The following table describes errors that you can fix in the transform configuration.

Reason for error	Description
Missing required options	When your job setup doesn't include the required parameters in the USPS License Information group, and you have DPV and/or LACSLink enabled in your job, the software issues a verification error.
Unwritable Log File directory	If the path that you specified for the USPS_LOG_PATH option in the Reference Files group isn't writable, the software issues an error.

Related Information

[LACSLink \[page 265\]](#)

11.5.7 LACSLink output fields

Several output fields are available for generated LACSLink processing results.

Ensure that you enable LACSLink and include the following output fields in your job so that the transform posts LACSLink information to output:

[LACSLINK_QUERY](#)
[LACSLINK_RETURN_CODE](#)
[LACSLINK_INDICATOR](#)

Related Information

[LACSLink \[page 265\]](#)

[USA Regulatory Address Cleanse output fields \[page 796\]](#)

11.5.8 Cache LACSLink directories

The transform performance improves when you cache the LACSLink directories.

To find the amount of disk space required to cache the directories, see the [Product Availability Matrix \(PAM\)](#)  on the SAP Support Portal.

For information about disk space for directories in the PAM appendix “Disk Space Requirements for Referential Data & Cleansing Packages.”

SAP Data Quality Management SDK issues a verification error message at runtime and the transform terminates under the following circumstances:

- You don't have adequate system memory to load the LACSLink directories.
- You set the *INSUFFICIENT_CACHE_MEMORY_ACTION* option to *ERROR*.

If you select to *Continue* for *INSUFFICIENT_CACHE_MEMORY_ACTION*, the transform attempts to continue LACSLink processing without caching.

1. Open the *Options* tab of the transform editor.
2. Expand the Transform Performance option group.
3. Set the *CACHE_LACSLINK_DIRECTORIES* option to *YES*.

Related Information

[LACSLink \[page 265\]](#)

11.5.9 USPS Form 3553

The USPS Form 3553 contains LACSLink record counts.

If you have LACSLink enabled in the job, the LACS/LACSLink field in the USPS Form 3553 shows the number of records that have a LACSLink Indicator of *Y* or *S*. If you don't enable LACSLink processing, the LACS/LACSLink field shows the count only for the LACS code.

Related Information

[LACSLink \[page 265\]](#)

11.6 USPS Residential Delivery Indicator (RDI)

When you use RDI to determine the number of residential addresses in your mailing list, you can decide whether to use a commercial shipping service or to ship your parcels with the USPS and save money.

RDI determines whether a given address is for a residence or non-residence. Use RDI in either CASS mode or non-certified mode. RDI doesn't require that you process with DPV (Delivery Point Validation).

Benefits of RDI

Many parcel shipping companies charge more for parcels going to U.S. residential addresses. The USPS does not charge extra for residential parcel delivery. Therefore, you can save money by using the USPS to ship parcels going to residential addresses.

[How RDI works \[page 275\]](#)

RDI uses the 11-digit postcode to determine that an address is residential.

[RDI directory files \[page 276\]](#)

SAP ships the RDI directory files with the U.S. National Directory update.

[RDI output field \[page 277\]](#)

The RDI feature outputs information to a single output field.

[RDI in reports \[page 277\]](#)

Data Quality Management SDK reports RDI information in the USPS Form 3553 and in statistics files.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

11.6.1 How RDI works

RDI uses the 11-digit postcode to determine that an address is residential.

Before you use the RDI feature in Data Quality Management SDK, install the USPS-supplied RDI directories. The 11-digit postcode contains the following elements:

- Postcode1
- Postcode2
- DPBC

i Note

Data Quality Management SDK can sometimes determine if an address is residential by using just the Postcode2.

Data Quality Management SDK indicates whether an address is for a residence in the output field, *RDI_INDICATOR*.

Related Information

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

11.6.2 RDI directory files

SAP ships the RDI directory files with the U.S. National Directory update.

Follow installation instructions for the RDI directory files in the directory letter.

RDI requires the directories described in the following table.

File	Description
<code>rts.hs11</code>	For 11-digit postcode lookups (Postcode2 plus DPBC). Data Quality Management SDK uses the <code>rts.hs11</code> file when an address contains an 11-digit postcode. RDI bases the residence determination on the delivery point.
<code>rts.hs9</code>	For 9-digit postcode lookups (Postcode2). The <code>rts.hs9</code> file is based on ZIP+4 codes. RDI is possible only when the addresses for the ZIP+4 are for residences or for non-residences.

Related Information

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

11.6.3 RDI output field

The RDI feature outputs information to a single output field.

Include the RDI output field in your field setup. Data Quality Management SDK populates the RDI field only when you set the [ENABLE_RDI](#) option in the Assignment Options group in the transform configuration to [YES](#).

Field	Description
RDI_INDICATOR	Populated with one of the following values: • Y: Address is for a residence. • N: Address isn't for a residence.

Related Information

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

11.6.4 RDI in reports

Data Quality Management SDK reports RDI information in the USPS Form 3553 and in statistics files.

When you process in CASS mode, Data Quality Management SDK populates the USPS Form 3553 with the number of residences. It also includes the information in the CASS header record.

Data Quality Management SDK reports RDI counts and percentages in the statistics files.

11.7 Z4Change

The Z4Change feature in the USA Regulatory Address Cleanse transform enhances automation compatibility and improves deliverability.

With Z4Change, Data Quality Management SDK adds ZIP+4 codes to addresses and updates incorrect ZIP+4 codes.

Z4Change is most cost-effective for databases that are large and fairly stable, such as databases of regular customers and subscribers. In our tests, based on files in which 5% of records were affected by a ZIP+4 change, total batch processing time was one third the normal processing time.

There are fewer benefits using Z4Change in real-time jobs.

[Z4Change and USPS rules \[page 278\]](#)

The USPS requires that mailing lists undergo a complete assignment process every 3 years.

[Z4Change directory \[page 278\]](#)

The USPS updates the Z4Change directory each month.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)
[USPS eLOT \[page 258\]](#)
[Early Warning System \(EWS\) \[page 259\]](#)
[SuiteLink \[page 261\]](#)
[LACSLink \[page 265\]](#)
[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)
[Suggestion lists \[page 279\]](#)
[USPS certifications \[page 282\]](#)
[NCOALink and Move Update \[page 292\]](#)
[Statistics for multiple data sources \[page 317\]](#)

11.7.1 Z4Change and USPS rules

The USPS requires that mailing lists undergo a complete assignment process every 3 years.

Use Z4Change only for updating a database that has previously undergone a full validation process.

Related Information

[Z4Change \[page 277\]](#)

11.7.2 Z4Change directory

The USPS updates the Z4Change directory each month.

If you've purchased the Z4Change option, access the Z4Change directory with the monthly USPS directory delivery.

The Z4Change directory contains a list of all the ZIP Codes and ZIP+4 codes in the country.

Related Information

[Z4Change \[page 277\]](#)

11.8 Suggestion lists

Use Suggestion List processing in transactional projects with the USA Regulatory Address Cleanse and the Global Address Cleanse transforms.

Use suggestion lists to complete and populate addresses that have minimal data. Suggestion lists offer suggestions for possible matches when the transform can't find an exact match.

For information about suggestion lists for the Global Address Cleanse transform, see the topics in [Global Address Cleanse transform reference \[page 395\]](#).

i Note

Suggestion lists aren't available for batch processing.

i Note

If you have suggestion lists enabled, you're not eligible for CASS discounts, and Data Quality Management SDK doesn't produce the required CASS documentation.

[How suggestion lists work \[page 280\]](#)

The USA Regulatory Address Cleanse transform presents address suggestions when it can't find an exact match to an address in the postal directories.

[When you need more information \[page 281\]](#)

When Data Quality Management SDK produces a suggestion list, you sometimes need more information before selecting a possible match.

[CASS and the suggestion list feature \[page 282\]](#)

Data Quality Management SDK doesn't produce a USPS Form 3553 when you enable suggestion lists.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[USPS certifications \[page 282\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

11.8.1 How suggestion lists work

The USA Regulatory Address Cleanse transform presents address suggestions when it can't find an exact match to an address in the postal directories.

Ideally, when the transform looks up an address in the USPS postal directories, it finds exactly one matching record with a matching combination of locality, region, and postcode. Then, during the look-up in the USPS national ZIP+4 directory, the transform finds exactly one record that matches the address. However, the transform doesn't always find an exact match.

Breaking ties

Sometimes it's impossible to match an input address to one matching record in the directory. At other times, the transform finds several directory records that are near matches to the input data.

When the transform is close to a match, but not quite close enough, it assembles a list of the near matches and presents them as suggestions. When you choose a suggestion, the transform tries again to assign the address.

❖ Example

Incomplete last line

Given the incomplete last line in the following table, the transform can't reliably choose one of the four localities. But if you choose one, the transform can proceed with the rest of the assignment process.

Input record	Possible matches in the City/ZCF directories
Line1 = 1000 vine Line2 = lacr wi	La Crosse, WI 54603 Lancaster, WI 53813 La Crosse, WI 54601 Larson, WI 54947

❖ Example

Missing directional

As with last lines, the transform can't reliably choose a match with incomplete address lines. A common problem is a missing directional. In the following table, there's an equal chance that the directional is North or South. The transform has no basis for choosing one way or the other.

Input record	Possible matches in the ZIP+4 directory
Line1 = 615 losey blvd Line2 = 54603	600–699 Losey Blvd N 600–698 Losey Blvd S

❖ Example

Missing suffix

A missing suffix would cause similar behavior as with a missing directional. In the following table, there's an equal chance that the suffix is Pl or St. The transform has no basis for choosing one or the other.

Input record	Possible matches in the ZIP+4 directory
Line1 = 121 dorn Line2 = 54601	100–199 Dorn Pl 101–199 Dorn St

❖ Example

Misspelled street names

A misspelled or incomplete street name can result in address suggestions as shown in the following table.

Input record	Possible matches in the ZIP+4 directory
Line1 = 4100 marl Line2 = minneapolis mn	4100–4199 Marshall 55421 4100–4199 Maryland 55427

Related Information

[Suggestion lists \[page 279\]](#)

11.8.2 When you need more information

When Data Quality Management SDK produces a suggestion list, you sometimes need more information before selecting a possible match.

❖ Example

The following list contains examples of additional information resources:

- Operators taking information over the phone can ask for more information from the customer to decide what suggestion to choose.
- Operators entering data from a consumer coupon that is a little smudged may be able to choose a suggestion based on the information that isn't smudged.

If you don't have additional resources to help you select a suggested address, don't guess. Leave the address unassigned until you have reliable information.

Related Information

[Suggestion lists \[page 279\]](#)

11.8.3 CASS and the suggestion list feature

Data Quality Management SDK doesn't produce a USPS Form 3553 when you enable suggestion lists.

CASS doesn't support suggestion lists for several reasons. One reason is that the USPS suspects that users are tempted to guess when presented with suggestions. An incorrect guess can result in misrouted mail that is expensive for the USPS to handle. Another reason is that the USPS Form 3553 is available only when you process in batch mode with the [DISABLE_CERTIFICATION](#) option set to [NO](#).

To produce a CASS-certified mailing using suggestion lists, perform two processes:

- Run the addresses through a real-time job with suggestions enabled.
- Run the addresses again through a batch process with [DISABLE_CERTIFICATION](#) set to [NO](#).

The second process in the batch job is CASS-compliant, and Data Quality Management SDK generates a USPS Form 3553 that covers the entire mailing database.

Related Information

[Suggestion lists \[page 279\]](#)

11.9 USPS certifications

The software is certified for USPS address match, move update, and address quality processes that comply with various USPS certifications.

Data Quality Management SDK is CASS certified. As long as you use Data Quality Management SDK to prepare mailings, and you enable all of the required options for certification, your mailing list is certified. That certification extends to when you integrate Data Quality Management SDK into your own applications.

Some companies may want to self-certify so that the company name appears in all USPS reports and in the USPS list of approved companies. However, keep in mind that when you self-certify, you may have to pay the USPS for extra charges or fees.

The USPS PostalPro Web site contains all of the certification forms and procedures in the following USPS postal publications:

- The CASS area of the PostalPro Web page is at <https://postalpro.usps.com/certifications/cass> .
- The NCOALink area of the PostalPro Web page is at <https://postalpro.usps.com/mailing-and-shipping-services/NCOALink> .

We maintain Data Quality Management SDK for the following USPS certifications:

- DPV
- LACSLink
- SuiteLink
- RDI (optional)

- eLOT (optional)
- NCOALink

[USPS CASS certification static directories \[page 283\]](#)

Only customers who are trying to self-certify for CASS with the USPS need static CASS directories.

[USPS CASS self certification \[page 285\]](#)

Even though SAP Data Services is fully certified for USPS CASS, some mailers go through the process of CASS self certification.

[USPS CASS self certification overview \[page 285\]](#)

SAP Data Services is already certified for USPS CASS, but we provide information about self certifying for CASS for mailers who want to self-certify.

[NCOALink self-certification \[page 289\]](#)

Become a certified NCOALink mailer by self-certifying using Data Quality Management SDK and the USA Regulatory Address Cleanse transform.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[NCOALink and Move Update \[page 292\]](#)

[Statistics for multiple data sources \[page 317\]](#)

11.9.1 USPS CASS certification static directories

Only customers who are trying to self-certify for CASS with the USPS need static CASS directories.

The static CASS directories should only be used for CASS self-certification and should never be used in production. The directory package includes static versions of the US National, eLOT, DPV, LACSLink, RDI, and SuiteLink directories.

For information about accessing and installing static CASS directories, see KBA [3321662](#) .

i Note

CASS self-certification is not required as the SAP software is already certified, but we provide the information for those mailers who want to self-certify themselves.

You may want to become a CASS-certified vendor so that your company name appears in all USPS reports and on the CASS vendor list.

There are two USPS Stage tests for CASS certification:

- Stage I: Optional. Practice test for which you view the results. Correct any inconsistencies with the USPS-expected results and take the Stage I test until you are satisfied with the results. You do not submit Stage I test results to the USPS.
- Stage II: Required. Does not contain answers like the Stage I test. Stage II tests the accuracy of the software.

After you take the Stage II test, forward the results to the USPS. You cannot see your results.

For complete information and resources for CASS, visit the PostalPro Web site at <https://postalpro.usps.com/certifications/cass> .

i Note

If you are self-certifying for CASS, use static directories instead of the active directories for consistent results that meet USPS expectations.

All USPS processing requires postal directories that change on a periodic basis. However, static directories do not change with the regular directory updates. The following are some of the benefits of using static directories for USPS certification:

- Static directories do not expire until the end of the current CASS cycle.
- Static directories ensure that your output is consistent between Stage I and Stage II tests.
- When you re-certify, use the same static directories as you used for original certification (as long as they haven't expired) for consistent results.

The following directories are available in static format:

- ZIP4 directories
- DPV directories
- RDI directories
- EWS directories
- SuiteLink directories
- LACSLink directories

The USPS provides the test input files that you will use during certifications.

i Note

The USPS provides static directories for CASS testing, however they are not in the proprietary format required by SAP software. Do not use the static directories from the USPS.

Only leverage the static directories provided by SAP.

Parent topic: [USPS certifications \[page 282\]](#)

Related Information

[USPS CASS self certification \[page 285\]](#)

[USPS CASS self certification overview \[page 285\]](#)

[NCOALink self-certification \[page 289\]](#)

[USPS certifications \[page 282\]](#)

11.9.2 USPS CASS self certification

Even though SAP Data Services is fully certified for USPS CASS, some mailers go through the process of CASS self certification.

Consider the following information before you decide to self certify for CASS:

- You'll receive the same certification that Data Services already has. Therefore, you do not have to self-certify for CASS if you are using Data Services to produce CASS-certified mailings.
- If you have self certified your data quality application for CASS, your end users, the ones that use your data quality application, don't have to obtain their own CASS certification. When you self-certify your product, users can produce CASS-certified mailing lists with your product.
- CASS certification proves that your application can assign and standardize addresses correctly.
- CASS certification is given to software programs, not the users of the software program.
- The CASS reports pertain to address lists, not mailers.

Parent topic: [USPS certifications \[page 282\]](#)

Related Information

[USPS CASS certification static directories \[page 283\]](#)

[USPS CASS self certification overview \[page 285\]](#)

[NCOALink self-certification \[page 289\]](#)

[USPS CASS self certification overview \[page 285\]](#)

11.9.3 USPS CASS self certification overview

SAP Data Services is already certified for USPS CASS, but we provide information about self certifying for CASS for mailers who want to self-certify.

You may want to become a CASS-certified vendor so that your company name appears in all USPS reports and on the CASS vendor list.

There are two USPS Stage tests for CASS certification: Stage I and Stage II:

- Stage I: Optional. Practice test for which you view the results. Correct any inconsistencies with the USPS-expected results and take the Stage I test until you are satisfied with the results. You do not submit Stage I test results to the USPS.
- Stage II: Required. Does not contain answers like Stage I test. Stage II tests the accuracy of the software. After you take the Stage II test, forward the results to the USPS. You cannot see your results.

For complete information and resources for CASS, visit the PostalPro Web site at <https://postalpro.usps.com/certifications/cass> .

[Preparing for CASS self-certification \[page 286\]](#)

Prepare to self-certify the software for CASS and NCOALink by completing all of the necessary USPS forms, obtaining an authorization code, and obtaining the license key code.

[CASS Stage testing \[page 287\]](#)

The USPS provides an optional practice test named Stage I and the final test named Stage II.

[USPS Form 3553 required options \[page 288\]](#)

Complete the required options in the CASS Report Options group in the USA Regulatory Address Cleanse transform to be compliant with CASS requirements.

[Using the SAP DQM SDK for CASS processing and self-certification \[page 288\]](#)

You can use APIs and configuration files in the SAP Data Quality Management SDK to perform CASS processing.

Parent topic: [USPS certifications \[page 282\]](#)

Related Information

[USPS CASS certification static directories \[page 283\]](#)

[USPS CASS self certification \[page 285\]](#)

[NCOALink self-certification \[page 289\]](#)

11.9.3.1 Preparing for CASS self-certification

Prepare to self-certify the software for CASS and NCOALink by completing all of the necessary USPS forms, obtaining an authorization code, and obtaining the license key code.

Before you certify your company for CASS, ensure that you completely understand the steps to take for certification from the USPS.

1. Obtain the USPS CASS forms from <https://postalpro.usps.com/certifications/cass> .

Forms include:

- CASS Order Form
- Terms and Conditions
- Electronic Product Fulfillment Web Access Request Form

2. Complete and submit the forms to the USPS.

After the USPS approves your application, they give you an authorization code.

3. Submit the USPS authorization code to SAP Technical Support.
4. Obtain the license key by following instructions provided by SAP Technical Support.

The license key enables the required functionality in the USPS Regulatory Address Cleanse transform.

Related Information

[USPS CASS self certification overview \[page 285\]](#)

[CASS Stage testing \[page 287\]](#)

11.9.3.2 CASS Stage testing

The USPS provides an optional practice test named Stage I and the final test named Stage II.

1. Optional. Download the CASS Stage I test from the USPS.
The USPS does not require that you take the Stage I test. It is a practice test. Use the results of the Stage I test to fine-tune your software so that the results compare favorably with the USPS. For details about obtaining the Stage I test from the USPS, see the USPS *CASS Technical Guide*.
2. When you are satisfied that your Stage I test results compare favorably with the USPS-expected results, request the Stage II test from the USPS.
3. Download and unzip the Stage II test file to an output area.
4. After you run the Stage II file, check that the totals on the USPS Form 3553 and the actual totals from the processed file match.
5. Follow USPS procedures to turn in the Stage II test results.

The USPS takes about two weeks to grade your test.

Related Information

[USPS CASS self certification overview \[page 285\]](#)

11.9.3.3 USPS Form 3553 required options

Complete the required options in the CASS Report Options group in the USA Regulatory Address Cleanse transform to be compliant with CASS requirements.

Option	Description
LIST_NAME	Retain the default setting of STAGE_FILE .
LIST_OWNER	Retain the default setting of USPS .
MAILER_ADDRESS(1-4)	Specify the name and address of your company, or the company that is becoming CASS certified (up to 29 characters per line).
COMPANY_NAME_CERTIFIED	Specify the name of the company that owns Data Quality Management SDK software that processes the mailing list. Reported in the Form 3553.
SOFTWARE_VERSION	Specify the name and version that you are using to receive CASS self certification.  Example ACE 14.03.00.0

Related Information

[USPS CASS self certification overview \[page 285\]](#)

11.9.3.4 Using the SAP DQM SDK for CASS processing and self-certification

You can use APIs and configuration files in the SAP Data Quality Management SDK to perform CASS processing.

The following table provides an overview of how to use the APIs and configuration files for CASS processing.

Requirement	Description
Send all records for processing from the input stage file in the form of a single collection.	Instantiate a MultiRecordTransform object with a U.S. Regulatory Address Cleanse XML configuration.

Requirement	Description
Configure the U.S. Regulatory Address Cleanse transform USPS Certification Testing Mode for CASS.	In the Assignment Options section of the U.S. Regulatory Address Cleanse configuration file, set the USPS_CERTIFICATION_TESTING_MODE option to CASS (instead of NONE).
Assure that only one output field, STAGE_TEST_RECORD , is included in the output schema.	The software outputs the test data to the STAGE_TEST_RECORD field.
Generate the PS FORM 3553 Report.	Use the Certified Report Generator class to generate the report. (For example, in the .Net implementation, the Certified Report Generator is in the TransformFactory class.)
Submit output file with PS FORM 3553 report for processing to the USPS.	The SDK produces a file that contains the data that you submit to the USPS.

11.9.4 NCOALink self-certification

Become a certified NCOALink mailer by self-certifying using Data Quality Management SDK and the USA Regulatory Address Cleanse transform.

Note

Self-certifying for NCOALink is optional. When you purchase NCOALink from SAP, the software is already certified for NCOALink. You may want to self-certify so that your company name appears on all reports and in the NCOALink vendor list.

The following steps provide a high-level overview of the tasks required to obtain NCOALink certification.

1. Complete the USPS Application and Self-Certification Statement Approval and submit to the USPS.
2. The USPS sends you an authorization code upon approval.
3. Contact SAP to purchase the necessary NCOALink functionality from SAP. SAP requires your USPS authorization code.
4. Take the required Stage II test.
5. Complete the USPS License Agreement and pay the USPS for NCOALink.

During certification, you process files from the USPS to prove that the software adheres to the requirements of your license agreement. NCOALink certification has two Stage tests:

- Stage I: Optional. Includes answers that allow you to troubleshoot and prepare for the Stage II test.
- Stage II: Required. Does not contain answers. Send results to the USPS for evaluation of the accuracy of your software configuration.

For information about the entire procedure, see the PostalPro Web site at <https://postalpro.usps.com/mailing-and-shipping-services/NCOALink>. Scroll to the Featured Resources and find the group of documents related to the type of license you are applying for:

- NCOALink End User Documents
- NCOALink Limited Service Provider Documents
- NCOALink Full Service Provider Documents

[NCOALink software product information \[page 290\]](#)

Complete the NCOALink Compliance Testing Product Information Form with the applicable information about the licensee and SAP software.

[NCOALink Directories and Daily Delete Files \[page 291\]](#)

For NCOALink certification, the USPS provides the NCOALink test data after you receive the USPS authorization code and after you obtain NCOALink software from Data Quality Management SDK.

[Use DQM SDK for NCOALink processing \[page 292\]](#)

Perform NCOALink certification processing using the APIs and configuration files from SAP Data Quality Management SDK .

Parent topic: [USPS certifications \[page 282\]](#)

Related Information

[USPS CASS certification static directories \[page 283\]](#)

[USPS CASS self certification \[page 285\]](#)

[USPS CASS self certification overview \[page 285\]](#)

[USPS certifications \[page 282\]](#)

[About NCOALink directories \[page 304\]](#)

11.9.4.1 NCOALink software product information

Complete the NCOALink Compliance Testing Product Information Form with the applicable information about the licensee and SAP software.

Obtain the Compliance Testing Product Information Form on the PostalPro Web site at <https://postalpro.usps.com/mailing-and-shipping-services/NCOALink>  .

Compliance Testing Product Information form	Description
Company Name & License Number	Specify the NCOALink licensee company name. The license number is the authorization code provided in your USPS approval letter.
Company's NCOALink Product Name	Enter Mover ID for NCOALink.
Platform or Operating System	Specify the NCOALink licensee information.
NCOALink Software Vendor	Enter SAP Americas, Inc.
NCOALink Software Product Name	Enter Mover ID.
NCOALink Software Product Version	Enter ACE.
Address Matching ZIP+4 Product Name Address Matching ZIP+4 Product Version	Contact SAP Support for this information.
Address Matching ZIP+4 System	Enter Closed.

Compliance Testing Product Information form	Description
Is Software Hardware Dependent?	Enter No.
DPV Product Name	Enter ACE.
DPV Product Version	Contact SAP Support.
LACSLink Product Name	Enter ACE.
LACSLink Product Version	Contact SAP Support.
NCOALink Software options: Integrated or Standalone checkboxes	Select Integrated.
ANLink Enhancement checkbox (applicable for Limited Service Providers and End Users)	Select if you purchased the ANLink option from SAP.
HASH—FLAT—BOTH checkboxes	Specifies the file format in which to receive the NCOALink information. SAP supports both formats.
NCOALink Level Option checkboxes	Specifies the NCOALink level for which you are testing.

Related Information

[NCOALink self-certification \[page 289\]](#)

11.9.4.2 NCOALink Directories and Daily Delete Files

For NCOALink certification, the USPS provides the NCOALink test data after you receive the USPS authorization code and after you obtain NCOALink software from Data Quality Management SDK.

After you are licensed for NCOALink, obtain the NCOALink directories from the USPS either weekly or monthly:

- Monthly for NCOALink end users
- Weekly for NCOALink full and limited service providers

Additionally, the USPS requires that you obtain the Daily Delete files daily, or on the days that you run your NCOALink jobs. Download the daily delete files from the USPS and copy it to the directory where your NCOALink directories are located.

Download the NCOALink directories using the EPF Downloader Manager, or your own tool if applicable. Then use the SAP NCOALink utility to extract and uncompress the directories.

The SAP NCOALink utility is a 64-bit application that you run from a command line. Find the utility in %LINK_DIR%\bin\ncoa\ncoutil.exe (Windows) or \$LINK_DIR/bin/ncoa/ncoutil (UNIX). If the utility is not there, download it from the SAP Support Portal at <https://support.sap.com/software/address-directories.html>.

For complete details about NCOALink directories and the daily delete files, visit the NCOALink pages in the PostalPro Web site at <https://postalpro.usps.com/mailling-and-shipping-services/NCOALink>.

Related Information

[NCOALink self-certification \[page 289\]](#)

11.9.4.3 Use DQM SDK for NCOALink processing

Perform NCOALink certification processing using the APIs and configuration files from SAP Data Quality Management SDK .

The following table provides an overview of how to use the SDK product for NCOALink processing.

Requirement	In SDK
Process at least 100 unique records.	Instantiate a <code>MultiRecordTransform</code> object with a USA Regulatory Address Cleanse XML configuration and process at least 100 unique records.
Enable the NCOALink option in configuration file.	In the Assignment Options group of the USA Regulatory Address Cleanse XML configuration file, set the Enable NCOALink option to Yes .
Set processing options in configuration file.	Complete options in the Processing Options group of the USA Regulatory Address Cleanse XML configuration file.
Obtain license keycode from SAP	Contact SAP Customer Support and request an NCOALink keycode. Enter the keycode in the <code>bojprods.key</code> file. Ensure that the keycode matches the setting configured in the Provider Level option in the USPS Licensee and Customer Information Options group in the USA Regulatory Address Cleanse XML configuration file.
Submit monthly statistics to the USPS.	Use the <code>StatisticsHandler</code> to collect statistics from each run and consolidate the results into a monthly CSL report. Name and format the report following USPS rules.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10 NCOALink and Move Update

NCOALink (National Change of Address) is a part of USPS Move Update, which ensures that your mailing list contains up-to-date address information.

Move Updating is the process of checking your mailing list against the National Change of Address (NCOA) database. Move Updating ensures that your data is updated with current addresses. When you process for

Move Update, Data Quality Management SDK updates records for individuals or businesses that have moved and have filed a Change of Address (COA) form with the USPS. Processing for Move Update qualifies your mailings for the discounts available for First-Class presorted mail.

In addition to supporting NCOALink, the USA Regulatory Address Cleanse transform supports other programs that are a part of Move Update: ANKLink and SuiteLink.

i Note

Mover ID is the name under which Data Quality Management SDK is certified with the USPS for NCOALink.

[The importance of move updating \[page 294\]](#)

Processing your mailing lists with Move Update is a good business practice plus you receive postal discounts for your mailing.

[How NCOALink works \[page 295\]](#)

To understand the effects of NCOALink on your mailing lists, it is important to understand how Data Quality Management SDK uses NCOALink.

[NCOALink service provider levels \[page 296\]](#)

NCOALink users fall into one of three categories of service providers.

[NCOALink brokers and list administrators \[page 297\]](#)

An NCOALink user can have a broker or list administrator who owns the lists they're processing.

[Address not known \(ANKLink\) \[page 298\]](#)

Enhance NCOALink data with an additional 30 months of data when you purchase ANKLink.

[NCOALink and software performance \[page 299\]](#)

Because of the additional processes required for NCOALink, processing speed is slower.

[Getting started with NCOALink \[page 300\]](#)

Prepare for using NCOALink by completing USPS applications and configuring Data Quality Management SDK for NCOALink processing.

[What to expect from the USPS and SAP \[page 300\]](#)

Understand the expectations and roles between you, the USPS, and vendors when you become an NCOALink licensee.

[NCOALink security requirements \[page 302\]](#)

The USPS imposes strict security procedures for software vendors who use NCOALink data and provide NCOALink processing.

[About NCOALink directories \[page 304\]](#)

Download the NCOALink directories with the USPS EPF Downloader Manager tool and extract and uncompress them with the SAP NCOALink Utility.

[About the NCOALink Daily Delete Files \[page 308\]](#)

The USPS requires that NCOALink service provider licensees download the daily delete files each day before they perform NCOALink processing.

[Output file strategies \[page 309\]](#)

By selecting specific Field Class Selections, configure your output file to meet your needs.

[Improving NCOALink processing performance \[page 310\]](#)

Many factors affect performance when processing NCOALink data.

[NCOALink log files \[page 314\]](#)

Use information from the StatisticsHandler to generate the USPS-required NCOALink log files.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

[Statistics for multiple data sources \[page 317\]](#)

[USA Regulatory Address Cleanse transform \[page 242\]](#)

11.10.1 The importance of move updating

Processing your mailing lists with Move Update is a good business practice plus you receive postal discounts for your mailing.

The USPS requires move updating on all First Class presorted mailings. To help mailers meet this requirement, the USPS offers certain options, including NCOALink.

Millions of people and businesses move every year. Many of these individuals and businesses report new address information to the USPS. Therefore, it makes sense to use the USPS data to keep your records up-to-date.

With NCOALink, you benefit in the following ways:

- Improve mail deliverability.
- Reduce the cost and time to forward mail.
- Meet the USPS move-updating requirement for presorted First Class mail.
- Prepare for the possible expansion of move-update requirements.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.2 How NCOALink works

To understand the effects of NCOALink on your mailing lists, it is important to understand how Data Quality Management SDK uses NCOALink.

With Move Update configured in your job, Data Quality Management SDK performs the following tasks:

- Parses and standardizes the input addresses.
- Searches the NCOALink database for records that match the input addresses.
- When it finds a match, updates the record with the move information from NCOALink, including the new address when available.
- Assigns postal and other codes to the move-updated addresses.
- Includes additional information to the output file based on the field class you select for output: Correct, Move-Updated, or Best. The following table describes the additional information based on the field class.

Field class	Description
Correct	Outputs the original address, completed and corrected using values found in the directories and based on the standardization options you set in the transform.
Move-Updated	Outputs the address components that have been updated with move-updated address data.
	Note The transform looks for the move-updated address information in the U.S. National Directories. When the transform does not find the move-updated address in the U.S. National Directories, it populates the Move Updated fields with information from the Move Update Directories only. The transform does not update the Move Updated fields that are populated as a result of standardizing against the U.S. National Directories.
Best	Outputs the move-updated address when it exists and when it matches in the U.S. National directories. Or, output contains the original data when a move does not exist or when the move does not match in the U.S. National Directories.

Based on the [APPLY_MOVE_TO_STANDARDIZED_FIELDS](#) option in the NCOALink group, standardized components can contain either original or move-updated addresses.

- Produces the reports and log files required for USPS compliance.

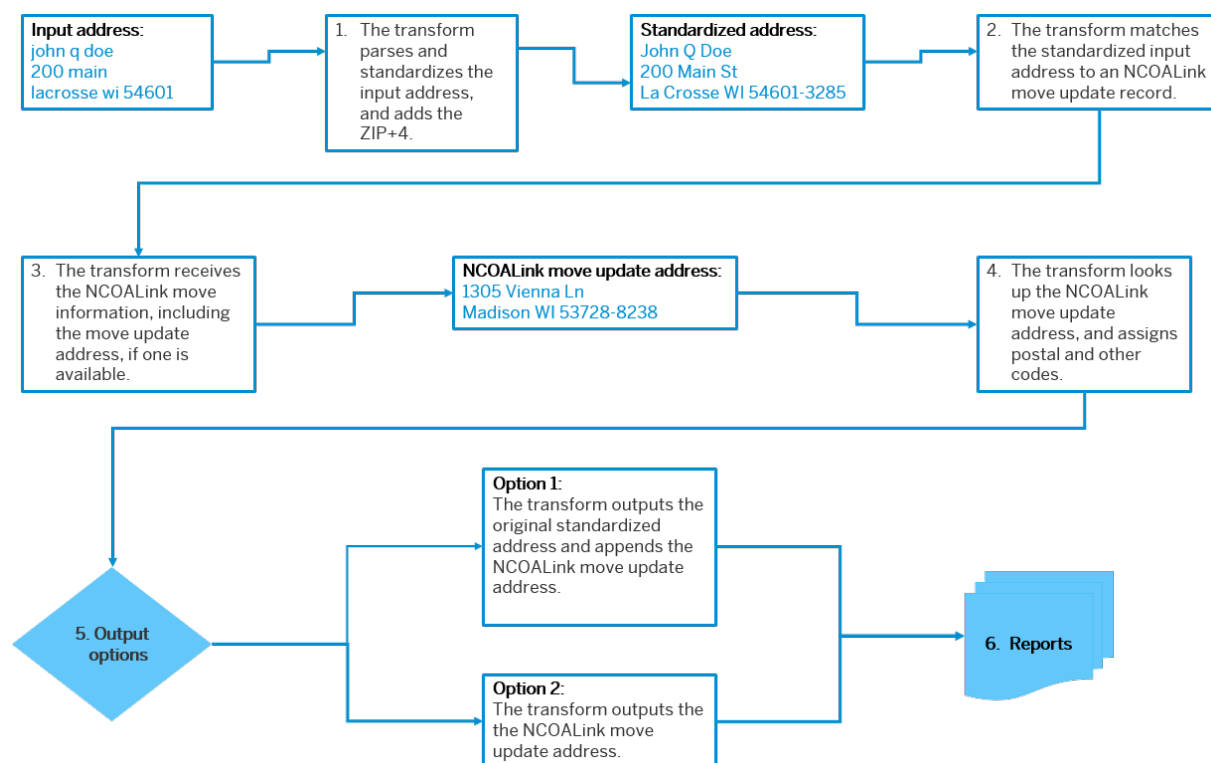
Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.2.1 Example of NCOALink process

To visualize the NCOALink process, view a flowchart with each step.

The following example follows an input address through the NCOALink process.



The following steps take you through the steps of the flowchart:

1. NCOALink requires parsed, standardized address data as input. Therefore, before NCOALink processing, Data Quality Management SDK performs its normal processing on the address data.
2. The transform searches the NCOALink database for a record that matches the parsed, standardized record.
3. The transform receives the move information, including the new address if one is available.
4. The transform looks up the move record that comes back from the NCOALink database, to assign postal and other codes.
5. At your option, the transform can either retain the old address and append the new, or replace the old address with the new.
6. The transform produces the reports and log files that the USPS requires for compliance.

11.10.3 NCOALink service provider levels

NCOALink users fall into one of three categories of service providers.

Specify the service provider level in the [PROVIDER_LEVEL](#) option under the USPS License Information group.

i Note

Only provider levels supported by your registered keycodes display in the selection list.

Provider Level	Description
FSP (Full Service Provider)	Provides NCOALink processing to third parties.
LSP (Limited Service Provider)	Provides NCOALink processing to third parties and internally.
EUM (End User Mailer)	Provides NCOALink processing to in-house lists only.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.4 NCOALink brokers and list administrators

An NCOALink user can have a broker or list administrator who owns the lists they're processing.

When there's a broker or list administrator involved, add contact information in the NCOALink group in the Contact Details options.

The following table describes the broker and list administrator roles.

Role	Description
Broker	Directs business to an NCOALink service provider.
List Administrator	Maintains and stores lists.

Brokers and list administrators are different in two ways:

- List administrators don't send move-updated files back to the list owner.
- List administrators can have an NCOALink license.

If a list administrator, a broker, or both are involved in your job, ensure that you duplicate the Contact Detail group and complete the options for each of them separately.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.5 Address not known (ANKLink)

Enhance NCOALink data with an additional 30 months of data when you purchase ANKLink.

NCOALink limited service providers and end users receive change of address data for the preceding 18 months. The ANKLink option enhances that information by providing extra data about moves that occurred in the previous months 19 through 48.

! Restriction

The additional 30 months of data that comes with ANKLink indicate only that a move occurred and the date of the move; it doesn't provide the new address.

When you have NCOALink through Data Quality Management SDK, you can use purchased ANKLink data in your jobs.

ANKLink data helps you make informed choices regarding a contact. If the data indicates that the contact has moved, you can choose to suppress that contact from the list or try to acquire the new address from an NCOALINK full service provider.

If an ANKLink match exists, Data Quality Management SDK outputs a code in the [ANKLINK_RETURN_CODE](#) field and in the NCOALink Processing Summary report.

→ Tip

If you're an NCOALink full service provider, you already have access to the full 48 months of move data, including the new addresses.

[ANKLink data \[page 298\]](#)

Request ANKLink data from the USPS National Customer Support Center (NCSC).

[ANKLink support for NCOALink provider levels \[page 299\]](#)

Data Quality Management SDK supports ANKLink for the three NCOALink provider levels defined by the USPS.

Related Information

[USA Regulatory Address Cleanse transform \[page 242\]](#)

11.10.5.1 ANKLink data

Request ANKLink data from the USPS National Customer Support Center (NCSC).

To request ANKLink data, either call the NCSC at 1-800-589-5766, or e-mail the NCSC at ncoalink@usps.gov. ANKLink data isn't available from SAP.

Data Quality Management SDK detects if you're using ANKLink data. Therefore, you don't have to specify whether you're using ANKLink in your job setup.

Related Information

[Address not known \(ANKLink\) \[page 298\]](#)

11.10.5.2 ANKLink support for NCOALink provider levels

Data Quality Management SDK supports ANKLink for the three NCOALink provider levels defined by the USPS.

The options in Data Quality Management SDK vary based on your provider level and the software package that you purchased. The following table shows the provider levels and support.

Provider Level	Service to Third Parties	COA Data Months	Data From USPS	ANKLink Support
FSP (Full Service Provider)	Yes. Requires at least 51% of services be for third-party processing.	48	Weekly	No
LSP (Limited Service Provider)	Yes. Can use services internally and for third parties.	18	Weekly	Yes
End User	No	18	Monthly	Yes

→ Tip

If you're an NCOALink end user, you can complete a Stop Processing Alternative application and enter into an agreement with the USPS. After USPS approval, purchase the stop processing alternative functionality from SAP, which allows DPV and LACSLink processing to continue after a false positive address record is detected.

Related Information

[Address not known \(ANKLink\) \[page 298\]](#)

[Stop Processing Alternative \[page 246\]](#)

11.10.6 NCOALink and software performance

Because of the additional processes required for NCOALink, processing speed is slower.

In our tests, Data Quality Management SDK ran slower with NCOALink enabled than with it disabled. Your processing speed depends on the computer running the software and the percentage of input records affected by a move. For example, more moves equals slower performance.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.7 Getting started with NCOALink

Prepare for using NCOALink by completing USPS applications and configuring Data Quality Management SDK for NCOALink processing.

Before you begin NCOALink processing, perform the following tasks:

1. Complete the USPS certification process, and obtain approval to become an NCOALink service provider or end user.
2. Learn about the available output strategies and performance optimization options.
3. Configure the USA Regulatory Address Cleanse transform for NCOALink processing.
4. Include the transform in a job to produce NCOALink output.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.8 What to expect from the USPS and SAP

Understand the expectations and roles between you, the USPS, and vendors when you become an NCOALink licensee.

NCOALink, and the license requirements that go with it, has created a new dimension in the relationship among mailers (you), the USPS, and vendors. It's important to be clear about what to expect from everyone.

[Move updating is a business decision \[page 301\]](#)

Because NCOALink offers options to replace individual or family addresses with new addresses, service providers select whether to accept move updates related to either family or individual moves.

[USPS responsibility for support \[page 301\]](#)

Obtain support from the USPS for problems with the NCOALink system.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.8.1 Move updating is a business decision

Because NCOALink offers options to replace individual or family addresses with new addresses, service providers select whether to accept move updates related to either family or individual moves.

The USPS recommends that you make these choices only after careful thought about your customer relationships. Consider the following examples:

❖ Example

You mail monthly account statements, checks, and other correspondence to your customers. You have a fiduciary responsibility for ensuring your customers receive the correct information in a timely manner. The USPS recommends that you verify each move by sending a double postcard, or other easy-reply piece before you change a financial record to a new address. As technology changes, keep in touch with the USPS to determine the best way to verify address changes with your customers.

❖ Example

You have a business relationship with a customer who is married. You do not have a business relationship with the spouse. Ensure that you handle move updates carefully with respect to divorce or separation. Consider taking the time to confirm each move before you permanently update a record.

Related Information

[What to expect from the USPS and SAP \[page 300\]](#)

11.10.8.2 USPS responsibility for support

Obtain support from the USPS for problems with the NCOALink system.

When you acquire an NCOALink license from the USPS, you are licensing a USPS product. The USPS develops and maintains the NCOALink database. Therefore, contact the USPS National Customer Support Center (NCSC) in Memphis as your first line of support for problems with the NCOALink system. The number is 800-589-5766.

Related Information

[What to expect from the USPS and SAP \[page 300\]](#)

11.10.9 NCOALink security requirements

The USPS imposes strict security procedures for software vendors who use NCOALink data and provide NCOALink processing.

NCOALink data is confidential and sensitive. The USPS requires security procedures to prevent users from abusing the information.

The USPS requires that each mailing list contain at least 100 unique records. Therefore, the USA Regulatory Address Cleanse transform checks your input files for at least 100 unique records. These checks make verification take longer, but, because it's a USPS requirement, the software must perform the check for each mailing list.

If Data Quality Management SDK finds that your data doesn't have 100 unique records, it issues an error and discontinues processing.

The process of checking for 100 unique records is a pre-processing step. So, if Data Quality Management SDK finds fewer than 100 unique records, it doesn't generate statistics and it doesn't perform any processing on the mailing list.

[How the software checks for 100 unique records \[page 302\]](#)

To satisfy the USPS requirement for 100 unique records for NCOALink processing, Data Quality Management SDK checks the input data for 100 unique records before it performs any processing on the data.

[Finding unique records in multiple threads \[page 303\]](#)

When your NCOALink job is multi-threaded, Data Quality Management SDK checks each thread for 100 unique records.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.9.1 How the software checks for 100 unique records

To satisfy the USPS requirement for 100 unique records for NCOALink processing, Data Quality Management SDK checks the input data for 100 unique records before it performs any processing on the data.

If Data Quality Management SDK finds 100 unique records, it continues and processes the job as usual. However, if it doesn't find 100 unique records, it issues an error stating that your input data doesn't have 100 unique records. Also, if your data has fewer than 100 records, Data Quality Management SDK issues an error stating that there aren't enough records to determine uniqueness.

Data Quality Management SDK concatenates all input fields for each record in the same order as the schema mapping. For a record to be considered as non-unique, each field of a record must be identical to each field in another record.

Example

Comparing records

The following table shows how Data Quality Management SDK concatenates the fields in each record and determines non-unique records. The first and last rows in this example aren't unique.

332	FRONT	STREET	NORTH	LACROSSE	WI	54601
332	FRONT	STREET	SOUTH	LACROSSE	WI	54601
331	FRONT	STREET	SOUTH	LACROSSE	WI	54601
332	FRONT	STREET	NORTH	LACROSSE	WI	54601

Related Information

[NCOALink security requirements \[page 302\]](#)

11.10.9.2 Finding unique records in multiple threads

When your NCOALink job is multi-threaded, Data Quality Management SDK checks each thread for 100 unique records.

The thread check doesn't necessarily check records in order. For example, Data Quality Management SDK checks thread #3 before thread #1. Therefore, each time you run the same multi-threaded job, there can be different unique record results.

Sometimes, Data Quality Management SDK issues an error even when there are 100 unique records per thread. An error can happen if one of the threads contains fewer than 100 records, and it's checked before other threads that contain at least 100 records. In this case, consider reducing the number of threads before you try to process the job again.

Low volume of data and DOP greater than 1

When an NCOALink job is set up with the DOP (Degree of Parallelism) greater than 1, Data Quality Management SDK does the following:

- Checks each thread for unique records within the first collection it processes.
- Shares knowledge of the unique records it finds with all other threads.

When Data Quality Management SDK finishes processing the collection from the first thread, it does the following:

- Counts the unique records found by all threads up to that point in time.
- Decides whether a thread has met the 100 record minimum.

❖ Example

Your list has 3,050 records and you set the DOP to 4.

If the number of records per collection is 1000, the job forms the following threads:

1. 1000 records
2. 1000 records
3. 1000 records
4. 50 records

Data Quality Management SDK is likely to finish processing the thread with 50 records sooner than others. Therefore, it makes the pass—fail decision before it encounters 100 unique records.

For this job to succeed, lower the DOP to 3 and try again.

Related Information

[NCOALink security requirements \[page 302\]](#)

11.10.10 About NCOALink directories

Download the NCOALink directories with the USPS EPF Downloader Manager tool and extract and uncompress them with the SAP NCOALink Utility.

The USPS makes the latest NCOALink directories available monthly for end user licensees. The USPS makes the directories available weekly for limited or full service providers.

The SAP NCOALink Utility is a 64-bit application that you run from a command line. Access the utility in one of the following ways:

- If you use an older version of the software that does not install the NCOALink Utility, download the application from the SAP Support Portal at <https://support.sap.com/software/address-directories.html>.
- If you use a version of the software that automatically installs the NCOALink Utility, the utility is in `$LINK_DIR\bin\ncoa\ncoutil.exe` (Windows) or `$LINK_DIR/bin/ncoa/ncoutil` (UNIX).

[NCOALink licensee requirements for directories \[page 305\]](#)

USPS licensing requires that limited and full service providers use the most current NCOALink directories available, either weekly or monthly depending on your license type.

[Installing NCOALink directories \[page 305\]](#)

After you download the directories using the EPF Downloader Manager or another download tool, use the SAP NCOALink utility to extract and install the directories.

[Extracting NCOALink directories with the utility \[page 307\]](#)

After you obtain the NCOALink directories from the USPS, use the NCOA utility to extract and uncompress the directories.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.10.1 NCOALink licensee requirements for directories

USPS licensing requires that limited and full service providers use the most current NCOALink directories available, either weekly or monthly depending on your license type.

Additionally, the USPS requires that limited or full service providers download the daily delete file and copy it to the same location as the NCOALink directories every day that they run NCOALink jobs.

NCOALink directories expire after 45 days.

Related Information

[About NCOALink directories \[page 304\]](#)

11.10.10.2 Installing NCOALink directories

After you download the directories using the EPF Downloader Manager or another download tool, use the SAP NCOALink utility to extract and install the directories.

Ensure that your system meets the following minimum requirements:

- At least 60 GB of available disk space (check the latest PAM for NCOALink directory space requirements)
- Sufficient RAM

Download the NCOALink directories with the EPF Downloader Manager or another download tool.

1. Open a command prompt and run the NCOALink Utility from the install location.

If SAP Data Quality Management SDK automatically installed the NCOALink Utility, it is located in:

- `$LINK_DIR\bin\ncoa\ncoutil.exe` (Windows)
- `$LINK_DIR/bin/ncoa/ncoutil` (UNIX)

Where `$LINK_DIR` is the installation directory.

2. Use the `ncoutil` command with the following command-line options:

Option		Description
Windows	UNIX	
/p:t	-p:t	Performs transfer to copy files from the source to the destination. Also specify the following: - Location of compressed NCOALink data files with /d or -d - Transfer destination location with /t or -t
/p:u	-p:u	Performs unpack to uncompress the files. Also specify the following: - Transfer destination location with the /d or -d option - Transfer destination location with /t or -t
/p:v	-p:v	Performs verification on the fields. Also specify the transfer destination location with /t or -t.
/d	-d	Specifies location of compressed NCOALink data files.
/t	-t	Specifies transfer destination location.
/nos	-nos	Does not stop on error (returns failure code as exit status).
/a	-a	Answers all warning messages with Yes.

You can combine p options. For example, if you want to transfer, unpack, and verify all in the same process, enter /p:tuv or -p:tuv.

After performing the p option specified, the program closes.

❖ Example

Your command line may look something like the following examples:

Windows

```
ncoautil /p:tuv /d D:\downloads\ncoa /t C:\Program Files (x86)\SAP
BusinessObjects\Data Services\DataQuality\reference_data
```

UNIX

```
ncoautil -a -nos -p:tuv -d /local/downloads/ncoa -t /local/dataservices/
DataQuality/reference_data
```

Related Information

[About NCOALink directories \[page 304\]](#)

11.10.10.3 Extracting NCOALink directories with the utility

After you obtain the NCOALink directories from the USPS, use the NCOA utility to extract and uncompress the directories.

This procedure applies after you have obtained your NCOALink license, for regular NCOALink processing.

Before you extract and uncompress the directories, ensure that your system meets the following minimum requirements:

- At least 60 GB of available disk space
 - Sufficient RAM
1. Run the NCOALink Utility from the location where it is installed.
If your NCOALink Utility was installed with the software, the utility is located in the following folder, where `LINK_DIR` is the software installation directory:
 - `%LINK_DIR%\bin\ncoa\ncoautl.exe` (Windows)
 - `$LINK_DIR/bin/ncoa/ncoautl` (UNIX)
 2. Use the `ncoautl` command with the following command-line options:

Option		Description
Windows	UNIX	
<code>/p:t</code>	<code>-p:t</code>	Perform transfer to copy files from the source to the destination. Also specify the following locations • Compressed NCOALink data files with <code>/d</code> or <code>-d</code> • Transfer destination location with <code>/t</code> or <code>-t</code>
<code>/p:u</code>	<code>-p:u</code>	Perform unpack to uncompress the files. When you use this command, include the following commands: • <code>/d</code> or <code>-d</code>: Transfer destination location • <code>/t</code> or <code>-t</code>: Transfer destination location
<code>/p:v</code>	<code>-p:v</code>	Perform verification on the fields. When you use this command, include the transfer destination location with <code>/t</code> or <code>-t</code> .
<code>/d</code>	<code>-d</code>	Specify location of compressed NCOALink data files.
<code>/t</code>	<code>-t</code>	Specify transfer destination location.
<code>/nos</code>	<code>-nos</code>	Do not stop on error (returns failure code as exit status).
<code>/a</code>	<code>-a</code>	Answer all warning messages with Yes.

Combine p options when you want to perform multiple tasks at once.

❖ Example

To transfer, unpack, and verify all in the same process, enter `/p:tuv` or `-p:tuv`.

After performing the p option specified, the NCOALink utility closes.

❖ Example

For Windows, the command may look like the following example:

```
ncoautil /p:tuv /d D:\downloads\ncoa /t C:\Program Files (x86)\SAP  
BusinessObjects\Data Services\DataQuality\reference_data
```

For UNIX, the command may look like the following example:

```
ncoautil -a -nos -p:tuv -d /local/downloads/ncoa -t /local/dataservices/  
DataQuality/reference_data
```

11.10.11 About the NCOALink Daily Delete Files

The USPS requires that NCOALink service provider licensees download the daily delete files each day before they perform NCOALink processing.

The daily delete file `dailydel.dat` contains records that are pending deletion from the NCOALink data.

❖ Example

Jane Doe files a change of address with the USPS, but then she doesn't move. Even though her address is in the NCOALink directories as a COA (Change of Address), the daily delete file informs the mailer that she didn't move.

The daily delete file informs mailers in the interim, until the USPS updates the record in the regular NCOALink directory.

Download the daily delete files and install them in the same directory as your NCOALink directories.

i Note

If you're an end user, you need the daily delete files only for processing Stage I or II test files. It isn't required for normal NCOALink processing.

The following list contains important information about the daily delete files:

- Data Quality Management SDK fails verification when you run an NCOALink certification stage test and you don't have the daily delete files installed.
- The USA Regulatory Address Cleanse transform supports only the ASCII version of the daily delete files.
- Don't rename the daily delete files from the original file name of `dhdr.txt` and `dailydel.dat`.
- Data Quality Management SDK issues a verification warning when the daily delete files are more than three days old.

[Installing the NCOALink Daily Delete Files \[page 309\]](#)

Install the NCOALink daily delete files each day that you plan to process mailing lists using NCOALink.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.11.1 Installing the NCOALink Daily Delete Files

Install the NCOALink daily delete files each day that you plan to process mailing lists using NCOALink.

To download and install the NCOALink daily delete files, perform the following steps:

1. Go to the **USPS Electronic Product Fulfillment** site at <https://epf.usps.gov/> .
2. Find the **Application Access** section and log in to the **USPS Electronic Product Fulfillment** site.
Use the e-mail and password that you established in the USPS Web Access Request form.
3. Download the NCOALink Daily Delete [TEXT] file to a location where you can extract the .tar file.
If your computer browser has pop-up blockers enabled, set your browser to accept popups.
4. Extract the `dailyDeletes.txt.tar` file.
5. Copy the `dhdr.txt` and `dailydel.dat` file to the same location as your NCOALink directories.

Related Information

[About the NCOALink Daily Delete Files \[page 308\]](#)

11.10.12 Output file strategies

By selecting specific Field Class Selections, configure your output file to meet your needs.

Select from the following Field Class Selections when you set up your output fields for NCOALink processing:

- **CORRECT:** Outputs the original address, completed and corrected using values found in the directories and based on the standardization options you set in the transform.
- **MOVE-UPDATED:** Outputs the address components that have been updated with move-updated address data.

i Note

The transform looks for the move-updated address information in the U.S. National Directories. When the transform doesn't find the move-updated address in the U.S. National Directories, it populates the Move Updated fields with information from the Move Update Directories only. The transform doesn't update the Move Updated fields that are populated as a result of standardizing against the U.S. National Directories.

- **BEST:** Outputs the move-updated address when it exists and when it matches in the U.S. National directories. Or, output contains the original data when a move doesn't exist or when the move doesn't match in the U.S. National Directories.

Based on the [APPLY_MOVE_TO_STANDARDIZED_FIELDS](#) option setting in the NCOALink group, standardized components contain either original or move-updated addresses. This option is set to **YES** by default. Therefore, Data Quality Management SDK updates standardized fields to contain details about the updated address available through NCOALink.

To retain the old addresses in the standardized components and append the new addresses to the output file, choose **NO** for the [APPLY_MOVE_TO_STANDARDIZED_FIELDS](#) option. Then use output fields such as [NCOALINK_RETURN_CODE](#) to determine whether a move occurred.

One way to set up your output file is to replicate the input file format, then append extra fields for move data. In the output records not affected by a move, most of the appended fields are blank. Alternatively, you can create a second output file specifically for move records. The following two approaches are possible:

- Output each record once, placing move records in the second output file and all other records in the main output file.
- Output move records twice; once to the main output file, and a second time to the second output file.

Both of these approaches require that you use an output filter to determine whether a record is a move-updated record.

Related Information

[NCOALink and Move Update \[page 292\]](#)

11.10.13 Improving NCOALink processing performance

Many factors affect performance when processing NCOALink data.

One of the most critical factors is the volume of disk access that occurs. Often the most effective way to reduce disk access is to have sufficient memory available to cache data. Other critical factors that affect performance include hard drive speed, seek time, and the sustained transfer rate. When the time spent on disk access is minimized, the performance of the CPU becomes significant.

[Operating systems and processors \[page 311\]](#)

The computation involved in both job processing and NCOALink processing is well suited to the microprocessors, such as the microcomputers made by Intel and AMD.

[Data storage \[page 311\]](#)

If possible, dedicate the hard drive for NCOALink data entirely to the NCOALink process, at least while your job is running.

[Memory \[page 312\]](#)

Because NCOALink processing uses many gigabytes of data, obtain as much memory as possible.

[Data format \[page 312\]](#)

To increase performance, consider your data format.

[Performance tips \[page 312\]](#)

Many factors can increase or decrease NCOALink processing speed.

11.10.13.1 Operating systems and processors

The computation involved in both job processing and NCOALink processing is well suited to the microprocessors, such as the microcomputers made by Intel and AMD.

RISC style processors, such as the processors in most UNIX systems, are slower for this type of computation. A common PC can often run a single job through Data Quality Management SDK and NCOALink about twice as fast as a common UNIX system.

A Windows server or a fast workstation are cost-effective tools for processing single jobs. Most UNIX systems have multiple processors that do a good job processing multiple jobs at once.

To maximize the processor or core usage on your system, consider increasing the degree of parallelism (DOP) in the data flow properties. Increasing the DOP depends on the complexity of the data flow.

Related Information

[Improving NCOALink processing performance \[page 310\]](#)

11.10.13.2 Data storage

If possible, dedicate the hard drive for NCOALink data entirely to the NCOALink process, at least while your job is running.

Any other process that competes for the same physical disk drive reduces NCOALink performance.

Explore the possibility of using a RAID system (redundant array of independent discs) to achieve even higher transfer rates.

When Data Quality Management SDK accesses NCOALink data directly rather than from a cache, the most significant hard drive feature is the average seek time.

Related Information

[Improving NCOALink processing performance \[page 310\]](#)

11.10.13.3 Memory

Because NCOALink processing uses many gigabytes of data, obtain as much memory as possible.

The exact amount of memory needed depends on your service provider level, the data format, and the specific release of the data from the USPS.

In general, obtain as much memory as possible under the following circumstances:

- Performance is critical.
- You are an NCOALink full service provider.
- You frequently run jobs with millions of records.

If you require it, cache the entire NCOALink data set. Caching the entire data set takes about 20 GB of RAM with enough memory left for the operating system.

You can manually adjust the amount of memory to use by setting the `NCOALINK_CACHING_MODE` option to `MANUAL`. Or set the `NCOALINK_CACHING_MODE` option to `AUTO` so that Data Quality Management SDK uses available memory for caching.

11.10.13.4 Data format

To increase performance, consider your data format.

Data Quality Management SDK supports both hash and flat file versions of NCOALink data.

If you can cache the entire hash file set, the hash format can provide the best performance. Because the flat file data is smaller, you can cache a larger share in a given amount of RAM. However, accessing the flat file data involves binary searches, which are slightly more time consuming than the direct access used with the hash file format.

Related Information

[Improving NCOALink processing performance \[page 310\]](#)

11.10.13.5 Performance tips

Many factors can increase or decrease NCOALink processing speed.

Some performance factors are within your control and others are inherent to your business. The following table contains some performance factors and discussion points to consider.

Factor	Description
Cache size	Using too little memory for NCOALink caching can cause unnecessary random file access and time-consuming hard drive seeks. Using too much memory can cause large files to be read from the disk into the cache even when only a tiny fraction of the data is used. The amount of cache that works best in your environment can require some testing to see what works best for your configuration and typical job size.
Directory location	Best practice is to put NCOALink directories on a local solid-state drive or a virtual RAM drive. Using a local solid-state drive or virtual RAM drive eliminates all I/O during NCOALink processing. If you have NCOALink directories on a hard drive, best practice is to use a defragmented local hard drive. Ensure that the hard drive isn't accessed for anything other than the NCOALink data while you're running your job.
Match rate	The more records you process that have forwardable moves, the slower your processing is. Retrieving and decoding the new addresses takes time. Therefore, updating a mailing list regularly improves the processing speed on that list.
Input format	Best practice is to input data to the transform with discrete fields for the pre, first, middle, last, and post name fields. If your input has only a name line, the transform takes time to parse the line before checking NCOALink data.
File size	The transform processes larger files faster than smaller files. There's overhead when processing any job, but if a job includes millions of records, a few seconds of overhead is insignificant.

Related Information

[Improving NCOALink processing performance \[page 310\]](#)

11.10.14 NCOALink log files

Use information from the StatisticsHandler to generate the USPS-required NCOALink log files.

Data Quality Management SDK generates the USPS-required log files automatically, and names them according to USPS requirements. Data Quality Management SDK creates one log file per license ID.

At the beginning of each month, Data Quality Management SDK starts new log files. The transform appends each log file with information about every NCOALink job processed that month for that specific license ID.

The USPS requires you to save the NCOALink log files for five years.

Data Quality Management SDK produces the following move-related log files:

- CSL (Customer Service log)
- PAF (Processing Acknowledgement Form) Customer Information log
- BALA (Broker/Agent/List Administrator) log

The PAF Customer Information log and the BALA log aren't required for end users.

The following table shows the log files required for Limited or Full Service Providers:

Log file	Required for	Description
CSL	• End users • Limited Service Providers • Full Service Providers	Contains one record for each list that you process. Each record details the results of change-of-address processing.
PAF Customer Information	• Limited Service Providers • Full Service Providers	Contains the information that you provide for the PAF. Lists each unique PAF entry. If a list is processed with the same PAF information, the information appears once in the log file. When contact information for the list administrator changes, the software writes the information for both the list administrator and the corresponding broker to the PAF log file.

Log file	Required for	Description
BALA	- Limited Service Providers - Full Service Providers	Contains the contact information that you entered for the broker or list administrator. Lists information for each broker or list administrator once. The USPS requires the Broker/Agent/List Administrator log file from service providers, even in jobs that don't involve a broker or list administrator. The software produces this log file for every job when you're a certified service provider.

For more information about NCOALink reports and log files, see the *Management Console Guide*.

[Log file retention and deletion \[page 315\]](#)

The USPS requires NCOALink users to retain the log files for several years, but Data Quality Management SDK deletes the log files after a short period of time.

[Log file names \[page 316\]](#)

Data Quality Management SDK follows a specific USPS file-naming convention for NCOALink log files.

11.10.14.1 Log file retention and deletion

The USPS requires NCOALink users to retain the log files for several years, but Data Quality Management SDK deletes the log files after a short period of time.

As an NCOALink provider, the USPS requires you to submit NCOALink log files every month. However, Data Quality Management SDK deletes log files after 30 days by default.

You can change the period of time by setting the *History Retention Period* option in the Central Management Console (CMC). To avoid losing monthly log information, set the option to more than 31 days; we recommend a setting of 50 days.

In addition to sending monthly log files to the USPS, you are required to have the data available for the USPS to examine for several years after you process the job. Make sure you are aware of current USPS rules for data retention by checking your USPS licensing agreement.

To ensure that you retain all required reports and logs before the data is deleted from the repository, we recommend that you export the required reports and logs to a local folder on a monthly basis.

Exporting the log files to a local folder also prevents the repository contents from becoming so large that the export process “times out” because the volume of statistics retained.

Related Information

[NCOALink log files \[page 314\]](#)

11.10.14.2 Log file names

Data Quality Management SDK follows a specific USPS file-naming convention for NCOALink log files.

Data Quality Management SDK follows a USPS file-naming scheme for the following log files:

- Customer Service log
- PAF Customer Information log
- Broker/Agent/List Administrators log

The following table describes the naming scheme for NCOALink log files.

Character 1		Characters 2 -5	Character 6		Characters 7-8	Extension
Log type		Platform ID	Month		Year	
B	Broker log	Exactly four characters long	1	January	Two characters, for example, 10 for 2010	.DAT
C	Customer service log		2	February		
P	PAF log		3	March		
			4	April		
			5	May		
			6	June		
			7	July		
			8	August		
			9	September		
			A	October		
			B	November		
			C	December		

❖ Example

P1234C10.DAT is a PAF Log file generated in December 2010 for a licensee with the ID 1234.

Related Information

[NCOALink log files \[page 314\]](#)

11.11 Statistics for multiple data sources

The USA Regulatory Address Cleanse transform can report statistics for jobs that contain multiple data sources, based on logical groupings.

An input database that contains a combination of several mailing lists can use unique list identifiers. The list identifiers track what list a record originates from. The unique list identifier can be a name or a number. The identifier resides in the same field across all of the combined lists.

For input field mapping, map your unique list identifier field to the Data Quality Management SDK input field [DATA_SOURCE_ID](#). Data Quality Management SDK collects statistics for each list using the [DATA_SOURCE_ID](#) input field.

Some of the generated reports contain a summary for the entire list, and a separate summary per list based on the value mapped to the [DATA_SOURCE_ID](#) field.

! Restriction

For compliance with NCOALink reporting restrictions, the USA Regulatory Address Cleanse transform doesn't support processing multiple mailing lists associated with different PAFs (Processing Acknowledgment Forms). Therefore, for NCOALink processing, Data Quality Management SDK considers all records in the input file to be a single mailing list. Also, Data Quality Management SDK reports the list as a single mailing list in the Customer Service Log (CSL) file.

! Restriction

Data Quality Management SDK doesn't support the [GATHER_STATISTICS_PER_DATA_SOURCE](#) functionality when you choose [YES](#) for the [ENABLE_PARSE_ONLY](#) or [ENABLE_GEO_ONLY](#) options in the Non Certified Options group.

[Physical Source Field and Cumulative Summary \[page 318\]](#)

Some reports don't include a Cumulative Summary section, but they do present lists sorted by the [DATA_SOURCE_ID](#) value.

Parent topic: [USA Regulatory Address Cleanse transform \[page 242\]](#)

Related Information

[USPS DPV \(Delivery Point Validation\) \[page 243\]](#)

[USPS eLOT \[page 258\]](#)

[Early Warning System \(EWS\) \[page 259\]](#)

[SuiteLink \[page 261\]](#)

[LACSLink \[page 265\]](#)

[USPS Residential Delivery Indicator \(RDI\) \[page 274\]](#)

[Z4Change \[page 277\]](#)

[Suggestion lists \[page 279\]](#)

[USPS certifications \[page 282\]](#)

11.11.1 Physical Source Field and Cumulative Summary

Some reports don't include a Cumulative Summary section, but they do present lists sorted by the [DATA_SOURCE_ID](#) value.

The following reports don't include a Cumulative Summary section:

- Address Standardization
- Address Information Code
- USA Regulatory Locking report

However, these reports include both of the following list types:

- **Physical Source Field:** A report per list based on the [DATA_SOURCE_ID](#) field. The report is identified in the report footer.
- **Cumulative Summary:** A summary of the entire list. The report is identified in the report footer.

i Note

When you enable NCOALink, Data Quality Management SDK reports a summary per list only for the following sections of the NCOALink Processing Summary Report:

- NCOALink Move Type Summary
- NCOALink Return Code Summary
- ANKLink Return Code Summary

Special circumstances

The following list describes circumstances when the report names “Cumulative Summary” and “Physical Source Field” don't appear in the report footer sections:

- When the [GATHER_STATISTICS_PER_DATA_SOURCE](#) option is set to *NO*
- When the [GATHER_STATISTICS_PER_DATA_SOURCE](#) option is set to *YES* and there's only one [DATA_SOURCE_ID](#) value present in the list, but it's empty.

[USPS Form 3553 and group reporting \[page 319\]](#)

The USPS Form 3553 includes a summary of the entire list and a report per list based on the [DATA_SOURCE_ID](#) field.

[Group statistics and cumulative statistics reports \[page 319\]](#)

There are various reports that include either a cumulative statistics section or a group statistics section, or both sections.

Related Information

[Statistics for multiple data sources \[page 317\]](#)

11.11.1.1 USPS Form 3553 and group reporting

The USPS Form 3553 includes a summary of the entire list and a report per list based on the [DATA_SOURCE_ID](#) field.

Cumulative Summary

The USPS Form 3553 designates the summary for the entire list with the words “Cumulative Summary.” The report name appears in the footer of the report. In addition, the Cumulative Summary contains the total number of lists in the job in Section B, under **Number of Lists**.

Physical Source Field

The USPS Form 3553 designates the summary for each Individual list with the words “Physical Source Field” followed by the [DATA_SOURCE_ID](#) value. It appears in the footer. The data in the report is for the list designated by the data source ID only.

Related Information

[Physical Source Field and Cumulative Summary \[page 318\]](#)

11.11.1.2 Group statistics and cumulative statistics reports

There are various reports that include either a cumulative statistics section or a group statistics section, or both sections.

Cumulative statistics are summaries for the entire mailing list. Group statistics are based on the Physical Source Field.

The following list contains the reports that include both cumulative and group statistics:

- Address Validation Summary
- Address Type Summary
- US Addressing

The following list contains the reports that include only a group statistics section but not a cumulative summary:

- Address Information Code Summary
- Address Standardization
- US Regulatory Locking

Related Information

[Physical Source Field and Cumulative Summary \[page 318\]](#)

12 USA Regulatory Address Cleanse transform reference

The USA Regulatory Address Cleanse transform XML configuration is grouped according to setting types.

The sample configurations installed with the software contain examples of the XML.

[System group \[page 322\]](#)

The System group controls high-level transform functions.

[USA Regulatory Address Cleanse Report and Analysis options \[page 323\]](#)

Configure the options in the *Report and analysis* option group to generate reports.

[USA Regulatory Address Cleanse transform performance options \[page 325\]](#)

Set options in the *Transform Performance* group to enhance performance for DPV, DSF2, LACSLink, NCOALink, RDI, and SuiteLink processing.

[USA Regulatory Address Cleanse reference files \[page 327\]](#)

Set the location for reference files that the USA Regulatory Address Cleanse transform uses to correct and standardize your data.

[USA Regulatory Address Cleanse assignment options \[page 330\]](#)

Set options in the Assignment Options group to enable the USPS processes that the transform performs on your data.

[USA Regulatory Address Cleanse standardization options \[page 333\]](#)

Set options for standardizing your address data in the Standardization group of options.

[USA Regulatory Address Cleanse USPS Z4 Change options \[page 344\]](#)

Set USPS Z4 Change options to enable Z4Change processing and to specify the last time the Postcode 2 was updated.

[USA Regulatory Address Cleanse USPS CASS Report Options \[page 345\]](#)

Complete the *CASS Report Options* group with USPS Form 3553 information as required by CASS.

[USA Regulatory Address Cleanse suggestion list options \[page 347\]](#)

The *Suggestion List* group contains options to configure how the software outputs suggestions lists.

[USA Regulatory Address Cleanse suggestion list components \[page 350\]](#)

Complete options in the *Suggestion List Components* to specify the address field components to include in the *Suggestion_List* output field.

[USA Regulatory Address Cleanse non-certified options \[page 353\]](#)

The Non-Certified options group contains settings that, when enabled, create a mailing list that is not compliant with CASS rules.

[USA Regulatory Address Cleanse USPS License Information \[page 356\]](#)

If you perform NCOALink, SuiteLink, LACSLink, or DPV processing, complete the required USPS license information.

[USPS License Information Required Options \[page 359\]](#)

The options in the USPS License Information group are required based on the type of service you provide and are licensed for.

Complete the options in the NCOALink group when you prepare your data for NCOALink certification.

12.1 System group

The System group controls high-level transform functions.

Syntax

```
<SYSTEM>
  <PLATFORM_SETTINGS>
    <SYSTEM_VARS>
      <LINK_DIR></LINK_DIR>
      <KEYFILE_LOCATION></KEYFILE_LOCATION>
    </SYSTEM_VARS>
    <RUNTIME_VARS>
      <JOB_TYPE></JOB_TYPE>
    </RUNTIME_VARS>
  <PLATFORM_SETTINGS>
```

Where

Option	Description
<i>LINK_DIR</i>	Specifies the relative path to the United States Regulatory Address Cleanse support files. The default Windows 32-bit location of these files is <code>SAPBusinessObjects\Data Quality Mgmt SDK\windows_32\DataQuality\reference_data</code>. Therefore, enter <code>SAPBusinessObjects\Data Quality Mgmt SDK\windows_32\DataQuality\reference_data</code>.
<i>KEYFILE_LOCATION</i>	Specifies the absolute path to the Data Quality Management SDK key file (<code>bojprods.key</code>).
<i>JOB_TYPE</i>	Specifies the job type. If you are running an NCOALink job, enter <i>BATCH</i>. Otherwise, leave the option blank and instantiate the transform as a multi-record transform.

Note

If an option exists in the XML, but it is not documented here, do not alter the contents of the option. Editing the contents could cause errors.

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12.2 USA Regulatory Address Cleanse Report and Analysis options

Configure the options in the *Report and analysis* option group to generate reports.

Syntax

```
<REPORT_AND_ANALYSIS>  
  <GENERATE_REPORT_DATA></GENERATE_REPORT_DATA>  
  <GATHER_STATISTICS_PER_DATA_SOURCE></GATHER_STATISTICS_PER_DATA_SOURCE>  
</REPORT_AND_ANALYSIS>
```

Report and Analysis group option descriptions

Option	Description
GATHER_STATISTICS_PER_DATA_SOURCE	Specifies whether to generate report data for each data source specified in the DATA_SOURCE_ID input field. • YES: If the GENERATE_REPORT_DATA option is enabled, generates report statistics for each data source specified in the DATA_SOURCE_ID input field. • NO: If the GENERATE_REPORT_DATA option is enabled, generates reports for the input database without generating statistics, based on the DATA_SOURCE_ID input field. i Note If you select YES, other setup requirements apply.
GENERATE_REPORT_DATA	Specifies whether to generate report data for this transform. • YES: Generates report data for this transform. • NO: Does not generate report data for this transform. i Note If you do not need to generate reports, such as for testing, set this option to NO to improve performance.

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12.3 USA Regulatory Address Cleanse transform performance options

Set options in the [Transform Performance](#) group to enhance performance for DPV, DSF2, LACSLink, NCOALink, RDI, and SuiteLink processing.

Syntax

```
<TRANSFORM_PERFORMANCE>
  <CACHE_DPV_DIRECTORIES>NO</CACHE_DPV_DIRECTORIES>
  <CACHE_LACSLINK_DIRECTORIES>NO</CACHE_LACSLINK_DIRECTORIES>
  <CACHE_RDI_DIRECTORIES>NO</CACHE_RDI_DIRECTORIES>
  <CACHE_SUITELINK_DIRECTORIES>NO</CACHE_SUITELINK_DIRECTORIES>
  <NCOALINK_CACHING_MODE>NONE</NCOALINK_CACHING_MODE>
  <NCOALINK_MEMORY_USAGE></NCOALINK_MEMORY_USAGE>
  <WINDOWS_USE_EXTENDED_MEMORY>NO</WINDOWS_USE_EXTENDED_MEMORY>
  <INSUFFICIENT_CACHE_MEMORY_ACTION>CONTINUE</
INSUFFICIENT_CACHE_MEMORY_ACTION>
</TRANSFORM_PERFORMANCE>
```

Transform Performance option group descriptions

Option	Description
CACHE_DPV_DIRECTORIES	Specifies whether the software caches DPV directories into memory. The software caches the directories only once and any other DPV threads that are running in the data flow use the same cache. • YES: Enables caching. • NO: Disables caching.
CACHE_LACSLINK_DIRECTORIES	Specifies whether the software caches the LACSLink directories into memory. • YES: Enables caching. • NO: Disables caching.
CACHE_RDI_DIRECTORIES	Specifies whether the software caches RDI directories into memory. • YES: Enables caching. • NO: Disables caching.
CACHE_SUITELINK_DIRECTORIES	Specifies whether the software caches the SuiteLink directories into memory. • YES: Enables caching. • NO: Disables caching.

Option	Description
<i>INSUFFICIENT_CACHE_MEMORY_ACTION</i>	If there is insufficient memory to cache the directories that you have set up for caching, specifies the action the software takes. <i>CONTINUE</i>: Attempts to continue initialization without caching. <i>ERROR</i>: Issues an error and terminates the transform.
<i>NCOALINK_CACHING_MODE</i>	Specifies the method for caching NCOALink directories. <i>AUTO</i>: The software uses available memory for caching. <i>MANUAL</i>: Enter a value in the <i>NCOALINK_MEMORY_USAGE</i> option. i Note Select <i>MANUAL</i> if you have a limited amount of memory available and you want to allocate a set amount of memory for caching. <i>NONE</i>: Disables caching. <i>NONE</i> is the default setting. i Note Select <i>NONE</i> for smaller jobs. Executing a smaller job could take less time than the time to cache the directories.
<i>NCOALINK_MEMORY_USAGE</i>	Specifies a value to allocate a set amount of memory for NCOALink directory caching. The software uses this value as the maximum amount of memory that it can use for caching the NCOALink directories. Required if you set the <i>NCOALINK_CACHING_MODE</i> to <i>MANUAL</i>. i Note This value is not for memory per thread. If the <i>DEGREE_OF_PARALLELISM</i> value is greater than one, the <i>NCOALINK_MEMORY_USAGE</i> value is the total to be allocated for all threads.

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12.4 USA Regulatory Address Cleanse reference files

Set the location for reference files that the USA Regulatory Address Cleanse transform uses to correct and standardize your data.

≡ Syntax

Example syntax does not show all options.

```
<REFERENCE_FILES>
  <ADDRESS_DIRECTORY1></ADDRESS_DIRECTORY1>
  <ADDRESS_DIRECTORY2></ADDRESS_DIRECTORY2>
  <ADDRESS_SHS_DIRECTORY></ADDRESS_SHS_DIRECTORY>
  . . .
```

Reference Files option group descriptions

Option	Description
<i>ADDRESS_DIRECTORY_1</i>	zip4us.dir Specifies the path to the National Directory. The National Directory is organized by ZIP Code. It lists street names, ranges of house numbers, postal, and other codes.
<i>ADDRESS_DIRECTORY_2</i>	*.dir Optional. Specifies the path to a directory that you customize for a special purpose. We do not provide a directory for this option. Most users should leave the <i>ADDRESS_DIRECTORY_2</i> option blank.

Option	Description
<i>ADDRESS_SHS_DIRECTORY</i>	zip4us.shs Required for standardization. Specifies the path to the SHS directory. The SHS directory enhances normal primary name lookups.
<i>CITY_DIRECTORY</i>	city##.dir Specifies the path to the City directory. The City directory contains a table of city names, states, and ZIP Codes organized by state and city.
<i>DPV_PATH</i>	Required for CASS certification. Specifies the path to the DPV (Delivery Point Validation) directory files. The DPV directory contains information that identifies whether a ZIP+4 coded address represents a known address and that the address is deliverable.
<i>ELOT_DIRECTORY</i>	eLOT.dir Specifies the path to the eLOT directory. The eLOT directory contains codes for the delivery point that represents the delivery route walk sequence for all mail carriers. Include only if the <i>ENABLE_ELOT</i> option in the Assignment Options group is set to <i>YES</i>.
<i>EWS_DIRECTORY</i>	ewsymmdd.dir Specifies the path to the EWS directory (Early Warning System). The transform lists the directory in the <i>Option Value</i> column so the transform finds the most current directory. EWS reduces the number of misdirected mail by providing valid delivery points that are created between national directory updates.
<i>LACSLINK_PATH</i>	Specifies the path to the LACSLink directory files. Required for CASS certification. The LACSLink directories update addresses when the physical address changes, not when residents move.
<i>NCOALINK_PATH</i>	Specifies the path to the NCOALink directory files. Required for NCOALink processing and certification. The NCOALink directories contain records of addresses changed through a change of address form filed with the USPS.

Option	Description
<i>POSTCODE_DIRECTORY</i>	zcf10.dir Specifies the path to the Postcode directory. The Postcode directory contains the same data as the City directory, but is organized by ZIP Code.
<i>POSTCODE_REVERSE_DIRECTORY</i>	revzip4.dir Specifies the path to the Postcode Reverse directory. The Reverse directory enables the software to assign more postal codes when the input data includes a unique ZIP Code and valid ZIP+4.
<i>RDI_PATH</i>	Specifies the path to the RDI directory. The RDI directory indicates whether an address is residential.
<i>REVERSE_SOUNDEX_ADDRESS_DIRECTORY</i>	zip4us.rev Specifies the path to the Reverse Soundex Address directory. The Reverse Soundex Address directory enhances primary name lookups.
<i>SUITELINK_PATH</i>	Required for CASS certification. Specifies the path to the SuiteLink directories. The SuiteLink directories contains specially indexed address information. For example, SuiteLink directories contain secondary numbers and unit designators for locations identified as high-rise business default buildings.
<i>USPS_LOG_PATH</i>	Specifies the path to the directory for NCOALink, DPV, and LACSLink log files. This directory must already exist and be writable. The transform determines the file names during processing as the USPS requires. i Note Use the same path to all jobs. If you have multiple clients, use the same log file directory for all clients so that the transform combines the log files.
<i>Z4_CHANGE_DIRECTORY</i>	z4change.dir Specifies the path to the Z4 Change directory. The Z4 Change directory lists all the ZIP Codes and ZIP+4 Codes in the country.

SAP provides a directory update letter that contains information about the current directory versions and how to obtain directories. Follow this link for the latest directory update letter: https://uacp2.hana.ondemand.com/viewer/p/ADDRESSING_DIRECTORIES.

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12.5 USA Regulatory Address Cleanse assignment options

Set options in the Assignment Options group to enable the USPS processes that the transform performs on your data.

Syntax

```
<ASSIGNMENT_OPTIONS>
  <ENABLE_DPV>YES</ENABLE_DPV>
  <ENABLE_ELOT>YES</ENABLE_ELOT>
  <ENABLE_EWS>NO</ENABLE_EWS>
  <ENABLE_LACSLINK>YES</ENABLE_LACSLINK>
  <ENABLE_NCOALINK>NO</ENABLE_NCOALINK>
  <ENABLE_RDI>NO</ENABLE_RDI>
  <ENABLE_REVERSE_SOUNDEX_SEARCH>YES</ENABLE_REVERSE_SOUNDEX_SEARCH>
  <ENABLE_SUITELINK>YES</ENABLE_SUITELINK>
  <DUAL_ADDRESS>POSITION</DUAL_ADDRESS>
  <USPS_CERTIFICATION_TESTING_MODE>NONE</USPS_CERTIFICATION_TESTING_MODE>
</ASSIGNMENT_OPTIONS>
```

Assignment group option descriptions

Option	Description
<i>ENABLE_DPV</i>	Specifies whether to perform DPV processing. <i>YES</i>: Enables DPV processing. <i>NO</i>: Disables DPV processing.
<i>ENABLE_ELOT</i>	Specifies whether to perform eLOT processing. <i>YES</i>: Enables eLOT processing. <i>NO</i>: Disables eLOT processing.
<i>ENABLE_EWS</i>	Specifies whether to perform EWS processing. <i>YES</i>: Enables EWS processing. <i>NO</i>: Disables EWS processing. If the transform cannot make an exact match using the zip4us.dir (Address Directory 1) directory, it searches the EWS directory to see if the address is a new delivery point. If the address is located in the EWS directory, the transform marks the record as an EWS match and does not attempt further assignment.
<i>ENABLE_LACSLINK</i>	Specifies whether to perform LACSLink processing. <i>YES</i>: Enables LACSLink processing. <i>NO</i>: Disables LACSLink processing.
<i>Enable NCOALink</i>	Specifies whether to perform NCOALink processing. <i>YES</i>: Enables NCOALink processing. <i>NO</i>: Disables NCOALink processing.
<i>ENABLE_RDI</i>	Specifies whether to perform RDI processing. <i>YES</i>: Enables RDI processing. <i>NO</i>: Disables RDI processing.
<i>ENABLE_REVERSE_SOUNDEX_SEARCH</i>	Specifies whether to use the zip4us.rev (Reverse Soundex) directory in an attempt to make address assignments. <i>YES</i>: Enables Reverse Soundex. <i>NO</i>: Disables Reverse Soundex.
<i>ENABLE_SUITELINK</i>	Specifies whether to perform SuiteLink processing. <i>YES</i>: Enables SuiteLink processing. <i>NO</i>: Disables SuiteLink processing.

Option	Description
<i>DUAL_ADDRESS</i>	Specifies the action to take when the transform encounters a dual address. • <i>POSITION</i>: Selects an address based on the arrangement of the input data. The transform attempts to validate the address that is closest to the lower left corner of the address block. That address might be the postal address (rural route or PO Box) or the street address based on how the data appears in the address. • <i>POSTAL</i>: The transform attempts to validate based on the postal address. If that fails, the transform attempts to validate based on the street address. • <i>STREET</i>: The transform attempts to validate based on the street address. If that fails, the transform attempts to validate based on the postal address.
<i>USPS_CERTIFICATION_TESTING_MODE</i>	Specifies the type of certification to use for processing so the software checks for the appropriate settings and issues warnings and errors when applicable. • <i>NONE</i>: Processes without any special settings for certification. <i>NONE</i> is the default setting. • <i>CASS</i>: Processes with the appropriate settings for CASS self certification.

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12.6 USA Regulatory Address Cleanse standardization options

Set options for standardizing your address data in the Standardization group of options.

Syntax

```
<STANDARDIZATION_OPTIONS>
  <STANDARDIZE_ASSIGNED_ADDRESS>YES</STANDARDIZE_ASSIGNED_ADDRESS>
  <STANDARDIZE_UNASSIGNED_ADDRESS>NO</STANDARDIZE_UNASSIGNED_ADDRESS>
  <USE_USPS_PRIMARY_NAME_ABBREVIATION>NO</
USE_USPS_PRIMARY_NAME_ABBREVIATION>
  <USE_USPS_LOCALITY_ABBREVIATION>NO</USE_USPS_LOCALITY_ABBREVIATION>
  <CAPITALIZATION>UPPER</CAPITALIZATION>
  <DIRECTIONAL_STYLE>SHORT</DIRECTIONAL_STYLE>
  <PRIMARY_TYPE_STYLE>SHORT</PRIMARY_TYPE_STYLE>
  <UNIT_DESCRIPTION>CONVERT</UNIT_DESCRIPTION>
  <RETAIN_POUND_SIGN_IN_UNIT_DESCRIPTION>NO</
RETAIN_POUND_SIGN_IN_UNIT_DESCRIPTION>
  <APPEND_PRIVATE_MAILBOX>YES</APPEND_PRIVATE_MAILBOX>
  <PRESERVE_DUAL_ADDRESS_ORDER>NO</PRESERVE_DUAL_ADDRESS_ORDER>
  <ADDRESS_LINE_ALIAS>CONVERT</ADDRESS_LINE_ALIAS>
  <PRESERVE_PLACE_NAMES>NO</PRESERVE_PLACE_NAMES>
  <ASSIGN_WITH_INPUT_LOCALITY>NO</ASSIGN_WITH_INPUT_LOCALITY>
  <INCLUDE_UNUSED_ADDRESS_LINE_DATA>NO</INCLUDE_UNUSED_ADDRESS_LINE_DATA>
  <MOVE_MULTILINE_DATA>NO</MOVE_MULTILINE_DATA>
  <COMBINE_MULTILINES>YES</COMBINE_MULTILINES>
  <MULTILINE_UPDATE_POSTCODE_1>UPDATE</MULTILINE_UPDATE_POSTCODE_1>
  <MULTILINE_UPDATE_POSTCODE_2>ERASE_THEN_UPDATE</
MULTILINE_UPDATE_POSTCODE_2>
  <ADD_FIRM_MATCH_SECONDARY>YES</ADD_FIRM_MATCH_SECONDARY>
</STANDARDIZATION_OPTIONS>
```

Standardization group option descriptions

Option	Description
<i>STANDARDIZE_ASSIGNED_ADDRESS</i>	Specifies whether to correct and standardize the assigned address line and lastline data. YES: Corrects and standardizes address line and lastline data. Select YES for CASS certification. NO: Does not standardize your address line or lastline data.
<i>STANDARDIZE_UNASSIGNED_ADDRESS</i>	Specifies whether to standardize unassigned data. YES: Attempts to parse and standardize any unassigned addresses. NO: Leaves unassigned addresses as entered on input.

Option	Description
<code>USE_USPS_PRIMARY_NAME_ABBREVIATION</code>	Specifies whether to abbreviate the primary name in a multiline or standardized last line field so that the entire output data does not exceed 30 characters. The abbreviation applies only when the multiline or standardized last line output data exceeds 30 characters. If the output address line is already 30 characters or less, the transform does not abbreviate the primary name. YES: Enables the transform to use the USPS form of the primary name abbreviation when available in the directory. Note Also affects the following multiline and standardized last line field styles: SUFFIX_STYLE: Uses Short style DIRECTIONAL_STYLE: Uses Short style ADDRESS_LINE_ALIAS: Overrides the PRESERVE setting. NO: Disables this option. When this option is set to YES, the software outputs an address line with more than 30 characters when there is no abbreviated form of the address in the USPS directory. If you require your data to fit exactly into 30 characters, we recommend setting the appropriate address output field lengths to 30. Intelligent truncation: When the number of characters in the output is greater than the length specified for the output field, the software uses intelligent truncation to fit the data in the field without eliminating vital address data. Using intelligent truncation, the software abbreviates data in the following ways: - The software uses the USPS abbreviation for the primary name when enabled. - The software uses the USPS abbreviation for the locality name when enabled. - If the resulting output data does not fit the field, the software truncates the data. Intelligent truncation is automatic and cannot be controlled with a setting.

Option	Description
<i>USE_USPS_LOCALITY_ABBREVIATION</i>	Specifies whether to use the USPS 13-character city name when available and when the full city name exceeds 13 characters. Applicable only when you use the multiline and standardized last line fields. YES: Uses the USPS 13-character city name when the full city name exceeds 13 characters for multiline and standardized last line fields. When the input city name is a non mailing city name, or it is not valid, the software relies on other settings in the job to determine what to output for city. NO: Disables this option. Intelligent truncation: When the number of characters in the output is greater than the length specified for the output field, the software uses intelligent truncation to fit the data in the field without eliminating vital address data. Using intelligent truncation, the software abbreviates data in the following ways: - The software uses the USPS abbreviation for the primary name when <i>USE_USPS_PRIMARY_NAME_ABBREVIATION</i> is enabled. - The software uses the USPS abbreviation for the locality name when <i>USE_USPS_LOCALITY_ABBREVIATION</i> is enabled. - If the resulting output data does not fit the field, the software truncates the data. Intelligent truncation is automatic and cannot be controlled with a setting.
<i>CAPITALIZATION</i>	Specifies the casing of your address data. LOWER: Converts data to all lowercase letters. For example, "Main Street South" becomes "main street south." MIXED: Converts data to initial capital letters. For example, "MAIN STREET SOUTH" becomes "Main Street South." UPPER: Converts data to all capital letters. For example, "Main Street South" becomes "MAIN STREET SOUTH." Note For consistent casing, ensure that this option and the <i>CAPITALIZATION</i> setting in the Data Cleanse transform are the same.

Option	Description
<i>DIRECTIONAL_STYLE</i>	Specifies whether to abbreviate directional data. • <i>LONG</i>: Uses fully spelled directionals such as "North," "South," "East," and "West." • <i>PRESERVE</i>: Preserves the style used in the input record. • <i>SHORT</i>: Uses abbreviated directionals such as "N," "S," "E," and "W." i Note When the <i>USE_USPS_PRIMARY_NAME_ABBREVIATION</i> option is set to <i>YES</i>, the software overrides a setting of <i>LONG</i> and <i>PRESERVE</i> for the <i>DIRECTIONAL_STYLE</i> option and outputs the short directional style.
<i>PRIMARY_TYPE_STYLE</i>	Specifies whether to abbreviate the street (primary) type. • <i>LONG</i>: Uses fully spelled primary types such as Street, Avenue, and Road. • <i>PRESERVE</i>: Preserves the style used in the input record. • <i>SHORT</i>: Uses abbreviated primary types such as St, Ave, and Rd. i Note When <i>USE_USPS_PRIMARY_NAME_ABBREVIATION</i> is set to <i>YES</i>, the software overrides a setting of <i>LONG</i> and <i>PRESERVE</i> for the <i>PRIMARY_TYPE_STYLE</i> option and outputs the short primary type style.
<i>UNIT_DESCRIPTION</i>	Specifies how to standardize the unit description. • <i>CONVERT</i>: Uses the unit description found in the postal directory, such as an apartment, suite, room, or floor. • <i>PRESERVE</i>: Preserves the unit description from the input record and corrects any spelling errors.

Option	Description
<i>RETAIN_POUND_SIGN_IN_UNIT_DESCRIPTION</i>	Specifies whether the software outputs the pound sign (#) to the <i>Unit_Description</i> output field when it is present on input. <i>YES</i>: Outputs the pound sign (#) unit designator to the <i>UNIT_DESCRIPTION</i> output field. <i>NO</i>: Outputs the pound sign (#) unit designator to the <i>EXTRANEOUS_SECONDARY_UNIT_NUMBER</i> or the <i>EXTRANEOUS_SECONDARY_NON_POSTAL</i> output fields, or to both fields. The option does not affect the following address situations: - Puerto Rico addresses - Military addresses - Rural Route addresses - Addresses without “#” in the address line - Addresses with remainder words
<i>APPEND_PRIVATE_MAILBOX</i>	Specifies whether the transform appends private mailbox data (PMB) to the address in the same field. <i>YES</i>: Appends the PMB with the address in the same field. <i>NO</i>: Outputs the PMB into the <i>NON_POSTAL_SECONDARY_ADDRESS</i>, <i>NON_POSTAL_UNIT</i>, and the <i>NON_POSTAL_UNIT_NUMBER</i> output fields. A PMB is like a post office box except that it is hosted by a private company.
<i>PRESERVE_DUAL_ADDRESS_ORDER</i>	Specifies whether to preserve or change the format of dual addresses. <i>YES</i>: Preserves the order when an input address contains both a street and mailing address. <i>NO</i>: Changes the order of input dual address lines. When the input address contains both a locality and mailing address, the transform moves the assigned address immediately above the locality and region.

Option	Description
<i>ADDRESS_LINE_ALIAS</i>	Specifies whether to convert an input address line alias on input to the preferred USPS address line on output. • <i>CONVERT</i>: Converts an input address line alias on input to the preferred USPS format on output. i Note To be compliant with CASS, set up your jobs to return the USPS preferred address. When you set <i>ADDRESS_LINE_ALIAS</i> to <i>CONVERT</i>, the software returns the USPS-preferred address, even when the input record has a base address or an alias address. The software does not produce a USPS 3553 when you select <i>PRESERVE</i> for <i>ADDRESS_LINE_ALIAS</i> • <i>PRESERVE</i>: Does not convert the input address line alias to the preferred USPS format, and outputs it as it was input. This option is not CASS compliant. The software does not produce a USPS 3553 form when you select this option.
<i>PRESERVE_PLACE_NAMES</i>	Specifies whether to preserve or change non-mailing city names (place names). • <i>YES</i>: Preserves the non-mailing city name. For example, when an input place name is Hollywood, the transform outputs Hollywood. • <i>NO</i>: Changes non mailing city names to city names preferred by the USPS. For example, when an input place name is Hollywood, the transform outputs Los Angeles. i Note When the <i>USE_USPS_LOCALITY_ABBREVIATION</i> option is set to <i>YES</i>, <i>PRESERVE_PLACE_NAMES</i> is set to <i>YES</i>, and <i>ASSIGN_WITH_INPUT_LOCALITY</i> is set to <i>NO</i>, the software may not preserve some place names over 13 characters and abbreviates them.

Option	Description
ASSIGN_WITH_INPUT_LOCALITY	Specifies whether to use the last-line index when assigning the locality (city) name. YES: Assigns the Locality1 based on the locality name that is input if it is valid for the Postcode 1. Does not change the Locality1 based on last line index. NO: Assigns the Locality1 based on the locality that is input if it is valid for the Postcode 1 and not a place name; otherwise, assigns Locality1 based on the last-line index of the address line. Produces a more geographically true Locality1. If you choose NO, the value you choose for the PRESERVE_PLACE_NAMES option does not matter; the software converts place names. i Note When the USE_USPS_LOCALITY_ABBREVIATION option is set to YES, PRESERVE_PLACE_NAMES is set to YES, and ASSIGN_WITH_INPUT_LOCALITY is set to NO, the software may not preserve some place names over 13 characters and abbreviates them.

Option	Description
<i>INCLUDE_UNUSED_ADDRESS_LINE_DATA</i>	Specifies whether to output the unused address line data for discrete and multiline fields. YES: Includes the unused address line data, including invalid secondary information, in the address line and in the <i>ADDRESS_LINE_REMAINDER1</i> field when included in the setup. NO: Does not output the unused address line data in the address line but outputs the information to certain output fields when included in the setup.

❖ Example

The input address in the following table contains the invalid secondary address of FL 9.

Input	YES	NO
SAP 332 FRONT ST S FL 9 LA CROSSE WI 54601	SAP 332 FRONT ST S FL 3 FL 9 LA CROSSE WI 54601-4025	SAP 332 FRONT ST S FL 3 LA CROSSE WI 54601-4025
Output fields		
<i>PRIMARY_SECONDARY_ADDRESSES</i>	332 FRONT ST S FL 3	332 FRONT ST S FL 3
<i>FULL_ADDRESS</i>	332 FRONT ST S FL 3 FL 9	332 FRONT ST S FL 3
<i>ADDRESS_LINE_REMAINDER1</i>	FL 9	FL 9
<i>SECONDARY_ADDRESS</i>	FL 3	FL 3
<i>PRE_SUITELINK_UNIT_DESCRIPTION</i>	FL	FL
<i>PRE_SUITELINK_UNIT_NUMBER</i>	9	9

Option	Description
<code>ADD_FIRM_MATCH_SECONDARY</code>	Specifies whether the software adds secondary address information when the SuiteLink match has a secondary address. YES: Adds secondary address information obtained from SuiteLink directories to the address line. Select YES for CASS certification. NO: Does not add secondary address information obtained from SuiteLink directories to the address line, but includes SuiteLink-found information reflected in the lastline ZIP+4 Code and in other output fields. NO: Doesn't add secondary address information obtained from SuiteLink directories to the address line, but outputs the ZIP+4 and 11-digit barcode from the valid secondary address. i Note When you select NO, SAP Data Quality Management SDK doesn't generate the USPS Form 3553. If you chose to generate the USPS Form 3553 and also select NO, SAP Data Quality Management SDK issues a warning and doesn't produce the USPS Form 3553. There are instances when the software ignores the NO setting for <code>ADD_FIRM_MATCH_SECONDARY</code>. For example, the software ignores the setting when it doesn't find a secondary address match in the National directories, but finds an exact match to the SuiteLink secondary address. The software doesn't consider the SuiteLink secondary address as a change to the input secondary address. The software also doesn't consider the SuiteLink secondary address as a change to the input secondary address when it finds an exact match to the SuiteLink secondary unit and or range.

Option	Description
MOVE_MULTILINE_DATA	Specifies whether the transform rearranges multiline data to conform to USPS guidelines, and the method for arranging blank lines. • NO: Keeps input multiline data in input order. • BOTTOM: Moves the primary address into position above the locality, region, postal code line, or lastline. Moves any blank lines to the top of the address, and shifts the data to the bottom of the block. • TOP: Moves the primary address into position above the locality, region, postal code line, or lastline. Moves any blank lines to the bottom of the block, and shifts the data to the top.

You do not have to standardize your data to enable this option.

❖ Example

Top option: All blank lines move to the bottom.

Line:	Input data:	Result of moving:
Line 1	100 Market Street	Sycamore Building
Line 2	Suite 202	Suite 202
Line 3	Sycamore Building	100 Market St
Line 4		Boston MA 02109
Line 5	Boston MA 02109	

When you choose **TOP** or **BOTTOM**, ensure that the length of all input fields mapped to Multiline fields are the same length. For example, When you set **MULTILINE1** to 60, set all other Multiline fields to 60.

COMBINE_MULTILINES	Specifies what to do with related fields input on separate lines. • YES: Combines related input fields from separate lines onto the same line. • NO: Does not combine input fields on separate lines.
------------------------------------	---

Option	Description
<i>MULTILINE_UPDATE_POSTCODE_1</i>	Specifies whether the transform assigns input postcode1 data with the USPS postcode data and outputs to Multiline. Affects data passed in and retrieved through multiline fields. • <i>DONT_UPDATE</i>: Assigns Postcode 1 to input, but does not output to Multiline. Outputs the original Postcode 1 and outputs the assigned Postcode 1 to other output fields. • <i>ERASE_THEN_UPDATE</i>: Assigns Postcode 1 and outputs the assigned Postcode 1 to the Multiline output fields. If the transform cannot assign input Postcode 1, does not output the original Postcode 1 to Multiline output fields. • <i>UPDATE</i>: Assigns Postcode 1 and outputs the assigned Postcode 1 to the Multiline output fields. Retains the input Postcode 1 when it does not assign Postcode 1.
<i>MULTILINE_UPDATE_POSTCODE_2</i>	Specifies whether the transform assigns input postcode2 data with the USPS postcode data and outputs to Multiline. Affects data passed in and retrieved through multiline fields. • <i>DONT_UPDATE</i>: Assigns Postcode 2 to input, but does not output to Multiline. Outputs the original Postcode 2 and outputs the assigned Postcode 2 to other output fields. • <i>ERASE_THEN_UPDATE</i>: Assigns Postcode 2 and outputs the assigned Postcode 2 to the Multiline output fields. If the transform cannot assign input Postcode 2, does not output the original Postcode 2 to Multiline output fields. • <i>UPDATE</i>: Assigns Postcode 2 and outputs the assigned Postcode 2 to the Multiline output fields. Retains the input Postcode 2 when it does not assign Postcode 2.

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12.7 USA Regulatory Address Cleanse USPS Z4 Change options

Set USPS Z4 Change options to enable Z4Change processing and to specify the last time the Postcode 2 was updated.

Syntax

```
<Z4_CHANGE_OPTIONS>
  <ENABLE_Z4_CHANGE>NO</ENABLE_Z4_CHANGE>
  <LAST_ZIP4_ASSIGN_DATE>06/2010</LAST_ZIP4_ASSIGN_DATE>
</Z4_CHANGE_OPTIONS>
```

Z4 Change Options group descriptions

Option	Description
<i>ENABLE_Z4_CHANGE</i>	Specifies whether the transform performs Z4Change processing. • YES: Performs Z4Change processing. • NO: Does not perform Z4Change processing.
<i>LAST_ZIP4_ASSIGN_DATE</i>	Specifies the month and year that you most recently updated the mailing list with ZIP4 information, either through a full address correction process or a previous Z4Change pass. Enter the date using the format MM/YYYY. The transform verifies that the date is within the 12-month period covered by the Z4Change file. If there is a date problem, the transform issues an execution error message.

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12.8 USA Regulatory Address Cleanse USPS CASS Report Options

Complete the *CASS Report Options* group with USPS Form 3553 information as required by CASS.

Syntax

```
<CASS_REPORT_OPTIONS>
  <LIST_NAME></LIST_NAME>
  <LIST_OWNER></LIST_OWNER>
  <MAILER_ADDRESS1></MAILER_ADDRESS1>
  <MAILER_ADDRESS2></MAILER_ADDRESS2>
  <MAILER_ADDRESS3></MAILER_ADDRESS3>
  <MAILER_ADDRESS4></MAILER_ADDRESS4>
  <COMPANY_NAME_CERTIFIED></COMPANY_NAME_CERTIFIED>
  <SOFTWARE_VERSION></SOFTWARE_VERSION>
  <LOT_CERTIFICATION>YES</LOT_CERTIFICATION>
</CASS_REPORT_OPTIONS>
```

CASS Report Options descriptions

Option	Description
<i>LIST_NAME</i>	Specifies the name of the mailing list (up to 20 characters).
<i>LIST_OWNER</i>	Specifies the name of your company (up to 19 characters).
<i>MAILER_ADDRESS_1</i>	Specifies the name and address of the person or organization for whom you are preparing the mailing (up to 29 characters per line).
<i>MAILER_ADDRESS_2</i>	
<i>MAILER_ADDRESS_3</i>	
<i>MAILER_ADDRESS_4</i>	

Option	Description
<i>COMPANY_NAME_CERTIFIED</i>	Specifies the name of the company that has the end-user CASS certification. If you rely on SAP as your CASS-certified company, leave this option blank. The software inserts “SAP” as the default value. If you have your own CASS end-user certification, enter your company name, up to 40 characters.
<i>SOFTWARE_VERSION</i>	Specifies the version number of the software that is certified to process CASS-certified mailing lists. If you rely on SAP as your CASS-certified company, leave this option blank. The software inserts the applicable software name and version number as the default value. If you used your own software for CASS certification, enter the applicable software name and version number.
<i>LOT_CERTIFICATION</i>	Specifies whether you or your company have LOT certification. • YES: LOT certification is in your or your company name, but not CASS certification. • NO: CASS certification is in your or your company name, but not LOT certification. i Note Setting <i>LOT_CERTIFICATION</i> to NO ensures that the LOT certification lines on the Form 3553 are blank.

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12.9 USA Regulatory Address Cleanse suggestion list options

The *Suggestion List* group contains options to configure how the software outputs suggestions lists.

Syntax

```
<SUGGESTION_LIST>
  <MAX_NUMBER_LASTLINES>15</MAX_NUMBER_LASTLINES>
  <MAX_NUMBER_ADDRESSES>50</MAX_NUMBER_ADDRESSES>
  <LASTLINES_MATCH_MINIMUM>60</LASTLINES_MATCH_MINIMUM>
  <ADDRESS_LINES_MATCH_MINIMUM>60</ADDRESS_LINES_MATCH_MINIMUM>
  <COMBINE_OVERLAPPING_RANGES>NONE</COMBINE_OVERLAPPING_RANGES>
  <ADDRESS_RANGE_WINDOW>0</ADDRESS_RANGE_WINDOW>
  <MATCH_RANGE>NO</MATCH_RANGE>
  <STYLE>XML</STYLE>
  <DELIMITER>***</DELIMITER>
  <FIELD_DELIMITER>|</FIELD_DELIMITER>
```

Suggestion List group option descriptions

Option/Option group	Description
<i>MAX_NUMBER_LASTLINES</i>	Specifies the maximum number of lastline suggestions that the transform generates. Maximum is 15. i Note If you set a low maximum, a viable suggestion could be left out of the suggestion list. ❖ Example - Use the setting to limit the size of the SOAP documents that the Web service sends. - Use the setting to limit the maximum number of suggestions that your users choose from.

Option/Option group	Description
<i>MAX_NUMBER_ADDRESSLINES</i>	Specifies the maximum number of address-line suggestions that the transform generates. Maximum is 100. i Note If you set a low maximum, a viable suggestion could be left out of the suggestion list. ❖ Example • Use the setting to limit the size of the SOAP documents that the Web service sends. • Use the setting to limit the maximum number of suggestions that your users choose from.
<i>LASTLINES_MATCH_MINIMUM</i>	Specifies the minimum similarity score the transform requires for lastline suggestions. From 0 to 80. A suggestion list candidate must meet the minimum threshold before the software presents it as a suggestion for the input lastline. A lower number may produce more suggestions.
<i>ADDRESS_LINES_MATCH_MINIMUM</i>	Specifies the minimum similarity score the transform requires for address line suggestions. From 0 to 80. A suggestion list candidate must meet the minimum threshold before the software presents it as a suggestion for the input address line. A lower number may produce more suggestions.
<i>COMBINE_OVERLAPPING_RANGES</i>	Specifies how the transform consolidates individual suggestions that have overlapping ranges. • <i>COMBINE_IGNOREING_GAPS</i>: Ignores gaps in ranges and overlaps the ranges. Selecting this option causes the transform to be more aggressive when overlapping ranges. • <i>COMBINE_PRESERVING_GAPS</i>: Preserves gaps in primary ranges, but still consolidates overlapping ranges. • <i>NONE</i>: Does not consolidate overlapping ranges.

Option/Option group	Description
<i>ADDRESS_RANGE_WINDOW</i>	Specifies a range around the primary address in which the transform selects suggestions. Use this option to limit the suggestions to be within a few blocks of your input. ❖ Example You set this option to 500. Input address is: 1000 Pine St. The software returns suggestions in a ranged from 750 to 1250 Pine Street. Enter "0" so there is no limit to the ranges that the transform returns.
<i>MATCH_RANGE</i>	Specifies whether the transform disregards an address-line suggestion when it does not match the primary range of the input address. • YES: Returns address-line suggestion only when it matches the primary range of the input address. • NO: Returns address-line suggestion when it doesn't have the same primary range as the input.
<i>STYLE</i>	Specifies the style of the output file. • DELIMITED: Outputs the suggestion list data in a delimited text format. Specify delimiter to use in the <i>DELIMITER</i> and <i>FIELD_DELIMITER</i> options. • XML: Outputs the suggestion list data as hierarchical XML. Specify XML when you integrate suggestion lists through the Web service. Then use the XML tools that you own to parse the suggestion list data.
<i>DELIMITER</i>	Specifies the delimiter to use between each suggestion for a delimited style suggestion list output file. Applicable when you chose DELIMITED for the <i>STYLE</i> option. Choose any character or string to separate each suggestion. Ensure that this value differs from the value you choose for the <i>FIELD_DELIMITER</i> value.
<i>FIELD_DELIMITER</i>	Specifies the delimiter to use between each suggestion list. Applicable when you chose DELIMITED for the <i>STYLE</i> option. Ensure that this value differs from the value you choose for the <i>DELIMITER</i> value.

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12.10 USA Regulatory Address Cleanse suggestion list components

Complete options in the *Suggestion List Components* to specify the address field components to include in the *Suggestion_List* output field.

```
<SUGGESTION_LIST_COMPONENTS>
  <SELECTION>YES</SELECTION>
  <LOCALITY1>YES</LOCALITY1>
  <REGION1>YES</REGION1>
  <POSTCODE>YES</POSTCODE>
  <PRIMARY_NUMBER_LOW>YES</PRIMARY_NUMBER_LOW>
  <PRIMARY_NUMBER_HIGH>YES</PRIMARY_NUMBER_HIGH>
  <PRIMARY_PREFIX1>YES</PRIMARY_PREFIX1>
  <PRIMARY_NAME1>YES</PRIMARY_NAME1>
  <PRIMARY_TYPE1>YES</PRIMARY_TYPE1>
  <PRIMARY_POSTFIX1>YES</PRIMARY_POSTFIX1>
  <PRIMARY_NAME_FULL1>NO</PRIMARY_NAME_FULL1>
  <POSTCODE2_ODD>YES</POSTCODE2_ODD>
  <POSTCODE2_EVEN>YES</POSTCODE2_EVEN>
  <PRIMARY_SIDE_INDICATOR>YES</PRIMARY_SIDE_INDICATOR>
  <FIRM>YES</FIRM>
  <UNIT_DESCRIPTION>YES</UNIT_DESCRIPTION>
  <UNIT_NUMBER_LOW>YES</UNIT_NUMBER_LOW>
  <UNIT_NUMBER_HIGH>YES</UNIT_NUMBER_HIGH>
  <SECONDARY_SIDE_INDICATOR>YES</SECONDARY_SIDE_INDICATOR>
</SUGGESTION_LIST_COMPONENTS>
```

i Note

When you leave a component blank and you set the suggestion list *STYLE* option to *XML*, the transform does not output a value to the *SUGGESTION_LIST* output field.

In the following options, select **YES** to include the item in the **SUGGESTION_LIST** output field.

Suggestion List Components option descriptions

Option/option group	Description
SELECTION	Specifies whether the transform outputs the selection number for multiple suggestions.
LOCALITY1	Specifies whether the transform outputs the Locality1 preferred by the USPS. Applicable for primary, secondary, and lastline address levels.
REGION1	Specifies whether the transform outputs the state, province, territory, or region. Applicable for primary, secondary, and lastline address levels.
POSTCODE	Specifies whether the transform outputs the five-digit Postcode1, excluding the ZIP4. Applicable for primary, secondary, and lastline address levels.
PRIMARY_NUMBER_LOW	Specifies whether the transform outputs the low portion of the primary number range. Applicable for primary and secondary address levels.
PRIMARY_NUMBER_HIGH	Specifies whether the transform outputs the high portion of the primary number range. Applicable for primary and secondary address levels.
PRIMARY_PREFIX1	Specifies whether the transform outputs the abbreviated directional, such as N, S, NW, and so on, that precedes a street name. Applicable for primary and secondary address levels.
PRIMARY_NAME1	Specifies whether the software outputs the street name description. Applicable for primary and secondary address levels.
i Note The transform does not output high or low primary numbers to this field.	
PRIMARY_TYPE1	Specifies whether the transform outputs the abbreviated street type, such as St, Ave, or Pl. Applicable for primary and secondary address levels.
PRIMARY_POSTFIX1	Specifies whether the transform outputs the abbreviated directional, such as N, S, NW, and so on, that follows a street name. Applicable for primary and secondary address levels.

Option/option group	Description
<i>PRIMARY_NAME_FULL1</i>	Specifies whether the transform outputs the primary address line, such as the street address or post office box. The output includes the following information: Primary Prefix1, Primary Name1, Primary Type1, and Primary Postfix1. Applicable for primary and secondary address levels. i Note The transform does not output high or low primary numbers to this field.
<i>POSTCODE2_ODD</i>	Specifies whether the transform outputs the four-digit ZIP4 code, odd numbers only. Applicable for primary and secondary address levels.
<i>POSTCODE2_EVEN</i>	Specifies whether the transform outputs the four-digit ZIP4 code, even numbers only. Applicable for primary and secondary address levels.
<i>PRIMARY_SIDE_INDICATOR</i>	Specifies whether the transform outputs “Odd” or “Even” for the primary side indicator. Applicable for primary and secondary address levels.
<i>FIRM</i>	Specifies whether the transform outputs the firm name for the secondary address.
<i>UNIT_DESCRIPTION</i>	Specifies whether the transform outputs the unit description, such as #, Apartment, or Flat. Applicable for secondary address level.
<i>UNIT_NUMBER_LOW</i>	Specifies whether the transform outputs the low portion of the unit number range. Applicable for secondary address level.
<i>UNIT_NUMBER_HIGH</i>	Specifies whether the transform outputs the high portion of the unit number range. Applicable for secondary address level.
<i>SECONDARY_SIDE_INDICATOR</i>	Specifies whether the transform outputs “Odd” or “Even” for the secondary side indicator. Applicable for secondary address level.

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12.11 USA Regulatory Address Cleanse non-certified options

The Non-Certified options group contains settings that, when enabled, create a mailing list that is not compliant with CASS rules.

Syntax

```
<NON_CERTIFIED_OPTIONS>
  <DISABLE_CERTIFICATION>NO</DISABLE_CERTIFICATION>
  <ASSIGN_WITH_INPUT_POSTCODE>NO</ASSIGN_WITH_INPUT_POSTCODE>
  <ACCEPT_INEXACT_POSTCODE_MOVE>NO</ACCEPT_INEXACT_POSTCODE_MOVE>
  <ASSIGN_POSTCODE2_NOT_DPV_VALIDATED>NO</
ASSIGN_POSTCODE2_NOT_DPV_VALIDATED>
  <ENABLE_PARSE_ONLY>NO</ENABLE_PARSE_ONLY>
  <ENABLE_SUGGESTION_LISTS>NO</ENABLE_SUGGESTION_LISTS>
</NON_CERTIFIED_OPTIONS>
```

Non-Certified group option descriptions

Option	Description
<i>DISABLE_CERTIFICATION</i>	Specifies whether the transform performs address cleansing without the restrictions of CASS certification rules. • YES: Disables CASS certification. Select Yes when you enable any of the other options in the Non-Certified group. With certification disabled, you can use the national directory for 14 months after the directory creation date. For CASS certified mailings, you can use the national directory only for 3 or 4 months after the creation date. • NO: Performs address cleansing under CASS certification rules. Ignores any enabled options in the Non-Certified group.
<i>ASSIGN_WITH_INPUT_POSTCODE</i>	Specifies whether the transform should use the last four digits of the input 9-digit post code to make a finer level of assignment than it could make under CASS rules. The last four digits of the 9-digit post code must be unique to a valid firm or secondary address. i Note The last four digits are usually the Postcode2, which is the ZIP+4. • YES: Enables this option. • NO: Disables this option. The transform ignores the last four digits of the input 9-digit postcode and makes a CASS certified assignment.
<i>ACCEPT_INEXACT_POSTCODE_MOVE</i>	Specifies whether the transform ignores non matching elements between the input data and the national directory. Transform uses the built-in matching thresholds to determine that the record matches the record in the national directory. Applicable when an input record has an obsolete postcode or a postcode move. • YES: Enables this option. • NO: Disables this option.

Option	Description
ASSIGN_POSTCODE2_NOT_DPV_VALIDATED	Specifies whether to output the Postcode2 data from the national directory even when you make the following settings: ENABLE_DPV in the Assignment Options group set to <i>NO</i> DISABLE_CERTIFICATION in this group to <i>YES</i> <i>YES</i>: The transform does not validate the address using DPV. <i>NO</i>: The transform leaves the output field <i>POSTCODE2</i> blank when it makes an assignment, and you set ENABLE_DPV to <i>NO</i>.
ENABLE_PARSE_ONLY	Specifies whether the transform only parses your data into their discrete components, and does not validate data against the postal directories. <i>YES</i>: Parses records into their discrete components, but does not perform a lookup in the postal directories. Processing is faster, but the parsed results are not verified against the postal directories. <i>NO</i>: Parses records into their discrete components and performs a lookup in the postal directories. Processing is slower, but the parsed results are verified against the postal directories.
ENABLE_SUGGESTION_LISTS	Specifies whether the software uses suggestion lists for records that do not assign. For transactional processes only. <i>YES</i>: Generates suggestion lists for addresses that do not assign so that you can choose an address to output. <i>NO</i>: Does not generate suggestion lists for unassigned addresses.

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12.12 USA Regulatory Address Cleanse USPS License Information

If you perform NCOALink, SuiteLink, LACSLink, or DPV processing, complete the required USPS license information.

Syntax

```
<USPS_LICENSE_INFORMATION>
  <LICENSEE_NAME></LICENSEE_NAME>
  <LIST_OWNER_NAICS_CODE></LIST_OWNER_NAICS_CODE>
  <LIST_ID></LIST_ID>
  <CUSTOMER_COMPANY_NAME>My Company</CUSTOMER_COMPANY_NAME>
  <CUSTOMER_COMPANY_ADDRESS>1234 Main St</CUSTOMER_COMPANY_ADDRESS>
  <CUSTOMER_COMPANY_URB></CUSTOMER_COMPANY_URB>
  <CUSTOMER_COMPANY_LOCALITY>Anywhere</CUSTOMER_COMPANY_LOCALITY>
  <CUSTOMER_COMPANY_REGION>CA</CUSTOMER_COMPANY_REGION>
  <CUSTOMER_COMPANY_POSTCODE1>12345</CUSTOMER_COMPANY_POSTCODE1>
  <CUSTOMER_COMPANY_POSTCODE2>6789</CUSTOMER_COMPANY_POSTCODE2>
  <CUSTOMER_COMPANY_PHONE>1234567890</CUSTOMER_COMPANY_PHONE>
  <LIST_PROCESSING_FREQUENCY>99</LIST_PROCESSING_FREQUENCY>
  <LIST_RECEIVED_DATE>YYYY/MM/DD</LIST_RECEIVED_DATE>
  <LIST_RETURN_DATE>YYYY/MM/DD</LIST_RETURN_DATE>
  <PROVIDER_LEVEL>NONE</PROVIDER_LEVEL>
  <IMB_MAILER_ID></IMB_MAILER_ID>
  <CUSTOMER_REGISTRATION_ID></CUSTOMER_REGISTRATION_ID>
</USPS_LICENSE_INFORMATION>
```

Provide USPS license information for the licensee and the customer.

- Licensee is the company performing the processing.
- Customer is the company for whom the licensee is performing the processing.

If you are performing the processing for yourself, you are the end user.

USPS License Information group option descriptions

Option	Description
<i>CUSTOMER_COMPANY_NAME</i>	Specifies the customer information including name and address. If you are an end user, leave these fields blank. However, complete the fields if you have an Alternate Stop Processing agreement from NCOALink and you have entered a special keycode into Data Services License Manager. The software includes the customer information in the NCOALink Processing Summary report and log files.
<i>CUSTOMER_COMPANY_ADDRESS</i>	
<i>CUSTOMER_COMPANY_URBANISATION</i>	
<i>CUSTOMER_COMPANY_LOCALITY</i>	
<i>CUSTOMER_COMPANY_REGION</i>	
<i>CUSTOMER_COMPANY_POSTCODE1</i>	
<i>CUSTOMER_COMPANY_POSTCODE2</i>	
<i>CUSTOMER_REGISTRATION_ID</i>	Required for USPS business customers who were assigned a <i>Customer Registration ID (CRID)</i> by the USPS. This number can be up to 15 digits long. Optional, if the USPS did not assign a CRID for your company. The software includes this information in the Processing Acknowledgement Form (PAF) monthly report log files.
<i>CUSTOMER_COMPANY_PHONE</i>	Optional.
<i>IMB_MAILER_ID</i>	Optional. Specify the unique Intelligent Mail Barcode (IMB) Mailer ID from the USPS, if applicable. The IMB Mailer ID is a unique code, 6- or 9-digits, assigned to mailers by the USPS based on their annual mail volumes. The software includes this information in the NCOALink Processing Acknowledgement Form (PAF).
<i>LICENSEE_NAME</i>	Required for NCOALink. Specifies the name of the company mentioned in the USPS license agreement. Maximum size is 30 characters. The software includes this information in the PAF log and NCOALink Processing Summary report.
<i>LIST_OWNER_NAICS_CODE</i>	Required. Specifies the North American Industry Classification System (NAICS) code, which identifies the business in which the list owner engages.

Option	Description
<i>LIST_ID</i>	Required for NCOALink. Specifies the 6-digit Customer or List ID for NCOALink limited and full service providers. If you are an end user, leave this field blank. The List ID is a unique ID assigned by the licensee to identify the customer or list owner. If you are the licensee and you do not have a naming scheme in place, we suggest that you create the 6-digit ID using the following information: • First 3 digits: Customer name or identifier • Last 3 digits: List name identifier
<i>LIST_PROCESSING_FREQUENCY</i>	Specifies the number of times per year the list is processed for NCOALink. Enter a 2-digit value from 1 to 52. If the list owner has other lists processed by the NCOALink licensee at different frequencies, enter 99.
<i>LIST_RECEIVED_DATE</i>	Specifies the date when the NCOALink licensee received the list. Use the yyyy/mm/dd format. If you are an end user, leave this field blank.
<i>LIST_RETURN_DATE</i>	Specifies the date when the list is to be returned to the customer. Use the yyyy/mm/dd format. If you are an end user, leave this field blank.
<i>PROVIDER_LEVEL</i>	Specifies the provider levels for which the licensee has a registered license keycode. The dropdown list contains the provider levels supported in your registered keycodes.

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Related Information

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[USPS License Information Required Options \[page 359\]](#)

12.13 USPS License Information Required Options

The options in the USPS License Information group are required based on the type of service you provide and are licensed for.

If you are processing NCOALink, DPV, SuiteLink, or LACSLink, the USPS License Information group contains options that you must complete. The following table indicates the USPS services that require a value for each option of the *USPS License Information* group.

Option	NCOALink Full Service Provider	NCOALink Limited Service Provider ¹	NCOALink End User	DSF2	DPV	LACSLink
<i>LICENSEE_NAME</i>	X	X	X	X		
<i>LIST_OWNER_NAICS_CODE</i>	X	X	X	X		
<i>CUSTOMER_REGISTRATION_ID</i>						
<i>CUSTOMER_COMPANY_URBANIZATION</i>						
<i>LIST_ID</i>	X	X	X	X		
<i>CUSTOMER_COMPANY_NAME</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_ADDRESS</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_LOCALITY</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_REGION</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_POSTCODE1</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_POSTCODE2</i>	X	X	X	X	X	X
<i>CUSTOMER_COMPANY_PHONE</i>						
<i>LIST_PROCESSING_FREQUENCY</i>	X	X	X			
<i>LIST_RECEIVED_DATE</i>	X	X		X		
<i>LIST_RETURN_DATE</i>	X	X		X		
<i>PROVIDER_LEVEL</i>	X	X	X			

¹ With NCOALink enabled

Option	NCOALink Full Service Provider	NCOALink Limited Service Provider ¹	NCOALink End User	DSF2	DPV	LACSLink
<i>IMB_MAILER_ID</i>						

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Related Information

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12.14 USA Regulatory Address Cleanse transform NCOALink options

Complete the options in the NCOALink group when you prepare your data for NCOALink certification.

Syntax

```
<NCOALink>
  <MAILING_LIST_NAME></MAILING_LIST_NAME>
  <PLATFORM_ID></PLATFORM_ID>
```

¹ With NCOALink enabled

Option	Description
<i>MAILING_LIST_NAME</i>	Specifies the name of the mailing list to be processed. Enter up to 30 characters. If this list is a master house list or your only mailing list, consider entering your company name here. This name appears in the log files.
<i>PLATFORM_ID</i>	Specifies the NCOALink licensee identification number. The Platform ID is the identification number assigned to the individual or company who is the NCOALink licensee. The ID is four characters.

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Related Information

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[USA Regulatory Address Cleanse USPS License Information \[page 356\]](#)

[USPS License Information Required Options \[page 359\]](#)

12.14.1 USPS NCOALink processing options

Specify the USPS processes to perform on the data by enabling processes in the Processing Options group.

⌘ Syntax

```
<PROCESSING_OPTIONS>
  <LIST_PROCESSING_MODE>CHANGE_OF_ADDRESS</LIST_PROCESSING_MODE>
  <RETRIEVE_MOVE_TYPES>INDIVIDUAL_AND_FAMILY_AND_BUSINESS</
RETRIEVE_MOVE_TYPES>
  <LIST_PROCESSING_OBJECTIVE>NORMAL</LIST_PROCESSING_OBJECTIVE>
  <HIGH_MATCH_RATE_EXPECTANCY>NONE</HIGH_MATCH_RATE_EXPECTANCY>
  <CONSIDER_MOVES_WITHIN_MONTHS></CONSIDER_MOVES_WITHIN_MONTHS>
```

```

<PROCESSING_FIRST_CLASS_MAIL>YES</PROCESSING_FIRST_CLASS_MAIL>
<PROCESSING_PERIODICALS>NO</PROCESSING_PERIODICALS>
<PROCESSING_STANDARD_MAIL>NO</PROCESSING_STANDARD_MAIL>
<PROCESSING_PACKAGE_SERVICE_MAIL>NO</PROCESSING_PACKAGE_SERVICE_MAIL>
<EXTERNAL_PROCESSES_UPDATING_LIST></EXTERNAL_PROCESSES_UPDATING_LIST>
</PROCESSING_OPTIONS>

```

Processing Options descriptions

Option	Description
<i>CONSIDER_MOVES_WITHIN_MONTHS</i>	Specifies whether the transform ignores change of address data older than the specified number of months. ❖ Example Enter 12 to use change of address data that has a move-effective date within the last 12 months. - If you are an end user or limited service provider, enter a value from 6 through 18. - If you're a full service provider or using ANKLink, enter a value from 6 through 48. - Leave the option blank to enable the transform to use all available data based on your license. Blank is the default setting.
<i>EXTERNAL_PROCESSES_UPDATING_LIST</i>	Specifies whether the transform performs additional processes on the list before or after the USA Regulatory Address Cleanse transform.

Option	Description
<i>HIGH_MATCH_RATE_EXPECTANCY</i>	Specifies whether you expect your data to has a high match rate. Helps the USPS distinguish between files that have a legitimate reason for a high percentage of NCOALink matches and files that do not, which are used to fraudulently create mover lists. • <i>NONE</i> or blank: You do not expect a high match rate. <i>NONE</i> is the default setting. • <i>ANKLINK_PROCESSED_LIST</i>: A legitimate reason for a high match rate. A file that you processed with ANKLink contains records for people who have moved but you don't yet have their new address. i Note <i>ANKLINK_PROCESSED_LIST</i> is available only to full service providers. • <i>STAGE_FILE</i>: A legitimate reason for a high match rate. A Stage file is for USPS testing. If you're performing Stage I or Stage II testing, ensure that the <i>LIST_PROCESSING_OBJECTIVE</i> option is set to a Stage option. • <i>RETURN_MAIL_LIST</i>: A legitimate reason for a high match rate. A returned mail list file contains records for mail that were returned to sender.

Option	Description
<i>LIST_PROCESSING_MODE</i>	Specifies a reason for processing the list. <i>CHANGE_OF_ADDRESS</i>: To update the list with the latest address data. <i>CHANGE_OF_ADDRESS</i> is the default setting. <i>STATISTICS_ONLY</i>: To analyze list statistics such as the number of records that have updated addresses or the total number of each type of move. i Note When you select <i>STATISTICS_ONLY</i>, the transform does not update your records with move-updated address information. <i>RETURN_CODES_ONLY</i>: For informational purposes. The transform posts return codes in the output components <i>NCOALINK_RETURN_CODE</i> or <i>ANKLINK_RETURN_CODE</i>. Examining the return codes can tell you whether matching records were found in the NCOALink directories and why or why not. i Note When you select <i>RETURN_CODES_ONLY</i>, the transform does not update your records with move-updated address information.
<i>LIST_PROCESSING_OBJECTIVE</i>	Specifies your reason for using NCOALink on this list: <i>EMPLOYEE_TRAINING</i>: Processing is a part of employee training. <i>INTERNAL_DATABASE_TESTING</i>: Processing a list owned by the licensee for testing a database. <i>MARKETING</i>: Processing to test an external customer list. <i>NORMAL</i>: Processing to update the list before a mailing. <i>NORMAL</i> is the default setting. <i>STAGE_I_AND_STAGE_II</i>: Processing to test the matching performance against a USPS test file. The USPS scores the Stage II test file. <i>SYSTEM_TESTING</i>: Processing as part of system testing. For example, test loading USPS file updates.
<i>PROCESSING_FIRST_CLASS_MAIL</i>	Specifies whether you are processing first class mail. Select <i>YES</i> or <i>NO</i>. <i>YES</i> is the default setting.
<i>PROCESSING_PERIODICALS</i>	Specifies whether you are processing periodical mail. Select <i>YES</i> or <i>NO</i>. <i>NO</i> is the default setting.

Option	Description
<i>PROCESSING_STANDARD_MAIL</i>	Specifies whether you are processing standard mail. Select <i>YES</i> or <i>NO</i> . <i>NO</i> is the default setting.
<i>PROCESSING_PACKAGE_SERVICE_MAIL</i>	Specifies whether you are processing package service mail. Select <i>YES</i> or <i>NO</i> . <i>NO</i> is the default setting.
<i>RETRIEVE_MOVE_TYPES</i>	Specifies what type of move information the software generates. <i>BUSINESS</i> <i>INDIVIDUAL</i> <i>INDIVIDUAL_AND_BUSINESS</i> <i>INDIVIDUAL_AND_FAMILY</i> <i>INDIVIDUAL_AND_FAMILY_AND_BUSINESS</i><i>INDIVIDUAL_AND_FAMILY_AND_BUSINESS</i> is the default setting.

12.14.2 USPS NCOALink Report Options

Set the option in the NCOALink Report Options group so the transform includes or excludes return code descriptions in the NCOALink Processing Summary report.

Syntax

```
<REPORT_OPTIONS>
  <GENERATE_RETURN_CODE_DESCRIPTIONS>NO</
GENERATE_RETURN_CODE_DESCRIPTIONS>
  <NCOA_SOFTWARE_NAME></NCOA_SOFTWARE_NAME>
  <NCOA_SOFTWARE_VERSION></NCOA_SOFTWARE_VERSION>
</REPORT_OPTIONS>
```

Report option description

Option	Description
<i>GENERATE_RETURN_CODE_DESCRIPTIONS</i>	Specifies whether the transform includes return code descriptions along with the return codes in the NCOALink Processing Summary report. <i>YES</i>: Includes return code descriptions with the return codes in the NCOALink Processing Summary report. <i>NO</i>: Does not include return code descriptions in the NCOALink Processing Summary report. <i>NO</i> is the default setting. Return codes indicate whether a record was affected by a move, how the NCOALink match was made, or why a match could not be made. Without this option enabled, the NCOALink Processing Summary report contains only a short summary of the return codes.

Option	Description	
<code>NCOA_SOFTWARE_NAME</code>	Specify your self-certified NCOALink software name	i Note If you did not self-certify your own NCOALink solution, leave these fields blank. The software will automatically populate with SAP's NCOALink software name and version.
<code>NCOA_SOFTWARE_VERSION</code>	Specify your self-certified NCOALink software version	

12.14.3 USPS NCOALink output option

Enable this output option so that the transform updates standardized output fields with NCOALink information.

Syntax

```
<OUTPUT_OPTIONS>
  <APPLY_MOVE_TO_STANDARDIZED_FIELDS>YES</
APPLY_MOVE_TO_STANDARDIZED_FIELDS>
</OUTPUT_OPTIONS>
```

NCOALink Output option description

Option	Description
<code>APPLY_MOVE_TO_STANDARDIZED_FIELDS</code>	Specifies whether the transform updates applicable standardized output fields with address data from NCOALink processing.

i Note
 This option does not affect component type output fields.

- YES:** Updates applicable standardized fields with NCOALink move update address data. **YES** is the default setting.
- NO:** Outputs standardized fields with only the standardized version of the input data.

12.14.4 USPS NCOALink PAF details

To complete the required NCOALink Processing Acknowledgement Form (PAF), complete options in the PAF Details group.

Syntax

```
<PAF_DETAILS>
```

```

<TYPE>INITIAL</TYPE>
<NAME_OF_PERSON_SIGNING></NAME_OF_PERSON_SIGNING>
<TITLE_OF_PERSON_SIGNING></TITLE_OF_PERSON_SIGNING>
<EMAIL_OF_PERSON_SIGNING></EMAIL_OF_PERSON_SIGNING>
<DATE_SIGNED_BY_CUSTOMER></DATE_SIGNED_BY_CUSTOMER>
<DATE_SIGNED_BY_LICENSEE></DATE_SIGNED_BY_LICENSEE>
<CUSTOMER_PARENT_COMPANY_NAME></CUSTOMER_PARENT_COMPANY_NAME>
<CUSTOMER_ALTERNATE_COMPANY_NAME></CUSTOMER_ALTERNATE_COMPANY_NAME>
<COMPANY_WEBSITE></COMPANY_WEBSITE>
<USING_ALTERNATIVE_PAF>NO</USING_ALTERNATIVE_PAF>
<USING_COOPERATIVE_DATABASE>NO</USING_COOPERATIVE_DATABASE>
</PAF_DETAILS>

```

End users do not have to complete the options in this group.

PAF Details option descriptions

Option	Description
<i>COMPANY_WEBSITE</i>	Optional. Specifies the company website address for the person signing the PAF.
<i>CUSTOMER_ALTERNATE_COMPANY_NAME</i>	Specifies an alternate name when the list owner company name is also known by a different name.
<i>CUSTOMER_PARENT_COMPANY_NAME</i>	Specifies the name of the parent company of the list owner company, if applicable.
<i>DATE_SIGNED_BY_LICENSEE</i>	Specifies the date that the NCOALink service provider signs the PAF. Use the following format: yyyy/mm/dd.
<i>DATE_SIGNED_BY_CUSTOMER</i>	Specifies the date the customer signs the PAF. Use the following format: yyyy/mm/dd.
<i>EMAIL_OF_PERSON_SIGNING</i>	Optional. Specifies the email address for the person who signs the PAF.
<i>NAME_OF_PERSON_SIGNING</i>	Specifies the name of the person signing the PAF. Limit of 50 characters.
<i>TITLE_OF_PERSON_SIGNING</i>	Specifies the job title of the person signing the PAF. Limit of 50 characters.
<i>TYPE</i>	Specifies the reason for this PAF. <i>INITIAL</i>: For initial authorization to process NCOALink for listed customer. <i>MODIFIED</i>: To update information that changed on an existing PAF. <i>RENEWAL</i>: To replace an existing PAF that is expired or is expiring.

Option	Description
<i>USING_ALTERNATIVE_PAF</i>	Required. Specifies whether you are using an alternative PAF that is not the official USPS PAF form. <i>YES</i>: Using an alternative PAF. i Note Requires specific permission from the USPS. <i>NO</i>: Not using the official USPS PAF form. <i>NO</i> is the default setting.
<i>USING_COOPERATIVE_DATABASE</i>	Required for Full and Limited Service Providers. Indicates whether the list is from a cooperative database. <i>YES</i>: List is from a cooperative database. i Note The USPS requires a PAF for each participant in the cooperative database. i Note When set to <i>YES</i>, the PAF log file contains a "C" to indicate that the processed list is from a cooperative database. <i>NO</i>: List is not from a cooperative database. <i>NO</i> is the default setting.

12.14.5 USPS NCOALink service provider options

Complete the information in the Service Provider group to provide required information about the service provider and the mailing list being processed.

Syntax

```

<SERVICE_PROVIDER_OPTIONS>
  <BUYER_COMPANY_NAME></BUYER_COMPANY_NAME>
  <POSTCODE_FOR_MAIL_ENTRY></POSTCODE_FOR_MAIL_ENTRY>
  <PRE_PROCESSES_PERFORMED>NO</PRE_PROCESSES_PERFORMED>
  <PRE_PROCESSED_DATA_MODIFIED>NO</PRE_PROCESSED_DATA_MODIFIED>
  <POST_PROCESSES_PERFORMED>NO</POST_PROCESSES_PERFORMED>
  <POST_PROCESSED_DATA_MODIFIED>NO</POST_PROCESSED_DATA_MODIFIED>
  <CONCURRENT_PROCESSES_PERFORMED>NO</CONCURRENT_PROCESSES_PERFORMED>
  <CONCURRENT_PROCESSED_DATA_MODIFIED>NO</
CONCURRENT_PROCESSED_DATA_MODIFIED>
  <OUTPUT_RETURNED>STANDARD</OUTPUT_RETURNED>
  <ADDITIONAL_NOTES>NONE</ADDITIONAL_NOTES>
  <INHOUSE_LIST_PROCESSING>NO</INHOUSE_LIST_PROCESSING>
</SERVICE_PROVIDER_OPTIONS>

```


End users do not have to complete the options in this group.

When you complete the options in this group, answer as if you have already processed the list.

Service Provider Options descriptions

Option	Description
<i>ADDITIONAL_NOTES</i>	Specifies whether the customer submitted a written request for an extension. • <i>NONE</i>: The customer did not submit a written request for an extension. <i>NONE</i> is the default setting. • <i>CUSTOMER_REQUESTED_EXTENSION</i>: The customer submitted a written request for an extension.
<i>BUYER_COMPANY_NAME</i>	Specifies the company or individual who is leasing, renting, or buying the list after you perform NCOALink processing on the list.
<i>CONCURRENT_PROCESSES_PERFORMED</i>	Specifies whether you processed this list concurrently with NCOALink processing. Select <i>YES</i> or <i>NO</i>.
<i>CONCURRENT_PROCESSED_DATA_MODIFIED</i>	Specifies whether concurrent processing you performed during NCOALink processing changed the data. • <i>NO</i>: Concurrent processes performed during NCOALink processing did not change the data. <i>NO</i> is the default setting. • <i>FROM_POSTAL_DATA</i>: Concurrent processes performed during NCOALink processing added postal data to the list. • <i>FROM_NON_POSTAL_DATA</i>: Concurrent processes performed during NCOALink processing added non postal data to the list. • <i>FROM_BOTH</i>: Concurrent processes performed during NCOALink processing added both postal and non postal data to the list.
<i>IN_HOUSE_LIST_PROCESSING</i>	Required for Full Service Providers. Specifies whether the list is an in-house or internal list. Select <i>YES</i> or <i>NO</i>. i Note When set to <i>YES</i>, the transform adds an "I" to the Customer Service log to indicate that the list is an in-house list.

Option	Description
<i>OUTPUT_RETURNED</i>	Specifies the type of output the NCOALink job returned to the client. • <i>STANDARD</i>: The job returned all required NCOALink output to the client. <i>STANDARD</i> is the default setting. • <i>MODIFIED</i>: The job returned output that was modified by one or more post processes. • <i>BOTH</i>: The job returned output that was modified by one or more post processes. The job also returned a separate file that contained all of the required output.
<i>POST_PROCESSES_PERFORMED</i>	Specifies whether you processed the list after NCOALink processing. Select <i>YES</i> or <i>NO</i>.
<i>POST_PROCESSED_DATA_MODIFIED</i>	Specifies whether post processes performed after NCOALink processing included postal or non postal changes to the data. • <i>NO</i>: Post processing performed after NCOALink processing did not include changes to the data. <i>NO</i> is the default setting. • <i>FROM_POSTAL_DATA</i>: Post processing performed after NCOALink processing included postal changes to the data. • <i>FROM_NON_POSTAL_DATA</i>: Post processing performed after NCOALink processing included non postal changes to the data. • <i>FROM_BOTH</i>: Post processing performed after NCOALink processing included both postal and non postal changes to the data.
<i>POSTCODE_FOR_MAIL_ENTRY</i>	Specifies the ZIP Code of the Business Mail Entry Unit (BMEU) or post office where you submitted the mail for mailing.
<i>PRE_PROCESSES_PERFORMED</i>	Specifies whether you performed processes on the list before you performed NCOALink processing. Select <i>YES</i> or <i>NO</i>.

Option	Description
<code>PRE_PROCESSED_DATA_MODIFIED</code>	Specifies whether preprocessing performed before NCOALink processing updated the data with postal or non postal data. <code>NO</code>: Preprocessing performed before NCOALink processing did not include changes to the data. <code>NO</code> is the default setting. <code>FROM_POSTAL_DATA</code>: Preprocessing performed before NCOALink processing included postal changes to the data. <code>FROM_NON_POSTAL_DATA</code>: Preprocessing performed before NCOALink processing included non postal changes to the data. <code>FROM_BOTH</code>: Preprocessing performed before NCOALink processing included both postal and non postal changes to the data.

12.14.6 USPS NCOALink Contact Detail Options

Complete the Contact Detail List options with information about the NCOALink broker, list administrator, or list owner.

Syntax

```
<CONTACT_DETAIL_LIST>
  <CONTACT_DETAILS>
    <TYPE>LIST_ADMINISTRATOR</TYPE>
    <LICENSEE_ASSIGNED_ID></LICENSEE_ASSIGNED_ID>
    <CONTACT_LEVEL></CONTACT_LEVEL>
    <NAICS_CODE></NAICS_CODE>
    <NAME></NAME>
    <ADDRESS></ADDRESS>
    <URBANIZATION></URBANIZATION>
    <LOCALITY></LOCALITY>
    <REGION></REGION>
    <POSTCODE1></POSTCODE1>
    <POSTCODE2></POSTCODE2>
    <PHONE></PHONE>
    <CONTACT_COMPANY_WEBSITE></CONTACT_COMPANY_WEBSITE>
    <PAF_SIGNING_DATE></PAF_SIGNING_DATE>
  </CONTACT_DETAILS>
</CONTACT_DETAIL_LIST>
```

The Contact Detail List options are not required for end users. Enter information about the list provider. A list provider can be any of the following:

- Broker: Directs business to the service provider.
- List Administrator: Stores and maintains address lists.
- Owner: Processes lists with no involvement from a broker or list administrator.

Contact Detail List option descriptions

Option	Description
<i>ADDRESS</i>	Specifies the address for the list provider.
<i>CONTACT_LEVEL</i>	Specifies the degree of separation the list provider is from you. Enter a value from 1 through 99. Example Enter 1 if you received the list from your broker. Enter a 2 if your broker received the list from a list administrator. Note The transform does not use this information in any log file.
<i>CONTACT_COMPANY_WEBSITE</i>	Optional. Specifies the website of the list provider.
<i>LOCALITY</i>	Specifies the locality (city name) of the list provider.
<i>TYPE</i>	Specifies the type of list provider. BROKER: List provider is a broker, who directs business to the service provider. LIST_ADMINISTRATOR: List provider is a list administrator, who stores and maintains address lists. OWNER: List provider is the owner, who processes the list with no involvement from a broker or list administrator.
<i>LICENSE_ASSIGNED_ID</i>	Specifies an identification that you assign to the list provider. Enter six characters.
<i>NAICS_CODE</i>	Specifies the numeric North American Industry Classification System code for the list provider. The NAICS code identifies the business in which the list provider engages. For more information about NAICS codes, see https://www.census.gov/naics.
<i>NAME</i>	Specifies the name of the list provider.
<i>PAF_SIGNING_DATE</i>	Specifies the date the list provider signed their PAF. Use the following format: yyyy/mm/dd.
<i>PHONE</i>	Specifies the phone number of the list provider.
<i>POSTCODE1</i>	Specifies the Postcode1 (ZIP Code) for the list provider.

Option	Description
<i>POSTCODE2</i>	Specifies the Postcode2 (ZIP+4) for the list provider.
<i>REGION</i>	Specifies the Region (State) for the list provider.
<i>URBANIZATION</i>	Specifies the urbanization for Puerto Rican addresses, if the information is known.

13 Global Address Cleanse transform

Use the Global Address Cleanse transform to identify, parse, validate, and correct your global address data.

Global Address Cleanse information

Characteristic	Description
Use	The Global Address Cleanse transform identifies, parses, validates, and corrects global address data, such as primary number, primary name, primary type, directional, secondary identifier, secondary number, locality, region, and postcode. The transform has options to enable generation of data quality statistics tables. These statistics provide insight into how the software cleansed and assigned your data by generating aggregated summary statistics and specific record-level statistics. Note The Global Address Cleanse transform does not support CASS certification or produce a USPS Form 3553. If you want to certify your U.S. address data, use the USA Regulatory Address Cleanse transform, which supports CASS. If you perform both address cleansing and data cleansing, the Global Address Cleanse transform typically comes before the Data Cleanse transform in the data flow.
Content objects	Sample XML configurations for a regular Global Address Cleanse job and one set for suggestion lists.

[Cleanse global address data \[page 375\]](#)

To cleanse your address data for any of the supported countries, use the Global Address Cleanse transform.

[Address assignment for buildings \[page 376\]](#)

The Global Address Cleanse transform uses building information in addresses to enhance the address assignment process and results.

[Process Japanese addresses \[page 379\]](#)

The Global Address Cleanse transform and the Global Address engine parses and normalizes Japanese addresses for address cleansing and data matching.

[Process Chinese addresses \[page 389\]](#)

Use the Global Address engine in the Global Address Cleanse transform to parse Chinese addresses.

13.1 Cleanse global address data

To cleanse your address data for any of the supported countries, use the Global Address Cleanse transform.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Select the engines to use in the Global Address Cleanse transform based on the goals for your processing:

- To generate the Canada Post certified SERP (Software Evaluation and Recognition Program), enable the Canada engine.
- To generate the Australia Post AMAS (Address Matching Approval System), enable the Global Address.
- To process a mix of countries, including USA addresses, enable the USA engine in combination with the other applicable engines.

Tip

Cleansing U.S. data with the USA Regulatory Address Cleanse transform can be faster than with the Global Address Cleanse transform and USA engine. Most of the time, this scenario is true even if you end up needing both transforms.

To create suggestion lists for the supported countries, create a real-time data flow and use the Global Address Cleanse transform with the Canada, USA, and Global Address engines.

The Global Address Cleanse transform supports many countries with the following limitations:

- The level of address correction varies by country and by the engine that you use. Complete coverage of all addresses in a country isn't guaranteed.
- For the Global Address engine, country support depends on the sets of postal directories you purchase.
- For Japan, the transform bases the assignment level on data provided by the Ministry of Public Management Home Affairs, Posts, and Telecommunications (MPT).
- The Country ID transform identifies many countries, but, the Global Address Cleanse transform engines don't always provide address correction for all countries.

Parent topic: [Global Address Cleanse transform \[page 374\]](#)

Related Information

[Address assignment for buildings \[page 376\]](#)

[Process Japanese addresses \[page 379\]](#)

[Process Chinese addresses \[page 389\]](#)

13.2 Address assignment for buildings

The Global Address Cleanse transform uses building information in addresses to enhance the address assignment process and results.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Addresses from various countries contain building information. For example, an address can contain a building name, building descriptor, building number, or building type. Additionally, some countries separate the base building name and the building descriptor in reference data, so the Global Address Cleanse transform accommodates the information in specific building output fields.

The Global Address Cleanse transform uses building information to make address assignments. For example, an input address has building information, but no street address. Or an input address has a partial street address and a building number.

When your data contains addresses with building information, configure the Global Address Cleanse transform to include output fields for building information.

The following table describes the output fields for building information.

Output field	Building information
<i>BUILDING1</i> <i>BUILDING2</i>	Contains the base name for a building. The transform separates the base name from the description for countries that separate the two in reference data.
❖ Example A building name in the Russian Federation is “корпус Б”. The transform outputs building data as follows: • <i>BUILDING1</i> contains the base name “Б”. • <i>BUILDING_DESCRIPTION1</i> contains the descriptor “корпус”. The <i>BUILDING2</i> field contains the base name for the second building in an address when there are two buildings in an address.	

Output field	Building information
<i>BUILDING_DESCRIPTION1</i> <i>BUILDING_DESCRIPTION2</i>	Contains the building descriptor. For countries that separate the base building name from the descriptor in reference data, the transform outputs the building description to this field. ❖ Example The transform outputs the building name “копный Б” from the Russian Federation as follows: • <i>BUILDING_DESCRIPTION1</i> contains the descriptor “копный” • <i>BUILDING1</i> contains the base name “Б” <i>BUILDING_DESCRIPTION2</i> contains the building descriptor for the second building in an address that contains two buildings.
<i>BUILDING_NAME1</i> <i>BUILDING_NAME2</i>	Contains the full building name, which includes the building name and the building descriptor. ❖ Example “White House” or “Empire Tower” <i>BUILDING_NAME2</i> contains the second building name for addresses with two buildings.
<i>ASSIGNMENT_LEVEL</i>	An input address contains the building name and a last line with other secondary information. Data Quality Management SDK makes an address assignment using the secondary information. For example, it outputs an <i>S</i> in the <i>ASSIGNMENT_LEVEL</i> output field because it made an assignment based on secondary information. An input address contains a primary street number, building information, and a lastline. Data Quality Management SDK makes an address assignment based on the primary street number. For example, it outputs a <i>PR</i> in the <i>ASSIGNMENT_LEVEL</i> output field because it made an assignment based on the primary number.
<i>PRIMARY_NUMBER_FULL</i>	When an input address includes a primary name and number, Data Quality Management SDK outputs the primary number without a descriptor. However, when there isn't a primary name on input, Data Quality Management SDK outputs the primary number and descriptor to the <i>PRIMARY_NUMBER_FULL</i> output field. Data Quality Management SDK also outputs the primary number and descriptor to the <i>PRIMARY_NUMBER_FULL</i> field when the input address contains a building and primary number, or only a primary number.

Swap or combine building numbers

When an input has two numbers in a row in the address line, Data Quality Management SDK combines the numbers and attempts to assign a building number. When an input address has two numbers in a row separated by a slash, period, or other delimiter, the transform swaps the number order and then combines the number to make an address assignment.

Standardize building descriptor

Use standardization options in the Standardize Group to control how the software outputs address components, including building information. The following list contains some settings you can make to control the format of your output data:

- **CAPITALIZATION:**
 - Set the option to *MIXED* to output the descriptor to the *BUILDING_DESCRIPTION1* output field. The descriptor is in lower case with the first letter capitalized.
 - Set the option to *UPPER* to output the descriptor to the *BUILDING_DESCRIPTION1* output field in all upper case letters.
- **PRIMARY_TYPE_STYLE:**
 - Set the output style of the primary number descriptor to *LONG*, *SHORT*, or *PRESERVE*. The *PRIMARY_TYPE_STYLE* setting affects the building descriptor, for example, in the *BUILDING_NAME* output field. The *BUILDING_NAME* output field contains the full building name, which includes the building name and the building descriptor.
- **SECONDARY_DESCRIPTION_STYLE:**
 - Set the output style of the *BUILDING_DESCRIPTION1* or *BUILDING_DESCRIPTION2* to output in long form, abbreviated form, or preserved to output the descriptor as it was input.

Parent topic: [Global Address Cleanse transform \[page 374\]](#)

Related Information

[Cleanse global address data \[page 375\]](#)

[Process Japanese addresses \[page 379\]](#)

[Process Chinese addresses \[page 389\]](#)

13.3 Process Japanese addresses

The Global Address Cleanse transform and the Global Address engine parses and normalizes Japanese addresses for address cleansing and data matching.

i Note

The Japanese engine supports only kanji and katakana data. The engine doesn't support Latin data.

A significant portion of the address parsing capability relies on the Japanese address database. SAP Data Services has data from the Ministry of Public Management, Home Affairs, Posts, and Telecommunications (MPT), and additional data sources. The enhanced address database consists of a government database that is updated regularly. The database includes regional postal codes mapped to localities.

[Standard Japanese address format \[page 380\]](#)

A Japanese address includes many components that are similar to USA addresses, and some components that are unique to Japan.

[Japanese address components \[page 380\]](#)

Japanese addresses have many components, some of which are subcomponents of bigger components.

[Common Japanese building name suffixes \[page 384\]](#)

Japanese building names can have one of many suffixes.

[Special Hokkaido regional formats \[page 385\]](#)

The Hokkaido region has two special address formats.

[Other special Japanese address formats \[page 386\]](#)

Japanese addresses have other special address formats such as accepted spellings and numbering.

[Sample Japanese address \[page 387\]](#)

The Global Address Cleanse transform processes the various parts of a Japanese address by parsing the address elements into address line fields, lastline fields, firm fields, and nonparsed fields.

Parent topic: [Global Address Cleanse transform \[page 374\]](#)

Related Information

[Cleanse global address data \[page 375\]](#)

[Address assignment for buildings \[page 376\]](#)

[Process Chinese addresses \[page 389\]](#)

13.3.1 Standard Japanese address format

A Japanese address includes many components that are similar to USA addresses, and some components that are unique to Japan.

In addition to the components in the following table, a Japanese address can also include building name, floor number, and room number.

The following table contains Japanese address components, Japanese and English translation, and the corresponding output field to use in the transform.

Address component	Japanese	English	Output field
Postal code	〒 654-0153	654-0153	POSTCODE_FULL
Prefecture	兵庫県	Hyogo-ken	REGION1_FULL
City	神戸市	Kobe-shi	LOCALITY1_FULL
Ward	須磨区	Suma-ku	LOCALITY2_FULL
District	南落合	Minami Ochiai	LOCALITY3_FULL
Block number	1 丁目	1 chome	PRIMARY_NAME_FULL1
Sub-block number	25 番地	25 banchi	PRIMARY_NAME_FULL2
House number	2 号	2 go	PRIMARY_NUMBER_FULL

13.3.2 Japanese address components

Japanese addresses have many components, some of which are subcomponents of bigger components.

❖ Example

A city in Japan is divided into wards. A ward is divided into districts.

The following table describes the Japanese address components.

Japanese address component	Description
Postal code	Japanese postal codes represent an area and a location type. Japanese postal codes use the following format: nnn-nnnn. - The first three digits represent the area. - The last four digits represent a location in the area: - District - Subdistrict - Block - Subblock - Building - Floor - Company Postal authorities require postal codes written with Arabic numbers. The post office symbol 〒 is optional. Before 1998, the postal code consisted of 3 or 5 digits. Some older databases can still reflect the old system.
Prefecture	Prefectures are regions. Japan has 47 prefectures. You can omit the prefecture for some well-known cities.
City	Japan city names have the suffix 市 (-shi). In some parts of the Tokyo and Osaka regions, people omit the city name. In some island villages, they use the island name with a suffix 島 (-shima) in place of the city name. In some rural areas, they use the county name with suffix 郡 (-gun) in place of the city name.
Ward	A city is divided into wards. The ward name has the suffix 区 (-ku). The ward component is omitted for small cities, island villages, and rural areas that don't have wards.
District	A ward is divided into districts. When there's no ward, the small city, island village, or rural area is divided into districts. The district name can have the suffix 町 (-cho/-machi), but it's sometimes omitted. 町 has two possible pronunciations, but only one is correct for a particular district. In small villages, people use the village name with suffix 村 (-mura) in place of the district. When a village or district is on an island with the same name, the island name is often omitted.

Japanese address component	Description
Sub-district	Primarily in rural areas, a district can be divided into sub-districts, marked by the prefix 字 (aza-). A sub-district can be further divided into sub-districts that are marked by the prefix 小字 (koaza-), meaning small aza. koaza can be abbreviated to aza. A sub-district can also be marked by the prefix 大字 (oaza-), which means large aza. Oaza can also be abbreviated to aza. Here are the possible combinations: • oaza • aza • oaza and aza • aza and koaza • oaza and koaza i Note Addresses frequently omit the characters 大字 (oaza-), 字(aza-), and 小字 (koaza-).
Sub-district parcel	A sub-district aza can be divided into numbered sub-district parcels, which are marked by the suffix 部 (-bu), meaning <i>piece</i>. The character 部 is frequently omitted. Parcels can be numbered in several ways: • Arabic numbers (1, 2, 3, 4, and so on) 石川県七尾市松百町 8 部 3 番地 1 号 • Katakana letters in iroha order (イ, ロ, ハ, ニ, and so on) 石川県小松市里川町ナ部 23 番地 • Kanji numbers, which are rare (甲, 乙, 丙, 丁, and so on) 愛媛県北条市上難波甲部 311 番地
Sub-division	A rural district or sub-district (oaza/aza/koaza) is sometimes divided into sub-divisions, marked by the suffix 地割 (-chiwari) which means division of land. The optional prefix is 第 (dai-). The following address examples show sub-divisions: 岩手県久慈市旭町 10 地割 1 番地 岩手県久慈市旭町第 10 地割 1 番地
Block number	A district is divided into blocks. The block number includes the suffix 丁目 (-chome). Districts usually have between 1 and 5 blocks, but they can have more. The block number can be written with a Kanji number. Japanese addresses don't include a street name. 東京都渋谷区道玄坂 2 丁目 2 5 番地 1 2 号 東京都渋谷区道玄坂二丁目 2 5 番地 1 2 号

Japanese address component	Description
Sub-block number	A block is divided into sub-blocks. The sub-block name includes the suffix 番地 (-banchi), which means numbered land. The suffix 番地 (-banchi) can be abbreviated to just 番 (-ban).
House number	Each house has a unique house number. The house number includes the suffix 号 (-go), which means number.

Block, sub-block, and house number variations

Block, subblock, and house number data can vary. The following table contains possible variations.

Variant	Description
Dashes	The suffix markers 丁目(chome), 番地 (banchi), and 号(go) can be replaced with dashes. 東京都文京区湯島 2 丁目 18 番地 12 号 東京都文京区湯島 2-18-12 Sometimes block, sub-block, and house number are combined or omitted. 東京都文京区湯島 2 丁目 18 番 12 号 東京都文京区湯島 2 丁目 18 番地 12 東京都文京区湯島 2 丁目 18-12
No block number	Sometimes the block number is omitted. For example, the following address is a ward of Tokyo. It has numbered districts and no block numbers. 東京都千代田区二番町 9 番地 6 号 二番町 means district number 2.
Building names	Building names can vary in the following ways: - Names of apartments or buildings are often included after the house number. - When a building name includes the name of the district, the district name is often omitted. - When a building is well known, the block, subblock, and house number are often omitted. - When a building name is long, it can be abbreviated or written using its acronym with English letters.

Variant	Description
Building numbers	- Numbers such as room numbers and apartment numbers follow the building name. - Building numbers can include the suffix 号室 (-gosh-itsu). - Floor numbers above ground level can include the suffix 階 (-kai) or the letter F. - Floor numbers below ground level can include the suffix 地下<n>階 (chika <n> kai) or the letters B<n>F (where <n> represents the floor number). - An apartment complex can include multiple buildings called Building A, Building B, and so on, marked by the suffix 棟 (-tou). The following address examples include building numbers: - Third floor above ground 東京都千代田区二番町 9 番地 6 号 パウエブタ 3 F - Second floor below ground 東京都渋谷区道玄坂 2-25-12 シティバンク地下 2 階 - Building A Room 301 兵庫県神戸市須磨区南落合 1-25-10 須磨パークヒルズ A 棟 301 号室 - Building A Room 301 兵庫県神戸市須磨区南落合 1-25-10 須磨パークヒルズ A-301

13.3.3 Common Japanese building name suffixes

Japanese building names can have one of many suffixes.

For Japanese addresses, names of apartments or buildings are often included after the house number. When a building name includes the name of the district, the district name is often omitted. When a building is well known, the block, sub-block, and house number are often omitted. When a building name is long, it may be abbreviated or written using its acronym with English letters.

The following table contains a list of the common Japanese building suffixes.

Suffix	Romanized	Translation
ビルディング	birudingu	building
ビルチング	birudingu	building
ビル	biru	building
センター	senta-	center

Suffix	Romanized	Translation
プラザ	puraza	plaza
パーク	pa-ku	park
タワー	tawa-	tower
会館	kaikan	hall
棟	tou	building (unit)
庁舎	chousha	government office building
マンション	manshon	condominium
団地	danchi	apartment complex
アパート	apa-to	apartment
荘	sou	villa
住宅	juutaku	housing
社宅	shataku	company housing
官舎	kansha	official residence

13.3.4 Special Hokkaido regional formats

The Hokkaido region has two special address formats.

The following table describes the address formats for the Hokkaido region.

Hokkaido regional format

Format	Description
Super block	A special super-block format exists only in the Hokkaido prefecture. A super-block, marked by the suffix 条 (-joh), is one level larger than the block. The super-block number or the block number can contain a directional 北 (north), 南 (south), 東 (east), or 西 (west). The following address example shows a super-block 4 Joh. 北海道札幌市西区二十四軒 4 条 4 丁目 1 3 番地 7 号

Format	Description
Numbered subdistricts	A subdistrict name can be marked with the suffix 線 (-sen) meaning number instead of the suffix 字 (-aza). When a subdistrict has a 線 suffix, the block can have the suffix 号 (-go), and the house number has no suffix. The following address contains first the sub-district 4 sen and then a numbered block 5 go: 北海道旭川市西神楽 4 線 5 号 3 番地 1 1

13.3.5 Other special Japanese address formats

Japanese addresses have other special address formats such as accepted spellings and numbering.

The following table describes the special address formats.

Format	Description
Accepted spelling	Names of cities, districts, and so on, can have multiple accepted spellings because there are multiple accepted ways to write certain sounds in Japan.
Accepted numbering	When the block, subblock, house number, or district contains a number, the number can be written in Arabic or Kanji. For example, 二番町 means district number 2, and in the following example it is for Niban-cho. 東京都千代田区二番町九番地六号
P.O. Box addresses	P.O. Box addresses contain the postal code, Locality1, prefecture, the name of the post office, the box marker, and the box number. i Note The Global Address Cleanse transform recognizes P.O. box addresses that are located in the Large Organization Postal Code (LOPC) database only. The address can be in one of the following formats: - Prefecture, Locality1, post office name, box marker (私書箱), and P.O. box number. - Postal code, prefecture, Locality1, post office name, box marker (私書箱), and P.O. box number. The following address example shows a P.O. Box address: The Osaka Post Office Box marker #1 大阪府大阪市大阪支店私書箱 1 号

Format	Description
Large Organization Postal Code (LOPC) format	The Postal Service can assign a unique postal code to a large organization, such as the customer service department of a major corporation. An organization can have up to two unique postal codes depending on the volume of mail it receives. The address can be in one of the following formats: - Address, company name - Postal code, address, company name The following is an example of an address in a LOPC address format. 100-8798 東京都千代田区霞が関 1 丁目 3 - 2 日本郵政 株式会社

13.3.6 Sample Japanese address

The Global Address Cleanse transform processes the various parts of a Japanese address by parsing the address elements into address line fields, lastline fields, firm fields, and nonparsed fields.

❖ Example

Input address

0018521 北海道札幌市北区北十条西 1 丁目 12 番地 3 号創生ビル 1 階 101 号室札幌私書箱センター

Address-line fields

<i>PRIMARY_NAME1</i>	1
<i>PRIMARY_TYPE1</i>	丁目
<i>PRIMARY_NAME2</i>	12
<i>PRIMARY_TYPE2</i>	番地
<i>PRIMARY_NUMBER</i>	3
<i>PRIMARY_NUMBER_DESCRIPTION</i>	号
<i>BUILDING_NAME1</i>	創生ビル
<i>FLOOR_NUMBER</i>	1
<i>FLOOR_DESCRIPTION</i>	階
<i>UNIT_NUMBER</i>	101

Address-line fields

<i>UNIT_DESCRIPTION</i>	号室
<i>PRIMARY_ADDRESS</i>	1 丁目 12 番地 3 号
<i>SECONDARY_ADDRESS</i>	創生ビル 1 階 101 号室
<i>PRIMARY_SECONDARY_ADDRESS</i>	1 丁目 12 番地 3 号 創生ビル 1 階 101 号室

Last line fields

<i>COUNTRY</i>	日本
<i>ISO_COUNTRY_CODE_3DIGIT</i>	392
<i>ISO_COUNTRY_CODE_2CHAR</i>	JP
<i>POSTCODE1</i>	001
<i>POSTCODE2</i>	8521
<i>POSTCODE_FULL</i>	001-8521
<i>REGION1</i>	北海
<i>REGION1_DESCRIPTION</i>	道
<i>LOCALITY1_NAME</i>	札幌
<i>LOCALITY1_DESCRIPTION</i>	市
<i>LOCALITY2_NAME</i>	北
<i>LOCALITY2_DESCRIPTION</i>	区
<i>LOCALITY3_NAME</i>	北十条西
<i>LASTLINE</i>	001-8521 北海道 札幌市 北区 北十条西

Firm

<i>FIRM</i>	札幌私書箱センター
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Non-parsed fields

<i>STATUS_CODE</i>	S0000
<i>ASSIGNMENT_TYPE</i>	F

Non-parsed fields

[ADDRESS_TYPE](#)

S

13.4 Process Chinese addresses

Use the Global Address engine in the Global Address Cleanse transform to parse Chinese addresses.

The primary purpose of the Global Address Cleanse transform and engine is to parse and normalize Chinese addresses for data matching and cleansing applications.

[Chinese address format \[page 389\]](#)

Chinese addresses have many components that identify various parts of an address.

[Common metro address components \[page 392\]](#)

Metro addresses in China include the district name, which is common for metropolitan areas in major cities.

[Chinese rural address components \[page 392\]](#)

Rural addresses in China include the village name as well as the town name and the county-level city name.

[Sample Chinese address \[page 393\]](#)

The Global Address Cleanse transform processes the various parts of a Chinese address by parsing the address elements into address line fields, lastline fields, and nonparsed fields.

Parent topic: [Global Address Cleanse transform \[page 374\]](#)

Related Information

[Cleanse global address data \[page 375\]](#)

[Address assignment for buildings \[page 376\]](#)

[Process Japanese addresses \[page 379\]](#)

13.4.1 Chinese address format

Chinese addresses have many components that identify various parts of an address.

Chinese addresses start with the postal code, followed by the largest administrative region, such as province, and continues down to the smallest unit, such as room number and mail receiver.

Mail sent between prefectures often includes the largest administrative region in the address. Buildings along the street are numbered sequentially, sometimes with odd numbers on one side and even numbers on the other side. In some instances, both odd and even numbers are on the same side of the street.

The following table contains descriptions of Chinese address components.

Format	Description
Postal Code	A 6-digit number to identify the target delivery point of the address. Postal codes often have the prefix 邮编.
Country/Region	中华人民共和国 is People's Republic of China, which is the full name for China. China often uses the words “中国 (PRC)” as an abbreviation of the country/region name. For mail delivered within China, the domestic addresses often omit the country/region name of the target address.
Province	Similar to a state in the United States. China has 34 province-level divisions, including the following divisions: - Provinces (省 shěng) - Autonomous regions (自治区 zìzhìqū) - Municipalities (直辖市 zhíxiáshì) - Special administrative regions (特别行政区 tèbié xíngzhèngqū)
Prefecture	Similar to what a state is in the US. Prefecture-level divisions are the second level of the administrative structure. China has 34 prefectures. Each prefecture name is followed by a suffix, which is one of the following four words: - 地区 dìqū: Prefectures - 自治州 zìzhìzhōu: Autonomous prefectures - 地级市 dìjīshì: Prefecture-level cities - 盟 méng: Leagues Some municipalities include the dominant minority name.
County	A sub-division of prefecture that includes the following: - 县 xiàn: Counties - 自治县 zìzhìxiàn: Autonomous counties - 县级市 xiànjíshì: County-level cities - 市辖区 shìxiáqū: Districts - 旗 qí: Banners - 自治旗 zìzhìqí: Autonomous banners - 林区 línqū: Forestry areas - 特区 tèqū: Special districts

Format	Description
Township	Township level division that includes the following: • 乡 xiāng: Townships • 民族乡 mínzúxiāng: Ethnic townships • 镇 zhèn: Towns • 街道办事处 jiēdàobànshìchù: Subdistricts • 区公所 qūgōngsuǒ: District public offices • 苏木 sūmù: Sumu • 民族苏木 mínzúsūmù: Ethnic sumu
Village	Villages include the following components: • 社区居民委员会 jūmínwēiyuánhùi: Neighborhood committees • 社区: Neighborhoods or communities • 村民委员会 cūnmínwēiyuánhùi: Village committees • 村民小组 cūnmínxiǎozǔ: Village groups • 行政村 xíngzhèngcūn: Administrative villages
Street information	Specifies the delivery point where the mail receiver can be found. In China, the street information often has the form of [Town name] -> [Village name] -> Street (Road) name -> [Block number]-> House number. The components within the brackets [] are optional components in the address, and can often be omitted. The following list describes the street information components: • Town name: The town name is followed by the suffix 镇, 乡, or 堡. • Village name: The village name is followed by the suffix 村, 新村. • Street (Road) name: The street (road) name is followed by one of these suffixes 路, 大道, 街, 大街, 里. • Block number: The block number is followed by one of these suffixes 弄, 巷, 厅, 胡同. • House number: The house number is followed by the suffix 号. The house name is a unique number within the block or street/road.
Residential community	Can be used for mail delivery, especially for some famous residential communities in major cities. The street name and house number can be omitted. The residential community doesn't have a naming standard. It isn't required to follow a residential community with a marker. However, the residential community is often followed by the typical suffixes, such as 新村, 小区, and so on.
Building name	Building name is often followed by a building marker such as 大厦 or 大楼, even though it isn't required. For example, 中华大厦. Building name in the residential communities is often represented by a number with a suffix of 号 or 幢. For example: 上海市浦东新区晨晖路 100 弄 10 号 101 室.

Format	Description
Common metro address	Includes the district name, which is common for metropolitan areas in major cities.
Rural address	Includes the village name, which is common for rural addresses.

13.4.2 Common metro address components

Metro addresses in China include the district name, which is common for metropolitan areas in major cities.

The following table contains the Chinese metro address components, Chinese and English translation, and the corresponding output field to use in the transform.

Address component	Chinese	English	Output field
Postcode	510030	510030	<i>POSTCODE_FULL</i>
Country/Region	中国	China	<i>COUNTRY</i>
Province	广东省	Guangdong Province	<i>REGION1_FULL</i>
City name	广州市	Guangzhou City	<i>LOCALITY1_FULL</i>
District name	越秀区	Yuexiu District	<i>LOCALITY2_FULL</i>
Street name	西湖路	Xihu Road	<i>PRIMARY_NAME_FULL1</i>
House number	99 号	No. 99	<i>PRIMARY_NUMBER_FULL</i>

13.4.3 Chinese rural address components

Rural addresses in China include the village name as well as the town name and the county-level city name.

The following table lists the Chinese rural address components, Chinese and English translation, and the corresponding output field to use in the transform.

Address component	Chinese	English	Output field
Postcode	5111316	5111316	<i>POSTCODE_FULL</i>
Country/Region	中国	China	<i>COUNTRY</i>
Province	广东省	Guangdong Province	<i>REGION1_FULL</i>
City name	广州市	Guangzhou City	<i>LOCALITY1_FULL</i>

Address component	Chinese	English	Output field
County-level City name	增城市	Zengcheng City	LOCALITY2_FULL
Town name	荔城镇	Licheng Town	LOCALITY3_FULL
Village name	联益村	Lianyi Village	LOCALITY4_FULL
Street name	光大路	Guangda Road	PRIMARY_NAME_FULL1
House number	99 号	No. 99	PRIMARY_NUMBER_FULL

13.4.4 Sample Chinese address

The Global Address Cleanse transform processes the various parts of a Chinese address by parsing the address elements into address line fields, lastline fields, and nonparsed fields.

❖ Example

The following input address is processed by the Global Address Cleanse transform. The transform parses address elements to address-line fields, lastline fields, and nonparsed fields.

Input address

510830 广东省广州市花都区赤坭镇广源路1号星辰大厦8层809室

Address-line fields

PRIMARY_NAME1	广源
PRIMARY_TYPE1	路
PRIMARY_NUMBER	1
PRIMARY_NUMBER_DESCRIPTION	号
BUILDING_NAME1	星辰大厦
FLOOR_NUMBER	8
FLOOR_DESCRIPTION	层
UNIT_NUMBER	809
UNIT_DESCRIPTION	室
PRIMARY_ADDRESS	广源路1号
SECONDARY_ADDRESS	星辰大厦8层809室
PRIMARY_SECONDARY_ADDRESS	广源路1号星辰大厦8层809室

Lastline fields

COUNTRY	中国
-------------------------	----

Lastline fields

<i>POSTCODE_FULL</i>	510168
<i>REGION1</i>	广东
<i>REGION1_DESCRIPTION</i>	省
<i>LOCALITY1_NAME</i>	广州
<i>LOCALITY1_DESCRIPTION</i>	市
<i>LOCALITY2_NAME</i>	花都
<i>LOCALITY2_DESCRIPTION</i>	区
<i>LOCALITY3_NAME</i>	赤坭
<i>LOCALITY3_DESCRIPTION</i>	镇
<i>LASTLINE</i>	510830 广东省广州市花都区赤坭镇

Non-parsed fields

<i>STATUS_CODE</i>	S0000
<i>ASSIGNMENT_TYPE</i>	S
<i>ADDRESS_TYPE</i>	S

14 Global Address Cleanse transform reference

The Global Address Cleanse transform XML configuration is grouped according to setting types.

i Note

The Global Address Cleanse transform does not support CASS certification or produce data for a USPS Form 3553. If you want to certify your U.S. address data, use the USA Regulatory Address Cleanse transform, which supports CASS.

If you perform both address cleansing and data cleansing in a job, SAP recommends placing the Global Address Cleanse transform before the Data Cleanse transform in a data flow.

The sample configurations installed with the software contain examples of the Global Address Cleanse XML.

[System group \[page 396\]](#)

The System group controls high-level transform functions.

[Global Address Cleanse Report and Analysis \[page 397\]](#)

To generate report data for the Global Address Cleanse transform, use the option in the Report and Analysis.

[Global Address Cleanse reference files \[page 398\]](#)

Set the location for the Global Address Cleanse reference files, which includes Global, USA, or Canada reference files as applicable.

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

[Global Address Cleanse Country ID Options \[page 404\]](#)

Specify how the Global Address Cleanse transform uses country ID processes in the Country ID Options group.

[Global Address Cleanse Engines group \[page 409\]](#)

Choose one or more engines to use in the Global Address Cleanse transform and set other country-specific options.

[Global Address Cleanse Country \[page 411\]](#)

Set a country name before you set options in the Standardization group.

[Global Address Cleanse Standardization Options \[page 412\]](#)

To standardize addresses for the selected country, set options in the Standardization Options group.

[Canada engine \[page 442\]](#)

Use the Canada engine to process your Canada address data with the Global Address Cleanse transform.

[Global Address Cleanse Canada group \[page 445\]](#)

Set Canada-specific processing options in the Canada group.

[Global Address engine \[page 450\]](#)

To standardize data from countries other than Canada and the USA, set options for the Global Address engine.

[Global Address Cleanse USA engine \[page 469\]](#)

To cleanse addresses from the United States and its territories, set USA-specific options in the Global Address Cleanse transform.

[Global Address Cleanse Suggestion List option \[page 473\]](#)

Instead of using the Global Suggestion List transform, configure the Global Address Cleanse transform's Suggestion List processing option.

14.1 System group

The System group controls high-level transform functions.

Syntax

```
<SYSTEM>
<PLATFORM_SETTINGS>
<SYSTEM_VARS>
<LINK_DIR>.\</LINK_DIR>
</SYSTEM_VARS>
</PLATFORM_SETTINGS>
</SYSTEM>
```

Note

If an option exists in the XML, but it is not documented here, do not alter the contents of the option. Editing the contents could cause errors.

System group option description

Option	Description
LINK_DIR	Specifies the relative path to the Global Address Cleanse support files. For example, the default location for a Windows 32-bit platform is \\SAPBusinessObjects\DataQualityManagementSDK\windows_32\DataQuality\reference_data.

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Related Information

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)
[Global Address Cleanse Country ID Options \[page 404\]](#)
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[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.2 Global Address Cleanse Report and Analysis

To generate report data for the Global Address Cleanse transform, use the option in the Report and Analysis.

Syntax

```
<REPORT_AND_ANALYSIS>  
  <GENERATE_REPORT_DATA>NO</GENERATE_REPORT_DATA>
```

Option	Description
<code>GENERATE_REPORT_DATA</code>	Specifies whether to generate report data for this transform. YES: Generates report data for this transform. NO: Does not generate report data for this transform. NO is the default setting. → Tip If you do not need to generate reports, such as during testing, set this option to NO to improve performance.

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[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)
[Global Address Cleanse Country ID Options \[page 404\]](#)

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14.3 Global Address Cleanse reference files

Set the location for the Global Address Cleanse reference files, which includes Global, USA, or Canada reference files as applicable.

Syntax

```
<REFERENCE_FILES>  
  <DIRECTORY_PATH> . . \DataQuality\gac\</DIRECTORY_PATH>  
</REFERENCE_FILES>
```

Reference files are directories required by the Global Address Cleanse transform to process your data.

Option	Description
<i>DIRECTORY_PATH</i>	Specifies the path to the location of your Global Address directory files. The default reference file location is the path relative to where Data Quality Management SDK is installed.

For information about downloading directories, see the latest Directories Update letters at https://uacp2.hana.ondemand.com/viewer/p/ADDRESSING_DIRECTORIES.

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Related Information

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[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.4 Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

The Data Quality Statistics Settings group appears in the Data Cleanse, Global Address Cleanse, and the Geocoder transforms.

Syntax

Data Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSE_STATISTICS_TABLE>NO</
GENERATE_CLEANSE_STATISTICS_TABLE>
  <GENERATE_CLEANSE_INFO_CODES_TABLE>NO</
GENERATE_CLEANSE_INFO_CODES_TABLE>
  <GENERATE_CLEANSE_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSE_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSE_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSE_COMPONENT_INFO_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

Syntax

Global Address Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSE_STATISTICS_TABLE>NO</
GENERATE_CLEANSE_STATISTICS_TABLE>
  <GENERATE_CLEANSE_INFO_CODES_TABLE>NO</
GENERATE_CLEANSE_INFO_CODES_TABLE>
  <GENERATE_CLEANSE_ADDRESS_RECORD_INFO_TABLE>NO</
GENERATE_CLEANSE_ADDRESS_RECORD_INFO_TABLE>
  <GENERATE_CLEANSE_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSE_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSE_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSE_COMPONENT_INFO_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

Syntax

Geocoder:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_GEOCODE_STATISTICS_TABLE>NO</
GENERATE_GEOCODE_STATISTICS_TABLE>
```

```
<GENERATE_GEOCODE_INFO_CODES_TABLE>NO</  
GENERATE_GEOCODE_INFO_CODES_TABLE>  
</DATA_QUALITY_STATISTICS_SETTINGS>
```

The following table contains the options for the [DATA_QUALITY_STATISTICS_SETTINGS](#) group. Because the group is applicable for more than one transform, the table lists the applicable transform for each option.

The software generates the following types of statistics for each transform:

- The Data Cleanse and the Global Address Cleanse transforms generate additional statistics and info codes about address cleansing results on the data.
- The Geocoder transform generates statistics and info codes that are unique to the Geocoding processes.

Data Quality Statistics options

Option	Transform	Description
GENERATE_CLEANSER_ADDRESS_RECORD_INFO_TABLE	Global Address Cleanse	Specifies whether the software generates the Cleanse Address Record Information Table. • YES: Generates the Cleanse Address Record Information Table. • NO: Does not generate the Cleanse Address Record Information Table. NO is the default setting. The Cleanse Address Record Information Table contains non-summary information about the results of the address cleanse process on the data. The information includes assignment information and assignment type for each address. The software generates this information only when you include the Global Address Cleanse transform for address data.

Option	Transform	Description
GENERATE_CLEANSER_CHANGE_INFORMATION_TABLE	Data Cleanser	Specifies whether the software generates the Cleanser Change Information Table. YES: Generates the Cleanser Change Information Table. NO: Does not generate the Cleanser Change Information Table. NO is the default setting. The Cleanser Change Information Table contains non-summary information. It identifies areas of concern from your source data by the number of significant changes.
	Global Address Cleanser	
GENERATE_CLEANSER_COMPONENT_INFORMATION_TABLE	Data Cleanser	Specifies whether the software generates the Cleanser Component Information Table. YES: Generates the Cleanser Component Information Table. NO: Does not generate the Cleanser Component Information Table. NO is the default setting. Note This option is available in Data Quality Management SDK version 4.2.6 and later. The Cleanser Component Information Table contains non-summary information. It contains position information for each specified data element when one or both of the following occurs: - The transform parses the data element from an input field. - The transform writes the data element to an output field.
	Global Address Cleanser	

Option	Transform	Description
GENERATE_CLEANSER_INFO_CODES_TABLE	Data Cleanse	Specifies whether the software generates Cleanse Information Codes Table. YES: Generates the Cleanse Information Codes Table. NO: Does not generate the Cleanse Information Codes Table. NO is the default setting. The Cleanse Information Codes table contains non-summary information. The information includes the distribution of information codes or a count of missing, suspect, or blank data.
	Global Address Cleanse	
GENERATE_CLEANSER_STATISTICS_TABLE	Data Cleanse	Specifies whether the software generates the Cleanse Statistics Table. YES: Generates the Cleanse Statistics Table. NO: Does not generate the Cleanse Statistics Table. NO is the default setting. The Cleanse Statistics Table contains summarized information for each unique Entity ID. It provides a high-level insight into what the software changed during the cleansing process.
	Global Address Cleanse	

Option	Transform	Description
GENERATE_GEOCODE_INFO_CODES_TABLE	Geocoder	Specifies whether to generate the Geocode Information Codes Table. YES: Generates the Geocode Information Codes Table. NO: Does not generate the Geocode Information Codes Table. NO is the default setting. The Geocode Information Codes Table contains non-summary information about why records were not assigned the highest level of latitude and longitude. The statistics are based on the information code data. Note Not all existing information codes apply to the data quality statistics.
GENERATE_GEOCODE_STATISTICS_TABLE	Geocoder	Specifies whether the software generates the Geocode Statistics Table. YES: Generates the Geocode Statistics Table. NO: Does not generate the Geocode Statistics Table. NO is the default setting. The Geocode Statistics Table contains totals for the following: - The number of rows in the data - The number of rows that were assigned a latitude and longitude.

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Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Parent topic: [Geocoder transform reference \[page 570\]](#)

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[Data quality statistics common requirements \[page 236\]](#)
[Data Cleanse Cleansing Package \[page 519\]](#)
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[Data Cleanse Other Standardization options \[page 531\]](#)
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[Geocoder transform options \[page 578\]](#)

14.5 Global Address Cleanse Country ID Options

Specify how the Global Address Cleanse transform uses country ID processes in the Country ID Options group.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Syntax

```
<COUNTRY_ID_OPTIONS>
  <SCRIPT_CODE>NONE</SCRIPT_CODE>
  <COUNTRY_ID_MODE>ASSIGN</COUNTRY_ID_MODE>
  <COUNTRY_NAME>NONE</COUNTRY_NAME>
  <USE_EXTENDED_LOCALITY_LIST>FALSE</USE_EXTENDED_LOCALITY_LIST>
  <ENABLE_SUGGESTION_LISTS>YES</ENABLE_SUGGESTION_LISTS>
```

```
<IDENTIFY_LOCALITIES>ALL</IDENTIFY_LOCALITIES>
</COUNTRY_ID_OPTIONS>
```

When the transform can't determine a best country from the parsed and ranked countries, it uses the settings in the Country ID group to determine the country.

This option group is required.

Country ID group option descriptions

Option	Description
<i>COUNTRY_ID_MODE</i>	Specifies whether the transform uses the specified country name from the <i>COUNTRY_NAME</i> option or runs Country ID processing using the Global Address engine. • <i>CONSTANT</i>: Assumes all of your input data is for the specified country in <i>COUNTRY_NAME</i> and doesn't run Country ID processing. Choose this option only when all of your input data is from one country. Selecting this option can save processing time. • <i>ASSIGNED</i>: Runs Country ID processing. Choose this option when the input data contains addresses from more than one country.
<i>COUNTRY_NAME</i>	Specifies the country of destination. Enter the applicable country or set to <i>NONE</i> Select <i>NONE</i> when you set the <i>COUNTRY_ID_MODE</i> to <i>ASSIGNED</i> and you don't want the transform to use a default country for records with unidentified countries. Instead, the transform uses the Global Address engine. Special considerations: • When you set the <i>COUNTRY_ID_MODE</i> to <i>CONSTANT</i>, ensure that you specify a country name. • If you set the <i>COUNTRY_ID_MODE</i> to <i>ASSIGNED</i>, and you specify a country for <i>COUNTRY_NAME</i>, the transform uses the specified country name instead of the Global Address engine when it can't identify a country.
<i>SCRIPT_CODE</i>	Specifies the ISO four-character script code for your data. • <i>CJKK</i>: Chinese, Japanese, and Korean • <i>CYRL</i>: Cyrillic • <i>GREK</i>: Greek • <i>LATN</i>: Latin • <i>NONE</i>: Attempts to identify the overall script of the input data as <i>CJKK</i>, <i>GREK</i>, <i>LATN</i>, or other.

Option	Description
<i>USE_EXTENDED_LOCALITY_LIST</i>	Specifies whether to use the extended locality list to determine the country name. <i>TRUE</i>: Searches the extended locality list. <i>FALSE</i>: Searches the major locality list to determine the country name. <i>FALSE</i> is the default setting. The transform behaves in the following ways when you set this option to <i>TRUE</i> and you make a setting in the <i>IDENTIFY_LOCALITIES</i> option: - Overrides a setting of <i>MAJOR</i> in the <i>IDENTIFY_LOCALITIES</i> option and uses the extended locality list. - Uses the setting in the <i>IDENTIFY_LOCALITIES</i> option when you set it to <i>EXTENDED</i> or <i>ALL</i>. The transform requires that you set the <i>COUNTRY_ID_MODE</i> option to <i>ASSIGN</i> to use the <i>USE_EXTENDED_LOCALITY_LIST</i> option.

Option	Description
IDENTIFY_LOCALITIES	Specifies the locality resource the transform uses to identify candidate countries for addresses with no country information. • MAJOR: Uses the major localities list in addition to the country dictionary, region dictionary, and parsing dictionary. MAJOR is the default setting. • EXTENDED: Uses the extended list of localities in addition to the country dictionary, region dictionary, and parsing dictionary. • ALL: Uses the Universal Postal Union (UPU) all-world directory in addition to the country dictionary, region dictionary, and parsing dictionary. i Note The major and extended localities lists are included with the installation at no extra cost. The UPU all-world directory isn't included with the installation and is available for purchase. If you select ALL, but you don't own the UPU all-world directory, or the directory isn't located in the directory path you specified in the REFERENCE_FILES group, the transform uses the EXTENDED option and issues a warning. The transform overrides a setting of TRUE for the USE_EXTENDED_LOCALITY_LIST option with one exception: When you set the IDENTIFY_LOCALITIES to MAJOR, the transform uses the extended list. The transform requires that you set the COUNTRY_ID_MODE option to ASSIGN to use the IDENTIFY_LOCALITIES option.

Option	Description
ENABLE_SUGGESTION_LISTS	Specifies whether the transform outputs a ranked list of candidate countries in the form of a suggestion list. • YES: Generates a ranked list of candidate countries in a suggestion list. After you select a country from the list, the transform continues to process the record using the selected country. • NO: Doesn't generate a ranked list of candidate countries in a suggestion list, but uses the setting in COUNTRY_NAME as follows: • When you select a country for the COUNTRY_NAME option, the transform assigns the record using that country. • When you set COUNTRY_NAME to NONE, the transform outputs the record as unassigned and generates the applicable information code that indicates the country is unknown. NO is the default setting. i Note Software performance can be affected when you enable this option and your data contains a high number of records that don't contain country information. The transform generates suggestion list information for the ENABLE_SUGGESTION_LISTS option in the following suggestion list components: • ISO_COUNTRY_CODE_2_CHAR: Contains the two-character ISO country code. • SUGG_CONFIDENCE_SCORE: Contains the confidence score for the two-character ISO country code. The transform requires you to set the COUNTRY_ID_MODE option to ASSIGN to use this option.

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Related Information

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[Global Address Cleanse Suggestion List option \[page 473\]](#)
[Global Address Cleanse information codes \[page 861\]](#)
[Global Address Cleanse Suggestion List Components \[page 475\]](#)

14.6 Global Address Cleanse Engines group

Choose one or more engines to use in the Global Address Cleanse transform and set other country-specific options.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Syntax

```

<ENGINES>
  <CANADA>NO</CANADA>
  <GLOBAL_ADDRESS>YES</GLOBAL_ADDRESS>
  <USA>NO</USA>
  <DYNAMIC_ENGINE_INIT>YES</DYNAMIC_ENGINE_INIT>
  <SUGG_SUPPRESS_OUTPUT>YES</SUGG_SUPPRESS_OUTPUT>
  <ASSIGN_COUNTRY_ID_RESULTS>SINGLE</ASSIGN_COUNTRY_ID_RESULTS>
</ENGINES>

```

Before you set options in this group, consider the countries represented in your data. For example, if your data doesn't contain addresses from Canada or USA, select the Global Address engine. The transform requires at least one engine.

This option group is required.

Engine group option descriptions

Option	Description
CANADA	Specifies whether the transform uses the Canada engine. YES: Uses the Canada engine for Global Address Cleanse processes. NO: Doesn't use the Canada engine. NO is the default setting.

Option	Description
DYNAMIC_ENGINE_INIT	Specifies whether engines that the transform uses are limited to the engines that it can initialize. • YES: Uses only the engines that it can successfully initialize. If an engine fails to initialize, the software issues a warning and the job continues. • NO: Uses all enabled engines. If an engine fails to initialize, the job fails. NO is the default setting.
ASSIGN_COUNTRY_ID_RESULTS	Specifies how the transform determines the country to assign when there are multiple suggested candidate countries. • SINGLE: Assigns the address using the country with the highest confidence score from the list of candidate countries. When multiple countries have the same high confidence score, the transform takes one of the following actions: • Generates a country suggestion list when country suggestion list is enabled. • Outputs the record as unassigned and generates an info code that indicates the country is unknown. SINGLE is the default setting. • MULTIPLE: Assigns an address for each candidate country using the applicable directory for each country. If the address assigns for only one of the candidate countries with a high confidence score, the transform assigns the address using that country. If the address assigns for multiple countries with equal high confidence scores, the transform takes one of the following actions: • Generates a country suggestion list when country suggestion list is enabled. • Outputs the record as unassigned and generates an info code that indicates the country is unknown. Requirements: • Candidate countries must have PAF-level address data to make complete assignments to the delivery point. • For countries that don't have PAF-level address data, Data Services uses the all world directory to make locality level assignments only. • You must place the applicable directories in the directory path set in the REFERENCE_FILES group. The transform requires that you set the COUNTRY_ID_MODE option in the COUNTRY_ID_OPTIONS group to ASSIGN to use this option.

Option	Description
GLOBAL_ADDRESS	Specifies whether the transform uses the Global Address engine. YES: Uses the Global Address engine for Global Address transform processes. NO: Doesn't use the Global Address engine.
USA	Specifies whether the transform uses the USA engine. YES: Uses the USA engine for Global Address transform processes. NO: Doesn't use the USA engine.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Country \[page 411\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.7 Global Address Cleanse Country

Set a country name before you set options in the Standardization group.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

<STANDARDIZATION_OPTIONS>

```
<COUNTRY>
  <COUNTRY_NAME>GLOBAL (APPLY TO ALL COUNTRIES)</COUNTRY_NAME>
```

Option	Description
<i>COUNTRY_NAME</i>	Specifies the country to which the standardization options apply. The default is <i>GLOBAL</i> .

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

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[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

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[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.8 Global Address Cleanse Standardization Options

To standardize addresses for the selected country, set options in the Standardization Options group.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

! Restriction

BEST_PRACTICE values are not supported as dynamic options in DQM SDK and Data Services.

≡ Syntax

```
<STANDARDIZATION_OPTIONS>
```

```

<OPTIONS>
  <CORRECT_ASSIGNED_DATA>YES</CORRECT_ASSIGNED_DATA>
  <FORMAT_ASSIGNED_DATA>YES</FORMAT_ASSIGNED_DATA>
  <CONVERT_LATIN_OUTPUT_TO_US_ASCII>NO</
CONVERT_LATIN_OUTPUT_TO_US_ASCII>
  <CORRECT_UNASSIGNED_DATA>YES</CORRECT_UNASSIGNED_DATA>
  <FORMAT_UNASSIGNED_DATA>YES</FORMAT_UNASSIGNED_DATA>
  <CAPITALIZATION>MIXED</CAPITALIZATION>
  <DIRECTIONAL_STYLE>SHORT</DIRECTIONAL_STYLE>
  <DIRECTIONAL_PUNCTUATION>NO</DIRECTIONAL_PUNCTUATION>
  <PRIMARY_TYPE_STYLE>SHORT</PRIMARY_TYPE_STYLE>
  <PRIMARY_TYPE_PUNCTUATION>NO</PRIMARY_TYPE_PUNCTUATION>
  <SECONDARY_DESCRIPTION_STYLE>SHORT</SECONDARY_DESCRIPTION_STYLE>
  <SECONDARY_DESCRIPTION_PUNCTUATION>NO</
SECONDARY_DESCRIPTION_PUNCTUATION>
  <SECONDARY_NUMBER_STYLE>PRESERVE</SECONDARY_NUMBER_STYLE>
  <POSTAL_PHRASE_STYLE>SHORT</POSTAL_PHRASE_STYLE>
  <POSTAL_PHRASE_PUNCTUATION>NO</POSTAL_PHRASE_PUNCTUATION>
  <REMOVE_ADDRESS_APOSTROPHES>NO</REMOVE_ADDRESS_APOSTROPHES>
  <ADDRESS_LINE_ALIAS>PRESERVE</ADDRESS_LINE_ALIAS>
  <ASSIGN_LOCALITY>PRESERVE</ASSIGN_LOCALITY>
  <LOCALITY_NAME_STYLE>PRESERVE</LOCALITY_NAME_STYLE>
  <INCLUDE_LOCALITY_ADDITION>PRESERVE</INCLUDE_LOCALITY_ADDITION>
  <REGION_STYLE>SHORT</REGION_STYLE>
  <EUROPEAN_POSTCODE_PREFIX>PREFERRED</EUROPEAN_POSTCODE_PREFIX>
  <INCLUDE_COUNTRY>YES</INCLUDE_COUNTRY>
  <COUNTRY_STYLE>NAME</COUNTRY_STYLE>
  <USE_POSTAL_COUNTRY_NAME>NO</USE_POSTAL_COUNTRY_NAME>
  <OUTPUT_COUNTRY_LANGUAGE>ENGLISH - LATIN</OUTPUT_COUNTRY_LANGUAGE>
  <INCLUDE_UNUSED_ADDRESS_LINE_DATA>NO</
INCLUDE_UNUSED_ADDRESS_LINE_DATA>
  <INCLUDE_UNUSED_LASTLINE_DATA>NO</INCLUDE_UNUSED_LASTLINE_DATA>
  <MOVE_MULTILINE_DATA>TOP</MOVE_MULTILINE_DATA>
  <EXTRA_LINES>REMOVE</EXTRA_LINES>
  <CHARACTER_WIDTH_STYLE>NORMAL_WIDTH</CHARACTER_WIDTH_STYLE>
  <STREET_NAME_STYLE>PRESERVE</STREET_NAME_STYLE>
  <USE_LOCAL_PRIMARY_TYPE_STYLE>NO</USE_LOCAL_PRIMARY_TYPE_STYLE>
</OPTIONS>
</COUNTRY>
</STANDARDIZATION_OPTIONS>

```

Set these options once to apply to all source records by selecting [GLOBAL](#) for [COUNTRY_NAME](#). Or, duplicate the Country group and specify a specific country for each group in the [COUNTRY_NAME](#) option.

The Standardization Options group is required and can be duplicated.

Standardization Options group

Option	Description
ADDRESS_LINE_ALIAS	Specifies how to standardize the address line when the address contains alias address elements. Engines supported: Canada, Global Address, and USA. • CONVERT: Converts address lines based on official address line components instead of delivery address line components. • PRESERVE: Retains non-preferred data in address lines unless the data is incorrect. i Note Since December 31, 2013, Korea does not officially support the old Korean address system. The Korea reference data contains the old addresses as aliases to the new addressing system. SAP recommends that you convert South Korean land-lot addresses to their new road name addresses. Set the ADDRESS_LINE_ALIAS and ASSIGN_LOCALITY options to CONVERT.
ASSIGN_LOCALITY	Specifies how to standardize the locality name. • CONVERT: Converts the locality name to the locality name preferred by the postal authority for the country. • PRESERVE: Preserves the input locality name unless it is incorrect. • VALID: Retains the input locality name unless it is not valid for mailing. If it is not valid for mailing, replaces it with the preferred locality name. i Note The TRANSLATE_MAJOR_LOCALITY option setting takes precedence over the setting in this option. i Note Since December 31, 2013, Korea does not officially support the old Korean address system. The Korea reference data contains the old addresses as aliases to the new addressing system. SAP recommends that you convert South Korean land-lot addresses to their new road name addresses. Set the ADDRESS_LINE_ALIAS and ASSIGN_LOCALITY options to CONVERT.

Option	Description
<i>CAPITALIZATION</i>	Specifies the casing of your address data. • <i>MIXED</i>: Converts data to initial capitals. For example, "MAIN STREET SOUTH" becomes "Main Street South." • <i>UPPER</i>: Converts data to full capitals. For example, "Main Street South" becomes "MAIN STREET SOUTH." • <i>BEST_PRACTICE</i>: Use initial or full capitals based on what the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic. i Note If you want consistent casing for your data, make sure that this option and the <i>CAPITALIZATION</i> setting in the Data Cleanse transform are the same.
<i>CHARACTER_WIDTH_STYLE</i>	Specifies whether to standardize half-width and full-width characters. This option applies only to Chinese, Japanese, and Korean data. • <i>NORMAL_WIDTH</i>: Converts full-width Latin characters to half width and converts half-width Chinese, Japanese, and Korean characters to full width. • <i>HALF_WIDTH</i>: Converts all characters to half width. • <i>FULL_WIDTH</i>: Converts all characters to full width.
<i>CORRECT_ASSIGNED_DATA</i>	Specifies whether to use the parsed or corrected data for the assigned output fields of type Best. • <i>YES</i>: Populates the Best components with corrected data. • <i>NO</i>: Populates the Best components with parsed data. i Note If you choose <i>NO</i> for this option, the <i>CAPITALIZATION</i> option is the only available Standardization option for your assigned data.
<i>CORRECT_UNASSIGNED_DATA</i>	Specifies whether the Global Address Cleanse transform standardizes your unassigned data. • <i>YES</i>: Populates the Best components with corrected data. • <i>NO</i>: Populates the Best components with parsed data. i Note If you choose <i>NO</i> for this option, the <i>CAPITALIZATION</i> option is the only available Standardization option for your unassigned data.

Option	Description
<i>COUNTRY_STYLE</i>	Specifies how the software standardizes country data. • <i>ISO_2CHAR</i>: Standardizes country data to the two-character ISO code, such as AU, CA, or US. • <i>ISO_3CHAR</i>: Standardizes country data to the three-character ISO code, such as AUS, CAN, or USA • <i>ISO_3DIGIT</i>: Standardizes country data to the three-digit ISO code, such as 038, 124, or 840. • <i>NAME</i>: Standardizes country data to the full country name, such as Australia, Canada, or United States. • <i>PRESERVE</i>: Attempts to retain the country data from the input record, otherwise uses the corrected country value on output.
<i>DIRECTIONAL_PUNCTUATION</i>	Specifies whether to use punctuation in the abbreviated directional data. • <i>YES</i>: Outputs directionals with punctuation. For example, N. or S.W.. • <i>NO</i>: Outputs directionals without punctuation. For example, N, SW. • <i>PRESERVE</i>: If parsed input component contains punctuation, retains it on output with corrections applied when the transform does not find the input in the dictionary. For example, "N/W" on input is updated to "N.W". • <i>BEST_PRACTICE</i>: Punctuates the abbreviated directional data based on what the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.
<i>DIRECTIONAL_STYLE</i>	Specifies whether to abbreviate directional data. • <i>LONG</i>: Uses fully spelled directionals such as "North," "South," "East," "West." • <i>PRESERVE</i>: Preserves the style used in the input record. • <i>SHORT</i>: Uses abbreviated directionals such as "N," "S," "E," "W." • <i>BEST_PRACTICE</i>: Abbreviates directional data based on what the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.

Option	Description
<i>EUROPEAN_POSTCODE_PREFIX</i>	For mail generated and distributed inside Europe, adds the one- to three-character European Postcode prefix, followed by a dash. YES: Adds the European Postcode prefix. NO: Does not add the European Postcode prefix. PRESERVE: If the transform finds a European Postcode prefix on input, retains the European Postcode prefix. Example In the following address, the D- is the European Postcode extension: Hallesches Ufer 32–38 D-10963 Berlin Germany Note The <i>EUROPEAN_POSTCODE_PREFIX</i> is for mail distributed from one European country to another European country.
<i>EXTRA_LINES</i>	Specifies what to do with extra lines of non address data. PRESERVE: Attempts to retain the extra line of non-address data in the general location in which it was input. REMOVE: Does not include any extra line of non-address data in the standardized lines or multiline fields. PREFERRED: Places all populated Extra fields above or below the multiline fields and standardized input lines based on the country data being processed. For example, Extra fields for Japan are located below the standardized lines.
<i>FORMAT_ASSIGNED_DATA</i>	Specifies whether to format your assigned data based on the preferred address format of the country. For example, the format for Germany is: <pre>{Primary_Name1} {Primary_Number} {Postcode1} {Locality} {Country}</pre> YES: Formats the assigned data. NO: Does not format the assigned data and leaves it in the location in which it was input. If data is added to the record, the transform bases the placement on the format string.

Option	Description
<i>FORMAT_UNASSIGNED_DATA</i>	Specifies whether to format unassigned data based on the preferred address format for the country. For example, the format for Germany is: <pre>{Primary_Name1} {Primary_Number} {Postcode1} {Locality} {Country}</pre> YES: Formats the unassigned data. NO: Does not format the unassigned data and leaves it in the line in which it was input.
<i>CONVERT_LATIN_OUTPUT_TO_US_ASCII</i>	If a character conversion is available for Latin script records, converts any extended ASCII characters in the Best component to the US ASCII character set. YES: Converts extended ASCII characters. NO: Does not convert extended ASCII characters. NO is the default setting. BEST_PRACTICE: Convert or not based on what the software sees as the most common for a country. For more information, see the “BEST_PRACTICE” topic. The transform leaves any extended ASCII characters for which there is no conversion as is. For example, the transform leaves the degree symbol or inverted exclamation and question marks as is. ❖ Example With the input street name “Østerbrogade”, preserve the local character or convert it to the international data format “Osterbrogade” in the cleansed output.
<i>INCLUDE_COUNTRY</i>	Specifies whether to include country names in standardized lines or multiline fields. YES: Includes country name. NO: Does not include country name. PRESERVE: If the transform finds the country name on input, retains the country name.

Option	Description
<code>INCLUDE_LOCALITY_ADDITION</code> (Germany only)	Specifies whether the <code>LOCALITY1_FULL</code> output field contains both the <code>LOCALITY1_NAME</code> and the <code>LOCALITY1_ADDITION</code> information in the standardized output. YES: Includes both the locality name and the locality addition information only when it is valid. NO: Does not include the locality addition. PRESERVE: If the transform finds a locality addition on input, includes the locality name and the locality addition in the standardized output only when it is valid. When there is no locality addition on input but the software finds a locality addition in the postal directory, the software includes the locality name and the locality addition in the standardized output only when it is valid. PRESERVE is the default setting. i Note When the software recognizes the locality addition, for example, the software finds it in the postal directory, but it is flagged as invalid by Deutsche Post, the software does not include the <code>LOCALITY1_ADDITION</code> in the standardized output regardless of a YES or PRESERVE setting. i Note If you set the <code>LOCALITY_NAME_STYLE</code> option to SHORT, the transform does not output the <code>LOCALITY1_ADDITION</code> to the standardized output, even if it is valid, regardless of a YES or PRESERVE setting. i Note If you set the <code>TRANSLATE_MAJOR_LOCALITY</code> option in the Global Address Country Options group to English, the transform does not output the <code>LOCALITY1_ADDITION</code> field to the standardized output, even if it is valid, regardless of a YES or PRESERVE setting.

Option	Description
<i>INCLUDE_UNUSED_ADDRESS_LINE_DATA</i>	Specifies whether to output the unused address line data for standardized lines and multiline fields. YES: Outputs the unused address line data in the remainder fields <i>ADDRESS_LINE_REMAINDER1</i> to 4. For example, when input is 100 Main St Red House, outputs "Red House" to the address line remainder output field. NO: Does not output the unused address line data. This option affects unused address data classified as remainder. This option does not affect unused address data classified as extra.
<i>INCLUDE_UNUSED_LASTLINE_DATA</i>	Specifies whether to output the unused last line data for standardized lines and multiline fields: YES: Outputs the unused last line data. NO: Does not output the unused last line data.
<i>LOCALITY_NAME_STYLE</i>	Specifies the format for locality data in the <i>LOCALITY1_NAME</i> output field for addresses. This option applies to German addresses. PRESERVE: Preserves the locality data format as it was input. PRESERVE is the default setting. SHORT: If the transform finds the locality data in the reference data, it outputs the locality data in the abbreviated version. i Note To use the short locality name style, set the <i>ADDRESS_LINE_ALIAS</i> option to CONVERT. i Note This option overrides the setting in the <i>INCLUDE_LOCALITY_ADDITION</i> option. i Note If the <i>TRANSLATE_MAJOR_LOCALITY</i> option is set to translate the <i>LOCALITY1</i> output field, the transform does not output short locality regardless of the setting in the <i>LOCALITY_NAME_STYLE</i> option.

Option

MOVE_MULTILINE_DATA

Description

Determines the position of blank lines in output addresses.

- BOTTOM**: Moves blank lines to the top and shifts the data to the bottom of the address block.
- NO**: Does not rearrange any lines, blank or otherwise.
- TOP**: If there are any blank lines, the transform moves them to the bottom of the address block and shifts the data to the top of the block.

Example

Top option: All blank lines move to the **bottom.**

Line:	Input data:	Result of moving:
Line 1	100 Market Street	Sycamore Building
Line 2	Suite 202	Suite 202
Line 3	Sycamore Building	100 Market St
Line 4		Boston MA 02109
Line 5	Boston MA 02109	

OUTPUT_COUNTRY_LANGUAGE

Specify which language and script to use on output for the country name, not the entire record.

PRESERVE: Preserves the input country name on output.

If you don't select **PRESERVE**, select one of the following option language in which to output the country name:

- CATALAN - LATIN
- CHINESE - HANI
- DANISH - LATIN
- DUTCH- LATIN
- ENGLISH- LATIN
- FINNISH - LATIN
- FRENCH - LATIN
- GREEK - GREEK
- GERMAN - LATIN
- HUNGARIAN - LATIN
- ITALIAN - LATIN
- JAPANESE - HANI
- JAPANESE - KANA
- KOREAN - HANG
- NORWEGIAN - LATIN
- POLISH - LATIN
- PORTUGUESE - LATIN
- RUSSIAN - CYRILLIC
- SPANISH - LATIN
- SWEDISH - LATIN

Option	Description
<i>POSTAL_PHRASE_PUNCTUATION</i>	If you choose <i>SHORT</i> for the <i>POSTAL_PHRASE_STYLE</i> option, this option specifies whether to use punctuation in the postal abbreviation. • <i>YES</i>: Includes punctuation in postal abbreviations. For example, "P.O. Box". • <i>NO</i>: Does not insert any punctuation for postal abbreviations. For example, "PO Box". • <i>PRESERVE</i>: If there is punctuation in the input parsed component, returns the corrected punctuation when the input is not found in the dictionary. • <i>BEST_PRACTICE</i>: Punctuates based on what the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.
<i>POSTAL_PHRASE_STYLE</i>	Specifies whether to abbreviate postal phrases. • <i>LONG</i>: Outputs the fully spelled out postal phrase. For example, Post Office Box. • <i>PRESERVE</i>: If the transform finds a postal phrase on input, it retains the style of the postal phrase on output. • <i>SHORT</i>: Outputs the abbreviated postal phrase. For example, PO Box. The transform uses the setting in the <i>POSTAL_PHRASE_PUNCTUATION</i> option to determine the punctuation for this option. • <i>BEST_PRACTICE</i>: Uses the style the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.
<i>PRIMARY_TYPE_PUNCTUATION</i>	If you choose <i>SHORT</i> for the <i>PRIMARY_TYPE_STYLE</i> option, this option specifies whether to use punctuation in the primary type abbreviations. • <i>YES</i>: Includes punctuation at the end of primary type abbreviations. For example, "St". • <i>NO</i>: Does not include punctuation at the end of primary type abbreviations, regardless of whether the input contains punctuation. For example, "St". • <i>PRESERVE</i>: If there is punctuation in the input parsed component, returns the corrected punctuation when the transform does not find the input in the dictionary. • <i>BEST_PRACTICE</i>: Punctuates based on what the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.

Option	Description
<i>PRIMARY_TYPE_STYLE</i>	Specifies the style for primary type address elements. • <i>LONG</i>: Uses fully spelled primary types such as Street, Avenue, Road, or Strasse. • <i>PRESERVE</i>: Retains the style used in the input record. • <i>SHORT</i>: Uses abbreviated primary type such as St, Ave, Rd, or Str. The transform determines the punctuation for this option by the setting in the <i>PRIMARY_TYPE_PUNCTUATION</i> option. • <i>BEST_PRACTICE</i>: Uses the style the software sees as the most common for a country. For more information, see the “BEST_PRACTICE” topic.
<i>REGION_STYLE</i>	Specifies whether to abbreviate the region name such as state or province. • <i>LONG</i>: Uses the fully spelled region name. For example, California or Ontario. • <i>PRESERVE</i>: Retains the style used in the input record. • <i>SHORT</i>: Abbreviates the region name. For example, CA or ON. • <i>BEST_PRACTICE</i>: Abbreviate the region name based on what the software sees as the most common for a country. For more information, see the “BEST_PRACTICE” topic.
<i>REMOVE_ADDRESS_APOSTROPHES</i>	Specifies whether to include apostrophes in certain street names such as DE L' or D'. • <i>YES</i>: Retains apostrophes in street names if it was present on input, for example, Rue D'Abbeville. • <i>NO</i>: Removes apostrophes in street names, for example, Rue D Abbeville.
<i>SECONDARY_DESCRIPTION_PUNCTUATION</i>	If you choose <i>SHORT</i> for the <i>SECONDARY_DESCRIPTION_STYLE</i> option, the <i>SECONDARY_DESCRIPTION_PUNCTUATION</i> option specifies whether to use punctuation in the abbreviation. • <i>YES</i>: Uses punctuation in the abbreviation, for example, “Apt.”. • <i>NO</i>: Does not use punctuation, for example, “Apt”. • <i>PRESERVE</i>: If there is punctuation in the input parsed component, returns the corrected punctuation when the transform does not find the input in the dictionary. • <i>BEST_PRACTICE</i>: Punctuates abbreviations based on what the software sees as the most common for a country. For more information, see the “BEST_PRACTICE” topic.

Option	Description
<i>SECONDARY_DESCRIPTION_STYLE</i>	Specifies whether to abbreviate the secondary description, such as a unit or an apartment. • <i>LONG</i>: Uses the fully spelled secondary description. For example, "Apartment". • <i>PRESERVE</i>: Retains the style used in the input record. • <i>SHORT</i>: Abbreviates the secondary description. For example, "Apt". The transform uses the setting in the <i>SECONDARY_DESCRIPTION_PUNCTUATION</i> option to determine whether to use punctuation for this option. • <i>BEST_PRACTICE</i>: Uses the style the software sees as the most common for a country. For more information, see the "BEST_PRACTICE" topic.
<i>SECONDARY_NUMBER_STYLE</i>	Applicable for Canada and New Zealand addresses. Specifies the format of the secondary number such as a suite or apartment number. • <i>ATTACHED</i>: Converts all secondary ranges to the attached format in which the secondary number is prepended to the primary number and separated with a delimiter. For example, for Canada addresses, places a dash between the secondary and primary range: 5–100 Main St. • <i>PRESERVE</i>: Preserves the style of the address as it was input. • <i>UNATTACHED</i>: Converts all secondary ranges to the unattached format. For example, for Canada, it places the unit designator at the end of the primary address: 100 Main St Suite 5.

Option	Description
<i>STREET_NAME_STYLE</i>	For the Netherlands only. Specifies the format for street data for addresses in the Netherlands. <i>PRESERVE</i>: Preserves the street data format as it was input. ❖ Example For example: Annelien Kappeyne Van de Coppellostr 2 Herten Limburg 6049 HD <i>SHORT</i>: Outputs street data in the format preferred by the postal authority. For the Netherlands, the Short option returns a street address with a maximum of 24 characters in mixed case. ❖ Example For example: A K vd Coppellostr 2 Herten Limburg 6049 HD i Note To use the short street name style, set the <i>ADDRESS_LINE_ALIAS</i> option to <i>CONVERT</i>. i Note The <i>CAPITALIZATION</i> option in the Global Address Cleanse transform overrides the setting in the <i>STREET_NAME_STYLE</i> option.
<i>USE_LOCAL_PRIMARY_TYPE_STYLE</i>	Applies to Austria, Germany, and Switzerland. Specifies whether to use the type style for primary address components that is present in the address data. <i>YES</i>: Uses the Primary Type Style that is present in the address data. <i>NO</i>: Uses the Primary Type Style specified in the <i>PRIMARY_TYPE_STYLE</i> option. When you set this option to <i>YES</i>, the transform ignores the setting in the <i>PRIMARY_TYPE_STYLE</i> option.

Option	Description
USE_POSTAL_COUNTRY_NAME	Specifies which country data is output for countries that receive their postal service from another country. • YES: Uses the postal country name. • NO: Uses the territory country name. ❖ Example When you use the USA engine for addresses from U.S. territories, the software populates the COUNTRY field with the name of the postal country, United States. The software bases the style of the COUNTRY field on the COUNTRY_STYLE option setting. If the country does not have a postal country, this setting does not change the output.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

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14.8.1 BEST_PRACTICE

The BEST_PRACTICE value (available for some Global Address Cleanse Standardization options) tells the software to use what it sees as the most common setting for a country.

The table shows what setting the software uses for each country when Global Address Cleanse Standardization options use the BEST_PRACTICE value.

! Restriction

BEST_PRACTICE values are not supported as dynamic options in DQM SDK and Data Services.

BEST_PRACTICE settings

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRI-MARY_PUNCTUATION	PRI-MARY_STYLE	REGION_STYLE	SECONDARY_DESCRIPTION_PUNCTUATION	SECONDARY_DESCRIPTION_STYLE
AD	Andorra	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
AE	United Arab Emirates	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AF	Afghanistan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AG	Antigua and Barbuda	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AI	Anguilla	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
AL	Albania	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AM	Armenia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AO	Angola	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AQ	Antarctica	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AR	Argentina	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
AS	American Samoa	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
AT	Austria	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
AU	Australia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
AW	Aruba	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
AX	Åland Islands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRIMARY_PUNCTUATION	PRIMARY_STYLE	REGION_STYLE	SECONDARY_DESCRIPTION_PUNCTUATION	SECONDARY_DESCRIPTION_STYLE
AZ	Azerbaijan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BA	Bosnia and Herzegovina	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BB	Barbados	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BD	Bangladesh	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BE	Belgium	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
BF	Burkina Faso	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BG	Bulgaria	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BH	Bahrain	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BI	Burundi	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BJ	Benin	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BL	Saint Barthélemy	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
BM	Bermuda	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
BN	Brunei Darussalam	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BO	Plurinational State of Bolivia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BQ	Bonaire, Sint Eustatius and Saba	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PUNCTUATION	POSTAL_PUNCTUATION_STYLE	PRIMARY_PUNCTUATION	PRIMARY_PUNCTUATION_STYLE	REGION_STYLE	SECONDARY_DESCRIPTION_PUNCTUATION	SECONDARY_DESCRIPTION_STYLE
BR	Brazil	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
BS	Bahamas	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BT	Bhutan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BV	Bouvet Island	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BW	Botswana	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BY	Belarus	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
BZ	Belize	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CA	Canada	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
CC	Cocos (Keeling) Islands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
CD	Democratic Republic of the Congo	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CF	Central African Republic	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CG	Congo	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CH	Switzerland	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
CI	Côte D'ivoire	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CK	Cook Islands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
CL	Chile	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CM	Cameroon	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CN	China	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPI- TALI- ZA- TION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TIO- NAL_P UNC- TUA- TION	DIRIC- TIO- NAL_S TYLE	POSTA L_PHR ASE_P UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
CO	Colom- bia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CR	Costa Rica	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CU	Cuba	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CV	Cabo Verde	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
CW	Cura- çao	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
CX	Christ- mas Is- land	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
CY	Cyprus	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
CZ	Czechia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
DE	Ger- many	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
DJ	Djibouti	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
DK	Den- mark	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
DM	Domin- ica	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
DO	Domini- can Re- public	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
DZ	Algeria	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
EC	Ecu- dor	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
EE	Estonia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
EG	Egypt	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
EH	West- ern Sa- hara	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ER	Eritrea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ES	Spain	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRIMARY_PUNCTUATION	PRIMARY_STYLE	REGION_STYLE	SECONDARY_PUNCTUATION	SECONDARY_STYLE
ET	Ethiopia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
FI	Finland	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
FJ	Fiji	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
FK	Falkland Islands (Malvinas)	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
FM	Federated States of Micronesia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
FO	Faroe Islands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
FR	France	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GA	Gabon	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GB	United Kingdom	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GD	Grenada	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GE	Georgia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GF	French Guiana	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GG	Guernsey	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GH	Ghana	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GI	Gibraltar	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GL	Greenland	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GM	Gambia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GN	Guinea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Coun- try Code (ISO)	Coun- try Name	CAP- ITALI- ZATION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TION_ UNC- TUA- TION	DIRIC- TION_ NAL_S TYLE	POSTA L_PHR UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
GP	Guade- loupe	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GQ	Equato- rial Guinea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GR	Greece	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GS	South Georgia and the South Sand- wich Is- lands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
GT	Guate- mala	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GU	Guam	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
GW	Guinea- Bissau	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
GY	Guyana	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
HK	Hong Kong	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
HM	Heard Island and McDo- nald Is- lands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
HN	Hondur- as	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
HR	Croatia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
HT	Haiti	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
HU	Hun- gary	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
ID	Indone- sia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
IE	Ireland	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
IL	Israel	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPI- TALI- ZA- TION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TIO- NAL_P UNC- TUA- TION	DIRIC- TIO- NAL_S TYLE	POSTA L_PHR ASE_P UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
IM	Isle of Man	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
IN	India	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
IO	British Indian Ocean Territory	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
IQ	Iraq	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
IR	Islamic Republic of Iran	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
IS	Iceland	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
IT	Italy	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
JE	Jersey	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
JM	Ja- maica	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
JO	Jordan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
JP	Japan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KE	Kenya	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KG	Kyrgyz- stan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KH	Cam- bodia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KI	Kiribati	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KM	Co- moros	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KN	Saint Kitts and Ne- vis	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Coun- try Code (ISO)	Coun- try Name	CAP- TALI- ZA- TION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TION_ UNC- TUA- TION	DIRIC- TION_ NAL_S TYLE	POSTA L_PHR UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
KP	Democ- ratic Peo- ple's Repub- lic of Korea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KR	Repub- lic of Korea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KW	Kuwait	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
KY	Cay- man Is- lands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
KZ	Ka- zakh- stan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LA	Lao Peo- ple's Demo- cratic Repub- lic	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LB	Leba- non	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LC	Saint Lucia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LI	Liech- ten- stein	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
LK	Sri Lanka	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
LR	Liberia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LS	Lesotho	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
LT	Lithua- nia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRI-MARY_PUNCTUATION	PRI-MARY_STYLE	REGION_STYLE	SEC-ON-DARY_DE-SCRIPTI-ON_PUNCTUATION	SEC-ON-DARY_DE-SCRIPTI-ON_STYLE
LU	Luxembourg	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
LV	Latvia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
LY	Libya	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MA	Morocco	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MC	Monaco	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
MD	Republic of Moldova	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ME	Montenegro	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MF	Saint Martin (French Part)	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
MG	Madagascar	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MH	Marshall Islands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
MK	North Macedonia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ML	Mali	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MM	Myanmar	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MN	Mongolia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MO	Macao	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MP	Northern Mariana Islands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRI-MARY_PUNCTUATION	PRI-MARY_STYLE	REGION_STYLE	SEC-ON-DARY_DE-SCRIPTI-ON_PUNCTUATION	SEC-ON-DARY_DE-SCRIPTI-ON_STYLE
MQ	Martinique	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
MR	Mauritania	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MS	Montserrat	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
MT	Malta	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MU	Mauritius	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MV	Maldives	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MW	Malawi	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
MX	Mexico	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
MY	Malaysia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
MZ	Mozambique	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NA	Namibia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NC	New Caledonia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
NE	Niger	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NF	Norfolk Island	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
NG	Nigeria	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NI	Nicaragua	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NL	Netherlands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
NO	Norway	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
NP	Nepal	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NR	Nauru	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
NU	Niue	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_Latin_OUS_ASCII	DIRECTIONAL_Punctuation	DIRCTIONAL_STYLE	POSTAL_Phrase_Punctuation	POSTAL_Phrase_Style	PRI-MARY_Punctuation	PRI-MARY_Style	REGION_Style	SEC-ON-DARY_DE-SCRIPTIION_PUNCTUA-TION	SEC-ON-DARY_DE-SCRIPTIION_Style
			OUT-PUT_T	UNC-TUA-TION		UNC-TUA-TION		TYPE_PUNCTUA-TION			RE-GION_PUNCTUA-TION	DE-SCRIPTIION_Style
NZ	New Zealand	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
OM	Oman	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PA	Panama	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PE	Peru	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PF	French Polynesia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
PG	Papua New Guinea	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PH	Philippines	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
PK	Pakistan	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
PL	Poland	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PM	Saint Pierre and Miquelon	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
PN	Pitcairn	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
PR	Puerto Rico	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	SHORT	NO	SHORT
PS	State of Palestine	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
PT	Portugal	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
PW	Palau	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
PY	Paraguay	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
QA	Qatar	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
RE	Réunion	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPI- TALI- ZA- TION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TIO- NAL_P UNC- TUA- TION	DIRIC- TIO- NAL_S TYLE	POSTA L_PHR ASE_P UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
RO	Roma- nia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
RS	Serbia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
RU	Russian Federa- tion	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
RW	Rwand a	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SA	Saudi Arabia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SB	Solo- mon Is- lands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SC	Sey- chelles	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SD	Sudan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SE	Swe- den	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SG	Singa- pore	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SH	Saint Helena, Ascen- sion and Tri- stan da Cunha	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SI	Slov- enia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SJ	Sval- bard and Jan Mayen	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SK	Slova- kia	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SL	Sierra Leone	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_Latin_OUS_ASCII	DIRECTIONAL_Punctuation	DIRCTIONAL_STYLE	POSTAL_Phrase_Punctuation	POSTAL_Phrase_Style	PRIMARY_Punctuation	PRIMARY_Style	REGION_Style	SECONDARY_Description_Punctuation	SECONDARY_Description_Style
			OUT_PUT_T	UNC-TUA-TION		UNC-TUA-TION					PUNC-TUA-TION	DE-SCRIPTION_STYLE
SM	San Marino	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SN	Senegal	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SO	Somalia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SR	Suriname	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SS	South Sudan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ST	Sao Tome and Principe	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SV	El Salvador	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SX	Sint Maarten (Dutch Part)	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
SY	Syrian Arab Republic	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
SZ	Eswatini	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TC	Turks and Caicos Islands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
TD	Chad	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TF	French Southern Territories	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
TG	Togo	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Coun- try Code (ISO)	Coun- try Name	CAP- ITALI- ZA- TION	CON- VERT_ LATIN_ OUT- PUT_T O_US_ ASCII	DIREC- TIO- NAL_P UNC- TUA- TION	DIRIC- TIO- NAL_S TYLE	POSTA L_PHR UNC- TUA- TION	POSTA L_PHR ASE_S TYLE	PRI- MARY_ PUNC- TUA- TION	PRI- MARY_ TYPE_ STYLE	RE- GION_ STYLE	SEC- ON- DARY_ DE- SCRIP- TION_ PUNC- TUA- TION	SEC- ON- DARY_ DE- SCRIP- TION_ STYLE
TH	Thailand	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	LONG
TJ	Tajikistan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TK	Tokelau	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
TL	Timor-Leste	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TM	Turkmenistan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TN	Tunisia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TO	Tonga	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TR	Turkey	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TT	Trinidad and Tobago	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TV	Tuvalu	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TW	Province of China Taiwan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
TZ	United Republic of Tanzania	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
UA	Ukraine	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
UG	Uganda	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
UM	United States Minor Outlying Islands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
US	United States	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT

Country Code (ISO)	Country Name	CAPITALIZATION	CONVERT_LATIN_OUTPUT_US_ASCII	DIRECTIONAL_PUNCTUATION	DIRECTIONAL_STYLE	POSTAL_PHRASE_PUNCTUATION	POSTAL_PHRASE_STYLE	PRI-MARY_PUNCTUATION	PRI-MARY_STYLE	REGION_STYLE	SEC-ON-DARY_DE-SCRIPTI-ON_PUNCTUATION	SEC-ON-DARY_DE-SCRIPTI-ON_STYLE
UY	Uruguay	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
UZ	Uzbekistan	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
VA	Holy See	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
VC	Saint Vincent and the Grenadines	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
VE	Bolivarian Republic of Venezuela	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
VG	British Virgin Islands	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
VI	U.S. Virgin Islands	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	SHORT	NO	SHORT
VN	Vietnam	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
VU	Vanuatu	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
WF	Wallis and Futuna	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
WS	Samoa	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
YE	Yemen	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
YT	Mayotte	UPPER	NO	NO	LONG	NO	SHORT	NO	LONG	LONG	NO	SHORT
ZA	South Africa	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT
ZM	Zambia	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

Country Code (ISO)	Country Name	CAPITULATION	CONVERT_LATIN_OUT_PUT_ASCII	DIRECTIONAL_Punctuation	DIRCTIONAL_STYLE	POSTAL_Phrase_Punctuation	POSTAL_Phrase_Style	PRIORITY_Punctuation	PRIORITY_Style	RE-GION_Style	SECONDARY_Punctuation	SECONDARY_Style
ZW	Zimbabwe	UPPER	NO	NO	SHORT	NO	SHORT	NO	SHORT	LONG	NO	SHORT

14.9 Canada engine

Use the Canada engine to process your Canada address data with the Global Address Cleanse transform.

The Canada engine includes specific options for processing Canada address data and suggestion lists.

[Global Address Cleanse Canada Report Options \[page 443\]](#)

To output the SERP required Statement of Address Accuracy report information, complete the options in the Canada engine report group.

[Global Address Cleanse Canada Suggestion List options \[page 443\]](#)

Enable suggestion lists and set how the transform presents suggestions for your Canada address data.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

[Global Address Cleanse Country \[page 411\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.9.1 Global Address Cleanse Canada Report Options

To output the SERP required Statement of Address Accuracy report information, complete the options in the Canada engine report group.

Syntax

```
<REPORT_OPTIONS>
  <CUSTOMER_COMPANY_NAME/>
  <MAILER_ADDRESS1/>
  <MAILER_ADDRESS2/>
  <MAILER_ADDRESS3/>
  <MAILER_ADDRESS4/>
  <CUSTOMER_CPC_NUMBER/>
</REPORT_OPTIONS>
```

Complete the options in this group when you prepare a mailing list for Canada Post certification.

Option	Description
<i>CUSTOMER_COMPANY_NAME</i>	Specifies the company name of the organization for whom you are preparing the mailing (up to 40 characters).
<i>MAILER_ADDRESS1</i>	Specifies the name and address of the person or organization for whom you are preparing the mailing (up to 40 characters per line).
<i>MAILER_ADDRESS2</i>	
<i>MAILER_ADDRESS3</i>	
<i>MAILER_ADDRESS4</i>	
<i>CUSTOMER_CPC_NUMBER</i>	Specifies the customer CPC number that is located on the Canada Post Contract (up to 15 characters).

14.9.2 Global Address Cleanse Canada Suggestion List options

Enable suggestion lists and set how the transform presents suggestions for your Canada address data.

Syntax

```
<SUGGESTION_LIST_OPTIONS>
  <ENABLE_SUGGESTION_LISTS>NO</ENABLE_SUGGESTION_LISTS>
  <MAX_NUMBER_LASTLINES>15</MAX_NUMBER_LASTLINES>
  <MAX_NUMBER_ADDRESS_LINES>50</MAX_NUMBER_ADDRESS_LINES>
  <LASTLINES_MATCH_MINIMUM>60</LASTLINES_MATCH_MINIMUM>
  <ADDRESS_LINES_MATCH_MINIMUM>60</ADDRESS_LINES_MATCH_MINIMUM>
  <COMBINE_OVERLAPPING_RANGES>NO</COMBINE_OVERLAPPING_RANGES>
  <ADDRESS_RANGE>0</ADDRESS_RANGE>
</SUGGESTION_LIST_OPTIONS>
```

Option	Description
<code>ADDRESS_LINES_MATCH_MINIMUM</code>	Specifies the similarity score required for address-line suggestions. Type a value from 0 through 80. The similarity score determines what suggestions the software returns in the list. A higher value requires the suggestion to be more similar to the input address than a lower value.
<code>ADDRESS_RANGE</code>	Specifies a span around the input primary address range for which to return suggestions. Maximum value is 5000. Enter 0 to set no limits to the ranges that the transform returns in suggestion lists. By setting an address range, you limit the suggestions returned to be within the range. ❖ Example <code>ADDRESS_RANGE</code> = 500 Input address: 1000 Pine St. Suggestion list range: 750–1250 Pine Street
<code>COMBINE_OVERLAPPING_RANGES</code>	Specifies whether individual suggestions with overlapping ranges are combined. • YES: Ignores gaps and overlaps in ranges and combines individual suggestions. • NO: Does not combine individual suggestions when they have gaps or overlapping ranges. Set this option to YES if you want to limit the number of total suggestions presented to your user. However, you might not see gaps of invalid ranges that would be apparent if you set the option to NO. ❖ Example The software presents the following suggestions when you set this option to NO: 1000–1099 Maple Ave 1100–1199 Maple Ave The software presents the following suggestions when you set this option to YES: 1000–1199 Maple Ave

Option	Description
<i>ENABLE_SUGGESTION_LISTS</i>	Specifies whether the software generates suggestion lists. <i>YES</i>: Generates suggestion lists when assignment candidates are present. <i>NO</i>: Does not generate suggestion lists.
<i>LASTLINES_MATCH_MINIMUM</i>	Specifies the minimum similarity score required for lastline suggestions. The score then determines which suggestions are returned in the list. Type a value from 0 through 80. A higher number indicates that the suggestion must be more similar to the input to be returned as a possible suggestion.
<i>MAX_NUMBER_ADDRESS_LINES</i>	Specifies the maximum number of address line suggestions that the software generates. Type a value from 2 through 50. Set this option, for example, to limit the size of SOAP documents that the Web service sends or to limit the maximum number of suggestions that users choose from.
<i>MAX_NUMBER_LASTLINES</i>	Specifies the maximum number of lastline suggestions that the software generates. Type a value from 2 through 15. Set this option, for example, to limit the size of SOAP documents that the Web service sends or to limit the maximum number of suggestions that users choose from.

14.10 Global Address Cleanse Canada group

Set Canada-specific processing options in the Canada group.

Syntax

```
<CANADA>
  <REFERENCE_FILES>
    <DIRECTORY_PATH>..\DataQuality\gac\</DIRECTORY_PATH>
  </REFERENCE_FILES>
  <OPTIONS>
    <DISABLE_CERTIFICATION>YES</DISABLE_CERTIFICATION>
    <PARSE_ONLY>NO</PARSE_ONLY>
    <OUTPUT_ADDRESS_LANGUAGE>PRESERVE</OUTPUT_ADDRESS_LANGUAGE>
    <UNIT_DESCRIPTION>DEFAULT</UNIT_DESCRIPTION>
    <DUAL_ADDRESS>STREET</DUAL_ADDRESS>
    <ENABLE_LVR_RULE>NO</ENABLE_LVR_RULE>
    <ENABLE_RURAL_RULE>NO</ENABLE_RURAL_RULE>
    <POSTCODE_ONLY_SEARCH>NO</POSTCODE_ONLY_SEARCH>
    <POSTCODE_NO_MATCH_SEARCH>NO</POSTCODE_NO_MATCH_SEARCH>
    <POSTCODE_PRIORITY_OVER_STREET>NO</POSTCODE_PRIORITY_OVER_STREET>
    <USE_FIRM_TO_ASSIGN>YES</USE_FIRM_TO_ASSIGN>
```

Canada processing options

Option	Description
<i>DISABLE_CERTIFICATION</i>	Specifies whether to disable Canada address certification. <i>YES</i>: Processes Canada addresses without including certification, and also processes with non-poc (Point of Call) directories for non-mailing purposes. <i>YES</i> is the default setting. i Note Does not print the SERP report. Any list that you create with certification disabled cannot be used for mailing. <i>NO</i>: Processes Canada addresses using processes required for SERP certification. Includes using POC directories and generating the SERP report. Any list that you create with certification enabled can be used for mailing. i Note If you set this option to <i>YES</i>, also set the following certification options to <i>YES</i>: <i>ENABLE_LVR_RULE</i> <i>ENABLE_RURAL_RULE</i> <i>POSTCODE_ONLY_SEARCH</i> <i>POSTCODE_NO_MATCH_SEARCH</i>
<i>DUAL_ADDRESS</i>	Specifies transform behavior when there is an input dual address. <i>POSITION</i>: Selects an address based on the position of the address in the address block. The Canada engine tries to validate the address that is closest to the lower left corner of the address block. The address could be the postal or the street address based on how the data is input. <i>POSITION</i> is required for SERP certification. <i>POSTAL</i>: Tries to validate based on the postal address. If that fails, tries again based on the street address. <i>STREET</i>: Tries to validate based on the street address. If that fails, tries again based on the postal address, such as rural route or PO box.

Option	Description
<i>ENABLE_LVR_RULE</i>	Specifies whether to enable the Canada Post LVR (Large Volume Receiver) postal code rule. The rule states that the LVR postal code is valid and cannot be changed to match other address components. Canada Post permits correction of LVR postal codes only when a unique address can be determined without changing the postal code. • YES: Regards any LVR address as assigned, even when the address line is so flawed that a match to the postal directory is impossible. Select YES to qualify for SERP certification. • NO: Disables the LVR postal code rule. Reports LVR addresses as unassigned when the address line is flawed.
<i>ENABLE_RURAL_RULE</i>	Specifies whether to enable the Canada Post Rural rule. The rule states that any address with a valid rural postal code is considered valid. Rural postal codes have a zero in the second position. • YES: Regards any rural address as valid, even if the address line is so flawed that a match to the postal directory is impossible. Select YES to qualify for SERP certification. • NO: Reports a rural address as invalid when the address line is bad. Canada Post states that you should not correct rural postal codes. However, the transform always attempts to correct all other address components except for the rural postal code. The rule applies under the following circumstances: • The address line is empty or contains bad data. • The address does not have a postal code or has an incorrect postal code. • The locality (city) has just one postal code associated with it that is a rural postal code.
<i>OUTPUT_ADDRESS_LANGUAGE</i>	Specifies the language for the output records. • CONVERT: Uses French for records in Quebec, and English for records in other regions or provinces. • ENGLISH: Converts output records to English. • FRENCH: Converts output records to French. • PRESERVE: Detects the input language and preserves that language upon output, ignoring the region or province. PRESERVE is the default setting.

Option	Description
<i>PARSE_ONLY</i>	Specifies whether the transform should parse input records into discrete components, but not perform a lookup in the postal directories. <i>YES</i>: Parses records into discrete components, but does not perform a lookup in the postal directories. When you select <i>YES</i>, processing is fast, but parsing results are unverified. <i>NO</i>: Parses records into discrete components and performs a lookup in the postal directories. When you select <i>NO</i>, processing may be slower, but parsing results are verified when the appropriate reference data is available.
<i>POSTCODE_NO_MATCH_SEARCH</i>	Specifies whether the transform should search the postal directories when the address line can be assigned, but it doesn't match the incoming postal code. <i>YES</i>: Searches the postal directories when the address line can be assigned, but it doesn't match the incoming postal code. Select <i>YES</i> to qualify for SERP certification. When you select <i>YES</i>, the transform searches the postal directories to determine the following: - If the incoming address line is a PO Box address, the postal code cannot be a valid postal code for an LVR (Large Volume Receiver), firm, or a civic (street) address, such as 100 Main St. - If the incoming address line is a civic (street) address, the postal code cannot be a valid postal code for an LVR PO Box address. SERP rules state that, if either one of these circumstances exists, Data Quality Management SDK cannot assign the address. <i>NO</i>: Does not search the postal directories when the address line can be assigned, but it doesn't match the incoming postal code. <i>NO</i> is the default setting.

Option	Description
<i>POSTCODE_ONLY_SEARCH</i>	Specifies whether the transform should search for the address based on postal code only, and attempt to find a street record that contains the range. YES: Performs a postcode-only search, and attempts to find a street record that contains the range. Select YES to qualify for SERP certification. NO Does not perform a post-code only search. In some cases, the result is a better address line. In other cases, the transform more reliably detects that it cannot assign an address line. This option affects assignment when the input address line is badly incomplete. For example, when the address includes a range but no street name. When you set this option to YES and the transform finds only one street record that contains the range, the transform assigns the address line from the postal code. A postal-code-only search is time consuming. Therefore, disabling this search could improve your processing time.
<i>POSTCODE_PRIORITY_OVER_STREET</i>	Specifies whether the transform weighs the postcode and street equally or whether street has priority over postcode. YES: Assigns by placing more weight on the address line when breaking a tie. This assignment makes sense when you consider how common data-entry errors are for postal codes. NO: Assigns by placing equal weight on the address line and the postal code when breaking a tie. Select NO to qualify for SERP certification. This option setting is important when the transform is trying to break a tie between two possible assignments: - A near match on address line. - An exact match on postal code.
<i>UNIT_DESCRIPTION</i>	Specifies the unit description in English: APARTMENT: Uses "Apartment" as the default unit designator. DEFAULT: Uses the default unit designator. UNIT: Uses "Unit" as the default unit designator.

Option	Description
<code>USE_FIRM_TO_ASSIGN</code>	Specifies whether the transform uses the firm information to make an assignment and whether to display the firm in suggestion lists. YES: Uses and displays the firm. YES is the default setting. NO: Does not use or display the firm.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

[Global Address Cleanse Country \[page 411\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.11 Global Address engine

To standardize data from countries other than Canada and the USA, set options for the Global Address engine.

Before you set options in this group, consider the countries represented in your data. For example, if your data does not contain addresses from Canada or USA, select the Global Address engine.

[Global Address Cleanse Country Options \[page 451\]](#)

To set options for specified countries, set country options in the Global Address group.

[Global Address Cleanse Suggestion List Options \[page 456\]](#)

Enable suggestion lists and set how the transform presents suggestions for your global addresses.

[Countries supported by lastline drilldown \[page 459\]](#)

When you set **ENABLE_LASTLINE_DRILLDOWN** to **YES**, indicate either a country name or the two-character ISO country code.

[Global Address Cleanse Report Options \[page 466\]](#)

Use the Report Options group to add required certification information to reports for Australia and New Zealand.

[Global Address Cleanse Australia report options \[page 467\]](#)

Set options to include the required Australia AMAS (Address Matching Processing Summary) information to reports.

[Global Address Cleanse New Zealand report options \[page 468\]](#)

Set options to include the required New Zealand Statement of Accuracy (SOA) information to reports.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

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[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.11.1 Global Address Cleanse Country Options

To set options for specified countries, set country options in the Global Address group.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

```
<COUNTRY_OPTIONS>
  <COUNTRY>
    <COUNTRY_NAME>GLOBAL (APPLY TO ALL COUNTRIES)</COUNTRY_NAME>
    <OPTIONS>
      <RETAIN_POSTCODE_IF_VALID_FORMAT>NO</RETAIN_POSTCODE_IF_VALID_FORMAT>
      <DUAL_ADDRESS>STREET</DUAL_ADDRESS>
      <DISABLE_CERTIFICATION>NO</DISABLE_CERTIFICATION>
      <OUTPUT_ADDRESS_SCRIPT>PRESERVE</OUTPUT_ADDRESS_SCRIPT>
      <TRANSLATE_MAJOR_LOCALITY>PRESERVE</TRANSLATE_MAJOR_LOCALITY>
      <USE_FIRM_TO_ASSIGN>YES</USE_FIRM_TO_ASSIGN>
```

```

        <PROCESSING_TIMEOUT>0</PROCESSING_TIMEOUT>
    </OPTIONS>
</COUNTRY>

```

Specify a country and then set address assignment options for the specified country.

Option	Description
<i>COUNTRY_NAME</i>	Specifies the country to apply the standardization options. <i>GLOBAL (APPLY TO ALL COUNTRIES)</i>: Applies country standardization options to all countries in your data. <i>Specific Country</i>: To apply Country Options to a specific country, enter a country name.
<i>RETAIN_POSTCODE_IF_VALID_FORMAT</i>	Specifies to process blank, valid, and invalid input postcodes. <i>YES</i>: Retains input postcode unless: - It is in an invalid format for the country - There's a single postcode match - There's a possible intelligent match <i>NO</i>: Updates output postcode if: - There's a single postcode match - There's a possible intelligent match Otherwise, retains input postcode. <i>NO</i> is the default setting i Note The transform populates the postcode when the input postcode is blank, and it finds a single postcode match.
<i>DUAL_ADDRESS</i>	Specifies the action to take when the Global Address engine encounters a dual address. <i>POSITION</i>: Selects an address based on the arrangement of the input data. The transform attempts to assign the first address found in the input data. If it can't assign the first address, it attempts to assign the next address it finds. <i>POSTAL</i>: Attempts to validate based on the postal address. If that fails, attempts again based on the street address. <i>STREET</i>: Attempts to validate based on the street address. If that fails, attempts again based on the postal address.

Option	Description
<code>DISABLE_CERTIFICATION</code>	Specifies whether to perform noncertified or certified processing of addresses for Australia, France, or New Zealand. If the country is Australia: • YES: Enables noncertified features and extends the directory expiration for nonmailing purposes. More information: • Extends the directory expiration period for up to 14 months from the date the directories were created. • Processing with expired directory data is allowed when you aren't planning to use the records for AMAS mailing purposes. • Processing with expired directory data is ideal for data warehousing industries, for example. However, when you select YES, you can't print the AMAS report. • You can't use any lists created with expired directories for postage discounts. • NO: Disables noncertified features and uses the most current directory information. Enables printing of the AMAS report. If the country is France: • YES: Enables noncertified processing of France addresses for nonmailing purposes. • NO: Disables noncertified processing of France addresses for mailing purposes. More information: • Removes all punctuation except for firm data. • Removes all accent characters (allows only A-Z and 0–9). • Returns address in 6 lines. Note To return the address in all upper case, set the <code>CAPITALIZATION</code> option to <code>UPPER</code>. If the country is New Zealand: • YES: Enables noncertified processing of New Zealand addresses and allows processing with expired directories for nonmailing purposes. Disables printing the SOA Report.

Option	Description
	i Note Don't use any list for mailing that you process with certification disabled. NO: Enables certified processing for New Zealand and enables printing of the SOA report. i Note If the reference data is missing or older than the certification requirements for any of the following countries, the software issues a warning: - Australia - France - New Zealand If the software issues the message for data that isn't required for the country you're processing, ignore the message. To avoid receiving this message, replicate the country options group for each country. Then set the DISABLE_CERTIFICATION option to NO only for the relevant country. When you set a Country Options group for each country, the software uses all other options in the country-specific group in place of the options you set in the Global group.
OUTPUT_ADDRESS_SCRIPT	Specifies how a native script is output. Supports the following countries: - Chinese when the input character script is CJKK - Russian when the input character script is CYRL Options include the following: LATIN: Transliterates the native script and outputs the data as Latin. PRESERVE: Outputs the data in the native script. i Note The setting in the TRANSLATE_MAJOR_LOCALITY option takes precedence over the setting in the OUTPUT_ADDRESS_SCRIPT option for the locality.

Option	Description
<i>TRANSLATE_MAJOR_LOCALITY</i>	Specifies whether to translate the <i>LOCALITY1</i> output field for major localities. • <i>ENGLISH</i>: Translates the output to English, if available. • <i>PRESERVE</i>: Outputs the locality data as it is input. <i>PRESERVE</i> is the default setting. i Note The setting in the <i>TRANSLATE_MAJOR_LOCALITY</i> option takes precedence over the setting in the <i>ASSIGN_LOCALITY</i> option. The setting in the <i>TRANSLATE_MAJOR_LOCALITY</i> option also takes precedence over the setting in the <i>OUTPUT_ADDRESS_SCRIPT</i> option for the locality.
<i>USE_FIRM_TO_ASSIGN</i>	Specifies whether to use the firm name to make an assignment and to display the firm name in a suggestion list. <i>YES</i>: Uses and displays the firm. <i>YES</i> is the default setting. <i>NO</i>: Doesn't use or display the firm.
<i>PROCESSING_TIMEOUT</i>	Specifies the timeout duration in milliseconds. After the duration is exceeded, the software stops processing and issues the information code 7000. Enter <i>0</i> to disable the timeout. <i>0</i> is the default setting. Enter a number greater than zero to set the amount of time to wait before the software stops processing. i Note SAP recommends that you set the <i>PROCESSING_TIMEOUT</i> at or above 100 milliseconds.
<i>SUPPRESS_SUGGESTION_LIST_WITH_NO_RANGE</i>	Specifies whether the software generates a suggestion list when it finds an "S/N" or no range on input, and the only records in the reference data are range records. • <i>YES</i>: Doesn't generate a suggestion list. Returns information code 3030 and makes a primary name level match assignment. • <i>NO</i>: Generates a suggestion list with the range values in the reference data record. <i>NO</i> is the default setting

14.11.2 Global Address Cleanse Suggestion List Options

Enable suggestion lists and set how the transform presents suggestions for your global addresses.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

```
<SUGGESTION_LIST_OPTIONS>
  <ENABLE_SUGGESTION_LISTS>NO</ENABLE_SUGGESTION_LISTS>
  <ENABLE_LASTLINE_DRILLDOWN>NO</ENABLE_LASTLINE_DRILLDOWN>
  <MAX_NUMBER_ADDRESS_LINES>50</MAX_NUMBER_ADDRESS_LINES>
  <MAX_NUMBER_LASTLINES>15</MAX_NUMBER_LASTLINES>
  <COMBINE_OVERLAPPING_RANGES>YES</COMBINE_OVERLAPPING_RANGES>
  <ADDRESS_RANGE>0</ADDRESS_RANGE>
</SUGGESTION_LIST_OPTIONS>
```

The settings that you make in the Suggestion List Options group are for global addresses only. The settings don't affect how the software processes addresses from the United States or Canada.

Option	Description
ENABLE_SUGGESTION_LISTS	Specifies whether the transform generates suggestion lists for global addresses. YES: Generates suggestion lists when assignment candidates are present. NO: Doesn't generate suggestion lists.

Option	Description
ENABLE_LASTLINE_DRILLDOWN	Specifies whether you can drill down to the next level of the lastline component in suggestion lists after selecting a parent component. YES: Enables drilling down to the next level of lastline components in suggestion lists after selecting a parent component. NO: Doesn't enable drilling down to the next level of lastline components in suggestion lists after selecting a parent component. ❖ Example Given an input country, a suggestion list provides a list of available regions. After you select one of the regions, a suggestion list provides a list of available locality1 options. After you select one of the localities, a suggestion list provides a list of available locality2 options. When you enable this option, ensure that you set the ENABLE_SUGGESTION_LISTS option to YES. ❖ Example The Canada and USA engines don't support the ENABLE_LASTLINE_DRILLDOWN option. A suggestion list using the ENABLE_LASTLINE_DRILLDOWN option is the same as a regular lastline suggestion list, except that the suggestion list doesn't include postcode or address type components. Also, the expected input for lastline drilldown is country. All other input fields are blank. For example, input files such as locality, region, postcode, firm, and multiline are blank. i Note Ensure that your input field combination complies with one of the accepted formats: Multiline, hybrid, and discrete. For a few countries, such as India, the suggestion list doesn't always return some localities. The data directory doesn't always link localities with regions for these countries. Performance can be slower when the software generates suggestion lists for countries and regions with a large number of localities.

Option	Description
	For a list of supported countries and country codes, see Countries supported by lastline drilldown [page 459]
<code>COMBINE_OVERLAPPING_RANGES</code>	Specifies whether the software combines individual suggestions with overlapping ranges. • YES: Ignores gaps and overlaps in ranges. • NO: Doesn't combine overlapping ranges. Set this option to YES to limit the number of total suggestions presented. However, it is possible that you won't see gaps of invalid ranges that would be apparent if this option was set to NO. ❖ Example The software presents the following suggestions when you set this option to NO: 1000–1099 Maple Ave 1100–1199 Maple Ave The software presents the following suggestions when you set this option to YES: 1000–1199 Maple Ave → Tip If you set this option to NO, increase the size of the <code>SUGGESTION_LIST</code> output field so that it can contain all of the suggestions.
<code>ADDRESS_RANGE</code>	Specifies a span around the input primary address range for which to return suggestions. Maximum value is 5000. Enter 0 to set no limits to the ranges that the transform returns in suggestion lists. By setting an address range, you limit the suggestions returned to be within the range. ❖ Example <code>ADDRESS_RANGE</code> = 500 Input address: 1000 Pine St. Suggestion list range: 750–1250 Pine Street

Option	Description
<i>MAX_NUMBER_ADDRESS_LINES</i>	Specifies the maximum number of address line suggestions that the software generates. Minimum setting: 2 Maximum setting: 10,000 Default setting: 100 Set this option, for example, to limit the size of the SOAP documents sent by the web service, or to limit the maximum number of suggestions to choose from.
<i>MAX_NUMBER_LASTLINES</i>	Specifies the maximum number of lastline suggestions that the software generates. • Minimum setting: 2 • Maximum setting: 10,000 • Default setting: 60 Set this option, for example, to limit the size of SOAP documents that the Web service sends or to limit the maximum number of suggestions that users choose from. i Note You can't set this option when you set the <i>ENABLE_LASTLINE_DRILLDOWN</i> is set to <i>YES</i>.

14.11.3 Countries supported by lastline drilldown

When you set [*ENABLE_LASTLINE_DRILLDOWN*](#) to *YES*, indicate either a country name or the two-character ISO country code.

i Note

Reference to "country" or "countries" in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Use either the English country name or the two-character ISO country code as input. For some countries, enter the country name in its native script. When the input country name is in its native script, the software also provides the drilldown list in the native script.

The Global engine supports the [*ENABLE_LASTLINE_DRILLDOWN*](#) option. Lastline drilldown isn't available for the Canada and USA engines.

Countries supported by lastline drilldown

Country name	Two-character ISO code	Country name in native script
Afghanistan	AF	
Åland Islands	AX	
Albania	AL	
Algeria	DZ	
American Samoa	AS	
Andorra	AD	
Angola	AO	
Antigua and Barbuda	AG	
Argentina	AR	
Armenia	AM	
Australia	AU	
Austria	AT	
Azerbaijan	AZ	
Bahamas	BS	
Bahrain	BH	
Bangladesh	BD	
Barbados	BB	
Belarus	BY	
Belgium	BE	
Belize	BZ	
Benin	BJ	
Bermuda	BM	
Bhutan	BT	
Bolivarian Republic of Venezuela	VE	
Bonaire, Sint Eustatius, and Saba	BQ	
Botswana	BW	
Brazil	BR	
British Virgin Islands	VG	
Brunei Darussalam	BN	
Bulgaria	BG	
Burkina Faso	BF	
Burundi	BI	
CABO VERDE	CV	
Cambodia	KH	

Country name	Two-character ISO code	Country name in native script
Cameroon	CM	
Canada	CA	
Cayman Islands	KY	
Central African Republic	CF	
Chad	TD	
Chile	CL	
China	CN	中国
Christmas Island	CX	
Cocos (Keeling) Islands	CC	
Colombia	CO	
Comoros	KM	
Congo	CG	
Costa Rica	CR	
Côte d'Ivoire	CI	
Croatia	HR	
Cuba	CU	
Curaçao	CW	
Cyprus	CY	
Czechia	CZ	
Democratic People's Republic of Korea	KP	
Denmark	DK	
Djibouti	DJ	
Dominica	DM	
Dominican Republic	DO	
Ecuador	EC	
Egypt	EG	
El Salvador	SV	
Equatorial Guinea	GQ	
Eritrea	ER	
Estonia	EE	
Ethiopia	ET	
Faroe Islands	FO	
Federated States of Micronesia	FM	
Fiji	FJ	

Country name	Two-character ISO code	Country name in native script
Finland	FI	
France	FR	
French Guiana	GF	
French Polynesia	PF	
Gabon	GA	
Gambia	GM	
Georgia	GE	
Germany	DE	
Ghana	GH	
Greece	GR	ΕΛΛΑΔΑ
Greenland	GL	
Grenada	GD	
Guadeloupe	GP	
Guam	GU	
Guatemala	GT	
Guernsey	GG	
Guinea	GN	
Guinea-Bissau	GW	
Guyana	GY	
Haiti	HT	
Holy See (Vatican City State)	VA	
Honduras	HN	
Hungary	HU	
Iceland	IS	
India	IN	
Indonesia	ID	
Iraq	IQ	
Ireland	IE	
Islamic Republic of Iran	IR	
Isle of Man	IM	
Italy	IT	
Jamaica	JM	
Japan	JP	日本
Jersey	JE	

Country name	Two-character ISO code	Country name in native script
Jordan	JO	
Kazakhstan	KZ	
Kenya	KE	
Kiribati	KI	
Kyrgyzstan	KG	
Lao People's Democratic Republic	LA	
Latvia	LV	
Lebanon	LB	
Lesotho	LS	
Liberia	LR	
Libya	LY	
Liechtenstein	LI	
Lithuania	LT	
Luxembourg	LU	
Madagascar	MG	
Malawi	MW	
Malaysia	MY	
Maldives	MV	
Mali	ML	
Malta	MT	
Marshall Islands	MH	
Martinique	MQ	
Mauritania	MR	
Mauritius	MU	
Mayotte	YT	
Mexico	MX	
Monaco	MC	
Mongolia	MN	
Montenegro	ME	
Montserrat	MS	
Morocco	MA	
Mozambique	MZ	
Myanmar	MM	
Namibia	NA	

Country name	Two-character ISO code	Country name in native script
Nauru	NR	
Nepal	NP	
Netherlands	NL	
New Caledonia	NC	
Nicaragua	NI	
Niger	NE	
Nigeria	NG	
Norfolk Island	NF	
Northern Mariana Islands	MP	
Norway	NO	
Oman	OM	
Pakistan	PK	
Palau	PW	
Panama	PA	
Papua New Guinea	PG	
Paraguay	PY	
Peru	PE	
Philippines	PH	
Plurinational State of Bolivia	BO	
Poland	PL	
Portugal	PT	
Puerto Rico	PR	
Qatar	QA	
Republic of Korea	KR	대한민국
Republic of Moldova	MD	
Réunion	RE	
Romania	RO	
Russian Federation	RU	РОССИЯ
Rwanda	RW	
Saint Barthélemy	BL	
Saint Helena, Ascension, and Tristan da Cunha	SH	
Saint Kitts and Nevis	KN	
Saint Lucia	LC	
Saint Martin (French part)	MF	

Country name	Two-character ISO code	Country name in native script
Saint Pierre and Miquelon	PM	
Saint Vincent and the Grenadines	VC	
Samoa	WS	
San Marino	SM	
Sao Tome and Principe	ST	
Saudi Arabia	SA	
Senegal	SN	
Seychelles	SC	
Sierra Leone	SL	
Sint Maarten (Dutch Part)	SX	
Slovakia	SK	
Slovenia	SI	
Solomon Islands	SB	
Somalia	SO	
South Africa	ZA	
South Sudan	SS	
Spain	ES	
Sri Lanka	LK	
Sudan	SD	
Suriname	SR	
Svalbard and Jan Mayen	SJ	
Swaziland	SZ	
Sweden	SE	
Switzerland	CH	
Syrian Arab Republic	SY	
Taiwan	TW	台湾
Tajikistan	TJ	
Thailand	TH	
The Democratic Republic of the Congo	CD	
Togo	TG	
Tonga	TO	
Trinidad and Tobago	TT	
Tunisia	TN	
Turkey	TR	

Country name	Two-character ISO code	Country name in native script
Turkmenistan	TM	
Turks and Caicos Islands	TC	
Tuvalu	TV	
U.S. Virgin Islands	VI	
Uganda	UG	
Ukraine	UA	
United Arab Emirates	AE	
United Kingdom	GB	
United Republic of Tanzania	TZ	
United States	US	
Uruguay	UY	
Uzbekistan	UZ	
Vanuatu	VU	
Viet Nam	VN	
Wallis and Futuna	WF	
Yemen	YE	
Zambia	ZM	
Zimbabwe	ZW	

14.11.4 Global Address Cleanse Report Options

Use the Report Options group to add required certification information to reports for Australia and New Zealand.

Country	Description
Australia	Set report options to complete the required Australia AMAS (Address Matching Processing Summary) Report required by Australia postal authorities.
New Zealand	Set report options to complete the required Statement of Accuracy (SOA) report required by New Zealand postal authorities.

14.11.5 Global Address Cleanse Australia report options

Set options to include the required Australia AMAS (Address Matching Processing Summary) information to reports.

Syntax

```
<REPORT_OPTIONS>
  <AUSTRALIA>
    <CUSTOMER_COMPANY_NAME/>
    <MAILER_ADDRESS1/>
    <MAILER_ADDRESS2/>
    <MAILER_ADDRESS3/>
    <MAILER_ADDRESS4/>
    <LIST_NAME/>
    <FILE_NAME/>
  </AUSTRALIA>
```

Completing this group of options is required only when you are preparing a mailing that needs an Australia AMAS report for postal certification.

Option	Description
<i>CUSTOMER_COMPANY_NAME</i>	Specify the customer company name. Maximum is 40 characters. The customer company is the entity for which you are preparing the list.
<i>MAILER_ADDRESS1</i>	Specify the name and address of the person or organization for which you are preparing the list. Maximum is 29 characters per line.  Example <i>MAILER_ADDRESS1</i> = ABCD Mailing, Inc. <i>MAILER_ADDRESS2</i> = 123 Main St. <i>MAILER_ADDRESS3</i> = West Beach SA 1234 <i>MAILER_ADDRESS4</i> = Australia
<i>MAILER_ADDRESS2</i>	
<i>MAILER_ADDRESS3</i>	
<i>MAILER_ADDRESS4</i>	
<i>LIST_NAME</i>	Specify the name of your database or mailing list. Maximum is 40 characters. For example, enter the file name, your title, or a formal name for the list.
<i>FILE_NAME</i>	Specify the input file name. Maximum is 40 characters. For example, australia.dbf.

14.11.6 Global Address Cleanse New Zealand report options

Set options to include the required New Zealand Statement of Accuracy (SOA) information to reports.

```
<NEW_ZEALAND>
  <CUSTOMER_NUMBER/>
  <CUSTOMER_COMPANY_NAME/>
  <MAILER_ADDRESS1/>
  <MAILER_ADDRESS2/>
  <MAILER_ADDRESS3/>
  <MAILER_ADDRESS4/>
  <MAILER_ADDRESS5/>
  <MAILER_ADDRESS6/>
  <SOA_ISSUER_NAME/>
  <FILE_NAME/>
```

Completing this group of options is required only when you are preparing a mailing that needs a New Zealand SOA report for postal certification.

Option	Description
<i>CUSTOMER_NUMBER</i>	Specifies the New Zealand Post number assigned to the customer for whom you are preparing the list. New Zealand post requires the customer number in the SOA report to qualify for postage discounts.
<i>CUSTOMER_COMPANY_NAME</i>	Specifies the customer company name. The customer company is the entity for which you are preparing the list. Maximum is 40 characters.
<i>MAILER_ADDRESS1</i> <i>MAILER_ADDRESS2</i> <i>MAILER_ADDRESS3</i> <i>MAILER_ADDRESS4</i> <i>MAILER_ADDRESS5</i> <i>MAILER_ADDRESS6</i>	Specifies the name and address of the person or organization for which you are preparing the list. Maximum is 29 characters per line. ❖ Example <i>MAILER_ADDRESS1</i> = ABCD Mailing, Inc. <i>MAILER_ADDRESS2</i> = 123 Main St. <i>MAILER_ADDRESS3</i> = Johnsonville <i>MAILER_ADDRESS4</i> = Wellington 6004 <i>MAILER_ADDRESS5</i> = NEW ZEALAND
<i>SOA_ISSUER_NAME</i>	Specifies the company name that prepared the list. Maximum of 40 characters.
<i>FILE_NAME</i>	Specifies the input file name. For example, <code>new_zealand.dbf</code>.

14.12 Global Address Cleanse USA engine

To cleanse addresses from the United States and its territories, set USA-specific options in the Global Address Cleanse transform.

Set options for address parsing and suggestion lists for addresses from the United States and territories.

[Global Address Cleanse USA Options \[page 469\]](#)

To determine how the transform standardizes addresses from the United States and territories in your data, set USA options.

[Global Address Cleanse USA Suggestion List Options \[page 471\]](#)

Set Suggestion List Options to enable suggestions for addresses from the United States and territories.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

[Global Address Cleanse Country \[page 411\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse Suggestion List option \[page 473\]](#)

14.12.1 Global Address Cleanse USA Options

To determine how the transform standardizes addresses from the United States and territories in your data, set USA options.

≡ Syntax

```
<USA>
...
<OPTIONS>
  <PARSE_ONLY>NO</PARSE_ONLY>
  <UNIT_DESCRIPTION>CONVERT</UNIT_DESCRIPTION>
  <DUAL_ADDRESS>STREET</DUAL_ADDRESS>
  <USE_FIRM_TO_ASSIGN>YES</USE_FIRM_TO_ASSIGN>
```

Option	Description
<i>PARSE_ONLY</i>	Specifies whether the software parses addresses and performs a lookup in the postal directory, or only parses addresses. <i>YES</i>: Parses records into discrete components, but does not perform a lookup in the postal directories. <i>PARSE_ONLY</i> is fast, but parsing results are unverified. <i>NO</i>: Parses records into discrete components and performs a lookup in the postal directories. Setting this option to <i>NO</i> may slow down processing, but parsing results are verified.
<i>UNIT_DESCRIPTION</i>	Specifies how to standardize unit descriptions including apartment, suite, room, or floor. <i>CONVERT</i>: Standardizes unit description based on what it finds in the postal directory (apartment, suite, room, or floor). <i>PRESERVE</i>: Preserves the unit description from the input record, but corrects any spelling errors.
<i>DUAL_ADDRESS</i>	Specifies what address line to validate when the input address is a dual address. <i>POSITION</i>: Validates the address that is closest to the lower left corner of the address block. The address may be the postal or street address based on how the data was entered on input. <i>POSTAL</i>: Validates the postal address. If the postal address (rural route or PO Box) validation fails, validates the street address. <i>STREET</i>: Validates the street address. If the street address validation fails, validates the postal address (rural route or PO Box).
<i>USE_FIRM_TO_ASSIGN</i>	Specifies whether the transform uses the firm to make an assignment and to display firm in a suggestion list. <i>YES</i>: Assigns address using the firm, and displays the firm in the suggestion list. <i>YES</i> is the default setting. <i>NO</i>: Does not assign address using the firm, and does not display the firm name in suggestion lists.

Related Information

[Global Address Cleanse Standardization Options \[page 412\]](#)

14.12.2 Global Address Cleanse USA Suggestion List Options

Set Suggestion List Options to enable suggestions for addresses from the United States and territories.

Syntax

```
<SUGGESTION_LIST_OPTIONS>
  <ENABLE_SUGGESTION_LISTS>YES</ENABLE_SUGGESTION_LISTS>
  <MAX_NUMBER_LASTLINES>15</MAX_NUMBER_LASTLINES>
  <MAX_NUMBER_ADDRESS_LINES>50</MAX_NUMBER_ADDRESS_LINES>
  <LASTLINES_MATCH_MINIMUM>60</LASTLINES_MATCH_MINIMUM>
  <ADDRESS_LINES_MATCH_MINIMUM>60</ADDRESS_LINES_MATCH_MINIMUM>
  <COMBINE_OVERLAPPING_RANGES>NO</COMBINE_OVERLAPPING_RANGES>
  <ADDRESS_RANGE>0</ADDRESS_RANGE>
</SUGGESTION_LIST_OPTIONS>
```

Option	Description
<i>ENABLE_SUGGESTION_LISTS</i>	Specifies whether the transform generates suggestion lists. <i>YES</i>: Generates suggestion lists when assignment candidates are present. <i>NO</i>: Does not generate suggestion lists.
<i>MAX_NUMBER_LASTLINES</i>	Specifies the maximum number of lastline suggestions that the transform can generate. Enter a value from 2 through 100. The default setting is 15.  Example Set the <i>MAX_NUMBER_LASTLINES</i> option to limit the size of SOAP documents sent by the web service. Or, set this option to limit the maximum number of suggestions that users have to choose from.
<i>MAX_NUMBER_ADDRESS_LINES</i>	Specifies the maximum number of lastline suggestions that the transform can generate. Enter a value from 2 through 100. The default setting is 50.  Example Set the <i>MAX_NUMBER_ADDRESS_LINES</i> option to limit the size of SOAP documents sent by the web service. Or, set this option to limit the maximum number of suggestions that users have to choose from.

Option	Description
<i>LASTLINES_MATCH_MINIMUM</i>	Specifies the similarity score required for last line suggestions. This score then determines which suggestions the transform returns in the list. Type a value from 0 through 80. A higher number indicates that the suggestion must be more similar to the input to be returned as a possible suggestion.
<i>ADDRESS_LINES_MATCH_MINIMUM</i>	Specifies the similarity score required for address line suggestions. This value then determines which suggestions the transform returns in the list. Type a value from 0 through 80. A higher number indicates that the suggestion must be more similar to the input to be returned as a possible suggestion.
<i>COMBINE_OVERLAPPING_RANGES</i>	Specifies whether the transform combines individual suggestions with overlapping ranges. • <i>YES</i>: Ignores gaps and overlaps in ranges. • <i>NO</i>: Does not combine overlapping ranges. You might set this option to <i>YES</i> to limit the number of total suggestions presented to your user. However, you might not see gaps of invalid ranges that would be apparent if this option was set to <i>NO</i>. ❖ Example A suggestion list shows the following suggestions when you set this option to <i>NO</i>: 1000-1099 Maple Ave 1100-1199 Maple Ave But the suggestion list shows the following suggestion if you set this option to <i>YES</i>: 1000-1199 Maple Ave

Option	Description
ADDRESS_RANGE	Specifies a span around the input primary address range for which to return suggestions. By using this option, you can limit the suggestions returned to be within a few blocks of your input. If you don't want to limit the ranges returned in suggestions, enter 0 for a value. ❖ Example ❖ Example You enter 500 for ADDRESS_RANGE. Then, you submit the following street address: 1000 Pine St. The software returns only suggestions in a range from 750 to 1250 Pine Street.

14.13 Global Address Cleanse Suggestion List option

Instead of using the Global Suggestion List transform, configure the Global Address Cleanse transform's Suggestion List processing option.

Use the suggestion list feature on extracted addresses from an automated process that aren't assigned. Based on the given input address, the Global Address Cleanse transform performs an error-tolerant search in the address directory and returns a list of possible matches. From the suggestion list returned, you can select the correct suggestion and update the database accordingly.

Note

If you use the Canada engine or Global Address engine for Australia and New Zealand, you can't certify your mailing for SERP, AMAS, or New Zealand certification.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)

[Global Address Cleanse reference files \[page 398\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Global Address Cleanse Country ID Options \[page 404\]](#)

[Global Address Cleanse Engines group \[page 409\]](#)

[Global Address Cleanse Country \[page 411\]](#)

[Global Address Cleanse Standardization Options \[page 412\]](#)

[Canada engine \[page 442\]](#)

[Global Address Cleanse Canada group \[page 445\]](#)

[Global Address engine \[page 450\]](#)

[Global Address Cleanse USA engine \[page 469\]](#)

14.13.1 Global Address Cleanse Suggestion List Format options

Select the format options for the suggestion list string that the transform outputs to the *SUGGESTION_LIST* output field.

Syntax

```
<SUGGESTION_LIST>
  <OUTPUT_STYLE>XML</OUTPUT_STYLE>
  <DELIMITER>***</DELIMITER>
  <FIELD_DELIMITER>|</FIELD_DELIMITER>
```

The Global Address Cleanse transform uses settings from each suggestion list group and the suggestion list components group for the content of the *SUGGESTION_LIST* output field.

Option	Description
<i>OUTPUT_STYLE</i>	Specifies the format for the output suggestion list string. <i>DELIMITED</i>: Outputs the suggestion list string in a delimited text format. Uses the delimiters that you specify in the <i>DELIMITER</i> and <i>FIELD_DELIMITER</i> options. <i>XML</i>: Outputs the suggestion list data as hierarchical XML. Use <i>XML</i> when you integrate suggestion lists via the web service. Use your own XML tools to parse the suggestion list data. <i>XML</i> is the default setting.

Option	Description
<i>DELIMITER</i>	Specifies a character to use to separate each suggestion in a suggestion list string. The value can be any character or string. Common delimiters include a pipe symbol (), or a string of multiple asterisks (***). Select a value that is different from the <i>FIELD_DELIMITER</i> value. The transform uses this value only when you set the <i>OUTPUT_STYLE</i> option to <i>DELIMITED</i>.
<i>FIELD_DELIMITER</i>	Specifies a character to use to separate each field in a single suggestion. Each suggestion can be made up of one or more fields. If you choose to retrieve multiple fields per suggestion, the software separates the fields with the character specified here. The value can be any character or string. The default value is a pipe symbol (). Select a delimiter that is different from the <i>DELIMITER</i> value. The transform considers this value only when you set the <i>OUTPUT_STYLE</i> option to <i>DELIMITED</i>.

Related Information

[Global Address Cleanse Suggestion Lists fields \[page 787\]](#)

14.13.2 Global Address Cleanse Suggestion List Components

Select the suggestion list components to include in the [*SUGGESTION_LIST*](#) output field.

```
<SUGGESTION_LIST_COMPONENTS>
  <SELECTION>YES</SELECTION>
  <LOCALITY1>YES</LOCALITY1>
  <LOCALITY2>YES</LOCALITY2>
  <LOCALITY3>YES</LOCALITY3>
  <LOCALITY4>YES</LOCALITY4>
  <LOCALITY1_OFFICIAL>YES</LOCALITY1_OFFICIAL>
  <LOCALITY2_OFFICIAL>YES</LOCALITY2_OFFICIAL>
  <LOCALITY3_OFFICIAL>YES</LOCALITY3_OFFICIAL>
  <LOCALITY4_OFFICIAL>YES</LOCALITY4_OFFICIAL>
  <POSTCODE>YES</POSTCODE>
  <POSTCODE1>YES</POSTCODE1>
  <POSTCODE2>YES</POSTCODE2>
  <REGION1>YES</REGION1>
  <REGION2>YES</REGION2>
```

```

<ISO_COUNTRY_CODE_2_CHAR>NO</ISO_COUNTRY_CODE_2_CHAR>
  <PRIMARY_NUMBER_LOW>YES</PRIMARY_NUMBER_LOW>
  <PRIMARY_NUMBER_HIGH>YES</PRIMARY_NUMBER_HIGH>
  <PRIMARY_NUMBER_DESCRIPTION>YES</PRIMARY_NUMBER_DESCRIPTION>
  <PRIMARY_NUMBER_EXTRA>YES</PRIMARY_NUMBER_EXTRA>
  <PRIMARY_NUMBER_FULL>YES</PRIMARY_NUMBER_FULL>
  <PRIMARY_PREFIX1>YES</PRIMARY_PREFIX1>
  <PRIMARY_PREFIX2>YES</PRIMARY_PREFIX2>
  <PRIMARY_NAME1>YES</PRIMARY_NAME1>
  <PRIMARY_NAME2>YES</PRIMARY_NAME2>
  <PRIMARY_NAME3>YES</PRIMARY_NAME3>
  <PRIMARY_NAME4>YES</PRIMARY_NAME4>
  <PRIMARY_TYPE1>YES</PRIMARY_TYPE1>
  <PRIMARY_TYPE2>YES</PRIMARY_TYPE2>
  <PRIMARY_TYPE3>YES</PRIMARY_TYPE3>
  <PRIMARY_TYPE4>YES</PRIMARY_TYPE4>
  <PRIMARY_POSTFIX1>YES</PRIMARY_POSTFIX1>
  <PRIMARY_POSTFIX2>YES</PRIMARY_POSTFIX2>
  <PRIMARY_NAME_FULL1>YES</PRIMARY_NAME_FULL1>
  <PRIMARY_NAME_FULL2>YES</PRIMARY_NAME_FULL2>
  <PRIMARY_NAME_FULL3>YES</PRIMARY_NAME_FULL3>
  <PRIMARY_NAME_FULL4>YES</PRIMARY_NAME_FULL4>
  <DELIVERY_INSTALLATION_NAME>YES</DELIVERY_INSTALLATION_NAME>
  <DELIVERY_INSTALLATION_QUALIFIER>YES</DELIVERY_INSTALLATION_QUALIFIER>
  <DELIVERY_INSTALLATION_TYPE>YES</DELIVERY_INSTALLATION_TYPE>
  <PRIMARY_SIDE_INDICATOR>YES</PRIMARY_SIDE_INDICATOR>
  <FIRM>YES</FIRM>
  <BUILDING_NAME>YES</BUILDING_NAME>
  <UNIT_DESCRIPTION>YES</UNIT_DESCRIPTION>
  <UNIT_NUMBER_LOW>YES</UNIT_NUMBER_LOW>
  <UNIT_NUMBER_HIGH>YES</UNIT_NUMBER_HIGH>
  <STAIRWELL_DESCRIPTION>YES</STAIRWELL_DESCRIPTION>
  <STAIRWELL_NAME>YES</STAIRWELL_NAME>
  <FLOOR_NUMBER_HIGH>YES</FLOOR_NUMBER_HIGH>
  <FLOOR_NUMBER_LOW>YES</FLOOR_NUMBER_LOW>
  <FLOOR_DESCRIPTION>YES</FLOOR_DESCRIPTION>
  <SECONDARY_SIDE_INDICATOR>YES</SECONDARY_SIDE_INDICATOR>
  <SUGG_FULL_ADDRESSLINE>NO</SUGG_FULL_ADDRESSLINE>
  <SUGG_FULL_LASTLINE>NO</SUGG_FULL_LASTLINE>
  <SUGG_SINGLE_ADDRESS>NO</SUGG_SINGLE_ADDRESS>
  <SUGG_CONFIDENCE_SCORE>NO</SUGG_CONFIDENCE_SCORE>
</SUGGESTION_LIST_COMPONENTS>

```

The components you select here apply to suggestion lists for Global, Canada, and USA engines. For each component, select **YES** or **NO**:

- Select **YES** to include the component in suggestion lists.
- Select **NO** to exclude the component from suggestion lists.

Component	Description
SELECTION	Returns a unique index number that distinguishes this suggestion from the other suggestions in the returned list. The suggestion selection number ranges from 1 to the number of suggestion selections in the suggestion list.
LOCALITY1 LOCALITY2 LOCALITY3 LOCALITY4	Returns the city, town, or suburb. Additional locality information goes in LOCALITY4 .

Component	Description
LOCALITY1_OFFICIAL LOCALITY2_OFFICIAL LOCALITY3_OFFICIAL LOCALITY4_OFFICIAL	Returns the locality name that the applicable postal authority prefers.
POSTCODE	Returns the postal code. For USA: Returns the 5-digit ZIP Code and ZIP+4.
POSTCODE1	Returns information specific to the country: • Australia: Returns 4-digit postcode • Canada: Returns the first three characters (FSA) of the postal code. • Global: Returns the postal code. • USA: Returns the 5-digit primary postal code (ZIP Code). Does not include the 4-digit secondary postal code (ZIP4).
POSTCODE2	Returns the secondary postal code. • Canada: Returns the last three characters (LDU) of the postal code. • USA: Returns the 4-digit ZIP Code (ZIP4). The ZIP4 follows the primary postal code with a hyphen placed between, such as 54601-1234.
REGION1 REGION2	Returns the state, province, or region.
ISO_COUNTRY_CODE_2_CHAR	Returns the two-character ISO country code.
PRIMARY_NUMBER_LOW PRIMARY_NUMBER_HIGH	Returns the low and high primary number. ❖ Example If the house number is a range, such as 100-102: • PRIMARY_NUMBER_LOW: 100 • PRIMARY_NUMBER_HIGH: 102 If the house number is not a range, both fields contain the house number: • PRIMARY_NUMBER_LOW: 100 • PRIMARY_NUMBER_HIGH: 100
PRIMARY_NUMBER_DESCRIPTION	Returns a description that precedes the primary number. For example, LOT (Australia) 99, Building 10, or blk. 101.

Component	Description
<i>PRIMARY_NUMBER_EXTRA</i>	Returns data that is near the parsed primary number in the address that the transform cannot identify or that does not belong in a standardized address.
<i>PRIMARY_NUMBER_FULL</i>	Returns the primary number, primary number description, and primary number extra, when they exist.
<i>PRIMARY_PREFIX1-2</i>	Returns the abbreviated or full directional that precedes a street name. For example, N, South, NW, SE. The transform uses your setting for <i>DIRECTIONAL_STYLE</i> to determine whether to return abbreviated or non-abbreviated directional.
<i>PRIMARY_NAME1</i>	Returns the street name description, such as a street name or box description.
<i>PRIMARY_NAME2</i>	Returns the second street name and description, such as a street name or box description.
<i>PRIMARY_NAME3</i> <i>PRIMARY_NAME4</i>	Returns the street name, delivery mode, and so on.
<i>PRIMARY_TYPE1-4</i>	Returns the primary name type. For example, rue, strasse, street, Ave, or Pl.
<i>PRIMARY_POSTFIX1-2</i>	Returns an abbreviated or non-abbreviated directional that follows a street name. For example, N, South, NW, SE. Abbreviated or non-abbreviated is based on the standardization setting for <i>DIRECTIONAL_STYLE</i>.
<i>PRIMARY_NAME_FULL1</i> <i>PRIMARY_NAME_FULL2</i>	Returns the primary name, primary type, primary prefix, and primary postfix.
<i>PRIMARY_NAME_FULL3</i> <i>PRIMARY_NAME_FULL4</i>	Returns the primary name and primary type.
<i>DELIVERY_INSTALLATION_NAME</i>	Returns the delivery installation city name, which is the same as the city name in most cases. If it is the same as the city name, the transform omits it from the address line.
<i>DELIVERY_INSTALLATION_QUALIFIER</i>	Returns the delivery installation qualifier.

❖ Example

Input: RR2 Vancouver Stn Main

Output: Main

Component	Description
<i>DELIVERY_INSTALLATION_TYPE</i>	Returns the delivery installation type. USA: • PO: Post Office • RPO: Retail Post Outlet • STN: Station • LCD: Letter Carrier Depot • CMC: Community Mail Center • CDO: Commercial Dealership Outlet France: • BDP: Bureau de Poste • CSP: Comptoir Service Postal • SUCC: Succursale • PDF: Poste de Facteurs • CPC: Centre Postal Communautaire • CC: Concession Commerciale
<i>PRIMARY_SIDE_INDICATOR</i>	Returns the valid primary side indicator. The value applies to Street and PO box. E: The record is even-numbered. O: The record is odd-numbered. B: The record covers both the even- and odd-numbered sides of the street and PO Boxes.
<i>FIRM</i>	Returns the firm, company, or organization name.
<i>BUILDING_NAME</i>	Returns the building name for the address, which in some countries is used in place of the primary number. ❖ Example Input address from the U.K.: White House, High Street With <i>BUILDING_NAME</i> set to <i>YES</i>, the output is "White House." When set to <i>NO</i>, the transform outputs the primary number, "100 High Street."
<i>UNIT_DESCRIPTION</i>	Returns the unit description within a building. For example, Room, Unit, Apt, and Suite.

Component	Description
UNIT_NUMBER_LOW UNIT_NUMBER_HIGH	If the unit number is a range, returns the low and high unit number. ❖ Example With a unit number range of 1-20: - UNIT_NUMBER_LOW = 1 - UNIT_NUMBER_HIGH = 20
STAIRWELL_DESCRIPTION	Returns the entrance or stairwell identifier for a building. For example, Stiege.
STAIRWELL_NAME	Returns the name or number of an entrance or stairwell for a building. For example, Stiege "1."
FLOOR_NUMBER_LOW FLOOR_NUMBER_HIGH	If the floor number is a range, returns the low and high number. ❖ Example with the range 20–22: - FLOOR_NUMBER_LOW = 20 - FLOOR_NUMBER_HIGH = 22 If the floor number is not a range, both fields contain the floor number. - FLOOR_NUMBER_LOW = 20 - FLOOR_NUMBER_HIGH = 20
FLOOR_DESCRIPTION	Returns the level description, such as Floor.
SECONDARY_SIDE_INDICATOR	Returns the secondary record side indicator. This value applies to floors and units: E: The secondary record is even-numbered. O: The secondary record is odd-numbered. B: The secondary record covers both the even and the odd-numbered values.
SUGG_RANGE_TYPE	Indicates the range type for each suggestion: PRIM = Suggestion represents a range of house numbers. UNIT = Suggestion represents a range of unit numbers. FLOOR = Suggestion represents a range of floor numbers. <i>blank</i> = Suggestion does not include a range.

Component	Description
<i>SUGG_FULL_ADDRESSLINE</i>	Returns the complete address line, including firm name, secondary address, and dual address (street and postal) as appropriate for the country. If the input address line uses a street name alias, includes the input street name alias in parentheses. ❖ Example Given the input 250 Bliss Rd: <i>SUGG_FULL_ADDRESSLINE</i> = 250 MAIN ST (BLISS RD) If the input address contains a firm name, includes the firm name in parentheses after the address line. ❖ Example Given the input: ABC, Inc Main St <i>SUGG_FULL_ADDRESSLINE</i> = 100 MAIN ST (ABC,INC)
<i>SUGG_FULL_LASTLINE</i>	Returns the locality, region, and postal code together in one component as appropriate for the country. Additionally, if you select <i>YES</i> for <i>ENABLE_SUGGESTION_LISTS</i> in the <i>COUNTRY_ID_OPTIONS</i> group, outputs the two-character ISO country code that you select from the list of candidate countries.
<i>SUGG_SINGLE_ADDRESS</i>	Returns the combined result of the full address line and full last line in the order appropriate for the country. Additionally, if you select <i>YES</i> for <i>ENABLE_SUGGESTION_LISTS</i> in the <i>COUNTRY_ID_OPTIONS</i> group, outputs the two-character ISO country code that you select from the list of candidate countries.
<i>SUGG_CONFIDENCE_SCORE</i>	Returns the confidence score for the suggestion entry. Values are from 0 through 100.

15 Data Cleanse transform

Use the Data Cleanse transform to parse and format custom or person and firm data as well as phone numbers, dates, e-mail addresses, and Social Security numbers.

Data Cleanse information

Characteristic	Description
Use	The Data Cleanse transform identifies and isolates specific parts of mixed data. It then parses and formats the data based on the referenced cleansing package and options that you set directly in the transform. Use Data Cleanse to assign gender and prenames to name data and to generate Match standards for all types of data. The transform has options to enable generation of data quality statistics tables in the repository. These statistics provide insight into how the software cleansed and assigned your data by generating aggregated summary statistics and specific record-level statistics. You specify a cleansing package in the transform. The cleansing package that you specify in the transform defines how the Data Cleanse transform parses and standardizes your data. Note The Social Security number algorithm is no longer updated by the Social Security Administration. The SSN validation performed in Data Cleanse is based on data through July, 2011. SAP builds the Social Security data into the cleansing package. Within a data flow, place the Data Cleanse transform after the address cleansing process and before the matching process.
Content objects	Sample XML file.

[Rank and prioritize parsing engines \[page 483\]](#)

Configure the Data Cleanse transform to use specific parsing engines and to specify the order in which the parsing engines run.

[Data parsing \[page 484\]](#)

The Data Cleanse transform parses fields by identifying and isolating a wide variety of data, ranging from person names to inventory items.

[Data standardization \[page 497\]](#)

The Data Cleanse transform standardizes input data so that it has a consistent format.

[Gender descriptions and prename assignment \[page 497\]](#)

The Data Cleanse transform assigns a gender description to each name and populates the generated output field *GENDER* with the assigned gender.

[Prepare records for matching \[page 499\]](#)

To increase the accuracy of the matching processing, process data with the Data Cleanse transform before the Match transform.

[Domains \[page 500\]](#)

Domains enhance the ability of Data Cleanse to parse and standardize data appropriately according to the cultural standards of a region.

[Output formats \[page 507\]](#)

The Data Cleanse transform uses specific output fields based on the specified domain, and formats the data in those fields according to the regional standards.

15.1 Rank and prioritize parsing engines

Configure the Data Cleanse transform to use specific parsing engines and to specify the order in which the parsing engines run.

Before you rank and prioritize parsing engines, consider your data content and carefully consider your parser requirements. For example, to improve parsing speed and reduce the chances for errors, turn off parsers that you don't need.

The Data Cleanse transform allows you to change parser order for a specific multiline input by modifying the corresponding parser sequence option in the *Parser Configuration* options group.

❖ Example

To change the order of parsers for the *MULTILINE1* input field, modify the *PARSER_SEQUENCE_MULTILINE1* option. Use a pipe (|) delimiter to separate the values of the parsers.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Data parsing \[page 484\]](#)

[Data standardization \[page 497\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

[Domains \[page 500\]](#)

[Output formats \[page 507\]](#)

15.2 Data parsing

The Data Cleanse transform parses fields by identifying and isolating a wide variety of data, ranging from person names to inventory items.

In the Data Cleanse transform, map the input fields in your data to the appropriate input fields in the transform.

Note

Map the following data to either discrete or multiline input fields as applicable:

- Person
- Firm
- Phone
- Date
- E-mail
- Social Security number

Example

The following example shows how the Data Cleanse transform parses product data from a multiline input field and displays it in discrete output fields. You can also configure the transform to display data in composite fields. For example, customize the composite field Standard_Description in Cleansing Package Builder to meet your needs.

Input data	Parsed data	
Glove ultra grip profit 2.3 large black synthetic leather elastic with Velcro Mechanix Wear	Product Category	Glove
	Size	Large
	Material	Synthetic Leather
	Trademark	Pro-Fit 2.3 Series
	Cuff Style	Elastic Velcro
	Palm Type	Ultra-Grip
	Color	Black
	Vendor	Mechanix Wear
	Standard Description	Glove - Synthetic Leather, Black, size: Large, Cuff Style: Elastic Velcro, Ultra-Grip, Mechanix Wear

Example

The following examples show how Data Cleanse parses name and firm data and displays it in discrete output fields. You can also display the data in composite fields that you customize in Cleansing Package Builder to meet your needs.

Input data	Parsed data		Composite data
Ms Mary Ann Smith Jones, CPA Account Mgr. Jones Inc.	Prename	Ms.	
	Given Name 1	Mary	Given_Name_Full
	Given Name 2	Anne	Mary Anne
	Family Name 1	Smith	Family_Name_Full
	Family Name 2	Jones	Smith Jones
	Honorary Postname	CPA	
	Title	Account Mgr.	
	Firm	Jones, Inc.	
Input data	Parsed data		
<First_last_name> Jr. xxx-xx-5678 e-mail@address 910-111-2131 Aug 20, 2003	Given Name 1	<first>	
	Family Name 1	<last>	
	Maturity Postname	Jr.	
	Social Security	xxx-xx-5678	
	Email Address	e-mail@address	
	Phone	910.111.2131	
	Date	August 20, 2003	

The Data Cleanse transform parses up to six names per record and two names per input field. For all six names the transform finds, it parses components such as prename, given names, family name, and postname. Then it sends the data to individual fields.

The transform also parses the following fields:

- Up to six job titles per record
- Up to six firm names per record
- One firm name per input field

[Parse phone numbers \[page 486\]](#)

When Data Cleanse parses a phone number, it outputs the individual components of the number into the appropriate output fields.

[Parse dates \[page 490\]](#)

The Data Cleanse transform recognizes dates in a variety of date formats and parses dates into components.

[Parse Social Security numbers \[page 490\]](#)

The Data Cleanse transform parses U.S. Social Security numbers (SSNs) that are either by themselves or on an input line surrounded by other text.

[Parse e-mail addresses \[page 492\]](#)

The Data Cleanse transform parses e-mail addresses that are input in discrete fields or combined with other data in a multiline field.

[Parse street addresses \[page 493\]](#)

The Data Cleanse transform doesn't identify and parse address components, therefore, process your data with either the Global Address Cleanse transform or the USA Regulatory Address Cleanse transform.

[Parse user-defined patterns \[page 493\]](#)

The Data Cleanse transform parses any alphanumeric sequence for which you define a pattern.

[Parse firm names \[page 493\]](#)

The Data Cleanse transform parses firm names that are alone in a field, or when they're together with other data.

[Parse name and title data \[page 494\]](#)

The Data Cleanse transform parses name and title data from input into discrete parts.

[One-to-one mapping \[page 494\]](#)

The Data Cleanse transform uses one-to-one mapping, where it outputs data parsed from certain discrete input fields to their corresponding output fields.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

[Data standardization \[page 497\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

[Domains \[page 500\]](#)

[Output formats \[page 507\]](#)

15.2.1 Parse phone numbers

When Data Cleanse parses a phone number, it outputs the individual components of the number into the appropriate output fields.

Phone numbering systems differ around the world. Data Cleanse parses both North American Numbering Plan (NANP) phone numbers and international phone numbers.

Parsing in Data Cleanse is most efficient when phone numbers have a valid country code that appears before the phone number. If the country code isn't present, or it isn't first in the string, Data Cleanse uses other resources to attempt to parse the phone number.

When a valid country code isn't present, Data Cleanse parses phone numbers by first searching internationally. To identify the country code, Data Cleanse uses ISO2 country codes, when available, along with the patterns defined in the cleansing package for each country. If Data Cleanse encounters a country that participates in the NANP, it automatically stops trying to parse the number as international. Then, it attempts to parse the phone number as a number from North America, comparing the phone number to commonly used patterns such as (234) 567-8901, 234-567-8901, and 2345678901.

Currently, the participating countries in the NANP include:

- AS - American Samoa
- AI - Anguilla
- AG - Antigua and Barbuda
- BS- The Bahamas
- BB - Barbados
- BM - Bermuda
- VG - British Virgin Islands
- CA - Canada
- KY - Cayman islands
- DM - Dominica
- DO - Dominican Republic
- GD - Grenada
- GU - Guam
- JM - Jamaica
- MS - Montserrat
- MP - Northern Mariana Islands
- PR - Puerto Rico
- KN - Saint Kitts and Nevis
- LC - Saint Lucia
- VC - Saint Vincent and the Grenadines
- SX - Saint Maarten
- TT - Trinidad and Tobago
- TC - Turks and Caicos Islands
- US - United States
- VI - United States Virgin Islands

Data Cleanse provides options for formatting North American numbers on output, such as your choice of delimiters. However, Data Cleanse outputs international numbers as they're input, without any formatting. Also, Data Cleanse doesn't cross-compare to the address to see whether the country and city codes in the phone number match the address.

[How Data Cleanse parses phone numbers \[page 487\]](#)

To align with how Data Cleanse parses phone data, set up international phone parsing in the Data Cleanse transform at specific levels.

[Adding ISO2 to your data \[page 490\]](#)

When your data is missing the ISO2 country code, use the Global Address Cleanse transform to add the code to your data.

15.2.1.1 How Data Cleanse parses phone numbers

To align with how Data Cleanse parses phone data, set up international phone parsing in the Data Cleanse transform at specific levels.

The specific levels, in order, include the following:

- Record
- Job file
- Global

If the level parse fails, Data Cleanse parses the phone number as a North American phone number.

During phone parsing, Data Cleanse attempts to parse any number that contains a North American Numbering Plan (NANP) country code as a North American phone number. It skips any remaining levels.

i Note

Ensure that you specify the country in the cleansing package so that Data Cleanse attempts to parse the phone numbers from that country.

Record level parsing

During record level parsing, Data Cleanse attempts to parse the phone number based on the ISO2 country code. If your phone data is missing the ISO2 country code, follow the steps in [Adding ISO2 to your data \[page 490\]](#) to add the code to your data.

i Note

Parsing at the record level is optional.

Advantages of record level parsing

If Data Cleanse parses a phone number at the record level, it continues parsing phone data from the next record. If the transform isn't able to parse phone data at the record level, it moves to the job file level to attempt to make a match and parse the phone data.

If the transform recognizes that the country code is from the NANP, it does the following:

- Stops international parsing
- Skips the job file and global levels
- Attempts to parse the phone data as a North American phone number

Job-file level parsing

Data Cleanse attempts to parse phone data using a sequence of countries in the *ISO2_COUNTRY_CODE_SEQUENCE* option in the *Phone Options* group.

i Note

Phone parsing at the job file level is optional.

Set up a country sequence to specify the order in which Data Cleanse searches for phone data. When you set a sequence:

- Data Cleanse attempts to parse phone data in the first country in the sequence.
- If Data Cleanse doesn't find a match in the first country, it moves to the next country in the sequence.
- Data Cleanse continues trying to match until it finds a matching country.

GLOBAL is the default setting for the sequence. Ensure that you set *GLOBAL* as the last entry in the country sequence, or, optionally delete *GLOBAL* from the sequence.

Advantages of job-file level parsing

If you know that your data is predominately from specific countries, set your sequence using those countries only, and remove *GLOBAL*. Paring down the list of countries reduces the number of countries that Data Cleanse has to search for a match.

If the country code is from the NANP, Data Cleanse stops the international parsing, skips the *GLOBAL* level, and attempts to parse the phone data as a North American phone number.

Data Cleanse attempts to parse the phone data in the record as a North American phone number under the following circumstances:

- You delete *GLOBAL* from the sequence.
- Data Cleanse didn't find a match to the countries in the sequence.

To skip this level of phone parsing, leave the *ISO2_COUNTRY_CODE_SEQUENCE* option blank. Data Cleanse then attempts to parse the phone data at the global level.

Global level parsing

Data Cleanse attempts to parse phone data using *GLOBAL* as the country.

i Note

Parsing phone data at the global level is optional.

There are two ways to set up for the global level:

- Use only *GLOBAL* in the *ISO2_COUNTRY_CODE_SEQUENCE*.
- Delete *GLOBAL* from the first *ISO2_COUNTRY_CODE_SEQUENCE* and then include *GLOBAL* in the next sequence.

Advantage of global level parsing

The global level helps when you're unsure of the countries represented in your data. Global is also helpful if your data doesn't have the ISO2 country codes. At the global level, the transform searches all countries that are listed in the cleansing package, except the NANP countries.

i Note

The global level doesn't include country codes as search criteria.

If the country code is from the NANP, Data Cleanse stops the global parsing and attempts to parse the phone data as a North American phone number.

If Data Cleanse still can't parse the phone data after the global level, it attempts to parse the phone data as a North American phone number.

15.2.1.2 Adding ISO2 to your data

When your data is missing the ISO2 country code, use the Global Address Cleanse transform to add the code to your data.

The Data Cleanse transform starts parsing phone data at the record level using the ISO2 country code. The Global Address Cleanse transform uses the record address to determine the ISO2 code, therefore, there is a good chance that the phone number is from the same country as the address.

To add the ISO2 country code to your data, perform the following steps:

1. Create a data flow and position a Global Address Cleanse transform before the Data Cleanse transform.
2. Open the Global Address Cleanse transform and set to assign the ISO2 country code to each record.
3. Include the output field *ISO_COUNTRY_CODE_2CHAR*, which contains the ISO2 country code.
4. Open the Data Cleanse transform and map the *ISO_COUNTRY_CODE_2CHAR* to the dynamic input field, *OPTION_COUNTRY*, in your input mapping.

15.2.2 Parse dates

The Data Cleanse transform recognizes dates in a variety of date formats and parses dates into components.

The Data Cleanse transform parses up to six dates from your defined record. That is, Data Cleanse identifies up to six dates in the input, breaks those dates into components, and makes dates available as output in either the original format or a user-selected standard format.

15.2.3 Parse Social Security numbers

The Data Cleanse transform parses U.S. Social Security numbers (SSNs) that are either by themselves or on an input line surrounded by other text.

The Data Cleanse transform outputs the individual components of a parsed Social Security number: The entire SSN, the area, the group, and the serial. The transform parses Social Security numbers in two steps:

1. Identifies a potential SSN by looking for patterns.
2. Performs a validity check on the first five digits only.

Pattern	Digits per grouping	Delimited by
nnnnnnnnnn	9 consecutive digits	N/A
nnn nn nnnn	3, 2, and 4 (for area, group, and serial)	spaces
nnn-nn-nnnn	3, 2, and 4 (for area, group, and serial)	all supported delimiters

The following table contains descriptions of the validity test outcomes.

Outcome	Description
Pass	Data Cleanse successfully parses the data and outputs the Social Security number to an SSN output field.
Fail	Data Cleanse doesn't parse the data because it isn't a valid Social Security number. It outputs the data as extra, unparsed data.

i Note
Data Cleanse uses the U.S. government definition of a Social Security number to determine validity.

Check validity

When performing a validity check, the Data Cleanse transform doesn't verify that a particular 9-digit Social Security number has been issued, or that it's the correct number for any named person. Instead, it validates only the first five digits (area and group). Data Cleanse doesn't validate the last four digits (serial), except to confirm that they are digits.

Output valid SSNs

The Data Cleanse transform outputs only Social Security numbers that pass its validation. If an SSN fails validation, the transform doesn't output the number. Instead, it passes the number as a parsed, but invalid, Social Security number.

Social Security Administration data

The Data Cleanse transform validates the first five digits based on a table from the Social Security Administration (http://www.socialsecurity.gov/employer/ssns/HGJune2411_final.txt). The rules and data that guide this check are available at <http://www.ssa.gov/history/ssn/geocard.html>. The Social Security number information that Data Cleanse references is included in the cleansing package.

i Note

The Social Security Administration no longer updates the table. The last time it was updated was in July 2011. Therefore, Social Security validation performed by the Data Cleanse transform is based on data through July 2011. For more information about the Social Security Administration number assignment process, see <http://www.ssa.gov/employer/randomizationfaqs.html>.

15.2.4 Parse e-mail addresses

The Data Cleanse transform parses e-mail addresses that are input in discrete fields or combined with other data in a multiline field.

After the Data Cleanse transform recognizes input data as an e-mail address, it parses the data and places the components into specific fields for output.

To parse e-mail data, the Data Cleanse transform looks at the relationships of the data components. It identifies the following data components in e-mail data, when present:

- User name
- Domain name
- Top domain and up to four more domains
- Host name

When parsing e-mail addresses, the Data Cleanse transform can take the following actions:

- Parse an e-mail address located either in a discrete field or combined with other data in a multiline field.
- Break down the domain name into subelements.
- Verify that an e-mail address is properly formatted.
- Flag the address as belonging to an internet service provider (ISP).

What the Data Cleanse transform doesn't verify

When parsing e-mail addresses, the Data Cleanse transform doesn't verify the following aspects of an e-mail address:

- Whether the domain name (the portion to the right of the @ sign) is registered.
- Whether an e-mail server is active at that address.
- Whether the user name (the portion to the left of the @ sign) is registered on that e-mail server (if any).
- Whether the personal name in the record can be reached at this e-mail address.

E-mail components

The output field where the Data Cleanse transform places the data depends on the position of the data in the record. The transform follows the Domain Name System (DNS) in determining the correct output field.

❖ Example

An input record contains the following data: expat@london.home.office.city.co.uk. During processing, Data Cleanse outputs the data as described in the following table.

Output field	Output value
Email	expat@london.home.office.city.co.uk

Output field	Output value
Email_User	expat
Email_Domain_All	london.home.office.city.co.uk
Email_Domain_Top	uk
Email_Domain_Second	co
Email_Domain_Third	city
Email_Domain_Fourth	office
Email_Domain_Fifth	home
Email_Domain_Host	london

15.2.5 Parse street addresses

The Data Cleanse transform doesn't identify and parse address components, therefore, process your data with either the Global Address Cleanse transform or the USA Regulatory Address Cleanse transform.

Place the Global Address Cleanse or USA Regulatory Address Cleanse transform to the left of, or upstream from, the Data Cleanse transform in the data flow.

i Note

If you send input address data through the Data Cleanse transform, the transform usually outputs the data to the Extra output fields.

15.2.6 Parse user-defined patterns

The Data Cleanse transform parses any alphanumeric sequence for which you define a pattern.

The transform uses the user-defined pattern matching (UDPM) parser to parse patterns in data, such as part numbers, invoice numbers, and driver licenses. The UDPM parser looks for the pattern across each entire field.

15.2.7 Parse firm names

The Data Cleanse transform parses firm names that are alone in a field, or when they're together with other data.

When a firm name ends in an "S", the Data Cleanse transform doesn't parse the word. However, if the firm name ends in an apostrophe and an "S", the transform parses the word as shown in the following example:

❖ Example

The Data Cleanse transform parses the firm name "Macy's" by breaking it into three tokens:

MACY

,

S

Data Cleanse then parses as follows:

- Combines the three tokens and looks up the full term “MACY'S”.
- If it can't find the term, it looks up the term “MACYS”.
- If it can't find the term, it keeps the three tokens together as “MACY'S”.

It then adds the FIRM_MISCELLANEOUS classification to the term.

Since the transform keeps words ending with S together automatically, you don't have to add the possessive form of firm names to the dictionary.

15.2.8 Parse name and title data

The Data Cleanse transform parses name and title data from input into discrete parts.

A name can consist of the following parts:

- Prenom
- Given name
- Family name
- Postname
- Other name parts

The Data Cleanse transform accepts up to two names and titles as discrete components. It also accepts name and title data together with other data or alone in a field. The name line or multiline field can contain one or two names per field.

15.2.9 One-to-one mapping

The Data Cleanse transform uses one-to-one mapping, where it outputs data parsed from certain discrete input fields to their corresponding output fields.

[ONE_TO_ONE_MAPPING](#) is an option in the Data Cleanse transform that controls how several parsers output the data.

The [ONE_TO_ONE_MAPPING](#) option is available for the following parsers:

- Date
- Email
- Phone

When you set the [ONE_TO_ONE_MAPPING](#) option to [YES](#) (enabled), the Data Cleanse transform outputs the data, parsed from certain discrete input fields, to their corresponding output fields. The output fields are “reserved” for parses from certain discrete input fields. When you enable the [ONE_TO_ONE_MAPPING](#) option,

SAP Data Services more clearly shows the mapping of the input field that held the original data based on the parsed data in the output field.

❖ Example

If the input data in *PHONE1-5* are blank, and the *PHONE6* field contains data, the transform outputs *PHONE1-5* as blank and outputs the parsed data to *PHONE6*.

When you enable *ONE_TO_ONE_MAPPING*, then all parsers that use this option are set to *YES*. For example, you can't set the *ONE_TO_ONE_MAPPING* option only for the phone parser.

When you set *ONE_TO_ONE_MAPPING* to *No*, the Data Cleanse transform parses and outputs the data in the order the data entered the parser. The transform doesn't necessarily output the data to the same field that you mapped it to on output. The transform outputs the data to the first available field in the category.

i Note

The following examples show date fields. The same behavior applies to phone and e-mail fields.

❖ Example

The following table compares date parsing when the *ONE_TO_ONE_MAPPING* option is set to *NO* and to *YES*.

Field	Input data	Output data when option is <i>No</i>	Output data when option is <i>Yes</i>
Date1	<blank>	1968/01/01	<blank>
Date2	1968/01/01	1968/02/02	1968/01/01
Date3	1968/02/02	1968/03/03	1968/02/02
Date4	<blank>	1968/04/04	<blank>
Date5	1968/03/03	<blank>	1968/03/03
Date6	1968/04/04	<blank>	1968/04/04

Multiline fields

The transform parses the discrete date, e-mail, and phone fields before the multiline fields, so that any unreserved fields can contain data from the multiline fields.

❖ Example

The following table compares discrete date parsing when the *ONE_TO_ONE_MAPPING* option is set to *NO* and to *YES*.

Field	Input data	Output data when option is <i>No</i>	Output data when option is <i>Yes</i>
Date1	<blank>	1968/01/01	<blank>

Field	Input data	Output data when option is <i>No</i>	Output data when option is <i>Yes</i>
Date2	1968/01/01	1968/02/02	1968/01/01
Date3	<blank>	1968/03/03	<blank>
Date4	1968/02/02	1968/04/04	1968/02/02
Date5	<blank>	<blank>	1968/03/03 (not reserved, so Multiline input can be added here)
Date6	<blank>	<blank>	1968/04/04 (not reserved, so Multiline input can be added here)
Multiline1	1968/03/03 1968/04/04	<blank>	<blank>

Extra fields

When the *ONE_TO_ONE_MAPPING* option is set to *YES* and the input field contains multiple sets of data, only the first set of data is placed in the corresponding output field. All other sets of data are put in the Extra field.

❖ Example

Field	Input data	Output data when option is <i>No</i>	Output data when option is <i>Yes</i>
Date1	1968/01/01 1968/02/02 1968/03/03 1968/04/04 1968/05/05 1968/06/06 1968/07/07 1968/08/08	1968/01/01	1968/01/01
Date2	<blank>	1968/02/02	<blank>
Date3	<blank>	1968/03/03	<blank>
Date4	<blank>	1968/04/04	<blank>
Date5	<blank>	1968/05/05	<blank>
Date6	<blank>	1968/06/06	<blank>
Extra	<blank>	1968/07/07 1968/08/08	1968/02/02 1968/03/03 1968/04/04 1968/05/05 1968/06/06 1968/07/07 1968/08/08

15.3 Data standardization

The Data Cleanse transform standardizes input data so that it has a consistent format.

Use the Cleansing Package Builder so that the Data Cleanse transform defines standard formats for individual variations of data. Additionally, the transform standardizes data to make the format more consistent.

The Data Cleanse transform standardizes data using the following data characteristics: Case, punctuation, and abbreviations.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

[Data parsing \[page 484\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

[Domains \[page 500\]](#)

[Output formats \[page 507\]](#)

15.4 Gender descriptions and prename assignment

The Data Cleanse transform assigns a gender description to each name and populates the generated output field **GENDER** with the assigned gender.

Each variation in a cleansing package has a gender associated with it. By default, the gender is unassigned. You can assign a gender to a variation in the Advanced mode of the Cleansing Package Builder. Variations in the SAP-supplied name and firm cleansing packages are already assigned genders.

The intelligence behind gender assignment lies partly in the application and partly in the cleansing package.

The gender descriptions output from the Data Cleanse transform include the following:

- Strong male
- Strong female
- Weak male
- Weak female
- Ambiguous

The Prenome output field always includes prenames that are part of the input name data. Additionally, when the **ASSIGN_PRENAMES** option is set to **YES**, Data Cleanse populates the **PRENAME** output field when a strong male or strong female gender is assigned to a variation.

Dual names

For dual names, the Data Cleanse transform offers four additional gender descriptions:

- Female multi-name
- Male multi-name
- Mixed multi-name
- Ambiguous multi-name

The transform generates the dual name genders based on the assigned genders of the two names. The following table shows how Data Cleanse assigns multi-name genders.

Dual name	First name gender	Second name gender	Dual name assigned gender
Bob and Sue Jones	strong male	strong female	mixed multi-name
Bob and Tom Jones	strong male	strong male	male multi-name
Sue and Sara Jones	strong female	strong female	female multi-name
Bob and Pat Jones	strong male	ambiguous	ambiguous multi-name

Chinese and Japanese given names

The Data Cleanse transform parses a give name by combining given name characters intelligently as opposed to including the given name as a variation in the cleansing package.

The transform generates the gender by combining the gender of the individual characters that make up the given name.

❖ Example

When the individual characters have the following gender assignment, the resulting gender is **Strong Female**:

- Strong female = strong female
- Strong male = ambiguous
- Weak female = strong female
- Weak male = ambiguous
- Ambiguous = strong female

The following table lists the gender followed by the information that the transform uses to determine the gender.

	Strong female	Strong male	Weak female	Weak male	Ambiguous
Strong female	strong female	ambiguous	strong female	ambiguous	strong female
Strong male	ambiguous	strong male	ambiguous	strong male	strong male
Weak female	strong female	ambiguous	weak female	ambiguous	weak female

	Strong female	Strong male	Weak female	Weak male	Ambiguous
Weak male	ambiguous	strong male	ambiguous	weak male	weak male
Ambiguous	strong female	strong male	weak female	weak male	ambiguous

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

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[Data standardization \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

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15.5 Prepare records for matching

To increase the accuracy of the matching processing, process data with the Data Cleanse transform before the Match transform.

Preprocessing with Data Cleanse standardizes your data consistently, which increases the accuracy of your matches.

The Data Cleanse transform generates Match Standards or alternates for many name and firm fields as well as all custom output fields. For example, Data Cleanse can identify Patrick and Patricia as potential matches for the name Pat. Match standards can help you overcome two types of matching problems:

- Alternate spellings such as Catherine and Katherine
- Nicknames such as Pat for Patrick

The following example shows how Data Cleanse prepares records for matching.

❖ Example

Data Source 1

Input record	Cleansed record	
Intl Marketing, Inc.	Given Name 1	Pat
Pat Smith, Accounting Mgr.	Match Standards	Patrick, Patricia
	Given Name 2	
	Family Name 1	Smith

Input record	Cleansed record	
	Title	Accounting Mgr.
	Firm	Intl. Mktg, Inc.

Notice that Data Source 1 has Patrick and Patricia as Match Standards.

Data Source 2

Input record	Cleansed record	
Smith, Patricia R.	Given Name 1	Patricia
International Marketing, Incorp.	Match Standards	
	Given Name 2	R
	Family Name 1	Smith
	Title	
	Firm	Intl. Mktg, Inc.

Notice that there is no Match Standards output for the multi-word firm name for Data Source 2.

When a cleansing package doesn't include an alternate, the Match Standards output field for that term is empty. When none of the variations in a multi-word output has an alternate, such as a firm name, the Match Standards output is empty. However, if at least one variation has an alternate associated with it, the cleansing package populates Match Standards with the variation alternate.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

[Data parsing \[page 484\]](#)

[Data standardization \[page 497\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

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15.6 Domains

Domains enhance the ability of Data Cleanse to parse and standardize data appropriately according to the cultural standards of a region.

Data Cleanse includes a Person-Firm cleansing package that supports a variety of regions, where each supported locale is a domain.

The following table contains examples that show how name parsing can vary by domain.

Domain	Name	Parsed Output			
		Given_Name1	Given_Name2	Family_Name1	Family_Name2
Spanish	Juan C. Sánchez	Juan	C.	Sánchez	
Portuguese	João A. Lopes	João		A.	Lopes
French	Jean Christophe Rousseau	Jean Christophe		Rousseau	
German	Hans Joachim Müller	Hans	Joachim	Müller	
English (U.S. and Canada)	James Andrew Smith	James	Andrew	Smith	

Data Cleanse automatically assigns each variation to the Global domain, which contains all domain variations and associated properties. Data Cleanse can also assign a variation to one or more other domains. The properties associated with a variation, such as standard form and classification, can be specific to a domain.

❖ Example

The variation "AG" has different meanings in German and in U.S. English.

- In German, "AG" is an abbreviation for "Aktiengesellschaft" (a firm type) and is cleansed as "AG"
- In English "AG" is an abbreviation for "Agriculture" and is cleansed as "Ag."

You control how Data Cleanse parses and standardizes your data by specifying the domain or domains to apply to data, and the order.

i Note

SAP Information Steward version 4.1 and higher supports multiple domains only in person and firm cleansing packages. For versions earlier than 4.1, Information Steward assigns variations in custom cleansing packages as well as person and firm cleansing packages only to the Global domain.

[Global domain \[page 502\]](#)

The Global domain is a special content domain that contains all variations and their associated properties.

[Cleanse domain-specific data \[page 502\]](#)

Control how Data Cleanse parses domain-specific data, such as name, gender, and title.

[Specify the content domain sequence \[page 504\]](#)

SAP Data Services allows you to set three levels of content domain sequence: Default, Intermediate, and Top.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

[Data parsing \[page 484\]](#)

[Data standardization \[page 497\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

[Output formats \[page 507\]](#)

15.6.1 Global domain

The Global domain is a special content domain that contains all variations and their associated properties.

If a variation isn't associated with domain-specific information, the Global domain serves as the default domain.

When you add a variation, the transform initially adds it to the Global domain. As necessary, you can add it to other domains to assign domain-specific information. Add a variation to a domain other than the Global domain only when the variation has properties such as gender, classification, and standard form. These properties differ between the Global and other domains.

If you delete a variation from the Global domain, the software also deletes the variation from all other associated domains.

Parent topic: [Domains \[page 500\]](#)

Related Information

[Cleanse domain-specific data \[page 502\]](#)

[Specify the content domain sequence \[page 504\]](#)

15.6.2 Cleanse domain-specific data

Control how Data Cleanse parses domain-specific data, such as name, gender, and title.

To control how Data Cleanse parses and standardizes domain-specific data, set the `CONTENT_DOMAIN_SEQUENCE` option.

❖ Example

Consider how gender is applied for the name “Jean” based on the information in the table.

Name	Domain	Gender
Jean	Global	AMBIGUOUS
Jean	French	STRONG_MALE
Jean	English (United States and Canada)	WEAK_FEMALE

To configure Data Cleanse so that it'sn't domain-specific, select only the [GLOBAL](#) domain.

❖ Example

For the name “Jean”, when you select only [GLOBAL](#), Data Cleanse assigns an ambiguous gender. With only Global, the transform doesn't consider the French or the English domain-specific information.

When you have data from a single domain region, specify a domain and then specify [GLOBAL](#).

❖ Example

For the name “Jean”, when you specify [EN_US](#) followed by [GLOBAL](#) ([EN_US|GLOBAL](#)), the name Jean is assigned a weak female gender.

When you have data from a multi-domain region, select the preferred sequence of domains ending with Global. For example, Benelux (Belgium, Netherlands, Luxembourg) includes the Dutch, French, and German domains.

Depending on your cleansing preference, order the domains in the desired sequence.

❖ Example

If you favor the Dutch domain, specify domains in the following order: [NL|FR|DE|GLOBAL](#). When the Data Cleanse transform encounters a variation in your data that has different properties in the selected content domains, Data Cleanse processes data in the following way:

- Uses the Dutch properties first when they exist in the data.
- If no Dutch properties exist, uses the French properties when they exist in the data.
- If no French properties exist, uses the German properties when they exist in the data.
- If none of the three domains have specific properties, then Data Cleanse uses the properties specified in the Global domain.

Another example of a multi-domain region is Switzerland. Switzerland includes German, French, and Italian domains. Depending on your cleansing preference, you can order the domains in the desired sequence. For example, if you favor the German domain, select [DE|FR|IT|GLOBAL](#).

Parent topic: [Domains \[page 500\]](#)

Related Information

[Global domain \[page 502\]](#)

[Specify the content domain sequence \[page 504\]](#)

15.6.3 Specify the content domain sequence

SAP Data Services allows you to set three levels of content domain sequence: Default, Intermediate, and Top.

Set the content domain sequence in the following transforms: Data Cleanse, Global Address Cleanse, and Query. The following table discusses the option for each level.

Level	Option
Default	In the Data Cleanse transform, the option is <code>CONTENT_DOMAIN_SEQUENCE</code>, and it's required. Data Cleanse uses the value in <code>CONTENT_DOMAIN_SEQUENCE</code> when there are errors or when you don't specify the intermediate and top levels.
Intermediate	In the Global Address Cleanse transform, include any of the following input fields in your field mapping: • <code>OPTION_COUNTRY</code> • <code>OPTION_LANGUAGE</code> • <code>OPTION_REGION</code> SAP Data Services populates and maps these input fields from the following Global Address Cleanse transform output fields: <code>ISO_COUNTRY_CODE_2CHAR</code>, <code>LANGUAGE</code>, and <code>REGION1</code>, respectively. If there are errors, or the top-level option isn't specified, the transform uses the intermediate setting. Using these input fields is optional.
Top	In the Query transform, include the following input field in your input field mapping: <code>OPTION_CONTENT_DOMAIN_SEQUENCE</code>. An upstream Query transform, or a different preprocessing step, populates the <code>OPTION_CONTENT_DOMAIN_SEQUENCE</code> field. Using this input field is optional.

❖ Example

During processing, Data Cleanse assigns the content domain sequence based on the top-level option, if the option is set and is valid. The following tables describe the process on each level.

Top level

Option	Output	Description
Input field <code>OPTION_CONTENT_DOMAIN_SEQUENCE = FR GLOBAL</code>	The content domain sequence is French. All data that isn't identified as French defaults to the Global domain.	The Top level overrides the Intermediate and Default level settings unless the Top-level option is invalid. For example, the Top level is invalid when a word is misspelled or the level isn't set, for example.

Intermediate level

Option	Output	Description
Input fields: <code>OPTION_COUNTRY = MX</code> <code>OPTION_LANGUAGE</code> isn't set. <code>OPTION_REGION</code> isn't set.	The content domain sequence is Spanish, ES_MX, which is local to Mexico. All of the data that isn't identified as Spanish defaults to the Global domain.	The Intermediate level overrides the Default level, unless the Intermediate level contains invalid data, or isn't set. In most cases, the transform can identify the content domain sequence based on the <code>OPTION_COUNTRY</code> input field only. However, there are certain countries that can use <code>OPTION_LANGUAGE</code> or <code>OPTION_REGION</code> to help determine the content domain sequence.

Default level

Option	Output	Description
Data Cleanse transform option: <code>CONTENT_DOMAIN_SEQUENCE = EN_US GLOBAL</code>	The content domain sequence is English and local to United States and Canada. All data that isn't identified as English defaults to the Global domain.	The <code>CONTENT_DOMAIN_SEQUENCE</code> option is required. The transform uses the setting when there are errors in the Top-level and Intermediate-level options.

When the Global Address Cleanse uses the Intermediate level, some countries use additional input data, if necessary, to determine the content domain sequence.

❖ Example

The following table shows an example of this behavior.

<i>OPTION_COUNTRY</i> input field	Additional input fields	Assigned content domain sequence	Notes
CA (Canada)	<i>OPTION_REGION</i> = QC	FR GLOBAL	The transform always assigns the content domain sequence to FR GLOBAL when <i>OPTION_REGION</i> is set to Quebec (QC). Acceptable data includes Quebec or QC in any type of casing. Any other valid region, such as BC for British Columbia, returns a content domain sequence of EN_US GLOBAL.
	<i>OPTION_REGION</i> = BC*	EN_US GLOBAL	
BE (Belgium)	<i>OPTION_LANGUAGE</i> = FR	FR GLOBAL	The transform always assigns the content domain sequence to FR GLOBAL when <i>OPTION_LANGUAGE</i> = FR. Any other specified language, such as NL for Dutch or EN_GB for British English, or blank, returns a content domain sequence of NL GLOBAL.
	<i>OPTION_LANGUAGE</i> = NL*	NL GLOBAL	
CH (Switzerland)	<i>OPTION_LANGUAGE</i> = FR	FR GLOBAL	The transform always assigns the content domain sequence to FR GLOBAL or IT GLOBAL when <i>OPTION_LANGUAGE</i> = FR or IT, respectively. Any other specified language, such as RU for Russian, or blank, returns a content domain sequence of DE GLOBAL.
	<i>OPTION_LANGUAGE</i> = IT	IT GLOBAL	
	<i>OPTION_LANGUAGE</i> = RU*	DE GLOBAL	

Note

*The content domain sequence is assigned independently from the output format.

Parent topic: [Domains \[page 500\]](#)

Related Information

[Global domain \[page 502\]](#)

[Cleanse domain-specific data \[page 502\]](#)

15.7 Output formats

The Data Cleanse transform uses specific output fields based on the specified domain, and formats the data in those fields according to the regional standards.

Based on the set domain and regional standards, Data Cleanse outputs name fields as compound given name, compound family name, or a composite in the Person output field.

Compound given name:

- Combines a compound given name and outputs it to the [GIVEN_NAME1](#) field.
- Outputs the first name of a compound give name to the [GIVEN_NAME1](#) field and the second name to the [GIVEN_NAME2](#) field.

Compound family name:

- Combines a compound family name and outputs it to the [FAMILY_NAME1](#) field.
- Outputs the first family name of a compound family name to the [FAMILY_NAME1](#) field and the second family name to the [FAMILY_NAME2](#) field.

Composite [PERSON](#) output field:

- Outputs a given name followed by the family name in the composite [PERSON](#) field.
- Outputs the family name followed by the given name in the composite [PERSON](#) field.

The Data Cleanse transform requires that you specify an output format, even when your data is truly global.

When you have data from a single-domain region, specify the domain. For example, for Germany select [DE](#), for China select [ZH](#). When you have data from a multi-domain region, select the preferred domain. Your data can be formatted differently depending on the domain you select.

❖ Example

Data from the Philippines can be output in English or Spanish output formats. As shown in the following table, Data Cleanse outputs name data to different fields depending on the selected output format.

Input name: Juan Carlos Sanchez Cruz

Output formats:

Output field	Output format	
	English	Spanish
GIVEN_NAME1	Juan	Juan
GIVEN_NAME2	Carlos	Carlos
FAMILY_NAME1	Sánchez Cruz	Sánchez
FAMILY_NAME2		Cruz
PERSON	Juan Carlos Sánchez Cruz	Juan Carlos Sánchez Cruz

❖ Example

For Benelux data, you can choose to output your data in Dutch, French, or German output formats. The Data Cleanse transform outputs name data in different fields based on the selected output format.

Input name: H.D. Budjhawan

Output formats:

Output field	Output format		
	Dutch	French	German
<i>GIVEN_NAME1</i>	H.D.	H. D.	H.
<i>GIVEN_NAME2</i>			D.
<i>FAMILY_NAME1</i>	Budjhawan	Budjhawan	Budjhawan
<i>FAMILY_NAME2</i>			
<i>PERSON</i>	H.D. Budjhawan	H. D. Budjhawan	H. D. Budjhawan

Modify the existing output format or add a new domain-specific output format by modifying the appropriate rules in your cleansing package.

Parent topic: [Data Cleanse transform \[page 482\]](#)

Related Information

[Rank and prioritize parsing engines \[page 483\]](#)

[Data parsing \[page 484\]](#)

[Data standardization \[page 497\]](#)

[Gender descriptions and prename assignment \[page 497\]](#)

[Prepare records for matching \[page 499\]](#)

[Domains \[page 500\]](#)

16 Data Cleanse transform reference

The Data Cleanse transform XML configuration is grouped according to setting types.

Use the Data Cleanse transform after the address cleansing process and before the matching process.

The sample configurations installed with the software contain examples of the Data Cleanse XML.

[System group \[page 510\]](#)

Set the option in the System group to control high-level transform functions.

[Data Cleanse Report and Analysis \[page 511\]](#)

Generate report data for the Data Cleanse transform by setting the option in the Report and Analysis group.

[Data quality statistics common requirements \[page 236\]](#)

When you generate data quality statistics, there are several requirements that apply to more than one type of statistics tables.

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

[Data Cleanse Cleansing Package \[page 519\]](#)

Specify a cleansing package and set related options in the Cleansing Package group.

[Data Cleanse Input Word Breaker \[page 522\]](#)

The option in the Input Word Breaker group controls how the parser breaks input data.

[Data Cleanse person standardization options \[page 524\]](#)

The options in the Person group control how the transform standardizes person-related output.

[Data Cleanse gender options \[page 528\]](#)

The gender options control what input fields Data Cleanse uses to assign gender.

[Data Cleanse Firm Standardization options \[page 530\]](#)

The Firm Standardization options control how the Data Cleanse transform standardizes firm-related output.

[Data Cleanse Other Standardization options \[page 531\]](#)

To control how the Data Cleanse transform standardizes other data types in your input data, set options in the Standardization group.

[Data Cleanse Date Options \[page 535\]](#)

To control how the Data Cleanse transform standardizes date data, set options in the Date Options group.

[Data Cleanse Phone Options group \[page 538\]](#)

To control how the Data Cleanse transform standardizes phone data on output, set options in the Phone Options group.

[Data Cleanse Parser configuration \[page 541\]](#)

Parser configuration options control the parsing engines that the Data Cleanse transform uses for parsing multiline fields and the order in which they are applied.

[Data Cleanse editor options \[page 543\]](#)

The Options group includes settings that control how the Data Cleanse transform parses and outputs data.

16.1 System group

Set the option in the System group to control high-level transform functions.

≡, Syntax

```
<SYSTEM>
  <PLATFORM_SETTINGS>
    <SYSTEM_VARS>
      <DS_COMMON_DIR></DS_COMMON_DIR>
```

Note

If an option exists in the XML, but it is not documented here, do not alter the contents of the option. Editing the contents could cause errors.

Option	Description
<i>DS_COMMON_DIR</i>	Specifies the relative path to the Data Cleanse support files. For example, the default Windows 32-bit location for the support files is \SAP BusinessObjects\Data quality Mgmt SDK\windows_32\DataQuality\dataacleanse.

Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Related Information

[Data Cleanse Report and Analysis \[page 511\]](#)

[Data quality statistics common requirements \[page 236\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Data Cleanse Cleansing Package \[page 519\]](#)

[Data Cleanse Input Word Breaker \[page 522\]](#)

[Data Cleanse person standardization options \[page 524\]](#)

[Data Cleanse gender options \[page 528\]](#)

[Data Cleanse Firm Standardization options \[page 530\]](#)

[Data Cleanse Other Standardization options \[page 531\]](#)

[Data Cleanse Date Options \[page 535\]](#)

[Data Cleanse Phone Options group \[page 538\]](#)

[Data Cleanse Parser configuration \[page 541\]](#)

[Data Cleanse editor options \[page 543\]](#)

16.2 Data Cleanse Report and Analysis

Generate report data for the Data Cleanse transform by setting the option in the Report and Analysis group.

Syntax

```
<REPORT_AND_ANALYSIS>
  <GENERATE_REPORT_DATA>NO</GENERATE_REPORT_DATA>
</REPORT_AND_ANALYSIS>
```

Option	Description
<i>GENERATE_REPORT_DATA</i>	Specifies whether to generate report data for this transform. <i>YES</i>: Generates report data for this transform. <i>NO</i>: Does not generate report data for this transform. Select <i>NO</i> when you don't need reports, such as during testing. Setting this option to <i>NO</i> improves performance.

Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Related Information

[System group \[page 510\]](#)

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16.3 Data quality statistics common requirements

When you generate data quality statistics, there are several requirements that apply to more than one type of statistics tables.

i Note

Generate data quality statistics tables in the Data Cleanse, Global Address Cleanse, and Geocoder transforms only.

Statistics tables and common requirements

Statistics table	Requirement
All data quality statistics tables: CLEANSE_ADDRESS_RECORD_INFO_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_INFO_CODES_TABLE CLEANSE_STATISTICS_TABLE GEOCODE_INFO_CODES_TABLE GEOCODE_STATISTICS_TABLE	Requires the Task Localization file, which you download from the SAP Customer Portal. The Task Localization file converts raw data that you extract from the data quality statistics tables into understandable language.
CLEANSE_STATISTICS_TABLE GEOCODE_STATISTICS_TABLE	Does not require the ROW_ID field output field. These statistic tables contain summary statistics and the ROW_ID field is required only for non-summary statistics tables. The software issues a warning when your job set up: - Includes the ROW_ID output field but is not configured to generate non-summary data quality statistics. - Excludes the ROW_ID output field but is configured to generate non-summary data quality statistics.

Statistics table	Requirement
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_ADDRESS_RECORD_INFO_TABLE GEOCODE_INFO_CODES_TABLE	Requires the ROW_ID output field. These statistics tables contain non-summary statistics and the ROW_ID field is required to join the transform input or output with the output statistics. The software issues a warning when your job set up: - Excludes the ROW_ID output field but is configured to generate non-summary data quality statistics. - Includes the ROW_ID output field but is not configured to generate non-summary data quality statistics. You can use ROW_ID to join the transform input or output with the non-summary statistics table information. For example, use ROW_ID to drill into the before and after changes in a specific component that was cleansed.
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_ADDRESS_RECORD_INFO_TABLE GEOCODE_INFO_CODES_TABLE	Requires that you map an input file primary key to the output file. Use the mapped primary key to join the input with any of the data quality statistics reports.
Applies to all data quality statistics tables: CLEANSE_ADDRESS_RECORD_INFO_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_COMPONENT_INFO_TABLE CLEANSE_INFO_CODES_TABLE CLEANSE_STATISTICS_TABLE GEOCODE_INFO_CODES_TABLE GEOCODE_STATISTICS_TABLE	If you generate data cleansing statistics, requires that you set up a field for data cleansing. If you do not set up a field for data cleansing, the software does not generate data cleansing statistics for that field. ❖ Example An input file contains firm, address, and phone information. You set up the firm and address fields for data cleansing, but you do not set up the phone field for data cleansing. The software generates statistics that include the firm name and address information, but not the phone information.
CLEANSE_INFO_CODES_TABLE CLEANSE_CHANGE_INFO_TABLE CLEANSE_STATISTICS_TABLE CLEANSE_COMPONENT_INFO_TABLE	Configure to generate in both the Data Cleanse and Global Address Cleanse transforms. Not applicable for the Geocoder transform.

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16.4 Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

The Data Quality Statistics Settings group appears in the Data Cleanse, Global Address Cleanse, and the Geocoder transforms.

Syntax

Data Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSER_STATISTICS_TABLE>NO</
GENERATE_CLEANSER_STATISTICS_TABLE>
  <GENERATE_CLEANSER_INFO_CODES_TABLE>NO</
GENERATE_CLEANSER_INFO_CODES_TABLE>
  <GENERATE_CLEANSER_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSER_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSER_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSER_COMPONENT_INFO_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

Syntax

Global Address Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSER_STATISTICS_TABLE>NO</
GENERATE_CLEANSER_STATISTICS_TABLE>
  <GENERATE_CLEANSER_INFO_CODES_TABLE>NO</
GENERATE_CLEANSER_INFO_CODES_TABLE>
  <GENERATE_CLEANSER_ADDRESS_RECORD_INFO_TABLE>NO</
GENERATE_CLEANSER_ADDRESS_RECORD_INFO_TABLE>
  <GENERATE_CLEANSER_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSER_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSER_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSER_COMPONENT_INFO_TABLE>
```

```
</DATA_QUALITY_STATISTICS_SETTINGS>
```

≡ Syntax

Geocoder:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_GEOCODE_STATISTICS_TABLE>NO</
GENERATE_GEOCODE_STATISTICS_TABLE>
  <GENERATE_GEOCODE_INFO_CODES_TABLE>NO</
GENERATE_GEOCODE_INFO_CODES_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

The following table contains the options for the [DATA_QUALITY_STATISTICS_SETTINGS](#) group. Because the group is applicable for more than one transform, the table lists the applicable transform for each option.

The software generates the following types of statistics for each transform:

- The Data Cleanse and the Global Address Cleanse transforms generate additional statistics and info codes about address cleansing results on the data.
- The Geocoder transform generates statistics and info codes that are unique to the Geocoding processes.

Data Quality Statistics options

Option	Transform	Description
GENERATE_CLEANSER_ADDRESS_RECORD_INFO_TABLE	Global Address Cleanse	Specifies whether the software generates the Cleanse Address Record Information Table. • YES: Generates the Cleanse Address Record Information Table. • NO: Does not generate the Cleanse Address Record Information Table. NO is the default setting. The Cleanse Address Record Information Table contains non-summary information about the results of the address cleanse process on the data. The information includes assignment information and assignment type for each address. The software generates this information only when you include the Global Address Cleanse transform for address data.

Option	Transform	Description
<code>GENERATE_CLEANSER_CHANGE_INFO_TABLE</code>	Data Cleanse Global Address Cleanse	Specifies whether the software generates the Cleanse Change Information Table. YES: Generates the Cleanse Change Information Table. NO: Does not generate the Cleanse Change Information Table. NO is the default setting. The Cleanse Change Information Table contains non-summary information. It identifies areas of concern from your source data by the number of significant changes.
<code>GENERATE_CLEANSER_COMPONENT_INFO_TABLE</code>	Data Cleanse Global Address Cleanse	Specifies whether the software generates the Cleanse Component Information Table. YES: Generates the Cleanse Component Information Table. NO: Does not generate the Cleanse Component Information Table. NO is the default setting. Note This option is available in Data Quality Management SDK version 4.2.6 and later. The Cleanse Component Information Table contains non-summary information. It contains position information for each specified data element when one or both of the following occurs: - The transform parses the data element from an input field. - The transform writes the data element to an output field.

Option	Transform	Description
GENERATE_CLEANSER_INFO_CODES_TABLE	Data Cleanse	Specifies whether the software generates Cleanse Information Codes Table. YES: Generates the Cleanse Information Codes Table. NO: Does not generate the Cleanse Information Codes Table. NO is the default setting. The Cleanse Information Codes table contains non-summary information. The information includes the distribution of information codes or a count of missing, suspect, or blank data.
	Global Address Cleanse	
GENERATE_CLEANSER_STATISTICS_TABLE	Data Cleanse	Specifies whether the software generates the Cleanse Statistics Table. YES: Generates the Cleanse Statistics Table. NO: Does not generate the Cleanse Statistics Table. NO is the default setting. The Cleanse Statistics Table contains summarized information for each unique Entity ID. It provides a high-level insight into what the software changed during the cleansing process.
	Global Address Cleanse	

Option	Transform	Description
GENERATE_GEOCODE_INFO_CODES_TABLE	Geocoder	Specifies whether to generate the Geocode Information Codes Table. YES: Generates the Geocode Information Codes Table. NO: Does not generate the Geocode Information Codes Table. NO is the default setting. The Geocode Information Codes Table contains non-summary information about why records were not assigned the highest level of latitude and longitude. The statistics are based on the information code data. Note Not all existing information codes apply to the data quality statistics.
GENERATE_GEOCODE_STATISTICS_TABLE	Geocoder	Specifies whether the software generates the Geocode Statistics Table. YES: Generates the Geocode Statistics Table. NO: Does not generate the Geocode Statistics Table. NO is the default setting. The Geocode Statistics Table contains totals for the following: - The number of rows in the data - The number of rows that were assigned a latitude and longitude.

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Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Parent topic: [Geocoder transform reference \[page 570\]](#)

Related Information

[System group \[page 396\]](#)

[Global Address Cleanse Report and Analysis \[page 397\]](#)
[Global Address Cleanse reference files \[page 398\]](#)
[Global Address Cleanse Country ID Options \[page 404\]](#)
[Global Address Cleanse Engines group \[page 409\]](#)
[Global Address Cleanse Country \[page 411\]](#)
[Global Address Cleanse Standardization Options \[page 412\]](#)
[Canada engine \[page 442\]](#)
[Global Address Cleanse Canada group \[page 445\]](#)
[Global Address engine \[page 450\]](#)
[Global Address Cleanse USA engine \[page 469\]](#)
[Global Address Cleanse Suggestion List option \[page 473\]](#)
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[Geocoder Report and Analysis \[page 571\]](#)
[Geocoder transform options \[page 578\]](#)

16.5 Data Cleanse Cleansing Package

Specify a cleansing package and set related options in the Cleansing Package group.

Syntax

```
<CLEANSING_PACKAGE>
  <CLEANSING_PACKAGE_NAME>PERSON_FIRM</CLEANSING_PACKAGE_NAME>
  <CONTENT_DOMAIN_SEQUENCE>EN_US|GLOBAL</CONTENT_DOMAIN_SEQUENCE>
  <OUTPUT_FORMAT>EN_US</OUTPUT_FORMAT>
</CLEANSING_PACKAGE>
```

Data Cleanse cleansing package option descriptions

Option	Description
CLEANSING_PACKAGE_NAME	Specifies the cleansing package to use for processing. Enter the name of the cleansing package. The transform uses the location that you set in the DS_COMMON_DIR option in the SYSTEM_GROUP as the location for the cleansing package.
CONTENT_DOMAIN_SEQUENCE	Specifies the content domain to use for processing. Specify more than one content domain. Be sure to add GLOBAL as the last domain in the sequence. Content domains include the following: - GLOBAL - Global - AR - Arabic - ZH - Chinese - CS - Czech - DA - Danish - NL - Dutch - EN_US - English (United States & Canada) - EN_GB - English (United Kingdom & Ireland) - EN_AU - English (Australia & New Zealand) - EN_IN - English (India) - FR - French - DE - German - HU - Hungarian - ID - Indonesian - IT - Italian - JA - Japanese - MS - Malay - NO - Norwegian - PL - Polish - PT_BR - Portuguese (Brazil) - PT_PT - Portuguese (Portugal) - RO - Romanian - RU - Russian - SK - Slovak - ES_MX - Spanish (Latin America) - ES_ES - Spanish (Spain) - SV - Swedish - TR - Turkish The Global domain is a special content domain that contains all variations and their associated properties. If a variation is not associated with domain-specific information, the Global domain serves as the default domain. Note You can set this option as a dynamic input field.

Option	Description
<code>OUTPUT_FORMAT</code>	Specifies the format for output. Data Cleanse uses certain output fields based on the specified domain in the output format. The transform formats the data in those fields according to the regional standards. Note You can set this option as a dynamic input field. Valid values for this option are: - AR Arabic - ZH Chinese - CS Czech - DA Danish - NL Dutch - EN_US English (United States & Canada) - EN_GB English (United Kingdom & Ireland) - EN_AU English (Australia & New Zealand) - EN_IN English (India) - FR French - DE German - HU Hungarian - ID Indonesian - IT Italian - JA Japanese - MS Malay - NO Norwegian - PL Polish - PT_BR Portuguese (Brazil) - PT_PT Portuguese (Portugal) - RO Romanian - RU Russian - SK Slovak - ES_MX Spanish (Latin America) - ES_ES Spanish (Spain) - SV Swedish - TR Turkish

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16.6 Data Cleanse Input Word Breaker

The option in the Input Word Breaker group controls how the parser breaks input data.

Syntax

```
<INPUT_WORD_BREAKER>  
<BREAK_ON_WHITESPACE_ONLY>NO</BREAK_ON_WHITESPACE_ONLY>  
</INPUT_WORD_BREAKER>
```

Option	Description
<i>BREAK_ON_WHITESPACE_ONLY</i>	Specifies whether the Data Cleanse transform breaks input data on white space only or on white space, punctuation, alphanumeric transitions, and script transitions. YES: Breaks input data on white space only. NO: Breaks input data on white space, punctuation, and alphanumeric transitions, and script transitions. For data in the CJK and Kana scripts, input data breaks on each character. NO is the default setting. Enables the Data Cleanse transform to recognize alphanumeric strings, such as product codes, as entries in a custom cleansing package. ❖ Example Input data: AF302 <i>BREAK_ON_WHITESPACE_ONLY = NO</i> - Parser breaks input into two tokens: AF and 302. <i>BREAK_ON_WHITESPACE_ONLY = YES</i> - Parser recognizes input as a single entry AF302 Note This option typically applies to custom cleansing packages. The out-of-the-box person and firm cleansing packages are designed to use the parsing strategy that breaks data on white space, punctuation, alphanumeric transitions, and script transitions.

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16.7 Data Cleanse person standardization options

The options in the Person group control how the transform standardizes person-related output.

≡ Syntax

```
<PERSON>
  <ASSIGN_PRENAMES>YES</ASSIGN_PRENAMES>
  <COMBINE_COMPOUND_NAMES>YES</COMBINE_COMPOUND_NAMES>
  <NAME_ORDER>UNKNOWN</NAME_ORDER>
  <ASSOCIATE_NAME_TITLE>YES</ASSOCIATE_NAME_TITLE>
  <ENABLE_PRESUMPTIVE_NAME_PARSING>YES</
ENABLE_PRESUMPTIVE_NAME_PARSING>
  <GENDER_OPTIONS>
    <USE_GIVEN_NAME2_TO_ASSIGN_GENDER>YES</
USE_GIVEN_NAME2_TO_ASSIGN_GENDER>
    <USE_FAMILY_NAME_TO_ASSIGN_GENDER>NO</
USE_FAMILY_NAME_TO_ASSIGN_GENDER>
  </GENDER_OPTIONS>
  <PARSE_DISCRETE_INPUT>NO</PARSE_DISCRETE_INPUT>
</PERSON>
```

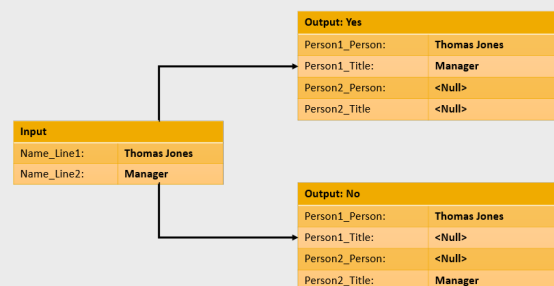
Person standardization options

Option	Description
<i>ASSIGN_PRENAMES</i>	Specifies whether the transform includes assigned prenames, such as Mr., Mrs., Miss, or Ms., in the <i>PRENAME</i> output field. <i>YES</i>: Assigns prenames based on the gender of the value in the <i>GIVEN_NAME1</i> field, or on gender option settings. <i>NO</i>: Does not assign prenames. The <i>PRENAME</i> output field always includes prenames that are part of the name input data. Additionally, the Data Cleanse transform adds prenames based on the gender of the name (<i>STRONG_MALE</i> or <i>STRONG_FEMALE</i>) in the <i>GIVEN_NAME1</i> field. When the gender of <i>GIVEN_NAME1</i> is not strong, the transform assigns prenames based on the gender option settings in <i>USE_GIVEN_NAME2_TO_ASSIGN_GENDER</i> and <i>USE_FAMILY_NAME_TO_ASSIGN_GENDER</i>.

Option	Description
ASSOCIATE_NAME_TITLE	Specifies whether the transform should consider data in separate name and occupational title fields as one person. YES: Consider data in separate name and occupational title fields as one person. NO: Do not consider data in separate name and occupational title fields as one person.

❖ Example

The following diagram shows the difference in the output based on the setting you choose for [ASSOCIATE_NAME_TITLE](#).



! Restriction

If you generate data quality statistics tables in the Data Cleanse transform, set this option to **NO**.

COMBINE_COMPOUND_NAMES	Specifies how the software standardizes compound family names when the family name includes a PRE_FAMILY_NAME that is also a PRE_FAMILY_NAME_COMBINE. YES: Combines compound family names. For example, the transform combines the family name Mc Donald to McDonald. NO: Preserves the input format of compound family names. For example, the family name Mc Donald remains Mc Donald.
--	--

i Note

This option has no impact on a [PRE_FAMILY_NAME](#) that is not classified as a [PRE_FAMILY_NAME_COMBINE](#).

❖ Example

The software does not combine the name Van Helsing into VanHelsing because "Van" is not classified with both [PRE_FAMILY_NAME](#) and [PRE_FAMILY_NAME_COMBINE](#).

Option	Description
<code>ENABLE_PRESUMPTIVE_NAME_PARSING</code>	Specifies whether to use presumptive name parsing on <code>NAME_LINE</code> input fields. • YES: Turns on presumptive name parsing. The transform treats data in the <code>NAME_LINE</code> input field as a name. • NO: Turns off presumptive name parsing. Data in the <code>NAME_LINE</code> input field that does not parse as a name remains unparsed and the transform outputs it to the <code>EXTRA</code> field. ❖ Example If data contains an automobile brand and model in a <code>NAME_LINE</code> input field, the transform tries to parse the information as a name based on rules in the cleansing package. • If the option is set to NO and Data Cleanse cannot assign the data, it outputs the unparsed data to the <code>EXTRA</code> field. • If the option is set to YES, Data Cleanse assigns the data as a name.
<code>NAME_ORDER</code>	Defines how the transform applies parsing rules to determine the content of the <code>GIVEN_NAME</code> and <code>FAMILY_NAME</code> output fields. • GIVEN_FAMILY_NAME_STRICT: Specifies the order of given name and family name in the input file. The transform considers only parsing rules that follow the strict name definition order to determine the rule to apply to the input string. Select when the order of given name and family name is consistent in the input data. • FAMILY_GIVEN_NAME_STRICT: Specifies the order of given name and family name in the input file. The transform considers only parsing rules that follow the strict name definition order to determine the rule to apply to the input string. Select when the order of family name and given name is consistent in the input data. • GIVEN_FAMILY_NAME_SUGGEST: Specifies the parsing order of given name and family name in the input file. The transform uses this order as a tie breaker when two rules have the same confidence score. • FAMILY_GIVEN_NAME_SUGGEST: Specifies the parsing order of family name, given name in the input file. The transform uses this order as a tie breaker when two rules have the same confidence score. • UNKNOWN: Specifies to use the order of the rule with the highest confidence score based on information in the dictionary and rule file.

Option	Description
<code>PARSE_DISCRETE_INPUT</code>	Specifies how the transform maps discrete person data to output fields. NO: Maps discrete input fields directly to the corresponding output fields without parsing. YES: Combines discrete input fields to one field, parses the data, and outputs data to discrete output fields. ❖ Example Input data: PERSON1_GIVEN_NAME1 = Mr John T PERSON1_FAMILY_NAME1 = Smith III Output: Option = NO PERSON1.PRENAME = <i>blank</i> PERSON1.GIVEN_NAME1 = Mr John T PERSON1.GIVEN_NAME2 = <i>blank</i> PERSON1.FAMILY_NAME1 = Smith III PERSON1.MATURITY_POSTNAME = <i>blank</i> Output: Option = YES PERSON1.PRENAME = Mr PERSON1.GIVEN_NAME1 = John PERSON1.GIVEN_NAME2 = T PERSON1.FAMILY_NAME1 = Smith PERSON1.MATURITY_POSTNAME = III

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16.8 Data Cleanse gender options

The gender options control what input fields Data Cleanse uses to assign gender.

Syntax

```
<GENDER_OPTIONS>
    <USE_GIVEN_NAME2_TO_ASSIGN_GENDER>YES</
USE_GIVEN_NAME2_TO_ASSIGN_GENDER>
    <USE_FAMILY_NAME_TO_ASSIGN_GENDER>NO</
USE_FAMILY_NAME_TO_ASSIGN_GENDER>
</GENDER_OPTIONS>
```

Note

The options in this group apply only to person and firm cleansing packages.

Option	Description
USE_GIVEN_NAME2_TO_ASSIGN_GENDER	Specifies how the transform assigns gender when the input gender in PRENAME or GIVEN_NAME1 is unassigned or ambiguous. YES: Assigns ambiguous gender data using the gender of the parsed PRENAME or GIVEN_NAME2 fields. NO: Does not assign gender data using the gender of the parsed PRENAME or GIVEN_NAME2 field.  Example Input: Pat Robert Smith Parsed: GIVEN_NAME1 = Pat GIVEN_NAME2 = Robert USE_GIVEN_NAME2_TO_ASSIGN_GENDER = YES. Gender: GIVEN_NAME1 is ambiguous. Therefore, Data Cleanse uses the data in GIVEN_NAME2, which is not ambiguous, to determine that the gender is STRONG_MALE.

Option	Description
<i>USE_FAMILY_NAME_TO_ASSIGN_GENDER</i>	Specifies how the transform assigns gender when the input gender in <i>PRENAME</i>, <i>GIVEN_NAME1</i>, or <i>GIVEN_NAME2</i> is unassigned or ambiguous. • YES: Assigns gender based on the parsed <i>FAMILY_NAME1</i> or <i>FAMILY_NAME2</i> fields when it can't determine gender using the <i>PRENAME</i> or <i>GIVEN_NAME1 GIVEN_NAME2</i> fields. • Uses <i>FAMILY_NAME1</i> when gender is assigned and is not ambiguous. • Uses <i>FAMILY_NAME2</i> when unable to use <i>FAMILY_NAME1</i>. • NO: Does not assign gender of the parsed <i>FAMILY_NAME1</i> or <i>FAMILY_NAME2</i> fields. ❖ Example Input data: N. Albiantsev Parsed: Given_Name1 = N. <i>USE_FAMILY_NAME_TO_ASSIGN_GENDER</i> = NO Results: No gender assigned because <i>GIVEN_NAME1</i> is ambiguous. Parsed: Family_Name1 = Albiantsev <i>USE_FAMILY_NAME_TO_ASSIGN_GENDER</i> = YES Results: Assigned gender is STRONG_MALE because value of <i>FAMILY_NAME1</i>, Albiantsev, is STRONG_MALE

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16.9 Data Cleanse Firm Standardization options

The Firm Standardization options control how the Data Cleanse transform standardizes firm-related output.

Syntax

```
<FIRM>
    <ENABLE_PRESUMPTIVE_FIRM_PARSING>YES</
ENABLE_PRESUMPTIVE_FIRM_PARSING>
</FIRM>
```

Note

The options in this group apply only to person and firm cleansing packages.

Option	Description
ENABLE_PRESUMPTIVE_FIRM_PARSING	Specifies whether the transform uses presumptive firm parsing on FIRM_LINE input fields. YES: Uses presumptive firm parsing. Assigns data in the FIRM_LINE input field as firm data regardless of the content. NO: Does not use presumptive firm parsing. Assigns data in the FIRM_LINE input field as firm data. If it cannot assign data as firm data, outputs unparsed data to the EXTRA field. Presumptive firm parsing tells the transform to process any information in a Firm field as a firm, regardless of the type of information in the field.  Example If your input data has a given name and family name in a FIRM_LINE input field, the Data Cleanse transform tries to parse the information as a firm based on rules in the cleansing package. - If the ENABLE_PRESUMPTIVE_FIRM_PARSING is set to NO and Data Cleanse is not able to assign the data, it outputs the unparsed data to the EXTRA field. - If the option is set to YES, Data Cleanse assigns the data as a firm.

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16.10 Data Cleanse Other Standardization options

To control how the Data Cleanse transform standardizes other data types in your input data, set options in the Standardization group.

Syntax

```
<OTHER>
  <CAPITALIZATION>MIXED</CAPITALIZATION>
  <REMOVE_PUNCTUATION>NO</REMOVE_PUNCTUATION>
  <REMOVE_DIACRITICAL_CHARACTERS>NO</REMOVE_DIACRITICAL_CHARACTERS>
  <CHARACTER_WIDTH_STYLE>NORMAL_WIDTH</CHARACTER_WIDTH_STYLE>
  <ONE_TO_ONE_MAPPING>YES</ONE_TO_ONE_MAPPING>
  <SSN_DELIMITER>DASH</SSN_DELIMITER>
</OTHER>
```

Standardization options

Option	Description
<i>CAPITALIZATION</i>	Specifies the casing of your output. <i>LOWER</i>: Converts the output to lowercase. For example, john mckay. <i>MIXED</i>: Preserves the casing for the standard form as defined within the cleansing package. If a standard form is not defined, the output is converted to mixed case.  Example If you define the standard form as JOHN MCKAY, the transform preserves the form. If you do not define a standard form, the transform converts the output to mixed case as in John McKay. <i>PRESERVE</i>: Preserves the input casing. Input is JOHN MCKAY, stays as JOHN MCKAY. <i>UPPER</i>: Converts the output to uppercase. For example, converts john mckay to JOHN MCKAY.

Option	Description
<code>CHARACTER_WIDTH_STYLE</code>	Specifies the character width used in output fields. Useful when you process Japanese or mixed language data. <code>NORMAL_WIDTH</code>: Output field width reflects the normalized character width based on the script type. Some output columns contain halfwidth characters and other columns contain fullwidth characters. <code>NORMAL_WIDTH</code> does not require special processing and therefore is the most efficient setting. ❖ Example For <code>NORMAL_WIDTH</code>, the transform standardizes all fullwidth Latin characters to their halfwidth forms. The transform standardizes all halfwidth katakana characters to their fullwidth forms. <code>FULL_WIDTH</code>: Converts characters from halfwidth form to fullwidth form in all output fields. If characters do not have fullwidth forms, leaves characters in halfwidth form. <code>HALF_WIDTH</code>: Converts characters from fullwidth forms to halfwidth forms in all output fields. If characters do not have halfwidth forms, leaves the characters in fullwidth form. → Tip Since the output width is based on the normalized width for the character type, the output data may be larger than the input data. Consider increasing the column width in the target table. For template tables, selecting <code>USE_NVARCHAR_FOR_VARCHAR_COLUMNS_IN_SUPPORTED_DATABASES</code> changes the VARCHAR column type to NVARCHAR and allows for increased data size.

Option	Description
<code>ONE_TO_ONE_MAPPING</code>	Specifies whether the transform places the input data into the corresponding output field for the following parsers: Phone, Email, Date. YES: Places the parsed data into the corresponding output field. NO: Places the parsed data into the first available output field in the category. ❖ Example Input: Date1 = <i>blank</i> Date2 = <i>blank</i> Date3 = 10/2018 Output when you set <code>ONE_TO_ONE_MAPPING</code> to YES: The output is the same as input. Output when you set <code>ONE_TO_ONE_MAPPING</code> to NO: Date1 = 10/2018 Date2 = <i>blank</i> Date3 = <i>blank</i>
<code>REMOVE_DIACRITICAL_CHARACTERS</code>	Specifies whether the software removes diacritical characters and replaces them with the ASCII equivalents. YES: Replaces diacritical characters such as accent marks and umlauts, with the ASCII equivalent. NO: Standardizes and retains diacritical characters. ❖ Example <code>REMOVE_DIACRITICAL_CHARACTERS = NO</code> Input: María Hernández Output: María Hernández <code>REMOVE_DIACRITICAL_CHARACTERS = YES</code> Input: María Hernández Output: Maria Hernandez

Option	Description
<i>REMOVE_PUNCTUATION</i>	Specifies whether the transform removes all punctuation from standardized data, with the exception of hyphens between names. <i>YES</i>: Removes punctuation except for hyphens between names. <i>NO</i>: Does not remove punctuation. ❖ Example <i>REMOVE_PUNCTUATION = YES</i> Input: X.L. Output: XL <i>REMOVE_PUNCTUATION = NO</i> Input: X.L. Output: X.L.
<i>SSN_DELIMITER</i>	Specifies the character to use for standard U.S. Social Security number (SSN) output delimiters. <i>BACKSLASH</i>: Uses backward slashes as the delimiter in the SSN. For example, 799\45\6789. <i>DASH</i>: Uses dashes as the delimiter in the SSN. For example, 799-45-6789. <i>SLASH</i>: Uses forward slashes as the delimiter in the SSN. For example, 799/45/6789. <i>NONE</i>: Does not add a delimiter to the SSN. For example, 799456789. <i>PERIOD</i>: Uses periods as the delimiter in the SSN. For example, 799.45.6789. <i>SPACE</i>: Uses spaces as the delimiter in the SSN. For example, 799 45 6789.

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16.11 Data Cleanse Date Options

To control how the Data Cleanse transform standardizes date data, set options in the Date Options group.

Syntax

```
<DATE_OPTIONS>
  <CENTURY_THRESHOLD>25</CENTURY_THRESHOLD>
  <INPUT_MONTH_BEFORE_DAY>YES</INPUT_MONTH_BEFORE_DAY>
  <INPUT_YEAR_FIRST>NO</INPUT_YEAR_FIRST>
  <DATE_FORMAT>MONTH_DAY_YEAR</DATE_FORMAT>
  <DATE_DELIMITER>SLASH</DATE_DELIMITER>
  <NUMERIC_FORMAT>ARABIC_NUMBERS</NUMERIC_FORMAT>
  <ENABLE_ZERO_PAD>NO</ENABLE_ZERO_PAD>
  <MONTH_FORMAT>NUMERIC</MONTH_FORMAT>
  <YEAR_FORMAT>FULL_YEAR</YEAR_FORMAT>
</DATE_OPTIONS>
```

Date option descriptions

Option group	Description
<i>CENTURY_THRESHOLD</i>	Specifies a number for which the transform considers a two-digit date as part of the 20th or 21st century. Specify a two-digit integer that represents the first year that a parsed two-digit year is considered part of the 21st century (20xx). All two-digit years greater than the specified integer are considered part of the 20th century (19xx).

Example

CENTURY_THRESHOLD = 11:

- Input = 8
- Output = 2008
- Input = 11
- Output = 2011
- Input = 73
- Output = 1973

Option group	Description
<i>DATE_DELIMITER</i>	Specifies what character to use for standard date output delimiters. • <i>BACKSLASH</i>: Uses backward slashes as the delimiter for the date. For example, 04\01\2010. • <i>DASH</i>: Uses dashes as the delimiter for the date. For example, 04-01-2010. • <i>SLASH</i>: Uses forward slashes as the delimiter for the date. For example, 04/01/2010. • <i>NONE</i>: Does not add a delimiter to the date. For example, 04012010 • <i>PERIOD</i>: Uses periods as the delimiter for the date. For example, 04.01.2010. • <i>SPACE</i>: Uses spaces as the delimiter for the date. For example, 04 01 2010. • <i>CHINESE_JAPANESE</i>: Uses the following Chinese or Japanese characters as delimiters: • 月 always follows the month • 日 always follows the day • 年 always follows the year ❖ Example Arabic numbers with Chinese or Japanese delimiters: 04 月 01 日 2010 年 ❖ Example Chinese or Japanese Numbers with Chinese or Japanese delimiters: 四月 一 日 二零一十年
<i>DATE_FORMAT</i>	Specifies how the transform standardizes dates on output. • <i>YEAR_MONTH_DAY</i>: For Example, August 16, 2012 is 2012-08-16 • <i>YEAR_DAY_MONTH</i>: For Example, August 16, 2012 is 2012-16-08 • <i>MONTH_DAY_YEAR</i>: For Example, August 16, 2012 is 08-16-2012 • <i>DAY_MONTH_YEAR</i>: For Example, August 16, 2012 is 16-08-2012
<i>ENABLE_ZERO_PAD</i>	Specifies whether the transform places a zero in front of one-digit days and months on input. • <i>YES</i>: Places a zero in front of one-digit day and month data. For example, outputs July 4 as 07/04. • <i>NO</i>: Does not place a zero in front of one-digit day and month date. For example, outputs July 4 as 7/4.

Option group	Description
<i>INPUT_MONTH_BEFORE_DAY</i>	Specifies whether input dates use a pattern with the month listed first. <i>YES</i>: Lists the month first in a date string. For example, 11/12/2004 is November 12, 2004. <i>NO</i>: Does not list the month first in a date string.
<i>INPUT_YEAR_FIRST</i>	Specifies whether input dates use a pattern with the year listed first. <i>YES</i>: Lists the year first in a date string. For example, 03/02/04 is 2003 February 4. <i>NO</i>: Does not list the year first in a date string.
<i>MONTH_FORMAT</i>	Specifies how the transform standardizes month output components in a date output. <i>FULL_TEXT</i>: Outputs month component using the full form of the spelled word. For example, outputs Aug as August. Outputs 12 as December. <i>NUMERIC</i>: Outputs month component using the numeric form of the month. For example, outputs September as 9. Outputs December as 12. <i>SHORT_TEXT</i>: Outputs month component using the standard abbreviation of the month. For example, outputs December as Dec Outputs March as Mar.
<i>NUMERIC_FORMAT</i>	Specifies what numeric format the transform uses for date values. <i>ARABIC_NUMBERS</i>: Returns numeric date values in Arabic. <i>CHINESE_JAPANESE_NUMBERS</i>: Returns numeric date values in Chinese or Japanese.
<i>YEAR_FORMAT</i>	Specifies how the transform standardizes year components in a date output. <i>FULL</i>: Outputs the year using four digits. For example, outputs 04 as 2004. <i>SHORT</i>: Outputs the year using 2 digits. For example, outputs 2004 as 04.

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16.12 Data Cleanse Phone Options group

To control how the Data Cleanse transform standardizes phone data on output, set options in the Phone Options group.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Syntax

```
<PHONE_OPTIONS>
  <ISO2_COUNTRY_CODE_SEQUENCE>GLOBAL</ISO2_COUNTRY_CODE_SEQUENCE>
  <NORTH_AMERICAN_PHONE_PARENS_AREA>YES</
NORTH_AMERICAN_PHONE_PARENS_AREA>
  <NORTH_AMERICAN_PHONE_DELIMITER_AFTER_AREA>NO</
NORTH_AMERICAN_PHONE_DELIMITER_AFTER_AREA>
  <NORTH_AMERICAN_PHONE_DELIMITER>DASH</
NORTH_AMERICAN_PHONE_DELIMITER>
  <PHONE_EXTENSION_TEXT>Ext</PHONE_EXTENSION_TEXT>
</PHONE_OPTIONS>
```

Option	Description
<i>ISO2_COUNTRY_CODE_SEQUENCE</i>	Specifies a sequence of countries using the ISO two-character code. Enter the two-character ISO country code for each country in the search. Separate the country codes with a pipe character. - The codes you include determine the countries in which the transform searches for phone matches. - The order in which you place the codes in the sequence determines the order in which Data Cleanse searches for phone information. GLOBAL is the default value. It's optional. GLOBAL contains the international regular expressions that are set in the cleansing package. If you include GLOBAL in your sequence, place it last in the sequence. Note Even if you place GLOBAL first in the sequence, the transform always searches GLOBAL last. Example If most of your data contains records from Germany and Australia, make sure that the ISO2 code for Germany comes first, followed by Australia: DE AU. If you're sure that there are no other countries represented in your data, exclude GLOBAL to save processing time. If you don't know what countries are represented in your data, include only GLOBAL in your sequence. If you know that some countries are represented in your data, but not all countries, include the countries that you're sure of and then include Global: DE AU GLOBAL. For this example sequence, the transform searches for phone matches in the following way: - Searches Germany phone data to find a match. - If the data doesn't parse as German phone data, the transform searches Australia phone data to find a match. - If the data doesn't match either Germany or Australia phone data, the transform searches the global phone data.

Option	Description
	If there's still no match, the transform searches for a match in the North American Phone Parens Area Plan (NANP).
NORTH_AMERICAN_PHONE_PARENS_AREA	Specifies whether the transform includes parentheses around the area code of phone numbers belonging to the NANP countries. YES: Includes the parentheses. For example, (123) 555-1234. NO: Omits the parentheses. For example, 123 555-1234.
NORTH_AMERICAN_PHONE_DELIMITER_AFTER_AREA	Specifies whether the transform follows the rule for delimiter placement after the area code following the NANP. To use this option, also set a delimiter in the NORTH_AMERICAN_PHONE_DELIMITER option. YES: Includes the specified delimiter after the area code in the phone data output. For example, 123-555-1234. NO: Doesn't include a delimiter after the area code in the phone data output. For example, 123 555-1234.
NORTH_AMERICAN_PHONE_DELIMITER	Specifies what character the transform uses to delimit phone output following the NANP. Note To use this option, set the NORTH_AMERICAN_PHONE_DELIMITER_AFTER_AREA to YES. BACKSLASH: Uses backward slashes as the delimiter in the phone number. For example, 123\555\1234. DASH: Uses dashes as the delimiter in the phone number. For example, 123-555-1234. SLASH: Uses forward slashes as the delimiter in the phone number. For example, 123/555/1234. NONE: Doesn't add a delimiter to the phone number. For example, 1235551234. PERIOD: Uses periods as the delimiter in the phone number. For example, 123.555.1234. SPACE: Uses spaces as the delimiter in the phone number. For example, 123 555 1234.
PHONE_EXTENSION_TEXT	Specifies the standard text the transform outputs as the phone extension indicator. For example, Ext.

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16.13 Data Cleanse Parser configuration

Parser configuration options control the parsing engines that the Data Cleanse transform uses for parsing multiline fields and the order in which they are applied.

Syntax

```
<PARSER_CONFIGURATION>
  <PARSER_SEQUENCE_MULTILINE1>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE1>
  <PARSER_SEQUENCE_MULTILINE2>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE2>
  <PARSER_SEQUENCE_MULTILINE3>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE3>
  <PARSER_SEQUENCE_MULTILINE4>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE4>
  <PARSER_SEQUENCE_MULTILINE5>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE5>
  <PARSER_SEQUENCE_MULTILINE6>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE6>
  <PARSER_SEQUENCE_MULTILINE7>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE7>
  <PARSER_SEQUENCE_MULTILINE8>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE8>
  <PARSER_SEQUENCE_MULTILINE9>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE9>
  <PARSER_SEQUENCE_MULTILINE10>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE10>
  <PARSER_SEQUENCE_MULTILINE11>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE11>
  <PARSER_SEQUENCE_MULTILINE12>UDPM|EMAIL|SSN|DATE|PHONE|
PERSON_OR_FIRM</PARSER_SEQUENCE_MULTILINE12>
</PARSER_CONFIGURATION>
```

If you do not include a particular parser, Data Cleanse does not look for that type of data in the input field.

Data Cleanse Parser options

Option	Description
PARSER_SEQUENCE_MULTILINE1	Specifies the parser or parsers the transform uses, and the order in which to use them. ADDRESS: Parses data as an address. DATE: Parses data as a date. EMAIL: Parses data as an e-mail address. FIRM: Parses data as a firm name. LAST_LINE: Parses data as a last line. PERSON: Parses data as a person name. PERSON_OR_FIRM: Parses data as a person or a firm name. PHONE: Parses data as a phone number. i Note The transform parses any non-North American numbers first. Phone numbers may be output to the PHONE and INTERNATIONAL_PHONE fields. The transform parses North American numbers later using the North American Numbering Plan (NANP). The data may be output to PHONE and NORTH_AMERICAN_PHONE output fields. Define phone number patterns that do not follow the NANP in Cleansing Package Builder.
PARSER_SEQUENCE_MULTILINE2	
PARSER_SEQUENCE_MULTILINE3	
PARSER_SEQUENCE_MULTILINE4	
PARSER_SEQUENCE_MULTILINE5	
PARSER_SEQUENCE_MULTILINE6	
PARSER_SEQUENCE_MULTILINE7	
PARSER_SEQUENCE_MULTILINE8	
PARSER_SEQUENCE_MULTILINE9	
PARSER_SEQUENCE_MULTILINE10	
PARSER_SEQUENCE_MULTILINE11	
PARSER_SEQUENCE_MULTILINE12	
	SSN: Parses data as U.S. Social Security number. UDPM: Parses data using user-defined patterns created in Cleansing Package Builder. i Note Order is important. To order your parsers, enter the parsers separated with a pipe character in the applicable order in which the transform should parse your data.
	❖ Example For PARSER_SEQUENCE_MULTILINE1, include EMAIL, SSN, and DATE parsers as follows: <pre><PARSER_SEQUENCE_MULTILINE1>EMAIL SSN DATE</ PARSER_SEQUENCE_MULTILINE1></pre> The transform parses data in the MULTILINE1 field using the e-mail parser first, then the Social Security Number parser, and finally the Date parser.

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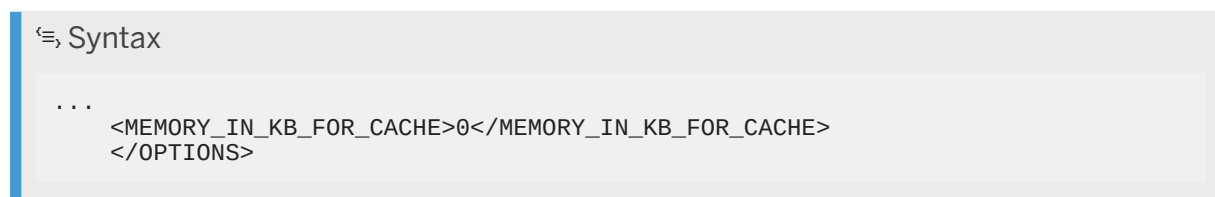
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16.14 Data Cleanse editor options

The Options group includes settings that control how the Data Cleanse transform parses and outputs data.



Option	Description
<i>MEMORY_IN_KB_FOR_CACHE</i>	(Optional) Enter a value of memory to allocate per thread for processing the Data Cleanse transform. Setting this option to 0 indicates that the internal default value (65536) should be used.

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17 Geocoder transform

Use the Geocoder transform to append latitude, longitude, and US census data to your data.

Geocoder information

Characteristic	Description
Use	The Geocoder transform uses geographic coordinates expressed as latitude and longitude, addresses, and point-of-interest (POI) data. Append addresses, latitude and longitude, US census data, and other information to your data. i Note Data Quality Management SDK uses the World Geodetic System (WGS84) for latitude and longitude results. WGS84 is defined and maintained by the US National Geospatial-Intelligence Agency (NGA). The transform has options to enable generation of data quality statistics tables in the repository for address geocoding mode. i Note Not applicable for reverse geocoding and point of interest (POI) textual search. These statistics provide insight into how the software cleansed and assigned your data by generating aggregated summary statistics and specific record-level statistics. The Geocoder transform has three modes of geocode processing: • Address geocoding: The Geocoder transform assigns geographic data. Based on the completeness of the input address data, the transform returns multiple levels of latitude and longitude data. Appending different levels of latitude and longitude information to your data may help your organization to target certain population sizes and other regional geographical data. • Reverse geocoding: The Geocoder transform identifies the closest address or point of interest based on an input reference location. • POI textual search: The Geocoder transform uses address fields and POI name or type fields as search cri-

Characteristic	Description
	teria to match with points of interest. The transform outputs the results in the Result_List XML output field. Best practice is to use the Geocoder transform with the Global Address Cleanse or USA Regulatory Address Cleanse transform.
Content objects	Sample XML file.

[Address geocoding \[page 546\]](#)

In address geocoding mode, the Geocoder transform assigns geographic data to the address.

[Reverse geocoding \[page 552\]](#)

In reverse geocoding mode, the Geocoder transform identifies the closest address or point of interest based on an input reference location.

[Point of Interest \(POI\) and Address Geocoding \[page 559\]](#)

The Geocoder transform returns a list of locations that match within an area of the input address or geographical coordinates when available.

[Point of Interest \(POI\) and Reverse Geocoding \[page 560\]](#)

Using reverse geocoding, the Geocoder transform returns the closest address or POI when you input reference location.

[POI textual search \[page 565\]](#)

In the POI (point of interest) textual search mode, the Geocoder transform uses address fields and POI name or type fields as search criteria to match with points of interest.

[Geocoder directories \[page 569\]](#)

The Geocoder transform provides two directory formats based on the type of geocoding processing you perform.

Related Information

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

17.1 Address geocoding

In address geocoding mode, the Geocoder transform assigns geographic data to the address.

Based on the completeness of the input address data, the Geocoder transform can return multiple levels of latitude and longitude data. To help your organization target certain population sizes and other regional geographical data, append different levels of latitude and longitude information.

Prepare records for geocoding

The Geocoder transform works best when it has standardized and corrected address data. To obtain the most accurate information, place an address cleanse transform before the Geocoder transform in a data flow.

17.1.1 Latitude and longitude levels

The Geocoder transform returns latitude and longitude levels for primary number, postcode centroid, and locality centroid.

Primary number level

If your input data has a complete address that includes the primary number, the Geocoder transform returns latitude and longitude coordinates to the exact location. The transform bases the exact location on the directory type, either range-based or parcel-based, that you subscribe to.

Primary Range Interpolated (PRI) assignment level

The PRI assignment level is based on range-type directories. In general, the Geocoder transform uses geocoding directories to calculate latitude and longitude values for a house. Geocoder interpolates between the beginning and ending point of a line segment, where the line segment represents a range of houses. The latitude and longitude values can be slightly offset from the exact location from where the house actually exists.

i Note

If you want an address-level assignment, ensure that you map the *PRIMARY_NUMBER* input field and that it isn't blank.

Primary Range Exact (PRE) assignment level

The PRE assignment level is based on the parcel-type directories. Parcel directories contain the most precise and accurate latitude and longitude values available for addresses, depending on the available country data. Parcel data is stored as points. Rather than getting you near the house, it takes you to the exact door.

Postcode Centroid level

For the Postcode Centroid level, if an address has a postcode, the Geocoder transform outputs coordinates in the appropriate postcode area. The transform supports the following assignment levels based on the available country data:

- [POSTCODE_FULL](#) (PF)
- [POSTCODE2_PARTIAL](#) (P2P)
- [POSTCODE1](#) (P1)

Locality Centroid level

For the Locality Centroid level, if an address has a locality, the Geocoder transform outputs coordinates in the appropriate locality area. The transform supports the following locality assignment levels based on the available country data:

- [LOCALITY1](#) (I1)
- [LOCALITY2](#) (I2)
- [LOCALITY3](#) (I3)
- [LOCALITY4](#) (I4)

17.1.2 Address geocoding field mapping

For the address geocoding mode, the Geocoder transform has few requirements for mapping input and output fields, however, the more input data that you provide, the better the results.

i Note

Input and output fields for geocoding have categories. Therefore, we list input and output fields by category.

Input field mapping

The only requirement for input field mapping for address geocoding mode is that you provide at least one input field from the Address field category.

The following table contains the input field categories, and the field requirements for each category.

Input field category	Address geocoding mode
Address	Required (At least one input field from the Address category.)

Input field category	Address geocoding mode
Address POI	Optional
Latitude/Longitude	n/a
Max Records	n/a
Search Filter	n/a

Output field mapping

The address geocoding mode doesn't require any output fields. However, include optional fields so that the Geocoder transform enriches your data with additional information.

The following table contains the output field categories, and the requirements for the output fields in each category.

Output field category	Address geocoding mode
Address	n/a
Address POI	n/a
Assignment Level	Optional
Census	Optional
Distance	n/a
Info Code	Optional
Latitude/Longitude	Optional
Population	Optional
Results	n/a
Side of Street	Optional
Status Code	Optional

17.1.3 Example: Address geocoding with latitude and longitude

Assign latitude and longitude to data by mapping input fields from the address and POI (point of interest) categories.

❖ Example

Input record:

332 Front St.
La Crosse, WI 54601-4023
US

The following table shows the mapped input values.

Input field name	Input value
<i>COUNTRY</i>	US
<i>POSTCODE1</i>	54601
<i>POSTCODE2</i>	4023
<i>PRIMARY_NAME1</i>	Front
<i>PRIMARY_NUMBER</i>	332
<i>PRIMARY_TYPE1</i>	St.

The output field *ASSIGNMENT_LEVEL* shows that the Geocoder transform used the parcel-type directory for the Primary Range Exact (PRE) assignment level. The transform also outputs the latitude and longitude information for the primary address.

The following table shows the output values after the Geocoder transform processes the input record.

Output field name	Output value
<i>ASSIGNMENT_LEVEL</i>	PRE
<i>LATITUDE</i>	43.811616
<i>LONGITUDE</i>	-91.256695

17.1.3.1 Address geocoding input fields

For the address geocoding example for latitude and longitude, map the *COUNTRY* input field from the Address category.

To enhance the results, also include optional input fields from the Address and Address POI categories.

The following table lists all the possible input fields to map for this example. The input fields listed with the asterisks indicate the fields in the example.

Category	Input field name
Address	* <i>COUNTRY</i> (required)
	<i>LOCALITY1-4</i>
	* <i>POSTCODE1-2</i>
	* <i>PRIMARY_NAME1-4</i>
	* <i>PRIMARY_NUMBER</i>
	<i>PRIMARY_POSTFIX1</i>
	<i>PRIMARY_PREFIX1</i>
	* <i>PRIMARY_TYPE1-4</i>
Address POI	<i>REGION1-2</i>
	<i>POI_NAME</i>
	<i>POI_TYPE</i>

For descriptions of all Geocoder input fields, see [Geocoder input fields \[page 714\]](#).

17.1.3.2 Address geocoding output fields

For the address geocoding example for latitude and longitude, output fields are optional, however, select optional fields from the output field categories.

The following table lists the output fields to map for this example. The output fields listed with the asterisks indicate the fields in the example.

Category	Output field name
Assignment Level	* <i>ASSIGNMENT_LEVEL</i>
	<i>ASSIGNMENT_LEVEL_LOCALITY</i>
	<i>ASSIGNMENT_LEVEL_POSTCODE</i>
Census	<i>CENSUS_TRACT_BLOCK</i>
	<i>CENSUS_TRACT_BLOCK_GROUP</i>
	<i>GOV_COUNTRY_CODE</i>
	<i>GOV_LOCALITY1_CODE</i>
	<i>GOV_REGION1_CODE</i>
	<i>METRO_STAT_AREA_CODE</i>
	<i>MINOR_DIV_CODE</i>
	<i>STAT_AREA_CODE</i>

Category	Output field name
Info Code	<i>INFO_CODE</i>
Latitude/Longitude	<i>*LATITUDE</i> <i>LATITUDE_LOCALITY</i> <i>LATITUDE_POSTCODE</i> <i>LATITUDE_PRIMARY_NUMBER</i> <i>*LONGITUDE</i> <i>LONGITUDE_LOCALITY</i> <i>LONGITUDE_POSTCODE</i> <i>LONGITUDE_PRIMARY_NUMBER</i>
Population	<i>POPULATION_CLASS_LOCALITY1</i>
Side of Street	<i>SIDE_OF_PRIMARY_ADDRESS</i>
Status Code	<i>STATUS_CODE</i>

For descriptions of all Geocoder output fields, see [Geocoder output fields \[page 718\]](#).

17.2 Reverse geocoding

In reverse geocoding mode, the Geocoder transform identifies the closest address or point of interest based on an input reference location.

The Geocoder transform processes reverse geocoding in four modes:

- Address input and single record output
- Address input and multiple record output
- Latitude and longitude input and single record output
- Latitude and longitude input and multiple record output

The mode that the transform uses is based on the input mapping as described in the following table.

Mode	Description
Address input and single record output	Determines a unique reference point. Input: • Include the required field plus additional fields from the Address category. • Include the optional fields from the Address POI category. Output: The single closest record that matches the input field from the Search Filter category. → Remember If don't map fields from the Search Filter category, the Geocoder transform defaults to the nearest point of interest or address.
Address input and multiple record output	Determines unique reference points. Input: • Include the required field plus additional fields from the Address category. • Include the optional fields from the Address POI category. Output: • Reference point data fields. • RESULT_LIST: Include for the multiple closest records that match the fields from the Search Filter category. To limit the number of multiple records, enter a value in the MAX_RECORDS option or the MAX_RECORDS input field.
Latitude and longitude input and single record output	Determines the single closest record. Input: • Latitude/Longitude category fields (as the reference point) • Search Filter category fields Output: The single closest record that matches the fields from the Search Filter category. → Remember If don't map fields from the Search Filter category, the Geocoder transform defaults to the nearest point of interest or address.

Mode	Description
Latitude and longitude input and multiple record output	Determines multiple closest records. Input: - Latitude/Longitude category fields (as the reference point) - Search Filter category fields Output: RESULT_LIST: Contains multiple closest records that match the fields from the Search Filter category. To limit the number of multiple records, enter a value in the MAX_RECORDS option. Or, if you map the MAX_RECORDS input field, the transform uses the value to determine the number of multiple records. → Remember If don't map fields from the Search Filter category, the Geocoder transform defaults to the nearest points of interest or addresses.

i Note

- Mapping the **RADIUS** and **DISTANCE_UNIT** input fields lets you define the distance from the specified reference point and identify an area in which matching records are located.
- To find one or more locations that can be points of interest, addresses, or both, set the **SEARCH_FILTER_NAME** or **SEARCH_FILTER_TYPE** input field. Setting these fields limits the output matches to your search criteria.
- To return a non-POI address only, enter **ADDR** in the **SEARCH_FILTER_TYPE** input field.
- To return a POI only, enter the POI name or type in the **SEARCH_FILTER_TYPE** input field.
- If you don't set a search filter, the transform returns both addresses and points of interest.

17.2.1 Reverse geocoding field mapping

The fields that you map for the input mapping determine the reverse geocoding mode in which the Geocoder transform processes your data.

i Note

Input and output fields for reverse geocoding have categories. Therefore, we list input and output fields by category.

The following tables specify the input and output fields that are required, optional, or can't be mapped in each of the reverse geocoding modes.

Input field mapping

Consider including optional input fields because, the more input data you provide, the better the results.

The following table lists the input field mapping requirements for each input field category by reverse geocoding mode.

Input field category	Reverse geocoding mode			
	Address input single output	Address input multiple output	Latitude/longitude input single output	Latitude/longitude input multiple output
Address	At least one required	At least one required	n/a	n/a
Address POI	Optional	Optional	n/a	n/a
Latitude/Longitude	n/a	n/a	At least one required	At least one required
Max Records	n/a	At least one required	n/a	At least one required
Search Filter	At least one required	At least one required	Optional	Optional

Output field mapping

The following table lists the output field mapping requirements for each output field category by reverse geocoding mode.

Output field category	Reverse geocoding mode			
	Address input single output	Address input multiple output	Latitude/longitude input single output	Latitude/longitude input multiple output
Address	Optional	n/a	Optional	n/a
Address POI	Optional	n/a	Optional	n/a
Assignment Level	Optional	Optional	Optional	n/a
Census	Optional	n/a	Optional	n/a
Distance	Optional	n/a	Optional	n/a
Info Code	Optional	Optional	Optional	Optional
Latitude/ Longitude	Optional	Optional	Optional	n/a
Population	Optional	n/a	Optional	n/a
Results	n/a	Optional	n/a	Optional
Side of Street	Optional	Optional	Optional	n/a

17.2.2 Example: Reverse geocoding using latitude and longitude

The following example uses latitude and longitude and a search filter to output a single point of interest closest to the input latitude and longitude.

❖ Example

Return the address for the input latitude and longitude for the firm name Business Objects.

Input: The following table shows the input record parsed into input fields.

Input field name	Input value
<i>Latitude</i>	43.811616
<i>Longitude</i>	-91.256695
<i>Search_Filter_Name</i>	BUSINESS OBJECTS

Output: After running the job, the transform outputs a matching address into the mapped output fields as shown in the following table. You can see how the additional data added to the record enriches your data.

Output field name	Output value
<i>Address_Line</i>	332 FRONT ST
<i>Assignment_Level</i>	PRE
<i>Country_Code</i>	US
<i>Distance</i>	1.3452
<i>Locality1</i>	LA CROSSE
<i>Postcode</i>	54601-4023
<i>Postcode1</i>	54601
<i>Postcode2</i>	4023
<i>Primary_Name1</i>	FRONT
<i>Primary_Number</i>	332
<i>Primary_Type1</i>	ST
<i>POL_Name</i>	BUSINESS OBJECTS
<i>POL_Type</i>	5800
<i>Region1</i>	WI

[Reverse geocoding using latitude and longitude: Input fields \[page 557\]](#)

For the reverse geocoding example, the input fields that you map are based on whether you use the single-record or multiple-record modes.

[Reverse geocoding using latitude and longitude: Available output fields \[page 557\]](#)

Even though all output fields are optional, there are several categories of fields that you can include in output field mapping for the reverse geocoding example.

17.2.2.1 Reverse geocoding using latitude and longitude: Input fields

For the reverse geocoding example, the input fields that you map are based on whether you use the single-record or multiple-record modes.

Required: The following table lists the required input fields for the single-record and multiple-record result modes of reverse geocoding.

Category	Single-record results input field name	Multiple-record results input field name
Latitude/Longitude	<i>LATITUDE</i>	<i>LATITUDE</i>
	<i>LONGITUDE</i>	<i>LONGITUDE</i>
Max Records	n/a	<i>MAX_RECORDS</i>

Optional: The following table lists the optional input fields for the single-record and multiple-record result modes of reverse geocoding.

Category	Single-record results input field name	Multiple-record results input field name
Search Filter	<i>RADIUS</i>	<i>RADIUS</i>
	<i>SEARCH_FILTER_NAME</i>	<i>SEARCH_FILTER_NAME</i>
	<i>SEARCH_FILTER_TYPE</i>	<i>SEARCH_FILTER_TYPE</i>

For descriptions of all Geocoder input fields, see [Geocoder input fields \[page 714\]](#).

Parent topic: [Example: Reverse geocoding using latitude and longitude \[page 556\]](#)

Related Information

[Reverse geocoding using latitude and longitude: Available output fields \[page 557\]](#)

17.2.2.2 Reverse geocoding using latitude and longitude: Available output fields

Even though all output fields are optional, there are several categories of fields that you can include in output field mapping for the reverse geocoding example.

The following table lists the field categories and output fields for both single-record and multiple-record results.

Category	Single-record results	Multiple-record results
	Output field name	Output field name
Address	<i>ADDRESS_LINE</i>	n/a
	<i>COUNTRY_CODE</i>	
	<i>POSTCODE</i>	
	<i>LOCALITY1-4</i>	
	<i>POSTCODE1-2</i>	
	<i>PRIMARY_NAME1-4</i>	
	<i>PRIMARY_NUMBER</i>	
	<i>PRIMARY_POSTFIX1</i>	
	<i>PRIMARY_PREFIX1</i>	
	<i>PRIMARY_RANGE_HIGH</i>	
	<i>PRIMARY_RANGE_LOW</i>	
	<i>PRIMARY_TYPE1-4</i>	
	<i>REGION1-2</i>	
Address POI	<i>POI_NAME</i>	n/a
	<i>POI_TYPE</i>	
Assignment Level	<i>ASSIGNMENT_LEVEL</i>	n/a
	<i>ASSIGNMENT_LEVEL_LOCALITY</i>	
	<i>ASSIGNMENT_LEVEL_POSTCODE</i>	
Census	<i>CENSUS_TRACT_BLOCK</i>	n/a
	<i>CENSUS_TRACT_BLOCK_GROUP</i>	
	<i>GOV_COUNTY_CODE</i>	
	<i>GOV_LOCALITY1_CODE</i>	
	<i>GOV_REGION1_CODE</i>	
	<i>METRO_STAT_AREA_CODE</i>	
	<i>MINOR_DIV_CODE</i>	
	<i>STAT_AREA_CODE</i>	
Distance	<i>DISTANCE</i>	n/a
Info Code	<i>INFO_CODE</i>	<i>INFO_CODE</i>

Category	Single-record results	Multiple-record results
	Output field name	Output field name
Latitude/Longitude	<i>LATITUDE</i>	n/a
	<i>LATITUDE_LOCALITY</i>	
	<i>LATITUDE_POSTCODE</i>	
	<i>LATITUDE_PRIMARY_NUMBER</i>	
	<i>LONGITUDE</i>	
	<i>LONGITUDE_LOCALITY</i>	
	<i>LONGITUDE_POSTCODE</i>	
	<i>LONGITUDE_PRIMARY_NUMBER</i>	
Population	<i>POPULATION_CLASS_LOCALITY1</i>	n/a
Results	n/a	<i>RESULT_LIST</i>
		<i>RESULT_LIST_COUNT</i>
Side of Street	<i>SIDE_OF_PRIMARY_ADDRESS</i>	n/a

For descriptions of all Geocoder output fields, see [Geocoder output fields \[page 718\]](#).

Parent topic: [Example: Reverse geocoding using latitude and longitude \[page 556\]](#)

Related Information

[Reverse geocoding using latitude and longitude: Input fields \[page 557\]](#)

17.3 Point of Interest (POI) and Address Geocoding

The Geocoder transform returns a list of locations that match within an area of the input address or geographical coordinates when available.

To output geographic data, run the Geocoder transform in address geocoding mode. Based on the completeness of the input address data, the Geocoder transform can return multiple levels of latitude and longitude data. Latitude and longitude information may help your organization to target certain population sizes and other regional geographical data.

If you have a complete address with the primary number as input data, the Geocoder transform returns the latitude and longitude coordinates to the exact location. If you have an address that has only a locality or Postcode, the Geocoder transform returns coordinates in the locality or Postcode area, respectively.

17.4 Point of Interest (POI) and Reverse Geocoding

Using reverse geocoding, the Geocoder transform returns the closest address or POI when you input reference location.

The reference location can include one of the following pieces of information:

- Latitude and longitude
- Address
- Point of interest

When you map the following input fields, you define the distance from the specified reference point and identify an area in which matching records are located:

- [Option_Radius](#)
- [Option_Distance_Unit](#)

When you map the following input fields, you limit the results to points of interest, addresses, or both:

- [Search_Filter_Name](#)
- [Search_Filter_Type](#)

By including the filter fields, you limit the output matches to your search criteria. If you don't set a search filter, the transform returns both addresses and points of interest.

❖ Example

To return only an address, set the input field [Search_Filter_Type](#) to ADDR.

❖ Example

To return only a point of interest, enter the POI name in the [Search_Filter_Name](#) field or enter the POI type in the [Search_Filter_Type](#) field.

17.4.1 Example: Reverse geocoding using POI

To output a single POI (point of interest) closest to the input address, use an input address and a search filter.

❖ Example

The request is to find the closest bank to the input location and to also output the latitude and longitude information for the single POI.

i Note

The POI type for banks is 6000.

Input location:

BUSINESS OBJECTS
332 Front St.

La Crosse, WI 55601-4023

US

The following table lists the data from the input record mapped to the input fields.

Input field name	Input value
Country	US
Postcode1	55601
Postcode2	4023
Primary_Name1	Front
Primary_Number	332
Primary_Type1	St.
POI_Name	BUSINESS OBJECTS
Search_Filter_Type	6000

Output: The mapped output fields contain the results listed in the following table.

Output field name	Output value
Address_Line	201 MAIN ST
Assignment_Level	PRE
Country_Code	US
Distance	0.4180
Locality1	LA CROSSE
POI_Name	US BANK
POI_Type	6000
Postcode1	54601
Primary_Name1	MAIN
Primary_Number	201
Primary_Type1	ST
Region1	WI

[Reverse geocoding using POI input fields \[page 562\]](#)

For the reverse geocoding, the input fields that you map are based on whether you use the single-record or multiple-record modes.

[Reverse geocoding using POI: Available output fields \[page 563\]](#)

Even though all output fields are optional, there are several categories of fields that you can include in output field mapping for the reverse geocoding example.

17.4.1.1 Reverse geocoding using POI input fields

For the reverse geocoding, the input fields that you map are based on whether you use the single-record or multiple-record modes.

Required:

For required input fields, at least one input field in each category must be mapped. The following table lists the required input fields for each category for single-record or multiple-record results.

Category	Single-record results input field name	Multiple-record results input field name
Address	<i>ADDRESS_LINE</i>	<i>LATITUDE</i>
	<i>COUNTRY</i>	<i>LONGITUDE</i>
	<i>LOCALITY1-4</i>	
	<i>POSTCODE</i>	
	<i>POSTCODE1-2</i>	
	<i>PRIMARY_NAME1-4</i>	
	<i>PRIMARY_NUMBER</i>	
	<i>PRIMARY_POSTFIX1</i>	
	<i>PRIMARY_PREFIX1</i>	
	<i>PRIMARY_RANGE_HIGH</i>	
	<i>PRIMARY_RANGE_LOW</i>	
	<i>PRIMARY_TYPE1-4</i>	
	<i>REGION1_2</i>	
Max Records	n/a	<i>MAX_RECORDS</i>
Search Filter	<i>RADIUS</i>	<i>RADIUS</i>
	<i>SEARCH_FILTER_NAME</i>	<i>SEARCH_FILTER_NAME</i>
	<i>SEARCH_FILTER_TYPE</i>	<i>SEARCH_FILTER_TYPE</i>

Optional:

The following table lists the optional input fields for each category for single-record or multiple-record results.

Category	Single-record results	Multiple-record results
	Input field name	Input field name
Address POI	<i>POI_NAME</i>	<i>POI_NAME</i>
	<i>POI_TYPE</i>	<i>POI_TYPE</i>

For descriptions of all Geocoder input fields, see [Geocoder input fields \[page 714\]](#).

Parent topic: [Example: Reverse geocoding using POI \[page 560\]](#)

Related Information

[Reverse geocoding using POI: Available output fields \[page 563\]](#)

17.4.1.2 Reverse geocoding using POI: Available output fields

Even though all output fields are optional, there are several categories of fields that you can include in output field mapping for the reverse geocoding example.

For a single-record result, the output fields are the results for the spatial search.

For multiple-record results, the output fields in the Assignment Level and Latitude/Longitude categories are the results for the reference address assignment. Output fields in the Results category are the results for the reverse geocoding.

The following table contains the output fields available by category for reverse geocoding.

Category	Single-record results Output field name	Multiple-record results Output field name
Address	<i>ADDRESS_LINE</i>	n/a
	<i>COUNTRY_CODE</i>	
	<i>LOCALITY1-4</i>	
	<i>POSTCODE</i>	
	<i>POSTCODE1-2</i>	
	<i>PRIMARY_NAME1-4</i>	
	<i>PRIMARY_NUMBER</i>	
	<i>PRIMARY_POSTFIX1</i>	
	<i>PRIMARY_PREFIX1</i>	
	<i>PRIMARY_RANGE_HIGH</i>	
	<i>PRIMARY_RANGE_LOW</i>	
	<i>PRIMARY_TYPE1-4</i>	
	<i>REGION1-2</i>	
Address POI	<i>POI_NAME</i>	n/a
	<i>POI_TYPE</i>	

Category	Single-record results Output field name	Multiple-record results Output field name
Assignment Level	<i>ASSIGNMENT_LEVEL</i>	<i>ASSIGNMENT_LEVEL</i>
	<i>ASSIGNMENT_LEVEL_LOCALITY</i>	<i>ASSIGNMENT_LEVEL_LOCALITY</i>
	<i>ASSIGNMENT_LEVEL_POSTCODE</i>	<i>ASSIGNMENT_LEVEL_POSTCODE</i>
Census	<i>CENSUS_TRACT_BLOCK</i>	n/a
	<i>CENSUS_TRACT_BLOCK_GROUP</i>	
	<i>GOV_COUNTY_CODE</i>	
	<i>GOV_LOCALITY1_CODE</i>	
	<i>GOV_REGIONAL1_CODE</i>	
	<i>METRO_STAT_AREA_CODE</i>	
	<i>MINOR_DIV_CODE</i>	
	<i>SAT_AREA_CODE</i>	
Distance	<i>DISTANCE</i>	n/a
Info Code	<i>INFO_CODE</i>	<i>INFO_CODE</i>
Latitude/Longitude	<i>LATITUDE</i>	<i>LATITUDE</i>
	<i>LATITUDE_LOCALITY</i>	<i>LATITUDE_LOCALITY</i>
	<i>LATITUDE_POSTCODE</i>	<i>LATITUDE_POSTCODE</i>
	<i>LATITUDE_PRIMARY_NUMBER</i>	<i>LATITUDE_PRIMARY_NUMBER</i>
	<i>LONGITUDE</i>	<i>LONGITUDE</i>
	<i>LONGITUDE_LOCALITY</i>	<i>LONGITUDE_LOCALITY</i>
	<i>LONGITUDE_POSTCODE</i>	<i>LONGITUDE_POSTCODE</i>
	<i>LONGITUDE_PRIMARY_NUMBER</i>	<i>LONGITUDE_PRIMARY_NUMBER</i>
Population	<i>POPULATION_CLASS_LOCALITY1</i>	n/a
Results	n/a	<i>RESULT_LIST</i>
		<i>RESULT_LIST_COUNT</i>
Side of Street	<i>SIDE_OF_PRIMARY_ADDRESS</i>	<i>SIDE_OF_PRIMARY_ADDRESS</i>

For descriptions of all Geocoder output fields, see [Geocoder output fields \[page 718\]](#).

Parent topic: [Example: Reverse geocoding using POI \[page 560\]](#)

Related Information

[Reverse geocoding using POI input fields \[page 562\]](#)

17.5 POI textual search

In the POI (point of interest) textual search mode, the Geocoder transform uses address fields and POI name or type fields as search criteria to match with points of interest.

The transform outputs the results to the [RESULT_LIST XML](#) output field.

The transform determines the number of multiple records by the [MAX_RECORDS](#) input field if it is mapped and populated, or the [MAX_RECORDS](#) option.

[POI textual search field mapping \[page 565\]](#)

For POI textual search, the Geocoder transform uses input and output fields that are required, optional, or not applicable.

[Example: POI textual search \[page 566\]](#)

Use an address and POI (point of interest) information to identify a list of potential POI matches.

17.5.1 POI textual search field mapping

For POI textual search, the Geocoder transform uses input and output fields that are required, optional, or not applicable.

i Note

Input and output fields for POI textual search have categories. Therefore, we list input and output fields by category.

Input field mapping

In the POI textual search mode, you must map at least one input field from the Address POI category, which is the key difference between the POI textual search and address geocoding modes.

Input field category	Address geocoding mode
Address	At least one required
Address POI	At least one required
Latitude/Longitude	n/a
Max Records	At least one required
Search Filter	n/a

Output field mapping

The following table lists the requirements for output field mapping based on the output field category. Even though none of the output fields are required, use the output fields to enrich your data.

Output field category	Address geocoding mode
Address	n/a
Address POI	n/a
Assignment Level	n/a
Census	n/a
Distance	n/a
Info Code	Optional
Latitude/Longitude	n/a
Population	n/a
Results	Optional
Side of Street	n/a

Parent topic: [POI textual search \[page 565\]](#)

Related Information

[Example: POI textual search \[page 566\]](#)

17.5.2 Example: POI textual search

Use an address and POI (point of interest) information to identify a list of potential POI matches.

❖ Example

Use POI textual search to identify a list of potential POI matches for all records for BUSINESS OBJECTS on Front Street.

Input:

The following table shows the input mapping for one input record.

Input field name	Input value
<i>Country</i>	US
<i>Postcode1</i>	54601

Input field name	Input value
<i>Postcode2</i>	4023
<i>Primary_Name1</i>	Front
<i>Primary_Type1</i>	St.
<i>POI_Name</i>	BUSINESS OBJECTS
<i>Max_Records</i>	10

i Note

The number of records is determined by the *Max_Records* input field or the *Max Records* option. The setting for this example is 10.

Output:

The following table shows the processed output results for the input record in the mapped output fields.

Output field name	Output value
<i>Result_List</i>	Output as XML
<i>Result_List_Count</i>	1

Result_List XML: The XML result for this example has one record.

```
<RESULT_LIST>
  <RECORD>
    <ADDRESS_LINE>332 FRONT ST</ADDRESS_LINE>
    <ASSIGNMENT_LEVEL>PRE</ASSIGNMENT_LEVEL>
    <COUNTRY_CODE>US</COUNTRY_CODE>
    <DISTANCE>0.3340</DISTANCE>
    <LATITUDE>43.811616</LATITUDE>
    <LOCALITY1>LA CROSSE</LOCALITY1>
    <LONGITUDE>-91.256695</LONGITUDE>
    <POI_NAME>BUSINESS OBJECTS</POI_NAME>
    <POI_TYPE>5800</POI_TYPE>
    <POSTCODE>56001-4023</POSTCODE>
    <POSTCODE1>56001</POSTCODE1>
    <POSTCODE2>4023</POSTCODE2>
    <PRIMARY_NAME1>FRONT</PRIMARY_NAME1>
    <PRIMARY_NUMBER>332</PRIMARY_NUMBER>
    <PRIMARY_TYPE1>ST</PRIMARY_TYPE1>
    <RANKING>1</RANKING>
    <REGION1>WI</REGION1>
  </RECORD>
</RESULT_LIST>
```

Parent topic: [POI textual search \[page 565\]](#)

Related Information

[POI textual search field mapping \[page 565\]](#)

17.5.2.1 POI textual search input and output fields

When configuring a POI textual search, use input and output fields based on your input data and your output requirements.

Input:

Map at least one input field from each of the listed categories in the following table. You must map the [COUNTRY](#) field from the Address category. The more input data you can provide, the better results you will obtain.

Category	Input field name
Address	COUNTRY (required)
	LOCALITY1-4
	POSTCODE1-2
	PRIMARY_NAME1-4
	PRIMARY_NUMBER
	PRIMARY_POSTFIX1
	PRIMARY_PREFIX1
	PRIMARY_TYPE1-4
	REGION1-2
Address POI	POI_NAME
	POI_TYPE
Max Records	MAX_RECORDS

Output: All output fields are optional.

Category	Output field name
Info Code	INFO_CODE
Results	RESULT_LIST
	RESULT_LIST_COUNT

For descriptions of all Geocoder input and output fields, see [Geocoder fields \[page 714\]](#).

17.6 Geocoder directories

The Geocoder transform provides two directory formats based on the type of geocoding processing you perform.

The Geocoder directories are designed specifically to use with the Geocoder transform. SAP offers two directory formats:

- **Basic**: Supports the address geocoding mode. It returns centroid-level and address-level latitude and longitude information.
- **Advanced**: Supports the address geocoding, reverse geocoding, and POI textual search modes. It returns range-based and centroid-level latitude and longitude information and point-of-interest information

Directory updates

A quarterly vendor-specific directory update is available on the SAP Help Portal and is also included in each directory package. The directory update provides instructions for installing the directories and information about the directory contents. It describes the package names, files names, supported countries, directory format, geocoding mode, centroid level, and census information. It also lists the available POI types supported by the vendor.

The Geocoder transform is flexible enough to accept new country directory data immediately after the directory data is released. There is no need to wait for the next SAP Data Quality Management SDK release to begin using new country directory data.

For information about downloading directories, see the latest Directories Update letters at https://uacp2.hana.ondemand.com/viewer/p/ADDRESSING_DIRECTORIES.

18 Geocoder transform reference

The Geocoder transform XML configuration is grouped according to setting types.

Use the Geocoder transform after an address cleanse transform such as the USA Regulatory Address Cleanse or Global Address Cleanse transforms.

The Data Quality Management SDK installation includes a sample Geocoder XML configuration. The XML contains the following groups for the Geocoder transform:

- System group
- Report and Analysis group
- Geocoder statistics settings and info codes
- Geocoder transform settings

[System group \[page 570\]](#)

To set the location of supporting files and other system settings, complete the options in the System group.

[Geocoder Report and Analysis \[page 571\]](#)

Generate report data for the Geocoder transform by setting the option in the Report and Analysis group.

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

[Geocoder transform options \[page 578\]](#)

To specify the assignment level options for the Geocoder transform, set options in the Geocoder options group.

18.1 System group

To set the location of supporting files and other system settings, complete the options in the System group.

Syntax

```
<SYSTEM>
  <PLATFORM_SETTINGS>
    <SYSTEM_VARS>
      <LINK_DIR>../../../../</LINK_DIR>
    </SYSTEM_VARS>
  </PLATFORM_SETTINGS>
  <SHARED_OBJECT>dqgeocodertransformu</SHARED_OBJECT>
</SYSTEM>
```

The SYSTEM group contains PLATFORM_SETTINGS and the SYSTEM_VARS groups.

Option	Description
LINK_DIR	Specifies the relative path to the Geocoder support files. Enter the relative path to the default location, unless you changed to a different location. The default locations are: - Windows 32-bit: \SAP BusinessObjects\Data quality Mgmt SDK\windows_32\DataQuality\geocoder - UNIX: <DQM SDK install location>/DataQuality/geocoder.
SHARED_OBJECT	Specifies the name of the transform that the platform loads to create the Geocoder transform. The value is set in the XML sample file and you should not change it. For Geocoder, the value is always <i>dqgeocodertransformu</i>.

Parent topic: [Geocoder transform reference \[page 570\]](#)

Related Information

[Geocoder Report and Analysis \[page 571\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Geocoder transform options \[page 578\]](#)

18.2 Geocoder Report and Analysis

Generate report data for the Geocoder transform by setting the option in the Report and Analysis group.

Syntax

```
<TRANSFORM_SETTINGS>
  <REPORT_AND_ANALYSIS>
    <GENERATE_REPORT_DATA>NO</GENERATE_REPORT_DATA>
```

REPORT_AND_ANALYSIS is the beginning part of the TRANSFORM_SETTINGS group.

Option	Description
<code>GENERATE_REPORT_DATA</code>	Specifies whether to generate report data for this transform. <code>YES</code>: Generates report data for this transform. <code>NO</code>: Turns off report data generation To improve performance, set this option to <code>NO</code> when you do not need to generate statistics. For example, during testing. Note This option is available only for the address geocoding mode. If you set this value to <code>YES</code> for the reverse geocoding and POI textual search modes, the software generates a warning and does not generate a report.

Parent topic: [Geocoder transform reference \[page 570\]](#)

Related Information

[System group \[page 570\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

[Geocoder transform options \[page 578\]](#)

18.3 Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms

To enable the generation of data quality statistics, make settings in the Data Quality Statistics Settings group.

The Data Quality Statistics Settings group appears in the Data Cleanse, Global Address Cleanse, and the Geocoder transforms.

Syntax

Data Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSE_STATISTICS_TABLE>NO</
GENERATE_CLEANSE_STATISTICS_TABLE>
  <GENERATE_CLEANSE_INFO_CODES_TABLE>NO</
GENERATE_CLEANSE_INFO_CODES_TABLE>
  <GENERATE_CLEANSE_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSE_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSE_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSE_COMPONENT_INFO_TABLE>
```

```
</DATA_QUALITY_STATISTICS_SETTINGS>
```

Syntax

Global Address Cleanse:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_CLEANSE_STATISTICS_TABLE>NO</
GENERATE_CLEANSE_STATISTICS_TABLE>
  <GENERATE_CLEANSE_INFO_CODES_TABLE>NO</
GENERATE_CLEANSE_INFO_CODES_TABLE>
  <GENERATE_CLEANSE_ADDRESS_RECORD_INFO_TABLE>NO</
GENERATE_CLEANSE_ADDRESS_RECORD_INFO_TABLE>
  <GENERATE_CLEANSE_CHANGE_INFO_TABLE>NO</
GENERATE_CLEANSE_CHANGE_INFO_TABLE>
  <GENERATE_CLEANSE_COMPONENT_INFO_TABLE>NO</
GENERATE_CLEANSE_COMPONENT_INFO_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

Syntax

Geocoder:

```
<DATA_QUALITY_STATISTICS_SETTINGS>
  <GENERATE_GEOCODE_STATISTICS_TABLE>NO</
GENERATE_GEOCODE_STATISTICS_TABLE>
  <GENERATE_GEOCODE_INFO_CODES_TABLE>NO</
GENERATE_GEOCODE_INFO_CODES_TABLE>
</DATA_QUALITY_STATISTICS_SETTINGS>
```

The following table contains the options for the [DATA_QUALITY_STATISTICS_SETTINGS](#) group. Because the group is applicable for more than one transform, the table lists the applicable transform for each option.

The software generates the following types of statistics for each transform:

- The Data Cleanse and the Global Address Cleanse transforms generate additional statistics and info codes about address cleansing results on the data.
- The Geocoder transform generates statistics and info codes that are unique to the Geocoding processes.

Data Quality Statistics options

Option	Transform	Description
GENERATE_CLEANSER_ADDRESS_REC ORD_INFO_TABLE	Global Address Cleanse	Specifies whether the software generates the Cleanse Address Record Information Table. YES: Generates the Cleanse Address Record Information Table. NO: Does not generate the Cleanse Address Record Information Table. NO is the default setting. The Cleanse Address Record Information Table contains non-summary information about the results of the address cleanse process on the data. The information includes assignment information and assignment type for each address. The software generates this information only when you include the Global Address Cleanse transform for address data.
GENERATE_CLEANSER_CHANGE_INFO_ TABLE	Data Cleanse Global Address Cleanse	Specifies whether the software generates the Cleanse Change Information Table. YES: Generates the Cleanse Change Information Table. NO: Does not generate the Cleanse Change Information Table. NO is the default setting. The Cleanse Change Information Table contains non-summary information. It identifies areas of concern from your source data by the number of significant changes.

Option	Transform	Description
<code>GENERATE_CLEANSER_COMPONENT_INFORMATION_TABLE</code>	Data Cleanse Global Address Cleanse	Specifies whether the software generates the Cleanse Component Information Table. YES: Generates the Cleanse Component Information Table. NO: Does not generate the Cleanse Component Information Table. NO is the default setting. Note This option is available in Data Quality Management SDK version 4.2.6 and later. The Cleanse Component Information Table contains non-summary information. It contains position information for each specified data element when one or both of the following occurs: - The transform parses the data element from an input field. - The transform writes the data element to an output field.
<code>GENERATE_CLEANSER_INFORMATION_CODES_TABLE</code>	Data Cleanse Global Address Cleanse	Specifies whether the software generates Cleanse Information Codes Table. YES: Generates the Cleanse Information Codes Table. NO: Does not generate the Cleanse Information Codes Table. NO is the default setting. The Cleanse Information Codes table contains non-summary information. The information includes the distribution of information codes or a count of missing, suspect, or blank data.

Option	Transform	Description
<code>GENERATE_CLEANSER_STATISTICS_TABLE</code>	Data Cleanse Global Address Cleanse	Specifies whether the software generates the Cleanse Statistics Table. YES: Generates the Cleanse Statistics Table. NO: Does not generate the Cleanse Statistics Table. NO is the default setting. The Cleanse Statistics Table contains summarized information for each unique Entity ID. It provides a high-level insight into what the software changed during the cleansing process.
<code>GENERATE_GEOCODE_INFO_CODES_TABLE</code>	Geocoder	Specifies whether to generate the Geocode Information Codes Table. YES: Generates the Geocode Information Codes Table. NO: Does not generate the Geocode Information Codes Table. NO is the default setting. The Geocode Information Codes Table contains non-summary information about why records were not assigned the highest level of latitude and longitude. The statistics are based on the information code data. Note Not all existing information codes apply to the data quality statistics.

Option	Transform	Description
GENERATE_GEOCODE_STATISTICS_TABLE	Geocoder	Specifies whether the software generates the Geocode Statistics Table. YES: Generates the Geocode Statistics Table. NO: Does not generate the Geocode Statistics Table. NO is the default setting. The Geocode Statistics Table contains totals for the following: - The number of rows in the data - The number of rows that were assigned a latitude and longitude.

Parent topic: [Global Address Cleanse transform reference \[page 395\]](#)

Parent topic: [Data Cleanse transform reference \[page 509\]](#)

Parent topic: [Geocoder transform reference \[page 570\]](#)

Related Information

[System group \[page 396\]](#)
[Global Address Cleanse Report and Analysis \[page 397\]](#)
[Global Address Cleanse reference files \[page 398\]](#)
[Global Address Cleanse Country ID Options \[page 404\]](#)
[Global Address Cleanse Engines group \[page 409\]](#)
[Global Address Cleanse Country \[page 411\]](#)
[Global Address Cleanse Standardization Options \[page 412\]](#)
[Canada engine \[page 442\]](#)
[Global Address Cleanse Canada group \[page 445\]](#)
[Global Address engine \[page 450\]](#)
[Global Address Cleanse USA engine \[page 469\]](#)
[Global Address Cleanse Suggestion List option \[page 473\]](#)
[System group \[page 510\]](#)
[Data Cleanse Report and Analysis \[page 511\]](#)
[Data quality statistics common requirements \[page 236\]](#)
[Data Cleanse Cleansing Package \[page 519\]](#)
[Data Cleanse Input Word Breaker \[page 522\]](#)
[Data Cleanse person standardization options \[page 524\]](#)

[Data Cleanse gender options \[page 528\]](#)
[Data Cleanse Firm Standardization options \[page 530\]](#)
[Data Cleanse Other Standardization options \[page 531\]](#)
[Data Cleanse Date Options \[page 535\]](#)
[Data Cleanse Phone Options group \[page 538\]](#)
[Data Cleanse Parser configuration \[page 541\]](#)
[Data Cleanse editor options \[page 543\]](#)
[System group \[page 570\]](#)
[Geocoder Report and Analysis \[page 571\]](#)
[Geocoder transform options \[page 578\]](#)

18.4 Geocoder transform options

To specify the assignment level options for the Geocoder transform, set options in the Geocoder options group.

Syntax

```

<OPTIONS>
  <BEST_ASSIGNMENT_LEVEL>PREFERRED</BEST_ASSIGNMENT_LEVEL>
  <LOCALITY_ASSIGNMENT_THRESHOLD>PREFERRED</LOCALITY_ASSIGNMENT_THRESHOLD>
  <POSTCODE_ASSIGNMENT_THRESHOLD>PREFERRED</POSTCODE_ASSIGNMENT_THRESHOLD>
  <OFFSET_COORDINATES>YES</OFFSET_COORDINATES>
  <DISTANCE_UNIT>KILOMETERS</DISTANCE_UNIT>
  <MAX_RECORDS></MAX_RECORDS>
  <RADIUS></RADIUS>
</OPTIONS>

```

The following option group is required. The options support all countries.

Each option supports a directory format based on the mode. The mapped input fields determine the mode. The following table contains the directory format and related mode and returns.

Directory formats, modes, and returns

Format	Mode	Returns
Basic	Mode: Address geocoding mode.	Returns: - Centroid level latitude and longitude information - Address level latitude and longitude information

Format	Mode	Returns
Advanced	Mode: • Address geocoding mode • Reverse geocoding mode • POI textual search mode	Returns: • Range based latitude and longitude • Centroid level latitude and longitude • Point of interest

The following table contains descriptions for the Geocoder transform options, and it includes the supported directory format.

Option	Description	Directory Formats
<i>BEST_ASSIGNMENT_LEVEL</i>	Specifies the depth of assignment for the latitude and longitude output fields. • <i>PREFERRED</i>: Assigns to the finest depth, which is to the primary number level by default. • <i>PRIMARY_NUMBER</i>: Assigns to the primary number level. • <i>POSTCODE</i>: Assigns to the post-code level. • <i>LOCALITY</i>: Assigns to the locality, city, or suburb level. • <i>SMALLEST_AREA</i>: First attempts to assign to the primary number. If no primary number, assigns based on postcode or locality information, depending on the level that is the smaller area. ❖ Example For example, the French post-code 75014 is a smaller area than the locality of Paris. i Note If you set the <i>LOCALITY_ASSIGNMENT_THRESHOLD</i> or <i>POSTCODE_ASSIGNMENT_THRESHOLD</i> options to <i>NONE</i> for the <i>PREFERRED</i>, <i>PRIMARY_NUMBER</i>, and <i>SMALLEST_AREA</i> assignment level values, the software uses locality or postcode centroid assignment.	Basic and Advanced

Option	Description	Directory Formats
<i>DISTANCE_UNIT</i>	Specifies the unit of distance used for the radius. • <i>KILOMETERS</i> • <i>MILES</i> The value of the <i>OPTION_DISTANCE_UNIT</i> input field takes precedence over the value of the <i>DISTANCE_UNIT</i> option. The transform uses the value of the <i>DISTANCE_UNIT</i> option only under the following circumstances: • You do not map the <i>OPTION_DISTANCE_UNIT</i> input field • The <i>OPTION_DISTANCE_UNIT</i> input field is invalid • The <i>OPTION_DISTANCE_UNIT</i> input field is blank • The <i>OPTION_DISTANCE_UNIT</i> input field is NULL	Advanced

Option	Description	Directory Formats
<i>MAX_RECORDS</i>	Specifies the maximum number of records that can be returned. Enter a number up to 100. The value of the <i>OPTION_MAX_RECORDS</i> input field takes precedence over the value of the <i>MAX_RECORDS</i> option. The software uses the value of the <i>MAX_RECORDS</i> option only when you do not map the <i>OPTION_MAX_RECORDS</i> input field or it is blank.	Advanced
<i>OFFSET_COORDINATES</i>	Specifies whether the software returns the offset values of latitude and longitude when the side of the street is known. Use this option for address geocoding mode, and also for reverse geocoding with address mode. • <i>YES</i>: Returns the offset values. • <i>NO</i>: Returns the center value regardless of whether the side of the street is known.	Advanced

Option	Description	Directory Formats
<i>POSTCODE_ASSIGNMENT_THRESHOLD</i>	Limits the level of postcode centroid assignment. • <i>POSTCODE_FULL</i>: Assigns to the entire extended postcode. ❖ Example In the USA, <i>POSTCODE_FULL</i> assigns to the 5-digit postcode and all four digits of the ZIP+4. • <i>POSTCODE1</i>: Assigns to the city or postcode area. ❖ Example In the USA, <i>POSTCODE1</i> assigns to the 5-digit ZIP Code. • <i>POSTCODE2_PARTIAL</i>: Assigns to the first few characters of the extended postcode. ❖ Example In the USA, <i>POSTCODE2_PARTIAL</i> assigns the 5-digit postcode and the first two digits of the ZIP+4. • <i>PREFERRED</i>: Assigns to the finest depth at the postcode level. • <i>NONE</i>: Skips the specific assignment level. Use this setting when you do not want to return an assignment threshold on postcode.	Basic and Advanced

Option	Description	Directory Formats
RADIUS	The distance from a specified reference point. The reference point identifies an area in which matching records are located. The default value is 1 Kilometer. In reverse geocoding mode, the maximum radius distance is 111 kilometers or 68.97 miles. The value of the OPTION_RADIUS input field takes precedence over the value of the RADIUS option. The software uses the value of the RADIUS option only when the OPTION_RADIUS input field is not mapped or is blank.	Advanced

Parent topic: [Geocoder transform reference \[page 570\]](#)

Related Information

[System group \[page 570\]](#)

[Geocoder Report and Analysis \[page 571\]](#)

[Data Quality Statistics for Data Cleanse, Global Address Cleanse, and Geocoder transforms \[page 399\]](#)

19 Match transform

Use the Match transform to find duplicate records in your data sources and process them based on defined business rules.

Match information

Characteristic	Description
Use	Set rules for matching records, and establish thresholds and scores that the software uses to determine near matches and matching records. After the transform completes processing, it sends matching and unique records on to the next transform in the data flow. For best results, process the data in which you are attempting to find matches through data cleanse and Address Cleanses standardization processes. Best practice is to include other Data Quality transforms before the Match transform in your data flow.
Content objects	Sample XML file.

The Match transform is only one tool for you to use in your matching strategy. For more information about matching concepts and other transforms you can use to achieve the results you are looking for, see the Match section of the *Designer Guide*.

[Develop a match strategy \[page 587\]](#)

A match strategy defines the type of matching to perform on your data based on the data content and your intended results, such as individual, family, or household matching.

[Match components \[page 588\]](#)

When you prepare for matching, you configure components on many levels, especially match sets, match levels, and match criteria.

[Track physical and logical sources \[page 592\]](#)

Tracking your data sources is essential for producing informative match statistics and reports.

[Control comparisons \[page 595\]](#)

To save processing time and increase match quality, control the tables and the records that the Match transform compares.

[Prepare data for matching \[page 598\]](#)

To achieve the best possible match results, preprocess your data with other transforms, such as with Data Cleanse or with Global Address Cleanse transforms.

[Compare tables \[page 601\]](#)

Use compare tables to compare records within sources, across sources, or a combination of both.

[Data Salvage \[page 602\]](#)

Data salvaging prevents the Match transform from making matches between records that you don't want to match.

[Match criteria \[page 604\]](#)

Use match criteria in each match level to determine the threshold scores for matching and to define how the Match transform treats various types of data.

[Matching methods \[page 606\]](#)

Matching methods are ways to set up and order match criteria to get desired matching results.

[Matching business rules \[page 611\]](#)

Business rules establish how the Match transform handles various forms of data and differences in your data.

[Group statistics \[page 625\]](#)

Add the *GROUP_STATISTICS* post-match operation after any match level and post-match operation for which you generate statistics about your match groups or input sources.

[Input source select records \[page 626\]](#)

To flag specific record types for evaluation, add an *INPUT_SOURCE_SELECT_RECORD_GROUP* operation to each match level for post match processing.

19.1 Develop a match strategy

A match strategy defines the type of matching to perform on your data based on the data content and your intended results, such as individual, family, or household matching.

Develop a match strategy before you use the Match transform. To help develop your match strategy, answer the following questions:

- What is my data content? For example, does my data contain customer data, or household data? Are records from a single country, or multiple countries?
- What fields do I compare? For example, do I compare the person fields or firm fields?
- What are the relative strengths and weaknesses of the data in the fields I've chosen to compare?

→ Tip

If you process your data through an address cleanse transform before the Match transform, you get better results. Also, profiling your data helps determine the strengths and weaknesses of your data.

- What is my intended end result after matching? For example, do I want one record per family, or do I want one record per family member?

The following table describes the match strategies supported by SAP Data Services.

Strategy	Description
Simple match	Single match criteria that identifies relationships in consumer, business, or product data.
Consumer Householding	Multiple levels of consumer relationships, such as residential matches, family matches, and individual matches.
Corporate Householding	Multiple levels of corporate relationships, such as corporate matches, subsidiary matches, and contact matches.

Strategy	Description
Multinational consumer match	Multiple countries, with different business rules for each country.
Identify a person multiple ways	Multiple match criteria that identifies relationships and overlaps between all relationships.

Parent topic: [Match transform \[page 586\]](#)

Related Information

[Match components \[page 588\]](#)

[Track physical and logical sources \[page 592\]](#)

[Control comparisons \[page 595\]](#)

[Prepare data for matching \[page 598\]](#)

[Compare tables \[page 601\]](#)

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[Match criteria \[page 604\]](#)

[Matching methods \[page 606\]](#)

[Matching business rules \[page 611\]](#)

[Group statistics \[page 625\]](#)

[Input source select records \[page 626\]](#)

19.2 Match components

When you prepare for matching, you configure components on many levels, especially match sets, match levels, and match criteria.

The following are the basic components to prepare for matching:

- Match sets: Each set represented by a Match transform.
- Match levels: Determines the broad category of matching, such as family or corporate.
- Match criteria: Determines the fields on which to match.

Parent topic: [Match transform \[page 586\]](#)

Related Information

[Develop a match strategy \[page 587\]](#)

[Track physical and logical sources \[page 592\]](#)

[Control comparisons \[page 595\]](#)

[Prepare data for matching \[page 598\]](#)

[Compare tables \[page 601\]](#)

[Data Salvage \[page 602\]](#)

[Match criteria \[page 604\]](#)

[Matching methods \[page 606\]](#)

[Matching business rules \[page 611\]](#)

[Group statistics \[page 625\]](#)

[Input source select records \[page 626\]](#)

19.2.1 Match sets

The Match transform represents a match set, with its own match criteria and business rules.

A Match set performs tasks, such as the following:

- Match certain records.
- Separate records.
- Match records independently.

❖ Example

Create a match set that matches records from the United States. Create a different match set that matches international records.

A match set allows the Match transform to do the following:

- Group data that complies with a given set of match criteria for comparison.

❖ Example

Ensure that the only records allowed in a match set have populated, non blank, Social Security number fields.

- Stack related match scenarios to create a multi-level match set.
- Consider multiple match sets for association in an Associate match set.

Match criteria and strategies

For each match set that you create, select the match criteria. Match criteria is the data to compare between records in a break group. Break groups are smaller groups of records that are more likely to match.

SAP Data Services works independently in each match set and in each break group.

i Note

Data Services doesn't compare the records in one break group against records in any other break group.

Strategies

The following describe examples of strategies:

- **Single match set with multiple fields:** Find the data that matches all fields that you select for matching.
- **Multiple match sets with two fields per match set:** Find the data that matches only in a specific combination of fields. Select a combination of fields that show the overlapping information, such as last name and the Social Security number (SSN).

❖ Example

Multiple match sets with two fields per match set:

Your source data contains “snowbird” or seasonal addresses, where an individual has two valid addresses. To separate seasonal addresses, create two match sets with the following match criteria:

Set 1: Full Name and SSN.

Set 2: Street Address and City State.

The following table contains data from two records. Because the **Full Name** and **SSN** fields have a high confidence score based on similarity, Data Services determines that each address is valid. Based on your business rules, you determine to keep both records as unique, but you place the records in separate output tables.

Record	Full Name	Street Address	City State	SSN
1	R. Carson	1239 Whistle Lane	Columbus, Ohio	xxx-xx-4333
2	Robert T. Carson	52 Sunbird Suites	Tampa, Florida	xxx-xx-4333

Select the criteria:

Match Set	Match on	Compare Using
Set 1	Full Name	<i>Word similarity</i>
	SSN	<i>Field difference</i>
Set 2	City State	<i>Word similarity</i>
	Street Address	<i>Word similarity</i>

19.2.2 Match levels

A match level indicates the type of matching that the Match transform performs, such as individual, family, resident, and firm.

A match level refers to the broad category of matching instead of specific criteria.

Use as many match levels in a Match transform as you need. You can define each match level in a match set in a way that is increasingly more strict. Multi-level matching feeds only the records that match from match level to match level, such as resident to family to individual, for comparison.

The following table describes the match levels supported in Data Services.

Match level	Description
Family	Determines whether two people are members of the same family based on their record data. The transform compares the last name and the address data of two records. A match means that the two records represent members of the same family. The result of the match is one record per family.
Individual	Determines whether two records represent the same person based on their record data. The transform compares the first name, last name, and address data. A match means that the two records represent the same person. The result of the match is one record per individual.
Resident	Determine whether two records are considered members of the same residence based on their record data. The transform compares the address data. A match means that the two records represent members of the same household. Contrast this match type with the family match type, which compares address data plus last-name data. The result of the match is one record per residence.
Firm	Determines whether two records are from the same firm. The transform compares the firm name and address data of two records. A match means that the two records represent the same firm. The result of the match is one record per firm.
Firm-Individual	Determines whether two records are for the same person at the same firm based on the record data. The transform compares the first name, last name, firm name, and address data. A match means that the two records reflect the same person at the same firm. The result of the match is one record per individual per firm.

19.2.3 Match criteria

Match criteria refers to the fields on which to match.

Use match criteria options to specify business rules for matching each of the fields. The options allow you to control how close to exact the data has to be before the transform considers the records as matching.

❖ Example

A match criteria for an individual match level has the following requirements:

- First names must be at least 85% similar.
- Allow a first name initial to match a spelled-out first name.
- Allow a first name to match a middle name.

The following table contains suggested record elements to include in your match criteria for each match type.

Match type	Suggested elements
Family	Last (family) name and address.
	Last (family) name and telephone number.
Individual	Full name and address.
	Full name and Social Security number.
	Full name and e-mail address.
Firm	Firm name and address.
	Firm name and Standard Industrial Classification Code (SIC)
	Firm name and Data Universal Numbering system (DUNS) number.

19.3 Track physical and logical sources

Tracking your data sources is essential for producing informative match statistics and reports.

Track physical and logical sources for each record by designating a field that contains source identification for each record. Base data sources on an input source or some data element in the records that the transform reads.

SAP Data Services tracks the following types of sources:

- **Physical source:** The actual table or file in which the record originates.
- **Logical source:** A group of records that span multiple input sources or a subset of records from a single input source.

To track the physical or logical source in a specific output field, use a field that contains the source file name or a value attributed to the input source.

❖ Example

To count source statistics in the Match transform, such as for the Match Source Statistics Summary report, you need a source ID field.

The following table describes how to track the physical and logical sources in your match jobs.

Source Type	How to track
Physical	After processing jobs with multiple sources, or merging multiple sources into a Match or Associate transform, SAP Data Services tracks the record source through the source ID field. When you view reports and statistics, each record reported contains the physical source ID that identifies where the record is from. i Note If your source is a flat file, you can use the <i>Include file name</i> option in the file format to automatically generate a column containing the file name.
Logical	The logical source field indicates the various logical sources within a Reader for reporting purposes. The transform uses the field in the following ways: • In a reader for reporting. • In the Group Statistics operation of the Match transform to generate the source statistics. • In compare tables, so that you can specify sources to compare.

[Source types \[page 594\]](#)

Identify a source as one of three types: Normal, suppression, or special.

[Source groups \[page 595\]](#)

The source group capability adds a higher level of source management.

Parent topic: [Match transform \[page 586\]](#)

Related Information

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19.3.1 Source types

Identify a source as one of three types: Normal, suppression, or special.

The Match transform process your records differently depending on their source type.

Source	Description
Normal	A group of records that you consider to be good, eligible records.
Suppress	A grouping of records based on a specific data characteristic that you can identify. For example: • DMA Mail Preference File • American Correctional Association prisons/jails sources • No pandering or non-responder sources • Credit card or bad-check suppression sources
Special	Like a Normal source, with one exception. The transform does not count a Special source when determining whether a match group is single-source or multi-source. A Special source contributes records, but the transform does not count the records toward multi-source status. ❖ Example ABC, Inc uses a source that contains seed names. Seed names are for people who report to the company when they receive advertisements from the company in the mail. ABC, Inc uses the information to measure mail delivery timing.

The reason for identifying the source type is so the transform can set that identity for each of the records that are members of the source. Source type plays an important role in how the transform processes matching records, like members of a match group. And also how the transform produces output, such as whether it includes or excludes a record from the output.

Related Information

[Track physical and logical sources \[page 592\]](#)

19.3.2 Source groups

The source group capability adds a higher level of source management.

Source groups primarily affect reports. However, you can also use source groups to select records from multiple sources based on the number of source groups in which a value occurs.

❖ Example

You rent five files from two brokers. Then, you define each file as a source in a Match data flow. The Match transform uses the sources for ranking the records. To configure the transform to break down match statistics by broker, you define a source group for each broker based on the broker name.

You can't use source groups in the same way that you use physical sources. For example, you can't give one source group priority over another.

19.4 Control comparisons

To save processing time and increase match quality, control the tables and the records that the Match transform compares.

When you configure record comparisons, the transform groups only records that have a potential to match, which returns better results.

Also use compare tables feature to include or exclude records for comparison by logical source.

[Break groups and break keys \[page 596\]](#)

The Match transform creates break groups based on a field or fields that match, which increases the likelihood that the records are the same or similar.

Parent topic: [Match transform \[page 586\]](#)

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19.4.1 Break groups and break keys

The Match transform creates break groups based on a field or fields that match, which increases the likelihood that the records are the same or similar.

❖ Example

Designate the postcode field as the basis on which Match selects records to form a break group. The chances of the Match transform finding a matching record with a different postcode are small. Therefore, a postcode break group ensures that the Match transform never wastes time comparing records from different postcodes.

Break keys

The Match transform uses break keys as the basis for forming break groups. A break key is a field that consists of parts of other fields, or a single field, which is then used to group together records based on similar data.

❖ Example

Create a break key that consists of the following combined elements:

- Five digits of the `POSTCODE1` field
- First three characters of the `ADDRESS_PRIMARY_NAME` field.

The following table shows a break key generated from elements from the two fields. the elements of the field are defined as **pos** and **length**:

- **Pos** is position, which designates where the transform starts the string in the field. A position of 1 indicates to start at the first character of a field.
- **Length** is how many characters to include in the string. A length of 5 means to include 5 characters in the string.

Field (Start pos:length)	Data in field	Generated break key
<code>POSTCODE1</code> (1:5)	10101	10101Mai
<code>ADDRESS_PRIMARY_NAME</code> (1:3)	Main	

The Match transform places all records that match the generated break key in the same break group and compares each record against each other.

Break groups

Break groups consist of driver and passenger records:

- **Driver:** The first record in the break group.
- **Passenger:** All other records in the break group.

The driver record is the record that “drives” the comparison process in matching. The transform compares the driver to all of the passengers first.

❖ Example

The break key in the following illustration consists of the first three digits (1:3) of the postcode.

Record 1	00501
Record 2	54601
Record 3	54602
Record 4	50005
Record 5	23032
Record 6	23031
Record 7	00501
Record 8	23032
Record 9	50005
Record 10	00501
Record 11	50005
Record 12	54602

Break Group 1

Driver	Record 1	005	01
Passenger	Record 7	005	01
Passenger	Record 10	005	01

Break Group 2

Driver	Record 2	546	01
Passenger	Record 3	546	02
Passenger	Record 12	546	02

Break Group 3

Driver	Record 4	500	05
Passenger	Record 9	500	05
Passenger	Record 11	500	05

Break Group 4

Driver	Record 5	230	32
Passenger	Record 6	230	31
Passenger	Record 8	230	32

Record sorting in the break group

The transform sorts records based on the break key field. To specify the driver records, add a Group Prioritization operation after the Break Groups operation.

→ Remember

Order is important! If you create break groups using records from a Suppress-type source, be sure that the suppression records are the drivers in the break group.

19.5 Prepare data for matching

To achieve the best possible match results, preprocess your data with other transforms, such as with Data Cleanse or with Global Address Cleanse transforms.

The following table describes several prematch data preparation methods.

Method	Description
Data correction and standardization	Accurate matches depend on good data coming into the Match transform. Perform data and address cleansing before you input data to the Match transform. Transforms: Data Cleanse, and Global Address Cleanse, for example.
Filter empty records	For better match performance, filter empty records out of your databases before you perform matching.
Noise words	Remove or replace words that are meaningless to the matching process. For example, for a firm match job, output words, such as Inc., Corp., or Ltd. into extraneous fields that you don't pass into the Match transform.
Remove punctuation	To maximize your matching process, standardize data using the Data Cleanse transform. For example, output data in all upper case with punctuation removed.

Method	Description
Break groups	To improve match performance, use break groups, which are smaller groups of records. To create a break group, you sort and combine records into smaller groups that are more likely to match. i Note Create break groups before you input data into the Match transform.
Match standards	Establish match standards. For example, include variations of name or firm data in your match standards to help ensure a match. For example, if the transform compares first names but determines they don't match, it then compares the variations. If the variations match, the transform continues trying to match the two records instead of failing them, because it didn't consider the original first names a match.
Custom Match Standards	Create custom match standards by using Data Cleanse. Match custom Data Cleanse output fields and associated aliases. Then map the custom output fields from Data Cleanse into the Match transform.

[Fields to include for matching \[page 600\]](#)

Create and include match-specific fields to enhance record matching, such as source identification fields and break key value fields.

Parent topic: [Match transform \[page 586\]](#)

Related Information

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19.5.1 Fields to include for matching

Create and include match-specific fields to enhance record matching, such as source identification fields and break key value fields.

The following table describes the match-specific fields to create, and how the Match transform uses the information from the field.

i Note

The items listed in the **Field** column don't reflect the actual field name. Use your own naming conventions for the fields.

Field	Content	Use
Logical source name	Name of a logical source, which is a group of records that span multiple input sources or a subset of records from a single input source.	In the Group Statistics operation and in compare tables.
Physical source name	Name of the table or file in which a record originates.	In the Match transform options, Candidate Selection operation, and the Associate transform.
Break key	Value of the break key that creates a break group.	Improves the performance of break group creation. Saves the Match transform from doing the parsing of multiple fields to create the break key.
Criteria fields	Data on which to match or compare.	Determines the types of operations to perform and the match threshold scores.
Count flags	Yes or No value.	Specifies whether a logical source is counted in a Group Statistics operation.
Record priority	A value that signifies a record as having priority over another when ordering records.	Group Prioritization operations.
Apply blank penalty	Yes or No value to specify whether Match applies a blank penalty to a record.	Group Prioritization operations.
Starting unique ID value	Value that increments by 1 every time the transform assigns a unique ID.	Unique ID operation.

The table isn't a complete list. Depending on the features you use, you can include many other fields for the Match transform.

19.6 Compare tables

Use compare tables to compare records within sources, across sources, or a combination of both.

Compare tables are sets of rules that define what records to compare. Compare tables are almost an additional way to create break groups. You use your logical source values to determine what records the Match transform compares or doesn't compare.

The Match transform uses a compare table to determine what record pairs qualify for comparisons, and what sources to compare. If you don't include a Compare table in the Match transform, the transform uses a driver record to compare with all remaining passenger records in the data collection.

If you have many physical or logical sources in your project, it can be easier to specify what *not* to compare, as opposed to what to compare.

❖ Example

For example, you have 10 sources named A through J. You want to compare all but A and B. Set the `DEFAULT_ACTION` option to `COMPARE`. Then set up a table row for both source A and source B. Set the Action options for source A and B to `NO_MATCH`.

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19.7 Data Salvage

Data salvaging prevents the Match transform from making matches between records that you don't want to match.

With data salvaging, the transform temporarily copies data from a passenger record to the driver record after it compares the two records. The data that's copied is data that is missing or incomplete in the driver record.

❖ Example

The following table shows a match group that contains three records. Without data salvaging, Match includes the records in the same match group. The driver record has a blank Name field, but the Address and Postcode fields match the passenger records.

Record	Name	Address	Postcode
1 (driver)		123 Main St.	54601
2	John Smith	123 Main St.	54601
3	Jack Hill	123 Main St.	54601

With data salvaging enabled, the transform temporarily copies the Name from record 2 to the driver record. The transform doesn't create a match group from the three records, however, because of the following factors:

- Record 1 matches Record 2.
- Record 1 doesn't match Record 3 because of the Name field: John Smith doesn't match Jack Hill.

Record	Name	Address	Postcode
1 (driver)	John Smith (copied from record 2)	123 Main St.	54601
2	John Smith	123 Main St.	54601
3	Jack Hill	123 Main St.	54601

❖ Example

You have a suppression source, which is a list of no-pandering addresses. You set the suppression source with the highest priority, and you don't enable data salvaging.

The transform suppresses all records that match the suppression source records. A suppression record of 123 Main St matches 123 Main St #2 and 123 Main St Apt C. Therefore, the transform suppresses both addresses.

[Data salvaging and initials \[page 603\]](#)

When a driver record has a Name field with an initial instead of a full name, the Match transform can temporarily borrow the full name of a matching record for data salvaging.

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19.7.1 Data salvaging and initials

When a driver record has a Name field with an initial instead of a full name, the Match transform can temporarily borrow the full name of a matching record for data salvaging.

By retaining more information for the driver record, data salvaging helps improve the quality of your matching results.

Note

The transform performs data salvaging for initials only for the given name and family name fields.

Example

The following three records represent potential matches. In the following table, the transform groups the records into a break group based on the address and ZIP Code (not shown).

Record	First name	Last name	Address	Notes
357	J	L	123 Main	Driver
391	Juanita	Lopez	123 Main	
839	Joanne	London	123 Main	Lowest ranking record

The first match comparison is between the driver record, 357, and the next highest ranking record, 391. The transform calls the two records a match. Juanita and Lopez are temporarily copied to the first and last name fields of record 357.

The next match comparison is between record 357 and the next lower ranking record, 839. With data salvaging enabled, the data in the driver record now contains Juanita Lopez, borrowed from the first comparison. Therefore, the transform concludes that record 839 isn't a match to record 357.

Initials and suppress-type records

If the driver record is a suppress-type record, turning off data salvaging can be beneficial.

❖ Example

The following table contains records in a suppression source, with data salvaging turned off.

Record	First name	Last name	Address	Notes
357	J	L	123 Main	Driver
391	Juanita	Lopez	123 Main	
839	Joanne	London	123 Main	Lowest ranking record

The transform makes the first match comparison between the driver record, 357, and the next- highest ranking record, 391. The transform considers the two records a match because the JL in the driver record matches Juanita Lopez.

The transform makes the next comparison between the driver record, 357, and the next lower ranking record, 839. The transform considers the two records a match because driver JL in the driver record matches Joanne London.

Since both records 391 and 839 matched the suppress-type driver record, 357, the transform designates both records as suppress matches and doesn't include the records in the output.

19.8 Match criteria

Use match criteria in each match level to determine the threshold scores for matching and to define how the Match transform treats various types of data.

Match criteria represent the fields to compare.

❖ Example

To match on the first ten characters of a given name and the first 15 characters of the family name, you create two criteria that specify these requirements.

Criteria provide a way to let the Match transform know what kind of data is in the input field and, therefore, what types of operations to perform on that data.

Pre-defined vs. custom criteria

The following table details the two types of criteria supported in Match.

Criteria	Description
Predefined	Set criteria for fields that you typically use for matching such as first name, family name, and address components. With field criteria, the transform identifies the type of data in the field, and optimizes the data for matching without altering the actual input data.
Data Cleanse Custom	Set criteria for custom output fields such as user-defined and non-party data. Map the custom output fields from the Data Cleanse transform to the Match transform. The custom fields are then available in the Match Criteria input fields group. Set criteria for other types of fields such as part numbers or proprietary data, for which there's no pre-defined criteria. The transform matches on abbreviations, substrings, and numeric data, but not on certain cross-field comparisons such as name match.

Match criteria pre-comparison options

Use the Address Cleanse and Data Cleanse transforms for most of the pre-match tasks to standardize data. However, the Match transform also performs some preprocessing per criteria to provide more accurate matches.

Note

The transform doesn't actually alter your data when it performs the preprocessing.

Find the following options to control pre-comparison standardization in the `MATCH_CRITERIA_KEY_LAYOUT` group:

Option	Description
<code>CONVERT_DIACRITICAL_CHARACTERS</code>	Specifies whether to include diacritical characters in the matching process.
<code>CONVERT_TEXT_TO_NUMBERS</code>	Specifies whether numbers represented as text, such as one, two, three, are converted to numbers 1,2,3.
<code>CONVERT_TO_UPPER_CASE</code>	Specifies whether to convert all data to uppercase for matching purposes only.
<code>REMOVE_PUNCTUATION</code>	Specifies whether to remove punctuation from your data to help provide more accurate matches.
<code>LOCALE</code>	Specifies the locale setting for this criterion field.

For complete descriptions of all match comparison options, see the *Reference Guide*.

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19.9 Matching methods

Matching methods are ways to set up and order match criteria to get desired matching results.

The following table contains descriptions of matching methods that you can use in the Match transform. Each of these methods has advantages and disadvantages, so consider them carefully.

Match method	Description
Rule-based	You control the criteria that the transform uses to determine a match by setting Match and No Match scores.
Weighted-scoring	Assign importance, or weight, to any criteria. The transform evaluates every rule before it determines a match, which may affect processing time.
Combination method	Use a combination of matching methods based on your data and your matching goals.

The similarity score plays an important role in the matching methods. The similarity score is the percent that your data is alike. The transform calculates the similarity score based on your configuration in the *CRITERIA_DEFINITION_GROUP* group.

[Similarity score \[page 607\]](#)

The similarity score is the percentage that your data is alike.

[Rule-based matching method \[page 608\]](#)

In rule-based matching, the transform relies on the Match and No-match scores to determine matches within the criteria.

[Weighted-scoring method \[page 609\]](#)

With weighted scoring, the Match transform relies on the total contribution score to determine matches instead of the match and no-match scores alone.

[Combination method \[page 610\]](#)

For the combination method, the Match transform combines the rule-based and weighted scoring methods of matching.

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19.9.1 Similarity score

The similarity score is the percentage that your data is alike.

The similarity score plays an important role in the matching methods. The Match transform calculates the similarity score based on your configuration in the `CRITERIA_DEFINITION_GROUP` group.

❖ Example

An example of how the Match transform determines similarity scores.

The following comparison table is an example. It doesn't illustrate how the matching process works.

Comparison	No match score	Match score	Similarity score	Match
Smith > Smith	72	95	100%	Yes
Smith > Smitt	72	95	80%	Depends on other criteria
Smith > Smythe	72	95	72%	No

Comparison	No match score	Match score	Similarity score	Match
Smith > Jones	72	95	20%	No

Considerations:

- Match considers only the first comparison a match because the similarity score meets or exceeds the match score.
- In the second comparison, Match can't determine a match using a single criteria. Therefore, it moves to the next criteria, if possible.
- Match considers the last comparison a no-match because the similarity score is less than the no-match score.

19.9.2 Rule-based matching method

In rule-based matching, the transform relies on the Match and No-match scores to determine matches within the criteria.

The following example illustrates the rule-based matching method.

❖ Example

Criteria	Record A	Record B	No match score	Match score	Similarity score
GIVEN_NAME1	Mary	Mary	82	101	100
FAMILY_NAME	Smith	Smitt	74	101	80
EMAIL	msmith@sap.com	mary.smith@sap.com	79	80	91

- **A Match score of 101 eliminates the possibility of a match:** Two fields can't be more than 100% alike. The match score of 101 for the first two criteria, [GIVEN_NAME1](#) and [FAMILY_NAME](#), causes Match to never consider these criteria fields for determining a match. However, the transform can use the fields to determine a no-match.
- **Don't leave a gap between the No match score and the Match score: Any comparison that reaches the last criteria is either a Match or a No match because there'sn't a gap between the scores.** The [MATCH_SCORE](#) of 80 and the [NO_MATCH_SCORE](#) of 79 for the [EMAIL](#) criteria means that the e-mails in two records are either a match or not a match.

→ Remember

Order is important! For performance reasons, ensure that you order criteria with the criteria that is most likely to make the match or no-match decisions first in your order of criteria. Ordering can help reduce the number of criteria comparisons.

19.9.3 Weighted-scoring method

With weighted scoring, the Match transform relies on the total contribution score to determine matches instead of the match and no-match scores alone.

Contrast weighted scoring to the rule-based matching method. For rule-based matching, the transform gives all of the criteria the same amount of importance or weight. That is, if any criteria fails to meet the specified match score, the transform determines that the records don't match.

Contribution values

Set contribution values to assign weight to individual criteria. The higher the contribution value, the more weight that criteria carries in determining matches. Generally, the following fields carry more weight than others:

- Account numbers
- Social Security numbers
- Customer numbers
- Postcode1
- Address

i Note

Ensure that the total of all contribution value totals is 100. Keep in mind, however, that you don't have to set a contribution value to all criteria.

Define contribution values in the [CONTRIBUTION_TO_WEIGHTED_SCORE](#) option in the `CRITERIA_DEFINITION_GROUP` group.

Contribution and total contribution score

The Match transform generates the contribution score for each criteria by multiplying the contribution value you assign with the similarity score (the percentage alike). Match then adds these individual contribution scores to get the total contribution score.

Weighted match score

In the weighted scoring method, Match determines matches only by comparing the total contribution score with the weighted match score.

- If the total contribution score is equal to or greater than the weighted match score, the transform considers the records a match.
- If the total weighted score is less than the weighted match score, the transform considers the records a no-match.

Set the weighted match score in the [WEIGHTED_MATCH_SCORE](#) option of the [COMPARE_MATCH_CRITERIA_GROUP](#).

❖ Example

The following table is an example of how to set up weighted scoring. In this example, we used the following logic for setting some scores:

- To ensure that the transform doesn't find a match or a no-match, we set the [NO_MATCH_SCORE](#) option to -1, and the [MATCH_SCORE](#) option to 101.
- We set the highest contribution value to the E-mail criteria to give it the most importance.

Criteria	Record A	Record B	No match	Match	Similarity score	Contribution value	Contribution score (similarity X contribution value)
FIRST_NAME	Mary	Mary	-1	101	100	25	25
LAST_NAME	Smith	Smitt	-1	101	80	25	20
EMAIL	ms@ sap.com	msmith@ sap.com	-1	101	84	50	42
							Total contribution score: 87

If the weighted match score is 87, then the transform considers any comparison whose total contribution score is 87 or greater a match. In this example, the comparison is a match because the total contribution score is 87.

19.9.4 Combination method

For the combination method, the Match transform combines the rule-based and weighted scoring methods of matching.

❖ Example

The following table shows an example of how to use the rule-based method with the weighted score method to determine matching records. The following table shows that the two records don't match based on the combined method.

Criteria	Record A	Record B	No match	Match	Sim score	Contribution value	Contribution score (actual similarity X contribution value)
First Name	Mary	Mary	59	101	100	25	25
Last Name	Smith	Hope	59	101	22	N/A (No Match)	N/A
E-mail	ms@ sap.com	msmith@ sap.com	49	101	N/A	N/A	N/A
							Total contribution score: N/A

Related Information

[Matching methods \[page 606\]](#)

19.10 Matching business rules

Business rules establish how the Match transform handles various forms of data and differences in your data.

❖ Example

Every field in **Record A** matches every field in **Record B** with one exception: **Record B** has a blank value in one of the fields.

Should the transform consider **Record A** and **Record B** as matches?

Figuring out the software behavior in this situation is a part of defining your business rules. Define most of your business rules in the **CRITERIA_DEFINITION_GROUP** group.

[Multiple field \(cross-field\) comparisons \[page 612\]](#)

Multiple field comparisons allow you to match data in one field against data in multiple fields that might contain the same data.

[Matching on strings, abbreviations, and initials \[page 615\]](#)

Part of defining your business rules is to set options for matching strings, abbreviations, and initials.

[Extended abbreviation matching \[page 616\]](#)

The extended abbreviation options allow matching for situations that aren't covered by the string, substring, and initials options.

[Name matching \[page 617\]](#)

The Match transform provides many ways to ensure that variations on names or multiple names, are considered.

[Match compound family names \[page 618\]](#)

Use the *APPROX_SUBSTRING_ADJUSTMENT_SCORE* option to assist in setting up comparisons of compound family names.

[Numeric data matching \[page 619\]](#)

Specify how the Match transform matches data that consists of a mixture of numbers and letters.

[Blank field matching \[page 622\]](#)

Control how the transform compares fields when one or both of the fields are blank in the *CRITERIA_DEFINITION_GROUP* group.

Parent topic: [Match transform \[page 586\]](#)

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19.10.1 Multiple field (cross-field) comparisons

Multiple field comparisons allow you to match data in one field against data in multiple fields that might contain the same data.

In most cases, you compare data in one field against data in another field; a one to one ratio.

❖ Example

One to one:

Field1 in the first record is compared with **Field1** in the second record.

However, there are situations when comparing one field to multiple fields is useful; a one to many ratio.

❖ Example

One to many:

Match values in a **Phone** field against values in fields such as **Fax**, **Mobile**, and **Home**.

By default, Match performs multiple field comparisons on match standard type fields. Match doesn't require additional configuration to perform multiple field comparison against match standard fields.

❖ Example

Match automatically compares *PERSON1_GIVEN_NAME1* to *PERSON1_GIVEN_NAME_MATCH_STD1*.

[How Match compares selected fields to all selected fields \[page 613\]](#)

When the transform compares each selected field to all selected fields in other records, it compares all fields that you define in the match criteria to each other.

[How Match compares selected fields to the same fields in other records \[page 614\]](#)

When the Match transform compares selected fields to the same fields in other records, it considers two records as matching when just one field in the other record matches the selected field.

19.10.1.1 How Match compares selected fields to all selected fields

When the transform compares each selected field to all selected fields in other records, it compares all fields that you define in the match criteria to each other.

The transform compares selected fields to all selected fields in the following ways:

- If one or more field comparisons meet the settings for *MATCH_SCORE*, the transform considers the two rows being compared as matches.
- If one or more field comparisons exceed the *NO_MATCH_SCORE*, the transform considers other defined criteria or weighted scoring to determine if the two rows are matches.

❖ Example

You have the following input data in the Firm1 and Firm2 fields:

Row ID	Firm1	Firm2
1	Firstlogic	Postalsoft
2	SAP	Firstlogic

With the *MATCH_SCORE* set to *100* and *NO_MATCH_SCORE* set to *99*, the transform determines the two records as matching. The transform performs the following comparison process to come up with the match decision:

- First, the transform compares Row 1, Firm1 (Firstlogic) to Row 2, Firm1 (SAP). Without multi-field comparison enabled, the transform would make a No Match decision. However, with multi-field comparison, it doesn't make the No Match decision yet.

- Next, the transform compares Row 1, Firm2 to Row 2, Firm2. It continues with comparisons until it has made all other comparisons between all fields in all rows. Because Row 1, Firm1 (Firstlogic) and Row 2, Firm2 (Firstlogic) are 100% similar, the transform considers the two records as matches.

19.10.1.2 How Match compares selected fields to the same fields in other records

When the Match transform compares selected fields to the same fields in other records, it considers two records as matching when just one field in the other record matches the selected field.

When you set the Match transform to compare the same fields between records, you set an OR condition within a match criteria.

Match uses each field to determine a match.

❖ Example

If Field_1, or Field_2, or Field_3 passes the match criteria, the records match.

The *NO_MATCH_SCORE* for one field doesn't automatically fail the criteria when you use multi-field comparison.

→ Remember

"Selected" fields include the criteria field and the other fields you define for the criteria.

❖ Example

Your input data contains a phone, fax, and cell phone field. If the data in any one of these input fields is the same between the rows, the transform determines the records to be matches.

Row ID	Phone	Fax	Cell
1	608-555-1234	608-555-0000	608-555-4321
2	608-555-4321	608-555-0000	608-555-1111

If you define the Phone and Cell fields in different match criteria, they fail the following match criteria settings:

MATCH_SCORE = 100

NO_MATCH_SCORE = 99

However, because the match criteria is set up for comparing the selected fields to all selected fields in other records, and the Fax field values match exactly, the transform determines the Row 1 and Row 2 to be matches.

19.10.2 Matching on strings, abbreviations, and initials

Part of defining your business rules is to set options for matching strings, abbreviations, and initials.

String data

Configure the Match transform to match longer strings to shorter strings using the [SUBSTRING_ADJUSTMENT_SCORE](#) option in the CRITERIA_DEFINITION_GROUP group.

❖ Example

When you set the [SUBSTRING_ADJUSTMENT_SCORE](#), the transform matches the string "Mayfield Painting and Sand Blasting" with "Mayfield painting."

Approximate substring

Configure the transform to match leftover portions of a string that didn't match during the substring matching. Set the [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#) and the [SUBSTRING_ADJUSTMENT_SCORE](#) options so the transform assigns a score to all of the leftover words and spaces that don't match during the substring matching.

Abbreviations

Use the [ABBREVIATION_ADJUSTMENT_SCORE](#) option to enable the transform to match whole words to abbreviations.

❖ Example

The transform matches "International Health Providers" to "Intl Health Providers."

Initials and acronyms

Use the [INITIALS_ADJUSTMENT_SCORE](#) option to enable the transform to match initials to whole words.

❖ Example

The transform matches "International Health Providers" to "IHP."

19.10.3 Extended abbreviation matching

The extended abbreviation options allow matching for situations that aren't covered by the string, substring, and initials options.

Extended abbreviation matching offers functionality for situations like the following:

- You've localities information in your data such as New Mexico and New York. However, your data lists these localities both with the spaces and without the spaces as in NewMexico and NewYork. Under normal matching, the Match transform doesn't match words with spaces to the same words without spaces at 100%. Normally, the score for New Mexico and NewMexico is 94%. The score for New York and NewYork is 93%.

Note

Actual scores can differ.

- Your data has Metropolitan Life and MetLife (an abbreviation and combination of Metropolitan Life) in your data. The [ABBREVIATION_ADJUSTMENT_SCORE](#) option doesn't detect the combination of the two words.

If you are concerned about either of these cases in your data, use the [EXT_ABBREVIATION_ADJUSTMENT_SCORE](#) option. When you set the [EXT_ABBREVIATION_ADJUSTMENT_SCORE](#) option in the [CRITERIA_DEFINITION_GROUP](#) group, the transform recognizes the example situations as matches.

How the extended abbreviation adjustment score works

The score you set in the [EXT_ABBREVIATION_ADJUSTMENT_SCORE](#) option tunes your similarity score to consider these types of abbreviations and combinations in your data.

The adjustment score adds a penalty for the non-matched part of the words. The higher the number, the lower the penalty. A score of 100 means no penalty and score of 0 means maximum penalty.

❖ Example

The following table shows the assigned similarity score when you set the [EXT_ABBREVIATION_ADJUSTMENT_SCORE](#) option to 0, 50, and 100.

Ext abbreviation adjustment score					
String 1	String 2	0	50	100	Notes
MetLife	Metropolitan Life	58	79	100	
MetLife	Met Life	93	96	100	

Ext abbreviation adjustment score					
String 1	String 2	0	50	100	Notes
MetLife	MetropolitanLife	60	60	60	The similarity score is the same for all three settings. Each string is one word, therefore, the transform doesn't use the EXT_ABBREVIATION_ADJUSTMENT_SCORE setting.

i Note
 Actual scores can differ.

Related Information

[Matching business rules \[page 611\]](#)

19.10.4 Name matching

The Match transform provides many ways to ensure that variations on names or multiple names, are considered.

Part of creating your business rules is to define how you want names handled in the matching process.

i Note

Unlike other business rules, you set name matching options in the `CRITERIA_DEFINITION_GROUP` group, because they affect all appropriate name-based match criteria.

Two names; two persons

With the [NUMBER_OF_NAMES_THAT_MUST_MATCH](#) option, control how the transform performs matching on match keys with more than one name. For example, comparing "John and Mary Smith" to "Dave and Mary Smith." Choose whether the transform matches only one name for the records to be identified as a match, or whether the transform disregards any persons other than the first name it parses.

With this method you can require either one or both persons to match for the record to match.

Two names; one person

With the [COMPARE_GIVEN_NAME1_TO_GIVEN_NAME2](#) option, compare the [GIVEN_NAME1](#) data (first name) from one record with the [GIVEN_NAME2](#) data (middle name) from the second record. With this option enabled, the transform correctly identifies matching records such as the two partially shown in the following table. Typically, these record pairs represent sons or daughters named for their parents, but known by their middle name.

❖ Example

Record #	First name	Middle name	Last name	Address
170	Leo	Thomas	Smith	225 Pushbutton Dr
198	Tom		Smith	225 Pushbutton Dr

Hyphenated family names

With the [MATCH_ON_HYPHENATED_FAMILY_NAME](#) option, control how the transform performs matching when a [FAMILY_NAME](#) (last name) field contains a hyphenated family name. For example, comparing "Smith-Jones" to "Jones." Choose whether the transform requires both names to match, or just one of the names to match to consider the records a match.

Related Information

[Matching business rules \[page 611\]](#)

19.10.5 Match compound family names

Use the [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#) option to assist in setting up comparisons of compound family names.

The transform assigns the [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#) to the words that do not match to other words in a compared string. The [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#) option loosens some of the requirements of the [SUBSTRING_ADJUSTMENT_SCORE](#) option in the following ways:

- First words in a string do not have to match exactly.
- The transform uses initials and abbreviation adjustments for the words that match. (For example, Rodriguez and RDZ.)
- Matching words have to be in the same order, but there can be non-matching words before or after the matching words.

- The transform assigns the leftover words and spaces in the compared string to the [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#).

The [APPROX_SUBSTRING_ADJUSTMENT_SCORE](#) option increases the score for some matches found when using the [SUBSTRING_ADJUSTMENT_SCORE](#).

❖ Example

The following table shows how settings in the substring options affect the calculated similarity score when the transform compares CRUZ RODRIGUEZ and GARCIA CRUZ DE RDZ.

Similarity score	SUBSTRING_ADJUSTMENT_SCORE	ABBREVIATION_ADJUSTMENT_SCORE	APPROX_SUBSTRING_ADJUSTMENT_SCORE
48	None	None	None
66	80	80	None
91	None	80	80

Related Information

[Matching business rules \[page 611\]](#)

19.10.6 Numeric data matching

Specify how the Match transform matches data that consists of a mixture of numbers and letters.

To set numeric data requirements, use the [NUMERIC_WORDS_MATCH_EXACTLY](#) option.

Select a value so the transform requires data with a mixture of numbers and letters to match exactly in any position, or in the same position, for example. The [NUMERIC_WORDS_MATCH_EXACTLY](#) option applies most often to address data and custom data such as a part number.

The Match transform processes data with numeric and letter data as follows:

1. Breaks the string into words using punctuation and spacing.
2. Assigns a numeric attribute to the words.

i Note

A numeric word is any word that contains at least one number from 0 to 9. For example, 4L is considered a numeric word, whereas FourL is not.

3. Performs numeric matching based on settings in the [NUMERIC_WORDS_MATCH_EXACTLY](#) option.

Numeric matching option values and how they work

The following table describes the options for *NUMERIC_WORDS_MATCH_EXACTLY*.

Option value	Description
<i>ANY_POSITION</i>	Requires that numeric words match exactly. The position of the word is not important. ❖ Example The transform determines that the following street addresses are a match: 4932 Main St # 101 # 101 4932 Main St The transform determines that the following street addresses are not a match: 4932 Main St # 101 # 102 4932 Main St The transform determines that the following part numbers are a match: ACCU 1.4L 29BAR ACCU 29BAR 1.4L
<i>SAME_POSITION</i>	Requires that numeric words match exactly. This value differs from the <i>ANY_POSITION</i> value in that the position of the word is important. ❖ Example The transform determines that, for the phone number 608-123-4567: 608-123-4567 is a match 123-608-4567 is not a match

Option value	Description
<i>ANY_POSITION_CONSIDER_PUNCTUATION</i>	The transform performs word breaking on all punctuation and spaces, but not on decimal separators (period or comma). Therefore, the transform does not break any decimal numbers into words. ❖ Example The transform considers the string 123.456 as a single numeric word instead of two numeric words. The position of the numeric word is not important, however, decimal separators impact the matching process. ❖ Example The transform determines that the following strings are a match: ACCU 29BAR 1.4L ACCU 1.4L 29BAR The transform determines that the following strings are not a match because there are different decimal indicators between the 1 and the 4 in each line: ACCU 1,4L 29BAR ACCU 29BAR 1.4L The transform determines that the following strings are not a match because there are different decimal indicators between the 25 and the 435 in each line: 25,435 25.435

Option value	Description
ANY_POSITION_IGNORE_PUNCTUATION	This value behaves similarly to the ANY_POSITION_CONSIDER_PUNCTUATION value, except that decimal separators do not impact the matching process. ❖ Example The transform determines that the following strings are a match: ACCU 29BAR 1.4L ACCU 1.4L 29BAR The transform determines that the following strings are a match, even though there is a different decimal indicator between the 1 and the 4: ACCU 1.4L 29BAR ACCU 29BAR 1.4L The transform determines that the following strings are a match, even though there is a different decimal indicator between the 25 and the 435: 25.435 25.435

Related Information

[Matching business rules \[page 611\]](#)

19.10.7 Blank field matching

Control how the transform compares fields when one or both of the fields are blank in the `CRITERIA_DEFINITION_GROUP` group.

❖ Example

Record #2 in the following table is missing a first name. For your business rules, do you want the Match transform to consider these records as matches? What if the first name field is blank in both records?

Record #1	Record #2
John	
Doe	Doe

Record #1	Record #2
204 Main St	204 Main St
La Crosse WI	La Crosse WI
54601	54601

The following options in the Match transform enable you to control the way the transform compares records with missing fields:

- [BOTH_FIELDS_BLANK_OPERATION](#)
- [BOTH_FIELDS_BLANK_SCORE](#)
- [ONE_FIELD_BLANK_OPERATION](#)
- [ONE_FIELD_BLANK_SCORE](#)

Blank field operations

The [BOTH_FIELDS_BLANK_OPERATION](#) and [ONE_FIELD_BLANK_OPERATION](#) options have the following value choices:

- [EVAL](#): The transform scores the comparison using the score you enter for the "score" options.
- [IGNORE](#): Ignores comparison criteria and distributes criteria contribution to the weighted score proportionally among the remaining criteria.

Blank field scores

The [BOTH_FIELDS_BLANK_SCORE](#) and [ONE_FIELD_BLANK_SCORE](#) options control how the Match transform scores field comparisons when the field is blank in one or both records. Enter any value from 0 to 100.

To help you decide what score to enter, determine if the Match transform should consider a blank field:

- 0% similar to a populated field or another blank field
- 100% similar to a populated field or another blank field
- Somewhere in between 0% and 100%.

Your answer probably depends on what field you're comparing. Giving a blank field a high score might be appropriate if you're matching on a first or middle name or a company name, for example.

The following examples may help you understand how your settings of the blank matching options affect the overall scoring of records.

❖ Example

[ONE_FIELD_BLANK_OPERATION](#) = [IGNORE](#)

Input field: [GIVEN_NAME1](#)

i Note

Note that when you set the blank options to *IGNORE*, the Match transform redistributes the contribution allotted for this field to the other criteria and recalculates the contributions for the other fields.

Fields compared	Record A	Record B	% alike	Contribution	Score (per field)
Postcode	54601	54601	100	20 (or 22)	22
Address	100 Water St	100 Water St	100	40 (or 44)	44
Family_Name	Hamilton	Hammilton	94	30 (or 33)	31
Given_Name1	Mary		—	10 (or 0)	—
					Weighted score: 97

❖ Example

ONE_FIELD_BLANK_OPERATION = EVAL

ONE_FIELD_BLANK_SCORE = 0

Input field: *GIVEN_NAME1*

Fields compared	Record A	Record B	% alike	Contribution	Score (per field)
Postcode	54601	54601	100	20	20
Address	100 Water St	100 Water St	100	40	40
Family_Name	Hamilton	Hammilton	94	30	28
Given_Name1	Mary		0	10	0
					Weighted score: 88

❖ Example

ONE_FIELD_BLANK_OPERATION = EVAL

ONE_FIELD_BLANK_SCORE = 100

Input field: *GIVEN_NAME1*

Fields compared	Record A	Record B	% alike	Contribution	Score (per field)
Postcode	54601	54601	100	20	20
Address	100 Water St	100 Water St	100	40	40
Family_Name	Hamilton	Hammilton	94	30	28
Given_Name1	Mary		100	10	10
					Weighted score: 98

Related Information

[Matching business rules \[page 611\]](#)

19.11 Group statistics

Add the [GROUP_STATISTICS](#) post-match operation after any match level and post-match operation for which you generate statistics about your match groups or input sources.

[GROUP_STATISTICS](#) can also count statistics from logical input sources that you have already identified with values in a field (pre-defined) or from logical sources that you specify in the [INPUT_SOURCES](#) operation.

The [GROUP_STATISTICS](#) operation also enables you to exclude certain logical sources based on your criteria.

i Note

If you choose to count input source statistics in the [GROUP_STATISTICS](#) operation, the transform also counts basic statistics about your match groups.

Group statistics fields

When you include a [GROUP_STATISTICS](#) operation in your Match transform, the transform generates the following fields by default:

- GROUP_COUNT
- GROUP_ORDER
- GROUP_RANK
- GROUP_TYPE

In addition, if you choose to generate source statistics, the transform also generates the following fields for output:

- SOURCE_COUNT
- SOURCE_ID
- SOURCE_ID_COUNT
- SOURCE_TYPE_ID

Parent topic: [Match transform \[page 586\]](#)

Related Information

[Develop a match strategy \[page 587\]](#)

[Match components \[page 588\]](#)
[Track physical and logical sources \[page 592\]](#)
[Control comparisons \[page 595\]](#)
[Prepare data for matching \[page 598\]](#)
[Compare tables \[page 601\]](#)
[Data Salvage \[page 602\]](#)
[Match criteria \[page 604\]](#)
[Matching methods \[page 606\]](#)
[Matching business rules \[page 611\]](#)
[Input source select records \[page 626\]](#)

19.12 Input source select records

To flag specific record types for evaluation, add an *INPUT_SOURCE_SELECT_RECORD_GROUP* operation to each match level for post match processing.

Adding this operation generates the Select_Record output field for you to include in your output schema. The transform populates the Select_Record output field with a Y or N depending on the type of record you select in the operation.

The results appear in the Match Input Source Output Select report. In that report, determine what records came from what source or source group and how many of each type of record were output per source or source group.

Select Source types

Record type	Description
Unique	Record that is not a member of any match group for which the transform did not find any matching records. Unique records can be from sources with a normal or special source.
<i>Single source masters</i>	Highest ranking member of a match group whose members all came from the same source. Single source masters can be from normal or special sources.
<i>Single source subordinates</i>	A record that came from a normal or special source and is a subordinate member of a match group.
<i>Multiple source masters</i>	Highest ranking member of a match group whose members came from two or more sources. Multiple source masters can be from normal or special sources.
<i>Multiple source subordinates</i>	A subordinate record of a match group that came from a normal or special source whose members came from two or more sources.
<i>Suppression matches</i>	Subordinate member of a match group that includes a higher-priority record that came from a suppress-type source. Suppression matches can be from normal or special source.
<i>Suppression uniques</i>	Record that came from a suppress source for which the transform did not find any matching records.
<i>Suppression masters</i>	Record that came from a suppress source and is the highest ranking member of a match group.

Record type	Description
<i>Suppression subordinates</i>	Record that came from a suppress-type source and is a subordinate member of a match group.

Parent topic: [Match transform \[page 586\]](#)

Related Information

[Develop a match strategy \[page 587\]](#)
[Match components \[page 588\]](#)
[Track physical and logical sources \[page 592\]](#)
[Control comparisons \[page 595\]](#)
[Prepare data for matching \[page 598\]](#)
[Compare tables \[page 601\]](#)
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[Matching business rules \[page 611\]](#)
[Group statistics \[page 625\]](#)

20 Match transform reference

The Match transform XML configuration is grouped according to setting types.

The Match transform is responsible for performing matching based on the business rules you define. The transform then sends matching and unique records on to the next transform.

For best results, cleanse the data in which you are attempting to find matches. To cleanse data, include the applicable Data Quality transforms before the Match transform.

The sample configurations installed with the software contain examples of the Match XML.

[System group \[page 629\]](#)

The System group contains subgroups for high-level transform functions such as memory size for comparison buffer and the default location for support files.

[Report and Analysis \[page 631\]](#)

The Report and Analysis group contains a setting to enable the generation of statistics about the match process.

[Match Control \[page 632\]](#)

The Match Control group contains settings to set the type of matching and the levels for which the transform searches for matches.

[Match Criteria Key Layout group \[page 634\]](#)

The Match Criteria Key Layout group defines fields on which to match and the business rules for matching on each of these fields.

[Compare Table Group \[page 645\]](#)

Use the Compare Table Group options to create rules that define the records to compare.

[Compare Match Criteria Group \[page 648\]](#)

In the Match Criteria Group, set up a comparison between two records to determine if they match.

[Post Match Processing Group \[page 665\]](#)

Add operations to the Post Match Processing Group.

[Group Statistics Group \[page 666\]](#)

Use the Group Statistics Group when you do not set input sources in the Input Sources Group in a configuration.

[Input sources \[page 670\]](#)

The INPUT_SOURCES group contains settings that define the input sources.

[Field algorithm numeric difference group \[page 676\]](#)

The Match transform uses the numeric difference algorithm to compare numeric information between records, and to form groups of records with values that are close enough to be considered duplicates.

[Field algorithm numeric percent difference group \[page 679\]](#)

The transform uses the numeric percent difference to form groups of records with values that are close enough to be considered duplicates.

[Field algorithm geo proximity group \[page 681\]](#)

The Match transform uses the geo proximity group algorithm to compare distance information between records to form groups of records with values that are close enough to be considered duplicates.

[Input Source Group Statistics Group \[page 682\]](#)

Use the Input Source Group Statistics Group to generate statistics posted with output fields.

[Input Source Select Record Group \[page 684\]](#)

To select types of records to flag on output based on each of the input sources, complete options in the Input Source Select Record Group.

20.1 System group

The System group contains subgroups for high-level transform functions such as memory size for comparison buffer and the default location for support files.

Syntax

```
<SYSTEM>
  <PLATFORM_SETTINGS>
    <MatchSettings>
      <MemoryInKBForComparisons></MemoryInKBForComparisons>
      <PreserveRecordOrder></PreserveRecordOrder>
    </MatchSettings>
  <SYSTEM_VARS>
    <LINK_DIR></LINK_DIR>
  </SYSTEM_VARS>
</PLATFORM_SETTINGS>
<SHARED_OBJECT>dqmatchtransformu</SHARED_OBJECT>
</SYSTEM>
```

Note

If an option exists in the XML, but it is not documented here, do not alter the contents of the option. Editing the contents could cause errors.

Where

MATCH_SETTINGS

Option	Description
<i>MemoryInKBForComparisons</i>	Specifies the maximum number of Kilobytes in the memory buffer for comparisons. Enter the value in Kilobytes. The default value is 10240 KB. Use the <i>MemoryInKBForComparisons</i> value with the largest break group size to fine-tune performance of the match process. If the largest break group size is smaller than the number of records that can fit in the compare buffer, then the transform stores and accesses the records from memory. Storing and accessing the records from memory makes the match process faster. However, if the largest break group is bigger than the number of records that can fit in the compare buffer, then the transform uses caching, which may slow down processing. To improve performance in this case: • Change your breaking strategy to form smaller break groups. • Increase the buffer size. i Note Making the compare buffer too large can also degrade performance.
<i>PreserveRecordOrder</i>	Specifies whether the transform completes each record in a break group order of records in the break group. ❖ Example Transform processes record1 first, then record2, then record3, and so on. • YES: The transform completes each record in break group order. • NO: The transform does not complete each record in break group order. NO is the default setting.

SYSTEM_VARIABLES

Option	Description
LINK_DIR	Specifies the relative path to the match support files. Enter the relative path. The default locations are: - Windows 32-bit: \SAP BusinessObjects\Data quality Mgmt SDK\Windows_32\DataQuality\ - UNIX: <code><DQM SDK install location>/DataQuality/</code>

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[Report and Analysis \[page 631\]](#)

[Match Control \[page 632\]](#)

[Match Criteria Key Layout group \[page 634\]](#)

[Compare Table Group \[page 645\]](#)

[Compare Match Criteria Group \[page 648\]](#)

[Post Match Processing Group \[page 665\]](#)

[Group Statistics Group \[page 666\]](#)

[Input sources \[page 670\]](#)

[Field algorithm numeric difference group \[page 676\]](#)

[Field algorithm numeric percent difference group \[page 679\]](#)

[Field algorithm geo proximity group \[page 681\]](#)

[Input Source Group Statistics Group \[page 682\]](#)

[Input Source Select Record Group \[page 684\]](#)

20.2 Report and Analysis

The Report and Analysis group contains a setting to enable the generation of statistics about the match process.

Syntax

```
<REPORT_AND_ANALYSIS>
  <GENERATE_REPORT_DATA></GENERATE_REPORT_DATA>
</REPORT_AND_ANALYSIS>
```

The `REPORT_AND_ANALYSIS` is a required group and you cannot repeat it.

REPORT_AND_ANALYSIS group

Option	Description
GENERATE_REPORT_DATA	Specifies whether the transform generates the statistics output for the StatsHandler. • YES: Generates statistics for this transform. • NO: Does not generate statistics for this transform.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)

[Match Control \[page 632\]](#)

[Match Criteria Key Layout group \[page 634\]](#)

[Compare Table Group \[page 645\]](#)

[Compare Match Criteria Group \[page 648\]](#)

[Post Match Processing Group \[page 665\]](#)

[Group Statistics Group \[page 666\]](#)

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[Field algorithm numeric difference group \[page 676\]](#)

[Field algorithm numeric percent difference group \[page 679\]](#)

[Field algorithm geo proximity group \[page 681\]](#)

[Input Source Group Statistics Group \[page 682\]](#)

[Input Source Select Record Group \[page 684\]](#)

20.3 Match Control

The Match Control group contains settings to set the type of matching and the levels for which the transform searches for matches.

Syntax

The following code block contains sample settings.

```
<MATCH_CONTROL>
  <NAME>NameAddr</NAME>
  <MATCH_LEVELS_GROUP>
    <MATCH_LEVELS>
      <MATCH_LEVEL_NAME>Individual</MATCH_LEVEL_NAME>
    </MATCH_LEVELS>
  </MATCH_LEVELS_GROUP>
</MATCH_CONTROL>
```


The [MATCH_CONTROL](#) group contains the following subgroups:

- [MATCH_LEVELS_GROUP](#): Define a match level group. Required group. Do not duplicate.
 - [MATCH_LEVELS](#): Define a match level. Required group. Duplicate for a maximum of 255 match levels.

Required group. Do not duplicate.

MATCH_CONTROL group option descriptions

Option	Description
NAME	Specifies a logical name for the transform object. Enter a name from 1-15 characters. This option is required and cannot be repeated.
MATCH_LEVELS_GROUP	
MATCH_LEVELS	
MATCH_LEVEL_NAME	Specifies the name of this match level. This option is required and cannot be repeated.

Match Levels group

Use the [MATCH_LEVELS_GROUP](#) to specify the individual levels on which to match. [MATCH_LEVELS_GROUP](#) is required and you cannot duplicate it.

Within the [MATCH_LEVELS_GROUP](#) is the [MATCH_LEVELS](#) group. Specify one match level for the the [MATCH_LEVELS_GROUP](#). Duplicate the [MATCH_LEVELS](#) group for up to 255 levels.

MATCH_LEVELS_GROUP option description

Option	Description
MATCH_LEVEL_NAME	Specifies the name of the match level. For example, if the purpose of the match transform is to match on consumer household data, this match level name could be Residence . Other MATCH_LEVELS groups could be named Family and Individual . Required option. Do not repeat.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)

[Report and Analysis \[page 631\]](#)
[Match Criteria Key Layout group \[page 634\]](#)
[Compare Table Group \[page 645\]](#)
[Compare Match Criteria Group \[page 648\]](#)
[Post Match Processing Group \[page 665\]](#)
[Group Statistics Group \[page 666\]](#)
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[Field algorithm numeric difference group \[page 676\]](#)
[Field algorithm numeric percent difference group \[page 679\]](#)
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[Input Source Group Statistics Group \[page 682\]](#)
[Input Source Select Record Group \[page 684\]](#)

20.4 Match Criteria Key Layout group

The Match Criteria Key Layout group defines fields on which to match and the business rules for matching on each of these fields.

≡ Syntax

```

<MATCH_CRITERIA_KEY_LAYOUT>
  <MATCH_ENGINE>LATIN1</MATCH_ENGINE>
  <DEFAULT_PERFORM_DATA_SALVAGE>YES</DEFAULT_PERFORM_DATA_SALVAGE>

```

Match criteria represents the fields you want to compare. Criteria provides a way to let the Match transform know what kind of data is in the input field and what types of operations to perform on that data.

The *MATCH_CRITERIA_KEY_LAYOUT* group contains the following subgroups:

- *INPUT_SOURCE_ATTRIBUTES*: Specifies the name or names of the input sources and whether to perform data salvage.
- *INPUT_FIELDS*: Specifies an input field group in which you designate an input field based on the match criteria standard keys. Required group. May not be duplicated.
 - *FIELDS*: Name a specific input field and set pre match processes to prepare the field for matching.

The *MATCH_CRITERIA_KEY_LAYOUT* group affects the following aspects of the Match transform:

- *COMPARE_MATCH_CRITERIA*: Uses the specifications that you set in the *MATCH_CRITERIA_KEY_LAYOUT* group.
- All records of each data collection use the *MATCH_CRITERIA_KEY_LAYOUT* settings.

The group is optional. Do not duplicate.

The following table contains option descriptions for the base *MATCH_CRITERIA_KEY_LAYOUT* group.

MATCH_CRITERIA_KEY_LAYOUT group option descriptions

Option	Description
MATCH_ENGINE	Specifies the match engine the transform uses. Select the matching engine based on your data. • CHINESE: Specifies that the Match transform is processing Chinese data in Chinese script. • JAPANESE: Specifies that the Match transform is processing Japanese data in Japanese script. • KOREAN: Specifies that the Match transform is processing Korean data in Korean script. • LATIN1: Specifies that the Match transform is processing Latin1 data. In general, Latin1 is the data used throughout the Americas, Western Europe, Oceania, and much of Africa. If you attempt to process non-latin1 data with the Latin1 engine, the results are unpredictable. • OTHER_NON_LATIN1: Specifies that the Match transform is processing non-Latin1 data, other than Chinese, Japanese, Korean, and Taiwanese, such as Russian, Greek, Hebrew, Arabic, and others. • TAIWANESE: Specifies that the Match transform is processing Taiwanese data in Taiwanese script. For optimum accuracy and performance, filter your multinational data to separate match transforms and enable the applicable match engine.
DEFAULT_PERFORM_DATA_SALVAGE	Specifies whether to perform data salvage when the driver record does not contain a value in a field but the passenger record contains a value for the field. • YES: Performs the data salvage on the driver record after it matches a passenger record. • NO: Does not perform the data salvage on the driver record after it matches a passenger record. NO is the default setting. The transform performs data salvage when the driver record has no data for this field and the passenger record does. Data is "salvaged" from the passenger record to the driver record. i Note Keep in mind that the setting in DEFAULT_PERFORM_DATA_SALVAGE affects only the sources for which you do not select PERFORM_DATA_SALVAGE in the SOURCE group under INPUT_SOURCE_ATTRIBUTES.

[Match Criteria Key Layout: Input Source Attributes \[page 636\]](#)

Use the options in the Input Source Attributes group to assign the [PERFORM_DATA_SALVAGE](#) option to one or more input sources.

[Match Criteria Key Layout: Input Fields \[page 638\]](#)

Use the Input Fields group to define the input fields from the specified source to use for match processing.

[Match criteria standard keys \[page 642\]](#)

Match criteria refers to the fields on which to match.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)
[Report and Analysis \[page 631\]](#)
[Match Control \[page 632\]](#)
[Compare Table Group \[page 645\]](#)
[Compare Match Criteria Group \[page 648\]](#)
[Post Match Processing Group \[page 665\]](#)
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[Input sources \[page 670\]](#)
[Field algorithm numeric difference group \[page 676\]](#)
[Field algorithm numeric percent difference group \[page 679\]](#)
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[Input Source Group Statistics Group \[page 682\]](#)
[Input Source Select Record Group \[page 684\]](#)

20.4.1 Match Criteria Key Layout: Input Source Attributes

Use the options in the Input Source Attributes group to assign the [PERFORM_DATA_SALVAGE](#) option to one or more input sources.

≡ Syntax

```
<INPUT_SOURCE_ATTRIBUTES>
  <SOURCE>
    <NAME></NAME>
    <PERFORM_DATA_SALVAGE></PERFORM_DATA_SALVAGE>
  </SOURCE>
</INPUT_SOURCE_ATTRIBUTES>
```

The group contains the name or names of the input sources and whether to perform data salvage. Do not complete the [INPUT_SOURCE_ATTRIBUTES](#) group when you enable the [PERFORM_DATA_SALVAGE](#) option for individual input fields in the source.

Optional group. Do not duplicate.

INPUT_SOURCE_ATTRIBUTES group option descriptions

Option	Description
SOURCE	Group. Specifies the source name and the data salvage option for the input source. Before you define your input sources, ensure that your data has a mapped field that contains the source name. Required group. Repeat this subgroup for each source you define for this match transform. Maximum is 10,000.
NAME	Specifies the source name. Ensure that the value that you enter here matches the values that you specify for the NAME option in the following groups: INPUT_SOURCES group. (For more information, see Input sources [page 670]) SOURCE_GROUP group (For more information, see Input Sources: Source Groups [page 673]) Do not specify an input source more than once. You can also use the same name for the NAME attribute as a NAME attribute of a defined Source Group. Matching the NAME attribute to the NAME attribute of a Source Group is the equivalent of specifying that all the input sources belong to that Source Group. If you specify an input source both individually and in a SOURCE_GROUP, the individual setting takes precedence.
PERFORM_DATA_SALVAGE	Specifies whether the transform performs data salvaging on the specified source. YES: Performs the data salvage on the driver record in the source after it matches a passenger record. NO: Does not perform the data salvage on the driver record in the source after it matches a passenger record. i Note The setting in DEFAULT_PERFORM_DATA_SALVAGE affects the input sources that you do <i>not</i> specify in this group.

Related Information

[Match Criteria Key Layout group \[page 634\]](#)

20.4.2 Match Criteria Key Layout: Input Fields

Use the Input Fields group to define the input fields from the specified source to use for match processing.

Syntax

```
<INPUT_FIELDS>
  <FIELD>
    <MAPPED_NAME></MAPPED_NAME>
    <EXTRA_DATA>
      <INCLUDE_SEPARATOR></INCLUDE_SEPARATOR>
      <FIELD_GROUP>
        <FIELD>
          <MAPPED_NAME></MAPPED_NAME>
        </FIELD>
      </FIELD_GROUP>
    </EXTRA_DATA>
    <FIELD_TYPE></FIELD_TYPE>
    <STANDARD_KEY_NAME></STANDARD_KEY_NAME>
    <CUSTOM_KEY_NAME></CUSTOM_KEY_NAME>
    <KEY_LENGTH></KEY_LENGTH>
    <LOCALE></LOCALE>
    <REMOVE_PUNCTUATION></REMOVE_PUNCTUATION>
    <CONVERT_TO_UPPERCASE></CONVERT_TO_UPPERCASE>
    <CONVERT_DIACRITICAL_CHARACTERS></CONVERT_DIACRITICAL_CHARACTERS>
    <CONVERT_TEXT_TO_NUMBERS></CONVERT_TEXT_TO_NUMBERS>
  </FIELD>
</INPUT_FIELDS>
</MATCH_CRITERIA_KEY_LAYOUT>
```

The *INPUT_FIELDS* group contains the following sub groups:

- *FIELD*: Specify one input field for the match processing. Use the input field to define a key field to hold data to compare, or to make decision while processing. For example, decide whether to perform data salvaging on a per record basis. Required group. Do not duplicate.
- *FIELD_GROUP*: Defines the fields to concatenate. Optional group. Do not duplicate.
 - *FIELD*: Defines one input field to concatenate. Required group. Can duplicate for a maximum of ten fields.

Option descriptions

Option	Description
	<i>FIELD</i> group
<i>MAPPED_NAME</i>	Specifies the name of the input field to use. Ensure that the name you specify matches a <i>MAPPED_NAME</i> attribute in the <i>INPUT_FIELDS</i> group. Required option. Do not duplicate.
<i>EXTRA_DATA</i>	Specifies additional data to store in the key. If you do not enter a value, the transform does not store extra data in the key. Optional option. Do not duplicate.

Option	Description
<i>INCLUDE_SEPARATOR</i>	Specifies whether a separator should be placed between each data field specified in this section. <i>YES</i>: If the data type of the input field is Character, the transform places a space between each field that is concatenated. <i>NO</i>: Concatenates fields without a separator. Required option. Do not duplicate.
<i>FIELD_GROUP</i> group	
<i>FIELD</i>	
<i>MAPPED_NAME</i>	Specifies the mapped name of the field. Ensure that you enter a name that is also in the <i>MAPPED_NAME</i> option in the <i>INPUT_FIELDS</i> group. Required option. Do not duplicate.
<i>FIELD_TYPE</i>	Specifies the type of data in the specified input field. <i>KEY</i>: Specifies that the FIELD group defines one key field for the transform to compare, along with a mapped input field and all formatting options connected to the mapped input field. <i>PERFORM_DATA_SALVAGE</i>: Specifies that the input field holds a value in the <i>PERFORM_DATA_SALVAGE</i> option. Required option. Do not duplicate.
<i>STANDARD_KEY_NAME</i>	Specifies the logical name for the standard key that you are defining.
<i>CUSTOM_KEY_NAME</i>	Specifies a logical name for the custom key defined in this group. The transform ignores the setting in the <i>CUSTOM_KEY_NAME</i> option unless you set the <i>STANDARD_KEY_NAME</i> to <i>CUSTOM</i> instead of a named field.
<i>KEY_LENGTH</i>	Specifies the number of characters of the field to use in the comparison field. Enter a value from 1 through 255.

Option	Description
<code>LOCALE</code>	Specifies the locale setting for the defined field. If you leave this option blank, or you set <code>DEFAULT</code> as the value, the transform uses the system default locale. If you plan to enable the <code>CONVERT_TEXT_TO_NUMBERS</code> option, we recommend that you set a value for <code>LOCALE</code> instead of leaving it blank or setting it to <code>DEFAULT</code>.
<code>REMOVE_PUNCTUATION</code>	Specifies whether the transform removes punctuation from data in the specified field before matching to provide more accurate matches. <code>YES</code>: Removes punctuation from the data in the specified field. <code>NO</code>: Does not remove punctuation from the data in the specified field. <code>NO</code> is the default setting. Before you set this option, be aware of the following information: - This option is valid only for Latin1 data. - The transform does not remove a dash when there is a dash present in a <code><Family_Name*></code> field. ⚠ Caution Setting this option to <code>YES</code> and the <code>CONVERT_TEXT_TO_NUMBERS</code> option to <code>YES</code> may produce undesirable results. ❖ Example Suppose that you have 1.23 as data in a criteria field. If you set <code>REMOVE_PUNCTUATION</code> to <code>YES</code>, the transform converts this number to 123. The number 123 could then match another value of 123. In the case of enabling <code>CONVERT_TEXT_TO_NUMBERS</code>, 123 could match a value of “one hundred twenty-three.” i Note Enabling this option does not alter your actual data.

Option	Description
<code>CONVERT_TO_UPPERCASE</code>	Specifies whether the transform converts all data to uppercase before matching to provide more accurate matches. YES: Converts the output data to uppercase for the comparison process. NO: Preserves the case of the data for the comparison process. For example, the transform compares "A" to "a" and determines the values are different. NO is the default setting i Note If you use the same mapped input field for other criteria, changes you make here affect all other criteria related to the mapped input field. Therefore, before you set this option, check that there are no other criteria using the same mapped input field. i Note Enabling this option does not alter your actual data.
<code>CONVERT_DIACRITICAL_CHARACTERS</code>	Specifies whether the transform converts diacritical characters to the closest English ASCII equivalent before matching to provide more accurate matches. YES: Converts diacritical characters to the closest English ASCII equivalent for matching purposes. For example, "ä" converts to "a". NO: Preserves diacritical characters in the matching process, treating "ä" and "a" as not identical characters. NO is the default setting. Before you set this option, be aware of the following information: - This option works only on upper Latin1 characters, values from 128 through 255. - This option is valid for all match engine options i Note Enabling this option does not alter your actual data.

Option	Description
CONVERT_TEXT_TO_NUMBERS	Specifies whether the transform converts numbers represented as text (one, two, three) to numbers before matching to provide more accurate matches. YES: Converts numbers represented as text to numbers. Numbers are in cardinal format: one = 1. Or, numbers are in ordinal format: first = 1st. NO: Leaves any numerical text intact. NO is the default setting. ⚠ Caution Setting this option and the REMOVE_PUNCTUATION option to YES may produce undesirable results. ❖ Example Suppose that you have 1.23 as data in a criteria field. If you set REMOVE_PUNCTUATION to YES, the transform converts this number to 123. The number 123 could then match another value of 123. In the case of enabling CONVERT_TEXT_TO_NUMBERS, 123 could match a value of "one hundred twenty-three." i Note Enabling this option does not alter your actual data.

Related Information

[Match Criteria Key Layout group \[page 634\]](#)

20.4.3 Match criteria standard keys

Match criteria refers to the fields on which to match.

When you select input fields for your match criteria, select fields that best reflect the data in the field you want to compare. Then use the criteria options to specify business rules for matching on each of the applicable fields.

The following table contains input field names for fields you can choose for match criteria. The fields are listed by category.

Category	Fields
<i>Geographic</i>	<i>ADDRESS_DATA1-5</i>: Use for address data that is not accounted for in other address-based criteria in the Geographic category. Also use the criteria for fields that you know contain address data, but you're not sure what type of address data. Or, use the criteria for international data that has not been parsed. <i>ADDRESS_POST_OFFICE_BOX</i>: Post Office box number. <i>ADDRESS_PRIMARY_NAME</i>: Street name data. <i>ADDRESS_PRIMARY_NUMBER</i>: Street number data. <i>ADDRESS_PRIMARY_POSTFIX</i>: Address data that comes at the end of a street name, such as a directional. <i>ADDRESS_PRIMARY_PREFIX</i>: Address data that comes at the beginning of a street name, such as a directional. <i>ADDRESS_PRIMARY_TYPE</i>: Data that tells what type of street it is (street, boulevard, lane, and so on). <i>ADDRESS_PRIVATE_MAIL_BOX</i>: A private mail box (PMB) number. PMBs are mail boxes that are run by non-postal authorities. <i>ADDRESS_RURAL_ROUTE_BOX</i>: Rural-route box number (number only, without "Box" prefix). <i>ADDRESS_RURAL_ROUTE_NUMBER</i>: Rural route number. <i>ADDRESS_SECONDARY_NUMBER</i>: The number of a unit, building, floor, or room. <i>COUNTRY</i>: Country name. <i>LOCALITY</i>: City, town, locality, or suburb. <i>LATITUDE_LONGITUDE</i>: Latitude and longitude. <i>POSTCODE1</i>: Primary postal code. <i>POSTCODE2</i>: Secondary postal code. <i>REGION</i>: Region data, such as state or province.
<i>Firm</i>	<i>FIRM</i>: Firm name. <i>FIRM_DATA1-3</i>: Use for firm data that is not accounted for in other firm-based criteria. Also use the criteria for fields that you know contain firm data, but you're not sure which type it contains. Also use the criteria for international data. <i>FIRM_MATCH_STD1-6</i>: Firm match standards. The Data Cleanse transform or other pre match transform generates the Firm_Match_Std1-6 fields. <i>FIRM_LOCATION</i>: A location within a company or organization.

Category	Fields
Person	<i>FIRM_LOCATION_MATCH_STD1-6</i> Match standards for a location within a company or organization.
	<i>NAME_DATA1-3</i> : Use for name data that is not accounted for in other name-based criteria. Also use the criteria for fields that you know contain name data, but you're not sure which type it contains.
	<i>PERSON1-3_GIVEN_NAME1</i> : The given name1 (first name) of the persons.
	<i>PERSON1_GIVEN_NAME1_MATCH_STD1-6</i> : Given_Name1 (first name) match standards for the first person.
	<i>PERSON2_GIVEN_NAME1_MATCH_STD1-6</i> : Given_Name1 (first name) match standards for the second person.
	<i>PERSON3_GIVEN_NAME1_MATCH_STD1-6</i> : Given_Name1 (first name) match standards for the third person.
	<i>PERSON1-3_GENDER</i> : Gender.
	<i>PERSON1-3_FAMILY_NAME1</i> : Family (last) name.
	<i>PERSON1-3_FAMILY_NAME1_MATCH_STD1-6</i> : Family_Name1 match standards for the persons in your data record.
	<i>PERSON1-3_FAMILY_NAME2</i> : Family (last) name. Use this field when family name data is split into two fields. For example, some cultures store the paternal family name and the maternal family name in different fields.
	<i>PERSON1-3_FAMILY_NAME2_MATCH_STD1-6</i> : Family_Name2 match standards for the persons in your data record.
	<i>PERSON1-3_MATURITY_POSTNAME</i> : Maturity postname. For example, Sr. or Jr. (one standard per person).
	<i>PERSON1-3_MATURITY_POSTNAME_MATCH_STD1-6</i> : Maturity postname match standards for the persons in your data record.
	<i>PERSON1-3_GIVEN_NAME2</i> : Given name2 (middle name).
	<i>PERSON1_GIVEN_NAME2_MATCH_STD1-6</i> : Given_Name2 (middle name) match standards for the first person.
	<i>PERSON2_GIVEN_NAME2_MATCH_STD1-6</i> : Given_Name2 (middle name) match standards for the second person.
	<i>PERSON3_GIVEN_NAME2_MATCH_STD1-6</i> : Given_Name2 (middle name) match standards for the third person.
	<i>PERSON1-3_HONORARY_POSTNAME</i> : Honorary postname for up to three persons indicating certification, academic degree, or affiliation. For example, CPA.

Category	Fields
	PERSON1-3_HONORARY_POSTNAME_MATCH_STD1-6: Honorary postname match standards. PERSON1-3_PRENAME: Prenom (for example, Mr. or Mrs.) for up to three persons. PERSON1-3_PRENAME_MATCH_STD1-6: Prenom match standards. PERSON1-3_TITLE: Job or occupational title of each person. For example, Manager. PERSON1-3_TITLE_MATCH_STD1-6: Title match standards for each person. SOCIAL_SECURITY_NUMBER1-3: Social Security numbers for up to three people in a record.
Other	DATE1-3: Date data. For example, birthdate data. PHONE: Phone number.
Custom	Use custom fields to match data that does not qualify for any of the specifically named criteria. If you prepared fields for matching through a custom Universal Data Cleanse solution, the Data Cleanse dictionary names appear in the Custom category.

Related Information

[Match Criteria Key Layout group \[page 634\]](#)

20.5 Compare Table Group

Use the Compare Table Group options to create rules that define the records to compare.

```
<COMPARE_TABLE_GROUP>
  <COMPARE_TABLE>
    <NAME></NAME>
    <ID_SOURCE></ID_SOURCE>
    <DEFAULT_ID></DEFAULT_ID>
    <DEFAULT_ACTION></DEFAULT_ACTION>
    <COMPARE></COMPARE>
    <TABLE_ENTRY_GROUP>
      <TABLE_ENTRY>
        <ACTION></ACTION>
        <DRIVER_ID></DRIVER_ID>
        <PASSENGER_ID></PASSENGER_ID>
      </TABLE_ENTRY>
    </TABLE_ENTRY_GROUP>
    <ID_FIELD>
      <MAPPED_NAME></MAPPED_NAME>
    </ID_FIELD>
```

```
</COMPARE_TABLE>
</COMPARE_TABLE_GROUP>
```

The *COMPARE_TABLE_GROUP* contains the following subgroups:

- *COMPARE_TABLE*: Specify all sources and fields plus configure options for the compare table.
- *TABLE_ENTRY_GROUP*: Specifies the name of the table that contains the *TABLE_ENTRY* configuration items.
 - *TABLE_ENTRY*: Defines an entry into the compare table. If a driver and passenger pair match an entry in the compare table, then the transform performs the action specified in the *ACTION* option. Required group. May be duplicated.
- *ID_FIELD*: Defines the ID field. The transform ignores the *ID_FIELD* group when you set the *ID_SOURCE* option to *INPUT_SOURCES*. Required group when *ID_SOURCE* is set to *ID_FIELD*. Otherwise, optional. Do not duplicate.

Optional group. Do not duplicate.

COMPARE_TABLE_GROUP

Option	Description
<i>COMPARE_TABLE</i>	
<i>NAME</i>	Specifies a logical name for the compare table. Any compare object with a COMPARE option may reference the name.
<i>ID_SOURCE</i>	Specifies the source for the ID values. • <i>ID_FIELD</i>: Takes ID values from the input field specified in the <i>ID_FIELD</i> group. <i>ID_FIELD</i> is the default setting. • <i>INPUT_SOURCES</i>: Takes ID values from a predefined input source. You define predefined input sources in the <i>NAME</i> option of the <i>INPUT_SOURCES > SOURCE</i> group. The transform also obtains the driver and passenger ID values from the <i>NAME</i> option in an <i>INPUT_SOURCES > SOURCE_GROUPS > SOURCE_GROUP</i> section (a predefined Source Group). Specifying a Source Group eliminates the need to specify all the input sources that belong to a given source group. If you specify an Input Source both individually and by a Source Group, the individual source group settings takes precedence.
<i>DEFAULT_ID</i>	Specifies the ID the transform should use when the input ID is blank.
<i>DEFAULT_ACTION</i>	Specifies what action the transform takes when it does not find an ID in the compare table. • <i>COMPARE</i>: Performs the comparison. • <i>NO_MATCH</i>: Considers the two records unique.

Option	Description
<i>COMPARE</i>	Specifies what Compare the transform performs when the <i>DEFAULT_ACTION</i> is set to <i>COMPARE</i>. Specify a name that matches a value in the <i>NAME</i> option of a Compare Match Criteria object. If the <i>DEFAULT_ACTION</i> is set to <i>COMPARE</i> or if an entry in the table is set to <i>COMPARE</i>, this <i>COMPARE</i> option is required. Otherwise this option is optional. Do not duplicate.
<i>TABLE_ENTRY_GROUP</i>	
<i>TABLE_ENTRY</i> group	
<i>ACTION</i>	Specifies what action the transform takes when a driver—passenger pair matches this entry. <i>COMPARE</i>: Performs the comparison. <i>NO_MATCH</i>: Considers the two records unique.
<i>DRIVER_ID</i>	Specifies the ID value that the driver record must have to match this entry. If you do not set this option, the transform matches any driver ID value to this entry.
<i>PASSENGER_ID</i>	Specifies the ID value that the passenger record must have to match this entry. If you do not set this option, the transform matches any passenger ID value to this entry.
<i>ID_FIELD</i>	
<i>MAPPED_NAME</i>	Specifies the mapped name of the input field. When you select <i>ID_FIELD</i> for the <i>ID_SOURCE</i> option value, the transform uses the ID values from the input field that you specify here for <i>MAPPED_NAME</i>. Required option. Do not duplicate.

❖ Example

The following code shows how to disable comparisons between records from a house list, which presumably has no matches within it.

```
<COMPARE_TABLE>
  <NAME>House List Check</NAME>
  <ID_SOURCE>ID_FIELD</ID_SOURCE>
  <DEFAULT_ID>Unknown</DEFAULT_ID>
  <DEFAULT_ACTION>COMPARE</DEFAULT_ACTION>
  <COMPARE>NormalMatch</COMPARE>
  <TABLE_ENTRY>
    <ACTION>NO_MATCH</ACTION>
    <DRIVER_ID>H</DRIVER_ID>
    <PASSENGER_ID>H</PASSENGER_ID>
  </TABLE_ENTRY>
</ID_FIELD>
```

```
<MAPPED_NAME>ListID</MAPPED_NAME>
</ID_FIELD>
</COMPARE_TABLE>
```

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)
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20.6 Compare Match Criteria Group

In the Match Criteria Group, set up a comparison between two records to determine if they match.

Syntax

```
<COMPARE_MATCH_CRITERIA_GROUP>
  <COMPARE_MATCH_CRITERIA>
    <NAME></NAME> ...
```

The *COMPARE_MATCH_CRITERIA_GROUP* group contains the following subgroups:

- *COMPARE_MATCH_CRITERIA*: Specifies the match options and match criteria.
 - *STANDARD_KEY_MATCH_OPTIONS*: Specify specialty match options applicable to standard keys. Optional group. Do not duplicate.
- *CRITERIA_DEFINITION_GROUP*: Specify the match criteria for the match process. Required group. Do not duplicate.
 - *CRITERIA_DEFINITION*: Specifies one match criteria and includes one or more key fields to compare and the matching options for comparing specified key fields. Required group. Duplicate for up to 255 definitions.

Optional group. Do not duplicate.

COMPARE_MATCH_CRITERIA group option description

Option	Description
<i>NAME</i>	Specifies a logical name for the <i>COMPARE_MATCH_CRITERIA</i> group. Any Compare object, like <i>Match_Level</i> or <i>Match_Table</i>, that has a <i>COMPARE</i> option may reference the name you specify here. Enter from 1 through 256 characters. Required option. Do not duplicate.

[Standard Key Match Options \[page 649\]](#)

Set specialty match options that apply to specified standard keys.

[Criteria Definition Group \[page 654\]](#)

Define match criteria by establishing key fields and setting options for comparing key fields.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)

[Report and Analysis \[page 631\]](#)

[Match Control \[page 632\]](#)

[Match Criteria Key Layout group \[page 634\]](#)

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[Field algorithm numeric percent difference group \[page 679\]](#)

[Field algorithm geo proximity group \[page 681\]](#)

[Input Source Group Statistics Group \[page 682\]](#)

[Input Source Select Record Group \[page 684\]](#)

20.6.1 Standard Key Match Options

Set specialty match options that apply to specified standard keys.

Syntax

```
<STANDARD_KEY_MATCH_OPTIONS>  
    <COMPARE_GIVEN_NAME1_TO_GIVEN_NAME2></  
COMPARE_GIVEN_NAME1_TO_GIVEN_NAME2>
```

```

    <MATCH_ON_HYPHENATED_FAMILY_NAME></MATCH_ON_HYPHENATED_FAMILY_NAME>
    <IGNORE_FAMILY_NAME_WHEN_FEMALE></IGNORE_FAMILY_NAME_WHEN_FEMALE>
    <NUMBER_OF_NAMES_THAT_MUST_MATCH></NUMBER_OF_NAMES_THAT_MUST_MATCH>
    <IGNORE_FIRM_IF_NAMES_MATCH></IGNORE_FIRM_IF_NAMES_MATCH>
    <IGNORE_STREET_RR_IF_POBOX_MATCH></IGNORE_STREET_RR_IF_POBOX_MATCH>
    <ADDR_BLANK_OP_IGNORE_IF_FIRMS_MATCH></
ADDR_BLANK_OP_IGNORE_IF_FIRMS_MATCH>
    <UNIQUE_ON_RESIDENT_RR_NO_BOX></UNIQUE_ON_RESIDENT_RR_NO_BOX>
    </STANDARD_KEY_MATCH_OPTIONS>
    <WEIGHTED_MATCH_SCORE></WEIGHTED_MATCH_SCORE>

```

Optional group. Do not duplicate.

STANDARD_KEY_MATCH_OPTIONS group option descriptions

Option	Description
<i>COMPARE_GIVEN_NAME1_TO_GIVEN_NAME2</i>	Specifies whether the transform compares the <i>GIVEN_NAME1</i> (first name) of one record to the <i>GIVEN_NAME2</i> (middle name) of another record. YES: Compares <i>GIVEN_NAME1</i> of one record to <i>GIVEN_NAME2</i> of another record. Before you set this option to YES, ensure that you configure one or more of the following criteria: - Person1_Given_Name1 and Person1_Given_Name2 - Person2_Given_Name1 and Person2_Given_Name2 - Person3_Given_Name1 and Person3_Given_Name2 NO: Does not compare <i>GIVEN_NAME1</i> of one record to <i>GIVEN_NAME2</i> of another record. NO is the default setting.

❖ Example

the transform considers the following records as duplicates when you set this option to **YES**.

GIVEN_NAME1	GIVEN_NAME2	FAMILY_NAME1
John		Smith
R	John	Smith

Before you set this option to **YES**, ensure that you configure one or more of the following criteria:

- Person1_Given_Name1 and Person1_Given_Name2
- Person2_Given_Name1 and Person2_Given_Name2
- Person3_Given_Name1 and Person3_Given_Name2

Option	Description						
MATCH_ON_HYPHENATED_FAMILY_NAME	Specifies whether a single family name (last name) in one record matches a hyphenated family name in another record. YES: Considers the records matching when the single family name in one record matches one of the hyphenated family names in another record. NO: Considers the hyphenated family name as a single family name and assigns a low similarity score. If the similarity score does not meet your family name criteria, the transform considers the records as a no-match. NO is the default setting. The software performs this comparison only when one field has a hyphen and the other field does not. ❖ Example ❖ Example When you set this option to YES, the transform compares the records in the following table. <table> <tr> <th>Given</th><th>Family</th></tr> <tr> <td>Laura</td><td>Smith</td></tr> <tr> <td>Laura</td><td>Albers-Smith</td></tr> </table> Ensure that you have the following criteria: Family_Name1, Family_Name2, or Family_Name3.	Given	Family	Laura	Smith	Laura	Albers-Smith
Given	Family						
Laura	Smith						
Laura	Albers-Smith						
IGNORE_FAMILY_NAME_WHEN_FEMALE	Specifies whether the transform makes an adjustment for family names when the given name is female. YES: Ignores the Family_Name1 criteria when the given name gender is a female (gender code Gender1=5). For example, Laura Smith may match Laura Albers. NO: Does not use gender when determining family name matches. NO is the default setting. Ensure that you have at least the following three criteria: - Given_Name1 - Family_Name1 - Gender1						

Option	Description
<i>NUMBER_OF_NAMES_THAT_MUST_MATCH</i>	Specifies the number of names that must match. This option requires that you have criteria of Person1_Given_Name1, Person1_Family_Name1, and so on. <i>ONE</i>: Specifies that records are a match when at least one of the names meets the criteria (default value). <i>ALL</i>: Specifies that records are a match only when all of the names meet the criteria.
<i>IGNORE_FIRM_IF_NAMES_MATCH</i>	Specifies criteria for firm name matching when firm name contains odd abbreviations or spellings. <i>YES</i>: If the transform finds matching names at the same address, consider the records as duplicates even when the firm names do not match exactly because of odd abbreviations or spellings. <i>NO</i>: Does not consider records as duplicates when the address is the same but the firm names do not match exactly. <i>NO</i> is the default setting. ❖ Example When you set the option to <i>YES</i>, the transform considers the following records as duplicates: Rita Terranova ETI 100 Bren Rd 55343 Rita Terranova Eco Technologies 100 Bren Rd 55343 When you set the option to <i>NO</i>, the transform may not consider the records as duplicates. i Note In setting this option to <i>YES</i>, the transform assumes that you configure the job to match on real names, firm, and address line, and to break or match on City or ZIP Code.
<i>IGNORE_STREET_RR_IF_POBOX_MATCH</i>	Specifies whether the transform considers records as duplicates when the PO Boxes match, or the address and rural route address pass the match criteria settings. <i>YES</i>: Considers records as duplicates when the PO Boxes match. Also considers records as duplicates when the PO Boxes do not match, but the address and rural route addresses pass the match criteria settings. <i>NO</i>: Considers records as duplicates only when all forms of the address match. Includes street, rural route, and PO Box. <i>NO</i> is the default setting.

Option	Description
ADDR_BLANK_OP_IGNORE_IF_FIRMS_MACH	Specifies whether the transform allows blank matching for all address components when the firm data matches and neither firm field is blank. use this option to control blank matching when the transform compares firm records within the same break group. • YES: Allows blank matching for all address components when the firm data matches and neither firm field is blank. The transform ignores the blank match settings in the CRITERIA_DEFINITION group. • NO: Does not allow blank matching for all address components when the firm data matches and neither firm field is blank, unless you have enabled blank matching in one of the settings in the CRITERIA_DEFINITION group. NO is the default setting.
UNIQUE_ON_RESIDENT_RR_NO_BOX	Specifies whether the transform places records into the same dupe group when both records contain a resident-type name such as Current Resident or Occupant, and a matching rural route address with no box number. • YES: Places all records with a resident-type name and the same rural route address with no box number into the same dupe group. • NO: Treats this type of record the same as other records. NO is the default setting.
WEIGHTED_MATCH_SCORE	Specifies the weighted match score for this match level. When your matching method includes weighted scoring, the transform considers records as matches when the total contribution score is greater than or equal to the value you enter for the weighted match score. The default value is 101. Optional option.

Related Information

[Compare Match Criteria Group \[page 648\]](#)

20.6.2 Criteria Definition Group

Define match criteria by establishing key fields and setting options for comparing key fields.

Syntax

```
<CRITERIA_DEFINITION_GROUP>
  <CRITERIA_DEFINITION>
    <CRITERIA_NAME></CRITERIA_NAME>
    <STANDARD_KEY_NAME></STANDARD_KEY_NAME>
    <CUSTOM_KEY_NAME></CUSTOM_KEY_NAME>
    <COMPARE_FIELD_GROUP>
      <COMPARE_FIELD>
        <STANDARD_KEY_NAME></STANDARD_KEY_NAME>
        <CUSTOM_KEY_NAME></CUSTOM_KEY_NAME>
      </COMPARE_FIELD>
    </COMPARE_FIELD_GROUP>
    <COMPARE_FIELD_COMPARISON_OVERRIDE_GROUP>
      <COMPARE_FIELD_COMPARISON_OVERRIDE>
        <DRIVER_KEY_NAME></DRIVER_KEY_NAME>
        <PASSENGER_KEY_NAME></PASSENGER_KEY_NAME>
      </COMPARE_FIELD_COMPARISON_OVERRIDE>
    </COMPARE_FIELD_COMPARISON_OVERRIDE_GROUP>
    <FIELD_ALGORITHM></FIELD_ALGORITHM>
    <ENABLE_INTERSCRIPT_MATCHING></ENABLE_INTERSCRIPT_MATCHING>
    <ONE_FIELD_BLANK_OPERATION></ONE_FIELD_BLANK_OPERATION>
    <ONE_FIELD_BLANK_SCORE></ONE_FIELD_BLANK_SCORE>
    <BOTH_FIELDS_BLANK_OPERATION></BOTH_FIELDS_BLANK_OPERATION>
    <BOTH_FIELDS_BLANK_SCORE></BOTH_FIELDS_BLANK_SCORE>
    <MATCH_SCORE></MATCH_SCORE>
    <NO_MATCH_SCORE></NO_MATCH_SCORE>
    <CONTRIBUTION_TO_WEIGHTED_SCORE></CONTRIBUTION_TO_WEIGHTED_SCORE>
    <USE_IN_WEIGHTED_SCORE_IF_GREATER_THAN></
USE_IN_WEIGHTED_SCORE_IF_GREATER_THAN>
    <ZERO_WEIGHTED_SCORE_IF_LESS_OR_EQUAL></
ZERO_WEIGHTED_SCORE_IF_LESS_OR_EQUAL>
    <COMPARE_ALGORITHM></COMPARE_ALGORITHM>
    <CHECK_FOR_TRANSPOSED_LETTERS></CHECK_FOR_TRANSPOSED_LETTERS>
    <INITIALS_ADJUSTMENT_SCORE></INITIALS_ADJUSTMENT_SCORE>
    <SUBSTRING_ADJUSTMENT_SCORE></SUBSTRING_ADJUSTMENT_SCORE>
    <ABBREVIATION_ADJUSTMENT_SCORE></ABBREVIATION_ADJUSTMENT_SCORE>
    <EXT_ABBREVIATION_ADJUSTMENT_SCORE></
EXT_ABBREVIATION_ADJUSTMENT_SCORE>
    <NUMERIC_WORDS_MATCH_EXACTLY></NUMERIC_WORDS_MATCH_EXACTLY>
  </CRITERIA_DEFINITION>
</CRITERIA_DEFINITION_GROUP>
```

The [CRITERIA_DEFINITION_GROUP](#) group defines one match criteria. A match criteria specifies one or more key fields to compare. It also specifies what match options to use when comparing the specified key fields and how well the fields must match in order to be considered either a match or a "no match."

The [CRITERIA_DEFINITION_GROUP](#) group contains the following subgroups:

- [CRITERIA_DEFINITION](#): Specifies one match criteria and includes one or more key fields to compare and the matching options for comparing specified key fields. Required group. Duplicate for up to 255 definitions.
- [COMPARE_FIELD_GROUP](#): Specifies a compare group. Use this group in addition to the [STANDARD_KEY_NAME](#) and [CUSTOM_KEY_NAME](#) options under [CRITERIA_DEFINITION](#) or instead of them. Defining more than one Compare Field is a way of performing cross-field matching. The transform compares all the fields that you define. The transform selects the compare with the highest score as the result. Optional group. Duplicate the [COMPARE_FIELD](#) subgroup for up to 255 compare field groups.

- [*COMPARE_FIELD_COMPARISON_OVERRIDE_GROUP*](#): Defines one comparison to be performed between two Compare Fields. If you do not include set this group, the transform compares all the compare fields that you define against each other. However, if you define one or more of the [*COMPARE_FIELD_COMPARISON_OVERRIDE_GROUP*](#) groups, the transform performs only the comparisons defined in these groups. The verifier should issue a warning when there is a Compare Field defined that is not referenced in a [*COMPARE_FIELD_COMPARISON_OVERRIDE*](#) group.
Optional group. Duplicate the [*COMPARE_FIELD_COMPARISON_OVERRIDE*](#) subgroup for up to 255 groups.

Required group. Do not duplicate.

CRITERIA_DEFINITION_GROUP option descriptions

Option	Description
<i>CRITERIA_DEFINITION</i>	
<i>CRITERIA_NAME</i>	Specifies a logical name for the criteria definition.
<i>STANDARD_KEY_NAME</i>	Specifies a logical name for the standard criteria being defined. For a list of standard keys, see Match criteria standard keys [page 642] .
<i>CUSTOM_KEY_NAME</i>	Specifies a logical name for the custom criteria being defined. The transform ignores the <i>CUSTOM_KEY_NAME</i> unless you set the <i>STANDARD_KEY_NAME</i> to <i>CUSTOM</i>. Required if you set the <i>STANDARD_KEY_NAME</i> to <i>CUSTOM</i>. Otherwise, the option is optional. Do not duplicate.
<i>COMPARE_FIELD_GROUP</i>	
<i>COMPARE_FIELD</i> group	
<i>STANDARD_KEY_NAME</i>	Specifies a logical name for the standard criteria being defined. For a list of standard keys, see Match criteria standard keys [page 642] .
<i>CUSTOM_KEY_NAME</i>	Specifies a logical name for the custom criteria being defined. The transform ignores the <i>CUSTOM_KEY_NAME</i> unless you set the <i>STANDARD_KEY_NAME</i> to <i>CUSTOM</i>. Required if you set the <i>STANDARD_KEY_NAME</i> to <i>CUSTOM</i>. Otherwise, the option is optional.
<i>COMPARE_FIELD_COMPARISON_OVERRIDE_GROUP</i>	
<i>COMPARE_FIELD_COMPARISON_OVERRIDE</i> group	
<i>DRIVER_KEY_NAME</i>	Specifies the name of a key field in the driver record to compare.
<i>PASSENGER_KEY_NAME</i>	Specifies the name of a key field in the passenger record to compare.

Option	Description
<i>FIELD_ALGORITHM</i>	Specifies a Field Algorithm Object that the transform uses to compare the defined Compare Fields. If you do not specify a field algorithm, the transform uses the current SIMIL algorithm. When you specify a field algorithm, ensure that the field algorithm that you specify matches the <i>NAME</i> option of a field algorithm object such as <i>FIELD_ALGORITHM_NUMERIC_DIFFERENCE</i>. Also, when you complete <i>FIELD_ALGORITHM</i>, the transform ignores the following options from the <i>CRITERIA_DEFINITION</i> group: • COMPARE_ALGORITHM • CHECK_FOR_TRANSPOSED_LETTERS, INITIALS_ADJUSTMENT_SCORE • SUBSTRING_ADJUSTMENT_SCORE • ABBREVIATION_ADJUSTMENT_SCORE • EXT_ABBREVIATION_ADJUSTMENT_SCORE • NUMERIC_WORDS_MATCH_EXACTLY • APPROX_SUBSTRING_ADJUSTMENT_SCORE
<i>ENABLE_INTERSCRIPT_MATCHING</i>	Specifies whether the transform should attempt to use transliterated data in addition to normal Unicode matching. • <i>YES</i>: Attempts to use transliterated data in addition to normal Unicode matching. The transform transliterates the data to Latin script before it compares the data. • <i>NO</i>: Does not attempt to use transliterated data in addition to normal Unicode matching. <i>NO</i> is the default setting. Enable this option when you have the same data in different scripts (writing systems). For example, if one record has Latin1 and the other has Katakana, or one has Latin and the other has Cyrillic. When you set the option to <i>YES</i>, the comparison is an extra comparison. The transform performs the comparison before the normal Unicode comparison. The transform keeps the comparison with the highest score. The transform does not need the LOCALE setting for script conversion. Optional option. Do not duplicate.

Option	Description
<i>ONE_FIELD_BLANK_OPERATION</i>	Specifies whether the transform uses the defined criteria when one record has a populated field and the other record has a blank field. • <i>EVAL</i>: Uses the score set in the <i>ONE_FIELD_BLANK_OPERATION</i> option as the similarity score for this criteria. • <i>IGNORE</i>: Ignores this criteria in the comparison process and distributes the criteria contribution to the weighted score proportionally among the remaining criteria. The distribution negates any impact the contribution score may have had.
<i>ONE_FIELD_BLANK_SCORE</i>	Specifies the similarity score that the transform uses when one of the fields is blank and you set the <i>ONE_FIELD_BLANK_OPERATION</i> option to <i>EVAL</i>. Enter a value from 0 through 100.
<i>BOTH_FIELDS_BLANK_OPERATION</i>	Specifies whether the transform uses the defined criteria when both records have blank field for this criteria. • <i>EVAL</i>: Uses the score set in the <i>BOTH_FIELDS_BLANK_SCORE</i> option as the similarity score for this criteria. • <i>IGNORE</i>: Ignores this criteria in the comparison process and distributes the criteria contribution to the weighted score proportionally among the remaining criteria. The distribution negates any impact the contribution score may have had.
<i>BOTH_FIELDS_BLANK_SCORE</i>	Specifies the similarity score that the transform uses when both of the fields are blank and you set the <i>BOTH_FIELDS_BLANK_OPERATION</i> option to <i>Eval</i>. Enter a value from 0 through 100.
<i>MATCH_SCORE</i>	Specifies the minimum similarity score needed for the transform to consider the records as matching. Enter a value from 0 through 101. Entering 101 disables this option.
<i>NO_MATCH_SCORE</i>	Specifies the maximum similarity score for the records for the transform to consider the records as not matching. Enter a value from -1 through 100. Entering -1 disables this option.

Option	Description
<i>CONTRIBUTION_TO_WEIGHTED_SCORE</i>	Specifies the contribution value, when you use the weighted or the combination scoring method. If no single criteria decides a match or no-match, the transform calculates the contribution score by summing the products of each criteria score by each criteria weight. Enter a value from 0 through 100. 0 is the default value.
<i>USE_IN_WEIGHTED_SCORE_IF_GREATER_THAN</i>	Specifies the minimum similarity score needed for the transform to contribute this criteria to the Weighted Match Score. -1 is the default value. A setting of -1 implies that the transform always uses the criteria in the weighted score. ❖ Example You set the <i>USE_IN_WEIGHTED_SCORE_IF_GREATER_THAN</i> option to 59 for a given name criteria. The given names between two records are less than 60% similar. The transform ignores the given name criteria and distributes the contribution value proportionally among the remaining criteria.
<i>ZERO_WEIGHTED_SCORE_IF_LESS_OR_EQUAL</i>	Specifies the minimum similarity score needed for the transform to qualify this criteria for contributing a value, other than zero, to the weighted match score. ❖ Example You set the <i>ZERO_WEIGHTED_SCORE_IF_LESS_OR_EQUAL</i> option to 59 for the given name criteria. The given names between two records are less than 60% similar. Therefore, the given name criteria contributes zero toward the weighted match score. Zero is the default setting. A value of 0 means that the criteria does not receive a 0 weighted score unless the score is 0.

Option	Description
<i>COMPARE_ALGORITHM</i>	Specifies how the transform processes fields when more than one word commonly exists. Only the options that are appropriate for the chosen value appear in the Comparison Rules table. <i>FIELD</i>: Compares all the data in the field as a single string. The transform processes the algorithm more efficiently when the fields have only one word. <i>WORD</i>: Compares using the <i>FIELD</i> method first and then compares the words in the field. The transform keeps the best score between the comparisons. Many criteria options require that you set the <i>COMPARE_ALGORITHM</i> option to <i>WORD</i>, such as <i>SUBSTRING_ADJUSTMENT_SCORE</i> and <i>INITIALS_ADJUSTMENT_SCORE</i>. We include the restriction in the option description when applicable.
<i>CHECK_FOR_TRANSPOSED_LETTERS</i>	Specifies whether the transform adjusts the match score for any transposed characters encountered. <i>YES</i>: Deducts half as many points for transposed characters as it deducts for other non-matching characters. ❖ Example Compare: Smith and Simth Finding: Words differ by one transposition Penalty: 1 correction Percent alike: 90% <i>NO</i>: Handles transposed characters the same way as for any non-matching characters. <i>NO</i> is the default setting. ❖ Example Compare: Smith and Simth Findings: Words differ by two corrections: One insert and one delete. Percent alike: 80%

Option	Description
<i>INITIALS_ADJUSTMENT_SCORE</i>	Specifies whether initials or acronyms match whole words. For example, the firm name International Health Providers could match IHP. Enter a value from 0 through 100. A value of 0 disables this option. 0 is the default setting. ♣ Example The firm name International Health Providers could match IHP. Keep in mind the following information when you use this option: • The first initial must match the first letter of the word. • The letters that match score 100. The remaining letters score the value that you specify (1-100). • The transform combines the two scores proportionally to determine the overall score. If there are other words in the field that are not shortened, the transform determines the score in the usual way. For example, New York Police Department may be shortened to New York PD and still match. i Note For this option to work for multiple-word abbreviations, such as International Health Providers = IHP, set the <i>COMPARE_ALGORITHM</i> option to <i>WORD</i>. For this option to work for single-word abbreviations, such as Maria = M, set the <i>COMPARE_ALGORITHM</i> option to either <i>WORD</i> or <i>FIELD</i>.

Option	Description
<i>SUBSTRING_ADJUSTMENT_SCORE</i>	Specifies whether the transform allows matching longer strings of words to shorter strings. ❖ Example A company name "Mayfield Painting and Sand Blasting" could be shortened to "Mayfield Painting". Enter a value from 0 through 100. 0 is the default setting. Keep in mind the following information before you set this option: • Enter a value of 0 to disable substring checking. • Enter a value of 100 to consider substrings compared with longer strings as a perfect match. For this option to work, ensure that you set the <i>COMPARE_ALGORITHM</i> to <i>WORD</i>. After you set this option to a value other than 0, the software assigns scores as follows: • The transform assigns a score of 100 for matching letters. • The transform assigns the score that you specify for the remaining letters. • The transform combines the two scores proportionally to determine the overall score. To qualify as a substring match, the shorter string must exactly match the first part of the longer string. ❖ Example The following strings are matching substrings: • Mayfield • Mayfield Painting • Mayfield Painting and Sand The following substrings do not match: • Mayfield Sand Blasting • Painting and Sand Blasting The transform disqualifies substrings with alternate spellings in any of the words. For example, the following substring does not match: "Murphy Painting and Sand Blasting." The transform assigns a similarity score of 85% when this option is disabled (set to 0).

Option	Description
APPROX_SUBSTRING_ADJUSTMENT_SCORE	Specifies the score for the words that do not match to the other string. Enter a value from 0 through 100. 0 is the default setting. 0 disables this option. ❖ Example The following two strings do not match exactly: • CRUZ RDZ • SMITH CRUZ DE RODRIQUEZ The first words do not match, nor do the words match consecutively. However, with this option and the ABBREVIATION_ADJUSTMENT_SCORE option enabled, the transform accepts the strings and assigns a score. Then, the transform gives the leftover words, SMITH DE, and the spaces the adjustment score set in this option (APPROX_SUBSTRING_ADJUSTMENT_SCORE) option.

Option	Description						
<i>ABBREVIATION_ADJUSTMENT_SCORE</i>	Specifies how the transform matches whole words to abbreviations. The value you enter is the score given to the letters that are in the longer word but not the shorter word. Enter a value from 0 through 100: • A value of 0 disables abbreviation checking. 0 is the default setting. • Enter a value of 100 for the transform to consider abbreviations and longer words a perfect match. For this option to work, ensure that you set the <i>COMPARE_ALGORITHM</i> option to <i>WORD</i>. ♣ Example Long firm names such as International Health Providers, are sometimes abbreviated to Intl Health Providers. Other possible abbreviations include the abbreviations in the following table. <table> <tr> <th>Full word</th><th>Possible abbreviations</th></tr> <tr> <td>Business</td><td>Bus, Bsnss, Bss</td></tr> <tr> <td>Database</td><td>Dat, Db, Dse</td></tr> </table> For the Match transform, abbreviation means: • The first letter of the shorter word matches the first letter of the longer word. • All remaining letters of the shorter word appear in the longer word in the same order as in the shorter word.	Full word	Possible abbreviations	Business	Bus, Bsnss, Bss	Database	Dat, Db, Dse
Full word	Possible abbreviations						
Business	Bus, Bsnss, Bss						
Database	Dat, Db, Dse						

Option	Description
<i>EXT_ABBREVIATION_ADJUSTMENT_SCORE</i>	Specifies the score for an extended version of the <i>ABBREVIATION_ADJUSTMENT_SCORE</i> option. Enter a value from 0 to 100. 0 is the default setting and disables this option. For this option to work, ensure that you set the <i>COMPARE_ALGORITHM</i> option to <i>WORD</i>. Before you set this option, keep in mind the following information: • The first letter of the short word must match the first letter of the first word in the multiple-word string. • The remaining letters of the short word must also be in the multiple-word string in the same order. • The transform assigns a score of 100 to matching letters. • The transform assigns the score that you specify for this option to the remaining letters. • The transform combines the two scores proportionally to determine the overall score.
<i>NUMERIC_WORDS_MATCH_EXACTLY</i>	Specifies how the transform matches data that contain both numbers and letters. • <i>NONE</i>: Does not require numeric words to match exactly to be considered a match. <i>NONE</i> is the default setting. • <i>ANY_POSITION</i>: Does not require numeric words to be in the same position in two different strings to be considered a match. • <i>ANY_POSITION_CONSIDER_PUNCTUATION</i>: Same as <i>ANY_POSITION</i> but considers the position of the decimal separator, such as a comma or period, within the numeric words. • <i>ANY_POSITION_IGNORE_PUNCTUATION</i>: Same as <i>ANY_POSITION</i> but ignores the decimal separators such as a comma or a period. • <i>SAME_POSITION</i>: Requires numeric words to match exactly and be in the same position in the string to be considered a match. For this option to work, ensure that you set the <i>COMPARE_ALGORITHM</i> option to <i>WORD</i>.

Related Information

[Compare Match Criteria Group \[page 648\]](#)

20.7 Post Match Processing Group

Add operations to the Post Match Processing Group.

Note

The following XML contains sample data.

```
<POST_MATCH_PROCESSING_GROUP>
  <POST_MATCH_PROCESSING>
    <NAME>IndividualPMP</NAME>
    <OPERATIONS>
      <OP>
        <NAME>IndGroupStats</NAME>
      </OP>
    </OPERATIONS>
  </POST_MATCH_PROCESSING>
</POST_MATCH_PROCESSING_GROUP>
```

The *POST_MATCH_PROCESSING_GROUP* contains the following subgroups:

- *POST_MATCH_PROCESSING*: Optional group. May be duplicated.
- *OPERATIONS*: Defines one or more post processing operations to perform. Required option. Do not duplicate.
 - *OP*: Defines one post processing operation to perform. Required option. May be duplicated.

Optional group. Do not duplicate.

POST_MATCH_PROCESSING_GROUP group option descriptions

Option	Description
<i>POST_MATCH_PROCESSING</i> group	
<i>NAME</i>	Specifies a logical name for the post match object. This name should correspond to a name referenced in the <i>MATCH_LEVEL</i> object. Required option. Do not duplicate.
<i>OPERATIONS</i>	
<i>OP</i> group	

Option	Description
<i>NAME</i>	Specifies the name of the object that performs the operation. This name should correspond to a name referenced in a <i>GROUP_STATISTICS</i>, <i>INPUT_SOURCE_GROUP_STATISTICS</i>, or an <i>INPUT_SOURCE_SELECT_RECORD</i> group. Required option. Do not duplicate.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)
[Report and Analysis \[page 631\]](#)
[Match Control \[page 632\]](#)
[Match Criteria Key Layout group \[page 634\]](#)
[Compare Table Group \[page 645\]](#)
[Compare Match Criteria Group \[page 648\]](#)
[Group Statistics Group \[page 666\]](#)
[Input sources \[page 670\]](#)
[Field algorithm numeric difference group \[page 676\]](#)
[Field algorithm numeric percent difference group \[page 679\]](#)
[Field algorithm geo proximity group \[page 681\]](#)
[Input Source Group Statistics Group \[page 682\]](#)
[Input Source Select Record Group \[page 684\]](#)

20.8 Group Statistics Group

Use the Group Statistics Group when you do not set input sources in the Input Sources Group in a configuration.

i Note

The following XML contains sample data.

```
<GROUP_STATISTICS_GROUP>
  <GROUP_STATISTICS>
    <NAME>IndGroupStats</NAME>
    <SOURCE_TYPE_DEFINITIONS>
```

```

<DEFINITION>
  <INCLUDE_IN_SOURCE_COUNT>YES</INCLUDE_IN_SOURCE_COUNT>
  <SOURCE_TYPE_ID>YES</SOURCE_TYPE_ID>
</DEFINITION>
<DEFINITION>
  <INCLUDE_IN_SOURCE_COUNT>NO</INCLUDE_IN_SOURCE_COUNT>
  <SOURCE_TYPE_ID>NO</SOURCE_TYPE_ID>
</DEFINITION>
</SOURCE_TYPE_DEFINITIONS>
</GROUP_STATISTICS>
</GROUP_STATISTICS_GROUP>

```

The *GROUP_STATISTICS_GROUP* defines a group statistics group for configurations that do not have input sources set up with the *INPUT_SOURCES_GROUP*. The *GROUP_STATISTICS_GROUP* contains the following subgroups:

- *GROUP_STATISTICS*: Defines one group for statistics. Optional group. May be duplicated.
- *SOURCE_TYPE_DEFINITIONS*: Defines the source types for this object. Required group if you include any of the following output fields, otherwise optional:
 - GROUP_TYPE
 - SOURCE_COUNT
 - SOURCE_ID_COUNT
 - SOURCE_ID
 - SOURCE_TYPE_ID

Do not duplicate.

- *DEFINITION*: Defines one source type. Required section. May be duplicated.
- *AUTO_GENERATE_SOURCES*: Specifies the information that the transform requires to generate sources automatically. If you omit this group from the group, the transform does not automatically generate sources. Optional group. Do not duplicate.
- *SOURCE_TYPE_ID_FIELD*: Specifies the input field that holds the source type ID. Optional group. Do not duplicate.
- *PRE_DEFINED_SOURCES*: Defines one or more predefined sources. Assign a source type to each predefined source. Complete this group when the only source information in the record is the source ID. Optional group. Do not duplicate.
 - *SOURCE*: Defines one source. Required group. Do not duplicate.

Optional group. Do not duplicate.

GROUP_STATISTICS_GROUP group option descriptions

Option	Description
<i>GROUP_STATISTICS</i>	
<i>NAME</i>	Specifies a name for the group statistics operation. If you include more than one GROUP_STATISTICS group in this Match transform, ensure that the name is unique. Enter a value from 1 through 15 characters. The transform references the group statistics name in the Post Match Processing object. Required option. Do not duplicate.

Option	Description
<i>SOURCE_TYPE_DEFINITIONS</i>	
<i>DEFINITION</i> group	
<i>SOURCE_TYPE_ID</i>	Specifies the source type ID: • <i>N</i>: Normal. Group of records considered to be good, eligible records. • <i>S</i>: Special. Like a normal source with one exception. The transform does not count the source when it determines whether a match group is single-source or multi-source. A special source can contribute records but it is not counted towards multi-buyer status. • <i>P</i>: Purge or suppress. Contains records that could be ineligible for use. For example, a mailer seed list. This option is required and cannot be repeated.
<i>INCLUDE_IN_SOURCE_COUNT</i>	Specifies whether the transform includes the source type in the source count. • <i>YES</i>: Includes the source type in the source count. • <i>NO</i>: Does not include the source type in the source count. ❖ Example The special source type and suppress source type are not included in the source (buyer) count. Required option. Do not duplicate.
<i>SOURCE_ID_FIELD</i>	
<i>MAPPED_NAME</i>	Specifies the mapped name of the source ID input field. Required option. Do not duplicate.
<i>AUTO_GENERATE_SOURCE</i>	
<i>SOURCE_TYPE_ID_FIELD</i> group	
<i>MAPPED_NAME</i>	Specifies the mapped name of the source type ID input field. Optional group. Do not duplicate.
<i>PRE_DEFINED_SOURCES</i>	
<i>SOURCE</i> group	
<i>SOURCE_ID</i>	Specifies the source ID. The transform compares the value you enter here against the value in the source ID input field. Required option. Do not duplicate.

Option	Description
SOURCE_TYPE_ID	Specifies the source type for this source. Ensure that the value you enter matches a value SOURCE_TYPE_ID option under ▶ SOURCE_TYPE_DEFINITIONS ▶ DEFINITION ▾. Required option. Do not duplicate.
DEFAULT_SOURCE_ID	Specifies a default source ID. The transform uses this value when the source ID input field is not defined or is blank. Additionally, the transform uses this value when the AUTO_GENERATE_SOURCES group is not completed and the source ID input field value does not match any of the predefined sources. Also ensure that this value matches a predefined source when the AUTO_GENERATE_SOURCES group is not completed. Required option when you use any of the following output fields: • GROUP_TYPE • SOURCE_COUNT • SOURCE_ID_COUNT • SOURCE_ID • SOURCE_TYPE_ID Otherwise, optional option. Do not duplicate.
DEFAULT_SOURCE_ID_TYPE	Specifies a default source type ID that the transform uses when the DEFAULT_SOURCE_ID does not match any ID defined so far (either predefined or auto). The transform also uses this value when the SOURCE_TYPE_ID option is blank or holds an invalid value. Ensure that the value for this option matches a predefined type. Required option when the DEFAULT_SOURCE_ID option is complete and when the DEFAULT_SOURCE_ID does not match a predefined ID. This option is also required if auto-generate IDs is enabled. Otherwise, optional option. Do not duplicate.

The Group statistics group can post to the following output fields:

- GROUP_COUNT
- GROUP_ORDER
- GROUP_RANK
- GROUP_TYPE
- SOURCE_COUNT
- SOURCE_ID_COUNT
- SOURCE_ID
- SOURCE_TYPE_ID

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[Input Source Select Record Group \[page 684\]](#)

20.9 Input sources

The `INPUT_SOURCES` group contains settings that define the input sources.

Syntax

```
<INPUT_SOURCES>
  <DEFAULT_SOURCE_NAME></DEFAULT_SOURCE_NAME>
  <SOURCES>
    <SOURCE>
      <NAME></NAME>
      <VALUE></VALUE>
      <TYPE></TYPE>
    </SOURCE>
  </SOURCES>
  <AUTO_SOURCES>
    <DEFAULT_TYPE></DEFAULT_TYPE>
  </AUTO_SOURCES>
  ...
</INPUT_SOURCES>
```

Before you define your input sources, map a field that contains the value that identifies the input source.

The `INPUT_SOURCES` group contains the following subgroups:

- **`SOURCES`**: Defines the sources for the match process.
 - **`SOURCE`**: Defines one predefined source. An input record tries each source group sequentially until it finds the source group to which it belongs. Define at least one **`SOURCE`** group.

Required group. Duplicate for up to 500 sources.

- **AUTO_SOURCES**: Specifies settings for when the transform automatically defines a source. When you include the **AUTO_SOURCES** group, the transform pulls source definitions from the input fields. When you do not include the **AUTO_SOURCES** group, the transform does not define the sources from input fields. The transform starts processing using the source or sources defined in the **SOURCE** group. The transform processes the record as follows:
 - Determines if the record belongs to a predefined source.
 - If the record belongs to a predefined source, the transform assigns the record to the predefined source.
 - If the record does not belong to a predefined source, the transform determines if the record belongs to an auto-defined source.
 - If the record belongs to an auto-defined source, the transform uses the auto-defined source.
 - If the Input source is not defined, the transform adds the definition to the list of defined sources.
 - If it has reached the maximum number of source definitions, the transform uses the default source.

Ensure that you define the **VALUE** input field before you define this group. Optional group. Do not duplicate.

Optional group. Do not duplicate.

INPUT_SOURCES group

Option	Description
INPUT_SOURCES	
DEFAULT_SOURCE_NAME	Specifies a source that the transform uses when an input record does not belong to any predefined sources and any one of the following occurs: • The sources are not automatically generated. • The VALUE input field is blank. • The maximum of 10,000 sources is reached. Enter a value of up to 255 characters. Ensure that the value you enter matches a predefined source. Required option. Do not duplicate.
SOURCES	
SOURCE group	
NAME	Specifies a symbolic source name. Enter a name from 1 through 255 characters. The transform uses this name in other locations of the XML and the name also appears in the source statistics. Required option. Do not duplicate.

Option	Description
<i>VALUE</i>	Specifies a source value. Enter a value from 1 through 255 characters. The transform compares the content of the <i>VALUE</i> option against the input record <i>VALUE</i> field. If the contents match, the transform assigns the source definition to the input record. The comparison is case-sensitive. The transform trims all leading and trailing spaces before attempting to match the values. Required option when this source is not the default source. Otherwise optional. Do not duplicate.
<i>TYPE</i>	Specifies a source type. <i>NORMAL</i>: Contains good or eligible records. <i>SUPPRESS</i>: Contains records that disqualify a record from use. ❖ Example - a <i>SUPPRESS</i> type removes records from your data based on specific criteria. Examples: - DMA Mail Preference File - American Correctional Association prisons/jails lists - No pandering or non-responder lists - Credit card or bad-check suppression lists <i>SPECIAL</i>: Like a <i>NORMAL</i> source with one exception. A <i>SPECIAL</i> source contributes records. However, the transform does not use a <i>SPECIAL</i> source when determining whether a match group is single- or multi-source. ❖ Example Some companies use a source of seed names. Seed names are names of people who report when they receive advertising mail, so that the mailer can measure mail delivery. Appearance on the seed source should not be counted toward multi-source status. Required option. Do not duplicate.
<i>AUTO_SOURCES</i>	
<i>DEFAULT_TYPE</i>	Specifies the default source type. <i>NORMAL</i> <i>SUPPRESS</i> <i>SPECIAL</i> The transform uses this setting when you do not define the <i>TYPE</i> input field, or when the <i>TYPE</i> input field does not have a value of N (Normal), P (Purge or Suppress), or S (Special). Required option. Do not duplicate.

[Input Sources: Source Groups \[page 673\]](#)

A Source Group consists of one or more input sources.

[Input Sources: Input Fields \[page 675\]](#)

To define the input fields for the specified input source, complete the Input Fields group.

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20.9.1 Input Sources: Source Groups

A Source Group consists of one or more input sources.

Syntax

```
<SOURCE_GROUPS>
  <UNDEFINED_ACTION/></UNDEFINED_ACTION>
  <DEFAULT_NAME></DEFAULT_NAME>
  <SOURCE_GROUP_GROUP>
    <SOURCE_GROUP>
      <NAME></NAME>
      <SOURCES>
        <SOURCE>
          <NAME></NAME>
        </SOURCE>
      </SOURCES>
    </SOURCE_GROUP>
  </SOURCE_GROUP_GROUP>
</SOURCE_GROUPS>
```

The *SOURCE_GROUPS* group is a part of the *INPUT_SOURCES* group. The settings enable you combine input sources logically.

❖ Example

You want to put all the rented input sources into one source group, and the in-house input sources into another source group. You do not have to put all of your input sources into a source group. Therefore, the record count for the source groups may be lower than the record count for the input sources. If you leave out the `SOURCE_GROUPS` group, the transform does not generate source group statistics and it does not output any data to the source group output fields.

Adding a `SOURCE_GROUPS` group provides additional statistics in certain Match reports.

The `SOURCE_GROUPS` group contains the following subgroups:

- `SOURCE_GROUP_GROUP`:
 - `SOURCE_GROUP`: Configures a source group.
 - `SOURCES`: Defines one source group. Required group when the `UNDEFINED_SOURCE_GROUP_ACTION` is set to `DEFAULT`. Otherwise this group is optional. May be duplicated.

Optional group. Do not duplicate.

`SOURCE_GROUPS` group option descriptions

Option	Description
<code>UNDEFINED_ACTION</code>	Specifies what the transform does when an input source does not appear in a source group. <code>IGNORE</code>: Ignores the input source. <code>DEFAULT</code>: Assigns the input source to the default source group defined in the <code>DEFAULT_SOURCE_GROUP</code> option. <code>AUTO</code>: • If the <code>SOURCE_GROUP</code> option is not defined, the transform creates a source group using the same name as the input source, and assigns the input source to it. If the source group already exists, the transform assigns the input source to the existing group. • If the <code>SOURCE_GROUP</code> option is defined, then the transform assigns the input source to the defined source group. If the source group does not exist, the transform creates it using the defined <code>SOURCE_GROUP</code> value. • If the <code>SOURCE_GROUP</code> option is blank, the transform ignores the input source.

Option	Description
<i>DEFAULT_NAME</i>	Specifies the default Source Group name. Enter a name from 1 to 255 characters. Ensure that the value you enter matches a <i>NAME</i> option in a <i>SOURCE_GROUP</i> group. Required when you set the <i>UNDEFINED_ACTION</i> option to <i>DEFAULT</i>. Do not duplicate.
<i>SOURCE_GROUP_GROUP</i>	
<i>SOURCE_GROUP</i> group	
<i>NAME</i>	Specifies a symbolic name for the Source Group. Enter a name from 1 to 255 characters. The transform uses this name in the source statistics. Required option. Do not duplicate.
<i>SOURCES</i>	Specifies one source that belongs to this Source Group. Required when the value in <i>DEFAULT_NAME</i> does not match the <i>NAME</i> value in the <i>SOURCE_GROUP</i> sub group of the <i>SOURCE_GROUPS</i> group. Can be duplicated.
<i>NAME</i>	Specifies an Input Source name that matches a name of a predefined source. The name can be from 1 through 255 characters.

Related Information

[Input sources \[page 670\]](#)

20.9.2 Input Sources: Input Fields

To define the input fields for the specified input source, complete the Input Fields group.

 Syntax

```
<INPUT_FIELDS>
  <FIELD>
    <MAPPED_NAME></MAPPED_NAME>
    <FIELD_TYPE></FIELD_TYPE>
  </FIELD>
</INPUT_FIELDS>
```

```
</INPUT_SOURCES>
```

Required group only when you define the AUTO_SOURCES group. Otherwise optional.

Do not duplicate.

Input Fields group option descriptions

Option	Description
<i>FIELD</i>	Group. Specifies an input field for the input source. Optional group. Duplicate for up to 3 fields.
<i>MAPPED_NAME</i>	Specifies the mapped name of the input field. Enter a name that is also listed as a mapped name in the <i>INPUT_FIELDS</i> group. Required option. Do not duplicate.
<i>FIELD_TYPE</i>	Specifies the type of data the input field holds. • <i>VALUE</i>: Select <i>VALUE</i> when any of the Input source definitions have the SOURCE / VALUE option enabled or if you use auto sources. All the other input field types are optional. • <i>TYPE</i>: Provides a source type when auto-generating input sources. • <i>SOURCE_GROUP_NAME</i>: Provides a Source Group name for a record that does not belong to a predefined Source Group. Optional field. Do not duplicate.

Related Information

[Input sources \[page 670\]](#)

20.10 Field algorithm numeric difference group

The Match transform uses the numeric difference algorithm to compare numeric information between records, and to form groups of records with values that are close enough to be considered duplicates.

≡, Syntax

```
<FIELD_ALGORITHM_NUMERIC_DIFFERENCE_GROUP>
```

```

<FIELD_ALGORITHM_NUMERIC_DIFFERENCE>
  <NAME></NAME>
  <MAX_DIFFERENCE></MAX_DIFFERENCE>
  <MAX_DIFFERENCE_SCORE></MAX_DIFFERENCE_SCORE>
</FIELD_ALGORITHM_NUMERIC_DIFFERENCE>
</FIELD_ALGORITHM_NUMERIC_DIFFERENCE_GROUP>

```

The [FIELD_ALGORITHM_NUMERIC_DIFFERENCE_GROUP](#) group contains one subgroup, [FIELD_ALGORITHM_NUMERIC_DIFFERENCE](#).

The [FIELD_ALGORITHM_NUMERIC_DIFFERENCE](#) enables you to specify one field algorithm numeric difference for the group. Optional group. May be duplicated.

FIELD_ALGORITHM_NUMERIC_DIFFERENCE_GROUP option descriptions

Options	Description
FIELD_ALGORITHM_NUMERIC_DIFFERENCE	
NAME	Specifies a logical name for the field algorithm. The COMPARE_MATCH_CRITERIA group in a FIELD_ALGORITHM option may reference the name you enter here. Required option. Do not duplicate.
MAX_DIFFERENCE	Specifies the maximum difference that the transform uses to consider records a match. Enter a numeric value from 0 through 100. - When the calculated difference is larger than the set MAX_DIFFERENCE, the transform assigns a MAX_DIFFERENCE_SCORE of 0. - When the calculated difference is equal to the set MAX_DIFFERENCE, the transform assigns a score equal to the setting in MAX_DIFFERENCE_SCORE. - When the calculated difference is less than the set MAX_DIFFERENCE, the transform assigns a proportional score between the MAX_DIFFERENCE_SCORE and 100. Required option. Do not duplicate.

Options	Description
<i>MAX_DIFFERENCE_SCORE</i>	Specifies the score the transform uses when the numeric difference is the same as the value in <i>MAX_DIFFERENCE</i>. Enter a value from 0 through 100. - When the calculated difference is larger than the set <i>MAX_DIFFERENCE</i>, the transform assigns a <i>MAX_DIFFERENCE_SCORE</i> of 0. - When the calculated difference is equal to the set <i>MAX_DIFFERENCE</i>, the transform assigns a score equal to the setting in <i>MAX_DIFFERENCE_SCORE</i>. - When the calculated difference is less than the set <i>MAX_DIFFERENCE</i>, the transform assigns a proportional score between the <i>MAX_DIFFERENCE_SCORE</i> and 100. Required option. Do not duplicate.

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20.11 Field algorithm numeric percent difference group

The transform uses the numeric percent difference to form groups of records with values that are close enough to be considered duplicates.

Syntax

```
<FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE_GROUP>
  <FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE>
    <NAME></NAME>
    <MAX_PERCENT_DIFFERENCE></MAX_PERCENT_DIFFERENCE>
    <MAX_PERCENT_DIFFERENCE_SCORE></MAX_PERCENT_DIFFERENCE_SCORE>
  </FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE>
</FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE_GROUP>
```

The *FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE_GROUP* has one subgroup: *FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE*. Use the subgroup to define one numeric percent difference for the group. Required group. May be duplicated.

Optional group. Do not duplicate.

FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE_GROUP option descriptions

Options	Description
<i>FIELD_ALGORITHM_NUMERIC_PERCENT_DIFFERENCE</i>	
<i>NAME</i>	Specifies a logical name for the field algorithm. The <i>COMPARE_MATCH_CRITERIA</i> group in a <i>FIELD_ALGORITHM</i> option may reference the name you enter here. Required option. Do not duplicate.
<i>MAX_PERCENT_DIFFERENCE</i>	Specifies the maximum percent difference allowed. The value is a percent of the absolute driver value. Enter a value from 0 through 100. Required option. Do not duplicate.

Options	Description
<i>MAX_PERCENT_DIFFERENCE_SCORE</i>	Specifies the score to generate when the difference is the same as <i>MAX_PERCENT_DIFFERENCE</i>. Enter a value from 0 through 100. - When the calculated difference is larger than the set <i>MAX_PERCENT_DIFFERENCE</i>, the transform assigns a <i>MAX_PERCENT_DIFFERENCE_SCORE</i> of 0. - When the calculated difference is equal to the set <i>MAX_PERCENT_DIFFERENCE</i>, the transform assigns a score equal to the setting in <i>MAX_PERCENT_DIFFERENCE_SCORE</i>. - When the calculated difference is less than the set <i>MAX_PERCENT_DIFFERENCE</i>, the transform assigns a proportional score between the <i>MAX_PERCENT_DIFFERENCE_SCORE</i> and 100. Required option. Do not duplicate.

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20.12 Field algorithm geo proximity group

The Match transform uses the geo proximity group algorithm to compare distance information between records to form groups of records with values that are close enough to be considered duplicates.

Syntax

```
<FIELD_ALGORITHM_GEO_PROXIMITY_GROUP>
  <FIELD_ALGORITHM_GEO_PROXIMITY>
    <DISTANCE_UNIT></DISTANCE_UNIT>
    <MAX_DISTANCE></MAX_DISTANCE>
    <MAX_DISTANCE_SCORE></MAX_DISTANCE_SCORE>
  </FIELD_ALGORITHM_GEO_PROXIMITY>
</FIELD_ALGORITHM_GEO_PROXIMITY_GROUP>
```

The *FIELD_ALGORITHM_GEO_PROXIMITY_GROUP* contains one subgroup: *FIELD_ALGORITHM_GEO_PROXIMITY*. Use the subgroup to specify one geo proximity for the group. Required group. May be duplicated.

Optional group. Do not duplicate.

FIELD_ALGORITHM_GEO_PROXIMITY_GROUP option descriptions

Options	Description
<i>FIELD_ALGORITHM_GEO_PROXIMITY</i>	
<i>NAME</i>	Specifies a logical name for the geo proximity algorithm. Any <i>FIELD_ALGORITHM</i> attribute of a Compare Match Criteria Object may reference it. Required option. Do not duplicate.
<i>DISTANCE_UNIT</i>	Specifies the distance unit for the <i>MAX_DISTANCE</i> tolerance calculation. <i>FEET</i> <i>MILES</i> <i>METERS</i> <i>KILOMETERS</i> Required option. Do not duplicate.
<i>MAX_DISTANCE</i>	Specifies the maximum distance allowed. Enter a value from 0 through the maximum double value. - A distance larger than the <i>MAX_DISTANCE</i> setting receives a score of 0. - A distance equal to <i>MAX_DISTANCE</i> receives the score set for <i>MAX_DISTANCE_SCORE</i>. - A distance less than <i>MAX_DISTANCE</i> receives a proportional score between <i>MAX_DISTANCE_SCORE</i> and 100. Required option. Do not duplicate.

Options	Description
<code>MAX_DISTANCE_SCORE</code>	Specifies the score when the distance is the same as <code>MAX_DISTANCE</code>. Enter a value from 0 through 100. Required option. Do not duplicate.

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20.13 Input Source Group Statistics Group

Use the Input Source Group Statistics Group to generate statistics posted with output fields.

⌘ Syntax

```
<INPUT_SOURCE_GROUP_STATISTICS_GROUP>
  <INPUT_SOURCE_GROUP_STATISTICS>
    <NAME></NAME>
  </INPUT_SOURCE_GROUP_STATISTICS>
</INPUT_SOURCE_GROUP_STATISTICS_GROUP>
```

The functionality of the `INPUT_SOURCE_GROUP_STATISTICS_GROUP` object is similar to the `GROUP_STATISTICS` object. The main difference is the following:

- `INPUT_SOURCE_GROUP_STATISTICS_GROUP` object generates statistics for the input sources that you define.

- [GROUP_STATISTICS](#) object is a stand-alone group and independent of defined input sources.

If you use the [INPUT_SOURCE_GROUP_STATISTICS_GROUP](#) object, then ensure that you also define the [INPUT_SOURCE](#) object.

The [INPUT_SOURCE_GROUP_STATISTICS_GROUP](#) has one subgroup: [INPUT_SOURCE_GROUP_STATISTICS](#). This subgroup is a required group and it may be duplicated.

Optional group. Do not duplicate.

INPUT_SOURCE_GROUP_STATISTICS_GROUP option descriptions

Option	Description
INPUT_SOURCE_GROUP_STATISTICS	
NAME	Specifies a logical name for the input source group object. Enter a name from 1 through 15. The name may be referenced in the POST_MATCH_PROCESSING group. Required option. Do not duplicate.

The [INPUT_SOURCE_GROUP_STATISTICS](#) group can post the following output fields:

- GROUP_COUNT
- GROUP_SOURCE_TYPE
- GROUP_ORDER
- GROUP_RANK
- GROUP_SOURCE_APPEARANCE
- GROUP_SOURCE_ORDER
- GROUP_SOURCE_GROUP_APPEARANCE
- GROUP_SOURCE_GROUP_ORDER
- MULTI_SOURCE_COUNT
- SOURCE_COUNT
- SOURCE_GROUP_COUNT

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20.14 Input Source Select Record Group

To select types of records to flag on output based on each of the input sources, complete options in the Input Source Select Record Group.

Syntax

```

<INPUT_SOURCE_SELECT_RECORD_GROUP>
  <INPUT_SOURCE_SELECT_RECORD>
    <NAME></NAME>
    <SELECT_UNIQUE_RECORDS></SELECT_UNIQUE_RECORDS>
    <SELECT_SINGLE_SOURCE_MASTERS></SELECT_SINGLE_SOURCE_MASTERS>
    <SELECT_SINGLE_SOURCE_SUBORDINATES></
SELECT_SINGLE_SOURCE_SUBORDINATES>
    <SELECT_MULTIPLE_SOURCE_MASTERS></SELECT_MULTIPLE_SOURCE_MASTERS>
    <SELECT_MULTIPLE_SOURCE_SUBORDINATES></
SELECT_MULTIPLE_SOURCE_SUBORDINATES>
    <SELECT_SUPPRESS_SOURCE_MATCHES></SELECT_SUPPRESS_SOURCE_MATCHES>
    <SELECT_SUPPRESS_SOURCE_UNIQUES></SELECT_SUPPRESS_SOURCE_UNIQUES>
    <SELECT_SUPPRESS_SOURCE_MASTERS></SELECT_SUPPRESS_SOURCE_MASTERS>
    <SELECT_SUPPRESS_SOURCE_SUBORDINATES></
SELECT_SUPPRESS_SOURCE_SUBORDINATES>
  </INPUT_SOURCE_SELECT_RECORD>
</INPUT_SOURCE_SELECT_RECORD_GROUP>

```

The *INPUT_SOURCE_SELECT_RECORD_GROUP* group contains one subgroup: *INPUT_SOURCE_SELECT_RECORD*. This subgroup is required and it may be duplicated.

The *INPUT_SOURCE_SELECT_RECORD_GROUP* group generates the *SELECT_RECORD* output field.

Note

You may want to flag these records so that they are available for writing to output.

Optional group. Do not duplicate.

INPUT_SOURCE_SELECT_RECORD_GROUP option descriptions

Record type	Description
	<i>INPUT_SOURCE_SELECT_RECORD</i>

Record type	Description
<i>NAME</i>	Specifies a unique name for this operation that enables you to identify it in a report and in your Select_Record output field. The transform references the name in the <i>POST_MATCH_PROCESSING_GROUP</i>. ❖ Example You have two input source select record operations for this match level: DMA_MATCHES and MAIL_LIST. The transform generates the following output fields: • <operation name>_DMA_Matches_Select_Record • <operation name>_Mail_List_Select_Record
<i>SELECT_UNIQUE_RECORDS</i>	Specifies to include records that are not members of any match group. Unique records can be from sources with a normal or special-type source. • <i>YES</i>: Includes unique records. • <i>NO</i>: Does not include unique records.
<i>SELECT_SINGLE_SOURCE_MASTERS</i>	Specifies to include the highest ranking records of a match group whose members are from the same source. Single source masters can be from normal or special-type sources. • <i>YES</i>: Includes single source masters. • <i>NO</i>: Does not include single source masters.
<i>SELECT_SINGLE_SOURCE_SUBORDINATES</i>	Specifies to include subordinate records from the same match group. Single source subordinates can be from a normal or suppress-type source. • <i>YES</i>: Includes single source subordinates. • <i>NO</i>: Does not include single source subordinates.
<i>SELECT_MULTIPLE_SOURCE_MASTERS</i>	Specifies to include the highest ranking records from a match group whose members came from two or more sources. Multiple source masters can be from normal or special-type sources. • <i>YES</i>: Includes multiple source masters. • <i>NO</i>: Does not include multiple source masters.
<i>SELECT_MULTIPLE_SOURCE_SUBORDINATES</i>	Specifies to include subordinate records of a match group whose members came from two or more sources. Multiple source subordinates can be from normal or suppress-type sources. • <i>YES</i>: Includes multiple source subordinates. • <i>NO</i>: Does not include multiple source subordinates.

Record type	Description
<i>SELECT_SUPPRESS_SOURCE_MATCHES</i>	Specifies to include subordinate members of match groups that include a higher-priority record that came from a suppress -type source. Suppress source matches can be from normal or special-type source. • <i>YES</i>: Includes suppress source matches. • <i>NO</i>: Does not include suppress source matches.
<i>SELECT_SUPPRESS_SOURCE_UNIQUES</i>	Specifies to include records that came from a suppress-type source, and for which no matching records are found. • <i>YES</i>: Includes suppress source uniques. • <i>NO</i>: Does not include suppress source uniques.
<i>SELECT_SUPPRESS_SOURCE_MASTERS</i>	Specifies to include records that came from a suppress-type source and which are the highest ranking members of their match group. • <i>YES</i>: Includes suppress source masters. • <i>NO</i>: Does not include suppress source masters.
<i>SELECT_SUPPRESS_SOURCE_SUBORDINATES</i>	Specifies to include records that came from a suppress-type source and are subordinate member of their match group. • <i>YES</i>: Includes suppress source subordinates. • <i>NO</i>: Does not include suppress source subordinates.

Parent topic: [Match transform reference \[page 628\]](#)

Related Information

[System group \[page 629\]](#)
[Report and Analysis \[page 631\]](#)
[Match Control \[page 632\]](#)
[Match Criteria Key Layout group \[page 634\]](#)
[Compare Table Group \[page 645\]](#)
[Compare Match Criteria Group \[page 648\]](#)
[Post Match Processing Group \[page 665\]](#)
[Group Statistics Group \[page 666\]](#)
[Input sources \[page 670\]](#)
[Field algorithm numeric difference group \[page 676\]](#)
[Field algorithm numeric percent difference group \[page 679\]](#)
[Field algorithm geo proximity group \[page 681\]](#)
[Input Source Group Statistics Group \[page 682\]](#)

21 About Data Quality fields

Many Data Quality Management SDK transforms require mapped input and generated output fields.

Map Data Quality fields in the *INPUT_FIELDS* and *OUTPUT_FIELDS* groups.

For detailed information about the input and output fields applicable for each transform, see the individual transform topics.

[Data Quality support for native data types \[page 687\]](#)

The Data Quality transforms in Data Quality Management SDK process most incoming data types as character data, with some exceptions.

[Output field definition options \[page 688\]](#)

In the Output Fields group, configure the properties for each output field to include in the generated output.

[Data type support \[page 691\]](#)

Each programming language for the SAP Data Quality Management SDK supports specific data types.

[Data Cleanse fields \[page 692\]](#)

Map fields from upstream objects and transforms to Data Cleanse fields so that the transform knows the type of data in each field and how to process the data.

[Geocoder fields \[page 714\]](#)

Map fields from upstream objects and transforms to Geocoder fields so that the transform knows the type of data in each field and how to process the data.

[Global Address Cleanse fields \[page 730\]](#)

Map fields from upstream objects and transforms to the Global Address Cleanse fields so that the transform knows the type of data in each field and how to process the data.

[USA Regulatory Address Cleanse fields \[page 793\]](#)

Map fields from upstream objects and transforms to USA Regulatory Address Cleanse fields so that the transform knows the type of data in each field and how to process the data.

[Match fields \[page 821\]](#)

There is no single list of Match input fields, rather, input field descriptions appear in the various matching strategy topics in the Match Transform topics. Output fields vary based on the match setup.

21.1 Data Quality support for native data types

The Data Quality transforms in Data Quality Management SDK process most incoming data types as character data, with some exceptions.

When you map your fields, you set a data type. If you map a non-character data type on input, the transform converts the data to a character string before passing it through the Data Quality transform for processing.

The software recognizes some data types and processes them using the same data type as the input data type.

❖ Example

The software recognizes date type data. Therefore, when your input field is date type, map the field to the Data Quality date data type.

When you map your input fields using specific Data Quality data type fields, the software recognizes and sorts the incoming data using the specific data type. Mapping applicable input fields to Data Quality data type fields increases efficiency because there are fewer fields the transform has to convert to character data.

For more information about data types, see the *Reference Guide*.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match fields \[page 821\]](#)

21.2 Output field definition options

In the Output Fields group, configure the properties for each output field to include in the generated output.

≡ Syntax

```
<OUTPUT_FIELDS>
  <FIELD>
    <PARENT_COMPONENT></PARENT_COMPONENT>
    <GENERATED_FIELD_NAME></GENERATED_FIELD_NAME>
    <GENERATED_FIELD_CLASS>STANDARDIZED</GENERATED_FIELD_CLASS>
    <DATA_RECORD_FIELD_NAME></DATA_RECORD_FIELD_NAME>
    <DATA_TYPE>CHARACTER_UCS2</DATA_TYPE>
    <DATA_LENGTH></DATA_LENGTH>
    <DATA_SCALE>0</DATA_SCALE>
  </FIELD>
</OUTPUT_FIELDS>
```

Option	Description
PARENT_COMPONENT	Specifies the output field name. For example, PERSON1 , EMAIL1 , FIRM1 , or NORTH_AMERICAN_PHONE1 .

Option	Description
<i>GENERATED_FIELD_NAME</i>	Specifies the field name (discrete component) that represents the generated data the transform outputs for the specified parent component. For example, to output the match standard the transform used for <i>PERSON1</i> , assign the <i>PRENAME_MATCH_STD1</i> generated field name.
<i>GENERATED_FIELD_CLASS</i>	Specifies the form of discrete component to output. For example, with a <i>GENERATED_FIELD_NAME</i> of <i>DATE</i> and a <i>GENERATED_FIELD_CLASS</i> of <i>STANDARDIZED</i>, the transform outputs the standardized form of the date. Options vary based on the transform. i Note Ensure that you use a generated field class that corresponds to the generated field name. For example, a generated field of <i>STATUS_CODE</i> can have a generated field class of <i>ASSIGNMENT_INFO</i> but not <i>STANDARDIZED</i>. The <i>ASSIGNMENT_INFO</i> field class for <i>STATUS_CODE</i> contains the status code for the record.
<i>DATA_RECORD_FIELD_NAME</i>	Specifies the name of the field from the data source.
<i>DATA_TYPE</i>	Identifies the type of data in the field. Valid values include: • <i>CHARACTER_UCS2</i>: <i>CHARACTER_UCS2</i> is the default setting. • <i>INTEGER32</i> • <i>DOUBLE</i> • <i>DECIMAL</i> • <i>DATE</i> • <i>TIME</i> • <i>DATE_TIME</i>

Option	Description
DATA_LENGTH	Specifies the length of the output field. The length setting provides insight into how much data the transform can expect from a field. Keep in mind the following information about setting data lengths: • CHARACTER_UCS2: For version backward compatibility, the data length defaults to unlimited when you do not set a length. • INTEGER32, DOUBLE, and DATE: The transform ignores a length setting for these fixed-length fields. • DECIMAL: Required. The length you enter represents the precision or number of digits before the decimal point. • TIME and DATE_TIME: You may leave the length blank. The transform calculates the length by the value in DATA_SCALE. • The calculated TIME length is the sum of the DATA_SCALE plus 6 (HHMMSS). • The DATE_TIME length is the sum of the DATA_SCALE plus 14 (YYYYMMDDHHMMSS).
DATA_SCALE	Specifies the number of digits to the right of a decimal point for DECIMAL data types. Required option for DECIMAL, TIME, and DATE_TIME field types: • For DECIMAL data types, the value specifies the number of digits to the right of the decimal. • For TIME and DATE_TIME data types, the transform uses the DATA_SCALE value to calculate the DATA_LENGTH.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

21.3 Data type support

Each programming language for the SAP Data Quality Management SDK supports specific data types.

Character data (UCS-2)

Language	Data type
C++	uint_16 char
.NET	.NET String
Java	Java String

Integer data (32-bit)

Language	Data type
C++	int32_t
.NET	.NET System.Int32 (32-bit signed integer)
Java	Java int data type (signed 32 bit two's complement)

Double data

Language	Data type
C++	Double
.NET	.NET System.Double data type. (double precision 64-bit floating point number).
Java	Java double data type. (64-bit IEEE 754 floating point)

Decimal

Language	Data type
C++	Native C++ double
.NET	.NET System.Decimal
Java	Java java.math.BigDecimal

Date

Language	Data type
C++	Date class
.NET	.NET System.DateTime
Java	Java java.util.Date

Time

Language	Data type
C++	Time class
.NET	.NET System.DateTime
Java	Java java.util.Date

Date-Time

Language	Data type
C++	DateTime class
.NET	.NET System. DateTime
Java	Java java.util.Calendar

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match fields \[page 821\]](#)

21.4 Data Cleanse fields

Map fields from upstream objects and transforms to Data Cleanse fields so that the transform knows the type of data in each field and how to process the data.

The Data Cleanse transform requires that you map fields on input and output. The process of mapping occurs when you configure the input and output fields in the Input Fields group and Output Fields group.

[Data Cleanse input fields \[page 693\]](#)

Map fields from upstream objects to Data Cleanse input fields so the transform knows the type of data to expect on input.

[Data Cleanse output fields \[page 698\]](#)

Map Data Cleanse output fields to output Standardized and corrected data, generated data, and additional data.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match fields \[page 821\]](#)

21.4.1 Data Cleanse input fields

Map fields from upstream objects to Data Cleanse input fields so the transform knows the type of data to expect on input.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The following table contains an alphabetical list of the recognized input fields to use in the input mapping for the Data Cleanse transform.

Data Cleanse Input Fields

Name	Description
DATA_SOURCE_ID	Specifies the name of the data source that the record comes from. → Tip If you choose to generate data quality statistics tables, use this field as part of your mapping strategy, along with primary key fields and the output field ROW_ID. This mapping identifies the source of individual records in any of the non-summary data quality statistics tables, thus enabling you to trace problem data to the data source. This input field is applicable for the following transforms: • Data Cleanse • Global Address Cleanse • Geocoder

Name	Description
<i>DATE1-6</i>	Contains date data. For example, 08/16/2004. Use up to six date fields for mapping multiple dates.
<i>EMAIL1-6</i>	Contains e-mail address data. Use up to six e-mail fields for mapping multiple e-mail addresses.
<i>FIRM_LINE1-6</i>	Contains firm name, firm location, or both. Use up to six firm lines for mapping multiple firm names.
<i>FIRM_LOCATION1-2</i>	Contains location within a company or organization, such as a department, mail stop, room, or building. Use up to two firm location fields.
<i>FIRM_NAME1-2</i>	Contains the name of a company or organization. Use up to two firm name fields.
<i>MULTILINE1-12</i>	Contains multiline data. The transform parses data from this input field in the order that you set in the PARSER_SEQUENCE_MULTILINE option, including parsers from custom cleansing packages. Use up to 12 multiline fields.
<i>NAME_LINE1-6</i>	Contains a whole name such as first and last name, or multiple whole names. Can include job title. Use up to six name line fields.
<i>NAME_OR_FIRM_LINE1-6</i>	Contains the name of a person or organization. Use up to six name or firm line fields.

Name	Description
OPTION_COUNTRY (Dynamic input field)	Contains an ISO2 country code to help parse phone data and to determine the content domain sequence. Obtain ISO2 country code data either from an input file that contains the codes, or from an upstream transform such as Global Address Cleanse that is set up to output ISO2 codes. For phone parsing: Data Cleanse uses the ISO2 country code to help determine the country code for parsing phone data when: • The transform populates the OPTION_COUNTRY dynamic input field with an ISO2 country code. • The country is included in the cleansing package. Map the ISO_COUNTRY_CODE_2CHAR field from the Global Address Cleanse transform to the OPTION_COUNTRY input field. For setting domain sequence: Data Cleanse uses the ISO2 code to determine content domain sequence when the OPTION_COUNTRY dynamic input field is populated with an ISO2 code. The transform automatically generates the output format based on the ISO2 country code and values in the OPTION_LANGUAGE and OPTION_REGION fields.

Name	Description
<i>OPTION_LANGUAGE</i>	Contains the language and region.
<i>OPTION_REGION</i> (Dynamic input fields)	The transform uses the values in <i>OPTION_LANGUAGE</i> and <i>OPTION_REGION</i>, along with the value in the <i>OPTION_COUNTRY</i> field, to determine the content domain and output format. There are a few countries where <i>OPTION_LANGUAGE</i> and <i>OPTION_REGION</i> data is helpful to determine the most appropriate content domain and output format. The two fields are applicable only to certain countries. For example, use <i>OPTION_LANGUAGE</i> and <i>OPTION_REGION</i> for Switzerland, Belgium, or Canada. The transform uses the <i>OPTION_LANGUAGE</i> and <i>OPTION_REGION</i> fields only to determine the most appropriate content domain and output format. Map the <i>OPTION_COUNTRY</i>, <i>OPTION_LANGUAGE</i>, and <i>OPTION_REGION</i> input fields from the following Global Address Cleanse output fields in this order: • <i>ISO_COUNTRY_CODE_2CHAR</i> • <i>LANGUAGE</i> • <i>REGION1</i> These input fields work together to determine the <i>OPTION_CONTENT_DOMAIN_SEQUENCE</i> and-or <i>OPTION_OUTPUT_FORMAT</i>.
<i>OPTION_CONTENT_DOMAIN_SEQUENCE</i> (Dynamic input field)	Contains the content domain sequence. The valid predefined values are: AR, CS, DA, DE, EN_AU, EN_GB, EN_IN, EN_US, ES_MX, ES_ES, FR, HU, ID, IT, JA, MS, NL, NO, PL, PT_BR, PT_PT, RO, RU, SK, SV, ZH, and GLOBAL.
<i>OPTION_OUTPUT_FORMAT</i> (Dynamic input field)	Contains the format for output specified as an abbreviation of the domain. The valid predefined values are: AR, CS, DA, DE, EN_AU, EN_GB, EN_IN, EN_US, ES_MX, ES_ES, FR, HU, ID, IT, JA, MS, NL, NO, PL, PT_BR, PT_PT, RO, RU, SK, SV, and ZH.
<i>PERSON1_FAMILY_NAME1</i>	Contains a discrete family name or last name such as Smith.
<i>PERSON2_FAMILY_NAME1</i>	Map up to two family names.

Name	Description
<i>PERSON1_FAMILY_NAME2</i>	Contains a second discrete family name.
<i>PERSON2_FAMILY_NAME2</i>	Can be useful for cultures where people are known by both paternal and maternal family names. If your input data contains two family names, map the first to <i>PERSON1_FAMILY_NAME2</i> and the second to <i>PERSON2_FAMILY_NAME2</i> .
<i>PERSON1_GIVEN_NAME1-2</i>	Contains discrete given names, for example, "John" or "B". Map up to two given names for person 1, and map up to two given names for person 2.
<i>PERSON2_GIVEN_NAME1-2</i>	
<i>PERSON1_HONORARY_POSTNAME</i>	Contains an honorary postname indicating certification, academic degree, or affiliation, such as CPA. Map an honorary postname for up to two people.
<i>PERSON2_HONORARY_POSTNAME</i>	
<i>PERSON1_MATURITY_POSTNAME</i>	Contains a maturity postname indicating heritage, such as Jr., Sr., III. Map maturity postnames for up to two people.
<i>PERSON2_MATURITY_POSTNAME</i>	
<i>PERSON1_PRENAME</i>	Contains a discrete prename, such as Mr., Mrs., Dr., or Lt. Col. Map a discrete prename for up to two people.
<i>PERSON2_PRENAME</i>	
<i>PERSON1_TITLE</i>	Contains a discrete job title, such as Accountant or Software Engineer. Map a discrete job title for up to two people.
<i>PERSON2_TITLE</i>	
<i>PHONE1-6</i>	Contains a phone number. Map up to six phone numbers.
<i>SSN1-6</i>	Contains a U.S. Social Security Number. Map up to six Social Security Numbers.
<i>TITLE_LINE1-6</i>	Contains a job title, such as Accountant, or Assistant. Map up to six title lines.
<i>UDPM1-4</i>	Contains information associated with patterns and rules that you define in the user-defined type of reference Data in the Cleansing Package Builder utility. Map up to four user-defined fields.

Parent topic: [Data Cleanse fields \[page 692\]](#)

Related Information

[Data Cleanse output fields \[page 698\]](#)

[Domains \[page 500\]](#)

21.4.2 Data Cleanse output fields

Map Data Cleanse output fields to output Standardized and corrected data, generated data, and additional data.

The following table contains an alphabetical list of recognized output fields that you use in output mapping for the Data Cleanse transform. By default, the [EXTRA](#), [INFO_CODE](#), and [STATUS_CODE](#) fields are always available. Map other fields based on the mapped input fields and the selected parser sequence multiline options. Additionally, the [IGNORE](#) field may be available.

Data Cleanse output fields

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
DATE	Contains a date.	Parsed Standardized	Date
DATE_DAY	Contains the day that the transform parses from the date.	Parsed Standardized	Date
DATE_MONTH	Contains the month that the transform parses from the date.	Parsed Standardized	Date
DATE_YEAR	Contains the year that the transform parses from the date.	Parsed Standardized	Date
DUAL_NAME	Contains name components from an input string that contains two names separated by a connecting word such as "and" or "or."	Parsed Standardized	None
❖ Example Input: Terry and Kris Johnson Output: Terry Johnson and Kris Johnson ❖ Example Input: Terry Johnson or Kris Adams Output: Terry Johnson or Kris Adams			
EMAIL	Contains a full e-mail address.	Parsed Standardized	Email

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>EMAIL_DOMAIN_ALL</i>	Contains the domain portion of an e-mail address. The domain portion includes all information after the @ symbol. For example, in "joex@sap.com", "sap.com" is the domain.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_FIFTH</i>	Contains the fifth to last domain from an e-mail address with multiple domains listed.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_FOURTH</i>	Contains the fourth to last domain from an e-mail address with multiple domains listed.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_HOST</i>	Contains the host portion of an e-mail address. The host portion is the first string after the @ symbol. For example, in "joex@sap.com", "sap" is the host.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_SECOND</i>	Contains the second to last domain in an e-mail address with multiple domains listed.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_THIRD</i>	Contains the third to last domain in an e-mail address with multiple domains listed.	Parsed Standardized	Email
<i>EMAIL_DOMAIN_TOP</i>	Contains the last domain in an e-mail address with multiple domains listed. For example, in jsmith@one.two.three.com, ".com" is the last domain.	Parsed Standardized	Email
<i>EMAIL_IS_ISP</i>	Indicates whether the e-mail address is a known ISP (internet service provider) or e-mail domain name listed in the e-mail data type of Reference Data in Cleansing Package Builder.	Standardized	Email

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>EMAIL_USER</i>	Contains the user name of the e-mail address. For example, in "joex@sap.com", "joex" is returned.	Parsed Standardized	Email
<i>EXTRA</i>	Contains data that the transform does not parse using any of the active parsers, or that the transform does not recognize as applicable for one of the other output fields.	Parsed	None
<i>FAMILY_NAME1</i>	Contains the family name portion of a name. For example, Smith.	Parsed Standardized	Family_Name
<i>FAMILY_NAME1_MATCH_STD1-6</i>	Contains the match standard for family names. Use this field only with cleansing packages that include name data in more than one script. The match standards include the name as it is written in alternate script types. For example, for a family name included in the Japanese dictionary in Kanji script, the match standards include Kana renditions of the name. If the dictionary does not have an alias entry, the field is empty. Compare with <i>MATCH_FAMILY_NAME</i>.	Standardized	Family_Name1_Match_Std
<i>FAMILY_NAME2</i>	Contains the second family name. Use to output the paternal and maternal family names to separate fields.	Parsed Standardized	Family_Name 2

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
FAMILY_NAME2_MATCH_STD1-6	Contains the match standard for second family names. Use this field only with cleansing packages that include name data in more than one script. The match standards include the name as it is written in alternate script types. For example, for a family name included in the Japanese dictionary in Kanji script, the match standards include Kana renditions of the name. If the dictionary does not have an alias entry, the field is empty. Compare with MATCH_FAMILY_NAME.	Standardized	Family_Name2_Match_Std
FAMILY_NAME_FULL	Contains a combination of the FAMILY_NAME1 and FAMILY_NAME2 fields. ❖ Example FAMILY_NAME1 = Smith FAMILY_NAME2 = Jones FAMILY_NAME_FULL = Smith Jones	Parsed Standardized	Family_Name_Full
FIRM	Contains the firm, company, or organization name.	Parsed Standardized	Firm
FIRM_LOCATION	Contains a location within a company or organization, such as a department. For example, Mailstop.	Parsed Standardized	Firm_Location

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>FIRM_MATCH_STD1-6</i>	Contains the match standard for the firm, company, or organization name. For example, HP is the match standard or alias for Hewlett Packard. If the cleansing package does not have an alias entry, the output field is empty. Compare with <i>MATCH_FIRM</i>.	Standardized	Firm_Match_Std
<i>FIRM_LOCATION_MATCH_STD1-6</i>	Contains the match standard for firm locations. For example, MS is the match standard or alias for mailstop. If the dictionary does not have an alias entry, the transform outputs an empty field.	Standardized	Firm_Location_Match_Std

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
GENDER	Contains the gender description. The gender description can be one of the following values: • AMBIGUOUS: The name does not reliably indicate a gender. The name could be either male or female. For example, Pat. • MALE_STRONG: High confidence that the person is male. That is, the name belongs to someone who is almost certainly a male. For example, John. • MALE_WEAK: Some confidence that the person is male. That is, the name belongs to someone who is probably male. For example, Terry. • FEMALE_STRONG: High confidence that the person is female. That is, the name belongs to someone who is almost certainly a female. For example, Mary. • FEMALE_WEAK: Some confidence that the person is female. That is, the name belongs to someone who is probably a female. For example, Lynn. For dual names, the following output is also available: • MULTI_NAMES_AMBIGUOUS: At least one of the names does not reliably	Standardized	None

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
	indicate a gender. For example, Pat and John. • MULTI_NAMES_FEMALE: Some or high confidence that both of the names belong to people who are female. For example, Mary and Lynn. • MULTI_NAMES_MALE: Some or high confidence that both of the names belong to people who are male. For example, John and Terry. • MULTI_NAMES_MIXED: Some or high confidence that one of the names belongs to a person who is female, and the other name belongs to a person who is male. For example, Lynn and John.		
GENDER_ID	A numeric value that corresponds to the gender description: • 0: Unassigned • 1: Male_Strong • 2: Male_Weak • 3: Ambiguous • 4: Female_Weak • 5: Female_Strong • 6: Multi_Names_Mixed • 7: Multi_Names_Male • 8: Multi_Names_Female • 9: Multi_Names_Ambiguous	Standardized	None
GIVEN_NAME1	Contains the given name, for example, Robert.	Parsed Standardized	Given_Name

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>GIVEN_NAME1_MATCH_STD</i> 1-6	Contains the match standard for given names. For example, the transform can tell you that Patrick and Patricia are potential matches for the given name Pat. Match standards help you overcome two types of matching problems: • Alternate spellings such as Catherine and Katherine • Nicknames such as Pat and Patty. Compare with <i>MATCH_GIVEN_NAME1</i>.	Standardized	Given_Name1_Match_Std
<i>GIVEN_NAME2</i>	Contains the second given name.	Parsed Standardized	Given_Name2
<i>GIVEN_NAME2_MATCH_STD</i> 1-6	Contains the match standard for second given names. For example, the transform can tell you that Patrick and Patricia are potential matches for the given name Pat. Compare with <i>MATCH_GIVEN_NAME2</i>.	Standardized	Given_Name2_Match_Std
<i>GIVEN_NAME_FULL</i>	Contains the combined values of <i>GIVEN_NAME1</i> and <i>GIVEN_NAME2</i>. ♣ Example <i>GIVEN_NAME1</i> = Jonathan <i>GIVEN_NAME2</i> = Peter <i>GIVEN_NAME_FULL</i> = Jonathan Peter	Parsed Standardized	Given_Name_Full

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>HONORARY_POSTNAME</i>	Contains the honorary postname. Honorary postname indicates certification, academic degree, or affiliation. For example, CPA.	Parsed Standardized	Postname
<i>HONORARY_POSTNAME_MATCH_STD1-6</i>	Contains the match standard for honorary postname. For example, M.B.A. is the match standard or alias for MBA. If the dictionary does not have an alias entry, the field is empty.	Standardized	Postname_Match_Std
<i>INFO_CODE</i>	Contains the code that identifies the rows that may require manual review because the data is suspect.	Assignment_Info	None
<i>INTERNATIONAL_PHONE</i>	Contains the entire international phone number, including extra items such as the country code.	Parsed Standardized	Phone
<i>INTERNATIONAL_PHONE_COUNTRY_CODE</i>	Contains the country code of an international phone number.	Parsed Standardized	Phone
<i>INTERNATIONAL_PHONE_COUNTRY_NAME</i>	Contains the name of the country of origin of an international phone number.	Parsed Standardized	Phone
<i>INTERNATIONAL_PHONE_LINE</i>	Contains the portion of the international phone number that is not the country code or the city code.	Parsed Standardized	Phone
<i>INTERNATIONAL_LOCALITY_CODE</i>	Contains the locality code of the international phone number.	Parsed Standardized	Phone

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
MATCH_FAMILY_NAME	Contains the combined Standardized form of FAMILY_NAME1 and FAMILY_NAME2 that the software used in the Match transform during the comparison process. The transform separates the two fields with a space. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space. Removes pre family name data. Compare with FAMILY_NAME1_MATCH_ST D1-6 and FAMILY_NAME2_MATCH_ST D1-6.	Standardized	Family_Name1
MATCH_FIRM	Contains a form of Firm that may be used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space, and removes data that is extraneous for matching purposes. This extraneous data includes business types such as Ltd. and GmbH, and noise words such as The, And, and Of. i Note Some words are classified to be removed from all domains, while others are language-specific and are classified to be removed in specific cultural domains. Compare with FIRM_MATCH_STD1-6.	Standardized	Firm

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
MATCH_GIVEN_NAME1	Contains the Standardized form of GIVEN_NAME1 used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space. Removes pre given name data. Compare with GIVEN_NAME1_MATCH_STD1-6.	Standardized	Given_Name1
MATCH_GIVEN_NAME2	Contains the Standardized form of GIVEN_NAME2 used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space. Removes pre given name data. Compare with GIVEN_NAME1_MATCH_STD1-6.	Standardized	Given_Name2
MATCH_MATURITY_POSTNAME	Contains the Standardized form of MATURITY_POSTNAME used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space. Compare with MATURITY_POSTNAME_MATCH_STD1-6.	Standardized	Postname

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
MATCH_PERSON	Contains a form of PERSON used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with a single space, and removes data that is extraneous for matching purposes. Extraneous data includes pre-given name, pre-family name, and pre-name as well as honorary and maturity post names and name designators.	Standardized	Person
MATCH_PHONE	Contains a form of PHONE used in the Match transform during the comparison process. Data is output as a string of digits. The transform removes spaces, punctuation, alphabetical characters, and leading zeros.	Standardized	Phone
MATCH_PRENAME	Contains the Standardized form of PRENAME used in the Match transform during the comparison process. The transform outputs data in uppercase, removes apostrophes, and replaces other punctuation with spaces. Compare with PRENAME_MATCH_STD1-6 .	Standardized	Prename
MATURITY_POSTNAME	Contains the MATURITY_POSTNAME that indicates heritage, such as Jr., Sr., III.	Standardized	Postname
MATURITY_POSTNAME_MATCH_STD1-6	Contains the match standard for MATURITY_POSTNAME . For example, Sr. is a match standard or alias for Senior. If the cleansing package does not have an alias entry, the field is empty.	Standardized	Postname_Match_Std
NAME_CONNECTOR	Contains the connector component of a dual name. For example, "and" or "or."	Parsed Standardized	None

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>NAME_DESIGNATOR</i>	Contains the name designator such as "Attn" or "c/o".	Parsed Standardized	None
<i>NAME_SPECIAL</i>	Contains the generic description of a person such as "occupant" or "current resident".	Parsed Standardized	None
<i>NORTH_AMERICAN_PHONE</i>	Contains an entire North American Numbering Plan (NANP) phone number.	Parsed Standardized	Phone
<i>NORTH_AMERICAN_PHONE_AREA_CODE</i>	Contains the area code Parsed from the phone number.	Parsed Standardized	Phone
<i>NORTH_AMERICAN_PHONE_EXTENSION</i>	Contains the extension Parsed from the phone number.	Parsed Standardized	Phone
<i>NORTH_AMERICAN_PHONE_LINE</i>	Contains the last four numbers (excluding the extension) Parsed from a phone number. For example, in (123) 456-7890, the transform returns 7890.	Parsed Standardized	Phone
<i>NORTH_AMERICAN_PHONE_PREFIX</i>	Contains the middle three numbers Parsed from a phone number. For example, in (123) 456-7890, the transform returns 456.	Parsed Standardized	Phone
<i>NORTH_AMERICAN_PHONE_TYPE</i>	Contains the type of phone number Parsed from the input phone number. For example, "Home" or "Work."	Parsed Standardized	Phone
<i>PERSON</i>	Contains combined components that define a single person. For example, Thomas Williams-Doyle Sr., M.D.	Parsed Standardized	None

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
PHONE	Contains the phone number that the transform identified as either North American or International. <i>i</i> Note Data Cleanse standardizes North American phone numbers but it does not standardize or enhance international phone numbers. Data Cleanse outputs international phone numbers as they are input.	Parsed Standardized	Phone
PRENAME	Contains the prename such as Mr. or Mrs.	Parsed Standardized	Prename
PRENAME_MATCH_STD1-6	Contains the match standard for a prename. For example, Mr. is the match standard or alias for Mister. If the cleansing package does not have an alias entry, the field is empty. Compare with MATCH_PRENAME.	Standardized	Prename_Match_Std

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
<i>ROW_ID</i>	Contains an internal ID generated by the transform that uniquely identifies a row processed by that transform. Use this field to join the input and output to the applicable non-summary data quality statistics table. If you select to generate any data quality statistics tables in the transform, and you don't include the <i>ROW_ID</i> output field, the software issues a warning. If you map this output field and don't select to generate any non-summary data cleanse statistics tables in the transform, the software also issues a warning. Applicable for the following transforms: • Data Cleanse • Global Address Cleanse • Geocoder	Standardized	Statistics
<i>RULE_LABEL</i>	Contains the rule that the transform uses to parse the indicated field.	Standardized	None
<i>SCORE</i>	Contains the confidence score for the Parsed field.	Standardized	None
<i>SSN</i>	Contains the entire Social Security number.	Parsed Standardized	SSN
<i>SSN_AREA</i>	Contains the first three numbers of the Social Security number.	Parsed Standardized	SSN
<i>SSN_GROUP</i>	Contains the fourth and fifth numbers of a Social Security number.	Parsed Standardized	SSN
<i>SSN_SERIAL</i>	Contains the last four numbers in a Social Security number.	Parsed Standardized	SSN

Field name	Description	GENERATED_FIELD_CLASS	GENERATED_FIELD_NAME
STATUS_CODE	Contains the code describing the following: - When the job is set for standardization, the code indicates how the transform Standardized the data. - When the job is not set for standardization, the code indicates how the transform could standardize the data.	Assignment_Info	None
TITLE	Contains the job or occupational title of a person, such as Manager or President.	Parsed Standardized	Title
TITLE_MATCH_STD1-6	Contains the match standard for occupational title. For example, CFO is the match standard or alias for Chief Financial Officer. If the dictionary does not have an alias entry, the field is empty.	Standardized	Title_Match_Std
UDPM	Contains the attribute field defined in the user-defined pattern rules in Cleansing Package Builder Reference Data.	Standardized	None
UDPM_SUBCOMPONENT1-5	Contains the subcomponents of the UDPM attribute field defined in the user-defined pattern rule.	Standardized	None

Parent topic: [Data Cleanse fields \[page 692\]](#)

Related Information

[Data Cleanse input fields \[page 693\]](#)

21.5 Geocoder fields

Map fields from upstream objects and transforms to Geocoder fields so that the transform knows the type of data in each field and how to process the data.

The Geocoder transform requires that you map fields on input and output. The process of mapping occurs when you configure the input and output fields in the Input Fields group and Output Fields group.

[Geocoder input fields \[page 714\]](#)

Map fields from your input data to Geocoder input fields so the transform knows the type of data to expect on input.

[Geocoder output fields \[page 718\]](#)

Map Geocoder output fields to output standardized and corrected data, generated data, and additional data.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match fields \[page 821\]](#)

21.5.1 Geocoder input fields

Map fields from your input data to Geocoder input fields so the transform knows the type of data to expect on input.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The following table contains an alphabetical list for all of the recognized input fields for the Geocoder transform. It includes the applicable category for each field.

The table also shows the input field availability based on currently supported directory formats. Supported directory formats are:

- [Basic](#): Supports the address geocoding mode. It returns centroid-level and address-level latitude and longitude information.

- **Advanced:** Supports the address geocoding, reverse geocoding, and POI textual search modes. It returns range-based and centroid-level latitude and longitude information and point-of-interest information.

The input fields support all countries.

Geocoder input fields

Name	Category	Description	Directory format
COUNTRY	Address	Contains the two-character ISO country code.	Basic, Advanced
DATA_SOURCE_ID	N/A	Specifies the name of the data source that the record comes from.	N/A
→ Tip If you choose to generate data quality statistics tables, use this field as part of your mapping strategy, along with primary key fields and the output field ROW_ID. This mapping identifies the source of individual records in any of the non-summary data quality statistics tables, thus enabling you to trace problem data to the data source. This input field is applicable for the following transforms: • Data Cleanse • Global Address Cleanse • Geocoder			
LATITUDE	Latitude/ Longitude	Contains the relative distance north or south of the equator, measured in 0-90 degrees.	Advanced
LOCALITY1-4	Address	Contains the city, town, or suburb and any additional related information.	Basic, Advanced
i Note For China geocoding, locality description is required. Use LOCALITY1_FULL, LOCALITY2_FULL from the GAC output fields for mapping since they contain a locality description.			
LONGITUDE	Latitude/ Longitude	Contains the relative distance east or west of the Greenwich meridian, measured in 0-180 degrees.	Advanced

Name	Category	Description	Directory format
OPTION_DISTANCE_UNIT	Search Filter	Contains the distance unit of measure used for the radius. KILOMETERS MILES The value of the OPTION_DISTANCE_UNIT input field takes precedence over the value of the DISTANCE_UNIT Geocoder option. The transform uses the value of the DISTANCE_UNIT option only when the OPTION_DISTANCE_UNIT input field isn't mapped, is invalid, blank, or NULL. The OPTION_DISTANCE_UNIT field is a dynamic input field. If you change the setting, you don't have to terminate and reinitialize the transform for the transform to recognize the new configuration.	Advanced
OPTION_MAX_RECORDS	Max Records	Contains the maximum number of records that the transform returns. Enter a number from 0 through 100. A value greater than 0 outputs multiple results as XML to the RESULT_LIST output field rather than to individual output fields. The value of the OPTION_MAX_RECORDS input field takes precedence over the value of the MAX_RECORDS Geocoder option. The transform uses the value of the MAX_RECORDS option only when the OPTION_MAX_RECORDS input field isn't mapped, is blank, or NULL. The OPTION_MAX_RECORDS field is a dynamic input field. If you change this setting, you don't have to terminate and reinitialize the transform for the transform to recognize the new configuration.	Advanced

Name	Category	Description	Directory format
<i>OPTION_RADIUS</i>	Search Filter	Contains the distance from a specified reference point to identify an area in which matching records are located. In reverse geocoding mode, the maximum radius distance is 111 kilometers or 68.97 miles. The value of the <i>OPTION_RADIUS</i> input field takes precedence over the value of the <i>RADIUS</i> Geocoder option. The transform uses the value of the <i>RADIUS</i> option only when the <i>OPTION_RADIUS</i> input field isn't mapped, is blank, or NULL. This field is a dynamic input field. If you change this setting, you don't have to terminate and reinitialize the transform for the transform to recognize the new configuration.	Advanced
<i>POI_NAME</i>	Address POI	Contains the name of a point of interest, such as the Washington Monument.	Advanced
<i>POI_TYPE</i>	Address POI	Contains the point-of-interest type expressed as a number. For example, for one vendor 5999 is a historical monument. To return multiple point-of-interest types in POI textual search mode, concatenate POI type codes using a colon as a delimiter. ❖ Example To return all schools (type 8211) and libraries (type 8231) within a defined area, you enter: 8211:8231 The POI types and their corresponding codes differ depending on the data vendor that you use. For a detailed list of available POI types, see the vendor-specific directory update.	Advanced
<i>POSTCODE1-2</i>	Address	Contains the postal code and a secondary postal code, when available.	Basic, Advanced
<i>PRIMARY_NAME1-4</i>	Address	Contains the street name.	Basic, Advanced
<i>PRIMARY_NUMBER</i>	Address	Contains the premise number.	Basic, Advanced
<i>PRIMARY_POSTFIX1</i>	Address	Contains the abbreviated directional, such as N, S, NW, SE, that follows a street name.	Basic, Advanced
<i>PRIMARY_PREFIX1</i>	Address	Contains the abbreviated directional, such as N, S, NW, SE, that precedes a street name.	Basic, Advanced
<i>PRIMARY_TYPE1-4</i>	Address	Contains the abbreviated type of primary name, such as St., Ave., or Pl.	Basic, Advanced

Name	Category	Description	Directory format
REGION1-2	Address	Contains the region symbol of the state, province, or territory.	Basic, Advanced
SEARCH_FILTER_NAME	Search Filter	Contains the search criteria for a point-of-interest name.	Advanced
SEARCH_FILTER_TYPE	Search Filter	Contains the search criteria for a point-of-interest type, expressed as a four-digit number. For example, for one vendor, 5999 is a historical monument. To return only an address, enter ADDR. To return multiple point-of-interest types, concatenate POI codes using a colon as a delimiter. ❖ Example To return all schools (type 8211) and libraries (type 8231) within a defined area, you enter: 8211 : 8231 To return a point-of-interest type and its address, enter: 5999 : ADDR The POI types and their corresponding codes differ depending on the data vendor that you use. For a detailed list of available POI types, see the vendor-specific directory update.	Advanced

Parent topic: [Geocoder fields \[page 714\]](#)

Related Information

[Geocoder output fields \[page 718\]](#)

21.5.2 Geocoder output fields

Map Geocoder output fields to output standardized and corrected data, generated data, and additional data.

The following table contains an alphabetical list of recognized output fields to use in the output mapping for the Geocoder transform. The table also shows the output field availability based on supported countries and directory formats.

Supported countries include the following:

- All

- **USA**

Supported directory formats include the following:

- **Basic:** Supports address geocoding mode. Returns centroid-level and address-level latitude and longitude information.
- **Advanced:** Supports address geocoding, reverse geocoding, and POI textual search modes. Returns range-based and centroid-level latitude and longitude information and point-of-interest information.

Geocoder output fields

Name	Category	Description	Supported countries	Directory Format
ADDRESS_LINE	Address	Contains the primary address. The primary address can contain components such as the primary range, primary name, post-directional, pre-directional, and suffix.	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
ASSIGNMENT_LEVEL	Assignment Level	Contains the level to which the transform matches the address to the data in the reference files (directories). • PRE: Primary Range Exact. Assigns to the exact location of the address, for example, 123 Main St. PRE is the most precise level of assignment. To obtain the PRE, map either the POI_TYPE input field or the PRIMARY_NAME and PRIMARY_NUMBE R input fields. • PRI: Primary Range Interpolated. Assigns to the level of the address range, for example, 100–500 Main St. • L1–4: Assigns to the level of city, town, or suburb. • P1: Postcode1. Assigns to the level of Postcode1. • P2P: Postcode2 Partial. Assigns the full Postcode1 and the first few characters of Postcode2. • PF: Postcode Full. Assigns to the level of Postcode1	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
		and Postcode2, when available.		
ASSIGNMENT_LEVEL_LOCALITY	Assignment Level	Contains the level to which the transform assigns the locality. L1-4: Returns up to four locality levels. L1 is the most general and L4 is the most specific.	All	Basic, Advanced
ASSIGNMENT_LEVEL_POSTCODE	Assignment Level	Contains the level to which the transform assigns the postcode. P1: Postcode1. Assigns to the level of Postcode1. P2P: Postcode2 Partial. Assigns the full Postcode1 and the first few characters of Postcode2. PF: Postcode Full. Assigns to the level of Postcode1 and Postcode2, when available.	All	Basic, Advanced
CENSUS_TRACT_BLOCK	Census	Contains the census tract code as defined by the government for reporting census information. Census tracts are small, relatively permanent statistical subdivisions of a county.	USA	Basic, Advanced
CENSUS_TRACT_BLOCK_PREV	Census	Contains the census tract code in the previous version of census data.	N/A	N/A

Name	Category	Description	Supported countries	Directory Format
CENSUS_TRACT_BLOCK_GROUP	Census	Contains the census tract and block group code as defined by the government for reporting census information. The government uses census tract and block group codes for matching with demographic-coding databases. In the USA, the code represents the following elements: • The first six digits contain the tract number, for example, 002689. • The first of the last four digits contain the BG number within the tract. The BG is a cluster of census blocks having the same first digit within a census tract. For example, BG 6 includes all blocks numbered from 6000 to 6999.	USA	Basic, Advanced
CENSUS_TRACT_BLOCK_GROUP_PREV	Census	Contains the census tract block group code in the previous version of census data.	N/A	N/A
COUNTRY_CODE	Address	Contains the two-character ISO country code.	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
<i>DISTANCE</i>	Distance	Contains the distance from the input address, geographical coordinates, or point of interest to the closest address or point of interest.	All	Advanced
<i>GOV_COUNTY_CODE</i>	Census	Contains a unique county code as defined by the government for reporting census information. For example, in the USA, the code is a Federal Information Processing Standard (FIPS) three-digit county code.	USA	Basic, Advanced
<i>GOV_LOCALITY1_CODE</i>	Census	Contains a unique code for an incorporated municipality such as a city, town, or locality, as defined by the government for reporting census information.	USA	Basic
<i>GOV_REGION1_CODE</i>	Census	Contains a unique region code as defined by the government for reporting census information. For example, in the USA, the code is a Federal Information Processing Standard (FIPS) two-digit state code.	USA	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
INFO_CODE	Info Code	Contains a three-character code that provides information about the geocoding results: • Third character indicates the status for address and point-of-interest geocoding assignment. • Second and third characters indicate the status for reverse geocoding assignment. If assigned to the best level, the INFO_CODE field is blank. The first character is reserved for future use.	All	Basic, Advanced
LATITUDE	Latitude/ Longitude	Contains the latitude at the best assigned level, which is 0–90 degrees north or south of the equator. The transform standardizes the latitude to six decimals in the format 45.801357.	All	Basic, Advanced
LATITUDE_LOCALITY	Latitude/ Longitude	Contains the latitude at the locality level centroid of the city, town, locality, or suburb. The transform standardizes the latitude to six decimals in the format 45.801357.	All	Basic, Advanced
LATITUDE_POSTCODE	Latitude/ Longitude	Contains the latitude at the postcode level centroid of the postcode. The transform standardizes the latitude to six decimals in the format 45.801357.	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
LATITUDE_PRIMARY_NUMBER	Latitude/ Longitude	Contains the latitude at the primary number level. The transform standardizes the latitude to six decimals in the format 45.801357.	All	Basic, Advanced
LOCALITY1-4	Address	Contains the city, town, or suburb and any additional related information.	All	Basic, Advanced
LONGITUDE	Latitude/ Longitude	Contains the longitude at the best assigned level, which is 0–180 degrees east or west of Greenwich meridian. The transform standardizes the longitude to six decimals in the format 123.458331.	All	Basic, Advanced
LONGITUDE_LOCALITY	Latitude/ Longitude	Contains the longitude at the locality level centroid of the city, town, locality, or suburb. The transform standardizes the longitude to six decimals in the format 123.458331.	All	Basic, Advanced
LONGITUDE_POSTCODE	Latitude/ Longitude	Contains the longitude at the postcode level centroid of the postcode. The transform standardizes the longitude to six decimals in the format 123.458331.	All	Basic, Advanced
LONGITUDE_PRIMARY_NUMBER	Latitude/ Longitude	Contains the longitude at the primary number level. The transform standardizes the longitude to six decimals in the format 123.458331.	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
<i>METRO_STAT_AREA_CODE</i>	Census	Contains the metropolitan statistical area. A metropolitan statistical area has a large population with a high degree of social and economic integration with the core of the area. The government defines the area for reporting census information. In the USA, the 0000 code indicates that the address does not lie in a metropolitan statistical area; usually a rural area.	USA	Basic
<i>METRO_STAT_AREA_CODE_PREV</i>	Census	Contains the metropolitan statistical area in the previous version of census data.	N/A	N/A
<i>MINOR_DIV_CODE</i>	Census	Contains the minor civil division or census county division code when the minor civil division is not available. The minor civil division designates the primary government and or administrative divisions of a county such as a civil township or precinct. Census county division is defined in a state or province that does not have a well-defined minor civil division. The government defines the area for reporting census information.	USA	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
MINOR_DIV_CODE_PREV	Census	Contains the minor civil division or census county division code in the previous version of census data.	N/A	N/A
POI_NAME	Address POI	Contains the point of interest name such as the Washington Monument.	All	Advanced
POI_TYPE	Address POI	Contains the point of interest type expressed as a four-digit number, such as 5999, which is a historical monument.	All	Advanced
POPULATION_CLASS_LOCALITY1	Population	Contains a code that indicates the population falls within a certain size. • 0: Undefined. The population may be too large or small to provide accurate data. • 1: Over 1 million. • 2: 500,000–999,999. • 3: 100,000–499,999. • 4: 50,000–99,999. • 5: 10,000–49,999. • 6: Less than 10,000.	All	Advanced
POSTCODE	Address	Contains the post code.	All	Basic, Advanced
POSTCODE1-2	Address	Contains the post code and a secondary post code, when available.	All	Basic, Advanced
PRIMARY_NAME1-4	Address	Contains the street name.	All	Basic, Advanced
PRIMARY_NUMBER	Address	Contains the premise number.	All	Basic, Advanced

Name	Category	Description	Supported countries	Directory Format
<i>PRIMARY_POSTFIX1</i>	Address	Contains the abbreviated directional, such as N, S, NW, SE, that follows a street name.	All	Basic, Advanced
<i>PRIMARY_PREFIX1</i>	Address	Contains the abbreviated directional, such as N, S, NW, SE, that precedes a street name.	All	Basic, Advanced
<i>PRIMARY_RANGE_LO W</i>	Address	Contains the high value of a primary number range.	All	Basic, Advanced
<i>PRIMARY_RANGE_HIG H</i>	Address	Contains the low value of a primary number range.	All	Basic, Advanced
<i>PRIMARY_TYPE1-4</i>	Address	Contains the abbreviated type of primary name, such as St., Ave., or Pl.	All	Basic, Advanced
<i>REGION1-2</i>	Address	Contains the region symbol of the state, province, or territory.	All	Basic, Advanced
<i>RESULT_LIST</i>	Results	Contains the XML output when multiple records are returned for a search.	All	Advanced
<i>RESULT_LIST_COUNT</i>	Results	Contains the number of results in the <i>RESULT_LIST</i> output field.	All	Advanced

Name	Category	Description	Supported countries	Directory Format
ROW_ID	Statistics	Contains an internal ID generated by the transform that uniquely identifies a row processed by that transform. Use this field to join the input and output to the applicable non-summary data quality statistics table. If you select to generate any data quality statistics tables in the transform, and you don't include the ROW_ID output field, the software issues a warning. Applicable transforms are Data Cleanse, Global Address Cleanse, and Geo-coder.	N/A	N/A
SIDE_OF_PRIMARY_ADDRESS	Side of Street	Indicates that the primary address is on the L (left) or R (right) side of the street when moving north, northeast, northwest, or east.	All	Advanced
STAT_AREA_CODE	Census	Contains a core-based statistical area code where an area has a high degree of social and economic integration within the core that the area surrounds. The government defines the area for reporting census information.	USA	Basic
STAT_AREA_CODE_PREV	Census	Contains the statistical area code in the previous version of census data.	N/A	N/A

Name	Category	Description	Supported countries	Directory Format
STATUS_CODE	Status Code	Contains a four-character status code that always starts with an S. This code represents the fuzzy matches made to the address/point of interest (POI) component during address geocoding processing.	All	Advanced, Basic

Parent topic: [Geocoder fields \[page 714\]](#)

Related Information

[Geocoder input fields \[page 714\]](#)

21.6 Global Address Cleanse fields

Map fields from upstream objects and transforms to the Global Address Cleanse fields so that the transform knows the type of data in each field and how to process the data.

The Global Address Cleanse transform requires that you map fields on input and output. The process of mapping occurs when you configure the input and output fields in the Input Fields group and Output Fields group.

[Global Address Cleanse input fields \[page 731\]](#)

Map fields from your input data to Global Address Cleanse input fields so the transform knows the type of data to expect on input.

[Global Address Cleanse NW input fields \[page 740\]](#)

Use the NW input fields for working with data from SAP applications.

[Global Address Cleanse output fields \[page 745\]](#)

The Global Address Cleanse transform provides output fields for your processed data and output fields for additional generated data.

[Global Address Cleanse NW_PO_Box output fields \[page 786\]](#)

The Global Address Cleanse transform populates the [NW_PO_BOX_](#) output fields only when you map them to NW input fields.

[Global Address Cleanse Suggestion Lists fields \[page 787\]](#)

Use the suggestion list input and output fields when you enable the suggestion list options in the Global Address Cleanse transform.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match fields \[page 821\]](#)

21.6.1 Global Address Cleanse input fields

Map fields from your input data to Global Address Cleanse input fields so the transform knows the type of data to expect on input.

Before you start to map your input data to SAP Data Services input fields, read the topic “Preparing your address-line input data”.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The following table contains an alphabetical list of the Global Address Cleanse input fields. The table also shows the engines applicable for each field. Engines include the following:

- Canada (C)
- Global Address (G)
- USA (U)

Input field	Description	Engine
ADDRESS_LINE	Contains the delivery address line, for example, "123 Main Street, Unit 4." China: ADDRESS_LINE contains the following address components: • Street and street number • Building, floor, and unit • Residential community ❖ Example 晨晖路 123 号中华大厦 12 楼 1201 室 宝山新村 100 号 201 室 Japan: ADDRESS_LINE contains the following address components: • Block (chome, kumi, Hokkaido go), sub-block (banchi, gaiku, tochi kukaku), and house number (go). • The building name, building floor, and building room. • The P.O. Box portion of the address, if applicable. South Korea: ADDRESS_LINE contains one of the following address components: • Land-oriented lot number address, neighborhood or village, lot number, and additional address information. For example, 효자동 200 • Road name address, street, house number, and additional address information. For example, 월미로 377-1	All engines
COUNTRY	Contains the identified country name of the address.	All engines

Input field	Description	Engine
<i>DATA_SOURCE_ID</i>	Specifies the name of the data source that the record comes from. → Tip If you choose to generate data quality statistics tables, use this field as part of your mapping strategy, along with primary key fields and the output field <i>ROW_ID</i>. This mapping identifies the source of individual records in any of the non-summary data quality statistics tables, thus enabling you to trace problem data to the data source. This input field is applicable for the following transforms: • Data Cleanse • Global Address Cleanse • Geocoder	All engines
<i>FIRM</i>	Contains the name of a company or organization. In some countries, large firms have their own postal code. If you include a <i>FIRM</i> field in your input, the Global Address Cleanse transform can assign more specific postal codes. China: China doesn't support Firm assignment. There's no Firm data for China. However, if firm data is available on input, the transform places it in the <i>FIRM</i> field. Japan: Map all Firm data for addresses in Japan to the <i>FIRM</i> field.	All engines
<i>LASTLINE</i>	Contains the locality, region (when it's included in the data), and postal code on one line.	All engines

Input field	Description	Engine
LOCALITY1	Contains the city, town, or suburb. China: Contains the following Prefecture-level localities when present on input: • Prefectures (地区 diqu) • Autonomous prefectures (自治州 zizhizhou) • Prefecture-level cities (地级市 di-jishi) • Leagues (盟 meng) • Provincial countries (省直辖县 shengzhixiaxian) Japan: Contains the following information when present on input: • City (shi) • Island (shima) • Ward (ku) • County (gun) • District (machi) • Village (mura)	All engines

Input field	Description	Engine
LOCALITY2	Contains any additional city, town, or suburb information. China: Contains county level localities when present on input: - Counties (县 xian) - Autonomous counties (自治县 zizhixian) - County-level cities (县级市 xianjishi) - Districts (市辖区 shixiaqu) - Banners (旗 qi) - Autonomous banners (自治旗 zizhiqi) - Forestry areas (林区 linqu) - Special districts (特区 tequ) Japan: Contains any additional ward, district, village, or sub-district (aza, bu, chiwari, sen) when present on input. ! Restriction South Korea: For land-oriented lot number address input, don't include neighborhood or village in LOCALITY2. Instead, map neighborhood, village, and lot number, to ADDRESS_LINE or MULTILINE1-12. USA: Contains the Puerto Rico urbanization when present on input.	G, U

Input field	Description	Engine
LOCALITY3	Contains any additional city, town, or suburb information. China: Contains the following information when present on input: - Township level localities - Townships (乡 xiang) - Ethnic townships (民族乡 minzuxiang) - Towns (镇 zhen) - Subdistricts (街道办事处 jiedao-banshichu) - District public offices (区公所 qu-gongsuo) - Sumu (苏木 sumu) - Ethnic sumu (民族苏木 minzsumu). Japan: Contains any additional district, village, sub-district (aza, bu, chiwari, sen, donchi, and tori), or super block (joh). ! Restriction South Korea: For land-oriented lot number address input, don't include neighborhood or village in LOCALITY2. Instead, map neighborhood, village, and lot number, to ADDRESS_LINE or MULTILINE1-12.	G

Input field	Description	Engine
LOCALITY4	Contains any additional city, town, or suburb information. China: Contains village and neighborhood level localities when present on input: • Administrative villages (行政村) • Neighborhood committees (社区居民委员会) • Neighborhoods or communities (社区) • Village committees (村民委员会). Japan: Contains any additional district, village, sub-district (aza, bu, chiwari, sen, and tori), or super block (joh).	G
MULTILINE1-12	Contains any data. The type of data in these fields can vary from record to record. Japan: Contains any data with the following restrictions: • If the input is address data, the address in total has to be in the traditional order of a Japanese address. • If the input is block data, place the block (chome, kumi, Hokkaido go), subblock (banchi, gaiku, tochi kukaku), and house number (go) in one line on input.	All engines
NW_FIELDS	Contains fields for SAP application data. Use with other NW input fields for cleansing data. Use NW fields as a group only, not individually or with non-NW input fields. For descriptions of NW input fields, see the "Global Address Cleanse NW input fields" topic.	
POSTCODE	Contains the postal code. USA: Contains the five-digit ZIP Code and the ZIP+4.	All engines

Input field	Description	Engine
<i>REGION1</i>	Contains the state, province, or region. China: Contains the province-level regions when present on input: • Provinces (省 sheng) • Autonomous regions (自治区 zizhiqu) • Municipalities (直辖市 zhixiashi) • Special administrative regions (特别行政区 tebie xingzhengqu) Japan: Contains the prefecture (to, do, fu, ken). A prefecture is similar to a state in the U.S.	All engines
<i>REGION2</i>	Contains the state, province, or region.	All engines

Input field	Description	Engine
SUGGESTION_REPLY1-6	Contains the index number that corresponds to a specific last line suggestion, an address line suggestion, or secondary list suggestion. Can also contain a street primary range or a street secondary range. i Note To use these fields, enable suggestion lists in the transform configuration. To use one suggestion reply input field to hold all of the suggestion replies rather than using all six reply fields, use the SUGGESTION_REPLY1 field and separate the replies with a pipe (). When using the SUGGESTION_REPLY1-6 fields for SAP applications for street and PO Box addresses, insert the following symbols to indicate whether the user has accepted changes made to the street address and when they're done with the street address: - asterisk plus (*+): User accepts the changes made to the street address up to the specified point and is done with the street address. - asterisk minus (*-): User doesn't accept the changes made to the street address up to the specified point and is done with the street address.	All engines

Parent topic: [Global Address Cleanse fields \[page 730\]](#)

Related Information

[Global Address Cleanse NW input fields \[page 740\]](#)

[Global Address Cleanse output fields \[page 745\]](#)

[Global Address Cleanse NW_PO_Box output fields \[page 786\]](#)

21.6.2 Global Address Cleanse NW input fields

Use the NW input fields for working with data from SAP applications.

Ensure that you do not use non-NW fields with NW fields. Use the fields properly to avoid unexpected results in your data.

Keep in mind that the transform processes NW input fields as multiline fields, even though they appear as discrete fields. The software maps NW fields internally to *MULTILINE1-12* before it performs normal Global Address Cleanse processing. If you exclude an NW input field from your mapping, the software maps the multiline that would have been mapped to the missed NW field to the next available NW input field.

The following table contains an alphabetical list of the NW fields and the applicable engines. Engines include the following:

- Canada (C)
- Global Address (G)
- USA (U)

NW Fields and Applicable Engines

Input field	Description	Engine
<i>NW_BUILDING</i>	Contains the building information. Requirement: When you map this input field, also map the <i>NW_STREET</i> field.	All engines
<i>NW_CITY1</i>	Contains the locality. Required field. The transform requires that you map the <i>NW_CITY1</i> and <i>NW_CITY2</i> input fields in sequence.	All engines
<i>NW_CITY2</i>	Contains additional locality or district information.	All engines
<i>NW_COUNTRY</i>	Contains the country name. Required field.	All engines

Input field	Description	Engine
NW_FIRM	Contains the input firm name. Including a firm name in some addresses helps the software better assign and verify addresses. Also, some postal authorities require a firm name for specific discounts. For example, a firm name is required for France PO Box addresses that use the CEDEX postal code numbering system.	G
NW_FLOOR_NUM	Contains the floor number. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_HOME_CITY	Contains additional locality information.	All engines
NW_HOUSE_NUM1	Contains the house number. The transform requires that you map the NW_HOUSE_NUM1 and NW_HOUSE_NUM2 fields in sequence. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_HOUSE_NUM2	Contains additional house number information. The transform requires that you map the NW_HOUSE_NUM1 and NW_HOUSE_NUM2 fields in sequence. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_LOCATION	Contains additional street information. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_PO_BOX_CITY	Contains the locality. When you use one NW_PO_BOX_-type field, the transform requires that you include all NW_PO_BOX_-type fields in your mapping.	All engines

Input field	Description	Engine
NW_PO_BOX_COUNTRY	Contains the country. When you use one NW_PO_BOX-type field, the transform requires that you include all NW_PO_BOX-type fields in your mapping.	All engines
NW_PO_BOX_POSTCODE	Contains the postcode. When you use one NW_PO_BOX-type field, the transform requires that you include all NW_PO_BOX-type fields in your mapping.	All engines
NW_PO_BOX_REGION	Contains the state, province, or region. When you use one NW_PO_BOX-type field, the transform requires that you include all NW_PO_BOX-type fields in your mapping.	All engines
NW_PO_BOX	Contains the PO Box number. When you use one NW_PO_BOX-type field, the transform requires that you include all NW_PO_BOX-type fields in your mapping.	All engines
NW_POSTCODE	Contains the postcode. Required field.	All engines
NW_REGION	Contains the state, province, or region. Required field.	All engines
NW_ROOM_NUM	Contains the room number. Required field.	All engines
NW_STR_SUPPL1	Contains additional street information. The transform requires that you map the NW_STR_SUPPL1-3 fields in sequence. Requirement: When you map this input field, also map the NW_STREET field.	All engines

Input field	Description	Engine
NW_STR_SUPPL2	Contains additional street information. The transform requires that you map the NW_STR_SUPPL1-3 fields in sequence. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_STR_SUPPL3	Contains additional street information. The transform requires that you map the NW_STR_SUPPL1-3 fields in sequence. Requirement: When you map this input field, also map the NW_STREET field.	All engines
NW_STREET	Contains the primary street name. Required field.	All engines

[Mapping Global Address Cleanse NW input fields \[page 743\]](#)

To avoid unexpected results when you use NW input fields, ensure that you follow the specific rules.

Parent topic: [Global Address Cleanse fields \[page 730\]](#)

Related Information

[Global Address Cleanse input fields \[page 731\]](#)

[Global Address Cleanse output fields \[page 745\]](#)

[Global Address Cleanse NW_PO_Box output fields \[page 786\]](#)

[Global Address Cleanse Suggestion Lists fields \[page 787\]](#)

21.6.2.1 Mapping Global Address Cleanse NW input fields

To avoid unexpected results when you use NW input fields, ensure that you follow the specific rules.

The following table contains rules or restrictions to know when you map the Global Address Cleanse NW input fields.

Rule or restriction	More information
Use NW input fields as a group.	Use the NW input fields with other NW input fields when cleansing address data from SAP applications. Do not mix with non-NW fields.
Do not map multiline or address line fields with NW fields.	Although the NW input fields appear discrete, they behave like multiline fields, and the transform processes them as multiline fields. The software maps NW fields internally to <i>MULTILINE1-12</i> before it performs normal Global Address Cleanse processing. If you do exclude an NW input field from your mapping, the software maps the multiline that would have been mapped to the missed NW field to the next available NW input field.
The transform requires that you use specific NW input fields for every NW input field mapping.	Required NW input fields include the following: • <i>NW_CITY1</i> • <i>NW_COUNTRY</i> • <i>NW_POSTCODE</i> • <i>NW_REGION</i> • <i>NW_STREET</i>
Map multiple numbered NW fields in sequence.	The transform requires that you map the following NW input fields in sequence: • <i>NW_CITY1-2</i> • <i>NW_HOUSE_NUM1-2</i> • <i>NW_STR_SUPPL1-3</i>
Requirement for mapping <i>NW_PO_BOX_-</i> -type input fields.	When you map any of the <i>NW_PO_BOX_-</i> -type fields, include all <i>NW_PO_BOX_-</i> -type fields in your mapping.
When you map any of the <i>NW_PO_BOX_-</i> -type input fields, include specific NW street-level fields.	Include the following street-level input fields when you use any of the <i>NW_PO_BOX_-</i>-type input fields: • <i>NW_CITY1</i> • <i>NW_COUNTRY</i> • <i>NW_POSTCODE</i> • <i>NW_REGION</i> • <i>NW_STREET</i>

→ Tip

To obtain a change significance of *N* in the *NW_PO_BOX_CHANGE_SIGNIFICANCE* output field, ensure that you set the *FORMAT_ASSIGNED_DATA* option to *NO*.

CJK script and NW fields

For CJK script input, the transform concatenates several NW input fields into the last multiline field for internal processing.

If the transform finds a descriptor in [NW_HOUSE_NUM2](#), the transform processes the multiline as follows (the + sign in the following code indicates that the fields are concatenated with no space between them):

```
NW_STREET+NW_HOUSE_NUM1+NW_HOUSE_NUM2+<DESCRIPTOR>NW_ROOMNUMBER+<DESCRIPTOR>NW_FLOOR+NW_BUILDING
```

If the transform doesn't find a descriptor in [NW_HOUSE_NUM2](#), the transform processes the multiline as follows (the + sign in the following code indicates that the fields are concatenated with no space between them):

```
NW_STREET+NW_HOUSE_NUM1+NW_ROOMNUMBER<DESCRIPTOR>+NW_FLOOR<DESCRIPTOR>+NW_BUILDING NW_HOUSE_NUM2
```

21.6.3 Global Address Cleanse output fields

The Global Address Cleanse transform provides output fields for your processed data and output fields for additional generated data.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The following table contains an alphabetical list of the Global Address Cleanse output fields.

The table also indicates the applicable engines for each field. The engines include the following:

- Canada (C)
- Global Address (G)
- USA (U)

Output field name	Description	Engine
<i>ADDITIONAL_INFO1</i>	Contains additional information specific to the country. • Austria: The PAC code of the current valid address when: • You preserve the alias address on output • The PAC codes for the master record and the alias record are different. • Belgium: The NIS code. • Canada: The official 13-character abbreviation of the city name, or the full spelling when the city name is less than 13 characters including spaces. • France: The INSEE code. • Germany: A portion of the German freightcode (Frachtleitcode). • Liechtenstein: The postal service district (Botenbezirke) when it is present in the data. • Poland: The community name (gmina). • Russia: A tax code (IFNSUL) and territory code (TERRIFNSUL) for the level that the address validates. The two codes are separated with a pipe symbol () as in IFNSUL TERRIFNSUL. • South Korea: The 25-digit administration number. • Spain: The INE 91 section code. • Switzerland: The postal service district (Botenbezirke) when it is available in the data.	C, G

Output field name	Description	Engine
<i>ADDITIONAL_INFO2</i>	Contains additional information specific to the country: • Austria: The City ID (OKZ). • Canada: The official 18-character abbreviation of the city name, or the full spelling if the city name is less than 18 characters including spaces. • Germany: The District Code. • Liechtenstein: Additional post-code. • Russia: An administrative territorial division code (OKATO) and municipality code (OKTMO) for the level that the address validates. The two codes are separated with a pipe symbol () as in OKATO OKTMO. • Spain: The INE Street code. • Switzerland: Additional postcode.	C, G
<i>ADDITIONAL_INFO3</i>	Contains additional information specific to the country: • Austria: Blank • Germany: The German City ID (ALORT). • Spain: The INE Town code.	G
<i>ADDITIONAL_INFO4</i>	Contains additional information specific to the country: • Austria: Blank • Germany: The German street name ID (StrSchl).	G
<i>ADDITIONAL_INFO5</i>	Contains additional information specific to the country: • Austria: The 7-digit SKZ Street Code. • Germany: The discount code for the freightcode.	G

Output field name	Description	Engine
ADDITIONAL_INFO6	Contains additional information specific to the country: Austria: The corner-house identification (1-digit). The value for a corner house is 1. Russia: An identifier (OBJECTID) and global unique identifier (OBJECTGUID) for the level that the address validates. The two codes are separated with a pipe symbol () as in OBJECTID OBJECTGUID.	G
ADDITIONAL_INFO7	Russia: A postcode for the level that the address validates.	All engines
ADDITIONAL_INFO8	Russia: A code of hierarchy. 0=Both 1=Administrative-Territorial 2=Municipal	All engines
ADDRESS_LINE_REMAINDER1-4	Contains extraneous data in the address line that the transform cannot identify or that does not belong in a standardized address. USA, ADDRESS_LINE_REMAINDER1-2: Complete secondary non-postal address such as Apt. 10, Ste 500, Box 34, Rm 7, 5th Flr.	All engines
ADDRESS_TYPE	Contains a one-character code that represents the type of address identified: P: Postal S: Street X: Unknown	All engines
AREA_NAME1	Contains an industrial area such as RICO INDUSTRIAL AREA.	G
ASSIGNMENT_INFO	Contains a code that indicates whether a record is valid, invalid, or corrected, based on the status and information codes. C: Corrected I: Invalid V: Valid B: Blank	All engines

Output field name	Description	Engine
<i>ASSIGNMENT_LEVEL</i>	Contains the level to which this transform matched the address to the data in the reference files (directories): • <i>C</i>: Country • <i>L1</i>: Locality1 • <i>L2</i>: Locality2 • <i>L3</i>: Locality3 • <i>L4</i>: Locality4 • <i>PN</i>: Primary name • <i>PR</i>: Primary range • <i>R</i>: Region • <i>S</i>: Secondary • <i>X</i>: Unknown, or the address was unassigned	All engines
<i>ASSIGNMENT_TYPE</i>	Contains a one- or two-character code that represents the type of address. • <i>BN</i>: Building name (Canada, Global Address) • <i>F</i>: Firm (Canada, Global Address, USA) • <i>G</i>: General delivery (Canada, Global Address, USA) • <i>H</i>: High-rise building (Canada, USA) • <i>HB</i>: House Boat (Global Address) • <i>L</i>: LOT (Global Address) • <i>M</i>: Military (Canada, USA) • <i>R</i>: Rural (Canada, USA) • <i>P</i>: Postal (Canada, Global Address, USA) • <i>PI</i>: Point of reference (Global Address) • <i>PS</i>: Packstation or Paketbox (Global Address) • <i>RP</i>: Postal Served by Route (Global Address) • <i>S</i>: Street (Canada, Global Address, USA) • <i>SR</i>: Street served by route (Canada, Global Address) • <i>U</i>: Uninhabited (Global Address) • <i>W</i>: Caravan (Global Address) • <i>X</i>: Unknown type or unassigned address (Canada, Global Address, USA)	Various engines
<i>BLOCK_DESCRIPTION</i>	Contains the block description.	G

Output field name	Description	Engine
<i>BLOCK_FULL</i>	Contains combined data from the following output fields: • <i>BLOCK_DESCRIPTION</i> • <i>BLOCK_NUMBER</i>	G
<i>BLOCK_NUMBER</i>	Contains the block number.	G
<i>BUILDING1</i>	Contains the base name for a building. The transform separates the base name from the description for countries that separate the two in reference data. ❖ Example A building name in the Russian Federation is “корпус Б.” The transform outputs building data as follows: • <i>BUILDING1</i> contains the base name “Б” • <i>BUILDING_DESCRIPTION1</i> contains the descriptor “корпус” For countries that consider the descriptor as part of the building name, the transform outputs both elements to the <i>BUILDING1</i> field. ❖ Example A building name in the United Kingdom is “Dover House.” The transform outputs the building data as follows: • <i>BUILDING1</i> contains “Dover House” • <i>BUILDING_DESCRIPTION1</i> is blank	G

Output field name	Description	Engine
<i>BUILDING2</i>	Contains the base name for the second building of an address that contains two buildings. The transform separates the base name from the description for countries that separate the two in reference data. For countries that consider the descriptor as part of the building name, the transform outputs both elements to the <i>BUILDING2</i> field. For examples, see the description for <i>BUILDING1</i>.	G
<i>BUILDING_DESCRIPTION1</i>	Contains the building descriptor. For countries that separate the base building name from the descriptor in reference data, the transform outputs the building description to this field. ❖ Example The transform outputs the building name “корпус Б” from the Russian Federation as follows: <i>BUILDING_DESCRIPTION1</i> contains the descriptor “корпус” <i>BUILDING1</i> contains the base name “Б” For countries that consider the descriptor as part of the building name, the transform outputs both elements to the <i>BUILDING1</i> field and leaves the <i>BUILDING_DESCRIPTION1</i> field blank. ❖ Example The transform outputs the building name “Dover House” from the United Kingdom as follows: <i>BUILDING_DESCRIPTION1</i> is blank <i>BUILDING1</i> contains “Dover House”	

Output field name	Description	Engine
<i>BUILDING_DESCRIPTION2</i>	Contains the building descriptor for the second building in an address that contains two buildings. For countries that separate the base building name from the descriptor in reference data, the transform outputs the building description to this field. For countries that consider the descriptor as part of the building name, the transform outputs both elements to the <i>BUILDING2</i> field and leaves the <i>BUILDING_DESCRIPTION2</i> field blank. For examples, see the description for <i>BUILDING_DESCRIPTION1</i>.	
<i>BUILDING_NAME1</i>	Contains the full building name, which includes the building name and the building descriptor. ❖ Example “White House” or “Empire Tower”	G
<i>BUILDING_NAME2</i>	Contains the full building name for the second building when an address contains two building names. The transform outputs the second building name to <i>BUILDING_NAME2</i>. ❖ Example “White House” or “Empire Tower”	G
<i>BUILDING_NAME1_2</i>	Contains combined data from the following output fields: <i>BUILDING_NAME1</i> <i>BUILDING_NAME2</i>	G

Output field name	Description	Engine
<i>BUILDING_PRIMARY_ADDR_DELIVERY_DUAL</i>	Contains combined data from the following output fields: <i>BUILDING_NAME1</i> <i>BUILDING_NAME2</i> <i>BUILDING_NAME1_2</i> <i>PRIMARY_ADDRESS_DELIVERY</i> <i>PRIMARY_ADDRESS_DELIVERY_DUAL</i> i Note <i>PRIMARY_ADDRESS_DELIVERY_DUAL</i> consists of the following combined output fields: <i>PRIMARY_ADDRESS1-4 (delivery)</i> <i>PRIMARY_ADDRESS1-4 (dual)</i>	All engines
<i>BUILDING_PRIMARY_SECONDARY_ADDR_DELIVERY_DUAL</i>	Contains combined data from the following output fields: <i>BUILDING_NAME1</i> <i>BUILDING_NAME2</i> <i>PRIMARY_SECONDARY_ADDRESS (delivery)</i> <i>PRIMARY_SECONDARY_ADDRESS (dual)</i>	All engines
<i>CERT_VALID</i>	Contains a valid certification indicator.	All engines

Output field name	Description	Engine
<i>CHANGE_SIGNIFICANCE</i>	Contains a code that indicates the level of change between the original input address and the output address. The transform considers differences in casing and diacritical characters to be a change. <i>H</i>: Highly significant change. <i>M</i>: Medium significant change. In Global Address Cleanse, this input address would produce a medium change significance due to the Locality Addition output field: Input: <pre>Multiline1 = WESTENDSTR 101 Multiline2 = FRANKFURT 60325 DE</pre> Output: <pre>COUNTRY = DE POSTCODE = 60325 LOCALITY1_ADDITION = am Main CHANGE_SIGNIFICANCE = M</pre> <i>L</i>: Low significant change. There is a difference between the input and output, but it is not considered highly significant. <i>N</i>: No changes between the original input and the output. <i>Blank</i> Some standardization options can affect the value of the change significance. The setting of some of the standardization options may never return an <i>N</i> for the change significance value. The input data dictates what options could affect the change significance. For example, if there are no European postcode prefixes, then the European Postcode Prefix setting has no effect. For best results, set the following options to <i>PRESERVE</i>: <i>ADDRESS_LINE_ALIAS</i> <i>ASSIGN_LOCALITY</i>	All engines

Output field name	Description	Engine
	COUNTRY_STYLE DIRECTIONAL_PUNCTUATION DIRECTIONAL_STYLE EUROPEAN_POSTCODE_PREFIX EXTRA_LINES INCLUDE_COUNTRY INCLUDE_LOCALITY_ADDITION LOCALITY_NAME_STYLE POSTAL_PHRASE_STYLE PRIMARY_TYPE_PUNCTUATION PRIMARY_TYPE_STYLE REGION_STYLE SECONDARY_DESCRIPTION_PUNCTUATION SECONDARY_DESCRIPTION_STYLE SECONDARY_NUMBER_STYLE STREET_NAME_STYLE Set the following options to <i>NO</i>: CONVERT_LATIN_OUTPUT_TO_US_ASCII FORMAT_ASSIGNED_DATA FORMAT_UNASSIGNED_DATA MOVE_MULTILINE_DATA REMOVE_ADDRESS_APOSTROPHES USE_POSTAL_COUNTRY_NAME	
COUNTRY	Contains the ISO country code or the country name of the input record. The parsed value of this component is the country data found in the input record.	All engines
COUNTRY_NAME	Contains the fully spelled country name in the languages specified in the OUTPUT_COUNTRY_LANGUAGE option.	All engines
COUNTY_NAME	Contains the fully spelled county name. USA : Addresses do not include county information.	U

Output field name	Description	Engine
DELIVERY_INSTALLATION_FULL	Contains combined data from the following output fields: DELIVERY_INSTALLATION_NAME DELIVERY_INSTALLATION_QUALIFIER DELIVERY_INSTALLATION_TYPE	C, G
DELIVERY_INSTALLATION_NAME	Contains the delivery installation city name, which is sometimes the same as the city name. If the delivery installation city name is the same as the city name, the transform omits the delivery installation city name from the address line. Canada: If the delivery installation name is different than the locality name, the transform outputs the delivery installation name to the secondary address fields. Japan: Returns the post office name.	C, G
DELIVERY_INSTALLATION_QUALIFIER	Contains the delivery Installation qualifier. For example, "Main" in "RR 2 Vancouver Stn Main".	C
DELIVERY_INSTALLATION_TYPE	Contains the delivery installation type. USA: PO: Post Office RPO: Retail Post Outlet STN: Station LCD: Letter Carrier Depot CMC: Community Mail Center CDO: Commercial Dealership Outlet France: BDP: Bureau de Poste CSP: Comptoir Service Postal SUCC: Succursale. PDF: Poste de Facteurs CPC: Centre Postal Communautaire CC: Concession Commerciale	C

Output field name	Description	Engine
DELIVERY_POINT	Contains the delivery indicator specific for the country of the address. • Australia: Eight-digit delivery point identifier, which is the primary component for generating a bar-code. The delivery point identifier is not printed on the mail pieces. • Austria: The PAC code, which is a unique identifier assigned by the Austrian postal authority. • New Zealand: A seven-character code that represents the delivery-point identifier. • United Kingdom: A two-character code that represents the delivery-point suffix.	G
EXTRA1-12	Contains any non-address data in the address block. Available only when the input data is presented through multi-line fields.	All engines
FIRM	Contains the firm name for the address. When you use a multiline format, the transform identifies the firm name based on the level of firm data available in the postal directories for each engine. Therefore results may be inconsistent. To avoid inconsistent identification of firm data, use the discrete FIRM field when you process multiline data. Canada and USA: The transform retrieves the firm name from the postal directory when the directory has the firm name; otherwise, the transform retrieves the firm name from the input record. Be aware that the postal directory might contain some unusual or shortened spellings that you may not find suitable for printing on mail pieces. If you prefer to retain your own firm data, retrieve the parsed component. Global Address: If the firm name is available on input or from reference data, the Global Address engine returns the firm name.	All engines

Output field name	Description	Engine
<i>FLOOR_DESCRIPTION</i>	Contains a building level description such as "Floor." Japan: The level description, such as kai.	All engines
<i>FLOOR_FULL</i>	Contains combined data from the following output fields: <i>FLOOR_DESCRIPTION</i> <i>FLOOR_NUMBER</i> <i>FLOOR_QUALIFIER</i>	All engines
<i>FLOOR_NUMBER</i>	Contains the building level number or floor information.	All engines
<i>FLOOR_QUALIFIER</i>	Contains any additional word that precedes or follows the floor information.	All engines
<i>ERROR</i>	Contains a value that represents the error status generated when the transform looks up a record and performs suggestion processing. Valid values are 0 through 6: <i>0</i>: No suggestion selection error. <i>1</i>: Blank suggestion selection/entry. <i>2</i>: Invalid suggestion selection. <i>3</i>: Invalid primary range. <i>4</i>: Invalid floor range. <i>5</i>: Invalid unit range. <i>6</i>: Too many possible matches to generate a suggestion list. Provide more information, such as a postal code, region, or locality.	All engines
<i>FULL_ADDRESS</i>	Contains the complete address line, including secondary address and dual address (street and postal).	All engines
<i>INFO_CODE</i>	Contains a four-character code for addresses that do not fully assign that describes why the transform could not assign the address. If the address is fully assigned, the field is blank.	All engines

Output field name	Description	Engine
<i>ISO_COUNTRY_CODE_2CHAR</i>	Contains the two-character ISO code that identifies a country, for example, DE is for Germany.	All engines
<i>ISO_COUNTRY_CODE_3CHAR</i>	Contains the three-character ISO-3166 code that identifies a country. For example, DEU is for Germany.	All engines
<i>ISO_COUNTRY_CODE_3DIGIT</i>	Contains the three-digit ISO code that identifies a country. For example, 276 is for Germany.	All engines
<i>ISO_SCRIPT_CODE</i>	Contains the four-character script code for an identified country. For example, LATN or KANA.	All engines
<i>LANGUAGE</i>	Contains the two-character ISO language code that represents the language of the address.	All engines
<i>LASTLINE</i>	Contains combined data from the following output fields: • <i>LOCALITY</i> (1–4 if available) • <i>REGION</i> • <i>POSTCODE</i> The transform includes the region only when the country requires it.	All engines
<i>LASTLINE_REMAINDER1-4</i>	Contains unused lastline remainder data.	G
<i>LOCALITY_CODE</i>	Contains a locality code, which some countries use to distinguish sections of a large locality. For example, in France, locality codes are called “arrondissements”.	G
<i>LOCALITY1_2_FULL</i>	Contains combined data from the following output fields: • <i>LOCALITY1_1_FULL</i> • <i>LOCALITY1_2_FULL</i>	All engines
<i>LOCALITY1_2_NAME</i>	Contains combined data from the following output fields: • <i>LOCALITY1_NAME</i> • <i>LOCALITY2_NAME</i>	All engines

Output field name	Description	Engine
<i>LOCALITY1_4_FULL</i>	Contains combined data from the following output fields: • <i>LOCALITY1_FULL</i> • <i>LOCALITY2_FULL</i> • <i>LOCALITY3_FULL</i> • <i>LOCALITY4_FULL</i>	All engines
<i>LOCALITY1_4_NAME</i>	Contains combined data from the following output fields: • <i>LOCALITY1_NAME</i> • <i>LOCALITY2_NAME</i> • <i>LOCALITY3_NAME</i> • <i>LOCALITY4_NAME</i>	All engines
<i>LOCALITY1_ADDITION</i>	Contains additional locality information.	G
<i>LOCALITY1_ALTERNATE</i>	Contains the preserved input locality when the postal authority recognizes it as a legitimate locality. The transform corrects any spelling errors.	C, U
<i>LOCALITY1_DESCRIPTION</i>	Contains the <i>LOCALITY1</i> descriptor. • Japan: For example, shi, shima. • China: For example, 市(Shi).	G
<i>LOCALITY1_FULL</i>	Contains combined data from the following output fields: • <i>LOCALITY1_NAME</i> • <i>LOCALITY_CODE</i> • <i>LOCALITY1_DESCRIPTION</i> • <i>LOCALITY1_QUALIFIER</i> The field may include <i>LOCALITY1_ADDITION</i>, depending on the standardization option settings of <i>LOCALITY_NAME_STYLE</i> and <i>INCLUDE_LOCALITY_ADDITION</i>.	All engines

Output field name	Description	Engine
LOCALITY1_NAME	Contains the city, town, locality, or sub-urb that is one of the following output fields based on the standardization option setting for ASSIGN_LOCALITY: LOCALITY1_ALTERNATE LOCALITY1_OFFICIAL Japan: Contains the city (shi), island (shima), ward (ku), county (gun), district (machi), or village (mura).	All engines
LOCALITY1_OFFICIAL	Contains the locality name preferred by the postal authority.	All engines
LOCALITY1_QUALIFIER	Contains the locality qualifier. Applicable only for France for Cedex.	G
LOCALITY2_4_FULL	Contains combined data from the following output fields: LOCALITY2_FULL LOCALITY3_FULL LOCALITY4_FULL	All engines
LOCALITY2_4_NAME	Contains combined data from the following output fields: LOCALITY2_NAME LOCALITY3_NAME LOCALITY4_NAME	All engines
LOCALITY2_CODE	Contains the LOCALITY2 code when an address has LOCALITY1 and LOCALITY2 information. Note Currently applicable to Czechia addresses only.	
LOCALITY2_DESCRIPTION	Contains the description of the subdivision of LOCALITY1 when it exists.	G
LOCALITY2_FULL	Contains the combined data from the following output fields: LOCALITY2_NAME LOCALITY2_DESCRIPTION	G, U

Output field name	Description	Engine
LOCALITY2_NAME	Contains additional locality information. USA: Contains urbanization (Puerto Rico addresses only).	G, U
LOCALITY2_OFFICIAL	Contains the locality name preferred by the postal authority.	G
LOCALITY2_QUALIFIER	Contains the locality qualifier.	G
LOCALITY3_4_FULL	Contains combined data from the following output fields: LOCALITY3_FULL LOCALITY4_FULL	All engines
LOCALITY3_4_NAME	Contains combined data from the following output fields: LOCALITY3_NAME LOCALITY4_NAME	All engines
LOCALITY3_DESCRIPTION	Contains the description of a subdivision of LOCALITY2 when it exists.	G
LOCALITY3_FULL	Contains combined data from the following output fields: LOCALITY3_NAME LOCALITY3_DESCRIPTION	G
LOCALITY3_NAME	Contains additional locality information.	G
LOCALITY3_OFFICIAL	Contains the locality name preferred by the postal authority.	G
LOCALITY3_QUALIFIER	Contains the locality qualifier.	G
LOCALITY4_DESCRIPTION	Contains the description of a subdivision of Locality3 when it exists.	G
LOCALITY4_FULL	Contains combined data from the following output fields: LOCALITY4_NAME LOCALITY4_DESCRIPTION	G
LOCALITY4_NAME	Contains additional locality information.	G
LOCALITY4_OFFICIAL	Contains the locality name preferred by the postal authority.	G
LOCALITY4_QUALIFIER	Contains the locality qualifier.	G

Output field name	Description	Engine
<i>MATCH_BLOCK_NUMBER</i>	Contains a form of <i>BLOCK_NUMBER</i> that the Match transform can use during the comparison process. The transform does not output block descriptions. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the transform uses normal width standardization for consistency. This field is only available as a Best component.	All engines
<i>MATCH_BUILDING_NAME</i>	Contains a form of <i>BUILDING_NAME1</i> that the Match transform may use during the comparison process. The transform outputs building descriptions in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the transform uses normal width standardization for consistency. Note For China, the transform removes building indicators. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_COUNTRY</i>	Contains a form of <i>ISO_COUNTRY_CODE_2CHAR</i> that the Match transform may use during the comparison process. The transform outputs data in upper-case and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines
<i>MATCH_FLOOR_NUMBER</i>	Contains a form of <i>FLOOR_NUMBER</i> that the Match transform may use during the comparison process. The transform does not output floor descriptions and qualifiers. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_LOCALITY</i>	Contains a form of <i>LOCALITY</i> that the Match transform may use during the comparison process. Standardization settings do not affect the output. The transform outputs <i>LOCALITY1_OFFICIAL</i> when a locality or better level assignment is made; otherwise, the transform outputs the <i>LOCALITY1_NAME</i> data. The transform does not output locality codes, qualifiers, or descriptions. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<code>MATCH_LOCALITY2</code>	Contains a form of <code>LOCALITY</code> that the Match transform may use during the comparison process. Standardization settings to not affect the output. The transform outputs <code>LOCALITY2_OFFICIAL</code> when a locality or better level assignment is made; otherwise, the transform outputs the <code>LOCALITY2_NAME</code>. The transform does not output locality codes, qualifiers, or descriptions. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. Note For China, Japan, South Korea, and Taiwan, the transform outputs <code>LOCALITY2-4_OFFICIAL</code> or <code>LOCALITY2-4_NAME</code>, if present. For all other countries or regions, the transform outputs only <code>LOCALITY2_OFFICIAL</code> or <code>LOCALITY2_NAME</code>. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_POSTCODE1</i>	Contains a form of <i>POSTCODE1</i> that the Match transform may use during the comparison process. The transform outputs data in upper-case and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines
<i>MATCH_PRIMARY_DIRECTIONAL</i>	Contains a form of <i>PRIMARY_PREFIX1</i> and <i>PRIMARY_POSTFIX1</i> that the Match transform may use during the comparison process. Standardization settings do not affect the output. The transform outputs the abbreviated form, if available. If a prefix and postfix are both present, the transform outputs them separated by a space. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_PRIMARY_NAME</i>	Contains a form of <i>PRIMARY_NAME1</i> that the Match transform may use during the comparison process. Standardization settings do not affect the output. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. The transform removes prefix, postfix, suffix, and type data. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_PRIMARY_NAME2</i>	Contains a form of <i>PRIMARY_NAME2</i> that the Match transform may use during the comparison process. Standardization settings do not affect the output. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. The transform removes prefix, postfix, suffix, and type data. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. Note For Brazil, China, and Japan, <i>PRIMARY_NAME2-4</i> are output, if present. For all other countries, only <i>PRIMARY_NAME2</i> is output. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_PRIMARY_NUMBER</i>	Contains a form of <i>PRIMARY_NUMBER</i> that the Match transform may use during the comparison process. The transform outputs data from only the <i>PRIMARY_NUMBER</i> and <i>PRIMARY_NUMBER_EXTRA</i> fields, and not the <i>PRIMARY_NUMBER_DESCRIPTION</i> field. The transform outputs data in upper-case and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_PRIMARY_TYPE</i>	Contains a form of <i>PRIMARY_TYPE</i> that the Match transform may use during the comparison process. Standardization settings do not affect the output. The transform outputs the abbreviated primary type, if available. The transform outputs data in upper-case and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. Note For Brazil, China, and Japan, <i>PRIMARY_NAME2-4</i> are output, if present. For all other countries, only <i>PRIMARY_NAME2</i> is output. The field is available only as a Best component.	All engines
<i>MATCH_REGION</i>	Contains a form of <i>REGION1_NAME</i> that the Match transform may use during the comparison process. The transform outputs data in upper-case and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_STAIRWELL_NAME</i>	Contains a form of <i>STAIRWELL_NAME</i> that the Match transform may use during the comparison process. The transform does not output stairwell descriptions. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines
<i>MATCH_UNIT_NUMBER</i>	Contains a form of <i>UNIT_NUMBER</i> that the Match transform may use during the comparison process. The transform does not output unit descriptions and qualifiers. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines

Output field name	Description	Engine
<i>MATCH_WING_NAME</i>	Contains a form of <i>WING_NAME</i> that the Match transform may use during the comparison process. The transform does not output wing descriptions. The transform outputs data in uppercase and removes diacritical characters and apostrophes. The transform replaces other punctuation and multiple spaces with a single space. For China, Russia, South Korea, and Taiwan: The transform transliterates non-Latin scripts to Latin to support inter-script matching. For other CJK-script countries or regions, the field uses normal width standardization for consistency. The field is available only as a Best component.	All engines
<i>MULTILINE1-12</i>	Contains a line of any data. The type of data in this line may vary from record to record.	All engines
<i>NW_FORMATTED_POSTCODE</i>	Contains the postcode in the format that SAP applications require.	All engines
<i>NW_POSTCODE_IN_SUPPORTED_FOR MAT</i>	Contains an indicator that specifies whether the <i>NW_FORMATTED_POSTCODE</i> output field is populated.	All engines
<i>PMB_FULL</i>	Contains private mailbox information.	All engines
<i>PNAME_SECONDARY_ADDR</i>	Contains the full primary name, with no associated primary number, and the full secondary address.	All engines
<i>POINT_OF_REFERENCE1-2</i>	Contains combined data from the following output fields: • <i>POINT_OF_REFERENCE1</i> • <i>POINT_OF_REFERENCE2</i>	All engines
<i>POINT_OF_REFERENCE1-2</i>	Contains a well known place or easily visible location to help locate an address. For example, "Opposite of Citibank ATM".	G

Output field name	Description	Engine
<i>POST_OFFICE_NAME</i>	Contains the name or numeric representation for a post office. For example, "01" BP 1012.	G
<i>POSTCODE_DESCRIPTION</i>	Contains a word that indicates a postal code. The transform populates this field only when the information is available on input. ❖ Example Brazil: CEP. CEP stands for Código de Endereçamento Postal, and is output as "CEP 52041-970". China: 邮编 Japan: 〒	G
<i>POSTCODE_FULL</i>	Contains a complete postcode based on country. Australia: Complete four-digit postal code. Canada: Complete six-character postal code, such as FSA + LDU. Global Address: Complete postal code. USA: The full ZIP Code with a hyphen (10 characters). Japan: The seven-digit postcode.al	All engines
<i>POSTCODE_IN_VALID_FORMAT</i>	Contains an indicator that the postcode is in the correct format as defined by the postal authority for that country.	All engines

Output field name	Description	Engine
<i>POSTCODE_PREFIX</i>	Contains the postcode prefix that some European countries use to differentiate one countries postcode from another. A prefix helps identify the country of the postcode. The prefix is useful for postal codes from countries that use the same format. For example, many countries use the four- or five-digit format. You can avoid confusion when sending mail to or from the European country when the postal code contains the country code as a prefix. The codes used are generally based on License plate codes (D for Germany or F for France) rather than ISO codes.	G
<i>POSTCODE1</i>	Contains the postcode, which varies by country. • Australia: Four-digit postcode. • Canada: First three characters (FSA) of the postal code. • Global Address: Postal code. • Japan: The first three digits of the postal code. • USA: Five-digit primary postal code (ZIP Code). Does not include the four-digit secondary postal code (ZIP4). • China: A software-generated default locality level postcode might be assigned when the transform cannot obtain an actual postcode . i Note When the transform assigns a software-generated default locality level postcode, it generates a status code for the record, with a sixth character of P or Q.	All engines

Output field name	Description	Engine
<i>POSTCODE2</i>	Contains the secondary postal code. • Canada: The last three characters (LDU) of the postal code. • Japan: The last four digits of the postal code. • USA: The four-digit ZIP Code, which follows the primary postal code on a mail piece with a hyphen placed between, for example, 54601-1234.	All engines
<i>PRIMARY_ADDRESS</i>	Contains the primary address line, such as the street address or post office box. Does not include secondary address information such as apartment. Japan: The full block data.	All engines
<i>PRIMARY_ADDRESS_DELIVERY_DUAL</i>	Contains combined data from the following output fields: • <i>PRIMARY_ADDRESS1-4 (DELIVERY)</i> • <i>PRIMARY_ADDRESS1-4 (DUAL)</i>	All engines
<i>PRIMARY_DELIVERY_MODE</i>	Contains the delivery mode for a street served by a route-type address, such as Rural Route.	C
<i>PRIMARY_DELIVERY_NUMBER</i>	Contains the delivery number for a street served by a route-type address, such as Rural Route.	C
<i>PRIMARY_NAME_FULL1</i>	Contains the primary name, primary type, primary prefix, and primary postfix.	All engines
<i>PRIMARY_NAME_FULL1_2</i>	Contains combined data from the following output fields: • <i>PRIMARY_NAME_FULL1</i> • <i>PRIMARY_NAME_FULL2</i>	All engines
<i>PRIMARY_NAME_FULL1_4</i>	Contains combined data from the following output fields: • <i>PRIMARY_NAME_FULL1</i> • <i>PRIMARY_NAME_FULL2</i> • <i>PRIMARY_NAME_FULL3</i> • <i>PRIMARY_NAME_FULL4</i>	All engines

Output field name	Description	Engine
<i>PRIMARY_NAME_FULL2</i>	Contains combined data from the following output fields: <i>PRIMARY_NAME2</i> <i>PRIMARY_TYPE2</i> <i>PRIMARY_PREFIX2</i> <i>PRIMARY_POSTFIX2</i>	G
<i>PRIMARY_NAME_FULL3_4</i>	Contains combined data from the following output fields: <i>PRIMARY_NAME_FULL3</i> <i>PRIMARY_NAME_FULL4</i>	All engines
<i>PRIMARY_NAME_FULL3-4</i>	Contains the primary name and primary type.	G
<i>PRIMARY_NAME1</i>	Contains the street name description, which is typically a street name or box description. Japan: Block (chome, kumi, Hokkaido go), subblock (banchi, gaiku, tochi kukaku). The Post office name description (yubinnyoku or siten).	All engines
<i>PRIMARY_NAME2</i>	Contains the second street and name description, typically a street name, or box description. Japan: Additional block and subblock information.	G
<i>PRIMARY_NAME3</i>	Contains the street name, delivery mode, and so on. Japan: Additional block and subblock information.	G
<i>PRIMARY_NAME4</i>	Contains the street name, delivery mode, and so on. Japan: Additional block and subblock information.	G
<i>PRIMARY_NUMBER</i>	Contains the premise number, rural route number, or PO Box number. In some cases, it may include a range.	All engines

Output field name	Description	Engine
<i>PRIMARY_NUMBER_DESCRIPTION</i>	Contains a description that precedes the primary number. For example, KM for Kilometer or Blk. for block. Japan: The postal number identifier 号 (go) or house number description 号 (go). China: The description after the street number. For example, 号 (hao).	G
<i>PRIMARY_NUMBER_EXTRA</i>	Contains data found near the parsed primary number, which in most cases cannot be identified or does not belong in a standardized address. Japan: The postal box identifier.	G
<i>PRIMARY_NUMBER_FULL</i>	Contains the primary number, primary number description, and primary number extra.	All engines
<i>PRIMARY_POSTFIX1</i>	Contains the abbreviated or non abbreviated directional, such as N, South, NW, that follows a street name. Japan: Directional that follows block or subblock.	All engines
<i>PRIMARY_POSTFIX2</i>	Contains the abbreviated or non abbreviated directional, such as N, South, NW, SE, that follows a street name. The transform uses the standardization setting for directional style to determine whether to output abbreviated or non abbreviated format. Japan: Directional that follows block or subblock.	G
<i>PRIMARY_PREFIX1</i>	Contains the abbreviated or non abbreviated directional, such as N, South, NW, SE, that precedes a street name. The transform uses the standardization setting for directional style to determine whether to output abbreviated or non abbreviated format. Japan: Directional that precedes block or subblock.	G, U

Output field name	Description	Engine
<i>PRIMARY_PREFIX2</i>	Contains the abbreviated or non abbreviated directional, such as N, South, NW, SE, that precedes a street name. The transform uses the standardization setting for directional style to determine whether to output abbreviated or non abbreviated format. Japan: Directional that precedes block or subblock.	G
<i>PRIMARY_SECONDARY_ADDR_DELIVERY_DUAL</i>	Contains combined data from the following output fields: <i>PRIMARY_SECONDARY_ADDRRESSES (DELIVERY)</i> <i>PRIMARY_SECONDARY_ADDRRESSES (DUAL)</i>	All engines
<i>PRIMARY_SECONDARY_ADDRESS</i>	Contains the combined data from the following output fields: <i>PRIMARY_ADDRESS</i> <i>SECONDARY_ADDRESS</i>	All engines
<i>PRIMARY_TYPE1</i>	Contains the type of primary name. For example, rue, strasse, street, Ave, or Pl.	All engines
<i>PRIMARY_TYPE2-4</i>	Contains the type of primary name. For example, rue, strasse, street, Ave, or Pl.	G
<i>QUALITY_CODE</i>	Contains a two-character code that provides additional information about the quality of the address. The quality of the address depends on the input data, the processing engine, country, information code, and status code when an information code is not generated. For more information about status codes, see Global Address Cleanse quality codes [page 864].	All engines
<i>REGION1</i>	Contains either the <i>REGION1_NAME</i> or <i>REGION1_SYMBOL</i> output based on the standardization setting for region style.	All engines

Output field name	Description	Engine
<i>REGION1_2_FULL</i>	Contains combined data from the following output fields: <i>REGION1_FULL</i> <i>REGION2_FULL</i> USA: Does not include <i>REGION2_FULL</i>.	All engines
<i>REGION1_2_NAME</i>	Contains combined data from the following output fields: <i>REGION1_NAME</i> <i>REGION2_NAME</i> USA: Does not include <i>REGION2_NAME</i>.	All engines
<i>REGION1_CODE</i>	Contains the region code, which may be the ISO region code.	All engines
<i>REGION1_DESCRIPTION</i>	Contains the region description.	G
<i>REGION1_FULL</i>	Contains combined data from the following output fields: <i>REGION1</i> <i>REGION1_DESCRIPTION</i>	All engines
<i>REGION1_NAME</i>	Contains the fully spelled Region name.	All engines
<i>REGION1_SYMBOL</i>	Contains the abbreviated form of the Region name.	All engines
<i>REGION2</i>	Contains either the <i>REGION2_NAME</i> or the <i>REGION2_SYMBOL</i> output data based on the standardization option you set for region style. USA: Contains the county name. Poland: The district name (powiat).	G, U
<i>REGION2_CODE</i>	Contains the region code, which may be the ISO region code.	All engines
<i>REGION2_DESCRIPTION</i>	Contains the region 2 description.	G
<i>REGION2_FULL</i>	Contains combined data from the following output fields: <i>REGION2</i> <i>REGION2_DESCRIPTION</i>	G
<i>REGION2_NAME</i>	Contains the fully spelled Region 2 name.	G

Output field name	Description	Engine
REGION2_SYMBOL	Contains the abbreviation of the Region 2 name.	G
ENGINE_NAME	Contains the name of the engine that you select in the transform configuration.	All engines
REMAINDER_EXTRA_PMB_FULL	Contains combined data from the following output fields: REMAINDER_FULL EXTRA1-2 PMB_FULL	All engines
REMAINDER_FULL	Contains all remainder information, including data from the ADDRESS_LINE_REMAINDER1-4 and LASTLINE_REMAINDER1-4 output fields.	All engines
ROOM_FULL	Contains combined data from the following output fields: UNIT_DESCRIPTION when it contains "room" or a variant UNIT_NUMBER	All engines
ROOM_NUMBER	Contains the unit number for units that are variations of "room." For example, RM, RMS, ROOM, ROOMS, RM., RMS, 号室, 室, 号.	All engines

Output field name	Description	Engine
ROW_ID	Contains an internal ID generated by the transform that uniquely identifies a row processed by that transform. Use this field to join the input and output to the applicable non-summary data quality statistics table. The software issues a warning if you select to generate any non-summary data quality statistics tables in the transform, and you don't include the ROW_ID output field. The software also issues a warning when you map this output field and don't select to generate any non-summary data quality statistics tables in the transform. Applicable for the following transforms: • Data Cleanse • Global Address Cleanse • Geocoder	All engines
SECONDARY_ADDRESS	Contains the block, floor, unit, stairwell, or wing data on one line.	All engines
SECONDARY_ADDRESS_NO_FLOOR_NO_ROOM	Contains combined data from all SECONDARY_FULL output fields except the following output fields: • FLOOR_FULL • ROOM_FULL	All engines
SECONDARY_ADDRESS_NO_FLOOR	Contains combined data from all SECONDARY_FULL output fields except FLOOR_FULL .	All engines
SECONDARY_ADDRESS_NO_ROOM	Contains combined data from all SECONDARY_FULL output fields except ROOM_FULL .	All engines
SINGLE_ADDRESS	Contains the full address and last line in one component.	All engines
STAIRWELL_DESCRIPTION	Contains the identifier for an entrance or stairwell of a building, such as, "Entrada" 1.	G

Output field name	Description	Engine
STAIRWELL_FULL	Contains combined data from the following output fields: • STAIRWELL_DESCRIPTION • STAIRWELL_NAME	All engines
STAIRWELL_NAME	Contains the name or number of an entrance or stairwell for a building, such as, Entrada "1."	G

Output field name	Description	Engine
STATUS	Contains the suggestion status generated as the result of looking up the current record and performing suggestion list processing. • A: Primary address-line suggestions available. • B: Primary and secondary ranges are invalid. • C: Address was not found. • CS: Country list is available. This is applicable when the ENABLE_SUGGESTION_LISTS option is set to YES. This is for the Country ID engine only. • F: Floor range is invalid. • L: Lastline suggestions available. • L1: Locality1 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • L2: Locality2 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • L3: Locality3 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • L4: Locality4 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • N: No suggestions available. • PC: Postcode suggestions available. • R: Primary range is invalid. • R1: Region1 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • R2: Region2 list available (when the Enable Lastline Drilldown option is Yes; Global Address engine only). • S: Unit range is invalid. • U: Secondary address-line suggestions available.	All engines

Output field name	Description	Engine
STATUS_CODE	Contains the six-character status code that always starts with an S. This code describes what parts of the address changed during processing. For more information about status codes, see USA Regulatory Address Cleanse status codes [page 871].	All engines
UNIT_DESCRIPTION	Contains the unit description, such as "Apartment" or "Flat." Japan: The unit description, such as gousitsu.	All engines
UNIT_FULL	Contains combined data from the following output fields: • UNIT_DESCRIPTION • UNIT_NUMBER • UNIT_QUALIFIER	All engines
UNIT_NUMBER	Contains the unit number, such as 100 in "Apartment 100."	All engines
UNIT_QUALIFIER	Contains the additional word that precedes or follows the unit information.	G
WING_DESCRIPTION	Contains the wing identifier within a building. For example, returns "Wing" from "West Wing."	G
WING_FULL	Contains combined data from the following output fields: • WING_DESCRIPTION • WING_NAME	All engines
WING_NAME	Contains the name or number of a wing within a building. For example, returns "West" from "West Wing."	G

Parent topic: [Global Address Cleanse fields \[page 730\]](#)

Related Information

[Global Address Cleanse input fields \[page 731\]](#)

[Global Address Cleanse NW input fields \[page 740\]](#)

[Global Address Cleanse NW_PO_Box output fields \[page 786\]](#)

21.6.4 Global Address Cleanse NW_PO_Box output fields

The Global Address Cleanse transform populates the *NW_PO_BOX_* output fields only when you map them to NW input fields.

Use *NW_PO_BOX_* output fields only for the PO Box address portion of SAP business suite software.

The following is a list of the available *NW_PO_BOX_* output fields. The content of each *NW_PO_BOX_* field is identical to its corresponding output field without the NW prefix.

❖ Example

To find the description for the *NW_PO_BOX_ASSIGNMENT_INFO* field, see the output field description for *ASSIGNMENT_INFO*.

- *NW_PO_BOX_ASSIGNMENT_INFO*
- *NW_PO_BOX_ASSIGNMENT_LEVEL*
- *NW_PO_BOX_ASSIGNMENT_TYPE*
- *NW_PO_BOX_DELIVERY_INSTALLATION_FULL*
- *NW_PO_BOX_DELIVERY_POINT*
- *NW_PO_BOX_INFO_CODE*
- *NW_PO_BOX_ISO_COUNTRY_CODE_2CHAR*
- *NW_PO_BOX_ISO_SCRIPT_CODE*
- *NW_PO_BOX_LOCALITY1_FULL*
- *NW_PO_BOX_MATCH_BLOCK_NUMBER*
- *NW_PO_BOX_MATCH_BUILDING_NAME*
- *NW_PO_BOX_MATCH_COUNTRY*
- *NW_PO_BOX_MATCH_FLOOR_NUMBER*
- *NW_PO_BOX_MATCH_LOCALITY*
- *NW_PO_BOX_MATCH_LOCALITY2*
- *NW_PO_BOX_MATCH_POSTCODE1*
- *NW_PO_BOX_MATCH_PRIMARY_DIRECTIONAL*
- *NW_PO_BOX_MATCH_PRIMARY_NAME*
- *NW_PO_BOX_MATCH_PRIMARY_NAME2*
- *NW_PO_BOX_MATCH_PRIMARY_NUMBER*
- *NW_PO_BOX_MATCH_PRIMARY_TYPE*
- *NW_PO_BOX_MATCH_REGION*
- *NW_PO_BOX_MATCH_STAIRWELL_NAME*
- *NW_PO_BOX_MATCH_UNIT_NUMBER*
- *NW_PO_BOX_MATCH_WING_NAME*
- *NW_PO_BOX_NW_FORMATTED_POSTCODE*
- *NW_PO_BOX_NW_POSTCODE_IN_SUPPORTED_FORMAT*

- [NW_PO_BOX_POSTCODE_FULL](#)
- [NW_PO_BOX_POSTCODE_IN_VALID_FORMAT](#)
- [NW_PO_BOX_PRIMARY_ADDRESS](#)
- [NW_PO_BOX_PRIMARY_NUMBER](#)
- [NW_PO_BOX_PRIMARY_SECONDARY_ADDRESS](#)
- [NW_PO_BOX_REGION1](#)
- [NW_PO_BOX_REGION1_FULL](#)
- [NW_PO_BOX_REGION2](#)
- [NW_PO_BOX_REGION2_FULL](#)
- [NW_PO_BOX_STATUS_CODE](#)

Parent topic: [Global Address Cleanse fields \[page 730\]](#)

Related Information

[Global Address Cleanse input fields \[page 731\]](#)

[Global Address Cleanse NW input fields \[page 740\]](#)

[Global Address Cleanse output fields \[page 745\]](#)

[Global Address Cleanse Suggestion Lists fields \[page 787\]](#)

21.6.5 Global Address Cleanse Suggestion Lists fields

Use the suggestion list input and output fields when you enable the suggestion list options in the Global Address Cleanse transform.

i Note

The Suggestion Lists feature does not support Chinese and Japanese addresses.

[Global Address Cleanse suggestion list input fields \[page 789\]](#)

The suggestion list feature in the Global Address Cleanse transform supports all Global Address Cleanse input fields in addition to the suggestion reply fields.

[Global Address Cleanse suggestion list output fields \[page 790\]](#)

Include the Global Address Cleanse output fields that have the *GENERATED_FIELD_CLASS* of SUGGESTION so that the transform outputs the suggestion list-specific information.

Parent topic: [Global Address Cleanse fields \[page 730\]](#)

Related Information

[Global Address Cleanse input fields \[page 731\]](#)

[Global Address Cleanse NW input fields \[page 740\]](#)

[Global Address Cleanse output fields \[page 745\]](#)

[Global Address Cleanse NW_PO_Box output fields \[page 786\]](#)

21.6.5.1 Global Address Cleanse suggestion list input fields

The suggestion list feature in the Global Address Cleanse transform supports all Global Address Cleanse input fields in addition to the suggestion reply fields.

Field name	Description
SUGGESTION_REPLY1-6	Contains the index number that corresponds to a specific last line suggestion, an address line suggestion, or secondary list suggestion. May also contain a street primary range or a street secondary range. i Note To use these fields, enable suggestion lists in the transform configuration. To use one suggestion reply input field to hold all of the suggestion replies rather than using all six reply fields, use the SUGGESTION_REPLY1 field and separate the replies with a pipe (). When using the SUGGESTION_REPLY1-6 fields for SAP applications for street and PO Box addresses, insert the following symbols to indicate whether the user has accepted changes made to the street address and when they are done with the street address: • asterisk plus (*+): User accepts the changes made to the street address up to the specified point and is done with the street address. • asterisk minus (*-): User does not accept the changes made to the street address up to the specified point and is done with the street address. For country suggestion lists The SUGGESTION_REPLY1-6 input field contains the two-character ISO country code prefixed with "CID:" under the following circumstances: • You enable the ENABLE_SUGGESTION_LISTS option in the COUNTRY_ID_OPTIONS group. • You enable the COUNTRY suggestion list component. The transform sends "CID:XX", where "XX" is the two-character ISO country code, to the SUGGESTION_REPLY1-6 input field. It then reassigns the address using the country code in this field.

21.6.5.2 Global Address Cleanse suggestion list output fields

Include the Global Address Cleanse output fields that have the [GENERATED_FIELD_CLASS](#) of SUGGESTION so that the transform outputs the suggestion list-specific information.

In the Global Address Cleanse transform, you enable components from the [SUGGESTION_LIST_COMPONENTS](#) group of options. The software outputs the suggestion information for each component to the [SUGGESTION_LIST](#) output field in an XML string. If you do not include the [SUGGESTION_LIST](#) output field, the software does not output the specific components that you choose.

Field name	Description
STATUS	Contains the suggestion status generated as a result of suggestion list processing. • A: Primary address line suggestions available. • AM: Follow-up primary address-line suggestions available. • B: Primary and secondary ranges are invalid. • C: Address not found. • F: Floor range is invalid. • L: Lastline suggestions available. • L1: Locality1 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine • L2: Locality2 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine. • L3: Locality3 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine. • L4: Locality4 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine. • N: No suggestions available. • PC: Postcode suggestions available. • R: Primary range is invalid. • R1: Region1 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine. • R2: Region2 list available. Applicable when you set the ENABLE_LASTLINE_DRILLDOWN option to YES. Applies only to Global Address engine. • S: Unit range is invalid. • U: Secondary address-line suggestions available. • UM: Follow up secondary address-line suggestions available.

Field name	Description
<i>ERROR</i>	Contains the error status code generated by suggestion list processing. Possible output values are 0- 6: • <i>0</i>: No suggestion selection error. • <i>1</i>: Blank suggestion selection or entry. • <i>2</i>: Invalid suggestion selection. • <i>3</i>: Invalid primary range. • <i>4</i>: Invalid floor range. • <i>5</i>: Invalid unit range. • <i>6</i>: Too many possible results to generate a suggestion list. Provide more information, such as a postal code, region, or locality.
<i>COUNT</i>	Contains a nonnegative number that represents the number of individual suggestion selections generated from querying the current record. If the transform does not generate a suggestion list for the input record, the value is 0. Your application developer uses this field to determine the number of suggestion selections to display to users of your custom application.
<i>SUGGESTION_LIST</i>	Contains a string of the combined suggestion list components that you enabled in the transform and the suggestion values. The format adheres to the <i>OUTPUT STYLE</i>, <i>DELIMITER</i>, and <i>FIELD_DELIMITER</i> values you set for <i>SUGGESTION_LIST_COMPONENTS</i> options.

Field name	Description
<i>MORE_SUGGESTIONS_AVAILABLE</i>	Contains an indicator that there are more suggestions to view: • <i>Y</i>: There are more suggestions to view. • <i>N</i>: There are no more suggestions to view. A <i>Y</i> value indicates that more than one page of data is available for the current address. Application developers use the reply fields to specify what page of data to retrieve by using the keyword GOTOPAGE(N), where “N” is the page to retrieve. It is not necessary to replace “N” with a page number in the reply fields, as the first page is always the default. ♣ Example When <i>MORE_SUGGESTIONS_AVAILABLE</i> = <i>Y</i>, the software enables a “page up” or “page down” mechanism to give the user access to all of the data. When the user selects page down, the address is reprocessed with a reply of GOTOPAGE(2). The actual number of pages available depends on the following factors: • The number of suggestions the software finds for the address. • Your settings in the transform options <i>MAX_NUMBER_ADDRESS_LINES</i> and <i>MAX_NUMBER_LASTLINES</i>.

21.7 USA Regulatory Address Cleanse fields

Map fields from upstream objects and transforms to USA Regulatory Address Cleanse fields so that the transform knows the type of data in each field and how to process the data.

The USA Regulatory Address Cleanse transform requires that you map fields on input and output. The process of mapping occurs when you configure the input and output fields in the Input Fields group and Output Fields group.

[USA Regulatory Address Cleanse input fields \[page 794\]](#)

The USA Regulatory Address Cleanse transform has recognized input fields to use in field mapping.

[USA Regulatory Address Cleanse output fields \[page 796\]](#)

Use the USA Regulatory Address Cleanse transform output fields for generated data and to add additional fields to your output.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[Match fields \[page 821\]](#)

21.7.1 USA Regulatory Address Cleanse input fields

The USA Regulatory Address Cleanse transform has recognized input fields to use in field mapping.

USA Regulatory Address Cleanse input field descriptions

Field name	Description
<i>ADDRESS_LINE</i>	Contains the delivery address line that includes information like the house number, street name, and unit information.
<i>CHECK_DIGIT</i>	Contains the check digit for an eleven-digit delivery-point bar code. Applicable only when the transform makes a full assignment. The transform provides the check digit for a 5-digit bar code when a 5-digit assignment is possible or when the address is undeliverable. When the address is unassigned, the transform basis the check digit on the unverified input Postcode1 (ZIP Code.)
<i>COUNTRY</i>	Contains the country name. i Note The USA Regulatory Address Cleanse transform processes only USA. Therefore, it does not attempt to make an assignment for addresses outside of the USA, USA possessions, territories, and protectorates.
<i>COUNTY_CODE</i>	Contains the three-digit county code. Numbers start at 001 within each state.

Field name	Description
DATA_SOURCE_ID	Contains an identification for the input source or list. Use this field to identify the source of an input set. If there are multiple lists on input, use this field to identify the list that an input record belongs to. The transform generates statistics for each unique value in this field when you include the field in your mapping and enable the GATHER_STATISTICS_PER_DATA_SOURCE option in the REPORT_AND_ANALYSIS group.
DELIVERY_POINT	Contains the two-digit DPBC code.
FAMILY_NAME1	Contains the family name, such as Smith or Jones.
FIRM	Contains the company name.
GIVEN_NAME1	Contains the given name, such as Robert or Catherine.
GIVEN_NAME2	Contains the second given name, also known as middle name.
LASTLINE	Contains the last line delivery information that can include all or some of the following fields: • LOCALITY1 • REGION1 • POSTCODE1 • POSTCODE2
LOCALITY1	Contains the city, town, or suburb.
LOCALITY2	Contains the urbanization information for Puerto Rico addresses.
LOT	Contains the Line of Travel information.
LOT_ORDER	Contains the one-character Line of Travel sortation order: • A: Ascending • D: Descending The transform requires LOT codes for non-automated Carrier Route (CART) presorting for Standard Mail and Enhanced Carrier Route (ECR) Subclass.
MULTILINE1-12	Contains a line of data from the input file. The data type can vary from record to record.
NAME	Contains the name of the person associated with the address.

Field name	Description
<i>POSTCODE_FULL</i>	Contains the complete postal code, either the ZIP10 with a hyphen or the ZIP9 without a hyphen.
<i>POSTCODE1</i>	Contains the five-digit primary ZIP Code. This field does not include the 4-digit ZIP4 Code.
<i>POSTCODE2</i>	Contains the four-digit portion of a ZIP Code, called ZIP4 or ZIP+4. The 4-digit ZIP Code is the code that follows the primary ZIP Code separated with a hyphen. For example, 54601-1234.
<i>POSTNAME</i>	Contains an honorary post name or a maturity post name. An honorary post name that indicates certification, academic degree, or affiliation. For example, CPA. A maturity post name indicates heritage. For example, Jr. and Sr.
<i>PRENAME</i>	Contains a prename, such as Mr., Ms., or Mrs.
<i>REGION1</i>	Contains the name of the state or province for this address.
<i>SORTCODE_ROUTE</i>	Contains the four-digit carrier route number.
<i>STAGE_ADDRESS_FLAG</i>	Contains information from the USPS stage file for NCOALink stage testing.
<i>STAGE_LASTLINE_FLAG</i>	
<i>STAGE_NAME_FLAG</i>	
<i>STAGE_RECORD_KEY</i>	
<i>SUGGESTION_REPLY1-5</i>	Contains an index number that corresponds to a specific last line suggestion or an address line suggestion. Optionally use these fields to input a street primary range or a street secondary range. With a value of 0, the transform ignores the suggestion list.
<i>SUGGESTION_START_SELECTION</i>	Contains the starting suggestion list number. If you leave the field blank, the transform uses the default value of 1.
<i>UNIT_NUMBER</i>	Contains the secondary address information such as the unit description or secondary number.

21.7.2 USA Regulatory Address Cleanse output fields

Use the USA Regulatory Address Cleanse transform output fields for generated data and to add additional fields to your output.

USA Regulatory Address Cleanse output field descriptions

Field name	Description
<i>ADDRESS_LINE</i>	Contains the complete standardized primary and secondary address line. Styles for suffixes, directionals, and unit designators depend on option settings in the transform. i Note If the output values don't fit within the length of the output field, the transform uses intelligent truncation.
<i>ADDRESS_LINE_REMAINDER1</i>	Contains extraneous data from the input address that the transform couldn't identify or that didn't belong in a standardized address line.

Field name	Description
<i>ADDRESS_TYPE</i>	Contains a two-character record-type indicator for the assigned address. Applicable for <i>DELIVERY</i> and <i>DUAL Generated Field Address Class</i>. The first character indicates the type of record in the address directory to which the address matched. The second character can be one of the following: • <i>D</i>: Default address type. If further information was available on input, the transform could make a finer level of address assignment. • <i>blank</i>: Input information was sufficient to make the finest level of address assignment. i Note When the transform can't assign an address, it provides an address-type indication based on the way it parsed the input data. This process isn't foolproof. The transform can indicate that it parsed a street, rural route, highway contract, general delivery, or PO Box. The following list contains descriptions for each first character with a <i>D</i> as the second character: • <i>FD</i>: Firm default. The transform didn't assign a firm-level Postcode2, but could do so if given more firm information. • <i>GD</i>: General delivery default. "General Delivery" was the only primary name listed for the Postcode1 in the postal directory. • <i>HD</i>: High-rise default. The transform assigned the Postcode2 for the entire building. If more information was available, the transform could make an assignment at the unit, floor, or wing level. ❖ Example The transform assigned the Postcode2 based on the building because the input suite or apartment number was out of range. • <i>RD</i>: Rural route or highway contract default. The transform assigned the Postcode2 for the entire route, but could make a finer assignment with the box number. • <i>SD</i>: Street default. The transform assigned a Postcode2 for the block, because there wasn't enough information on input, or there's not an existing Postcode2 for the building to make a finer level of assignment.

Field name	Description
	<i>UD</i>: Unique default. The transform assigned a unique Postcode1, because the owner of the Postcode1 didn't provide Postcode2 assignments or the transform couldn't find a match in the directories.
<i>ALIAS_TYPE</i>	Contains the alias-type indicator for the assigned address. Applicable for the output field attribute category <i>Generated Field Address Class</i>, type <i>OFFICIAL</i>. <i>A</i>: Input address matched an abbreviated street name. <i>B</i>: Input address matched the high-rise alternate default base record. <i>C</i>: Input address had a street name that was out of date. To get a new street name, convert the record to the preferred alias. <i>H</i>: Input address was an undesirable alternate that is subject to conversion to a USPS preferred street address (high-rise alternate). <i>O</i>: Input address had a street nickname or other alias. <i>P</i>: Input address was a preferred alias. <i>Blank</i>: Input address wasn't an alias or was unassigned. i Note <i>ALIAS_TYPE</i> describes the input address, not the output address.
<i>ANKLINK_RETURN_CODE</i>	Contains an ANKLink (attempted not known) return code. Valid values are: <i>77</i>: Transform found an ANKLink match. If <i>NCOALink_Return_Code</i> contains <i>A</i>, <i>91</i>, or <i>92</i>, obtain a new address from an NCOALink full service provider. <i>blank</i>: Transform didn't perform an NCOALink lookup or didn't find an ANKLink match. <i>ANKLINK_RETURN_CODE</i> is always blank for full service providers.
<i>AUDIT_DROPPED_SECONDARY</i>	Contains the dropped secondary address information. Use this field for audit testing or when the transform makes an ANKLink match.

Field name	Description
<i>AUDIT_PRENAME</i>	Contains the name data that the transform used to make an NCOALink match or an ANKLink match. Use for audit testing. In some cases, the name in the Audit name field isn't the same as the input name. For example, when the transform uses a nickname, alternate spelling, or initial. For firm matches, the Audit name fields contain a split version of the firm data.
<i>AUDIT_GIVEN_NAME1</i>	
<i>AUDIT_GIVEN_NAME2</i>	
<i>AUDIT_FAMILY_NAME1</i>	
<i>AUDIT_POSTNAME</i>	
<i>AUDIT_GENDER</i>	Contains the gender information. Use this field for audit testing or when the transform makes an ANKLink match.
<i>AUDIT_GENERAL</i>	Contains query data, result data, and hint bytes required by the USPS for Stage I and Stage II tests. Use this field for audit purposes only. This field is required for audits. The <i>AUDIT_GENERAL</i> field can also contain ANKLink return codes. For more information about the contents of the <i>AUDIT_GENERAL</i> field, see the Certification and Audit sections of the NCOALink User Technical Reference at https://postal-pro.usps.com/NCOALink_User_Tech_Info .
<i>AUDIT_PRIMARY_NAME</i>	Contains the primary name that NCOALink used for matching. Required for audits.
<i>AUDIT_RANGE</i>	Contains the range that NCOALink used for matching. Required for audits.
<i>AUDIT_SECONDARY_RANGE</i>	Contains the secondary range that NCOALink used for matching. Required for audits.
<i>AUDIT_TRUNCATED_GIVEN_NAME1</i>	Contains the truncated given name1 that NCOALink used for matching. Required for audits.
<i>AUDIT_TRUNCATED_GIVEN_NAME2</i>	Contains the truncated given name2 or middle name that NCOALink used for matching. Required for audits.
<i>AUDIT_UNIT</i>	Contains the unit data that NCOALink used for matching. Required for audits.

Field name	Description
<i>CARRIER_ROUTE_SORT_ZONE</i>	Contains the code that indicates the record eligibility for Standard Mail Automation Enhanced Carrier Route: • <i>A</i>: Eligible for carrier route rates. Merging is allowed. • <i>B</i>: Eligible for carrier route rates. Merging isn't allowed. • <i>C</i>: Not eligible for carrier route rates. Merging is allowed. • <i>D</i>: Not eligible for carrier route rates. Merging isn't allowed.
<i>CASS_ASSIGNMENT_TYPE</i>	Indicates the option that the transform used in making the assignment. • <i>0</i>: Non-CASS and DPV tie-break options were disabled or weren't used to make an assignment. • <i>1</i>: Inexact Postcode1 move assignment. • <i>2</i>: Input Postcode2 assignment. • <i>3</i>: Used DPV tie-breaking to make assignment. • <i>blank</i>: Transform couldn't assign input address.
<i>CASS_RECORD_TYPE</i>	Contains the record type required for posting on the CASS test. Populated for assigned records only. The valid record types include the following: • <i>F</i>: Firm • <i>G</i>: General delivery • <i>H</i>: High-rise • <i>P</i>: Post office box • <i>R</i>: Rural route or highway contract • <i>S</i>: Street
<i>CHECK_DIGIT</i>	Contains the check digit for delivery-point bar code. If the transform couldn't make a full postal code (ZIP+4) assignment, contains the check digit for a five-digit bar code.
<i>COUNT</i>	Contains the suggestion count generated when the transform looked up the current record. For example, if the current record didn't end processing with a suggestion list that needed resolution, then the value in this field is 0. The count is a non-negative value.
<i>COUNTY</i>	Contains the county name.

Field name	Description
COUNTY_CODE	Contains the Federal Information Processing Standard (FIPS) three-digit county code. Numbers are unique within states. Use count code information when you prepare a pre-sorted periodicals mailing.
COUNTY_NAME	Contains the full county name.
DELIVERY_POINT	Contains the two-digit Delivery Point Bar Code (DPBC).
DELIVERY_TYPE	Contains the type of postal facility: • A: Airport Mail Facility (AMF) • B: Branch Office • C: Community Post Office (CPO) • D: Area Distribution Center (ADC) • E: Sectional Center Facility (SCF) • F: Delivery Distribution • G: General Mail Facility (GMF) • K: Network Distribution Centers (NDC) • M: Money Order Unit • N: City/place name • P: Post Office (main) • S: Station • U: Urbanization (Puerto Rico only)
DISTRICT	Contains the district number for the U.S. House of Representatives.
DPV_CMRA	Contains the DPV Commercial Mail Receiving Agency (CMRA) component that the transform generated for this record. • L: Address triggered DPV locking. • N: Address isn't a CMRA. • Y: Address is a valid CMRA. • <i>blank</i>: Blank output value indicates that Enable DPV ValidationENABLE_DPV_VALIDATION option was set to NO, the software locked DPV processing, or the transform couldn't assign the input address.

Field name	Description
<i>DPV_DOOR_NOT_ACCESSIBLE</i>	Indicates that the address is designated as a Door Not Accessible (DNA) address. <i>Y</i>: Address is a DNA address. <i>N</i>: Address isn't a DNA address. <i>blank</i>: <i>ENABLE_DPV</i> is set to <i>NO</i>, <i>dpv_door_na.dir</i> isn't present, or there's an assignment error. The USPS uses DNA indicators to mark addresses where the carrier can't access the mail receptacle or door.
<i>DPV_DROP_INDICATOR</i>	Indicates that the input address is a DPV drop. <i>Y</i>: Input address is a DPV drop. <i>N</i>: Input address isn't a DPV drop. <i>blank</i>: <i>ENABLE_DPV</i> is set to <i>NO</i>, <i>dpv_drop_ind.dir</i> isn't present, or there's an assignment error.

Field name	Description
DPV_FOOTNOTE	Contains the DPV footnote or footnotes that are required for CASS certification. The footnotes indicate the results of the DPV assignment. The output field can contain one or more of the following footnotes (up to a maximum of 12 characters) based on the DPV assignment results: • AA: Input address matched to the ZIP+4 file. • A1: Input address didn't match to the ZIP+4 file. • BB: Both primary and secondary input address numbers matched to DPV/DSF2. Both primary and secondary input addresses were necessary to determine a valid delivery point. • CC: Input address primary number matched to DPV, but input secondary number didn't match. A secondary number isn't required. • C1: Input address primary number matched to DPV. Input secondary number didn't match to DPV. A secondary number is required. • F1: Input address matched to a military address. • G1: Input address matched to a general delivery address. • IA: Input address identified as an Informed Address. • M1: Input address primary number was missing. • M3: Input address primary number was invalid. • N1: Input address primary number matched to DPV. Input secondary number was missing but is required. • NL: NCOALink move address couldn't be DPV confirmed. NCOALink directory data didn't match DPV directory data exactly because the NCOALink directories are updated more frequently than the DPV directories. i Note The NL footnote is applicable for the MOVE_UPDATED field class for the output field attribute category GENERATED_FIELD_CLASS, type MOVE_UPDATED. • P1: Input address was missing the PO, RR, or HC Box number. • P3: Input address contained an invalid PO, RR, or HC Box number. • PB: Input address identified as a PBSA.

Field name	Description
	<i>R1</i>: Input address matched to a CMRA, but the PMB designator wasn't present (PMB or #). <i>RR</i>: Input address matched to a CMRA, and PMB designator was present (PMB or #). <i>R7</i>: Input address assigned to a phantom route of R777 or R779. <i>TA</i>: Input address primary number matched to DPV by dropping the trailing alpha. <i>UI</i>: Input address matched to a unique postcode. Note The transform always posts the DPV footnotes in the same order. The software combines footnotes to fully explain the DPV assignment. Therefore, the <i>DPV_FOOTNOTE</i> field isn't always 12 characters in length.
<i>DPV_NON_DELIVERY_DAYS</i>	Indicates whether there are days when there's no mail delivery. Values include the following: <i>Y</i>: There are days when there's no mail delivery. <i>N</i>: There are no non-delivery days. <i>blank</i>: DPV wasn't enabled or both of the directories for non-delivery days were missing. The following are the directories for non-delivery days: <code>dpv_non_delivery_days_flag.dir</code> <code>dpv_non_delivery_days.dir</code>
<i>DPV_NON_DELIVERY_DAY_SUN</i>	Contains a non-delivery day Indicator for the day of the week.
<i>DPV_NON_DELIVERY_DAY_MON</i>	
<i>DPV_NON_DELIVERY_DAY_TUE</i>	Valid values:
<i>DPV_NON_DELIVERY_DAY_WED</i>	<i>Y</i>: Indicates a non-delivery day.
<i>DPV_NON_DELIVERY_DAY_THU</i>	<i>N</i>: Indicates that mail is delivered on that particular day but not on all days.
<i>DPV_NON_DELIVERY_DAY_FRI</i>	<i>blank</i>: Indicates that the DPV option is disabled or both of the directory files for non-delivery days are missing.
<i>DPV_NON_DELIVERY_DAY_SAT</i>	

Field name	Description
<i>DPV_NO_SECURE_LOCATION</i>	Indicates that the address has a No Secure Location (NSL) indicator. <i>Y</i>: Address was a designated NSL. <i>N</i>: Address wasn't a designated NSL. <i>blank</i>: <i>ENABLE_DPV</i> was set to <i>NO</i>, The NSL directory wasn't present, or there was an assignment error. The USPS uses NSL indicators to mark addresses where there's no security. For example: - There's no one at the address to collect the mail. - The storefront doesn't have a mail receptacle. - The business is closed on certain days. The directory for NSL is <code>dpv_no_secure_loc.dir</code>.
<i>DPV_NOSTATS</i>	Contains a value that indicates whether the address is a vacant property, it receives mail as a part of a drop, or it doesn't have an established delivery yet. Output values include the following: <i>Y</i>: Input address was flagged as No Stat in DPV data. The value in the <i>DPV_NOSTATS_REASON_CODE</i> output field provides the reason for the No Stat status. <i>N</i>: Input address wasn't flagged as No Stat in DPV data. <i>blank</i>: The transform didn't look up the input address.

Field name	Description
DPV_NOSTATS_REASON_CODE	Contains the reason code that indicates why the output field DPV_NOSTATS returned a Y. The following are the possible reason codes: • 01: Input address was an Internal Drop Address (IDA), which is an address that doesn't receive mail delivery directly from the USPS. Instead, mail is delivered to a drop address that services the IDA. • 02: Input address was a Contract Delivery Service (CDS) address, which is either one of the following: • New construction for which delivery hasn't been established. • Rural/CDS/HCR where the delivery point has been unoccupied for more than 90 days. • 04: Input address is a College/Military Zone (CMZ) address, or other type of address where the ZIP+4 records have been incorporated into the data. • 05: Input address is a regular No Stat address, which isn't a possible delivery address because of one of the following reasons: • Address is on an R777 or R779 phantom route. • The PO Box has never been rented. • The PO Box has been declared unrentable. • 06: Secondary input address required. • Other: The No Stat reason isn't one of the other reasons.
DPV_PBSA	Indicates whether the address is a PBSA (Post Office Box Street Address) listed in the PBSA directory: • Y: Address is a PBSA. • N: Address isn't a PBSA. • <i>blank</i>: DPV isn't enabled or there was an error.

Field name	Description
<i>DPV_STATUS</i>	Contains a DPV status component that the transform generated for this record. <i>Y</i>: Both the primary and secondary (if present) input addresses DPV confirmed. <i>S</i>: Input address was DPV confirmed for the primary number only. The secondary number was present but invalid, or a single trailing alpha on a primary number was dropped to make a DPV match. <i>D</i>: Input address was DPV confirmed for the primary number only. The secondary number was missing. <i>N</i>: Input address failed to DPV confirm because the primary number was either missing or invalid. <i>L</i>: Input address triggered DPV locking. <i>blank</i>: Either DPV was disabled, the transform couldn't assign the input address, or DPV processing was disabled by the software because the input address was a false positive.
<i>DPV_STATUS_ENHANCED</i>	Contains a DPV enhanced status code generated for this record. <i>Y</i>: Input address was DPV confirmed for both primary and secondary numbers necessary to determine a valid delivery point. <i>S</i>: Input address was DPV confirmed for primary address only. Secondary data was present but didn't DPV confirm, or a single trailing alpha on a primary number was dropped to make a DPV match. Secondary information was required. <i>D</i>: Input address was DPV confirmed for the primary number only. Secondary information was missing. <i>N</i>: Both the input primary address and the secondary number (if present) failed to DPV confirm. <i>R</i>: Input address was DPV confirmed, but assigned to phantom route R777 or R779 and USPS delivery isn't provided. <i>L</i>: Input address triggered DPV locking. <i>blank</i>: Input address not presented to DPV.

Field name	Description
<i>DPV_THROWBACK_INDICATOR</i>	Indicates that the input address is a DPV PO Box Throwback. <i>Y</i>: Input address was a DPV PO Box Throwback. <i>N</i>: Input address wasn't a DPV PO Box Throwback. <i>blank</i>: <i>ENABLE_DPV</i> was set to <i>NO</i>, <i>dpv_po_box_throwback.dir</i> wasn't present, or there was an assignment error. The USPS uses the DPV PO Box Throwback indicator with both the DSF2 and DPV programs. The indicator identifies street addresses where the delivery point is a street address but mail delivery is to the customer PO Box address.
<i>DPV_VACANT</i>	Contains a vacant address indicator. <i>Y</i>: Input address was vacant. <i>N</i>: Input address wasn't vacant. <i>blank</i>: Input address wasn't looked up.
<i>ERROR</i>	Contains the error status code generated by the suggestion list process. <i>0</i>: No suggestion selection errors. <i>1</i>: Necessary selection information was blank. For example, the transform generated a last line suggestion list, but a last line selection input field was missing data. <i>2</i>: Suggestion selection invalid. For example, user selected 8 but there were only 5 suggestions. <i>3</i>: Suggestion entry in input field invalid. <i>4</i>: Suggestion range in input field invalid. <i>5</i>: Suggestion secondary range in input field invalid.
<i>EWS_Match</i>	Contains the results of the EWS (Early Warning System) match. <i>T</i>: True. Input address was in the EWS directory and was an EWS match. <i>F</i>: False. Input address wasn't in the EWS directory. <i>blank</i>: EWS wasn't enabled.
<i>EXTRA1-10</i>	Contains non-address data above or below the address data in the input address block. Available only for input data mapped to Multiline input fields.

Field name	Description
<i>EXTRANEOUS_SECONDARY_ADDRESS_DATA</i>	Contains data from the <i>EXTRANEOUS_SECONDARY_UNIT_NUMBER</i> output field and the <i>EXTRANEOUS_SECONDARY_NON_POSTAL</i> output field respectively. The transform also places additional pound unit designator “#” data in the remainder or extra components, including invalid secondary address data.
<i>EXTRANEOUS_SECONDARY_NON_POSTAL</i>	Contains data that the transform determines is private mail box data. Determined based on position in the address line and other information such as a pound unit designator “#”. Extraneous data can include invalid secondary address data.
<i>EXTRANEOUS_SECONDARY_UNIT_NUMBER</i>	Contains data that the transform determined is secondary range data based on position in the address line and other information in the address. Extraneous data can include invalid secondary address data.
<i>FAULT_CODE</i>	Contains a code that indicates why the transform couldn't assign the address. Field is blank when the transform assigned the address.
<i>FAULT_OR_STATUS_CODE</i>	Contains a fault code that indicates why the transform couldn't assign address, or a status code that indicates how the input address differed from the assigned address.
<i>FINANCE_AREA_POSTCODE</i>	Contains the lowest Postcode1 within a Finance Number.
	i Note Finance Numbers link data to a single post office or postmaster.
<i>FIRM</i>	Contains a firm name from the postal directory. If firm name wasn't in the postal directory, contains the firm name from the input record.
	i Note Not applicable for input data in multilines. If the firm name isn't in the postal directory, the transform can't reliably identify the firm name from input data. The postal directory can contain firm names that are shortened or spelled differently. If you don't want to use the postal directory firm names on your mail piece, retrieve the firm name from your input data.
<i>FOREIGN_CODE</i>	Specifies whether the address is foreign or domestic. <i>F</i>: Foreign address <i>blank</i>: Domestic (U.S.) address

Field name	Description
<i>FULL_ADDRESS</i>	Contains a complete address line, including secondary address and dual address line data (street and postal). i Note If you set the <i>INCLUDE UNUSED ADDRESS LINE DATA</i> option in the <i>STANDARDIZATION OPTIONS</i> group to <i>YES</i>, can contain invalid secondary address information. i Note If an output value doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>INTERMEDIATE_CODES</i>	Contains codes required by the USPS for NCOALink certification and audit testing.
<i>LACSCODE</i>	Contains LACS (Locatable Address Conversion System) indicator. <i>T</i>: Input address needs 9-1-1 conversion from box to street address and then needs to be submitted to a LACS vendor. <i>F</i>: Input address didn't need conversion. <i>blank</i>: Input address wasn't assigned.
<i>LACSLINK_INDICATOR</i>	Contains the conversion status of addresses processed with LACSLink. <i>Y</i>: Input address converted by LACSLink and assigned a LACSLink_Return_Code of A. <i>N</i>: Input address looked up but not converted by LACSLink. <i>F</i>: Input address was false-positive. <i>S</i>: Input address converted by LACSLink but secondary information was dropped. <i>blank</i>: No LACSLink lookup attempted.

Field name	Description
<i>LACSLINK_QUERY</i>	Contains the pre LACSLink-converted address, or is blank when there's no attempted LACSLink lookup. Can contain a pound sign (#) for the unit designator when the input address contains both a unit designator and a secondary unit. The software populates this field under the following circumstances: - You enable LACSLink in the transform. - The software attempts a LACSLink lookup. The Pre LACSLink-converted address is in the standard USPS Publication 28 format.
<i>LACSLINK_RETURN_CODE</i>	Contains the match status for LACSLink processing. A: LACSLink record matched and the address data fields contained a LACSLink address. 00: No LACSLink record matched and no converted address. 09: LACSLink record matched to an old address that is a "high-rise default" address. No new address provided. 14: LACSLink record matched but couldn't convert to a deliverable address. 92: LACSLink record matched only after dropping the secondary number from input address. <i>blank</i>: No LACSLink lookup attempted.
<i>LASTLINE</i>	Contains the locality, region, and postal code on one line. i Note If an output value doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>LOCALITY1</i>	Contains the locality. Canada and USA: Contains the locality preferred by the postal authority. Other countries: Contains the city, town, locality, or suburb. i Note If an output value doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.

Field name	Description
<i>LOCALITY1_ALTERNATE</i>	Contains the preserved input Locality1 when both of the following situations apply: - The postal authority doesn't recognize the Locality1 as valid for the input address line. - The transform changed the original Locality1 because of address assignment rules. Or, contains the default Locality1 for the assigned finance area under one of the following situations: - There wasn't an input Locality1. - There was an input Locality1 but it wasn't valid in the assigned finance area.
<i>LOCALITY1_LLIDX</i>	Contains the Locality1 (city) name that is more geographically precise than the <i>LOCALITY1_OFFICIAL</i>. LLIDX (last-line index) is a USPS number that ties a ZIP+4 record to a particular city, state, and ZIP. i Note If an output value doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>LOCALITY1_NAME</i>	Contains the City, town, or suburb name. The transform always preserves Locality1 names that are marked as invalid for mailing by the USPS, regardless of the values set for the <i>PRESERVE_PLACE_NAME</i> and <i>ASSIGN_WITH_INPUT_LOCALITY</i> options.
<i>LOCALITY1_OFFICIAL</i>	Contains the standardized locality name or converted locality name. The transform outputs a converted locality name when the input city name is a USPS-invalid city name, regardless of how you set the <i>PRESERVE_PLACE_NAME</i> option.
<i>LOCALITY1_OFFICIAL_ABBR</i>	Contains the official USPS abbreviation for the city name, when available. The field is blank when the full city name is fewer than 13 characters, or the full city name is longer than 13 characters but there's no official USPS abbreviation.
<i>LOCALITY2</i>	Contains the additional city, town, locality, or suburb. USA: Contains the Puerto Rico urbanization name.

Field name	Description
<i>LOCALITY2_OFFICIAL</i>	Contains the Puerto Rico urbanization name.
<i>LOT</i>	Contains the line of travel number.
<i>LOT_ORDER</i>	Contains the order in which the Line of travel is sorted. • <i>A</i>: Ascending • <i>D</i>: Descending
<i>MATCHED_ADDRESSLINE_INDICATOR</i>	Indicates if an address line matches a ZIP+4 record in the directories. • <i>T</i>: Address line matched a ZIP+4 record. • <i>F</i>: Address line didn't match a ZIP+4 record.
<i>MATCHED_LASTLINE_INDICATOR</i>	Indicates if a last line matches a City—ZCF record in the directories. • <i>T</i>: Last line matched a City—ZCF record. • <i>F</i>: Last line didn't match a City—ZCF record.
<i>MOVE_EFFECTIVE_DATE</i>	Contains the effective date of a move from NCOALink or ANKLink. Output in the format yyyy-mm. Required for USPS audit. → Tip To use this date in a function or to post it to an output file, you may have to convert the date format to mm/dd/yyyy.
<i>MOVE_TYPE</i>	Contains an indicator that the move type is for a business, family, or an individual. The transform populates this field when the input matches with information in NCOALink or ANKLink directories. • <i>B</i>: Business moved, matched by company name. • <i>F</i>: Family move, matched by last name. • <i>I</i>: Individual move, matched by first and last name.

Field name	Description
MULTILINE1-12	Contains any data type, which can vary from record to record. i Note Can contain invalid secondary address information when you set the INCLUDE_UNUSED_ADDRESS_LINE_DATA option in the STANDARDIZATION_OPTIONS group to YES. i Note If output is address data, and the output doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
NAME	Contains the name of a person associated with the address.
NCOALINK_HINT_BYTE	Used or audit testing only.
NCOALINK_RETURN_CODE	Contains the NCOALink return code. Also populated when the input record matches a record in ANKLink. To populate this field, set the LIST_PROCESSING_MODE to one of the three available options: • CHANGE_OF_ADDRESS • STATISTICS_ONLY • RETURN_CODES_ONLY The transform includes a brief description of the return codes in the NCOALink Processing Summary report. To include more detailed return code descriptions in the report, enable the GENERATE_RETURN_CODE_DESCRIPTIONS option in the NCOALING_REPORT_OPTIONS group.
STAGE_TEST_RECORD	Contains values for stage testing for NCOALink, CASS, DSF2 Augment, DSF2 Sequence, and DSF2 Invoice self-certifications. The transform populates the values of this field automatically to match the format required for stage testing.
NON_CASS_FIRM	Contains the firm name resulting from processing missing or invalid firm information using the input ZIP+4.

Field name	Description
<i>NON_CASS_SECONDARY_ADDRESS</i>	Contains the secondary address match resulting from processing missing or invalid secondary address with the input ZIP+4. i Note If output value doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>NON_CASS_UNIT</i>	Contains the unit designator match resulting from processing missing or invalid unit designator with the input ZIP+4.
<i>NON_CASS_UNIT_NUMBER</i>	Contains the unit designator match resulting from processing missing or invalid unit designator information with the input ZIP+4.
<i>NON_POSTAL_SECONDARY_ADDRESS</i>	Contains the complete non-postal secondary address. Non-Postal means that the mail is delivered through a private mailbox (PMB) company rather than the USPS. For example, PMB 10 or # 10.
<i>NON_POSTAL_UNIT</i>	Contains the non-postal unit designator. Non-Postal means that the mail is delivered through a private mailbox (PMB) company rather than the USPS. For example, output of "PMB" or "#".
<i>NON_POSTAL_UNIT_NUMBER</i>	Contains the non-postal secondary range, which is the private mailbox (PMB) number without the designator. Non-Postal means that the mail is delivered through a private mailbox company rather than the USPS.
<i>PARSED_FIRM</i>	Contains the firm name when the change of address is based on a firm name. Also applicable for ANKLink matches.
<i>PO_BOX_ONLY_POSTCODE</i>	Indicates whether the ZIP Code is listed in the postal directory as a ZIP Code for post office box delivery only. • <i>Y</i>: ZIP Code is for a PO Box only zone. • <i>N</i>: ZIP Code isn't for a PO Box only zone. • <i>Blank</i>: ZIP Code isn't assigned.
<i>POSTAL_BOX_NUMBER</i>	Contains the post office box number.
<i>POSTCODE_FULL</i>	Contains the complete ZIP10, which is the ZIP Code and the ZIP+4 separated with a hyphen. For example, 54601-1234.

Field name	Description
<i>POSTCODE_FULL_NO_HYPHEN</i>	Contains the complete ZIP9, which is the ZIP Code and ZIP+4 without a hyphen. For example, 546011234.
<i>POSTCODE_TYPE</i>	Contains the assigned ZIP Code type. • <i>M</i>: Military • <i>U</i>: Unique. Specific to a university, large firm, or other institution. • <i>Blank</i>: Ordinary ZIP Code or the ZIP Code wasn't assigned.
<i>POSTCODE1</i>	Contains the 5-digit ZIP Code. Doesn't include the 4-digit ZIP+4.
<i>POSTCODE1_CHANGE_IND</i>	Indicates whether postal code realignment affected the address. • <i>T</i>: True. The transform corrected the postal code and the locality, if applicable. • <i>F</i>: False. The transform didn't correct the postal code. • <i>Blank</i>: Address not corrected.
<i>POSTCODE2</i>	Contains the four-digit, ZIP+4 Code. Located after the primary postal code on a mail piece that is either preceded with a hyphen or not. For example, for the full ZIP Code 54601-1234, the value is "1234."
<i>PRE_SUITELINK_DELIVERY_POINT</i>	Contains the 2-digit delivery point barcode assigned before the transform performs SuiteLink processing.
<i>PRE_SUITELINK_POSTCODE1</i>	Contains the ZIP Code assigned before the transform performs SuiteLink processing.
<i>PRE_SUITELINK_POSTCODE2</i>	The ZIP+4 for a high-rise default or street default record assigned before the transform performs SuiteLink processing.
<i>PRE_SUITELINK_UNIT_DESCRIPTION</i>	Contains the unit designator assigned before the transform performs SuiteLink processing. <i>Blank</i> when the transform didn't assign secondary information.
<i>PRE_SUITELINK_UNIT_NUMBER</i>	Contains the secondary range information assigned before the transform performs SuiteLink processing. <i>Blank</i> when the transform didn't assign secondary information.

Field name	Description
<i>PRIMARY_ADDRESS</i>	Contains the primary address line, such as the street address or post office box. Doesn't include secondary address information such as apartment. If you enable the <i>USE_USPS_PRIMARY_NAME_ABBREVIATION</i> option, the transform uses the USPS Primary Name abbreviation first. i Note If output doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>PRIMARY_NAME1</i>	Contains the primary street name description. i Note If output doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>PRIMARY_NUMBER</i>	Contains the house or building number.
<i>PRIMARY_POSTFIX1</i>	Contains the abbreviated directional that follows the street name. For example, N, S, NW, or SE.
<i>PRIMARY_POSTFIX1_LONG</i>	Contains the full spelled-out directional that follows the street name. For example, North or South.
<i>PRIMARY_PREFIX1</i>	Contains the abbreviated directional that precedes a street name. For example, N, S, NW, or SE.
<i>PRIMARY_PREFIX1_LONG</i>	Contains the full spelled-out directional that precedes a street name. For example, North or South.

Field name	Description
<i>PRIMARY_SECONDARY_ADDRESS</i>	Contains the primary address and secondary address on one line. Doesn't include remainder data. The software outputs this line as if the <i>INCLUDE_UNUSED_ADDRESS_LINE_DATA</i> option is set to <i>NO</i>. When the <i>INCLUDE_UNUSED_ADDRESS_LINE_DATA</i> option is set to <i>NO</i>, the output doesn't include invalid secondary address line information. i Note If output doesn't fit within the length of the output field, the transform truncates the data using intelligent truncation.
<i>PRIMARY_TYPE1</i>	Contains the abbreviated street type, such as St, Ave, or Pl.
<i>PRIMARY_TYPE1_LONG</i>	Contains the full spelled-out street type, such as Street or Avenue.
<i>QSS_DEFAULT</i>	Indicates whether the record qualifies as a default match instead of a higher level of assignment. <i>T</i>: True. Record qualifies as a default match instead of a higher level of assignment. <i>F</i>: False. Record doesn't qualify as a default match instead of a higher level of assignment.
<i>RDI_INDICATOR</i>	Indicates whether the address is residential. <i>Y</i>: Residential address. <i>N</i>: Nonresidential address.
<i>REGION1</i>	Contains the state, province, territory, or region.
<i>RURAL_ROUTE_BOX_NUMBER</i>	Contains the rural route box number.
<i>RURAL_ROUTE_NUMBER</i>	Contains the rural route number.
<i>SECONDARY_ADDRESS</i>	Contains the building name, floor, and room number in one field.

Field name	Description
SORTCODE_POSTCODE	Contains the Federal Information Processing Standards (FIPS) code for state and county. Combines the 2-digit state code with the 3-digit county code. Note U.S. territories, possessions, or protectorates such as Puerto Rico, the U.S. Virgin Islands, or the Pacific Islands don't have FIPS state digits.
SORTCODE_ROUTE	Contains the four-digit carrier route.
STATUS	Specifies the suggestion status generated when the software looks up the record and processes suggestions. A: Suggestion processing ended because an address suggestion list needed resolution. L: Suggestion processing ended because a last line suggestion list needed resolution. N: No suggestion list processing and no generated suggestion lists. R: Primary range invalid for the selected address suggestion. S: Secondary range invalid for the selected address suggestion. U: Secondary address invalid for the selected address suggestion.
STATUS_CODE	Contains a code that indicates how the input address differs from the assigned address. <i>Blank</i> when the address is unsigned.
SUGGESTION_LIST	Contains a string of suggestion list component values. Include suggestion list components by selecting YES for the options in the Suggestion List Components SUGGESTION_LIST_COMPONENTS group.

Field name	Description
<i>SUITELINK_RETCODE</i>	Contains the match status code for SuiteLink processing. • <i>A</i>: SuiteLink record match. Secondary information exists and the transform assigned SuiteLink information to the record. • <i>00</i>: No SuiteLink record match. Transform attempted SuiteLink lookup but didn't find a matching record. • <i>Blank</i>: No SuiteLink lookup performed because of one of the following situations: • The address isn't a high-rise default according to CASS. • The address doesn't contain a firm.
<i>UNDELIVERABLE_INDICATOR</i>	Indicates whether the record has a deliverable address. • <i>T</i>: Address is tagged by the USPS as unsuitable for mail delivery. For example, address is for a cemetery. • <i>F</i>: Address was either not matched to a ZIP+4 record or was matched to a ZIP+4 record that indicates the address is suitable for mail delivery.
<i>UNIT_DESCRIPTION</i>	Contains the unit description, such as #, Apartment, or Flat.
<i>UNIT_DESCRIPTION_DIRECTORY</i>	Contains the unit designator from the ZIP+4 directory. The value is blank when the transform doesn't find a unit designator from the ZIP+4 directory.
<i>UNIT_NUMBER</i>	Contains the unit number, such as 100 in the unit APT 100.

21.8 Match fields

There is no single list of Match input fields, rather, input field descriptions appear in the various matching strategy topics in the Match Transform topics. Output fields vary based on the match setup.

Use input fields from the physical and logical sources connected to the match transform, and select them as you set up break groups and match criteria.

Before you start mapping Match fields, ensure that you have complete understanding about matching techniques and how the Match transform works.

[Match transform output fields \[page 822\]](#)

The Match transform generates output fields for various types of operations run either pre- or post-match.

Parent topic: [About Data Quality fields \[page 687\]](#)

Related Information

[Data Quality support for native data types \[page 687\]](#)

[Output field definition options \[page 688\]](#)

[Data type support \[page 691\]](#)

[Data Cleanse fields \[page 692\]](#)

[Geocoder fields \[page 714\]](#)

[Global Address Cleanse fields \[page 730\]](#)

[USA Regulatory Address Cleanse fields \[page 793\]](#)

[Match transform \[page 586\]](#)

21.8.1 Match transform output fields

The Match transform generates output fields for various types of operations run either pre- or post-match.

The following tables list the output field in the following groups:

- Match level
- Input source operation
- Group statistics post-match, generate source statistics from input sources
- Group statistics post-match, generate source statistics from source values
- Source group operation
- Group statistics operation
- Output flag election

21.8.1.1 Match-level output fields

Use match-level output fields when you map your output schema in the Match transform.

The following table contains descriptions for the match-level output fields generated by the transform.

Output field name (Match)	Description
<i>GROUP_NUMBER</i>	Contains the records that belong to the same match group and that share the same group number. The group numbers start with the number 1. Unique records have a blank group number. If you use association matching in your job, map <i>GROUP_NUMBER</i> on output because the Associate transform uses it.

Output field name (Match)	Description
<i>MATCH_CRITERION</i>	Contains the name of the criteria or pattern that decided that the records match. Applicable when <i>MATCH_TYPE</i> contains <i>R</i>. The field is blank if the <i>MATCH_TYPE</i> output field does not contain <i>R</i>. <i>i</i> Note Pattern matching currently not available.
<i>MATCH_LEVEL</i>	Contains the name of the match level the transform used for this record.
<i>MATCH_SCORE</i>	Contains one of the following types of match score. - The criteria or pattern similarity score when the <i>MATCH_TYPE</i> is <i>R</i>. - The total weighted score when the <i>MATCH_TYPE</i> is <i>W</i>. - Blank if the record is a driver record (<i>MATCH_TYPE</i> of <i>D</i>) or when the records are unique. <i>i</i> Note Pattern matching is not currently available.
<i>MATCH_STATUS</i>	Contains the status of the record in relationship to the match group. <i>D</i>: Record is a driver in a match group. <i>P</i>: Record is a passenger in a match group. <i>U</i>: Record is unique. <i>B</i>: Record is both a driver and a passenger. Applicable when you set the <i>MATCH_MODE</i> option to <i>MULTI_DRIVER</i>. <i>i</i> Note Currently, the <i>MATCH_MODE</i> option is not available.

Output field name (Match)	Description
<i>MATCH_TYPE</i>	Contains the type of match that brought the record into a match group. <i>blank</i>: Record is unique. <i>D</i>: Driver record. <i>R</i>: Rule (criteria) or pattern. <i>W</i>: Weighted score. The total weighted score met the Weighted match score.

i Note

Pattern matching is not currently available.

21.8.1.2 Input Source Operation output fields

Use input source output fields when you use the Input Source Operation in the Match transform.

The following table contains descriptions of the output fields available when you use an Input Source Operation in the transform.

Output field name (Match-Input Source)	Description
<i>SOURCE_GROUP_NAME</i>	Contains the name of the Source Group that the current record belongs to. If a record does not belong to any Source Group, then the field is empty.
<i>SOURCE_NAME</i>	Contains the name of the input source to which the current record belongs.
<i>SOURCE_TYPE</i>	Contains the type of source that the current record belongs to. <i>N</i>: Normal source. <i>P</i>: Suppress source. <i>S</i>: Special source.

21.8.1.3 Source statistics from input sources post-match output fields

Post-match output fields are available for group statistics when you enable the generation of source statistics from input sources.

The following table contains descriptions for the output fields for group statistics post-match operations when you enable [*GENERATE_SOURCE_STATISTICS_FROM_INPUT_SOURCES*](#).

The fields are in addition to the fields generated by the pre-match group statistics operation.

Output field name (Match–Input Source)	Description
<i>GROUP_SOURCE_APPEARANCE</i>	Contains the order in which the input source appears in this match group. The transform assigns the order number sequentially with the first group that appears in the match group receiving a 1. Records that come from the same input source receive the same order number. Unique records have a value of 0.
<i>GROUP_SOURCE_ORDER</i>	Contains the order of the records within the match group that have the same <i>GROUP_SOURCE_APPEARANCE</i> value. The transform assigns the order number sequentially with the first occurrence receiving a 1. The transform assigns a 0 to unique records.
<i>GROUP_SOURCE_TYPE</i>	Contains the type of source for the records in the match group. • <i>M</i>: Records are from more than one input source, excluding records from special sources. • <i>P</i>: Records are from a suppress source. • If the master record comes from a suppression source, then all records in the match group have a <i>P</i> • If the master record comes from a normal or special source, then the suppression record and all records after it, have a <i>P</i>, but the records before the suppression record have an <i>M</i> or <i>S</i>. • <i>S</i>: Records are from a single input source. • <i>blank</i>: Records are unique.
<i>MULTI_SOURCE_COUNT</i>	Contains the number of sources represented in the match group, excluding the following source types: • Special sources • Suppress sources • Normal sources that follow a suppress source in the match group The values of this field could range from 0 to the number of records in the match group. For unique records: • If a unique record is from a normal source, it has a value of 1. • If a unique record is from a special or suppression source, it has a value of 0.

Output field name (Match–Input Source)	Description
<i>SOURCE_COUNT</i>	Specifies the number of sources represented in the match group, regardless of the source types. The values of this output field can range from 1 to the number of records in the match group. Unique records have a value of 1.

21.8.1.4 Source Group Operation output fields

The software generates specific output fields for Source Group Operation in the Match transform.

The following table contains descriptions of Source Group Operation output fields.

Output field name (Match–Source Group)	Description
<i>GROUP_SOURCE_GROUP_APPEARANCE</i>	Contains the order in which the source group appears in this match group. The transform assigns 1 to the first group that appears in the match group. The numbering continues sequentially. Records that come from the same source group receive the same value. Unique records receive a value of 0. Records in a match group that are not assigned to a source group also receive a value of 0.
<i>GROUP_SOURCE_GROUP_ORDER</i>	Contains the order of the records within the match group that have the same <i>GROUP_SOURCE_GROUP_APPEARANCE</i> value. The transform assigns 1 to the first occurrence. The numbering continues sequentially. Unique records receive a value of 0. Records in a match group that are not assigned to a source group also receive a value of 0.
<i>SOURCE_GROUP_COUNT</i>	Contains the number of source groups represented in the match group. The transform does not count records in the match group that do not belong to a source group. Values range from 0 to the number of records in the match group. The transform assigns either 0 or 1 to unique records.
<i>SOURCE_GROUP_NAME</i>	Contains the name of the source group to which the current record belongs. If a record does not belong to any source group, the field is empty.

21.8.1.5 Group Statistics Operation output fields

The software generates specific output fields when you enable Group Statistics Operation in the Match transform.

The following table contains descriptions for the fields generated for Group Statistics Operations.

Output field name (Match-group statistics)	Description
<i>GROUP_COUNT</i>	Contains the total number of records in the match group. Unique records have a value of 1.
<i>GROUP_ORDER</i>	Contains the order number for the group in which the record belongs. • <i>1</i>: Master record. • <i>2</i> through the number of records in match group: Sub-ordinate records. • <i>0</i>: Unique record. Control the group order by including Prioritization operation in the Post Match processing group.
<i>GROUP_RANK</i>	Contains the record type the record has in the group. • <i>M</i>: Master record • <i>S</i>: Subordinate record • <i>blank</i>: Unique record
<i>GROUP_TYPE</i>	Specifies whether a record contributed to the source count, and if so, whether there were other sources represented in the match group. Note Records from multiple sources are not counted toward a multiple-source match group. • <i>M</i>: Multiple sources. • <i>S</i>: Single source. • <i>P</i>: At least one record from a Suppression source is included in the match group. • If the master record comes from a suppression source, then all records in the match group have a <i>P</i>. • If the master record comes from a normal or special source, then the suppression record, and all records after it, has a <i>P</i>, but the records before the suppression record have an <i>M</i> or <i>S</i>.

Output field name (Match-group statistics)	Description
<i>SOURCE_COUNT</i>	Contains the number of logical sources in this match group.
<i>SOURCE_ID</i>	Contains the logical source identification. In most cases, the logical source value is the input source value. In other cases, it is the default logical source value.
<i>SOURCE_TYPE_ID</i>	Contains the type of logical source. • <i>N</i>: Normal source • <i>P</i>: Suppress source • <i>S</i>: Special source

21.8.1.6 Output Flag Selection output fields

The software generates specific output fields when you include an output flag selection operation in the Match transform.

The following table contains descriptions of the output fields generated by the Match transform for output flag selection operation.

Output field name (Match-Output flag)	Description
<i>SELECT_RECORD</i>	Specifies whether the current record should be selected based on the settings in the object. • <i>Y</i>: Transform selects the record. • <i>N</i>: Transform doesn't select the record.

22 Data Quality codes

Each transform generates codes that you can use to learn more about the data quality processes on your data.

Codes appear in the applicable output fields.

[Data Cleanse codes \[page 829\]](#)

The Data Cleanse information codes have a specific format based on the type of information.

[Geocoder codes \[page 844\]](#)

Information codes and status codes for Geocoder.

[Global Address Cleanse codes \[page 849\]](#)

Global Address Cleanse codes consist of country ISO, information, quality, and status codes.

[USA Regulatory Address Cleanse fault codes \[page 869\]](#)

Fault codes describe why the USA Regulatory Address Cleanse transform could not assign an address.

[USA Regulatory Address Cleanse status codes \[page 871\]](#)

The USA Regulatory Address Cleanse transform assigns status codes to assigned addresses to indicate how the input address differs from the assigned address.

22.1 Data Cleanse codes

The Data Cleanse information codes have a specific format based on the type of information.

There are information codes for the following parsing information:

- Date
- Firm
- Input field
- Person
- Phone
- Record level

[Data Cleanse status codes \[page 830\]](#)

The Data Cleanse transform generates status codes that indicate how it standardized your data, or how a specific standard could be used to standardize your data.

[Data Cleanse person parser status codes \[page 832\]](#)

Person parser status codes indicate how the Data Cleanse transform standardized person information in your records.

[Data Cleanse firm parser status codes \[page 833\]](#)

Firm parser status codes indicate how the transform parsed firm information in your records.

[Data Cleanse Date, Phone, and SSN parser status codes \[page 834\]](#)

Date, Phone, and Social Security number parser status codes indicate how the transform standardized this information in your records.

[Data Cleanse custom parser status codes \[page 835\]](#)

Custom status codes indicate how the Data Cleanse transform parsed items that do not fall into the other categories.

[Data Cleanse information codes \[page 836\]](#)

The Data Cleanse transform issues information codes to inform you about data that may be suspect and require a manual review.

[Data Cleanse diacritical character conversions \[page 842\]](#)

In the Data Cleanse transform, there is an option where you can remove the diacritical characters from the data on output.

Parent topic: [Data Quality codes \[page 829\]](#)

Related Information

[Geocoder codes \[page 844\]](#)

[Global Address Cleanse codes \[page 849\]](#)

[USA Regulatory Address Cleanse fault codes \[page 869\]](#)

[USA Regulatory Address Cleanse status codes \[page 871\]](#)

22.1.1 Data Cleanse status codes

The Data Cleanse transform generates status codes that indicate how it standardized your data, or how a specific standard could be used to standardize your data.

To output the Data Cleanse status codes, include the `STATUS_CODE` output field in your job setup. The transform outputs status codes using the following format: `<parser name>_<output field name>_STD` or `<parser name>_<output field name>_ADD`. The value in the `<parser name>` variable can include one of the following parsers:

- custom
- date
- firm
- person
- phone
- SSN

The transform generates only the status codes for records that have standardized output.

Status code exceptions

The transform does not generate a status code for minor changes. The following table contains exceptions that explain why the transform does not generate status codes.

Status code exception descriptions

Exception	Description	Example
Trailing period	Transform does not generate a status code when the transform adds or removes a period after a piece of data.	❖ Example The transform does not generate a status code for an input of J Smith. The transform is set to add a period after single letters so it changes "J" to "J." ❖ Example The transform generates a status code: UK has a standard of "U.K." The transform generates a status code because there are more punctuation changes than a single trailing period.
Case	Transform does not generate a status code when the only difference between the standard and the original data is how the data is cased.	❖ Example The last name Mcdonald has a standard of "McDonald" but, the transform outputs it as "MCDONALD" because you have the <i>CAPITALIZATION</i> option set to <i>UPPER</i>. No status code.
Remove punctuation	Transform does not generate a status code when the only difference between the standard and the original data is that the punctuation is removed. Standardization option <i>REMOVE_PUNCTUATION</i> set to <i>YES</i> .	❖ Example UK has a standard of "U.K." However, the <i>REMOVE_PUNCTUATION</i> option is set to <i>YES</i>. No status code.
Diacritical characters	The transform does not generate a status code when the only difference between the standard and the original data is the diacritical characters are removed. Standardization option <i>REMOVE_DIACRITICAL_CHARACTER</i> <i>YES</i> .	❖ Example Hernandez has a standard of "Hernández". However, the <i>REMOVE_DIACRITICAL_CHARACTERS</i> option is set to <i>YES</i>. No status code. ❖ Example Hernandez has a standard of "Hernández". However, the <i>REMOVE_DIACRITICAL_CHARACTERS</i> option is set to <i>NO</i>. Transform generates a status code to show that a standard was used.

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Related Information

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[Data Cleanse Date, Phone, and SSN parser status codes \[page 834\]](#)

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[Data Cleanse diacritical character conversions \[page 842\]](#)

22.1.2 Data Cleanse person parser status codes

Person parser status codes indicate how the Data Cleanse transform standardized person information in your records.

Person parser status codes have the following formats:

- PERSON#_<output_field>_STD
- PERSON#_<output_field>_ADD

The transform replaces the pound sign (#) with the person parse.

Person parser status code descriptions

Status code	Description
PERSON#_NAME_DESIGNATOR_STD	Output <i>NAME_DESIGNATOR</i> using the Standardized field class.
PERSON#_PRENAME_ADD	Assigned <i>PRENAME</i>
PERSON#_PRENAME_STD	Output <i>PRENAME</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_GIVEN_NAME1_STD	Output <i>GIVEN_NAME1</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_GIVEN_NAME2_STD	Output <i>GIVEN_NAME2</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_FAMILY_NAME1_STD	Output <i>FAMILY_NAME1</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_FAMILY_NAME2_STD	Output <i>FAMILY_NAME2</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_HONORARY_POSTNAME_STD	Output <i>HONORARY_POSTNAME</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_MATURITY_POSTNAME_STD	Output <i>MATURITY_POSTNAME</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_TITLE_STD	Output <i>TITLE</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .
PERSON#_NAME_SPECIAL_STD	Output <i>NAME_SPECIAL</i> using the <i>STANDARDIZED GENERATED_FIELD_CLASS</i> .

The transform may assign multiple status codes for a record. The transform outputs multiple status codes separated by commas.

❖ Example

Status code output string:

```
PERSON1_PRENAME_STD, PERSON1_GIVEN_NAME1_STD, PERSON1_FAMILY_NAME1_STD, PERSON2_GIVE  
N_NAME2_STD, PERSON6_MATURITY_POSTNAME_STD
```


The status code string provides the following information:

- The Person1 parse used standards for *PRENAME*, *GIVEN_NAME1*, and *FAMILY_NAME1* output fields.
- The Person2 parse used standards for the *GIVEN_NAME2* output field.
- The Person6 parse used standards for the *MATURITY_POSTNAME* output field.

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22.1.3 Data Cleanse firm parser status codes

Firm parser status codes indicate how the transform parsed firm information in your records.

Firm parser status codes have the format *FIRM#_<output_field>_STD*. The transform replaces the pound sign (#) with the firm number parse.

Firm parser status code descriptions

Status code	Description
<i>FIRM#_FIRM_STD</i>	Output <i>FIRM</i> using the Standardized field class.
<i>FIRM#_FIRM_LOCATION_STD</i>	Output <i>FIRM_LOCATION</i> using the Standardized field class.

The transform may assign multiple status codes per record. The transform separates multiple status codes with commas.

❖ Example

The status code:

FIRM1_FIRM_STD, FIRM2_FIRM_LOCATION_STD, FIRM3_FIRM_STD, FIRM3_FIRM_LOCATION_STD provides the following information:

- The Firm1 parse used a *FIRM* standard.
- The Firm2 parse used a *FIRM_LOCATION* standard.
- The Firm3 parse used both a *FIRM* standard and a *FIRM_LOCATION* standard.

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22.1.4 Data Cleanse Date, Phone, and SSN parser status codes

Date, Phone, and Social Security number parser status codes indicate how the transform standardized this information in your records.

Date, Phone, and Social Security number (SSN) parser status codes have the following status code format: `<parser_name>#_<output_field>_STD`. The transform replaces the pound sign (#) with the parser number.

Date, Phone and Social Security number parser status codes

Status code	Description
<code>DATE#_DATE_STD</code>	Output <code>DATE</code> using the Standardized field class.
<code>PHONE#_PHONE_STD</code>	Output <code>PHONE_NUMBER</code> using the Standardized field class.
<code>SSN#_SSN_STD</code>	Output <code>SSN</code> using the Standardized field class.

The transform may assign multiple status codes per record. The transform separates multiple status codes with commas.

❖ Example

The Date status code, `DATE1_DATE_STD`, `DATE6_DATE_STD`, shows that both `DATE1` and `DATE6` were standardized.

`SSN1_SSN_STD`, `SSN5_SSN_STD` shows that both `SSN1` and `SSN5` were standardized.

`PHONE1_PHONE_STD`, `PHONE3_PHONE_STD` shows that both `PHONE1Phone1` and `PHONE3Phone3` were standardized.

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22.1.5 Data Cleanse custom parser status codes

Custom status codes indicate how the Data Cleanse transform parsed items that do not fall into the other categories.

Custom parser status codes have the following format: `<parser_name>#_<output_field>_STD`. The transform replaces the pound sign (#) with the parser number.

Custom parser status code description

Status code	Description
<code>PARSER NAME#_OUTPUT FIELD_STD</code>	Output the custom <code><output_field></code> using the Standard field class.

The transform may assign multiple status codes per record separated by commas.

❖ Example

The Automobile status code, `AUTOMOBILE1_COLOR_STD, AUTOMOBILE1_SIZE_STD, AUTOMOBILE2_COLOR_STD`, provides the following information:

- Automobile1 used a `COLOR` and `SIZE` size standard.
- Automobile2 used a `COLOR` standard.

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22.1.6 Data Cleanse information codes

The Data Cleanse transform issues information codes to inform you about data that may be suspect and require a manual review.

The transform outputs information codes to the *INFO_CODE* output field. The transform populates the *INFO_CODE* field with one or more codes separated by commas. Each code consists of a prefix that identifies the parser type, and the numeric code.

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Each information code consists of a prefix followed by the numeric code. The prefix identifies the parser type and the code indicates the reason why the record may be suspect and needs manual review.

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Each information code indicates a reason why a record is suspect and requires a manual review.

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22.1.6.1 Data Cleanse information code prefixes

Each information code consists of a prefix followed by the numeric code. The prefix identifies the parser type and the code indicates the reason why the record may be suspect and needs manual review.

The following table contains code prefixes that are two characters.

Two-character parser type prefixes

Parser type prefix	Description
D1## - D6##	<i>DATE1</i> through <i>DATE6</i> parse level information.
F1## - F6##	<i>FIRM1</i> through <i>FIRM6</i> parse level information.
P1## - P6##	<i>PERSON1</i> through <i>PERSON6</i> parse level information.

Parser type prefix	Description
T1## - T6##	PHONE1 through PHONE6 parse level information.

The following table contains code prefixes that are one character.

Single letter parser prefix

Parser type prefix	Description
I###	Input field level information.
R###	Record level information.

22.1.6.2 Data Cleanse information code descriptions

Each information code indicates a reason why a record is suspect and requires a manual review.

The following table shows more details about Data Cleanse information codes.

Information code	Description	Type
D#01	DATE# was not in the expected format.	Date parser
D#02	DATE# was converted from 2-digits to 4-digits. The transform applied the century threshold option.	Date parser
D#03	DATE# was in the Day_Month_Year format.	Date parser
D#04	DATE# was in the Month_Day_Year format.	Date parser
D#05	DATE# was in the Year_Month_Day format.	Date parser
D#06	DATE# was in the Year_Day_Month format.	Date parser
D#07	DATE# was in an ambiguous format. There is more than one possible format for the date. For example, 12/09/10 is valid for all of the formats, whereas 03/16/94 is valid only for the Month_Day_Year format.	Date parser
F#01	The FIRM# parse contained some data that was not found in the cleansing package.	Firm parser

Information code	Description	Type
F#02	The <i>FIRM#</i> parse had a close person parse. This is only applicable for <i>PERSON_FIRM</i> multiline parse when using the <i>PERSON_FIRM</i> multiline parser or <i>NAME_FIRM_LINE</i> when the data came from the input field <i>NAME_FIRM_LINE</i> .	Firm parser
F#03	The <i>FIRM#</i> parse was a presumptive firm parse based on reasonable evidence.	Firm parser
P#01	The <i>PERSON#</i> parse contained some data that was not found in the cleansing package. This information code does not report on title information, which is different from information code P#51.	Person parser
P#02	The <i>PERSON#</i> parse had a close <i>FIRM#</i> parse. This is only applicable for <i>PERSON_FIRM</i> multiline parse when using the <i>PERSON_FIRM</i> multiline parser or <i>NAME_FIRM_LINE</i> when the data came from the input field <i>NAME_FIRM_LINE</i> .	Person parser
P#03	The <i>PERSON#</i> parse was a presumptive name parse based on reasonable evidence.	Person parser
P#04	The <i>PERSON#</i> parse has no given name, or has a questionable given name.	Person parser
P#05	The <i>PERSON#</i> parse has no family name, or has a questionable family name.	Person parser
P#51	The <i>PERSON#</i> parse contained a title token that was not found in the cleansing package. This is different from information code P#01.	Person parser
P#52	The <i>TITLE#</i> parse has a close firm parse.	Person parser
P#53	The <i>TITLE#</i> parse was presumptive.	Person parser
T#01	The <i>PHONE#</i> parse did not have a North American area code.	Phone parser
T#02	The transform parsed phone data using a different ISO2 country code than the country listed in the <i>OPTION_COUNTRY</i> field.	Phone parser
T#03	The transform parsed phone data by prepending an ISO2 country code to the incoming phone data.	Phone parser

Information code	Description	Type
I111-I116	All of the input data in NAME_LINE# went to one or more EXTRA output fields; nothing was parsed for this input field.	Input field
I121-I126	There was no input data in Name_Line# .	Input field
I131-I136	Parsed some of the input data in NAME_LINE# , the rest of the data is in the EXTRA output field.	Input field
I151-I156	All of the input data in TITLE_LINE# went to one or more EXTRA output fields. Nothing was parsed for this input field.	Input field
I161-I166	There was no input data in TITLE_LINE# .	Input field
I171-I176	Parsed some of the input data in TITLE_LINE# , the rest of the data is in the EXTRA output field.	Input field
I211-I212	All of the input data in PERSON# discrete went to one or more EXTRA output fields. Nothing was parsed for this input field.	Input field
I221-I222	There was no input data in PERSON# discrete.	Input field
I231-I232	Parsed some input data from PERSON# discrete. Remaining data is in the EXTRA output field.	Input field
I311-I316	All of the input data in NAME_OR_FIRM_LINE# went to one or more EXTRA output fields; nothing was parsed for this input field.	Input field
I321-I326	There was no input data in NAME_OR_FIRM_LINE# .	Input field
I331-I336	Parsed some of the input data in NAME_OR_FIRM_LINE# , the rest of the data is in the EXTRA output field.	Input field
I351-I352	All of the input data in FIRM_LINE# went to one or more EXTRA output fields; nothing was parsed for this input field.	Input field
I361-I366	There was no input data in FIRM_LINE# .	Input field
I371-I372	Parsed some of the input data in FIRM_LINE# , the rest of the data is in the EXTRA output field.	Input field

Information code	Description	Type
I421-I422	There was no input data in <i>FIRM#</i> discrete.	Input field
I511-I516	All of the input data in <i>DATE#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I521-I526	There was no input data in <i>DATE#</i> .	Input field
I531-I536	Parsed some of the input data in <i>DATE#</i> , the rest of the data is in the <i>EXTRA</i> output field.	Input field
I711-I716	All of the input data in <i>EMAIL#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I721-I726	There was no input data in <i>EMAIL#</i> .	Input field
I731-I736	Parsed some of the input data in <i>EMAIL#</i> , the rest of the data is in the <i>EXTRA</i> output field.	Input field
I751-I756	All of the input data in <i>PHONE#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I761-I766	There was no input data in <i>PHONE#</i> .	Input field
I771-I776	Parsed some of the input data in <i>PHONE#</i> , the rest of the data is in the <i>EXTRA</i> output field.	Input field
I811-I816	All of the input data in <i>SSN#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I821-I826	There was no input data in <i>SSN#</i> .	Input field
I831-I836	Parsed some of the input data in <i>SSN#</i> , the rest of the data is in the <i>EXTRA</i> output field.	Input field
I851-I856	All of the input data in <i>UDPM#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I861-I866	There was no input data in <i>UDPM#</i> .	Input field
I871-I876	Parsed some of the input data in <i>UDPM#</i> , the rest of the data is in the <i>EXTRA</i> output field.	Input field
I901-I912	All of the input data in <i>DATE#</i> went to one or more <i>EXTRA</i> output fields; nothing was parsed for this input field.	Input field
I931-I942	There was no input data in <i>MULTILINE#</i> .	Input field

Information code	Description	Type
I951-I962	Parsed some of the input data in MULTILINE# , the rest of the data is in the EXTRA output field.	Input field
R001	All input field data went to one or more EXTRA output fields; nothing was parsed for the record.	Record level
R002	Parsed some of the input fields. One or more other input fields went to the EXTRA output field.	Record level
R003	Parsed part of the input fields. Some of the record data went to the EXTRA output field.	Record level
R004	All of the record input fields were parsed into appropriate output fields. None of the input data went to the EXTRA output field.	Record level
R400	The data in OPTION_CONTENT_DOMAIN_SEQUEN CE overrode the content domain sequence transform option.	Record level
R405	The data in OPTION_CONTENT_DOMAIN_SEQUEN CE was not recognized as a content domain sequence. The data in OPTION_COUNTRY or the data in the content domain sequence transform option was used to determine the content domain sequence.	Record level
R410	The data in OPTION_OUTPUT_FORMAT overrode the output format data cleanse transform option.	Record level
R415	The data in OPTION_OUTPUT_FORMAT was not recognized as a valid output format. The data in OPTION_COUNTRY or the output format transform option was used to determine the output format.	Record level
R420	The data in OPTION_COUNTRY was recognized as an ISO2 country code and overrode the content domain sequence transform option.	Record level
R421	The data in OPTION_COUNTRY was recognized as an ISO2 country code and overrode the output format transform option.	Record level

Information code	Description	Type
R425	The data in OPTION_COUNTRY was not recognized as an ISO2 country code. One or more situations occurred where OPTION_COUNTRY data was used to override the content domain sequence, or output format transform options, or both, but failed: - Attempted to use OPTION_COUNTRY to override the content domain sequence transform option. This occurs when the OPTION_CONTENT_DOMAIN_SEQUENCE data is invalid (R405) or the data is not supplied. No status code is generated when this occurs. The data was parsed using the content domain sequence transform option. - Attempted to use OPTION_COUNTRY to override the output transform option. This occurs when the OPTION_OUTPUT_FORMAT data is invalid (R415) or is not supplied. No status code is generated when this occurs. The data was parsed using the output format transform option.	Record level
R428	The cleansing package does not recognize the ISO2 country code provided in the OPTION_COUNTRY input field.	Record level

22.1.7 Data Cleanse diacritical character conversions

In the Data Cleanse transform, there is an option where you can remove the diacritical characters from the data on output.

The following table shows the diacritical characters and the ASCII equivalent letter that replaces the diacritical character.

Note

The diacritical character is listed first, separated by a comma and then the ASCII replacement letter.

Diacritical characters conversion chart

À,A	Ě,E	Ł,L	Ś,S	Ž,Z	ē,e	İ,I	Š,s
Á,A	Ě,E	Ĺ,L	Ŝ,S	Ž,Z	ě,e	ı,I	š,s

Â,A	Ē,E	Ĺ,L	Š,S	à,a	é,e	ı,I	ţ,t
Ã,A	Ĝ,G	Ł,L	Ț,T	á,a	ę,e	ň,n	ť,t
Ä,AE	Ğ,G	Ń,N	Ț,T	â,a	ğ,g	ń,n	ț,t
Å,A	Ġ,G	Ň,N	Ʀ,T	ă,a	ġ,g	ň,n	ù,u
Ă,A	Ģ,G	Ņ,N	Ù,U	ā,ae	ġ,g	ň,n	ú,u
Ǻ,A	Ĥ,H	Ň,N	Ú,U	ā,a	ġ,g	ò,o	û,u
Ȧ,A	Ĥ,H	Ò,O	Û,U	ā,a	ĥ,h	ó,o	ü,ue
Ç,C	Ì,I	Ó,O	Ü,UE	ǻ,a	ħ,h	ô,o	ÿ,u
Ć,C	Í,I	Ô,O	Ŭ,U	ą,a	ì,i	õ,o	ÿ,u
Ĉ,C	Î,I	Õ,O	Ū,U	ç,c	í,i	ö,oe	ÿ,u
Č,C	Ī,I	Ö,OE	Ů,U	ć,c	î,i	ø,o	ÿ,u
Ď,D	Ĵ,I	Ø,O	Ű,U	ĉ,c	ï,i	ō,o	ÿ,u
Đ,D	Ĵ,I	Ō,O	Ű,U	č,c	ī,i	ō,o	ŵ,w
Ď,D	Ĵ,I	Ō,O	Ű,U	č,c	ī,i	ō,o	ý,y
È,E	Ĵ,I	Ō,O	Ŵ,W	ď,d	ÿ,i	í,r	ÿ,y
É,E	Ĵ,I	Ř,R	Ý,Y	è,e	ĵ,i	ř,r	ÿ,y
Ê,E	Ķ,K	Ŗ,R	Ÿ,Y	é,e	ķ,k	ř,r	ž,z
Ě,E	κ,K	Ř,R	Ÿ,Y	ê,e	ĺ,l	š,s	ž,z
Ě,E	Ĺ,L	Š,S	Ž,Z	ë,e	ĵ,i	ś,s	ž,z

Example

This example shows the input data and how the diacritical characters are replaced on output.

Diacritical character removal example

Input	Output
María Hernández	Maria Hernandez
Geschäftsführer	Geschaeftsfuehrer
Telecomunicações São Paulo	Telecomunicacoes Sao Paulo

Parent topic: [Data Cleanse codes \[page 829\]](#)

Related Information

[Data Cleanse status codes \[page 830\]](#)

[Data Cleanse person parser status codes \[page 832\]](#)

[Data Cleanse firm parser status codes \[page 833\]](#)

[Data Cleanse Date, Phone, and SSN parser status codes \[page 834\]](#)

[Data Cleanse custom parser status codes \[page 835\]](#)

[Data Cleanse information codes \[page 836\]](#)

22.2 Geocoder codes

Information codes and status codes for Geocoder.

[Geocoder information codes \[page 844\]](#)

The transform assigns information codes to a record when you include the *INFO_CODE* output field in your field mapping.

[Geocoder status codes \[page 847\]](#)

The transform assigns status codes to a record when you include the *STATUS_CODE* output field in your field mapping.

Parent topic: [Data Quality codes \[page 829\]](#)

Related Information

[Data Cleanse codes \[page 829\]](#)

[Global Address Cleanse codes \[page 849\]](#)

[USA Regulatory Address Cleanse fault codes \[page 869\]](#)

[USA Regulatory Address Cleanse status codes \[page 871\]](#)

22.2.1 Geocoder information codes

The transform assigns information codes to a record when you include the *INFO_CODE* output field in your field mapping.

i Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The *INFO_CODE* output field is a three-character code that provides information about geocoding results. Each character of the code holds specific types of information:

- First character: Not currently used.
- Second and third character: Status for the reverse geocoding assignment.

- Third character: Status for the address geocoding and POI textual search modes.

If you set the assignment to the best level, the *INFO_CODE* field is blank.

Use the following table to understand the code assigned to the *INFO_CODE* output field.

Geocoder information codes

Information code	Description
001	Reference data isn't available for the input country. Verify that the directory is installed and the reference path to the directory is valid.
002	Address-level reference data isn't available for the input data. When Best Assignment Level is set to Primary Number and the address directory is unavailable or doesn't exist, this code is output. Verify that the directory is installed and the reference path to the directory is valid.
003	Centroid-level reference data isn't available for the input data. When the Best Assignment Level is set to Locality or Postcode, and the address directory is unavailable or doesn't exist, this code is output. Verify that the directory is installed and the reference path to the directory is valid.
004	Assignment is limited. The input data is insufficient or incorrect to match the reference data. When the Best Assignment Level fails, this code is output. The assignment may be made to a lower assignment level than the one specified. For example, if you set Best Assignment Level to Primary Number and the Primary Number field is blank, the assignment may be at the postcode or locality level, if the data is available. Verify your input data and input field mapping and make sure that the fields required for best assignment exist and are correctly mapped.
005	The input data doesn't match anything in the reference data. When the input record doesn't match the directory data for the Best Assignment Level or a lower assignment level, this code is output. Verify your input data and input field mapping and make sure that the fields required for best assignment exist and are correctly mapped.

Information code	Description
006	The input data assigns ambiguously in the reference data. There's a tie for the Best Assignment Level. The input record matches several records in the directory data and the software can't decide which one is the best. ❖ Example Your reference data has two records: Record 1: 100 Main St La Crosse WI 54650 Record 2: 100 Main St Bt Micts WI 54650 Record 2 doesn't have a valid locality name. Therefore, the software outputs 006 for ambiguous assignment. Verify your input data and input field mapping and make sure that the fields required for best assignment exist and are correctly mapped.
007	The input data is blank or invalid. For example, if the US Postcode1 is a five-digit string and your input data is a six-digit string, the 007 information code is output. Verify your input data and input field mapping and make sure that the fields required for best assignment exist and are correctly mapped.
008	The input data is insufficient. When the input data for the selected Best Assignment Level is blank, this code is output. For example, this code is output when you set the Best Assignment Level to Primary Number and the input data is blank for Primary Number.
009	The POI type provided on input is invalid. The point-of-interest type isn't correct. Verify your input POI type with the POI types described in the appropriate directory update letter.
050	Reverse geocoding. The input data doesn't match anything in the reference data.
060	Reverse geocoding. The input data assigns ambiguously in the reference data.
070	Reverse geocoding. The input data is blank or invalid.
090	Reverse geocoding. The POI type provided on input is invalid.
00A	The POI input data wasn't used. The POI name or type doesn't match the directory data. A PRI or PRE level assignment was made; however, the input POI name and POI type weren't used for the assignment.

Information code	Description
OB0	The input data wasn't found. This code only occurs when an address is input during reverse geocoding assignment. The input address doesn't match the directory data and reverse geocoding can't be performed based on the address.
OC0	Not all results returned for the input data, because the number of results exceeds the specified Max_Record. This code only occurs during reverse geocoding assignment. Increase the Max_Record value.
OD0	Not all results were returned for the input data, because the results exceed the field length available in the Result_List XML output field. This code only occurs during reverse geocoding assignment. The Geocoder transform allows a 60000 field length in the Result_List field.
OOE	The closest latitude and longitude to the input address is returned because the input house number doesn't exist in the geocoder directories. For example: Input address: 100 Main St La Crosse WI 54650 Directory data: 1–88 Main St La Crosse WI 54650 The software returns the latitude and longitude values for 88 Main St with the OOE information code to indicate that the house number doesn't match the directory data.
OFO	The feature enabled by the current configuration isn't supported by the installed basic directory. For example, a configuration requests reverse geocoding but it finds only the basic directory installed. Verify that the advanced directory format is installed and the reference path to the directory is valid. For more information about directories, see "Directory data" in the <i>SAP Data Services Installation Guide</i>.

22.2.2 Geocoder status codes

The transform assigns status codes to a record when you include the [STATUS_CODE](#) output field in your field mapping.

Status codes are four characters that represent the fuzzy matches made to the address—point of interest (POI) component during address geocoding processing. Each character represents an aspect of the status:

- The first character is always S (for Status).
- The second character is associated with any last-line fuzzy matches to the postal code, region, or locality.
- The third character is associated with any address-line fuzzy matches to the primary name prefix, primary name postfix, primary type, primary name, and primary range.

- The fourth character is associated with any POI data fuzzy matches.

i Note

The transform uses the [STATUS_CODE](#) output field only for the address geocoding process.

First character

The value of the first character is always S for Status.

Second character

The value of the second character depends on fuzzy matches to the postal code, region, or locality.

Second character status code descriptions

Value	Description
0	No significant difference between the input data and the reference data.
1	Fuzzy match on postal code.
2	Fuzzy match on region.
3	Fuzzy match on postal code and region.
4	Fuzzy match on locality.
5	Fuzzy match on postal code and locality.
6	Fuzzy match on region and locality.
7	Fuzzy match on postal code, region, and locality.

Third character

Third character status code descriptions

Value	Description
0	No significant difference between the input data and the reference data.
1	Fuzzy match on primary name prefix/primary name postfix.
2	Fuzzy match on primary type.
3	Fuzzy match on primary name prefix/primary name postfix and primary type.
4	Fuzzy match on primary name.
5	Fuzzy match on primary name prefix/primary name postfix and primary name.
6	Fuzzy match on primary type and primary name.

Value	Description
7	Fuzzy match on primary name prefix/primary name postfix, primary type, and primary name.
8	Fuzzy match on primary range.
9	Fuzzy match on primary name prefix/primary name postfix and primary range.
A	Fuzzy match on primary type and primary range.
B	Fuzzy match on primary name prefix/primary name postfix, primary type, and primary range.
C	Fuzzy match on primary name and primary range.
D	Fuzzy match on primary name prefix/primary name postfix, primary name, and primary range.
E	Fuzzy match on primary type, primary name, and primary range.
F	Fuzzy match on primary name prefix/primary name postfix, primary type, primary name, and primary range.

Fourth character

Fourth character status code descriptions

Value	Description
0	No significant difference between the input data and the reference data.
1	Fuzzy match on POI name.

22.3 Global Address Cleanse codes

Global Address Cleanse codes consist of country ISO, information, quality, and status codes.

[Country ISO codes and assignment engines \[page 850\]](#)

Use country ISO codes, assignment engine availability, and more to understand processing results.

[Global Address Cleanse information codes \[page 861\]](#)

Global Address Cleanse information codes explain why an address is unassigned.

[Global Address Cleanse quality codes \[page 864\]](#)

Global Address Cleanse quality codes relay additional information about address quality.

[Global Address Cleanse status codes \[page 865\]](#)

Global Address Cleanse status codes represent the corrections made to the address during processing.

Parent topic: [Data Quality codes \[page 829\]](#)

Related Information

[Data Cleanse codes \[page 829\]](#)

[Geocoder codes \[page 844\]](#)

[USA Regulatory Address Cleanse fault codes \[page 869\]](#)

[USA Regulatory Address Cleanse status codes \[page 871\]](#)

22.3.1 Country ISO codes and assignment engines

Use country ISO codes, assignment engine availability, and more to understand processing results.

Note

Reference to “country” or “countries” in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

The following table shows which engine (if any) provides address correction. Additionally, it lists the 2-character ISO code, the 3-character ISO code, the 3-digit ISO code, the European Postcode prefix, and the assignment level. The assignment level is based on the reference data that you own.

Table key

Engine	Assignment level
C = Canada	C = Country
G = Global Address	L = Locality
U = USA	Pn = Primary Name
	Pr = Premise
	S = Secondary

Country ISO codes and engines

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Afghanistan	AF	AFG	004		G	C, L
Åland Islands	AX	ALA	248	AX	G	C, L
Albania	AL	ALB	008		G	C, L
Algeria	DZ	DZA	012		G	C, L
American Sa- moa	AS	ASG	016		U	C, L, Pn, Pr, S
					G	C, L
Andorra	AD	AND	020	AND	G	C, L
Angola	AO	AGO	024		G	C, L
Anguilla	AI	AIA	660		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Antarctica	AQ	ATA	010		G	C
Antigua and Barbuda	AG	ATG	028		G	C, L
Argentina	AR	ARG	032		G	C, L
Armenia	AM	ARM	051		G	C, L
Aruba	AW	ABW	533		G	C, L
Australia	AU	AUS	036		G	C, L, Pn, Pr, S
Austria	AT	AUT	040	A	G	C, L, Pn, Pr, S
Azerbaijan	AZ	AZE	031		G	C, L
Bahamas	BS	BHS	044		G	C, L
Bahrain	BH	BHR	048		G	C, L
Bangladesh	BD	BGD	050		G	C, L
Barbados	BB	BRB	052		G	C, L
Belarus	BY	BLR	112		G	C, L
Belgium	BE	BEL	056	B	G	C, L, Pn, Pr
Belize	BZ	BLZ	084		G	C, L
Benin	BJ	BEN	204		G	C, L
Bermuda	BM	BMU	060		G	C, L
Bhutan	BT	BTN	064		G	C, L
Bolivia, Plurina- tional State of	BO	BOL	068		G	C, L
Bonaire, Sint Eustatius, and Saba	BQ	BES	535		G	C, L
Bosnia and Herzegovina	BA	BIH	070		G	C, L
Botswana	BW	BWA	072		G	C, L
Bouvet Island	BV	BVT	074		G	C
Brazil	BR	BRA	076		G	C, L, Pn, Pr
British Indian Ocean Territory	IO	IOT	086		G	C
British Virgin Islands	VG	VGB	092		G	C, L
Brunei Darus- salam	BN	BRN	096		G	C, L
Bulgaria	BG	BGR	100	BG	G	C, L
Burkina Faso	BF	BFA	854		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Burundi	BI	BDI	108		G	C, L
Cambodia	KH	KHM	116		G	C, L
Cameroon	CM	CMR	120		G	C, L
Canada	CA	CAN	124		C	C, L, Pn, Pr, S
					G	C, L
Cape Verde	CV	CPV	132		G	C
Cayman Is- lands	KY	CYM	136		G	C
Central African Republic	CF	CAF	140		G	C, L
Chad	TD	TCD	148		G	C, L
Chile	CL	CHL	152		G	C, L
China	CN	CHN	156		G	C, L, Pn, Pr
Christmas Is- land (Included in the Australia data package)	CX	CXR	162		G	C, L
Cocos (Keeling) Isles (Included in the Australia data package)	CC	CCK	166		G	C, L
Colombia	CO	COL	170		G	C, L
Comoros	KM	COM	174		G	C, L
Congo, Demo- cratic Republic of	CD	COD	180		G	C, L
Congo, Repub- lic of	CG	COG	178		G	C, L
Cook Islands	CK	COK	184		G	C, L
Costa Rica	CR	CRI	188		G	C, L
Côte d'Ivoire (Ivory Coast)	CI	CIV	384		G	C, L
Croatia (Hrvat- ska)	HR	HRV	191	HR	G	C, L
Cuba	CU	CUB	192		G	C, L
Curaçao	CW	CUW	531		G	C, L
Cyprus	CY	CYP	196	CY	G	C, L
Czechia	CZ	CZE	203	CZ	G	C, L, Pn, Pr

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Democratic People's Re- public of Korea	KP	PRK	408		G	C, L
Denmark	DK	DNK	208	DK	G	C, L, Pn, Pr
Djibouti	DJ	DJI	262		G	C, L
Dominica	DM	DMA	212		G	C, L
Dominican Re- public	DO	DOM	214		G	C, L
Ecuador	EC	ECU	218		G	C, L
Egypt	EG	EGY	818		G	C, L
El Salvador	SV	SLV	222		G	C, L
Equatorial Guinea	GQ	GNQ	226		G	C, L
Eritrea	ER	ERI	232		G	C, L
Estonia	EE	EST	233	EE	G	C, L, Pn, Pr
Ethiopia	ET	ETH	231		G	C, L
Falkland Islands	FK	FLK	238		G	C, L
Faroe Islands (Included in the Denmark data package)	FO	FRO	234	FO	G	C, L, Pn, Pr
Federated States of Mi- cronesia	FM	FSM	583		U G	C, L, Pn, Pr, S C, L
Fiji	FJ	FJI	242		G	C, L
Finland	FI	FIN	246	FI	G	C, L, Pn, Pr
France	FR	FRA	250	F	G	C, L, Pn, Pr, S
French Guiana (Included in the France data package)	GF	GUF	254		G	C, L, Pn, Pr
French Polyne- sia (Included in the France data package)	PF	PYF	258		G	C, L, Pn, Pr
French South- ern Territories	TF	ATF	260		G	C, L, Pn, Pr
Gabon	GA	GAB	266		G	C, L
Gambia	GM	GMB	270		G	C, L
Georgia	GE	GEO	268		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Germany	DE	DEU	276	D	G	C, L, Pn, Pr
Ghana	GH	GHA	288		G	C, L
Gibraltar	GI	GIB	292		G	C, L
Greece	GR	GRC	300	GR	G	C, L, Pn, Pr
Greenland (In- cluded in the Denmark data package)	GL	GRL	304	GL	G	C, L, Pn, Pr
Grenada	GD	GRD	308		G	C, L
Guadeloupe (Included in the France data package)	GP	GLP	312		G	C, L, Pn, Pr
Guam	GU	GUM	316		U G	C, L, Pn, Pr, S C, L
Guatemala	GT	GTM	320		G	C, L
Guernsey (In- cluded in the United King- dom data pack- age)	GG	GGY	831	G	G	C, L, Pn, Pr, S
Guinea	GN	GIN	324		G	C, L
Guinea-Bissau	GW	GNB	624		G	C, L
Guyana	GY	GUY	328		G	C, L
Haiti	HT	HTI	332		G	C, L
Heard Island and McDonald Islands	HM	HMD	334		G	C, L
Holy See (Vati- can City State) (Included in the Italy data pack- age)	VA	VAT	336		G	C, L, Pn, Pr
Honduras	HN	HND	340		G	C, L
Hong Kong	HK	HKG	344		G	C, L
Hungary	HU	HUN	348	H	G	C, L, Pn, Pr
Iceland	IS	ISL	352	IS	G	C, L
India	IN	IND	356		G	C, L, Pn, Pr
Indonesia	ID	IDN	360		G	C, L
Iraq	IQ	IRQ	368		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Ireland, Republic of	IE	IRL	372	IRL	G	C, L
Islamic Republic of Iran	IR	IRN	364		G	C, L
Isle of Man (Included in the United Kingdom data package)	IM	IMN	833		G	C, L, Pn, Pr, S
Israel	IL	ISR	376		G	C, L
Italy	IT	ITA	380	I	G	C, L, Pn, Pr
Jamaica	JM	JAM	388		G	C, L
Japan	JP	JPN	392		G	C, L, Pn, Pr, S
Jersey (Included in the United Kingdom data package)	JE	JEY	832		G	C, L, Pn, Pr, S
Jordan	JO	JOR	400		G	C, L
Kazakhstan	KZ	KAZ	398		G	C, L
Kenya	KE	KEN	404		G	C, L
Kiribati	KI	KIR	296		G	C, L
Kuwait	KW	KWT	414		G	C, L
Kyrgyzstan	KG	KGZ	417		G	C, L
Lao People's Democratic Republic	LA	LAO	418		G	C, L
Latvia	LV	LVA	428	LV	G	C, L, Pn, Pr
Lebanon	LB	LBN	422		G	C, L
Lesotho	LS	LSO	426		G	C, L
Liberia	LR	LBR	430		G	C, L
Libya	LY	LBY	434		G	C, L
Liechtenstein (Included in the Switzerland data package)	LI	LIE	438	FL	G	C, L, Pn, Pr
Lithuania	LT	LTU	440	LT	G	C, L, Pn, Pr
Luxembourg	LU	LUX	442	L	G	C, L, Pn, Pr
Macao	MO	MAC	446		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Macedonia (Now known as North Macedo- nia)	MK	MKD	807	MK	G	C, L
Madagascar	MG	MDG	450		G	C, L
Malawi	MW	MWI	454		G	C, L
Malaysia	MY	MYS	458	M	G	C,L
Maldives	MV	MDV	462		G	C, L
Mali	ML	MLI	466		G	C, L
Malta	MT	MLT	470		G	C, L
Marshall Is- lands	MH	MHL	584		U G	C, L, Pn, Pr, S C, L
Martinique (In- cluded in the France data package)	MQ	MTQ	474		G	C, L, Pn, Pr
Mauritania	MR	MRT	478		G	C, L
Mauritius	MU	MUS	480		G	C, L
Mayotte (In- cluded in the France data package)	YT	MYT	175		G	C, L, Pn, Pr
Mexico	MX	MEX	484		G	C, L
Moldova	MD	MDA	498	MD	G	C, L
Monaco (In- cluded in the France data package)	MC	MCO	492	F	G	C, L, Pn, Pr
Mongolia	MN	MNG	496		G	C, L
Montenegro	ME	MNE	499		G	C, L
Montserrat	MS	MSR	500		G	C, L
Morocco	MA	MAR	504		G	C, L
Mozambique	MZ	MOZ	508		G	C, L
Myanmar	MM	MMR	104		G	C, L
Namibia	NA	NAM	516		G	C, L
Nauru	NR	NRU	520		G	C, L
Nepal	NP	NPL	524		G	C, L
Netherlands	NL	NLD	528	NL	G	C, L, Pn, Pr

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
New Caledonia (Included in the France data package)	NC	NCL	540		G	C, L, Pn, Pr
New Zealand	NZ	NZL	554		G	C, L, Pn, Pr, S
Nicaragua	NI	NIC	558		G	C, L
Niger	NE	NER	562		G	C, L
Nigeria	NG	NGA	566		G	C, L
Niue	NU	NIU	570		G	C, L
Norfolk Island (Included in the Australia data package)	NF	NFK	574		G	C, L
Northern Ma- riana Islands	MP	MNP	580		U	C, L, Pn, Pr, S
					G	C, L
Norway	NO	NOR	578	N	G	C, L, Pn, Pr
Occupied Pal- estinian Terri- tory	PS	PSE	275		G	C
Oman	OM	OMN	512		G	C, L
Pakistan	PK	PAK	586		G	C, L
Palau	PW	PLW	585		U	C, L, Pn, Pr, S
					G	C, L
Panama	PA	PAN	591		G	C, L
Papua New Guinea	PG	PNG	598		G	C, L
Paraguay	PY	PRY	600		G	C, L
Peru	PE	PER	604		G	C, L
Philippines	PH	PHL	608		G	C, L
Pitcairn	PN	PCN	612		G	C, L
Poland	PL	POL	616	PL	G	C, L, Pn, Pr
Portugal	PT	PRT	620	P	G	C, L, Pn, Pr, S
Puerto Rico	PR	PRI	630		U	C, L, Pn, Pr, S
					G	C, L
Qatar	QA	QAT	634		G	C, L
Republic of Ko- rea	KR	KOR	410		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Réunion (In- cluded in the France data package)	RE	REU	638		G	C, L, Pn, Pr
Romania	RO	ROU	642	RO	G	C, L
Russian Feder- ation	RU	RUS	643	RUS	G	C, L
Rwanda	RW	RWA	646		G	C, L
Saint Barthé- lemy (Included in the France data package)	BL	BLM	652		G	C, L
Saint Helena, Ascension, and Tristan da Cu- nha	SH	SHN	654		G	C, L
Saint Kitts and Nevis	KN	KNA	659		G	C, L
Saint Lucia	LC	LCA	662		G	C, L
Saint Martin (French part) (Included in the France data package)	MF	MAF	663		G	C, L
Saint Pierre and Miquelon (Included in the France data package)	PM	SPM	666		G	C, L, Pn, Pr
Saint Vincent & Grenadines	VC	VCT	670		G	C, L
Samoa	WS	WSM	882		G	C, L
San Marino (In- cluded in the Italy data pack- age)	SM	SMR	674	SMR	G	C, L, Pn, Pr
Sao Tome and Principe	ST	STP	678		G	C, L
Saudi Arabia	SA	SAU	682		G	C, L
Senegal	SN	SEN	686		G	C, L
Serbia	RS	SRB	688		G	C, L
Seychelles	SC	SYC	690		G	C, L
Sierra Leone	SL	SLE	694		G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Singapore	SG	SGP	702		G	C, L
Sint Maarten (Dutch Part)	SX	SXM	534		G	C, L
Slovakia	SK	SVK	703		G	C, L, Pn, Pr
Slovenia	SI	SVN	705		G	C, L
Solomon Is- lands	SB	SLB	090		G	C, L
Somalia	SO	SOM	706		G	C, L
South Africa	ZA	ZAF	710		G	C, L
South Georgia and the South Sandwich Is- lands	GS	SGS	239		G	C, L
South Sudan	SS	SDN	728		G	C, L
Spain	ES	ESP	724	E	G	C, L, Pn, Pr
Sri Lanka	LK	LKA	144		G	C, L
Sudan	SD	SDN	736		G	C, L
Suriname	SR	SUR	740		G	C, L
Svalbard and Jan Mayen (In- cluded in the Norway data package)	SJ	SJM	744		G	C
Swaziland	SZ	SWZ	748		G	C, L
Sweden	SE	SWE	752	S	G	C, L, Pn, Pr
Switzerland	CH	CHE	756	CH	G	C, L, Pn, Pr
Syrian Arab Re- public	SY	SYR	760		G	C, L
Taiwan	TW	TWN	158		G	C, L
Tajikistan	TJ	TJK	762		G	C, L
Thailand	TH	THA	764		G	C, L
Timor-Leste	TL	TLS	626		G	C
Togo	TG	TGO	768		G	C, L
Tokelau	TK	TKL	772		G	C, L
Tonga	TO	TON	776		G	C, L
Trinidad and Tobago	TT	TTO	780		G	C, L
Tunisia	TN	TUN	788	TN	G	C, L

Country/ Region name	2-char ISO code	3-char ISO code	3-digit ISO code	European Post- code prefix	Engine	Assignment level
Turkey	TR	TUR	792	TR	G	C, L, Pn, Pr
Turkmenistan	TM	TKM	795		G	C, L
Turks and Cai- cos Islands	TC	TCA	796		G	C, L
Tuvalu	TV	TUV	798		G	C, L
U.S. Virgin Is- lands	VI	VIR	850		U G	C, L, Pn, Pr, S
Uganda	UG	UGA	800		G	C, L
Ukraine	UA	UKR	804	UK	G	C, L
United Arab Emirates	AE	ARE	784		G	C, L
United King- dom	GB	GBR	826	GB	G	C, L, Pn, Pr, S
United Republic of Tanzania	TZ	TZA	834		G	C, L
United States	US	USA	840		U G	C, L, Pn, Pr, S C, L
United States Minor Outlying Islands	UM	UMI	581		U G	C, L, Pn, Pr, S
Uruguay	UY	URY	858		G	C, L
Uzbekistan	UZ	UZB	860		G	C, L
Vanuatu	VU	VUT	548		G	C, L
Venezuela	VE	VEN	862		G	C, L
Viet Nam	VN	VNM	704		G	C, L
Wallis and Fu- tuna	WF	WLF	876		G	C, L, Pn, Pr
Western Sahara	EH	ESH	732		G	C, L
Yemen	YE	YEM	887		G	C, L
Zambia	ZM	ZMB	894		G	C, L
Zimbabwe	ZW	ZWE	716		G	C, L

22.3.2 Global Address Cleanse information codes

Global Address Cleanse information codes explain why an address is unassigned.

Information codes are output in the Global Address Cleanse Info_Code output field. Information codes have four characters. The following table explains the seven classification levels for Global Address Cleanse information codes.

Seven classification levels

Classification level	Represents
1000	Input record discrepancies
2000	Inconsistent lastline information
3000	Inconsistent address information
4000	Inconsistent secondary address information
5000	All other types of information
6000	Unclassified error
7000	Aborted process caused by a user-defined timeout

Each information code is available based on the engine or engines that you enable. The following table shows the code used for each engine.

Key for engines

Engine	Code
Canada	C
Global Address	G
USA	U
All engines	C, G, and U
Transform level (information code does not come from a specific engine)	T

Use the following table to determine the code assigned to the [INFO_CODE](#) output field.

Information code and supported engines

Information code	Description	Engine(s)
1020	Address validated in multiple countries or regions.	T
1030	No country or region found by Country ID or no country or region set for the record.	T
1040	Address contains at least one character that is not part of the character set supported by the engine.	T
1060	The country or region identified is not supported by any of the active engines.	T

Information code	Description	Engine(s)
1080	The script identified is not supported by any of the active engines.	T
2000	Unable to identify locality, region, and/or postcode information on input.	All engines
2010	Unable to identify locality and invalid postcode found.	All engines
2020	Unable to identify postcode. Invalid locality is preventing a possible address correction.	All engines
2030	Invalid locality and postcode are preventing a possible address correction.	All engines
2040	Invalid postcode is preventing a locality selection.	G, U
2050	Lastline matches are too close to choose one.	G
2070	Bad or no postcode for locality; multiple postcodes are available.	G
3000	Locality, region, and postcode are valid. Unable to identify the primary address line.	All engines
3010	Locality, region, and postcode are valid. Unable to match primary name to directory.	All engines
3020	Possible primary name matches are too close to choose one.	All engines
3030	Primary range is missing on input or not in the directory.	All engines
3050	An invalid or missing primary type is preventing a possible address match.	All engines
3060	A missing primary type and prefix/postfix (directional) are preventing a possible address match.	G, U
3070	An invalid or missing prefix/postfix (directional) is preventing a possible address match.	All engines
3080	An invalid or missing postcode is preventing a possible address match.	All engines
3090	An invalid or missing locality is preventing a possible address match.	G, U
3100	Possible address-line matches are too close to choose one.	All engines

Information code	Description	Engine(s)
3110	Address conflicts with postcode and the same primary name has a different postcode.	C
3200	The building name is missing on input or not in the directory.	G
3210	The building address is not in the directory.	G
3220	Possible building names are too close to choose one.	G
3250	The building name is missing on input or not in the directory, or the range is missing on input or invalid with the input building.	G
3300	The postcode only lookup returned multiple primary names.	G
4000	The secondary information is missing on input or not in the directory.	All engines
4010	Possible secondary address line matches are too close to choose one.	All engines
4500	The organization is missing on input or not in the directory.	G
4510	The organization's address is not in the directory.	G
4520	Possible organization names are too close to choose one.	G
5000	The address was valid, but the postal authority classified this address as undeliverable.	G, U
5010	The address does not reside in the specified country or region.	C, U
5020	The entire input address was blank.	T
5030	The country's postal authority will not permit assignment due to violation of an assignment rule.	G
5040	Assignment not made after lastline processing due to a violation of lastline assignment rules.	G
5050	The obsolete address can be mapped to multiple addresses.	G
6000	Unclassified error.	All engines

Information code	Description	Engine(s)
7000	The user-defined processing time-out threshold, set in the PROCESSING_TIMEOUT option, has been reached, preventing further corrective action on this address.	G

22.3.3 Global Address Cleanse quality codes

Global Address Cleanse quality codes relay additional information about address quality.

i Note

Reference to "country" or "countries" in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Quality codes appear in the [QUALITY_CODE](#) output field. There are six levels of quality codes based on these factors:

- The country of the input data
- The engine used for processing
- The information code
- The status code if there isn't an information code

Use the following table to determine the code assigned to the [QUALITY_CODE](#) output field.

Quality code	Description
Q1	Perfect address on input. All address components were validated without corrections.
Q2	Corrected address. All address components were validated after corrections were made.
Q3	Not all components of the address could be fully validated. There was insufficient information to make a final correction. However, the assessment of the record leads to the assumption that there's a "high" likelihood that this address is deliverable.
Q4	Not all components of the address could be fully validated. There was insufficient information to make a final correction. However, the assessment of the record leads to the assumption that there's a "fair" likelihood that this address is deliverable.
Q5	Not all components of the address could be fully validated. There was insufficient information to make a final correction. However, the assessment of the record leads to the assumption that there's a "small" likelihood that this address is deliverable.

Quality code	Description
Q6	Not all components of the address could be fully validated. There was insufficient information to make a final correction. However, the assessment of the record leads to the assumption that it is "highly unlikely" that this address is deliverable.

22.3.4 Global Address Cleanse status codes

Global Address Cleanse status codes represent the corrections made to the address during processing.

i Note

Reference to "country" or "countries" in this topic refers to a country, territory, or geographical area, like Antarctica, based on the context.

Status codes appear in the [STATUS_CODE](#) output field. They're five or six characters based on the engine used for processing.

Each position in the status code represents a specific category for which the code applies.

Position categories

Position	Category
First character	S for Status
Second character	Lastline corrections
Third character	Address line corrections
Fourth character	Secondary address line correction
Fifth character	Other primary or secondary address component changes (components that aren't considered basic address components)
Sixth character	Additional information about a record that isn't related to an address change

Second character

The value of the second character depends on corrections to the country, postcode, region, or locality.

Value	Description
0	No significant difference between the input data and the corrected data.
1	Corrected country.
2	Corrected postal code.

Value	Description
3	Corrected country and postal code.
4	Corrected region.
5	Corrected country and region.
6	Corrected postal code and region.
7	Corrected country, postal code, and region.
8	Corrected locality.
9	Corrected country and locality.
A	Corrected postal code and locality.
B	Corrected country, postal code, and locality.
C	Corrected region and locality.
D	Corrected country, region, and locality.
E	Corrected postal code, region, and locality.
F	Corrected country, postal code, region, and locality.

Third character

The value of the third character depends on corrections to the directionals, primary type, primary name, and primary range.

Value	Description
0	No significant difference between the input data and the corrected data.
1	Corrected directional.
2	Corrected primary type.
3	Corrected directional and primary type.
4	Corrected primary name.
5	Corrected directional and primary name.
6	Corrected primary type and primary name.
7	Corrected directional, primary type, and primary name.
8	Corrected primary range.
9	Corrected directional and primary range.
A	Corrected primary type and primary range.
B	Corrected directional, primary type, and primary range.
C	Corrected primary name and primary range.
D	Corrected directional, primary name, and primary range.

Value	Description
E	Corrected primary type, primary name, and primary range.
F	Corrected directional, primary type, primary name, and primary range.

Fourth character

The value of the fourth character depends on corrections to the unit description, unit number, firm, building name, floor description, floor number, stairwell description, stairwell name, wing description, and wing name.

Value	Description
0	No significant difference between the input data and the corrected data.
1	Corrected one or more secondary address components (unit description, floor description, stairwell description, or wing description).
2	Corrected one or more secondary address components (unit number, floor number, stairwell name, or wing name).
3	Corrected one or more secondary address components (unit description, unit number, floor description, floor number, stairwell description, stairwell name, wing description, or wing name).
4	Corrected building name.
5	Corrected one or more secondary address components (unit description, floor description, stairwell description, or wing description), and building name.
6	Corrected one or more secondary address components (unit number, floor number, stairwell name, or wing name), and building name.
7	Corrected one or more secondary address components (unit description, unit number, floor description, floor number, stairwell description, stairwell name, wing description, or wing name), and building name.
8	Corrected firm.
9	Corrected one or more secondary address components (unit description, floor description, stairwell description, or wing description), and firm.
A	Corrected one or more secondary address components (unit number, floor number, stairwell name, or wing name), and firm.

Value	Description
B	Corrected one or more secondary address components (unit description, unit number, floor description, floor number, stairwell description, stairwell name, wing description, or wing name), and firm.
C	Corrected building name and firm.
D	Corrected one or more secondary address components (unit description, floor description, stairwell description, or wing description), building name, and firm.
E	Corrected one or more secondary address components (unit number, floor number, stairwell name, or wing name), building name, and firm.
F	Corrected one or more secondary address components (unit description, unit number, floor description, floor number, stairwell description, stairwell name, wing description, or wing name), building name, and firm.

Fifth character

The value of the fifth character depends on changes to components that aren't considered basic address components, such as other primary address and secondary address information.

Other primary address components:

- [PRIMARY_DELIVERY_MODE](#)
- [PRIMARY_DELIVERY_NUMBER](#)

Other secondary address components:

- [DELIVERY_INSTALLATION_NAME](#)
- [DELIVERY_INSTALLATION_QUALIFIER](#)
- [DELIVERY_INSTALLATION_TYPE](#)

Value	Description
0	No significant change between the input data and the corrected data.
1	Changed the other primary address components.
2	Changed the other secondary address components.
3	Changed the other primary address and other secondary address components.
4	Changed the point of reference.
5	Changed the other primary address components and the point of reference.

Value	Description
6	Changed the other secondary address components and the point of reference.
7	Changed the other primary address, other secondary address components, and the point of reference.

Sixth character

The value of the sixth character indicates additional information about a record that isn't related to a change in the address.

Value	Description
A	Alias record used for assignment. Global Address engine.
B	Base record assignment. Global Address engine (New Zealand). A Bordering Locality. Global Address engine (Australia).
C	An Alias and a Bordering locality. Global Address engine (Australia).
D	Deleted record. Global Address engine (Austria and Germany).
E	An excluded postal code was found. Canada engine.
I	Record ignored. Global Address engine (New Zealand).
L	Large Volume Receiver (LVR) and Large Organization Postal Code. Global Address engine (Brazil and Japan)
P	A software-generated default locality level postcode was assigned. Global Address engine (China).
Q	An alias and a software generated default locality level postcode was assigned. Global Address engine (China).
U	Unique address. Global Address engine (New Zealand).

22.4 USA Regulatory Address Cleanse fault codes

Fault codes describe why the USA Regulatory Address Cleanse transform could not assign an address.

The transform outputs fault codes to the `FAULT_CODE` output field.

Fault Code	Description
E101	Last line is bad or missing.
E212	No locality and bad postal code.
E213	Bad locality, valid region, and no postal code.
E214	Bad locality and bad postal code.
E216	Bad postal code and can't determine which locality match to select.
E302	No primary address line parsed.
E412	Primary name not found in directory.
E413	Possible primary name matches are too close to choose one.
E420	Primary range is missing.
E421	Primary range is invalid for the street/route/building.
E422	Primary prefix needed, input is wrong or missing.
E423	Primary type needed, input is wrong or missing.
E425	Primary type and directional needed, input is wrong or missing.
E427	Primary postfix needed, input is wrong or missing.
E428	Bad postal code, can't select an address match.
E429	Bad locality, can't select an address match.
E430	Possible address-line matches too close to choose one.
E431	Locality2 needed, input is wrong or missing.
E439	Exact match made in EWS directory.
E500	Other error.
E501	Foreign address.
E502	Input record entirely blank.
E503	Postal code not in area covered by partial USPS directory.
E504	Overlapping ranges in USPS directory.
E505	Address does not exist in the USPS directories. Undeliverable address.
E600	Marked by USPS as unsuitable for delivery of mail.
E601	The primary address number did not DPV confirm, and the Postcode2 was removed.

Parent topic: [Data Quality codes \[page 829\]](#)

Related Information

[Data Cleanse codes \[page 829\]](#)

[Geocoder codes \[page 844\]](#)

[Global Address Cleanse codes \[page 849\]](#)

[USA Regulatory Address Cleanse status codes \[page 871\]](#)

22.5 USA Regulatory Address Cleanse status codes

The USA Regulatory Address Cleanse transform assigns status codes to assigned addresses to indicate how the input address differs from the assigned address.

The transform outputs status codes to the [STATUS_CODE](#) output field.

Digit	Description
1st	<i>A</i>: The transform truncated the ADDRESS_LINE to make it fit your field. <i>B</i>: The transform truncated both the ADDRESS_LINE and the LOCALITY1_NAME. <i>C</i>: The transform truncated the LOCALITY1_NAME to make it fit your field. <i>S</i>: No truncation occurred.

Digit	Description
2nd	0: Regarding the <i>LOCALITY1_NAME</i>, <i>REGION1</i>, <i>POSTCODE1</i>, and <i>POSTCODE2</i>, there is no significant difference between the input data and the data the transform assigned. 1: The transform assigned a different <i>POSTCODE1</i>. 2: The transform assigned a different <i>LOCALITY1_NAME</i>. 3: The transform assigned a different <i>LOCALITY1_NAME</i> and <i>POSTCODE1</i>. 4: The transform assigned a different <i>REGION1</i>. 5: The transform assigned a different <i>REGION1</i> and <i>POSTCODE1</i>. 6: The transform assigned a different <i>LOCALITY1_NAME</i> and <i>REGION1</i>. 7: The transform assigned a different <i>LOCALITY1_NAME</i>, <i>REGION1</i>, and <i>POSTCODE1</i>. 8: The transform assigned a different <i>POSTCODE2</i>. 9: The transform assigned a different <i>POSTCODE1</i> and <i>POSTCODE2</i>. A: The transform assigned a different <i>LOCALITY1_NAME</i> and <i>POSTCODE2</i>. B: The transform assigned a different <i>LOCALITY1_NAME</i>, <i>POSTCODE1</i>, and <i>POSTCODE2</i>. C: The transform assigned a different <i>REGION1</i> and <i>POSTCODE2</i>. D: The transform assigned a different <i>REGION1</i>, <i>POSTCODE1</i>, and <i>POSTCODE2</i>. E: The transform assigned a different <i>LOCALITY1_NAME</i>, <i>REGION1</i>, and <i>POSTCODE2</i>. F: The transform assigned a different <i>LOCALITY1_NAME</i>, <i>REGION1</i>, <i>POSTCODE1</i>, and <i>POSTCODE2</i>.

Digit	Description
3rd	<i>0</i>: Regarding the <i>PRIMARY_NAME</i>, <i>PRIMARY_PREFIX</i> or <i>PRIMARY_POSTFIX</i>, and <i>PRIMARY_TYPE</i>, there is no significant difference between the input and what the transform assigned. <i>1</i>: The transform assigned a different <i>PRIMARY_TYPE</i>. <i>2</i>: The transform assigned a different <i>PRIMARY_PREFIX</i>. <i>3</i>: The transform assigned a different <i>PRIMARY_TYPE</i> and <i>PRIMARY_PREFIX</i>. <i>4</i>: The transform assigned a different <i>PRIMARY_POSTFIX</i>. <i>5</i>: The transform assigned a different <i>PRIMARY_TYPE</i> and <i>PRIMARY_POSTFIX</i>. <i>6</i>: The transform assigned a different <i>PRIMARY_PREFIX</i> and <i>PRIMARY_POSTFIX</i>. <i>7</i>: The transform assigned a different <i>PRIMARY_PREFIX</i>, <i>PRIMARY_TYPE</i>, and <i>PRIMARY_POSTFIX</i>. <i>8</i>: The transform assigned a different <i>PRIMARY_NAME</i>. <i>9</i>: The transform assigned a different <i>PRIMARY_NAME</i> and <i>PRIMARY_TYPE</i>. <i>A</i>: The transform assigned a different <i>PRIMARY_PREFIX</i> and <i>PRIMARY_NAME</i>. <i>B</i>: The transform assigned a different <i>PRIMARY_PREFIX</i>, <i>PRIMARY_NAME</i>, and <i>PRIMARY_TYPE</i>. <i>C</i>: The transform assigned a different <i>PRIMARY_NAME</i> and <i>PRIMARY_POSTFIX</i>. <i>D</i>: The transform assigned a different <i>PRIMARY_NAME</i>, <i>PRIMARY_TYPE</i>, and <i>PRIMARY_POSTFIX</i>. <i>E</i>: The transform assigned a different <i>PRIMARY_PREFIX</i>, <i>PRIMARY_NAME</i>, and <i>PRIMARY_POSTFIX</i>. <i>F</i>: The transform assigned a different <i>PRIMARY_PREFIX</i>, <i>PRIMARY_NAME</i>, <i>PRIMARY_POSTFIX</i>, and <i>PRIMARY_TYPE</i>.

Digit	Description
4th	<i>0</i>: Regarding the <i>COUNTY_NUMBER</i>, <i>SORTCODE_ROUTE</i>, <i>DELIVERY_POINT</i>, and <i>UNIT_DESCRIPTION</i>, there is no significant difference between the input data and the data the transform assigned. <i>1</i>: The transform assigned a different <i>UNIT_DESCRIPTION</i>. <i>2</i>: The transform assigned a different <i>DELIVERY_POINT</i>. <i>3</i>: The transform assigned a different <i>DELIVERY_POINT</i> and <i>UNIT_DESCRIPTION</i>. <i>4</i>: The transform assigned a different <i>SORTCODE_ROUTE</i>. <i>5</i>: The transform assigned a different <i>SORTCODE_ROUTE</i> and <i>UNIT_DESCRIPTION</i>. <i>6</i>: The transform assigned a different <i>SORTCODE_ROUTE</i> and <i>DELIVERY_POINT</i>. <i>7</i>: The transform assigned a different <i>SORTCODE_ROUTE</i>, <i>DELIVERY_POINT</i>, and <i>UNIT_DESCRIPTION</i>. <i>8</i>: The transform assigned a different <i>COUNTY_NUMBER</i>. <i>9</i>: The transform assigned a different <i>COUNTY_NUMBER</i> and <i>UNIT_DESCRIPTION</i>. <i>A</i>: The transform assigned a different <i>COUNTY_NUMBER</i> and <i>DELIVERY_POINT</i>. <i>B</i>: The transform assigned a different <i>COUNTY_NUMBER</i>, <i>DELIVERY_POINT</i>, and <i>UNIT_DESCRIPTION</i>. <i>C</i>: The transform assigned a different <i>COUNTY_NUMBER</i> and <i>SORTCODE_ROUTE</i>. <i>D</i>: The transform assigned a different <i>COUNTY_NUMBER</i>, <i>SORTCODE_ROUTE</i>, and <i>UNIT_DESCRIPTION</i>. <i>E</i>: The transform assigned a different <i>COUNTY_NUMBER</i>, <i>SORTCODE_ROUTE</i>, and <i>DELIVERY_POINT</i>. <i>F</i>: The transform assigned a different <i>COUNTY_NUMBER</i>, <i>SORTCODE_ROUTE</i>, <i>DELIVERY_POINT</i>, and <i>UNIT_DESCRIPTION</i>.

Digit	Description
5th	0: Regarding the <i>LOT</i>, <i>LOT_ORDER</i>, and <i>LOCALITY2_OFFICIAL</i>, there is no significant difference between the input data and the data the transform assigned. 1: The transform assigned a different <i>LOT</i>. 2: The transform assigned a different <i>LOT_ORDER</i>. 3: The transform assigned a different <i>LOT</i> and <i>LOT_ORDER</i>. 4: The transform assigned a different <i>LOCALITY2_OFFICIAL</i>. 5: The transform assigned a different <i>LOCALITY2_OFFICIAL</i> and <i>LOT</i>. 6: The transform assigned a different <i>LOCALITY2_OFFICIAL</i> and <i>LOT_ORDER</i>. 7: The transform assigned a different <i>LOCALITY2_OFFICIAL</i>, <i>LOT</i>, and <i>LOT_ORDER</i>.
6th	Always outputs a zero (0).

Parent topic: [Data Quality codes \[page 829\]](#)

Related Information

[Data Cleanse codes \[page 829\]](#)

[Geocoder codes \[page 844\]](#)

[Global Address Cleanse codes \[page 849\]](#)

[USA Regulatory Address Cleanse fault codes \[page 869\]](#)

23 About ShowA and ShowL for USA addresses

Use the ShowA and ShowL utilities to question the postal directories about address assignment.

Find answers to questions such as the following:

- Why did the transform standardized the address in an unexpected way?
- Why didn't the transform assign the address?
- Why did the transform error code indicate a flaw in the directory?

Use ShowA to display or output information from the Address_1_Directory. Use ShowL to query the City_Directory and the Post_Code_Directory.

Run the ShowA and ShowL utilities from a DOS command line using specific command-line options. These options are listed when you enter the following command:

Windows USA ShowA: `showa /op`

UNIX USA ShowA: `showa -op`

Note

For addresses in Canada, use `cashowa`.

The Show configuration files

Each Show utility has a configuration file. The configuration files contain parameters for controlling how the utility works. The following tables contain the configuration file names and locations for each utility.

For USA addresses

Utility	Executable	File name	Location
ShowA	<code>showa.exe</code>	<code>showa.cfg</code>	<code><LINK_DIR>\dataquality\urac</code>
ShowL	<code>showl.exe</code>	<code>showl.cfg</code>	<code><LINK_DIR>\dataquality\urac</code>

For Canada addresses

Utility	Executable	File name	Location
ShowA	<code>cashowa.exe</code>	<code>cashowa.cfg</code>	<code>LINK_DIR\dataquality\gac</code>
ShowL	<code>cashowl.exe</code>	<code>cashowl.cfg</code>	<code>LINK_DIR\dataquality\gac</code>

Before you run the Show utilities, set both configuration files for the appropriate country directory. The configuration files contain instructions and detailed information about how to run the programs.

i Note

Run the ShowA and ShowL utilities in the same directory as the ShowA and ShowL configuration files. The utilities do not run when they are located outside of the default location.

[ShowA and ShowL query modes \[page 877\]](#)

Enter queries in the Show utilities using one of two modes: Prompts and commands.

[USA ShowA command-line options \[page 878\]](#)

Use commands to control how the ShowA utility behaves.

[USA ShowL command line options \[page 880\]](#)

Use commands to control how the ShowL utility behaves.

[Canada ShowA command line options \[page 881\]](#)

The ShowA utility has commands for queries about Canada addresses.

[Canada ShowL command line options \[page 882\]](#)

23.1 ShowA and ShowL query modes

Enter queries in the Show utilities using one of two modes: Prompts and commands.

Prompts

If you type the ShowA or ShowL command without any commands, the utility prompts you to enter your query data. The utility takes all other information and options from the configuration file.

After your first query, the utility prompts you to enter the next query. When you are finished entering queries, exit the utility by typing "Quit".

Commands

When you use commands in the ShowA and ShowL utilities, the utility uses information from the configuration file. For example, the configuration file contains information about the auxiliary files, the output file, display options, and search options. To override any value taken from the configuration file, use command-line options selectively. Depend on the configuration file for any other overrides. The only values that you cannot specify through a command line is the optional output fields.

i Note

If the utility detects any command, it determines that you are operating in command-line mode. Therefore, the utility does not prompt you to enter query data as it does when you use the prompts mode.

Parent topic: [About ShowA and ShowL for USA addresses \[page 876\]](#)

Related Information

[USA ShowA command-line options \[page 878\]](#)

[USA ShowL command line options \[page 880\]](#)

[Canada ShowA command line options \[page 881\]](#)

[Canada ShowL command line options \[page 882\]](#)

23.2 USA ShowA command-line options

Use commands to control how the ShowA utility behaves.

Use the following command to view a summary of command-line options in the utility:

Windows: `showa /op`

UNIX: `showa -op`

The following table contains the commands and descriptions for both Windows and UNIX.

UNIX	Windows	Description
-a	/a	Appends information to the output file (when it exists).
-alias	/alias	Includes preferred alias address lines.
-d	/d	Displays your query data on screen.
-fin	/fin	Expands the query to cover USPS finance area.
-op	/op	Displays the list of commands.
-p	/p	Pauses screen display every 22 lines.
-2:dpbc	/2:dpbc	Enter the DPBC code for dpbc .
-4:zip4	/4:zip4	Enter the postcode2 for zip4 .
-ad:file	/ad:file	Enter the Address-line dictionary addr ln. dct) and path name for file .
-c:cart	/c:cart	Enter the carrier route number for cart .

UNIX	Windows	Description
-f:file	/f:file	Enter the file path and name of the output file for file . The output file stores the information from the query instead of displaying it on screen.
-nd:file	/nd:file	Enter the National ZIP+4 directory path and name, zip4us.dir for file .
-pre:dir	/pre:dir	Enter the primary prefix, such as N, NE, E, SE, S, SW, W, NW, for dir .
-pos:dir	/pos:dir	Enter the primary postfix, such as N, NE, E, SE, S, SW, W, NW, for dir .
-s:street	/s:street	Enter the street primary name using quotes if there are multiple words, for street .
-sfx:suffix	/sfx:suffix	Enter the primary type, such as Ave, Blvd, St, Rd, for suffix .
-sh:range	/sh:range	Enter the street primary range high for range .
-sl:range	/sl:range	Enter the street primary range low or exact for range .
-t:type	/t:type	Enter the file type dBASE3, ASCII, or DELIMITED, for type .
-u:urb idx	/u:urb idx	Enter the urbanization Index for urb idx .
-z:lo-hi	/z:lo-hi	Enter the low and high range for Postcode1 for lo-hi .
-z:zip	/z:zip	Enter the Postcode1 for zip .

Parent topic: [About ShowA and ShowL for USA addresses \[page 876\]](#)

Related Information

[ShowA and ShowL query modes \[page 877\]](#)

[USA ShowL command line options \[page 880\]](#)

[Canada ShowA command line options \[page 881\]](#)

[Canada ShowL command line options \[page 882\]](#)

23.3 USA ShowL command line options

Use commands to control how the ShowL utility behaves.

Use the following command to view a summary of command line options in the utility:

Windows: `showl /op`

UNIX: `showl -op`

The following table contains the commands and descriptions for both Windows and UNIX.

UNIX	Windows	Description
-a	/a	Appends query information to an output file when it exists.
-d	/d	Displays query data on screen.
-op	/op	Displays the list of commands.
-p	/p	Pauses screen display every 22 lines.
-ab:query	/ab:query	Enter the abbreviated Locality1 for query .
-cd:file	/cd:file	Enter the City directory path and name (city04.dir) for file .
-cn:city	/cn:city	Enter Locality1 name, using quotes if there are multiple words, for city .
-dr:dir	/dr:dir	Enter the directory to search, City or ZCF, for dir .
-f:file	/f:file	Enter the output file path and name for file .
-ml:query	/ml:query	Enter the military Postcode1 for query .
-mz:query	/mz:query	Enter the multi-zone Locality1 for query .
-pn:query	/pn:query	Enter a place name for query .
-st:state	/st:state	Enter Region1 for state . For U.S., use USPS abbreviations or full state names.
-t:type	/t:type	Enter the output file type, dBASE3, ASCII, or DELIMITED, for type .
-un:query	/un:query	Enter the unique Postcode1 for query .
-z:zip	/z:zip	Enter the Postcode1 for zip .
-zd:file	/zd:file	Enter the ZCF directory path and name (zcf04.dir) for file .
-z:lo-hi	/z:lo-hi	Enter the Postcode1 range for lo-hi .

Parent topic: [About ShowA and ShowL for USA addresses \[page 876\]](#)

Related Information

[ShowA and ShowL query modes \[page 877\]](#)

[USA ShowA command-line options \[page 878\]](#)

[Canada ShowA command line options \[page 881\]](#)

23.4 Canada ShowA command line options

The ShowA utility has commands for queries about Canada addresses.

To view a summary of command line options, use this command:

Windows: **cashowa /op**

UNIX: **cashowa -op**

In addition to the commands listed for USA ShowA, there are commands specific for addresses in Canada.

Canada-specific commands

UNIX	Windows	Description
-accent	/accent	Displays French accented data.
-adct:file	/adct:file	Enter the Address line dictionary path and name (addrlnca.dct) for file .
-cdct:file	/cdct:file	Enter the Casing dictionary path and name (pwcasca.dct) for file .
-cdir:file	/cdir:file	Enter the Canada city directory path and name (cancity.dir) for file .
-ci:index	/ci:index	Enter the City index for index .
-di:index	/di:index	Enter the directory area index for index .
-f:file	/f:file	Enter the output file path and name for file .
-fdir:file	/fdir:file	Enter the Canada FSA directory path and name (canfsa.dir) for file .
-ldct:file	/ldct:file	Enter the Last Line Dictionary path and name (lastlnca.dct) for file .
-ndir:file	/ndir:file	Enter the Canada National directory path and name (canada.dir) for file .
-pc:pc	/pc:pc	Enter the postal code for pc .
-pdir:file	/pdir:file	Enter the Canada PCI directory path and name (canpci.dir) for file .

Parent topic: [About ShowA and ShowL for USA addresses \[page 876\]](#)

Related Information

[ShowA and ShowL query modes \[page 877\]](#)

[USA ShowA command-line options \[page 878\]](#)

[USA ShowL command line options \[page 880\]](#)

[Canada ShowL command line options \[page 882\]](#)

23.5 Canada ShowL command line options

To view a summary of command line options in the DOS window, use this command:

Windows: **cashowl /op**

UNIX: **cashowl -op**

The following table lists the command line options and command descriptions.

UNIX	Windows	Description
-a	/a	Appends query information to an output file (if it already exists).
-accent	/accent	Displays French accented data.
-d	/d	Displays query data on screen.
-op	/op	Displays this list of options.
-p	/p	Pauses screen display every 22 lines.
-aci:index	/aci:index	Enter the Alternate city index for index .
-adct:file	/adct:file	Enter the Address-line dictionary path and name (addrlnca.dct) for file .
-cdct:file	/cdct:file	Enter the Casing dictionary path and name (pwcasca.dct) for file .
-cdir:file	/cdir:file	Enter the Canada city directory path and name (cancity.dir) for file .
-ci:index	/ci:index	Enter the City index for index .
-cn:city	/cn:city	Enter region1 name (in quotes if multiple words) for city .
-di:index	/di:index	Enter the Directory area index for index .

UNIX	Windows	Description
-dr:dir	/dr:dir	Enter which directory to search, City or ZCF for dir .
-f:file	/f:file	Enter the output file path and name for file .
-fdir:file	/fdir:file	Enter the Canada FS directory path and name (canfsa.dir) for file .
-fsa:fsa	/fsa:fsa	Enter the first part of the postal code for fsa .
-ldct:file	/ldct:file	Enter the Last line dictionary path and name (lastlnca.dct) for file .
-ndr:file	/ndr:file	Enter the Canada National directory path and name (canada.dir) for file .
-pc:pc	/pc:pc	Enter the postcode1 for pc
-pdr:file	/pdr:file	Enter the Canada PCI directory path and name (canpci.dir) for file .
-pr:prov	/pr:prov	Enter the province (use two letter abbreviations only) for prov .
-t:type	/t:type	Enter the output file type (dBASE3, ASCII, or DELIMITED) for type .

Parent topic: [About ShowA and ShowL for USA addresses \[page 876\]](#)

Related Information

[ShowA and ShowL query modes \[page 877\]](#)

[USA ShowA command-line options \[page 878\]](#)

[USA ShowL command line options \[page 880\]](#)

[Canada ShowA command line options \[page 881\]](#)

24 Glossary

Address Cleanse	Transforms that produce a correct and complete standardized form of an input address. The transform can also assign codes for postal automation and append other useful address information.
address line	A line of data in an address that contains the primary and, possibly, secondary address. The primary address contains components such as the primary range, primary name, directionals (post- and pre-), and the suffix. The secondary address normally contains components such as the unit designator and the secondary range.
aggregated data	Data that results when a process combines elements. This data can be presented collectively or in summary form.
alias	Alternate form or name. • Aliases are alternate forms that could potentially be matched to the word. For example, Robert is a personal name alias for Bob. Alias data is output in the Match_Std fields. • In the Address Cleanse transforms, an alias is an alternative form of a primary address line. Aliases apply only to primary addresses (usually streets), not secondary addresses or last lines.
AMAS	Australia Post's Address Matching Approval System (AMAS). To receive postal discounts in Australia, you are required to file an AMAS report.
application	Another term for a software program.
attribute	A property created for a type of object.
Boolean expression	An expression that defines a logical relationship between two or more items. The expression is either TRUE or FALSE.
business rules	Settings within your Data Quality transforms that explain how you want to process your data. These include things like telling the Global Address Cleanse transform how to case output data, or setting up match criteria for a matching process.
case-sensitive	Pertaining to the differentiation between upper-case and lower-case letters. A case-sensitive program differentiates between upper-case and lower-case letters when evaluating a text string.
CASS	A United States Postal Service (USPS) certification that requires software vendors to go through a series of tests to prove that their software correctly codes addresses according to USPS requirements, and produces the required USPS reports. Long form: Coding Accuracy Support System
classifications	Indicators to Data Cleanse of the types of situations that apply to this word. For example, Hewlett is assigned the Firm_Name and Name_Weak_Family_Name classifications, because it can be used in both firm and personal names.
command	A directive given to a program to initiate an action.
constant	A data string that does not change from one record to the next.

contribution value	A value you assign to a match criteria that represents the importance (or weight) you place on that criteria's data. For example, your organization may place a high degree of importance on the customer number. For these types of criteria you would assign a higher contribution value to reflect a higher importance. The contribution value is part of weighted scoring.
Data Cleanse	A transform that identifies and isolates specific parts of mixed data, and then standardizes the data based on information stored in the parsing dictionary, business rules defined in the rule file, and expressions defined in the pattern file.
data record	A row of data that is constructed at runtime. The data remains in the form of the data record throughout the job.
data salvage	The process of temporarily copying data from a passenger record to the driver record after the two records are compared. The data that's copied is data that is found in the passenger record, but is missing or incomplete (initials, for example) in the driver record. Data salvaging prevents blank matching or initials matching from matching records that you may not want to match.
Data Services	A software system that allows users to build and execute applications with which they can create and maintain data warehouses. You can use Data Services to configure your Data Quality transforms and export the XML.
delivery point code	A two-digit number derived from the primary range (house number). This number is used in the generation of a DPBC barcode.
Delivery Point Validation (DPV)	A technology that assists you in validating the accuracy of your address information with the USA Regulatory Address Cleanse transform. With DPV, you can identify addresses that are undeliverable as addressed and determine whether or not an address is a Commercial Mail Receiving Agency (CMRA).
destination record	A location where you place your updated or "best" data when creating a best record. A destination record can be either a master record, a subordinate record, or both in a match group.
diacritical character	A character that contains an accent, dieresis (umlaut), tilde, cedilla, or other distinguishing marks (for example, ä or Ç). You can choose to have standardized data with these types of characters. The application uses the Latin-1 code page for assigning these accents.
dictionary	Relational database that contains a lexicon of words and phrases that the data cleansing package and the Data Cleanse transform use to identify, parse, and standardize data.
directional	A component of the address line that indicates direction. For example, North in "211 N. 115th St."
discrete field	Input or output data that has separate fields for each piece of information, such as addresses and names.
discrete format	Input source format in which pieces of data are parsed down to nearly the most distinct level. For example, a "first name" field would be discrete, whereas a "name" field that could contain first, middle, or last name information would not be discrete.
DPBC (Delivery Point Barcode)	A form of Postnet barcode, consisting of 62 bars and based on the combination of ZIP Code, ZIP+4, DPBC, and a check digit.

driver record	A record that drives the comparison process. Driver records are part of a break group and are compared with passenger records to determine matches. The driver record is the first record in the break group.
dual address	A dual address occurs when a record contains two address lines. Two combinations are typical: - PO box and street address: 1000 Main Street, Suite 51 PO Box 2342 - Rural route or Highway Contract and street address: RR 1 Box 345 12784 Old Columbus Road
dual names	Two names included on an address line, for example, John and Jane Doe.
Early Warning System (EWS)	A solution for matching valid delivery points that have been created between updates to the national ZIP+4 directory. EWS uses four months of rolling data found in an intermediate directory that is updated weekly with data from the USPS.
eLOT	Enhanced Line of Travel (eLOT) takes Line of Travel one step further in the presorting process. The original line of travel (LOT) narrowed down the mail carrier's delivery route to the block face level (ZIP+4 level) by discerning whether an address resided on the odd or even side of a street or thoroughfare. eLOT narrows the mail carrier's delivery route walk sequence to the house (delivery point) level. This allows you to sort your mailings to a more precise level.
fault code	A numeric value that is assigned to a record after the USA Regulatory Address Cleanse transform validation process that signifies that the particular record was not successfully validated. Each numeric value represent a different type of fault.
FSA (Forward Sortation Area)	The first three characters of a Canadian alphanumeric postal code. For example, K1A in the postal code for Canada Post's Ottawa headquarters, K1A 0B1.
gathering	Recombines terms that belong together, such as alphanumeric terms that you would look up together in the dictionary. For example, if Data Cleanse breaks 1st into "1" and "st", then gathering recombines them to 1st.
gender	A code that indicates the likelihood of a record being a certain gender. This code is derived from the name and has five possible values: strong male, strong female, weak male, weak female, ambiguous, and unassigned. For example, a record marked as "strong male" indicates a high likelihood that the person is male.
generated field	A field that is generated on output by a transform. For example, a postcode field generated by the Global Address Cleanse transform.
GeoCensus	A directory that contains latitude, longitude, census tract, and block information. That information sets the stage for mapping, demographic marketing, and other applications of your address data.

hybrid format	A format for records in which some fields are discrete, whereas others are in a multiline format.
intersource match	Match between records of different sources.
intrasource match	Match between records within a source.
key	A value used to identify a record in a database.
lastline	The lastline of an address contains components such as the locality, region, and postcode (and it may contain the country name).
line of travel (LOT)	A sorting sequence in which ZIP+4 codes are arranged in the order that they are served by the mail carrier. LOT sequencing is required for some bulk mailing discounts.
locality	A part of the address line of a record. Locality most often refers to the city or town. In some countries, such as the United Kingdom, locality can extend to include district.
Local Delivery Unit (LDU)	The last three characters of a Canadian alphanumeric postal code. For example, OB1 in the postal code for Canada Post's Ottawa headquarters, K1A 0B1.
Locatable Address Conversion System (LACS)	A database of addresses that have been permanently converted, usually due to 911 emergency system implementation. The changes often consist of conversion from rural-style addressing to standardized, city-style addressing, or renumbering of existing city-style addresses.
mail piece unit	Typically referred to as a version identifier for printers, it represents the unique characteristics of a portion of a mailing. Every segment within a Mail.dat must have at least one mail piece unit.
mapped field	A field in a specific transform, for which it has been defined which field it should read from upstream transforms.
master record	The first record in a match group. You can control which record is the master record by using the Group Prioritization operation in the Match transform.
match criteria	A group of options that determine the rules for matching on particular data.
match group	A group of records found to be matching with each other. A match group consists of a master record and subordinate records.
matching record	A group of records found to be matches based on the criteria and business rules you choose. The records do not necessarily have the same data.
match level	A Match level designates the level in "hierarchically" type matching. One Match set can have one or more match levels. Duplicates that are found at one level are passed to the next level, where they are compared based on that level's keys, and so on. For example, you could use multiple match levels if you wanted to detect duplicates at the household (residence), family, and individual level. The order of the match levels is important because duplicates are found at each level, and only the results are made available for the next level. Usually, you will define your "broadest" match levels first, followed by more specific match levels.
match set	A group of criteria used to perform matching on your data.

A typical setup might have only select data reaching each match set for comparison. For example, you might want to exclude blank SSNs (Social Security Numbers), certain foreign addresses, and so on from reaching a particular match set.

metadata	In the software, information acquired and maintained to describe tables in source and target databases. This information includes the names of tables and their columns, and the data types of the columns. In general, metadata typically includes a description of data models, a description of the layouts used in database design, the definition of the system of record, the mapping of data from the system of record to other places in the environment, and specific database design definitions.
multiline	The multiline format is a database record format in which address data is not consistently located in the same arrangement in all records. That is, data items “float” among fields. For example, an input source may have fields named Line1, Line2, Line3, and Line4 that contain various categories of name and address data, as well as non-address data.
multi-source	Records that appear on two or more sources. For example, let’s say you’re bringing together customer sources from several direct marketers or publishers. Your best prospects may be the people whose names appear on two or more sources, indicating they may be most receptive to your offer.
normal source	A source of records that the application should consider to be good, eligible records in a matching process.
North American Numbering Plan (NANP)	Telephone numbering plan shared by 19 North American countries. These countries include the United States and territories, Canada, Bermuda, Anguilla, Antigua & Barbuda, the Bahamas, Barbados, the British Virgin Islands, the Cayman Islands, Dominica, the Dominican Republic, Grenada, Jamaica, Montserrat, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Trinidad and Tobago, and Turks & Caicos.
null	The absence of a value within a database field for a given record. It does not mean zero because zero is a value.
option	Business rules that can be set for a Data Quality transform that specify how you want to process your data. Each Data Quality transform has a different set of available options.
option group	Contain a set of options that allow you to set different business rules for a transform.
other source	In a Match transform, a source of records that should be treated as transparent, such as seed sources. They are not counted in determining how to characterize a match group—for example, multi-source or single-source. For example, some mailers use a seed source of potential buyers who report back to the mailer when they receive a mail piece so that the mailer can measure delivery.
passenger record	The records that are compared against driver records in a break group. After a driver record has been compared with every passenger record in a break group, a passenger record can become the new driver record in the break group, or it can be found to be a match with a driver record. At this point it is taken out of the comparison process.
pattern file	User-defined patterns are stored in a pattern file. The pattern file is a plain text file and can be edited in any text editing program. The pattern file is used by the Data Cleanse transform.

PMB (Private mail box)	Private mail boxes are like post-office boxes but they are hosted by private companies. The USA Regulatory Address Cleanse and the Global Address Cleanse transforms can recognize certain forms of PMB data when it appears in an address line.
postal address	A delivery address that is a rural route or box number.
postal code	A system of letters and/or digits used for sorting mail. Examples include the ZIP Code used in the United States and the alphanumeric FSA LDU system used in Canada.
Postcode2	The secondary part of a postal code. For example, in the United States, a postcode is composed of two parts (54601-4051). The first five digits are followed by a hyphen and a four-digit code. The four-digit code is the Postcode2 for a US postcode.
postcode move	A valid postcode that has been split or moved, so only a portion of the area that had been covered by the one postcode now has two or more postcodes, including the original one, for the same area.
primary entry	A word or phrase in the dictionary that the data cleansing package and Data Cleanse transform use to identify parse, and standardize data.
primary key	A column that is guaranteed to contain unique values, and whose values identify all of the rows in a table.
reference file	A file of address data used by Data Quality Management SDK to match, assign, standardize, and verify addresses. Reference files are also referred to as postal directories. These files have a .dir extension.
rule matching	Matches the token classifications against defined rules.
secondary information	Assists Data Cleanse in determining how to process the word when it is used in different ways. Secondary information can include how Data Cleanse will standardize the output data for the word or alternate forms that could potentially be matched to the word .
SERP	Canada Post Corporation's Software Evaluation and Recognition Program. Data Quality is certified under this program, allowing you to receive postage discounts for mailings to and within Canada.
similarity score	A percentage that indicates how much two fields or values are considered alike. This percentage is calculated by the application after the comparison process. For example, Ron and Rob are considered 67% alike because two of the three characters are alike. Similarity scores are used in a number of situations— not just in the Match transform. For example, they can be used to determine which suggestions to return for suggestion lists. The similarity score is not always a direct result of a one-to-one comparison; it can be altered by some options, such as those defined in the Match transform, for example.
snowbird	A casual term to describe someone who has multiple residences. This term is derived from individuals who reside in a cooler-climate region during the summer, and relocate to a home in a warmer-climate region during the winter.
source	For the Match transform, the grouping of records on the basis of some data characteristic that you can identify. A source might be all records from one input file, or all records that contain a particular value in a particular field. Sources are abstract and arbitrary—there is no physical boundary line between sources. Source membership can cut across data sources as well as distinguish among records within a data source, based on how you define the source.

source group	A group of sources that you can use to prepare a second set of match statistics, combining the statistics for two or more regular sources. For example, suppose you define five sources—two house sources and three rented sources. You would get match statistics for each individual source. But suppose that you also wanted a summary for the house sources and a summary for the rented sources. You could create two source groups—one for the house sources and one for the rented sources. Source groups affect only the way that match statistics are reported. They do not affect matching.
source record	The location where the data you want to use to update or create your best record with resides. A source record can be the master or subordinate record of a match group.
standards	Define how Data Cleanse will standardize capitalization or other output formatting on data.
street address	A delivery address that is the street name and house number.
subordinate record	Records that are part of a match group, and are found to be matches with (and subordinate to) a master record. Subordinate records can contain data that may be used to update a master record and, thus, create a best record.
suggestion lists	Normally, when an address cleansing transform looks up an address in the postal directories, it finds one matching record. Sometimes, due to incomplete information, there may be two or more records (or suggestions) in the postal directories that could possibly be the correct record. Suggestion lists provide you with a list of “matching” addresses, so that you can choose which is the best address.
suppression source	A source that contains records of information that should be excluded from other output destinations. The records in the suppression source are used for matching in other sources. The records that match the suppression source could then be removed from further processing. For example, suppression sources may be your own bad-account file or no-mail sources provided by the government or direct-marketing association (DMA) to prevent wasted mailings and offending consumers.
table	A database table that the software reads data from or loads data into. The path and mechanisms for reading and loading data and apportioning the data among rows and columns are defined in the datastore that the table is associated with. Writing a data set to a database table means sending a combination of rows with appropriate operation code to the database table.
territory	The locale value for a geographical location (usually the country) where a locale language is used. The paring of a language with a territory determines factors such as date format, time format, decimal separator, currency format, and so on.
tokenization	Assigns specific meanings to each of the pieces that result from word breaking. Data Cleanse looks up each individual input word in the dictionary. A list of tokens is created using the classifications associated with each word in the dictionary.
transform	A step in a Data Quality job that acts on a data set.
Unicode	A standard that was designed to create a universal character set. It accomplishes this by providing a unique number for every character in every language.

The Unicode Standard describes more than 50,000 characters, including all the characters of the common character sets in use when Unicode was established around 1990, as well as many that have been added since then. Unicode is an open character set, meaning it can continue to incorporate characters as needed.

Unicode can handle letters, punctuation, and technical symbols—regardless of platform, program, writing system, or language.

unique identifier	In a Data Quality transform, an ID that is unique to a record or group of matching records. It is sequential, static, and will not change when records are updated or re-processed through the application.
unique record	Records that do not have any matching or subordinate records and, therefore, do not belong to any match group after the matching process is complete.
weighted scoring	A method of comparison that provides you with a greater degree of control in the matching process. This method allows you to use contribution values to place more or less importance on various match criteria.
word breaking	Breaks the input line down into smaller, more usable pieces. By default, Data Cleanse breaks an input line on white space, punctuation, and alphanumeric transitions. Terms such as 20GB, 4G, 1st, and U2 each break into two tokens at the alphanumeric transition. For example, "20GB" breaks into "20" and "GB" tokens.
XML	Extensible Markup Language. This markup language is like HTML (Hypertext Markup Language) in that it specifies a standard with which you can define your own markup languages with their own sets of tags. XML allows you to define various tags with various rules, such as tags that represent business rules, tags that represent data description, or tags that represent data relationships.
XML Schema	The XML format used by Data Quality Management SDK to support message processing that includes Web Services. XML Schemas describe the data structure of an XML file or message. You can use the same XML Schema to describe multiple XML sources or targets. XML Schema properties include: Name, Description, Imported from, Root element name, and Namespace.
Z4Change	The Z4Change directory lists all the ZIP and ZIP+4 Codes in the country. A record in this file is tagged if it has changed within the last 12 months. The change might be a postal-code change (ZIP, ZIP+4, or CART), or even a change in the standardized form of the address-line or city name.
ZCF	The ZIP-City File directory that is used by the USA Regulatory Address Cleanse transform when processing data from the U.S.
ZIP+4	A nine-digit number, consisting of the ordinary ZIP Code and a four-digit, add-on code.
ZIP Code	ZIP is an acronym that stands for "Zone Improvement Plan." This is a 3-, 5-, or 9-digit number that represents a geographic region of the United States. The ZIP Code is important in determining entry eligibility and presort containerization. Note that this code is different from a facility code.
zone	The ZIP-City File directory that is used by the USA Regulatory Address Cleanse transform when processing data from the U.S.

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